

Janne's Guide to

Russian Ground Equipment

Field reference for recognition and unit association

119 systems: armoured vehicles, fires, air defence, radars, electronic warfare, engineers, command and signals, logistics and unmanned aircraft. What each looks like, which units hold it, how far it reaches.

1

ARMoured
VEHICLES

2

FIRES

3

AIR DEFENCE

4

RADARS

5

ELECTRONIC
WARFARE

6

ENGINEERS

7

COMMAND
AND SIGNALS

8

LOGISTICS

9

UNMANNED
AIRCRAFT

Contents

About this guide	4
------------------	---

1 Armoured vehicles 6

MBTs and tank support

T-90	7
T-80B/BV/BVM, T-80U	8
T-64A/B/BV	9
T-62	9
T-54 / T-55	10
BMPT Terminator	11

Tracked IFVs, APCs and carriers

BMP-1	11
BMP-2	12
BRM-1K	13
BMD-4M	13
BMD-2	14
BTR-MDM Rakushka	15
BTR-D	16
MT-LB	17
BMO-T	18
Aleut and Plastun articulated and light tracked carriers	19

Wheeled APCs and armoured cars

BTR-80	20
BTR-90 Rostok	21
BRDM-2	22
Tigr and Vodnik armoured cars	22
Typhoon family (Typhoon-K, Typhoon-U, Typhoon-VDV, Linza)	23
Other armoured cars and utility vehicles (family)	24

2 Fires 25

MRLs and heavy flamethrowers

BM-21 Grad 122 mm MRL	26
Tornado-G	26
BM-27 Uragan	27
Uragan-1M	27
BM-30 Smerch	28
TOS-1A Sointsepek	29

Self-propelled guns and mortars

2S5 Giatsint-S	29
2S19 Msta-S	30
2S35 Koalitsiya-SV	30
2S7 Plon / 2S7M Malka	31
2S4 Tyulpan	31

Towed artillery

2A36 Giatsint-B 152 mm towed gun	32
2A65 Msta-B	32

Ballistic missiles

Iskander-M	33
------------	----

3 Air defence 34

Long and medium-range SAMs

S-400 Triumf	35
S-300V4	36
S-350 Vityaz	36
Buk-M1 / Buk-M2 / Buk-M3	37

Short-range SAMs and guns

Tor	38
Pantsir-S1	39
Tunguska	40
Strela-10 and Osa short-range SAM	40

4 Radars 41

Air surveillance radars

Nebo-M	41
96L6 all-altitude acquisition radar	42
Counter-battery and ground surveillance radars	
Zoopark-1 / Zoopark-1M	42
ARK-1M Rys	43
1B75 Penicillin	43
SNAR-10 / SNAR-10M1 Leopard	44

5 Electronic warfare 45

Radar and strategic jamming

Murmansk-BN	45
Krasukha-2 / Krasukha-4	46
Moskva-1	47

Communications jamming

R-330B Mandat-B and R-934BMV	47
Borisoglebsk-2	48
Palantin	49
R-330Zh Zhitel	50
Leer-2	51
Leer-3	52

GNSS, drone and IED jamming

Pole-21	53
Repellent-1	53
Infaua	54

6 Engineers 55

Crossing

TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge	55
PMP heavy folding pontoon bridge park	56
PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry	57
IRM Zhuk	57

Obstacle clearing and earthmoving

IMR-2 / IMR-2M / IMR-3M combat engineer vehicle	58
BAT-2	58
UR-77 Meteorit	59
BMR-3M	59
Uran-6	60
Military earthmoving and trenching machines	60

Mine laying

ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer	61
UMZ	61

Recovery

BREM and BTS armoured recovery vehicles	62
---	----

CBRN defence

CBRN reconnaissance vehicles (RKhM family)	62
Decontamination vehicles (ARS-14, TMS-65, TZ-74 family)	63
Smoke generator vehicles (TDA-2K, TDA-3, TDM-2K)	63

7 Command and signals 64

Command and staff vehicles

R-149 command and staff vehicles	64
R-145BM Chayka command and staff vehicle	65
BMP-1KSh command and staff vehicle	65
BMD-1KSh-A	66
VTS Ladoga	66

Artillery and air defence command

1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) . . .	67
Artillery command vehicles 1V18/1V19, 1V119, 1V152, 1V1003 . . .	67
PRP-4A Argus / PRP-4M Deytery . . .	68
Polyana-D4 . . .	68
PU-12M7 . . .	69
Barnaul-T . . .	69

Radio and signals

R-166 Artek . . .	70
R-161 Poisk HF radio station . . .	70
R-142N Deimos-N . . .	71
P-240I-4 Pereselenets . . .	71
R-416GM radio relay station . . .	72
Tropospheric scatter radio stations (R-410-7.5, R-412A/AM, R-417) . . .	72
R-439 Legenda-2 . . .	73
Redut-2US . . .	74
Strelets . . .	74

8 Logistics 75*Trucks*

KamAZ military trucks (4310 and Mustang family)	75
Ural-6370 / 63706 / 532301 trucks (newer Ural family)	76

9 Unmanned aircraft 77*Small reconnaissance drones*

ZALA 421-16E	77
Orlan-10	78
Supercam S350	79
Granat-1 / Granat-4	79
Eleron-3SV	80
Takhion	81
Zastava	81
Merlin-VR	82

Larger reconnaissance and strike drones

Orion	82
Forpost	83
Korsar	83
Mohajer-6	84

Loitering munitions and one-way attack

Lancet	85
KUB-BLA	86
Italmas	86
Geran-2	87
Gerbera	88
Molniya	88

FPV drones

FPV drones	89
----------------------	----

T Cross-reference tables 90

1 Master list of designations	90
1b Other names and designations	107
1c Cyrillic designations and names	114
2 Unit type to equipment	120
3a Reach of indirect fire	131
3b Reach of long-range missiles and one-way attack drones	133
3c Reach of direct fire	134
3d Reach of anti-tank weapons	138
3e Reach of air defence weapons	139
3f Reach of sensors	141
3g Reach of communications	146
3h Reach of electronic attack	148
3i Reach of UAVs	149

3j Deployment distances	152
3k Engineering equipment	153
4 Mobility and time into action	154
5 Source register	158
6 Picture credits	188

About this guide

"If you want it done good, do it yourself."

This is an open source (OSINT) guide to the equipment used by the Russian Ground Forces. It fills a need identified in Finnish Defence Forces reservist training exercises: lack of up-to-date, openly available guides to the equipment our most probable opponent uses that could be easily perused without Internet access, or even without any electronic devices.

It is made to be printed and carried, as an A5 booklet or on A4 sheets. For each system it shows what it looks like and how to tell it from similar ones, which kinds of units hold it, and how far it reaches. Entries are grouped by role, from armoured vehicles to drones, and the tables at the back let you look a system up by designation, NATO reporting name or Cyrillic name, see which unit types hold what, and compare ranges and mobility.

Everything in it comes from open sources: reference works, Russian official, industry and specialist publications, Western and Ukrainian analysis and defence journalism, and photo-documentation of equipment seen in the war in Ukraine. Each entry lists its sources by code at its foot, and the source register at the back gives them in full. Entries are researched and drafted with the help of AI tools, then checked claim by claim against their sources in a separate pass.

This is a private project, not a publication of the Finnish Defence Forces or any other authority, and it contains only openly published information.

Status of this proof. This proof holds 119 entries. 5 of them are research drafts that have not yet passed the verification step and carry a red mark. Figures may be wrong. Do not use this proof operationally.

Feedback. Corrections, better photographs and suggestions are welcome by email to jannesguide@pm.me or on Signal, username [jmkorhonen.32](#). Please give the edition, its date and the page number, all printed at the foot of every page, and a source if you have one: every change is checked against sources before it goes in. Send open-source material only: nothing classified, nothing about your own unit or where it is, and remove location data from photographs.

How to read an entry. The black box at top right is the mobility code. *Operated by* lists unit types that hold the equipment. *Numbers* gives inventory totals and confirmed losses where no holding unit is named. Each row ends with a tag of three parts, such as **D 2016 H**: basis, date and confidence. Basis is D for doctrinal tables of organisation, O for observed with an identified unit, R for reported or claimed. Confidence is H, M or L. *Recognise* bullets follow a fixed order and start with the first letter of the feature. Unknown values are printed INA.

Caution. Since 2022, army and district assets are often parcelled out to groupings of forces, new formations are under strength, and legacy or civilian vehicles appear in any unit. Equipment indicates a unit type. It does not prove it.

TR	Tracked, cross-country	TR-A	Tracked, amphibious
WH-AT	Wheeled all-terrain	WH-AT-A	Wheeled all-terrain, amphibious
WH-LT	Wheeled, limited off-road	RD	Roadbound, hard surface only
TW	Towed	MP	Man-portable
AIR-RW	Helicopter	AIR-FW	Fixed-wing aircraft
FX	Fixed installation		

1 Armoured vehicles

IN THIS CHAPTER

MBTs and tank support	7
Tracked IFVs, APCs and carriers	12
Wheeled APCs and armoured cars	20

LOOKING FOR THESE? THEY ARE IN OTHER CHAPTERS

SYSTEM	WHAT IT IS	CHAPTER	PAGE
BMP-1KSh command and staff vehicle	Command and staff vehicle for regimental and battalion commanders	7 Command and signals	65
BMD-1KSh-A	Airborne battalion chief-of-staff command and staff vehicle	7 Command and signals	66
R-149 command and staff vehicles	Tactical command and staff vehicles with HF/VHF radios and data terminals	7 Command and signals	64
R-145BM Chayka command and staff vehicle	Turretless radio command and staff vehicle for mobile command posts	7 Command and signals	65
VTS Ladoga	Armoured command, staff and reconnaissance vehicle for NBC conditions	7 Command and signals	66
1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)	Tracked command, observation and fire direction vehicles, SP howitzer battalion	7 Command and signals	67
PRP-4A Argus / PRP-4M Deytery	Mobile artillery reconnaissance post on BMP-1 chassis	7 Command and signals	68
Zoopark-1 / Zoopark-1M	Tracked radar that locates enemy guns, mortars, rockets and missile launchers	4 Radars	42
ARK-1M Rys	Artillery weapon-locating and fire-control radar	4 Radars	43
SNAR-10 / SNAR-10M1 Leopard	Tracked battlefield surveillance and artillery-adjustment radar	4 Radars	44
Borisoglebsk-2	Tactical HF/VHF/UHF radio intercept, direction-finding and jamming complex	5 Electronic warfare	48
Strela-10 and Osa short-range SAM	Short-range SAM	3 Air defence	40
Tunguska	Tracked short-range gun and missile air defence vehicle, 30 mm guns and SAMs	3 Air defence	40
Tor	Short-range tracked SAM with own radars	3 Air defence	38
IMR-2 / IMR-2M / IMR-3M combat engineer vehicle	Tank-based heavy engineer vehicle for obstacle and minefield clearing	6 Engineers	58
BMR-3M	Armoured mine-clearing vehicle on T-90 chassis with roller tawl	6 Engineers	59
UR-77 Meteorit	Tracked mine-clearing vehicle firing rocket-towed explosive line charges	6 Engineers	59
IRM Zhuk	Amphibious engineer reconnaissance of routes, water obstacles and minefields	6 Engineers	57
BREM and BTS armoured recovery vehicles	Armoured recovery and repair vehicles for tank and infantry vehicle units	6 Engineers	62
CBRN reconnaissance vehicles (RKhM family)	Radiation, chemical and biological reconnaissance	6 Engineers	62

T-90 T-90 «Валгумур»

TR

Main battle tank; 125 mm gun-launcher with autoloader. Derived from the T-72B. In service 1992

OPERATED BY tank battalion, tank companies; tank type not fixed by doctrine, 31 tanks per battalion of a motor rifle brigade; 41 with four companies (strength for tanks of any type; sources do not tie the T-90 to it) [D 2024 L](#)

motor rifle brigade, INA; T-90A elements geolocated in Luhansk oblast [O 2014 L](#) seen with 136th Guards Motor Rifle Brigade

motor rifle division [R 2020-04 M](#) seen with 2nd Guards Tamanskaya Motor Rifle Division, 1st Guards Tank Army

Unidentified unit, Central MD: tank units (Dec 2022); a motor rifle formation (May 2024), Central MD: T-90M to tank units Dec 2022; 23 T-90M to a motor rifle formation May 2024 (MoD claims)

[R 2024-05 M](#)

tank division, tank regiments (80th, 6th, 239th named as its regiments, not tied to T-90M), on paper about 300 tanks, mostly T-90M (analyst estimate quoted by TRENCH); IISS lists the division with 2 tank regiments [R 2025 L](#) seen with 90th Guards Tank Division

airborne division, INA; 7th, 76th and 106th Airborne Divisions each have one tank battalion (IISS), 30 T-90M across the Airborne Forces (IISS estimate); IISS does not say which formation holds them

[R 2024 L](#)

NUMBERS Holdings about 500 T-90M in service, held by about 20 units: battalion set of 30-40, company set of about 10, or a few [\(2025-09\)](#)

INDICATES T-90M went first to Guards formations of the 1st Guards Tank Army; by 2025 about 20 units held it, from battalion to platoon sets, and the Airborne Forces hold some. A T-90M alone does not show unit type or echelon.

RECOGNISE Hull Well sloped glacis with added armour. Exhaust outlet on left side of hull above last road wheel. Fuel drums and unditching beam often on hull rear.

Armament 125 mm gun; on T-90 and T-90A a box-shaped Shtora sensor head each side of the gun. 12.7 mm MG on right of turret roof.

Turret Rounded turret, ERA on front and sides, smoke dischargers each side. T-90A: welded turret. T-90M: new turret with rear ammunition box, remote MG.

Suspension Six road wheels each side, idler front, drive sprocket rear, three return rollers; upper run covered by skirts with large square armour panels at front.

T-90M add-ons T-90M: net screen around the lower turret, bar armour on hull and turret rear, armour plates on forward skirts. Exhaust in an over-track outlet.

NOT TO BE CONFUSED WITH T-72B / T-72B3 T-72B has an infra-red searchlight right of the gun where T-90 and T-90A carry the twin Shtora boxes (JANESRG describes the T-72B1; T-72B3 not covered). T-80B / T-80BV / T-80U [p. 8](#) T-80 has no exhaust outlet on the left of the hull; it has a distinctive rectangular air outlet on the hull rear.

VARIANTS T-90 Cast turret, Shtora-1 boxes beside the gun, Kontakt-5 ERA. 840 hp; early night sight was image-intensifier, not thermal. T-90K INA. Command tank: R-163-50K HF radio and TNA-4-3 navigation added; adopted 1994. T-90A Welded turret in place of the cast one. V-92S2 1,000 hp engine and ESSA thermal sight. T-90AK INA. Command tank with added radio, navigation and tactical control equipment. T-90S Export model; may lack the Shtora infra-red dazzlers beside the gun. Export model; figures for it are not domestic figures. A few in Russian service and lost in Ukraine.



Nickel nitride, CCO

MOBILITY Not amphibious. Fords 1.2 m unprepared and 5 m with snorkel after preparation.

Weight, combat 46 t T-90

also 46.5 t RWW GS31

Weight, combat 51 t T-90M

L × W × H 9.53 × 3.78 × 2.2 m

Road speed 60 km/h maximum

also 65 km/h GS31 WEG2015

Road range 550 km maximum

also 650 km WEG2015

Fording 1.2 m without preparation

also 1.7 m WEG2015

Fording 5.0 m with snorkel, after preparation

Ground pressure 0.94 kg/cm2

also 0.98 kg/cm2 WP-RU120

Bridge load class 51 46 t x 1.10, rounded up, estimated

Rate of fire, gun 7 rds/min maximum

also 6-8 rds/min GS31

Range, APFSDS 3BM42M 2,000-3,000 m effective, day

also 3,000-4,000 m WEG2015

Range, APFSDS 3BM42M 2,000-2,600 m effective, night

also 2,600 m WEG2015

Range, HEAT 3BK12M 4,000 m effective, day/night not stated

by WEG

Range, HE-Frag 30F19 4,000 m effective, day/night not stated

by WEG

also 5,000 m WEG2015

Range, ATGM 9M119M 5,000 m maximum effective, day

also 4,000-5,000 m WEG64

Range, 12.7 mm Kord MG 2,000 m effective, target and day/night

not stated by WEG

Range, 762 mm PKT coaxial 1,000 m effective, iron sights

also 1,500-2,000 m WEG2015

Detection range, Agava-2 2.5 km identification, target

identification, night

Detection range, Buran-PA 1.2-1.5 km identification, target

identification, night

also 800-1,300 m WEG2015

Detection range, T-90MS over 3,300 m tank target; T-90MS

thermal sight export variant

Crew 3

Main gun 1 x 2A46M smoothbore gun T-90

also 2A46M-2 GS31 WP-RU119

INA (disputed) T-90M

also 2A46M5-01 IZV7

SEEN WITH IMR-3M combat engineer vehicle; MTU-90 bridge/layer; BREM-1M armoured recovery vehicle; BMPT tank support vehicle (T-72 or T-90S chassis)

SOURCES ARS3 ATP GS31 IISS25 IZV7 JANES25 JANESRG MIL58 NI ORYX1 RWW TRENCH2 WEG64 WEG2015 WP-EN76 WP-RU119 WP-RU120 WWW111

T-80B/BV/BVM, T-80U T-80

NATO: M1989 **TR**

Main battle tank with gas turbine; 125 mm gun that also fires a guided missile In service 1978

OPERATED BY tank division, about 200 T-80 in the division (2019, Russian Wikipedia) **[O 2022-03 H]** seen with 4th Guards Tank Division motor rifle brigade, tank battalion, 26 T-80BVM in the brigade (2022) **[O 2022-03 H]** seen with 200th Separate Motor Rifle Brigade naval infantry brigade, IISS estimate 30 T-80BV/BVM in all Naval Infantry (2024) **[O 2024-10 H]** seen with 155th Naval Infantry Brigade (by 2024 the 55th Guards Naval Infantry Division), Pacific Fleet naval infantry (10 T-80BV reported 2019, state media) **tank army**, INA; 'majority' of the army's tanks were T-80U (2022) **[R 2022-03 M]** seen with 1st Guards Tank Army **motor rifle brigade**, 38th Bde: 40 (2019); others INA **[R 2019 L]** seen with 38th Guards Motor Rifle Brigade, 39th Separate Motor Rifle Brigade, 57th Guards Motor Rifle Brigade **naval infantry brigade** **[R 2019 L]** seen with 40th Naval Infantry Brigade

INDICATES T-80s were seen with 4th Guards Tank Division and the Arctic 200th Motor Rifle Brigade in 2022 and with 155th Naval Infantry Brigade in 2024; Rosgvardia also holds T-80BV. A T-80 may point to these formations, but Ukraine also fields T-80BV/BVM/U.

RECOGNISE **Hull** Sloped glacis with V splash board; turret centre, engine rear. Rectangular exhaust outlet in centre of hull rear, none on left side. Fuel drums often at rear.

Armament 125 mm gun with thermal sleeve and central fume extractor; 12.7 mm AA MG on roof (commander, right); smoke dischargers each side of gun; IR searchlight right of gun.

Turret Rounded turret, MG ammo boxes left, stowage box right, snorkel at rear. BV: Kontakt-1 ERA; U: new turret with Kontakt-5; BVM: Relikt ERA and slat screens.

Suspension Six large road wheels with gaps between 2nd/3rd and 4th/5th; idler front, sprocket rear; five return rollers, usually hidden by full-length skirts.

NOT TO BE CONFUSED WITH T-72 series (incl. T-72B3) Exhaust outlet on left hull side above last road wheel; three return rollers. T-90 p. 7 T-72 hull: exhaust outlet on left side towards the rear, not a rectangular outlet in the hull rear.

T-64 Road wheels much smaller than T-80's, four return rollers.

Ukrainian T-80BV/BVM/U Same vehicles; Ukraine holds about 100 (IISS estimate) and uses modernised T-80BV with air assault troops and marines.

T-80UD (Ukrainian diesel) Ukrainian diesel (6TD) version in place of the gas turbine; different engine decking.

SEEN WITH T-80BK / T-80BVK / T-80UK command tanks; BREM-80 armoured recovery vehicle

VARIANTS T-80B No ERA; IR searchlight right of gun; snorkel stowed at turret rear. 1A33 FCS with laser rangefinder; 9K112-1 Kobra gun-launched ATGM; 1,100 hp turbine from 1980. T-80BV Kontakt-1 ERA boxes added. T-80B with explosive reactive armour; 43.7 t. T-80BVK Additional antennas (stated for the T-80BK command tank). Command tank with additional communications equipment. T-80U New turret with built-in Kontakt-5 ERA; four smoke dischargers each side; armoured skirts. 1A45 Irtys FCS; 9K119 Refleks laser beam-riding ATGM; 1,250 hp GTD-1250 from 1990; 46 t. T-80UK INA. Command version: R-163-50K and R-163-U radios, TNA-4 navigation, Agava thermal sight.



Mike1979 Russia, CC BY-SA 4.0

MOBILITY Tracked, not amphibious. Fords 1.8 m unprepared, 5 m with snorkel. High fuel use limits road range to about 335 km without extra drums.

Weight, T-80B/BV 43.7 t combat; RUWIKI gives 43.7 t for T-80BV
Weight, T-80U 46 t combat
Weight, T-80BVM 45.7 t combat; Army-2020 display, claim
L × W × H 9.65 × 3.58 × 2.22 m T-80B/BV
Road speed 70 km/h maximum
Road range 335 km maximum, internal fuel; RUWIKI gives 335 km as the dirt-road figure and 500 km on road
Fording 1.8 m unprepared; 5.0 m with snorkel
Ground pressure 0.86 kg/cm2 T-80B; T-80U 0.92 kg/cm2 (WEG15)
Bridge load class 49 (T-80B/BV), 51 (T-80U) 43.7 t and 46 t x 110, rounded up, estimated

Range, APFSDS	2,000-3,000 m effective, day
Range, HEAT	3,000 m maximum effective, 3BK31 Start, day
Range, HE-Frag	5,000 m maximum effective, direct fire, day; RUWIKI's 5,000 m is 'greatest sighting range'
Range, 9M112 Kobra ATGM	4,000 m effective, T-80B; radio command guidance; RUWIKI: 4,000 m greatest sighting range for guided round
Minimum range, 9M112 Kobra ATGM	100 m effective, T-80B
Range, 9M119 Refleks ATGM	4,000-5,000 m operational, T-80U, T-80BVM; laser beam-riding; ATP: 4,000 m, 5 km against helicopters
Range, 762 mm PKT coax	1,000 m maximum effective, day; 850 m night
Range, 12.7 mm NSVT AA MG	2,000 m effective, ground targets; 1,500 m air targets
Detection range, TPN-3-49 night sight	1,300 m sighting, T-80B; active IR; 850 m passive
Detection range, T-80U gunner night sight	1,750 m night; classification 1,500 m
Detection range, Agava thermal sight	2,600 m acquisition, T-80UK, night
Crew	3
Main gun	125 mm 2A46M-1 T-80B built with 2A46-2 until 1982; T-80BVM 2A46M-4
Ready rounds in autoloader	28 carousel

T-64A/B/BV T-64S

TR

Main battle tank, 3-man crew with autoloader (older Kharkiv design) In service 1966

NUMBERS Holdings 50 T-64A/BV (IISS estimate) [2024]

INDICATES Few Russian formations use T-64s: former DPR/LPR formations. Ukraine holds 250+ T-64BM/BV/BV mod. 2017 (IISS est.), so most T-64s seen are Ukrainian; pro-Russian forces also use captured mod. 2017 tanks. Check markings.

RECOGNISE Hull Well-sloped glacis, driver top centre, distinctive V splash plate. Turret at hull centre, engine rear. Unlike T-72, no exhaust outlet on left side of hull.

Armament 125 mm smoothbore gun with thermal sleeve and fume extractor. Infra-red searchlight to the left of the gun.

Turret Circular turret, raised cupola with external 12.7 mm MG on right of roof. Two stowage boxes right side, one on rear (snorkel above), three MG ammo boxes left.

Suspension Six road wheels each side, much smaller than T-72, idler front, sprocket rear, four return rollers. Side skirts or gill panels often fitted over upper suspension.

NOT TO BE CONFUSED WITH T-72 T-72 has an exhaust outlet on the left hull side, larger road wheels, wider track, no IR searchlight left of the gun.

T-80 Rectangular air outlet at hull rear (not on T-64 series); larger road wheels with gaps between 2nd/3rd and 4th/5th.

SEEN WITH BREM-64 ARV

VARIANTS T-64A Smoke grenade dischargers either side of gun, side skirts. T-64B INA. T-64B1 INA.



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Tracked; fords 1.8 m unprepared, 5 m with snorkel (WEG15). Road range about 400 km, more with external tanks.

Combat weight 39.5 t T-64B

Road speed 60 km/h maximum, T-64B

Road range 400 km maximum, T-64B, internal fuel

Fording 1.8 m unprepared; 5.0 m with snorkel T-64B

Range, 125 mm APFSDS-T BM-42M 2,000-3,000 m max effective, day; night 800-1,300 m; maximum aimed range 3,000-4,000 m

Range, 125 mm HE-frag OF-26 5,000 m max effective, day; night 800-1,300 m; maximum aimed range 5,000 m

Range, 125 mm HEAT-MP BK-29M 3,000 m max effective, day; night 800-1,300 m; maximum aimed range 4,000 m

Range, 9M112 Kobra (AT-8 Songster) 4,000 m gun-launched, radio command guidance; day sight only unless target illuminated

SOURCES IISS25 JANESRG ORYX1 TRENCH WEG2015 WP-EN75 not in WEG

T-62 T-62

TR

Medium tank with 115 mm smoothbore gun; Cold War reserve type returned to combat in 2022 In service 1961

OPERATED BY tank brigade, platoon [R 2024-02 L]

motor rifle brigade, platoon; alternative attribution for the T-62MV obr. 2022 platoon filed under the 5th Tank Brigade [R 2024-02 L]

combined arms army, army units with T-62MV obr. 2022 (mine

ploughs, cages, RP-377), per Forbes as retold by UNIAN [R 2024-02 L]

INDICATES Russia has drawn T-62s from storage since 2022 to replace losses (UK MoD assessment: about 800 by March 2023). A T-62 suggests a unit refilled from reserve stocks, with lower frontline quality.

RECOGNISE Hull Well-sloped glacis with splash board midway; flat hull top; driver's hatch front left, turret centre, engine rear; flat fuel and oil tanks on right track guard.

Armament Long 115 mm gun with fume extractor about one third back from the muzzle; IR searchlight right of gun; 12.7 mm AA MG at loader's hatch on turret right.

Turret Circular cast turret, well sloped all round, set over the third road wheel pair; hatch for ejecting spent cases in turret rear; snorkel at left rear.

Suspension Five large road wheels (810 mm), no return rollers, idler front, sprocket rear; gaps widen between rear wheels; exhaust on left above fourth/fifth wheel.

NOT TO BE CONFUSED WITH T-54/T-55 p.10 T-55 has its bore evacuator just behind the muzzle and a distinct gap between first and second road wheels; T-62 evacuator is one third back, gaps widen towards the rear.

T-55AM, T-55MV Passive armour (AM) or ERA (MV) as on T-62M/MV. T-55: bore evacuator just behind the muzzle, gap between 1st and 2nd road wheels.



Nickel nitride, CCO

MOBILITY Not amphibious. Fords 1.4 m unprepared, 5.5 m with snorkel. Crosses 2.8 m trench, 0.8 m step. Add-on armour and cages lower power-to-weight.

Weight, combat 40,000 kg base T-62

L × W × H 6.63 × 3.30 × 2.40 m

Road speed 50 km/h maximum

Road range 450 km operational, internal and fixed external fuel, paved road

Range, APFSDS 1,870 m effective, APFSDS, 2 m high target, day; 2,260 m against 3 m high target

Range, APFSDS (night) 800 m effective, night, with IR night vision equipment; WEG does not name the ammunition

Range, HEAT 1,500 m maximum effective, BK-4 HEAT, day

Range, HE-Frag 4,000 m maximum aimed, OF-27 HE-Frag, day, direct fire

Range, 9M117 ATGM 4,000 m effective, 9K116-2 Shekna laser beam-riding missile, T-62M and T-62MV only, day only

SOURCES ATP ESPRESSO FAST IISS25 JANESRG MIL57 ORYX1 RSUPPLY RWW UKMOD UNIAN6 WEG62 WEG2015 WP-RU117

T-54 / T-55 T-54, T-55

TR

Medium tank; reserve stock, reactivated since 2023 In service 1946

OPERATED BY motor rifle division, one subunit, presumably of the 42nd Guards MRD, near Robotyne with unmodernised T-55s

O 2024-02 L seen with 42nd Guards Motor Rifle Division (presumed)

Second- and third-line units, used as mobile artillery **R 2024 M** naval infantry brigade, 30 T-55A across Naval Infantry (IISS estimate); not allocated by brigade **R 2024 M**

NUMBERS Holdings 70 T-55A (IISS estimate, Army); T-55A also among 2,900+ tanks in store (2024); Holdings about 100 T-55 reported moved out of storage (2024)

INDICATES From mid-2023 storage T-55s armed second- and third-line units and served in a limited mobile artillery role; in 2024 they joined assaults near Robotyne. IISS lists 30 with naval infantry. Seeing them suggests a unit short of modern tanks.

RECOGNISE Hull Well sloped glacis with splash board mid-way; flat hull top, driver's hatch front left, vertical rear. T-55A hull 6.2 m, T-54 6.04 m.

Armament 100 mm gun, no thermal sleeve, bore evacuator just behind muzzle; IR searchlight right of gun. KTD laser box over mantlet on units refitted from 1974.

Turret Circular, well sloped all round. Loader's cupola on right normally mounts 12.7 mm AA MG; T-55 had none until 1970 retrofit.

Suspension Five road wheels, idler front, sprocket rear; gap between wheels 1 and 2. No return rollers. Exhaust on left guard above wheels 4-5; fuel tanks right.

NOT TO BE CONFUSED WITH T-62 p. 9 Fume extractor about one-third along the gun from the muzzle; hull 6.63 m vs 6.04 m (T-54). TM-800 / TR-85 (Romanian) Six smaller road wheels each side, not five larger.

SEEN WITH T-54/55 armoured recovery vehicle; MT-55A bridgelayers; KMT-5 mine plough and roller set

VARIANTS T-54B As T-54A, gun with fume extractor; late production adds IR searchlight right of gun (Jane's: first with infra-red night equipment). D-10T2S gun with STP-2 two-plane stabiliser; IR night sights on late production. No NBC system on T-54 (Jane's). T-55 No dome ventilator front right of turret and no hull MG port (both on T-54). No AA MG on loader's hatch until 1970 retrofit. Basic PAZ NBC protection, V-55 engine, 43-round gun load. T-55A Hull 6.2 m (T-54 6.04 m); enlarged driver's hatch and turret hatch coamings; late ones have rubber skirts. Plastic lead anti-radiation lining, full PAZ/FVU filtration, PKT coax gun. T-55AM Add-on BDD armour with armour band round the turret, rubber side skirts, thermal sleeve on the gun. Volna fire control; fires 9M117 Bastion through the gun. Entered service 1983. Not listed in Russian holdings (IISS). T-55AMV As T-55AM but explosive reactive armour on glacis, turret and sides in place of the BDD armour. As T-55AM; entered service 1985. Not listed in Russian holdings (IISS).



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Fords 1.4 m unprepared, 5.5 m with snorkel; not amphibious. T-54 has no NBC system.

Weight, combat	41,000 kg	WEG figure; see open_questions
L × W × H	6.2 × 3.37 × 2.40 m	T-55 (RUWP), T-55A (ENWP), T-55AMV (WEG15). WEG 6.54 m also given; WEG figure treated as a transcription error (WEG 2015 agrees with the others; decision 1 Oct)
Road speed, maximum	50 km/h	T-55 (RUWP), T-54 (JANESRG), T-55AMV (WEG15). WEG figure treated as a transcription error (WEG 2015 agrees with the others; decision 1 Oct)
Road range	400 km	WEG figure; fuel load not stated
Fording	1.4 m unprepared; 5.5 m with snorkel	Not amphibious
Ground pressure	0.81 kg/cm2	Jane's figure is for T-54
Length, gun forward	9.00 m	Jane's figure is for T-54
Vertical obstacle	0.8 m	
Trench	2.85 m	

Rate of fire, main gun	4 rounds/min	average, Average; WEG max 5-6
Range, 100 mm AP, direct fire	1,080 m	grazing shot, Grazing shot (point-blank range) at 2 m high target
Range, 100 mm HE-frag, direct fire	1,100 m	grazing shot, Grazing shot at 2 m high target
Range, 100 mm HE	14,600 m	maximum, Indirect fire with side level and azimuth pointer; round type not named for this figure
Range, 762 mm PKT coaxial	800 m	effective, Day; T-55AMV
Range, 9M117 Bastion	4,000 m	effective, Only T-55M/AM/MV with Volna; day; laser beam-riding round fired through the gun
Crew	4	Commander, gunner, loader, driver
Main gun	D-10 Tank Gun, 100 mm	Designation is D-10T2S on T-55 per RUWP and WEG15
Ammunition, 100 mm	43 rounds	T-55 / T-55A. WEG 45 also given; WEG figure treated as a transcription error (WEG 2015 agrees with the others; decision 1 Oct)
Elevation, main gun	-5 to +18 deg	T-55; +9 deg before 1962 per RUWP. WEG figure treated as a transcription error (WEG 2015 agrees with the others; decision 1 Oct)
Penetration, 3BM-8 APDS	275 mm at 2 km	Round introduced 1967; armour type not stated. WEG text copies ENWP: one lineage
Anti-aircraft machine gun	12.7 mm DShKM	Fitted from 1970; WEG extract names a QJZ-89
Penetration, 9M117 Bastion	550 mm RHA	T-55AMV only; WEG variants tab

BMPT Terminator БМПТ «Терминатор»

TR

Tank support vehicle: twin 30 mm guns and ATGMs on a tank hull against infantry AT teams In service 2018

OPERATED BY tank division, BMPT company (first formed Dec 2021); a further subunit formed 2024, 9 per company [O 2024-12 H] tank regiment, separate BMPT company, 9 per company

[O 2021-12 M]

military district, Central Grouping of Forces, Sievierodonetsk axis; UK MoD assessment (state claim), at most 10 deployed; the only formation fielding BMPT [R 2022-05 L]

INDICATES Russia fields BMPT in few companies of nine, the first in a tank regiment of 90th Guards Tank Division. A BMPT most likely marks that division or a unit given one, attached to tanks for assaults on fortified or urban areas.

RECOGNISE Hull Tank hull built from T-90A components. BMPT: 30 mm grenade launcher mounts in niches above both tracks; BMPT-72 has none.

Armament Two slender 30 mm 2A42 barrels side by side; a pair of boxed Ataka-T missile tubes on each side of the turret, armoured covers since 2018.

Turret Low unmanned turret with commander's panoramic sight on roof; ERA on front and sides, slat armour at rear. 2025 batch: chain anti-drone screens at nose and rear.

Suspension T-90 running gear: six road wheels, idler front, drive sprocket rear; exhaust on left hull side above last road wheel (Jane's T-90). BMPT-72: three return rollers.

NOT TO BE CONFUSED WITH T-90A, T-72B3 tanks p. 7 Tanks have one long 125 mm gun; BMPT has two thin 30 mm barrels and box missile launchers on both turret sides.

BMPT-72 vs BMPT BMPT has grenade launcher housings above both tracks and five crew; BMPT-72 has none and three crew.

SOURCES ATP CNA DUA IISS25 INTFX IZV JANESRG MIL8 MIL9 NEWSRU ORYX1 PROF RUBR RWW TWZ3 WEG14 WP-EN19 WP-RU26 WP-RU27 WWW1A WWW1B



Nickel nitride, CCO

MOBILITY Tank-class mobility. Fords 1.2 m unprepared, 1.8 m prepared, 5 m with snorkel kit. Crosses 2.8 m trench, 0.85 m step.

Weight, combat

48 t BMPT, claim

Weight, combat, BMPT-72

44 t BMPT-72

L × W × H

7.2 × 3.8 × 3.44 m gun forward, claim

Road speed

60 km/h maximum, maximum, road

Range, 9M120-1/9M120-1F Ataka-T

6,000 m maximum, tandem HEAT (9M120-1) or thermobaric (9M120-1F); laser beam riding, claim

Range, 30 mm HE (2A42)

4,000 m maximum effective, personnel and unarmoured targets; WEG BMPT-72 sheet: maximum firing range, HEI

Range, 30 mm AP (2A42)

2,000 m effective, light armour, AP-T

Range, 30 mm (2A42) vs air

2,500 m capable of destroying, low-speed air targets

Range, 30 mm grenade (AG-17D)

1,700 m effective, Frag-HE grenade; BMPT only

SEEN WITH T-90A

BMP-1 БМП-1

NATO: M-1967

TR-A

Amphibious tracked infantry fighting vehicle, 73 mm gun and ATGM, 3 crew and 8 infantry In service 1966

OPERATED BY motor rifle battalion, motor rifle companies: 3 per platoon; battalion HQ also has 1 BMP-1K, 42 BMP-1/BMP-2 per BMP-equipped battalion (Soviet table) [D 1991 M]

motor rifle regiment, reconnaissance company, 3 BMP-1/BMP-2 with 1 BRM-1 (Soviet table) [D 1991 M]

INDICATES Usually indicates a motor rifle unit: Soviet tables gave BMP-equipped motor rifle battalions 42 BMPs. IISS lists 750 BMP-1/-1AM in service; the BMP-1AM upgrade reached motor rifle units from 2020 (state claim).

RECOGNISE Hull Pointed nose, almost horizontal ribbed glacis; driver front left, engine louvres right; four roof hatches over troop compartment; vertical rear with two bulged doors.

Armament 73 mm gun with coaxial MG to its right; Sagdger ATGM rail above the gun (BMP-1). BMP-1P: no rail, ATGM launcher on turret roof. Four firing ports each side.

Turret Low circular one-man turret with well-sloped sides, centre of hull; one hatch on left of roof. BMP-1AM: BTR-82A gun module, long 30 mm cannon.

Suspension Six road wheels each side, drive sprocket front, idler rear, three return rollers; upper track covered by skirts. WEG notes gaps between the 4th, 5th and 6th road wheels.

NOT TO BE CONFUSED WITH BMP-2 p. 12 Two-man turret with long-barrelled 30 mm cannon and muzzle brake, ATGM launcher on turret; two roof hatches over the troop compartment, not four.

BMP-1 p. 14 Same turret but smaller hull with five small road wheels each side and drive sprocket at the rear.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Swims with tracks after trim vane is raised; currents up to 1.2 m/s, waves up to 25 cm. Added armour may jeopardise swimming.

Weight, combat

13,500 kg BMP-1

L × W × H

6.735 × 2.94 × 2.15 m

Road speed

65 km/h maximum, maximum, road

Road range

600 km maximum, road; 600 km per WEG 2015 and Jane's, so WEG's 550 km is taken as a transcription error [kept as 'also']

Range, PG-15V HEAT-FS (3 mm 2A28)

1,300 m maximum effective, day, stationary; 600 m on the move, 800-1,000 m at night

Range, OG-15V HE-frag (73 mm 2A28)

4,500 m maximum, HE-frag; WEG gives 4,500 m as gun maximum, round not named

Range, 9M14M Maluytka (AT-3 Sagdger)

3,000 m effective, MCLOS wire-guided ATGM, day only; BMP-1

Minimum range, 9M14M Maluytka (AT-3 Sagdger)

500 m MCLOS wire-guided ATGM; BMP-1

SOURCES FM32 IISS25 JANESRG LENTA22 MIL7 ORYX1 RG20 RG23 RG222 RIA222 VPK1 WEG2015 WP-EN17 WP-RU22 WP-RU23

BMP-2 БМП-2

NATO: M1981

TR-A

Tracked amphibious IFV: carries a 7-man squad; 30 mm cannon and ATGM In service 1980

OPERATED BY motor rifle battalion, motor rifle companies, one vehicle per squad, 37 per BMP battalion (BMP-2 or BMP-3); 3 per platoon [D 2016 M](#)

Manoeuvre units, combat reconnaissance patrol, platoon-sized, task-organised from a manoeuvre unit (ATP example), 2 per patrol [D 2024 M](#)

motor rifle regiment, reconnaissance company, tracked platoon (Soviet TOE), 3 BMP-1/BMP-2 [D 1991 L](#)

motor rifle regiment, air defence battery (Soviet TOE), some regiments, 6 BMP-2, each carrying three SA-16 launchers [D 1991 L](#)

motor rifle brigade, motor rifle battalions [O 2023-06 H](#)

tank regiment, INA (BMP-2M Berezchok) [O 2023-02 M](#)

NUMBERS Holdings 50 in Airborne Forces (VDV) total (IISS estimate) (2024)

INDICATES Usually marks a motor rifle subunit: one per squad, about 37 per BMP battalion. Since 2022 also lost by tank divisions and regiments, and IISS lists it in Naval Infantry and Airborne holdings, so it does not fix the branch.

RECOGNISE Hull Pointed nose, near-flat ribbed glacis; driver front left, engine louvers right; two roof hatches over troop bay; two bulged rear doors. Exhaust top right (WEG).

Armament Long thin 30 mm cannon with muzzle brake; coaxial MG on its left; ATGM launcher on turret roof (JANESRG: centre rear; WEG: left side).

Turret Two-man turret, 'frying pan' shape (WEG), slightly aft of centre; three forward-firing smoke dischargers each side; large IR searchlight on top.

Suspension Six road wheels each side with gaps between 4th, 5th and 6th (WEG); sprocket front, idler rear; return rollers hidden by ribbed skirts.

Firing ports Firing ports: 4 left, 3 right, 1 in left rear door (WEG15; JANESRG: one in each rear door).

BMP-2M Same turret shape with twin Kornet launchers on the turret, 30 mm grenade launcher at turret rear, new combined sight; slat armour and skirts (ENWIKI).

NOT TO BE CONFUSED WITH BMP-1 [p. 11](#) Circular turret with 73 mm gun and Sagger launcher above, single gunner's hatch; four roof hatches over troop compartment instead of two; four firing ports each side.

BMP-1P [p. 11](#) BMP-1 with the Sagger removed and an AT-4 launcher on the turret roof, as on BMP-2; short 73 mm gun, four firing ports each side.

BMP-3 Larger box hull, 100 mm gun with 30 mm coaxial, bow machine guns; drive sprocket at rear.

BMD-2 [p. 14](#) Same 30 mm cannon but on a small airborne hull with five small road wheels and a bow MG.

BMO-1 Flamethrower squad carrier on BMP-2 base with a dummy turret and external storage racks.

VARIANTS BMP-2K Extra whip antennas on hull rear and a telescopic mast mount (WEG). Command vehicle with an additional HF radio. BMP-2D Spaced steel appliqué plates on hull sides and turret. Better protection; no longer swims; fords 1.2 m (RUWIKI). BMP-2 obr. 1984 'Kovrik' armour on turret front. Improved version; other changes INA. BMP-2 obr. 1986 INA. BPK-2-42 gunner's sight instead of BPK-1-42; Konkurs-M missile (RUWIKI). BMP-2M Berezchok Two twin Kornet launchers on the turret, 30 mm grenade launcher at turret rear, new sight heads. Four Kornet ATGM, AG-30M grenade launcher, thermal sight with laser rangefinder, commander's panoramic sight (manufacturer).



Nickel nitride, CDO

MOBILITY Swims with little preparation, track-propelled at up to 7 km/h; trim vane raised and bilge pumps on before entry. Add-on-armour BMP-2D cannot swim (fords 1.2 m).

Weight, combat	14,300 kg
L × W × H	6.735 × 3.15 × 2.45 m
Road speed	65 km/h maximum
Road range	600 km maximum
Fording	amphibious
Ground pressure	0.65 kg/cm ²
Bridge load class	16 14.3 t tracked x 1.10, rounded up, estimated
Cross-country speed	45–50 km/h maximum
Water speed	7 km/h maximum

Range, 30 mm AP-T	1,500 m effective, armour; day (1,300 m night, WEG15)
Range, 30 mm APDS	2,000 m maximum effective, day (1,300 m night)
Range, 30 mm HE	4,000 m effective, ground targets, Frag-HE; day (1,300 m night, WEG15)
Range, 30 mm, air targets	2,500 m effective, helicopters and slow aircraft; ammunition not stated
Range, 762 mm PKT	1,500 m effective, ground targets, 57-N-323S cartridge (WEG)
Range, 9M113 Konkurs	4,000 m maximum, SACLOS, wire-guided ATGM
Minimum range, 9M113 Konkurs	70 m SACLOS, wire-guided ATGM
Range, 9M111 Fagot	2,000 m maximum effective, SACLOS, wire-guided ATGM
Range, 9M133 Kornet (BMP-2M)	5,500 m maximum effective, Kornet-LR; day (3,500 m night)
Range, 30 mm AG-30M (BMP-2M)	1,700 m maximum effective, WEG15 figure for AGS-30 or AGS-17 grenade launcher; day and night
Detection range, BPK-1-42 sight	650 m vision range, night, passive starlight channel; 350 m with IR searchlight
Crew	3 + 7 dismounts
Main gun	30 mm 2A42, dual feed, stabilised (2E36-1)
Ammunition, 30 mm	500 rds (340 HE + 160 AP)
SEEN WITH 9P135M-series launcher for 9M113 Konkurs / 9M111 Fagot (WEG15: 9P135M1 or 9P135M3) (1 per vehicle, 4 missiles carried); B05Ya01 Berezchok combat module (1 per BMP-2M); BMO-1 flamethrower vehicle (BMP-2 based)	

SOURCES AR11 ATP D24 FM32 IISS25 JANESRG MILUA22 MILUA24 ORX1 ROSTEC23 ROSTEC25 RWW WEG13 WEG2015 WP-EN18 WP-RU25 WSPOT2 WSPOT3 WSPOT4 WSPOT5 WSPOT6

BRM-1K БРМ-1К

NATO: M1976/2

TR-A

Tracked amphibious combat reconnaissance vehicle on BMP-1 chassis In service 1972

OPERATED BY motor rifle division, reconnaissance battalion; two of its three troops operate BRM-1K, 3 per reconnaissance battalion (ARMYREC) [D 2019 M](#)

motor rifle regiment, reconnaissance company, 1 per company (Soviet era, WPEN) [D 1991 L](#)

INDICATES Points to a reconnaissance subunit of a motor rifle or tank formation, most likely a division reconnaissance battalion or a regiment reconnaissance company. A modernised 30 mm turret indicates a BRM-1KM.

RECOGNISE Hull BMP-1 hull: pointed nose, ribbed glacis, driver front left, engine louvres roof right. Two small rear roof hatches; log on right side, antenna mast at hull rear.

Armament 73 mm 2A28 Grom smoothbore gun with coaxial 762 mm PKT. Most vehicles have no Sagger launcher over the gun, unlike BMP-1; some had one.

Turret Wide, low two-man turret set well back. Three smoke dischargers in a row each side. Radar stows behind the turret roof, raised from a rear-opening hatch.

Suspension BMP-1 running gear: six road wheels each side, drive sprocket front, idler rear, three return rollers, upper track covered by skirts.

NOT TO BE CONFUSED WITH BMP-1 [p. 11](#) One-man turret near hull centre, Sagger launcher over the gun (not always fitted), four rear roof hatches against two on BRM-1K.

PRP-3 / PRP-4 artillery reconnaissance [p. 67](#) PRP-3: two-man turret set back too, but no 73 mm gun, only a PKT in a ball mount at turret front. PRP-4 adds optics housings on turret sides.



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Amphibious; swims on its tracks at 7 km/h (WEG).

BMP-1 base uses trim vane and bilge pumps before entering water.

Weight, combat 13.3 t
L × W × H 6.74 × 2.94 × 2.15 m
Speed, road 65 km/h
Speed, swimming 7 km/h

Range, 73 mm (munition not stated) 4,500 m maximum firing range, ammunition type not stated by WEG
Range, HEAT-FS 1,300 m maximum effective, day
Range, HEAT-FS 600 m maximum effective, firing on the move
Range, HEAT-FS 800–1,000 m maximum effective, night
Detection range, PSNR-5K 10,000 m maximum, target type not stated by WEG
Reconnaissance depth, BRM-1KM 50–80 km intended, Front-line reconnaissance, distance from own troops' positions, claim

SEEN WITH PSNR-5K (1RL133-1) Tall Mike radar (1 per vehicle)

SOURCES AR16 CNA ISS25 JANESRG MIL14 ORYX1 RGRU RIA23 WEG20 WEG2015 WP-EN21 WP-RU39 WSN4

BMD-4M БМД-4М

TR-A+AD

Air-droppable amphibious tracked IFV of the Airborne Forces (VDV) In service 2004

OPERATED BY airborne division, lost at Hostomel, 3 Mar 2022, as the 31st Guards Air Assault Brigade [O 2022-03 M](#)

airborne division, parachute battalion of a parachute regiment (battalion set), 31 per battalion (nominal battalion set) [R 2021-07 M](#)

air assault division, air assault battalion of an air assault regiment (battalion set), 31 per battalion (nominal battalion set) [R 2020-06 M](#)

INDICATES BMD-4M points to an Airborne Forces (VDV) parachute or air assault unit; it was delivered in battalion sets of about 31 vehicles with BTR-MDM, so a group of them suggests a VDV battalion. ATP gives a planned VDV total of 1,500.

RECOGNISE Hull BMD-3 layout: flat-topped aluminium hull, pointed nose, near-vertical sides; driver front centre, a hatch each side; engine at rear. New hull on BMD-4M.

Armament 100 mm 2A70 gun-launcher with a 30 mm 2A72 cannon and a 762 mm MG mounted coaxially to its right; ATGMs are fired through the 100 mm gun.

Turret Two-man Bakhcha-U turret behind the driver, with commander's panoramic sight; three smoke dischargers each side at the turret front.

Suspension Five dual rubber-tyred road wheels, idler front, drive sprocket rear, four return rollers. Hydropneumatic: ground clearance varies; about 130 to 530 mm.

NOT TO BE CONFUSED WITH BMP-3 / BMP-3M Six road wheels and three return rollers each side (BMD-4M: five and four); larger box hull with bow MGs.

BMD-3 Predecessor on the same layout: small BMP-2 type turret with a 30 mm cannon only and an ATGM launcher on the turret roof; no 100 mm gun.

SOURCES AR9 AR10 ARCH36 ATP GS8 GS9 ISS25 JANESRG MIL6 ORYX1 RDP2 RDP3 RG172 RWW WEG10 WEG2015 WP-EN16 WP-RU18 WP-RU19 WP-RU20 WSPOT1



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Swims with water jets after raising a trim vane. Air-droppable by parachute with crew inside. Light aluminium armour.

Weight, combat 13.5 t
L × W × H 6.1 × 3.1 × 2.5 m
Road speed 70 km/h maximum
Road range 500 km maximum

Range, 9M117M1 Arkan ATGM 5,500 m maximum, laser beam-riding, fired through 100 mm gun; tanks, claim
Minimum range, 9M117M1 100 m 3UBK23-3 round
Range, HE-Frag 4,000 m maximum effective, 100 mm, day; ammunition not stated by WEG
Range, 30 mm 2A72 2,000 m effective, ammunition not stated
Range, 762 mm PKT 1,500 m effective, ammunition not stated; WEG gives 100–1,500 m with sight adjustments

BMD-2 БМД-2

NATO: M1981/1, M1970

TR-A+AD

Air-droppable amphibious tracked IFV of the VDV; BMD-1 is the 73 mm predecessor In service 1969

OPERATED BY airborne division, airborne regiment, battalion, company: 3 BMD per platoon, one per squad (BMD-1 in source), over 100 BMD per regiment, BMD-1 as squad vehicle (Soviet structure)

D 1991 **M**

airborne division, parachute regiment (Kostroma); MoD claim via RIA

O 2021 **M**

airborne division, parachute regiment **O** 2011-09 **L**

INDICATES In Russia IISS lists the BMD-2 only under the Airborne Forces (VDV), pointing to a VDV formation. VDV units also field BMP-2 and BTR-82AM. Ukraine's Air Assault Troops also hold BMD-2.

RECOGNISE Hull Low hull with pointed nose and trim vane on glacis; hull sides vertical with chamfer along two-thirds of length; engine at rear, two waterjets at rear.

Armament BMD-2: long thin 30 mm cannon with muzzle brake, ATGM launcher on right of turret roof, one bow MG. BMD-1: 73 mm gun, Sagger rail above, two bow MGs.

Turret Small one-man turret, 'frying pan' shape, on forward hull roof. BMD-1 turret is the same as BMP-1's; BMD-2 has a new one-man turret.

Suspension Five small road wheels, idler front, drive sprocket rear, four return rollers (first and fourth support inside of track only); ride height adjustable.

NOT TO BE CONFUSED WITH BMD-4M ^{p. 13} Larger two-man turret with 100 mm gun and coaxial 30 mm cannon (BMD-2: one-man turret, 30 mm only); dual rubber-tired road wheels.

BMP-1 ^{p. 11} Same 73 mm turret as BMD-1 but larger hull with six road wheels, drive sprocket at front and three return rollers.



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Amphibious with waterjets after trim vane is raised.

Air-droppable from Il-76 by PRSM-916/PRSM-925 rocket-parachute systems from 500–1,500 m.

Weight, combat (BMD-2) 8,225 kg

Weight, combat (BMD-1) 7,500 kg

L × W × H 5.91 × 2.63 × 2.175 m gun forward

Road speed (BMD-2) 61 km/h maximum

Range, 30 mm (type not stated) 2,000 m effective, ammunition not stated by WEG

Range, 30 mm (type not stated) 4,000 m maximum, ammunition not stated by WEG

Range, 30 mm AP (BMD-2M) 2,000 m effective, armour-piercing, BMD-2M fire control, claim

Range, 30 mm HE (BMD-2M) 4,000 m effective, high-explosive, BMD-2M fire control, claim

Range, 762 mm PKT 1,500 m maximum effective (with sight adjustments), ball, coaxial

SEEN WITH BTR-D (INA (IISS: 150 BTR-D in VDV, estimate))

SOURCES AR8 AR9 ATP GS7 IISS25 JANESRG MIL5 ORYX1 ORYX2 PROFILE RBC W1RU WEG9 WEG2015 WMC3 WP-EN15 WP-RU16 WP-RU17 ZR2

BTR-MDM Rakushka БТР-МДМ «Ракетка»

TR-A+AD

Airborne tracked amphibious APC and carrier chassis for VDV In service 2016

OPERATED BY airborne division, Procurement plan, replacing BTR-D; not a holding, Plan (2015) of 2,500 by 2025; not met, IISS estimate 150 in 2024 [D 2015 L](#)

naval infantry brigade, INA; ATP says only 'also used by naval infantry' [D 2024 L](#)

air assault regiment, Air assault or parachute battalion, issued with BMD-4M in a battalion set, 8 per battalion set of 31 BMD-4M (June 2020); 6 in a 2021 set; some earlier sets 16 [O 2021-07 M](#) seen with 234th Guards Air Assault Regiment, 76th Guards Air Assault Division (sets Jan and Jun 2020), 137th Guards Airborne (parachute) Regiment, 106th Division (first battalion set)

airborne division, Battalion sets in regiments; some units mix BMD-4M and BTR-MDM companies [O 2021-07 M](#) seen with 98th Guards Airborne Division (Ivanovo); 6 BTR-MDM with 31 BMD-4M to one regiment, 2021-07-26 airborne division, 150 in Airborne Forces (IISS estimate) [R 2024 M](#)

INDICATES A BTR-MDM points to an airborne or air assault battalion issued new-generation vehicles, usually alongside BMD-4M. ATP also names naval infantry as a user, but IISS25 lists none there. Ukraine uses captured ones (2023).

RECOGNISE Hull Welded aluminium hull, no main turret; driver front centre, troop compartment amidships, engine at rear, inclined hatch at rear of troop compartment.

Armament Two 762 mm MGs: remote roof MG and one bow MG at front (right per ZRRU, left per GSBTR and NIKIT). Two banks of smoke grenade launchers.

Turret No main turret or cannon. Small remote MG turret on the left of the roof, with sights and an illuminator; ATP calls it an autonomous fire control turret.

Suspension Five road wheels and four support rollers each side, hydropneumatic suspension; ground clearance adjustable from 130 to 530 mm.

NOT TO BE CONFUSED WITH BTR-D [p. 16](#) BTR-D has six road wheels per side (Jane's); BTR-MDM has five per side.

BTR-MD Earlier carrier (Object 955) on the BMD-3 base; BTR-MDM uses the BMD-4M chassis. WEG mixes the two. Visible differences: INA. BMD-4M [p. 13](#) Same chassis family but carries a turret with a 100 mm gun and 30 mm cannon; BTR-MDM has none.

VARIANTS BTR-MDM with ZU-23-2 Twin 23 mm gun mounted on the carrier, as on BTR-ZD. Adds anti-aircraft and ground fire; shown in a September 2021 exercise. Fit seen in Ukraine: INA. BTR-MDM Rakushka-M 2024 delivery batch: Nakidka camouflage coating and upper-hemisphere protection kit, per Rostec release as reported by Army Recognition. Drone-suppression equipment and night vision devices added, per manufacturer claim. 9S931-2 MRU-D INA. Air surveillance radar vehicle for Verba MAMPADS teams; Russian Wikipedia only. 9S931-3 MP-D INA. Air defence command vehicle with Barnaul-T; Russian Wikipedia only. Andromeda INA. Artillery fire control vehicle on BTR-MDM, development work (ROC) reported by Concern Tractor Plants in 2015; service INA.



Komarov, CC BY-SA 4.0

MOBILITY Amphibious on two waterjets; sea trials left a landing ship in 4-5 sea state and returned in 3. Light armour limits use in direct assault.

Weight, combat
L × W × H

Speed, road

Speed, dry unpaved road

Range, road

Swimming speed

Fording
Ground clearance
Engine power

13.2 t NIKIT gives 13.2 t plus 2%, claim
6.1 × 3.1 × 2.5 m Both sources approximate and share one lineage, claim
70 km/h maximum, Same figure in RGRU2 (state media), claim
45-50 km/h average, Dry dirt road; RGRU2 (state media) gives up to 45-50 km/h, claim
500 km at least, On fuel; 350 km on dry ground per NIKIT, claim
10 km/h maximum, Two waterjets; at least 7 h afloat per NIKIT; VPK gives at least 10 km/h, claim
Amphibious
130-530 mm adjustable
500 hp (UTD-29) NIKIT 368 kW (500 hp); same figure in RGRU2 (state media); VPK names UTD-29 only, claim
also 450 hp WEG23

Range, 762 mm PKTM

Range, 762 mm PKTM

Range, 23 mm ZU-23-2 ammunition

Altitude, 23 mm ZU-23-2 ammunition

Crew
Passengers
Payload
Armament

Armour

1,500 m effective, Ground targets; remote roof MG; WEG BTR-MD sheet
4,000 m maximum, Ground targets; remote roof MG; WEG BTR-MD sheet
2,500 m claimed, Ground targets; ZU-23-2 fit on BTR-MDM, 2021 exercise, claim
1,500 m claimed, Air targets; ZU-23-2 fit on BTR-MDM, 2021 exercise, claim
2 (commander, driver)
13 paratroopers
2 t Cargo, not more than 2 t, claim
2 x 762 mm PKTM: remote roof MG and bow MG claim
also Roof turret 762 mm or 12.7 mm MG ATP
Welded aluminium; protects against small arms and shell splinters

SEEN WITH BMD-4M (31 per battalion set with 8 BTR-MDM (2020)); 9S931-3 MP-D; 9S931-2 MRU-D radar vehicle

BTR-D 5TP-A

NATO: M1979

TR+AD

Airborne tracked APC on a lengthened BMD-1 hull; base of a variant family In service 1974

OPERATED BY airborne division, 150 BTR-D in all Airborne Forces (IISS estimate); per-division count INA ^[R 2024-12 M]

air assault regiment, 42 in 247th regiment (2017 table); 26-38 per regiment in a 2000 listing ^[R 2017 L] seen with 247th Guards Air Assault Regiment, 51st, 104th, 108th, 137th, 217th, 234th, 331st airborne regiments (2000 listing)

air assault brigade, 57 in 2000 listing ^[R 2000 L] seen with 31st independent airborne brigade, Ulyanovsk

Unidentified unit, Airborne Forces troops with BMD-2s and BTR-Ds seen near Bucha, 27 Feb 2022; no unit named ^[R 2022-02 L]

INDICATES IISS25 lists Russian BTR-D only under the Airborne Forces, so a Russian one points to VDV or attached units. Ukraine's Air Assault Troops hold 30 and Russia uses some captured ones. 247th regiment's 2017 table shows ATGM, AA, fire-direction variants.

RECOGNISE Hull Turretless hull on the BMD-1 pattern, longer than the BMD-1, with a higher roof, small flat superstructure and roof hatch; upper front plate has a dual slope.

Armament Two bow machine guns kept from the BMD-1; no main gun. Pintle AGS-17/30/57 or MG may be fitted; ZU-23-2 may be towed or carried.

Turret None. Flat roof with hatches; firing ports in the sides and rear (sources differ on the number).

Suspension Six road wheels and five return rollers each side (BMD-1: five and four); hydropneumatic, ground clearance adjustable 100-450 mm.

NOT TO BE CONFUSED WITH BMD-1 p. 14 BMD-1 has the BMP-1 turret with 73 mm gun, five road wheels and four return rollers each side.

BMD-2 p. 14 BMD-2 has a new turret with 30 mm cannon; the BTR-D has none.

2S9 Nona-S Same six-wheel BTR-D-derived hull, but a large turret with sloped front and a 120 mm weapon level with the hull front.

VARIANTS BTR-RD Robot Pintle mount on right of hull; missile racks inside; two 9P135M-1 launchers carried. ATGM team carrier for Fagot and Konkurs missiles, fired dismounted or from the pintle. BTR-ZD Skrezhet Stowage rails for a MANPAD on the right of the roof; can carry a ZU-23-2 on the roof. Carries two MANPAD teams and spare missiles; no vehicle-mounted air-defence weapon unless ZU-23-2 is fitted. BMD-1KSh Soroka Clothes-horse antennas round the superstructure, two slim masts; bow guns and firing ports removed. Command post with R-123M, R-111 and R-130M or R-134 radios; ru Wikipedia says battalion level, FAS says regiment or division. BMD-1R Simitsa Heavy telescopic mast stowed on left of roof; slim mast at rear right. Command and signals vehicle with R-161A2M radio; WEG says division level, small numbers. 1V119 Reostat Turret fitted with observation devices and, per WEG, a PSNR-5K radar on the turret roof. Artillery battalion fire direction vehicle with three radios and rangefinders.



© M. Kuzmin, CC BY-SA 4.0

MOBILITY Amphibious, swims at about 10 km/h. Air-droppable by parachute from medium transport aircraft. Light aluminium armour.

Weight, combat	8,200 kg
	also 8.0 t WP-RU15
L × W × H	5.88 × 2.63 × 2.05 m
Road speed	61 km/h maximum
Road range	500 km
	WEG's 450 km treated as a transcription error [WEG 2015, FAS, ru Wikipedia give 500] also 450 km WEG22
Swimming speed	10 km/h maximum
Fording	Amphibious, no preparation stated
Ground pressure	0.50 kg/cm2
Bridge load class	MLC 10
	Tracked weight 8.2 t x 1.10, rounded up, estimated
Time into action	INA

Range, 762 mm API (bow machine gun)	1,000 m
	maximum effective, Day; 400-500 m on the move also 1,500 m FAS5
Crew	3 (driver, commander, bow machine gunner)
	WEG spec table '1 ea' treated as a transcription error; WEG text and WEG 2015 give three also 1 WEG22 FAS5
Troop capacity	10 troops
	WEG spec table '13 ea' treated as a transcription error; WEG text and WEG 2015 give ten also 12 passengers FAS5
Armament, bow machine guns	2 x 762 mm
Ammunition, 762 mm	Wikipedia says PKT or PKTM
Armour, upper hull front	2,000 rounds
	15 mm aluminium JANESRG gives 15 mm maximum hull for BMD-1, the parent hull
Engine power	240 hp
SEEN WITH	2S9 Nona-S 120 mm self-propelled mortar (18 in 247th regiment (2017 table)); 1V119 Reostat fire direction vehicle (8 in 247th regiment (2017 table)); BREM-D recovery vehicle; ZU-23-2 twin 23 mm gun

MT-LB MT-A5

NATO: M1970

TR-A

Light armoured amphibious tractor and APC; base for many armed field conversions In service 1964

OPERATED BY motor rifle battalion, motor rifle companies (squad carriers in BTR/MT-LB-based battalions), about 44 BTR-80/BTR-82/MT-LB per BTR- or MT-LB-based battalion [\[D 2016 M\]](#)

motor rifle brigade, motor rifle battalions, especially in northern (Arctic) brigades [\[D 2024 M\]](#)

motor rifle brigade, MANPADS battery of the brigade's anti-aircraft missile-artillery battalion (ATP fig. 4-19), 9 per MANPADS battery (27 launchers) [\[D 2024 M\]](#)

anti-tank battalion, anti-tank gun battery (prime mover for 100 mm MT-12) [\[D 2024 M\]](#)

motor rifle brigade, battalion tactical group sent to Ukraine; brigade under Navy Coastal Troops (MILU), 36 MT-LB left the brigade's base in May 2022 [\[O 2022-05 L\]](#) seen with 80th Separate Arctic Motor Rifle Brigade (Alakurtti)

naval infantry brigade, IISS estimate 250 MT-LB in Naval Infantry (6 brigades, 1 regiment) [\[R 2024 M\]](#)

INDICATES Several MT-LBs carrying infantry suggest an MT-LB-based motor rifle battalion (northern brigades, Naval Infantry). One towing a 100 mm MT-12 suggests a brigade anti-tank battery; towing a 122 mm D-30, a towed artillery battalion.

RECOGNISE Hull Low box hull. Crew compartment just behind glacis; troop compartment to rear with near-vertical sides, chamfered top, twin rear doors. Exhaust mid-hull, left.

Armament Basic vehicle: one 7.62 mm PKT only. Since 2022 often field-fitted with ZU-23-2, naval 25 mm or 14.5 mm mounts, S-60 or rocket launchers on the rear deck.

Turret Small flat-topped manual turret right of driver at front. MT-LBVM/VMK: open 12.7 mm MG mount instead. 6MA/6MB: conical BTR-80-type turret on rear roof.

Suspension Six equally spaced ribbed road wheels, drive sprocket front, idler rear, no return rollers. MT-LBV: wider tracks (565 mm). MT-LBu: seven road wheels.

NOT TO BE CONFUSED WITH MT-LBu (command, radar and EW carriers) Seven road wheels, hull 810 mm longer and 200 mm higher, no front MG turret, single rear door.

SNAR-10 (Big Fred) radar vehicle [p. 44](#) Same six-wheel hull, front MG turret and twin rear doors, but Big Fred radar on the rear half of the roof; antenna folds forward for travel.

2S1 Gvozdika Seven road wheels and a large enclosed turret with 122 mm howitzer towards the rear.

9K35 Strela-10 (SA-13 Gopher) [p. 40](#) MT-LB hull carrying a launcher with four missiles; no troop role.

9P149 Shturm-S MT-LB hull with an ATGM launcher that is raised from the hull when required.

VARIANTS MT-LBV Wide 565 mm tracks instead of 350 mm. Ground pressure 0.28 instead of 0.46 kg/cm2; better on snow and swamp. MT-LBVM, MT-LBVMK Open 12.7 mm machine-gun mount in place of the TKB-01-1 turret. 12.7 mm NSVT (VM) or Kord (VMK) instead of 7.62 mm PKT. MT-LBVM1K INA (modification of the MT-LBVMK). Mountain version: 310 hp YaMZ-238BL-1 engine, new radio and heater. MT-LBM 6MA BTR-80-type turret on the hull roof. 14.5 mm KPVT and coaxial 7.62 mm PKT. MT-LBM 6MB BTR-80A-type turret with 30 mm cannon on the hull roof. 30 mm 2A72 and 7.62 mm PKTM; ammunition stowed outside the hull; day/night gunner sight.



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Fully amphibious, propelled by its tracks. Very low ground pressure; wide-track MT-LBV for snow and swamp. Armour stops only small-arms fire and splinters.

Weight, combat **11,900 kg** MT-LB
also 9.7 t (12.2 t with load) WP-RU38

Weight, empty **9,700 kg** MT-LB
Weight, MT-LBu **11.5 t plus up to 4 t of mounted equipment**

L × W × H **6.454 × 2.86 × 1.865 m** MT-LB
Road speed **61.5 km/h** maximum, MT-LB, 240 hp
also 65 km/h ZR4

Road range **500 km** maximum, MT-LB
Swimming speed **5-6 km/h** maximum, propelled by tracks; 5-6 km/h per WEG 2015, Jane's and Russian Wikipedia, so WEG's 4.5 km/h is taken as a transcription error (kept as 'also')
also 4.5 km/h WEG41

Fording **amphibious**
Ground pressure **0.46 kg/cm2** standard 350 mm track; 0.28 with wide track

Range, 7.62 mm PKT **1,500 m** effective, 7.62x54R; WEG gives no target or light condition (WEG day-sight range also 1,500 m)
also 1,000 m day; 400-500 m on the move WEG2015

Range, 7.62 mm PKT **4,000 m** maximum, 7.62x54R, ground target

Range, 12.7 mm NSVT **1,500 m** maximum effective, MT-LB fitted with 12.7 mm NSVT ('some MT-LB; RWW); target not stated

Detection range, driver's TVN-2 IR periscope **about 40 m** approximate, night, with OU-3GK IR searchlight

Crew **2 + 11 passengers** MT-LB as APC
Armament **1 x 7.62 mm PKT in turret** basic MT-LB
Ammunition, 7.62 mm **2,000 rounds** basic MT-LB
also 2,500 rounds WP-EN53

Armour **7-14 mm steel** hull 7 mm sides and rear, 14 mm front and turret (WEG table, Russian Wikipedia infobox); printed over the 3-10 mm in WEG's notes and Jane's, editor decision 4 Oct
also 3-10 mm; 14 mm turret front JANESRG WEG41

SEEN WITH MT-12 Rapira 100 mm anti-tank gun (1 towed per MT-LB; 20 rounds carried in the MT-LB); MTP-LB technical support vehicle; MTO-UB1 maintenance vehicle (Ural-4320)

SOURCES ATP BAR15 FAS6 IISS25 JANESRG MIL42 MIL43 MIL44 ORYX1 RWW UKDI WEG41 WEG2015 WP-EN53 WP-RU38 WP-RU79 ZR4

BMO-T BMO-T

TR

Heavy flamethrower-crew carrier on T-72 chassis, carries RPO-A rocket flamethrowers In service 2001

OPERATED BY motor rifle brigade, flamethrower platoon, RKhBZ company, 3 per RKhBZ company flame platoon (ATP); BMP-2-based BMO-1 may be used instead (ATP, RWW) **D 2024 M**
tank brigade, flamethrower platoon, RKhBZ company, 3 per RKhBZ company flame platoon (ATP); BMP-2-based BMO-1 may be used instead **D 2024 L**

INDICATES Doctrine gives a brigade RKhBZ company's flamethrower platoon three BMO-T or BMP-2-based carriers; named holders are RKhBZ units and one tank brigade. A sighting suggests RPO teams supporting an assault on fortified or built-up positions.

RECOGNISE Hull T-72 hull, turret removed, with a box-shaped armoured superstructure over the crew and troop compartment; vehicle height 2.24 m. ERA on upper front.

Armament No main gun. 12.7 mm machine gun on a small commander's cupola, offset left of centre. Sources differ on type (Kord or NSVT) and remote control.

Turret No turret. Roof hatches over the troop compartment, an extra hatch in the rear plate, and 12 smoke grenade launchers (902A) at the front.

Suspension T-72 running gear: 6 road wheels and 3 return rollers per side, rear drive sprocket, hydraulic dampers on wheels 1, 2 and 5. Front dozer blade (TENC).

NOT TO BE CONFUSED WITH T-72B tank Same hull and running gear, but a tank has a turret and gun; BMO-T has a box superstructure and only a machine gun.

BMO-1 BMP-2-based flamethrower carrier, the doctrinal alternative in the same flame platoon; light BMP-2 hull, not a T-72 tank hull. Adopted 2001 with BMO-T.

SOURCES ARMYIN ATP BTVT GS10 IJSS25 ORYX1 RWW TENC UNIAN WP-RU21 not in WEG



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Tank-chassis mobility. Fords 1.2 m. Climbs 30 degrees, crosses 0.85 m wall and 2.6 to 2.8 m ditch.

Weight	43.9 t	Combat weight
L × W × H	7.22 × 3.787 × 2.24 m	Hull length
Road speed	60 km/h	
Road range	710 km	

Range, RPO-A rocket	400 m effective, 93 mm RPO-A thermobaric rocket
Dismount distance from enemy line	1.5 to 2.0 km practical, Doctrinal scenario: platoon attached to motor rifle battalion, then advances on foot to about 600 m
Crew	2 plus 7 dismounted flamethrower troops Driver and commander; dismounts ride in troop compartment
Rocket flamethrowers carried	30 RPO-A 93 mm RPO-A Shmel, on racks in troop compartment

Aleut and Plastun articulated and light tracked carriers TA3-3344-20

«Алеум»

TR

Two-unit unarmoured (Aleut) and light-armoured open-top (Plastun-SN) tracked carriers In service 2018

OPERATED BY Unidentified unit, Motor rifle formation of the Eastern MD combined arms army, Transbaikalia (unit not named)

[R 2021 L]

Unidentified unit, Eastern MD tank formation, Buryatia; source also says 'air defense forces' (garbled wording, unit not named) [R 2021 L]

NUMBERS Holdings 123 platforms reported delivered to northern ground forces, May 2019 (Russian media via ArmyInform) [2019-05]; Holdings 14 Plastun-SN (evacuation, reconnaissance, 82 mm mortar carrier) first delivered Oct 2023; 50 reported ordered [2023-11]

INDICATES Aleut was ordered for Arctic use and issued to Eastern MD formations in Transbaikalia and Buryatia; it has also appeared with combat columns in Ukraine. Plastun-SN is meant mainly for casualty evacuation from the front line.

RECOGNISE Hull Aleut: two tracked units joined by a steering joint. Front power module has a cab for 5 including driver; rear unit is a van body. Unarmoured.

Armament Aleut: pivot mount for a rifle-calibre machine gun ahead of a roof hatch on the front module. Plastun-SN shown with an 82 mm mortar or, for recce, a telescopic mast.

Turret None. Plastun-SN is a light-armoured tracked carrier with no hard roof.

Suspension Both tracked. Plastun-SN all-season track is 50 cm wide. Aleut: front cab side door, rear van door at the stern. Road wheel counts: INA.

NOT TO BE CONFUSED WITH DT-10 / DT-30 Vityaz Vityaz two-unit tracked carriers, held by the 80th and 200th brigades (CNA 2019); Za Rulem compares them with the GAZ-3344. Visible difference vs Aleut: INA.

GAZ-3344 (civilian) Civilian base vehicle, same two-unit layout. The army Aleut has a machine-gun pivot mount ahead of the front roof hatch; no other visible difference found.



Kirill Borisenko, CC BY-SA 4.0

MOBILITY Aleut is amphibious, 3.8 km/h afloat; ground pressure 0.2 to 0.21 kg/cm² (reported). Plastun-SN open-topped; fording INA. Both for difficult terrain.

Weight, Aleut	11 t	GAZ-3344-20 as shown at BIDE 2017; total weight, claim
Weight, Plastun-SN	3.8 t	Developer figure, claim
Road speed, Aleut	50 km/h	Maximum road speed, company figure for vehicle shown at BIDE 2017 arms fair, claim
Road speed, Plastun-SN	50 km/h	Maximum, developer figure, claim
Speed afloat, Aleut	3.8 km/h	Amphibious without preparation, maximum; BIDE 2017 arms fair figure, claim
Road range, Aleut	450 km	Maximum cruising range, company figure for vehicle shown at BIDE 2017 arms fair, claim
Ground pressure, Aleut	0.2 to 0.21 kg/cm ²	Army version, average; top.war.ru via Army Recognition also 0.2 kg/cm ² WP-RU10
Length, width, height	INA	

Crew and passengers, Aleut	Front cab 5 incl. driver; 12 persons total	GAZ-3344-20 as presented at BIDE 2017 arms fair; company figure, claim also Front cab 5 incl. driver; rear passenger module up to 15 AR4 ADU MILUA AINF22
Payload, Aleut	Front 750 kg; rear 2,250 kg	GAZ-3344-20 as presented at BIDE 2017 arms fair; company figure, claim also Front 500 kg; rear 2,000 kg to 2,500 kg AINF22 MILUA
Towed load, Aleut	3 t	Trailer mass, as reported by Ukrainian state media citing Russian sources, claim
Passengers, Plastun-SN	8; or 4 rescuers and 2 stretchers	Evacuation mode for the second figure, claim
Protection, Plastun-SN	Br3 and Br4	Br3 stops 9 mm pistol bullet; Br4 stops 5.45 mm enhanced penetration and 7.62 mm ball, claim
Engine, Aleut	YaMZ-5347-26 diesel	Vehicle shown at BIDE 2017 arms fair, claim also YaMZ-53402-10 diesel, 190 hp AR4

SEEN WITH Trailer, towed (Up to 3 t (Aleut))

VARIANTS GAZ-3344-20 Aleut Two-unit articulated carrier; transport configuration ordered by army. Carries 12 to 20 people or 2.5 t; amphibious; unarmoured. Rear module converts to command post, ambulance or sleeping unit. Plastun-SN Light tracked vehicle, open top (no hard roof), light armour. Carries 8; Br3 and Br4 protection; evacuation, recce or 82 mm mortar.

SOURCES ADU AINF22 AINF23 AR2 AR3 AR4 CNA MILUA RWW WP-RU10 WWW12 ZR not in WEG

BTR-80 BTP-80

NATO: M1978

WH-AT-A

Wheeled 8x8 amphibious APC; BTR-80A and BTR-82A carry a 30 mm cannon In service 1971

OPERATED BY motor rifle battalion, motor rifle companies, about 44 BTR-80/BTR-82/MT-LB per BTR-mounted battalion [D 2016 M]

motor rifle brigade, motor rifle battalions [O 2022-09 H] seen with 27th Separate Guards Motor Rifle Brigade, 15th Separate Guards Motor Rifle (Peacekeeping) Brigade

motor rifle regiment [O 2022-09 M] seen with 254th Motor Rifle Regiment, 752nd Motor Rifle Regiment

motor rifle division [O 2024-10 M] seen with 42nd Guards Motor Rifle Division

motor rifle brigade [O 2022-03 M] seen with 126th Separate Guards Coastal Defence Brigade (Black Sea Fleet)

naval infantry brigade, Naval Infantry branch total 200 BTR-82A, 100 BTR-80 (IISS estimate); IISS lists 6 naval infantry brigades and 1 regiment [R 2024 H]

NUMBERS Holdings 120 BTR-82AM in the Airborne Forces (VDV) (IISS estimate); IISS does not say which VDV units hold them (2024); Holdings about 20 BTR-82A with the Russian force in Syria (1 infantry battle group, 3 MP battalions; IISS estimate, late 2024) (2024)

INDICATES BTR-80/82 mark motor rifle units on wheels: a BTR battalion holds about 44 BTRs or MT-LBs. BTR-82A also serve naval infantry, airborne and Rosgvardia units, so the type alone does not fix the branch.

RECOGNISE Cab Commander and driver side by side at the front, windscreen hatches in the upper glacis and roof hatches above their seats; trim vane folds back onto the glacis.

Body Streamlined hull; roof slopes down at the far rear (BTR-70) or rises over the rear engine deck (BTR-80); small turret on roof in line with 2nd road wheel; engine at rear.

Axles Four large rubber-tyred road wheels each side, with a slightly larger gap between the second and third.

Details BTR-80/82: two-part side door between 2nd and 3rd wheels, lower half folds down as a step. BTR-80: higher rear roof; six smoke dischargers on turret rear.

Details BTR-70: only a small low door between 2nd and 3rd wheels; exhaust pipe each side at rear. BTR-80A/82A: turret with externally mounted 30 mm cannon.

Details Firing ports: three each side on the BTR-80 (JANESRG, WEG 2015 variants list); BTR-80A has one more on the right (WEG 2015). No rear doors.

NOT TO BE CONFUSED WITH BTR-60PB Square side door high in the hull above the 2nd/3rd axles, not low between the wheels; trim vane folds under the nose; exhaust pipes each side angled at 45°.

2S23 Nona-SVK Same BTR-80 hull, but a large central turret with 120 mm ordnance in a sloping front and no firing ports.

K1Sh1 (UNSh) command, signals and EW vehicles BTR-80 hull with a raised middle section and extra hatches; turret unarmed.

BRDM-2 p. 22 Same small turret as the BTR-70, but a short 4x4 hull.

VARIANTS BTR-70 Small low door between 2nd and 3rd wheels; BTR-60PB/BRDM-2 conical turret; exhaust each side at rear. Two 120 hp petrol engines; machine guns elevate only to +30°. BTR-70M BTR-70 with the rear hull of the BTR-80. Single 260 hp KamAZ-7403 diesel and BTR-80 transmission in place of two petrol engines. BTR-80 Two-part side door, higher rear roof with near-horizontal exhaust, six smoke dischargers on turret rear. Single diesel; KPVT elevates to +60° for use against slow aircraft and helicopters. BTR-80A New turret with externally mounted 30 mm cannon. 30 mm 2A72 and 7.62 mm PKT replace the KPVT; adopted 1994. BTR-82 Unified turret module (BPPU) as BTR-82A but with 14.5 mm KPVT. Stabilised KPVT and PKTM, 300 hp engine, spall liner.



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Amphibious, waterjet driven; trim vane raised before entering water. Climbs 60%, crosses 2 m trench and 0.5 m step. Central tyre pressure regulation.

Weight, combat (BTR-80) 13,600 kg

Weight, combat (BTR-70) 11,500 kg

Weight, combat (BTR-82A) 15,400 kg BTR-82A: mil.ru figure cited by Russian Wikipedia; Army Recognition text gives the same, claim

L × W × H 7.65 × 2.90 × 2.46 m

Road speed (BTR-80) 80 km/h maximum

Road speed (BTR-82A) 80 km/h maximum, BTR-82A

Road range (BTR-80) 600 km maximum

Road range (BTR-70) 600 km maximum

Swimming speed (BTR-80) 9 km/h maximum

Range, KPVT 14.5 mm API/API-T

2,000 m maximum effective, ground target, day

Range, KPVT 14.5 mm

4,000 m maximum, gun maximum firing range; round and target not named

Range, KPVT 14.5 mm, air target

1,000 m sight range, air target; BTR-80 1PZ-2 sight; round not named

Range, PKT 7.62 mm ball

1,500 m effective, ground target

Range, 2A72 30 mm, light armour (BTR-80A, BTR-82A)

2,000 m maximum, light armoured targets; round not named

Range, 2A72 30 mm, personnel (BTR-80A, BTR-82A)

4,000 m maximum (Wikipedia); effective (RWW), personnel (Wikipedia, round not named); RWW: HE-T and AP-T

Range, 2A72 30 mm, helicopters

1,500 m maximum, helicopters; round not named

Detection range, TKN-AI commander's sight (BTR-82A)

3 km ground targets, with laser rangefinder, claim

Crew (BTR-80, BTR-82A)

3 + 7 commander, driver, gunner + 7 troops

Crew (BTR-70)

2 + 9

Elevation, KPVT (BTR-80)

-4° to +60°

Elevation, KPVT (BTR-70)

-5° to +30°

Ammunition (BTR-70, BTR-80)

500 × 14.5 mm; 2,000 × 7.62 mm 7.62 mm figure per WEG 2015, WEG BTR-70 sheet, JANESRG and Wikipedia; WEG BTR-80 figure kept as "also" (transcription error)

Ammunition, 2A72 (BTR-80A)

300 × 30 mm

SEEN WITH 2S23 Nona-SVK; 1V152 [Kapustnik-B command and observation post]; BREM-K; RKM-4; R-149BMR Kushetka-B; RB-531B Infaua; Taifun-M

SOURCES AR19 ATP CINA DND GS14 GS15 GS16 IISS25 JANESRG KOLESA ORYX1 RWW WEG24 WEG2015 WP-EN13 WP-EN25 WP-RU42 WP-RU43 WP-RU44 WP-RU45 WSPOT9 WSPOT10 WSPOT11 WSPOT12

BTR-90 Rostok БТР-90 «Росток»

WH-AT

8x8 amphibious armoured personnel carrier with 30 mm turret In service 2008

OPERATED BY Unidentified unit, 38th Research Institute, Kubinka (storage) R 2023 L seen with 38th Research Institute, Kubinka

INDICATES The BTR-90 was never mass-produced and is not a standard fielded type. One seen near Avdiivka in Oct 2023 reportedly came from storage at the 38th Research Institute, Kubinka. It does not indicate a particular unit type.

RECOGNISE Cab Pointed nose; driver sits in mid-hull just ahead of the turret. Hull slightly larger and higher than BTR-80.

Body Welded steel hull, V-shaped belly; troop compartment behind driver and turret, engine at rear. Rostok turret is BMP-2 type, 30 mm gun with ATGM on turret.

Axles Eight wheels in two groups, two pairs at front and two at rear; split side doors between the groups; front four wheels steer.

Details Three smoke grenade launchers each side; AGS-17 on rear left of turret; periscopes in place of windscreens. Seen in Ukraine with white V marking.

NOT TO BE CONFUSED WITH BTR-80 / BTR-82A p. 20 Rostok has a BMP-2 type turret with ATGM launcher on the turret; hull larger and higher than BTR-80 (per WEG and Wikipedia).

VARIANTS BTR-90 Rostok BMP-2 type turret with 30 mm 2A42, PKTM, AGS-17 and Konkurs-M launcher. No fire control system or thermal imager (ru Wikipedia). The variant seen in Ukraine in Oct 2023.



Magdevans00, CC BY-SA 4.0

MOBILITY 8x8 wheeled, amphibious by two water jets without preparation (WEG); cross-country ability claimed comparable to tracked vehicles.

Weight	22,000 kg combat weight also 20.92 t WEG2015
L × W × H	8.05 × 3.06 × 3.05 m length as WEG gives it, undefined; 8.2 m is hull length
Road speed	100 km/h maximum, maximum also over 100 km/h WP-RU40 AR17
Road range	800 km WEG 2015 and Russian sources give 800 km; WEG online 850 km kept as 'also' (likely transcription error) also 850 km WEG21
Swimming	amphibious, no preparation; 9 km/h afloat maximum, two water jets also 12 km/h afloat WP-RU40 AR17 MIL15 DEFUA1
Bridge load class	21 wheeled metric 22 t × 0.94, rounded up, estimated
Ground clearance	0.51 m

Range, 9M113 Konkurs	4 km maximum effective, 9M113 Konkurs (WEG15; also AT-5B Konkurs-M, 4,000 m), wire-guided SACLOS, four missiles; WEG15 term; WEG online calls 70 m - 4 km 'operational range'
Minimum range, 9M113 Konkurs	70 m 9M113 Konkurs, lower bound of WEG 'operational range'
Range, 30 mm Frag-HE	4,000 m maximum effective, 2A42 gun; day [2,500 m night, WEG15]; WEG online gives 4,000 m maximum and 2,000 m effective, ammunition not stated also 2,000 m WEG21
Range, 30 mm APFSDS-T	2,500 m maximum effective, 2A42 gun; day and night
Range, 30 mm grenade (AGS-17)	800 to 1,700 m effective, AGS-17 (WEG: AG-17) automatic grenade launcher; ground targets also up to 1,700 m AR17
Crew	3 crew (driver, commander, gunner) plus 7 infantry
Main gun, 2A42 30 mm	500 rounds Rostok variant, BMP-2 type turret
Armament	30 mm 2A42, 7.62 mm PKT/PKTM coaxial, AGS-17 30 mm grenade launcher, four ATGM Rostok variant as seen in Ukraine; ATGM is Konkurs-M per Russian sources and ARECOG; WEG15 lists AT-5 Konkurs or AT-5B Konkurs-M
Armour	Defeats 14.5 mm over frontal arc welded steel hull; AP 7.62 mm and fragments over other arcs (DEFUA2)

SOURCES AR17 DEFBLG DEFUA1 DEFUA2 MIL15 WEG21 WEG2015 WP-EN22 WP-RU40

BRDM-2 БРДМ-2

WH-AT-A

Amphibious 4x4 armoured scout car for reconnaissance and patrol; base for NBC variants In service 1962

OPERATED BY reconnaissance battalion, reconnaissance

companies of division or brigade [D 2016 M]

RKhBZ battalion, BRDM-2RKH: divisional chemical defence battalion or company, regimental chemical defence company or platoon, reconnaissance battalion (Soviet structure) [D 1999 L]

Airborne Forces (VDV) units, Airborne Forces (VDV) units; no formation named [R 2023 L]

INDICATES IISS [late 2024] lists only 'some' BRDM-2/-2A with the Army and BRDM-2A with the National Guard. A turreted BRDM-2/-2A/-2MS suggests a reconnaissance subunit or a Rosgvardia unit; one with rear pole racks, NBC reconnaissance.

RECOGNISE Cab Boat-like bow; commander and driver side by side at front behind windscreen with armoured shutters; two circular roof hatches forward of turret are the only way in.

Body Box-like hull: vertical lower sides and rear, upper sides slope inwards, engine deck slopes down to rear. Small conical flat-topped turret, no hatch, on centre of roof.

Axles Two axles, 4x4, central tyre pressure; two pairs of powered belly wheels between the road wheels, raised when travelling on roads.

Details Trim vane folded under nose when not in water; single water-jet at hull rear. Vision blocks centrally on both hull sides.

NOT TO BE CONFUSED WITH BTR-60PB, BTR-70 p. 20 Same turret type, but 8x8 hulls with four axles.

FUG, FUG-70 (Hungary) Also rear-engined amphibious scout cars, but with twin water-jets; BRDM-2 has one.

BRDM-1 Engine at front and no turret; BRDM-2 has rear engine and centre roof turret.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Fully amphibious with water-jet, about 10 km/h afloat; belly wheels help cross trenches up to 1.25 m; unarmoured tyres. Crew exits only through roof hatches.

Weight, combat 7.0 t

L × W × H 5.75 × 2.35 × 2.31 m

Road speed 90 km/h maximum

Road range 750 km maximum

Range, 14.5 mm API/AP-T (KPVT)

Range, 7.62 mm ball (PKT)

2,000 m effective, ground target, day

100–1,500 m effective (with sight adjustments), ground target, day; FAS calls 1,500 m maximum aimed range

Detection range, TPKU-2B

commander's sight

Detection range, TKN-1S night sight

3,000 m observation range, day; 5x

periscope, target not stated

250–300 m observation range, night, with OU-3 infrared illuminator; target not stated

SOURCES D24A D24B FAS4 IISS25 JANESG ORYX1 RWW WEG17 WEG2015 WP-EN20 WP-RU31 WWW14

Tigr and Vodnik armoured cars ГАЗ-233014 «Турп», «Богучк»

WH-AT

Light 4x4 armoured cars for escort, reconnaissance, patrol and troop carriage In service 2006

OPERATED BY reconnaissance battalion [D 2016 M]

Rosgvardia unit, OMON; police SPM-1 and SPM-2 versions [R 2024 M]

Reconnaissance and special operations units, force protection for forward air controller detachments, Syria [R 2019 L]

Naval Infantry, branch; no unit named [R 2026 L]

INDICATES Tigr or Tigr-M suggests reconnaissance, airborne, Spetsnaz or National Guard elements; Army lists hold it without counts. Vodnik is rare, reported mainly with special forces.

RECOGNISE Cab Tigr: HMMWV-like layout, bonnet forward, crew cab in the middle, troop section behind. Vodnik 39371: nose pushed forward over the wheel arches, three-seat cab.

Body Tigr-M: one large square hatch replaces the first Tigr's two-leaf hatch (rear per WEG, roof turntable per ru-Wikipedia). Vodnik: wider hermetic hull, modular rear body.

Axles Both are 4x4. Tigr is not amphibious; Vodnik is amphibious and swims on its wheels.

Details Tigr roof mount: 7.62 or 12.7 mm MG or 30 mm AGL; Tigr-M may carry an Arbalet-DM remote station. Vodnik may have a BTR-80 type turret.

Details Tigr-M SpN with 2023 quick-mount kit: add-on armour panels over windshield, wheel arches, radiator grille and front doors.

NOT TO BE CONFUSED WITH BTR-80 (turret of Vodnik) p. 20 Vodnik with BPU-1 turret is a 4x4; the BTR-80 is an 8x8 carrying the same turret type.

Tigr versus Vodnik Vodnik is wider (2.6 m against 2.2 m, Wikipedia figures) with a hermetic amphibious hull.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Tigr fords 1.2 m and is not amphibious (WEG). Vodnik is amphibious, afloat speed 4 to 5 km/h (Wikipedia).

Weight, combat, Tigr 6 t GAZ-2975 baseline in WEG

Weight, Vodnik 7,500 kg combat

L × W × H

5.70 × 2.2 × 2.0 m WEG 2015 (GAZ-2330/Tigr-M page) and Wikipedia agree; WEG online 4.61 m kept as 'also' (likely transcription error)

Road speed, Tigr 140 km/h maximum

Detection range, Arbalet-

DM TV camera

Crew, Tigr

2.5 km claimed, target identification; thermal imager 1.5 km, claim

2 (driver, commander) WEG 2015 and Wikipedia agree; carries 5 to 10 personnel by variant (WEG 2015); WEG online crew 1 kept as 'also'

Crew, Vodnik

Armament, Tigr

2 7.62 or 12.7 mm MG, or 30 mm AGL, roof mounted crew-served

SOURCES AR55 AR56 AR57 ATP CNA IISS25 RWW WEG65 WEG2015 WP-EN77 WP-EN78 WP-RU122

Typhoon family (Typhoon-K, Typhoon-U, Typhoon-VDV, Linza)

KamAZ-63968, KamAZ-53949, KamAZ-4386, Урал-63095 «Тайфун»

WH-AT

Mine-protected 4x4 and 6x6 armoured trucks: troop carrier, recon, medical, weapon carrier In service 2014

OPERATED BY Spetsnaz brigade [O 2018 L] seen with 22nd Guards

Separate Spetsnaz Brigade (Rostov oblast), troop trials of 4x4 Typhoon-K; identification tentative

engineer-sapper brigade, assault and obstacle-clearing battalion

[O 2017 L] seen with Guards engineer-sapper brigade, Western Military District (Nizhny Novgorod oblast), field exercise Mar 2017

combined arms army, separate special-purpose (deep reconnaissance) company [R 2021 L]

military district [R 2021 L] seen with Southern Military District, Central Military District (first recipients of Linza)

motor rifle brigade, INA; first K-53949 built went to this brigade for trials [R 2021 L]

seen with 55th Separate Mountain Motor Rifle Brigade (41st Combined Arms Army, Kyzyl)

Military police units, sent to Syria with KamAZ-63968, early 2017

[R 2017 L]

INDICATES Typhoon-K went to spetsnaz, reconnaissance, engineer and military police units; first 4x4s were trialled by a mountain motor rifle brigade. Typhoon-VDV went first to reconnaissance and spetsnaz units. Named units are few: treat as a hint.

RECOGNISE Cab Typhoon-K 4x4: cab-forward, no bonnet; four doors in front part of hull, one in rear plate. Typhoon-VDV: three front doors, one rear. Typhoon-U: bonnet, three-seat cab.

Body Typhoon-K 6x6: separate armoured troop module behind cab, rear ramp plus side door, roof hatches. VDV: load-bearing hull, no frame. Linza: medical capsule.

Axles 2 axles: KamAZ-53949, Linza, Typhoon-VDV. 3 axles:

KamAZ-63968, Ural-63095. Height-adjustable hydropneumatic suspension (Linza: springs). 6x6: 16.00R20 run-flat tyres.

Details Typhoon-VDV: remote turret with 30 mm 2A42 gun and 762 mm gun, smoke launchers on sides. Typhoon-K: remote machine-gun station or none; V-shaped underside.

NOT TO BE CONFUSED WITH Tigr (GAZ-2330), against Typhoon-VDV p. 22. INA as a visible mark. RT expert: Tigr is a lighter class with weaker armour, mine protection and payload.

SEEN WITH Linza armoured ambulance (KamAZ-53949); 2S41 Drok 82 mm self-propelled mortar (Typhoon 4x4 chassis); Typhoon-PVO (KamAZ-4386) for man-portable SAM teams

VARIANTS KamAZ-53949 Typhoon-K 4x4 Short two-axle cab-forward vehicle; four doors in front part of hull, one rear door. Carries 2 crew and 8 troops (10 seats); remote machine-gun station; STANAG 4569 level III mine protection claimed (WEG), 3b (Russian Wikipedia). KamAZ-63968

Typhoon-K 6x6 Three axles; longer troop module with rear ramp and side door. Carries 14 to 16 troops; heavier (21 t curb).

Ural-63095 Typhoon-U Bonneted three-axle truck with armoured cab and separate armoured module. 26.5 t; 16 troops plus 3 in cab (design figures). Not in IIS25.

KamAZ-4386 Typhoon-VDV Frameless 4x4 with remote turret: 30 mm 2A42 gun and 762 mm PKTM; smoke launchers on turret sides. Air-droppable design; gun armament; 2 crew plus 6 troops. KamAZ-53949 Linza Two-seat cab with medical capsule in place of troop module. Casualty evacuation: 4 lying or 6 seated (Russian Wikipedia).



Alexey Vasilenko, CC BY-SA 4.0

MOBILITY Wheeled, all-wheel drive; 4x4 Typhoon-K not amphibious, fords 1.5 m unprepared. Typhoon-VDV designed for parachute delivery (system in development 2021).

Weight, Typhoon-K 4x4 (KamAZ-53949)

Weight, Linza

13,700kg curb weight, claim also more than 15 t gross WP-RU132

13,700kg WEG curb weight, identical to Typhoon-K 4x4; probably copied, claim also 15.2 t (ZSA-T), 15.4 t (ZSA-P) combat weight WP-RU131

Weight, Typhoon-K 6x6 (KamAZ-63968)

Weight, Typhoon-VDV

21 t curb weight; gross 24 t also 22 t combat weight WP-RU130

not more than 15 t gross weight, claim also 13.5 t gross RT5

Weight, Typhoon-U

Road speed, Typhoon-K 4x4

26.5 t gross, design figure

100 km/h claim

Road speed, Typhoon-K 6x6

also 105 km/h WP-RU132

105 km/h

Road speed, Typhoon-VDV

also 110 km/h WP-RU130

100 km/h claim

Road speed, Linza

also up to 130 km/h RT5

100 km/h maximum, WEG Linza page repeats Typhoon-K 4x4 figures, claim also 105 km/h WP-RU131

Crew and troops, Typhoon-K 4x4

2 crew, 8 troops claim

Crew and troops, Typhoon-K 6x6

3 crew, 14 troops Russian Wikipedia infobox; English Wikipedia gives up to 16 depending on configuration also 2 crew, up to 16 troops WP-EN84

Crew and troops, Typhoon-VDV

2 crew, 6 troops claim also capacity 7 persons RT5

Capacity, Linza

up to 10 wounded claim

Armament, Typhoon-K 4x4

also up to 4 lying or 6 seated WP-RU131 **stabilized RCWS; 7.62mm MMG or 12.7mm/14.5mm HMG** remote weapon station, claim

Armament, Typhoon-VDV

30 mm 2A42 gun with coaxial 7.62 mm PKTM; 300 rounds and 1,000 rounds Spitsa BM-30-D (32V01) module; elevation -10 to +60 degrees, claim

Mine protection, Typhoon-K 4x4

STANAG 4569 level III claimed: 8 kg TNT under floor, 10 kg under wheels, claim also level 3b WP-RU132

Mine protection, Typhoon-VDV

6 kg TNT under wheel, 4 kg under belly STANAG 4569 level 2a, developer figure, claim

Payload, Typhoon-K 4x4

2,000 kg claim

SOURCES BMPD10 IIS25 IZV9 RG213 RT5 WEG71 WP-EN84 WP-RU130 WP-RU131 WP-RU132 WP-RU133 WP-RU134 ZR6

Other armoured cars and utility vehicles (family)

WH-AT

Light armoured cars and MRAP-type utility vehicles for security and special units In service 1999

OPERATED BY Rosgvardia unit, Z-STs Akhmat: Chechen forces; vehicle also on National Guard list, INA (first 20 delivered July 2022)

R 2024-10 M seen with Chechen forces under Kadyrov (Akhmat)

Army, Vystrel: on Army AUV list; no unit or subunit given **R 2024-12 M**
naval infantry brigade, Vystrel on Naval Infantry equipment list; no unit named **R 2024-12 M**

Airborne Forces, Toros: on Airborne Forces AUV list; no unit named **R 2024-12 M**

INDICATES Most members point to Rosgvardia, special-purpose, reconnaissance or airborne users. Vystrel is also on Army, Naval Infantry and Border Guard lists. Z-STs Akhmat points to Chechen forces.

RECOGNISE Cab Engine at front, crew behind (Buran; Patrul is bonnet-type). Vystrel: separate engine and crew compartments. Buran: three doors per side with firing ports.

Body Length ranks them: Vystrel 5.3 m, Buran 5.81 m, VPK-Ural 6.0 to 6.5 m, Patrul 7.4 m, Z-STs 8.43 m. Sarmat-3 is the lightest at 3.5 t.

Axles 4x4 on Vystrel, VPK-Ural, KamAZ-435029 Patrul, Buran and Sarmat-3. Z-STs Akhmat (KamAZ-5350 chassis) is 6x6; Oryx also lists a Patrul 6x6.

Details Kord 12.7 mm in an armoured box (VPK-Ural); roof hatch with Arbalet-DM (Z-STs); 30 mm BM-30D turret on some Vystrel; no turret on Buran; open cargo area on Sarmat-3.

NOT TO BE CONFUSED WITH Tigr (GAZ-233014) p. 22 Buran is 685 mm longer and 140 mm narrower than Tigr; its body is built on a four-wheel-drive truck chassis.

Ural-VV (Ural-432009) Ural-432009 of the Ural plant, not the VPK-built AMN-590951 VPK-Ural; IISS lists Ural-VV for the National Guard. Visible difference INA.

SOURCES AR34 AR35 DBSAR DEFUA IISS25 LEXCN MIL34 ORYX1 WEG38 WP-EN46 WP-EN47 WP-RU73 WP-RU74



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Wheeled road-and-track vehicles, mostly 4x4; Z-STs and a Patrul version are 6x6. Z-STs fords 1.75 m; VPK-Ural is not amphibious (WEG). Other fording INA.

Weight, Vystrel (BPM-97)	10.5 t
Weight, VPK-Ural	14.5 t WEG says curb weight; Russian Wikipedia says gross weight
Weight, Z-STs Akhmat	18 t ARZSTS says "up to 18,000 kg"
Weight, Patrul (KamAZ-435029)	12.7 t Combat weight

Crew and seats, VPK-Ural	10 seats
Crew and passengers, Z-STs Akhmat	2 crew, 8 passengers
Crew and passengers, Vystrel (BPM-97)	2 crew, 8 to 12 passengers

Crew, Sarmat-3	4 crew; 900 kg cargo
Armament, Vystrel (BPM-97)	7.62 or 12.7 mm MG, 14.5 mm KPV, or 30 mm 2A42 turret Options by fit; BM-30D adds AG-30 grenade launcher

2

Fires

IN THIS CHAPTER

MRLs and heavy flamethrowers	26
Self-propelled guns and mortars	29
Towed artillery	32
Ballistic missiles	33

LOOKING FOR THESE? THEY ARE IN OTHER CHAPTERS

SYSTEM	WHAT IT IS	CHAPTER	PAGE
IV12 artillery fire control complex (IV13, IV14, IV15, IV16)	Tracked command, observation and fire direction vehicles, SP howitzer battalion	7 Command and signals	67
PRP-4A Argus / PRP-4M Deytery	Mobile artillery reconnaissance post on BMP-1 chassis	7 Command and signals	68
Artillery command vehicles IV18/IV19, IV119, IV152, IV1003	Artillery battery and battalion command, observation and fire direction vehicles	7 Command and signals	67
Zoopark-1 / Zoopark-1M	Tracked radar that locates enemy guns, mortars, rockets and missile launchers	4 Radars	42
ARK-1M Rys	Artillery weapon-locating and fire-control radar	4 Radars	43
SNAR-10 / SNAR-10M1 Leopard	Tracked battlefield surveillance and artillery-adjustment radar	4 Radars	44
1B75 Penicillin	Passive acoustic-thermal gun locator	4 Radars	43
Orlan-10	Catapult-launched recon UAV for artillery spotting and EW	9 Unmanned aircraft	78
Lancet	Loitering munition for precision strikes on artillery, air defence and armour	9 Unmanned aircraft	85

BM-21 Grad 122 mm MRL

BM-21 «Град»

NATO: M1964

WH-AT

Truck-mounted 40-tube 122 mm area-fire rocket launcher (divisional and brigade MRL) In service 1963

OPERATED BY motor rifle brigade, MRL battalion; 6 launchers per battery, 18 per battalion [D 2016 M](#)

artillery regiment, one Grad MRL battalion in division field artillery regiment, being replaced by Tornado-G [D 2019 M](#)

motor rifle regiment, BTG, attached MRL element, 3 in one BTG

[O 2022-04 M](#)

motor rifle brigade, BTG, attached BM-21 battery, 1 battery

[O 2022-04 M](#)

NUMBERS Holdings 36 in Airborne Forces (IISS estimate) (2024)

INDICATES A group of about 18 launchers points to a brigade MRL battalion; 6 launchers to one battery. A few launchers inside a battalion tactical group suggest an attached MRL element.

RECOGNISE Cab Unarmoured Ural 6x6 truck cab (Ural-375D; Ural-4320 on BM-21-1). The 40-tube pack can be turned away from the cab.

Body Rear pack of 40 tubes in four rows of ten, each tube 3 m long, on a welded cradle; elevates 0 to +55 degrees. Traverse 102 left, 70 right of the vehicle.

Axles Three axles, 6x6 truck chassis, wheeled; not amphibious.

Details Stabilizing jacks at the rear for firing (stated for the BM-21V; the text says the BM-21 has the same). Reload rockets carried on separate resupply trucks.

NOT TO BE CONFUSED WITH BM-21V Grad-V (9P125) 12 tubes on a GAZ-66B 4x4 truck, airborne version
9P138 Grad-1 36 tubes on a ZIL-131 6x6 truck
RM-70 (Czechoslovak) 40 tubes on an 8x8 Tatra 813 with an armoured cab and room behind the cab for 40 spare rockets

SOURCES ATP CNA IISS25 MISS6 RUSPL RWW WEG31 WEG2015 WP-EN32 WP-RU58



MILRU (Ministry of Defence of the Russian Federation), CC BY 4.0

MOBILITY Unarmoured truck chassis on wheels with central tyre inflation. Fords 1.5 m without preparation. Not amphibious.

Weight, combat 13.7 t BM-21 on Ural-375D, loaded with crew

L × W × H 7.35 × 2.4 × 3.09 m travelling

Road speed 75 km/h maximum, hard road, loaded

Road range 750 km

Range, 9M22U HE-frag

20.4 km maximum, 9M22U (M-210F) HE-fragmentation rocket

Minimum range, 9M22U HE-frag

5 km 9M22U (M-210F) HE-fragmentation rocket

Range, 9M28F HE-frag

15 km maximum, 9M28F HE-fragmentation rocket

Minimum range, 9M28F HE-frag

1.5 km minimum effective, 9M28F HE-fragmentation rocket

Tornado-G

9K51M «Торнадо-Г»

WH-AT

122 mm 40-tube truck MRL; automated Grad upgrade for brigade and division artillery In service 2012

OPERATED BY motor rifle division, artillery regiment, MRL battalion (replacing Grad battalion) [D 2019 H](#)

tank division, artillery regiment, MRL battalion [D 2019 H](#)

motor rifle brigade, brigade artillery, MRL battalion, INA for Tornado-G; it replaces the BM-21, 18 per brigade MRL battalion (RWW p.100); 18 delivered to one brigade 2012 (RBASE) [D 2019 M](#)

motor rifle brigade, 18 [O 2012-04 M](#) seen with 20th Guards Separate

Motor Rifle Brigade

INDICATES Tornado-G replaces the Grad in the MRL battalion of motor rifle and tank brigades and of division artillery regiments. A group of them suggests a manoeuvre brigade or division artillery group nearby.

RECOGNISE Cab Ural-4320 6x6 truck; cab seats driver and two. Launcher is laid and returned to travel position from inside the cab.

Body Single pack of 40 122 mm tubes, 10 × 4, on the rear body; traverses 102 deg left and 70 deg right, elevates 0 to 55 deg.

Axles Three axles, all driven (6x6); Ural-4320 has central tyre pressure regulation.

Details Rebuilt BM-21 Grad on Ural-4320 (about 20 rebuilt in 2019); additions are GLONASS navigation, a ballistic computer and a video monitor in the cab.

NOT TO BE CONFUSED WITH BM-21-1 Grad (Ural-4320) p. 26 Same truck and 40-tube pack. Tornado-G lays from the cab with no crew at the launcher; no reliable external mark found in text sources.

9A51 Prima 50 tubes instead of 40 on the same Ural-4320 chassis.

9K515 Tornado-S p. 26 12 tubes of 300 mm on an 8x8 MAZ-543M chassis, against 40 small tubes on a 6x6 Ural.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY 6x6 truck; WEG claims small-arms protection, WEG 2015 BM-21 page gives none. Fords about 1.5 to 1.75 m. Can fire from an unsurveyed position and leave straight after firing.

Weight, combat 14 t

L × W × H 7.35 × 2.5 × 3.1 m travelling

Road speed 85 km/h maximum

Road range 650 km cruising range

Range, 9M521 HE-frag

40 km maximum, 9M521 extended-range HE-frag rocket; attribution to 9M521 from RUWIKI (Karpenko), WEG, ZR and WWW1 give 40 km as system maximum

Range, 9M522 HE-frag

37.5 km maximum, 9M522 rocket, separable HE-frag warhead

Range, 9M538 HE-frag

20 km maximum, 9M538 rocket, enhanced monobloc HE-frag warhead, claim

Range, 9M539 HE-frag

20 km maximum, 9M539 rocket, separable HE-frag warhead; Tecmash statement at IDEX 2019, claim

SOURCES CNA D244 EDR3 GS33 IISS25 JF18 MAILRU3 ORYX1 RBASE RDP7 RWW TASS21 WEG68 WEG2015 WMC22 WP-RU126 WWW12 WZ5

BM-27 Uragan 9П140 «Ураган»

NATO: M1977

WH-AT

220 mm 16-tube MRL on ZIL-135LM 8x8 for area fire and remote minelaying at 10–35 km In service 1975

OPERATED BY rocket artillery regiment, three rocket launcher battalions, 54 per regiment (3 battalions of 18) [\[D 1991 M\]](#) rocket artillery brigade, rocket launcher battalions, 54 per brigade (3 battalions of 18) [\[D 1991 M\]](#) self-propelled artillery regiment, Uragan rocket artillery battalion, 8 per battalion [\[R 2017-12 M\]](#)

INDICATES A 9P140 points to army-level rocket or artillery brigade fires, or to a Uragan battalion (8 launchers) in a division artillery regiment; the MoD also planned such battalions for combined arms brigades. In the Airborne Forces it indicates the artillery brigade.

RECOGNISE Cab ZIL-135LM cab with a blast shield that is raised during firing.

Body 16 tubes in three banks at the rear: two lower rows of six, upper row of four. In travel the pack faces rearward; muzzles covered by a sheet-metal cover when loaded.

Axles Four axles, 8x8; only the front and rear axles steer. Two petrol engines, each driving the wheels on one side.

Details Two manually emplaced hydraulic jacks at the rear of the chassis are lowered for firing. Newer launchers sit on a 6x6 BAZ-69092 chassis.

NOT TO BE CONFUSED WITH 9T452 transloader Same ZIL-135 chassis but a flatbed with 16 rockets in two stacks either side and a loading device between them; no tube pack.

BM-21 Grad p. 26 40 tubes on a Ural-375 truck, against 16 larger tubes on an 8x8 ZIL-135.

9A52 Smerch 12 tubes of 300 mm on a MAZ-543A chassis, against 16 tubes on a ZIL-135.



Unknown author, CC BY 4.0

MOBILITY Unarmoured 8x8 petrol truck; rear jacks must be lowered to fire; fords 1.2 m unprepared; 60% gradient.

Weight, combat	20 t
L × W × H	9.63 × 2.8 × 3.23 m travel position
Road speed	65 km/h maximum
Road range	500 km cruising

Range, 9M27F HE-frag	35 km maximum, 9M27F HE-frag rocket
Minimum range, 9M27F HE-frag	10 km 9M27F HE-frag rocket

Range, 9M27K cluster	35 km maximum, 9M27K rocket, 30 9N210 HE-frag submunitions
Minimum range, 9M27K cluster	10 km 9M27K rocket, 30 9N210 HE-frag submunitions

Range, 9M27K2 AT mines	35 km maximum, 9M27K2 rocket, 24 PTM-1 anti-tank mines
Minimum range, 9M27K2 AT mines	10 km 9M27K2 rocket, 24 PTM-1 anti-tank mines

SEEN WITH 9T452 transloader; ZIL-135LM (1)

SOURCES CNA D245 FAS9 GS36 IISS25 IZ17 MIL63 MIL64 MIL110 MPER2 ORYX1 RUSIPL RWW W1R WEG75 WEG2015 WP-EN87 WP-EN88 WP-RU137

Uragan-1M 9K512 «Ураган-1М»

WH-AT

Heavy 220/300 mm MRL with swappable rocket pods; successor to 9K57 Uragan In service 2007

OPERATED BY self-propelled artillery regiment, rocket artillery battalion (planned replacement for 9K57 Uragan, MoD statement; no holding confirmed) [\[R 2017-12 L\]](#)

artillery brigade, MRL battalion (Uragan 'being succeeded' by Uragan-M1; no holding confirmed) [\[R 2019-10 L\]](#)

INDICATES Planned for divisional artillery regiments and army artillery brigades as the Uragan successor. Few were built and front-line use is unconfirmed, so a sighting is notable and should be checked against the 6x6 9K57-1.

RECOGNISE Cab Forward-control cab on the MZKT-7930 8x8 chassis, seating the driver and two passengers.

Body Launcher on a turntable at the rear carrying two launcher pods instead of a fixed tube rack.

Axles Four axles (8x8), all driven; same chassis family as the 9T249 transporter-loader.

Details 300 mm pods hold 6 tubes each in two layers of three; 220 mm pods hold 15 tubes each. Pods can be mixed, one of each calibre.

NOT TO BE CONFUSED WITH 9K57-1 Uragan-1 p. 27 6x6 BAZ-69092 chassis with armoured cab, three axles not four; Russian press also calls it Uragan-1M.

9P140 Uragan p. 27 Fixed rack of 16 tubes on a ZIL-135 8x8; no detachable pods.

9A52-2 Smerch p. 28 Fixed 12-tube block in 3 rows of 4 on a MAZ-543M 8x8; Uragan-1M has two separate 6-tube pods.

9A52-4 Tornado p. 28 KamaZ-63501 8x8 truck with a replaceable 6-rocket module (WEG 2015); Uragan-1M carries two pods on the larger MZKT-7930.



Ministerstwo Obrony Rosyjskiej Federacji | Министерство обороны Российской Федерации | Ministry of Defence of the Russian Federation, CC BY 4.0

MOBILITY Heavy 8x8 truck; fords 1.4 m, climbs 45% gradient, crosses 2.0 m trench.

Weight, combat	40 t
L × W × H	13.0 × 3.0 × 3.29 m travelling
Road speed	70 km/h maximum, maximum, road
Road range	1,000 km maximum, road

Range, 9M27F HE-frag 220 mm	35 km maximum effective, 9M27F unguided HE-frag rocket
Minimum range, 9M27F HE-frag 220 mm	10 km minimum effective, 9M27F unguided HE-frag rocket

Range, 9M27K cluster 220 mm	35 km maximum effective, 9M27K rocket, 30 9N210 fragmentation submunitions
Range, 9M55F HE-frag 300 mm	70 km maximum, 9M55F HE-frag rocket with trajectory correction

Minimum range, 9M55F HE-frag 300 mm	20 km 9M55F HE-frag rocket
-------------------------------------	----------------------------

SEEN WITH 9T249

SOURCES AR63 CNA D24A2 D24B2 IISS25 IZ17 ORYX1 RAND RWW U247 W1A3 W1C3 W1E2 WEG74 WEG2015 WP-EN86 WP-RU136 ZR5

BM-30 Smerch 9A52-2 «Смерч»

NATO: 280mm M1983

WH-AT

Heavy 300 mm 12-tube rocket launcher for deep area fire; Tornado-S adds guided rockets In service 1987

OPERATED BY rocket artillery brigade, three groups of 4 x 9A52-2 in ATP fig. 5-16; whether each group is a battalion or a battery is not readable from the text, 12 x 9A52-2 shown in ATP fig. 5-16 (3 groups of 4) D 2024 M

rocket artillery brigade, 7 delivered April 2017, 5 more announced

O 2017-04 M seen with rocket artillery brigade, Tver Oblast, Western MD (named as the 79th Guards Rocket Artillery Brigade by Wikipedia only)

rocket artillery brigade O 2026-09 M seen with 79th Guards Rocket Artillery Brigade (Zapad group), Tornado-S, 2026-09, 232nd Guards Rocket Artillery Brigade (Tsentr group), Tornado-S, 2026-08 and 2026-09

INDICATES *Smerch or Tornado-S points to an army-level rocket artillery brigade, not a manoeuvre brigade, whose MRL battalion holds BM-21. Expect deep fires out to 70-120 km and ammunition resupply by MAZ-543 transloaders.*

RECOGNISE Cab MAZ-543M 8x8 truck; driver cab front left, commander behind him; launch and fire-control cabin to the rear of the driver cab, one door each side, two small front windows.

Body Single package of 12 long 300 mm tubes over the rear deck, 3 rows of 4 (two banks of four with four more on top); traverses 30 deg each side, elevates to 55 deg.

Axles Four axles, 8x8; hydraulic jacks between the wheels of the 3rd and 4th axles are lowered to steady the rear of the launcher when firing.

Details Tornado-S (9A54): satellite navigation antenna on the roof of the fire control cab; otherwise laid out like the 9A52-2.

NOT TO BE CONFUSED WITH BM-27 Uragan (9P140) p. 27 16 tubes of 220 mm on a ZIL-135 chassis; Smerch has 12 larger tubes on the MAZ-543.

9A53 Uragan-1M p. 27 Two quick-change pods of six tubes on an MZKT-7930 8x8; Smerch has one fixed 12-tube package.

9A52-4 Tornado KamAZ-63501 8x8 truck with a single six-tube pod; Smerch has one fixed 12-tube package on the MAZ-543M.

9T234-2 and 9T255 transloaders Same MAZ-543A family chassis; carry 12 rockets and a crane (850 kg on the 9T255) with no launch tube package.

Vilkha (Ukraine) Ukrainian development that uses the same Smerch launcher vehicle; not separable by shape alone.

SEEN WITH MAZ-543M 8x8 chassis (1 per launcher); 9T234-2 transloader (TZM) (INA; carries 12 rockets); 9T255 transloader (Tornado-S) (INA; carries 12 rockets); 1K123 Vivary automated fire control system (up to 3 battalion and 18 battery vehicles per brigade system); 9S936 command and staff vehicle (Tornado-S); 1V168 ASUNO-S fire control; 1T12-2M survey vehicle; 1B44 meteorological complex; Zoopark counter-battery radar (INA; reported cueing Smerch fire 2024-03)

VARIANTS 9A54 Tornado-S Satellite antenna on the fire control cab roof; same MAZ-543M chassis and 12-tube package. GLONASS, automated laying and fire control; fires 9M544 and 9M549 guided rockets to 120 km; crew 3. 9A52-4 Tornado KamAZ-63501 8x8 truck with one six-tube 300 mm pod. Half the load; pod replaced after firing; lighter and more mobile. ISS25 lists none in Russian service.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Heavy 43.7 t 8x8; fords 1.1 m unprepared; not amphibious; rear jacks must be lowered to fire.

Weight, combat, 9A52-2 43.7 t loaded
Weight, empty 33.7 t 9A52-2 (WEG) and 9A54 (MISSTS)
L × W × H 12.1 × 3.05 × 3.05 m travelling
Road speed 60 km/h maximum
Road range 850 km 9A52-2; also 850 km on the WEG
Tornado-S page
Fording 1.1 m unprepared
Time into action 3 min from march to ready to fire
Time out of action 3 min
Ground pressure INA

Range, 9M55K cluster	70 km maximum, unguided, range-corrected rocket with fragmentation submunitions
Minimum range, 9M55K cluster	20 km unguided, range-corrected rocket with fragmentation submunitions
Range, 9M55F HE-frag	70 km maximum, unguided, separable HE-frag warhead
Minimum range, 9M55F HE-frag	20 km unguided, separable HE-frag warhead
Range, 9M528 HE-frag	90 km maximum, unguided extended-range rocket, separable HE-frag warhead, claim
Range, 9M544 and 9M549 guided	120 km maximum, Tornado-S guided cluster rockets, inertial and GLONASS, claim
Minimum range, 9M544 and 9M549 guided	20-40 km Tornado-S guided rockets; crew statement, claim
Employment range, Smerch	35-70 km practical, unguided rockets, Ukraine 2022, used at or near maximum range, observed
Employment range, Tornado-S	about 100 km practical, guided rockets against a command post (2026-08) and a fortified strongpoint (2026-09); MoD figures via state media, claim
Crew, 9A52-2	4
Crew, 9A54 Tornado-S	3
Tubes	12 x 300 mm 9A52-2 and 9A54
CEP, 9M55K cluster	150 m inertial range and attitude correction, no terminal guidance; also given for 9M528
CEP, 9M544 and 9M549 guided	5-10 m inertial and GLONASS; official claim, claim

TOS-1A Solntsepek TOC-1A «Солнценёк»

TR

Tracked 220 mm thermobaric and incendiary rocket launcher of the RKhBZ troops In service 1988

OPERATED BY RKhBZ brigade, flamethrower battalion of 3 batteries, each 6 BM-1 and 6 TZM-T; RWW places these battalions in NBC defence brigades and regiments and does not state the echelon, 18 per battalion (6 per battery) [D 2016 M](#)

RKhBZ brigade, TOS battalion of 3 companies of 6 launchers (ATP para 5-146), 18 per battalion [D 2024 M](#)

INDICATES TOS launchers belong to army or district RKhBZ units, not manoeuvre brigades. Seeing one suggests an attached flamethrower company or battalion supporting an attack or breach, since it must close to within about 6 km of the target.

RECOGNISE Hull T-72 tank hull with no gun turret; a large launcher box fills the hull roof. Paired with look-alike T-72-based TZM-T reload vehicles.

Armament 24 tubes of 220 mm in a heavy armoured box. Launchers shipped from Nov 2025 add a factory armour shell over the tubes that lifts open when the tubes elevate.

Turret No turret: the launcher box sits on a rotating platform and elevates to fire. Improvised anti-drone cover was fitted in the field before factory kits.

Suspension Standard T-72 tracked running gear with large-diameter road wheels. Road wheel and return roller counts not confirmed in cited sources.

NOT TO BE CONFUSED WITH TOS-1 Buratino 30 tubes, without the heavy armoured housing added on TOS-1A, which has 24 tubes. TZM-T reload vehicle Same T-72 hull but carries two rocket racks with a crane between them instead of a trainable launcher.



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Tracked T-72 chassis, not amphibious; fords 1.2 m; must close to within 6 km of the target to fire the longest-range thermobaric rocket.

Weight, combat	44.3 t
L × W × H	6.86 × 3.46 × 2.60 m
Road speed, maximum	60 km/h
Off-road speed, maximum	45 km/h

Range, thermobaric rocket	6,000 m maximum, MO.1.01.04M thermobaric rocket, 3.7 m, 217 kg
Range, incendiary rocket	6,000 m maximum effective, MO.1.01.04M.OP incendiary rocket, 3.7 m, 204 kg
Range, original short rocket	2,700 m maximum, MO.1.01.04 thermobaric rocket, 3.3 m, 173 kg, fired from TOS-1A
Range, minimum	600 m MO.1.01.04M rocket
Launch tubes	24 × 220 mm

SOURCES AR59 AR60 ATP CNA EMP GS34 MIL60 MIL61 MIL62 MISS14 RWW TASS22 TASS23 TASS24 TASS25 WEG69 WMC23 WP-EN79 WP-EN81 WP-EN82

2S5 Giatsint-S 2C5 «Гиацинт-С»

NATO: M1981 TR

Army-level 152 mm self-propelled gun for long-range counter-battery and deep fire In service 1975

OPERATED BY artillery brigade, gun battalion (152 mm), 18 per battalion (24 in forward armies); 2S5 or 2A36 or M-46 [D 1991 M](#)

Airborne Forces (VDV), Airborne Forces artillery; unit not named [R 2024 M](#)

INDICATES In Soviet doctrine the 2S5 marked the gun battalion of an army artillery brigade. Russian MoD reporting (2026) also places it with motor rifle brigade artillery, so alone it no longer points to army-level artillery.

RECOGNISE Hull Low tracked hull, engine and driver at front; front dozer blade; large spade or baseplate at rear, lowered to the ground before firing.

Armament Very long 152 mm barrel with five-part multi-baffle muzzle brake, held by a travel lock; with gun forward, length 8.95 m against a hull of 8.33 m.

Turret No turret: gun on an open mount at the rear of the hull roof, crew exposed; 15 deg traverse each side. Commander's cupola carries a 762 mm PKT and a searchlight.

Suspension Six rubber-tyred double road wheels and four return rollers per side; drive sprocket front, idler rear; torsion bars, track shoes 484 mm wide.

NOT TO BE CONFUSED WITH 2S3 Akatsiya Large enclosed turret at hull rear, 360 deg traverse, cupola offset left; double-baffle muzzle brake and fume extractor. 2S5: open mount, no turret, multi-baffle brake.

2S4 Tyulpan p. 31 Same chassis family, but a 240 mm mortar travels folded with muzzle forward, not reaching past the first road wheel; no long barrel over the front.



One half 3544, Public domain

MOBILITY Tracked; fords about 1 m; rear spade must be lowered to fire; gun traverse only 15 deg each side, so the hull is laid on the target line.

Weight, combat	28.2 t
L × W × H	8.33 × 3.25 × 2.76 m
Road speed	63 km/h maximum, road
Road range	500 km maximum, road

Range, HE-frag 30F29	28.4 km maximum, HE-frag 30F29, full charge
Range, HE-frag 30F29 reduced charge	21.5 km maximum, HE-frag 30F29 with reduced charge 4Zh48
Range, HE-frag 30F59	30.5 km maximum, HE-frag 30F59, improved aerodynamic shape, full charge
Range, rocket-assisted HE-frag 30F30	33 km maximum, rocket-assisted HE-frag
Range, Krasnopol guided 30F39	20 km maximum, laser-guided HE-frag Krasnopol
SEEN WITH	2A37 152 mm gun (1)

SOURCES BMPD2 FAS2 FM3 GS4 IISS25 IZ26 IZ233 JANESRG MF1 RIA22 RUSIPL RWW WEG3 WP-EN4 WP-RU7

2S19 Msta-S 2C19 «Mcma-C»

NATO: M1990 Farm

TR

152 mm tracked self-propelled howitzer of brigade, division and army artillery In service 1989

OPERATED BY motor rifle brigade, two self-propelled artillery battalions, each of three batteries of 6 howitzers; ATP notes 2S3 may be held instead, 18 per battalion (6 per battery), 36 per brigade; ATP: some battalions up to 24 systems [\[D 2024 M\]](#)

tank brigade, two howitzer (self-propelled artillery) battalions, three batteries of 6, 18 per battalion, 36 per brigade [\[D 2016 M\]](#)

INDICATES A battery of 6 or battalion of 18 Msta-S points to a self-propelled artillery battalion of a motor rifle or tank brigade, a division artillery regiment or an army artillery brigade. Wartime batteries may be smaller.

RECOGNISE Hull Tank-type tracked hull with T-72 hull geometry and T-80 running gear, built-in dozer blade, travel lock for the barrel. About 11.9 m long overall with the gun forward.

Armament Very long 152 mm L/47 barrel with double-baffle muzzle brake and a bore evacuator. 12.7 mm NSVT machine gun on the commander's cupola.

Turret Large turret over the hull centre; vehicle about 3 m high. Six 81 mm smoke launchers. 2S19M2: two stowage boxes on the right turret side.

Suspension T-80 type running gear: six road wheels and five return rollers per side, 580 mm tracks.

NOT TO BE CONFUSED WITH 2S3 Akatsiya (152 mm SP howitzer) Engine at front and turret toward the rear; much shorter L/27 barrel; four return rollers; about 28 t against 42 t. 2S5 Giatsint-S (152 mm SP gun) p. 29 152 mm 2A37 L/54 gun; most of the crew, all but the gunner, work outside the vehicle when firing; about 28 t.



Milru, CC BY 4.0

MOBILITY Heavy tracked vehicle, not amphibious. Fords 1.2 m unprepared; WEG gives 5 m with preparation. About 42 t, needs heavy bridging.

Weight, combat	42 t 2S19 and 2S19M1
Weight, combat, 2S19M2	43.24 t 2S19M2
L × W × H	11.917 × 3.584 m overall; 3.38 m without skirts × 2.985 m
Road speed, maximum	60 km/h

Range, HE-frag	24.7 km maximum, 30F45 (OF-45) HE-frag, full charge
Range, HE-frag base bleed	29 km maximum, 30F61 HE-frag base bleed (round 3V0F91)
Range, rocket-assisted	36 km HE rocket-assisted projectile (HERA/RAP), type not named
Range, guided	20 km maximum, Krasnopol laser-guided projectile; laser designation by an observer required

SOURCES AG5 ARMYT ATP CNA GS20 IIS25 MIL41 ORYX1 RWW TASS8 TASS9 WEG40 WP-EN4 WP-EN7 WP-EN51 WP-EN52 WP-UR78

2S35 Koalitsiya-SV 2C35 «Коалиция-СВ»

TR

152 mm self-propelled howitzers: tracked autoloader 2S35 and wheeled 8x8 2S43 In service 2020

OPERATED BY self-propelled artillery battalion, self-propelled howitzer battery, 6 per battery, 18 per battalion [\[D 2016 M\]](#)

self-propelled artillery regiment, 8 2S35 on troop trials

[\[R 2020-05 M\]](#)

artillery brigade, self-propelled artillery battalions, INA; planned replacement of the brigade's 2S19 by 2S35, reported 2017, not carried out by late 2024 [\[R 2019 L\]](#)

INDICATES 2S35 or 2S43 suggests new-build guns above manoeuvre-brigade level: 2S35 was planned for army artillery brigades and trialled in a division SP artillery regiment; 2S43 is reported with the 273rd Artillery Brigade, 34th Artillery Division. Rare (IIS: 8, 20).

RECOGNISE Hull 2S35: T-90-type tracked hull with reshaped front; driver, commander and gunner sit side by side in the front, three hatches across the hull front; engine at rear.

Armament 2S35: very long 152 mm 2A88 barrel (52 calibres) with muzzle brake, held by a travel lock on the glacis; Kord 12.7 mm remote weapon station on turret roof.

Turret 2S35: large unmanned turret mid-hull; 902 smoke dischargers on both sides and roof; laser warning sensors at front and rear of roof. From 2021 two boxes on turret front.

Suspension 2S35: six road wheels each side, idler front, drive sprocket rear, torsion bars; upper track covered by a skirt. Return roller count INA.

NOT TO BE CONFUSED WITH 2S19 Msta-S / 2S19M2 Msta-SM2 p. 30 2S19 has a crew of five with gun crew in the turret and a shorter 47-calibre barrel; 2S35 has three crew hatches across the hull front. 2S44 Giatsint-K Same BAZ 8x8 chassis as 2S43 but carries the gun of the 2A36 Giatsint-B towed gun.



Kirill Bortsenko, CC BY-SA 4.0

MOBILITY 2S35 tracked, 48–55 t, not amphibious. 2S43 wheeled 32 t, for road and off-road use, fits IL-76; fords 1.4 m.

Weight, 2S35	48–55 t combat
Weight, 2S43	32 t combat
L × W × H	9.63 × 3.78 × 2.2 m
Road speed, 2S35	65 km/h maximum

Range, 2S35 standard HE-frag	40 km maximum, standard (unassisted) rounds
Range, 2S35 guided or extended-range shell	80 km maximum, precision-guided or rocket-assisted projectile; munition not named, claim
Range, 2S43 HE-frag	24.5 km maximum, 2A64 gun; munition not stated by WEG or WPMALRU (Rostec: HE-frag more than 24 km)
Rate of fire, 2S35	16 rds/min maximum, claim
Rate of fire, 2S43	7 rds/min maximum, claim

SEEN WITH 2F66–1 transport-loader vehicle (KamAZ-6560), 92 rounds, loads 2S35 in 15 min

SOURCES AG2 ART1 ATP BMPD1 CNA IIS25 IZ23 IZMAL KZ1 KZ2 ORYX1 REDSTAR RG24 RGMAL RGMAL2 RGPAP RWW SHEP WEG1 WP-EN7 WP-EN8 WP-RU4 WP-RU5 WSN2

2S7 Pion / 2S7M Malka 2C7 «Пион»

NATO: M-1975 **TR**

203 mm heavy self-propelled gun for long-range fire on high-value targets In service 1975

OPERATED BY high-power artillery brigade, 2S7 battalions (45th bde: two battalions of eight, 2017), ATP fig. 5-18: 24 x 2S7 per brigade, 8 x 2S7 battalion block (OCR uncertain); 16 reported in 45th bde, 2017

D 2024-02 H

high-power artillery brigade, 16 x 2S7 Pion (with 8 x 2S4 Tyulpan)

O 2017-12 M

artillery brigade, 2S7 battalion (IZ17, 2017); 2S7M Malka (mil.ru via RDP18), 1 battalion; 12 x 2S7 (RUWIKI, 2019) **O 2018-04 M**

artillery brigade, 12 x 2S7 each (RUWIKI, 2019) **O 2019 L**

INDICATES A 203 mm tracked gun points to army-level artillery: a high-power artillery brigade or an army artillery brigade's heavy battalion, not a manoeuvre brigade's guns. Used on high-value targets, in urban fighting and, since 2022, for counter-battery fire.

RECOGNISE Hull Purpose-built tracked chassis using T-72/T-80 automotive parts. Four compartments: driver, power pack, crew, rear. Front dozer blade.

Armament Very long 203 mm 2A44 gun (about 12 m) on an open mount outside the hull; on the march it lies forward, overhanging the nose. No muzzle brake, no fume extractor.

Turret No turret. Gun on open external mount at hull rear. Large recoil spade at rear lowered to the ground when firing.

Suspension Seven twin rubber-tyred road wheels and six return rollers per side; drive sprocket front, idler rear, idlers lowered to the ground when firing. Torsion bar suspension.

NOT TO BE CONFUSED WITH 2S4 Tyulpan p.31 Also given the NATO name M-1975. A 240 mm self-propelled mortar, not a gun: no 12 m barrel overhanging the hull.



Boevaya mashina, CC BY-SA 3.0 de

MOBILITY Not amphibious. Needs firm ground to emplace; recoil spade and dozer blade used to dig in.

Weight, combat	46,500 kg
L × W × H	10.5 × 3.38 × 3.0 m
Road speed	51 km/h maximum
Range	500 km cruising

Range, HE-frag	37.5 km maximum, Conventional HE-frag, 110 kg projectile
Range, HE rocket-assisted	47.5 km maximum, RAP projectile, 103 kg

Rate of fire	1.5 rds/min (2S7); 2.5 rds/min (2S7M) maximum
--------------	---

Ammunition carried	4 (2S7); 8 (2S7M) On the vehicle
Crew	7 (2S7); 6 (2S7M) on the vehicle; 2S7 crew is 14 with the auxiliary vehicle crew (WIKI)

Elevation	0 to +60 deg; traverse 30 deg
-----------	-------------------------------

SEEN WITH Ammunition transport vehicle with the rest of the crew; carries the main load (40 rounds, RUWIKI; WIKI: 4); 1V12M command vehicle (2S7M)

SOURCES ATP CNA CNA21 DEF24 FAS3 IISS25 IZ17 IZ23B IZ232 LENTA16 LENTA23 MILN RDP1 TASS1 TWZ1 WEG4 WP-EN10 WP-RU8 WWW1

2S4 Tyulpan 2C4 «Тюльпан»

NATO: M-1975 **TR**

240 mm self-propelled breech-loading heavy mortar for destroying fortified targets In service 1971

OPERATED BY high-power artillery brigade, 2S4 battalion (ATP figure 5-18 shows two 2S4 groups; subunit name not given), ATP fig. 5-18: groups of 6 and 4 2S4 (OCR uncertain); 8 reported in 45th brigade (2017) **D 2024 H**

high-power artillery brigade, heavy mortar battalion (240-mm); brigade allocated to a front (Soviet RVGK) **D 1991 M**

INDICATES Held by high-power artillery brigades and by heavy battalions of army-level artillery brigades. A 2S4 sighting points to army-level heavy artillery tasked against fortified positions or built-up areas.

RECOGNISE Hull Turretless tracked hull; engine front right, driver front left, commander behind driver; ammunition drums in mid-hull; open, externally mounted mortar at rear.

Armament Very large 240 mm breech-loading mortar, barrel about 5 m, carried with its baseplate on the hull roof; in action it swings back onto a baseplate on the ground.

Turret No turret. Commander's rotating cupola behind the driver with an external 7.62 mm PKT and a searchlight.

Suspension Six rubber-tyred road wheels and four return rollers per side, drive sprocket front, idler rear, torsion bars. Road wheel size and spacing INA.

NOT TO BE CONFUSED WITH 2S3 Akatsiya Fully enclosed turret with 152 mm gun at the rear on a similar six-road-wheel chassis; 2S4 has no turret and a roof-carried mortar.

2S5 Giatsint-S p.29 Same chassis family, also turretless, but a long 152 mm gun overhangs the front; the 2S4 mortar muzzle is only level with the first road wheel, baseplate folded above it.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Tracked; heavy. Fords about 1 m unprepared. Mortar must be lowered onto a ground baseplate to fire.

Weight, combat	27.5 t
L × W × H	7.94 × 3.25 × 3.225 m travelling
Road speed	60 km/h maximum
Range	500 km

Range, 53-F-864 HE mine	9.65 km maximum, HE mine 130.7 kg, full charge
Minimum range, 53-F-864 HE mine	0.8 km HE mine, smallest charge

Range, 3F2 rocket-assisted HE mine	18 km maximum, 228 kg rocket-assisted HE mine; WEG and GS do not name the round, 3F2 per RUWIKI
------------------------------------	---

Minimum range, 3F2 rocket-assisted HE mine	0.8 km rocket-assisted HE mine
--	--------------------------------

Range, 3F5 Smel'chak laser-guided mine	9.2 km maximum, laser-guided mine, target designated by 1D15 from 0.2-5 km
--	--

SEEN WITH 6x6 truck or MT-LB carrying the rest of the crew

SOURCES AG3 ATP CNA CUL DB17 FM32 GS3 IISS25 IZ17 IZ232 JANESRG MK26 ORYX1 PRIME26 RG17 RG22 WEG2 WP-EN9 WP-RU6 VZ22

2A36 Giatsint-B 152 mm towed gun 2A36 «Гуацнм-Б»

NATO: M1976

TW

152 mm towed gun for long-range and counter-battery fire at army level In service 1976

OPERATED BY artillery brigade, gun battalion (152 mm) of the army artillery brigade; 2A36 or 2S5 or M-46, 36 per brigade (two 18-tube gun battalions) or 48 (four 24-tube battalions) [D 1991 M]
artillery brigade, gun brigade of the front artillery division; 2A36 or 2S7, 72 per brigade [D 1991 M]

Airborne Forces, INA; IISS lists the type with Airborne Forces without a count [R 2024-12 L]

artillery brigade [R 2026 L] seen with 244th Artillery Brigade (Baltic Fleet coastal troops, 11th Army Corps, Kaliningrad)

NUMBERS Holdings 150 in service, 350 in store, Ground Forces total (IISS estimate) [2024-12]; Holdings 50 across Naval Infantry (IISS estimate); holding unit not stated [2024-12]

INDICATES *ATP: towed artillery remains in some units. The 2A36 went to the gun battalions of artillery brigades and divisions at army and front level (FM 100-2-3, Victory Museum), so it indicates an artillery formation rather than a manoeuvre unit.*

RECOGNISE **Barrel** Long 152 mm barrel, 8197 mm with muzzle brake; multi-slot slit-type muzzle brake; recoil brake and recuperator sit above the barrel.

Shield Shield on the upper carriage protects the crew against bullets and fragments.

Carriage Split box-section welded trails; gun rests on a base plate for firing, wheels raised by a hydraulic jack.

Wheels Two axles, twin pneumatic-tyred wheels each side (four wheels), torsion-bar suspension.

NOT TO BE CONFUSED WITH 2A65 Msta-B p. 32 2A65 has a two-wheeled carriage (2A36 has four wheels), a shield sloping to the rear over the wheels and a caster wheel on each trail.

SOURCES ATP FM3 GS1 IISS25 ORYX1 RUSARMY RUSPL RWW VMUS WP-EN1 WP-EN2 WP-EN3 WP-EN4 WP-RUZ WSN1 not in WEG



Ministry of Defense of Ukraine, CC BY-SA 2.0

MOBILITY Towed by truck or tracked tractor; fired from a base plate with wheels raised, so needs a prepared firing position. Ground pressure and fording not found.

Weight, firing position	9,760 kg
L × W × H	12.92 × 2.788 × 2.76 m
Road speed, towed	80 km/h maximum, on road
Off-road speed, towed	35 to 45 km/h

Range, HE-frag 3VOF39	28.5 km maximum, HE-frag 30F29, full charge
Range, rocket-assisted 3VOF41	33.5 km maximum, active-rocket projectile 30F30
Minimum range, HE-frag 3VOF39	8.6 km ammunition not stated; listed with HE-frag maximum
Range, direct fire	1,530 m ammunition and target not stated in source
Rate of fire	6 rpm maximum
Crew	8

SEEN WITH KrAZ-260 6x6 truck (or KrAZ-255B, Ural-4320)

2A65 Msta-B 2A65 «Мсma-Б»

NATO: M1987

TW

Towed 152 mm howitzer for army-level artillery brigades and regiments In service 1986

OPERATED BY artillery brigade, artillery battalions (several per brigade; 2S19 or 2A65) [D 2019 M]

Naval Infantry (Coastal Troops of the Navy), Naval Infantry artillery; brigade not named [R 2024 M]

Russian forces in Syria, Russian deployment in Syria; unit not named [R 2024 M]

NUMBERS Holdings 50 2A65 with Naval Infantry (Coastal Troops of the Navy) (IISS 2024 figures) [2024]

INDICATES *A towed 152 mm howitzer points to an army-level artillery brigade or a division artillery regiment. CNA says replaced 2S19s and 2A65s move to division regiments and brigade battalions.*

RECOGNISE **Gun** Conventional towed howitzer, 152.4 mm, 47 calibres, monoblock barrel with a three-chamber muzzle brake and no bore evacuator.

Carriage Split trail with box-section trails, each with a caster wheel that swings up over the trail when firing. Hydraulic firing jack under the front.

Shield Gun shield on the upper carriage.

Running gear Sprung two-wheel running gear; springing cuts out automatically when the trails are spread.

Towing Towed by Ural-4320 6x6 or KamAZ-63501AT 8x8; in the travel position the base plate is raised and clamped to barrel and cradle.

NOT TO BE CONFUSED WITH 2S19 Msta-S p. 30 Self-propelled turret version of the same gun; the 2S19 barrel has a bore evacuator, the 2A65 does not.

2A36 Giatsint-B p. 32 Four-wheeled carriage, two wheels each side; the 2A65 has two-wheeled running gear.



One half 3544, public domain

MOBILITY Towed. Emplaced with hydraulic jack and spades. Towing speed 80 km/h on highway, 60 km/h on dirt roads, 20 km/h cross-country.

Weight, travel position	7,000 kg claim
L × W × H	12.7 × 2.5 × 2.95 m claim
Towing speed, road	80 km/h 20 km/h cross-country, 60 km/h dirt road, claim
Time into action	2 to 2.5 min claim

Range, OF45 HE-frag	24.7 km maximum, OF45 (30F45) HE-frag projectile, claim
Minimum range, OF45 HE-frag	6.5 km OF45 HE-frag projectile, claim
Range, OF61 HE-frag base bleed	29 km maximum, OF61 (30F61) HE-frag with base-bleed gas generator, round 3VOF91, claim
Rate of fire	up to 7 rounds/min maximum, claim

SOURCES ATP CNA FM3 GS2 IISS25 JANESRG ORYX1 RWW WP-EN2 WP-EN5 WP-EN6 WP-RU3 not in WEG

Iskander-M 9K720 «Искангеп-М»

NATO: SS-26 Stone

WH-AT

Army-level operational-tactical ballistic and cruise missile system for deep strike In service 2007

OPERATED BY missile brigade, 3 missile battalions of 2 batteries each; 2 launchers per battery, 12 launchers and 12 transloaders per brigade; 4 of each per battalion [D 2021 H](#)

combined arms army, army missile brigade, 1 missile brigade (12 launchers) per army [D 2024 H](#)

tank army, army missile brigade, 1 missile brigade (12 launchers)

[O 2021 M](#)

army corps, corps missile brigade, 1 missile brigade (12 launchers)

[O 2021 M](#)

combined arms army, army missile brigade, 12 launchers per brigade

[O 2023 M](#)

missile battalion, 4 launchers [O 2008 M](#)

INDICATES Iskander vehicles indicate an army-level missile brigade of 12 launchers. The army commander holds it as a reserve for priority strikes; it is not part of brigade artillery groups, so it does not by itself mark a manoeuvre brigade.

RECOGNISE Launcher Iskander-M: two ballistic missiles lie in the rear body on separate lift arms, raised to 85 deg to fire. Iskander-K: 2 x 9M728 or 4 x 9M729 in canisters.

Radar No radar is described on the launcher; it has satellite navigation and its own fire-control post. Target data comes from the 9S920 and 9S552 vehicles.

Chassis MZKT-7930 8x8 heavy truck, about 13 m long and 3 m wide, first two axles steer. Four hydraulic jacks lower to level the vehicle before launch.

Complex Launchers move with 9T250 transloaders on the same 8x8 chassis, which carry a crane. Command, data preparation, maintenance and living vehicles are KamAZ trucks.

NOT TO BE CONFUSED WITH 9T250 transporter-loader (same complex) Same MZKT-7930 chassis and two missiles, but has a loading crane and no launch equipment.

K-300P Bastion-P coastal launcher (MZKT-7930) Coastal troops system: two Oniks anti-ship missiles fired vertically; battery of four launchers with KamAZ command vehicles.

Bal coastal missile launcher (MZKT-7930) Carries eight Kh-35 anti-ship missiles, not two large missiles.

9A53 Uragan-1M MRL (MZKT-7930) [p. 27](#) Rear carries two rocket pods of six tubes each instead of two missiles.

SEEN WITH 9P78-1 self-propelled launcher, MZKT-7930 8x8 (2 per battery, 4 per battalion, 12 per brigade); 9T250 transporter-loader, MZKT-7930 8x8 with crane, 2 missiles (2 per battery, 4 per battalion, 12 per brigade); 9S552 command and staff vehicle, KamAZ truck (1 per battery; 11 per brigade); 9S920 information preparation point (mission planning), KamAZ truck (1 per brigade); Maintenance vehicle (MRT0), KamAZ truck (1 per brigade in the 51-vehicle brigade set); Life support vehicle, KamAZ truck, living space for up to 8 (14 per brigade); 9M723 ballistic missile (2 per launcher, 2 per transloader; 48 per brigade); 9M728 (R-500) and 9M729 cruise missiles (2 x 9M728 or 4 x 9M729 per launcher); MZKT-7930 8x8 chassis

VARIANTS 9K720 Iskander-M with 9M723 Baseline: two ballistic missiles about 7.3 m long and 0.92 m wide on lift arms. Quasi-ballistic missile, up to about 500 km; reported to release decoys during flight. Iskander-K with 9M728 (R-500) Same 9P78-1 launcher, loaded with two cruise missile canisters instead of two ballistic missiles. Low-flying subsonic cruise missile; official range up to 500 km. Used against Chernihiv in Aug 2023. Iskander launcher with 9M729 Special launcher configuration with four cruise missile canisters; the missile is 530 mm longer than 9M728. Range disputed: US assessed over 500 km in breach of the INF Treaty; Russia said it was within limits.



Vitaly V. Kuzmin, CC BY-SA 4.0



launcher with 9M728 cruise missiles. Boevaya mashina, CC BY-SA 3.0

MOBILITY Heavy 8x8 truck: 70 km/h on roads, about 40 km/h on dirt roads. Must halt and level on jacks to fire. Fords 1.4 m. A battery has been airlifted by An-124.

Weight, launcher loaded 42.85 t

Weight, 9T250 transloader 40 t

loaded

L × W × H

Road speed

Road range

Fording

Time into action

13.07 × 3.07 × 3.55 m

70 km/h

1,000 km

1.4 m

16 min to first launch from the march; 4 min from highest readiness claim

41 Derived: 42.85 t wheeled x 0.94,

rounded up, estimated

30 to 80 s to level on jacks; about

20 s to raise missiles

Military load class

Levelling and erecting

Range, 9M723 ballistic missile

Minimum range, 9M723

Range, 9M728 (R-500) cruise missile

Accuracy (CEP), 9M723

Warhead mass, 9M723

Warhead types, 9M723

Launch weight, 9M723

Missile length and diameter, 9M723

Missiles per launcher

Interval between launches

Crew

Brigade vehicle set

Range, 9M729 cruise missile

up to 500 km maximum, 9M723, domestic Iskander-M

50 km From the MIS Iskander-E (export) data table, max 280 km; domestic 9M723 minimum not confirmed

500 km official maximum, 9M728 cruise missile, Iskander-K, claim

10 to 30 m reported, With electro-optical terminal seeker

480 to 700 kg Depends on warhead type

Cluster, unitary HE, penetrator, fuel-air and others; nuclear capable

3,800 to 4,020 kg Depends on payload

7.3 m; 0.92 m

2 ballistic (9M723) or 2 cruise (9M728); 4 x 9M729 in special configuration claim

about 1 min Two missiles from one launcher

Launcher 3; transloader 2

51 vehicles 12 launchers, 12 transloaders, 11 command, 14 life support, 1 maintenance, 1 information preparation

INA maximum, 9M729 cruise missile

SOURCES AR27 ATP CNA CNA21 CSIS3 CSIS4 CSIS5 ESD IHS25 KI3 MDB MIL27 MISS7 RDP5 RUSI4 RUSIPL RWW TASS5 TE TH VPK3 WEG2015 WP-EN36 WP-EN37 WP-EN38 not in WEG

3 Air defence

IN THIS CHAPTER

Long and medium-range SAMs 35

Short-range SAMs and guns 38

LOOKING FOR THESE? THEY ARE IN OTHER CHAPTERS

SYSTEM	WHAT IT IS	CHAPTER	PAGE
Polyana-D4	Automated command post of an Army air defence missile brigade	7 Command and signals	68
PU-12M7	Air defence battery command post	7 Command and signals	69
Barnaul-T	Automated command and control set for tactical short-range air defence, battery...	7 Command and signals	69
Nebo-M	Mobile multiband air surveillance radar complex	4 Radars	41
96L6 all-altitude acquisition radar	3D C-band acquisition and target designation radar for S-300PM/PMU2 and S-400...	4 Radars	42

S-400 Triumf 40P6 «Триумф»

NATO: SA-21 Growler

WH-AT

Long-range SAM system against aircraft, cruise missiles and ballistic missiles In service 2007

OPERATED BY military district, SAM regiments of the district's Air Force and Air Defence Army, IISS late 2024, SAM regiments listed with S-400 in each district's Air Force and Air Defence Army: 45th (Leningrad MD) 5, plus 4 listed with S-300PS, S-300PM1 and S-400; 14th (Central MD) 4; 4th (Southern MD) 3; 11th (Eastern MD) 4 listed with S-300PS and S-400, plus 1 with S-300V and S-400; 6th (Moscow MD) none (Moscow area: 5 regiments under Aerospace Defence Command). Launchers per district INA [D 2024 H]

air defence missile regiment, about 12 launchers reported at the unit site; 4 launchers and 96L6 and 92N6 radars reported destroyed

[O 2025-T1 M]

air defence missile regiment, air defence missile battalion (division), usually 2 battalions of 8 launchers each (16 per regiment); 2012 press report relaying MoD press service [R 2012 M]

aerospace forces air defence regiment, IISS: 13 Aerospace Forces regts with S-400 and Pantsir-S1, plus 2 mixed with S-300 [R 2024 M]

air defence missile regiment, Navy: regiments under fleet air defence divisions, IISS: 64 S-400 in Navy; 3 SAM regts: 2 Baltic Fleet, 1 Pacific Fleet (naval aviation) [R 2024 M]

air defence missile battalion, IISS: 2 SAM bn in Belarus; 1 SAM bty in Syria [R 2024 M]

INDICATES S-400 belongs to Aerospace Forces or Navy SAM regiments, not ground-force brigades. A regiment usually has two battalions of eight launchers; launchers imply a 92N6 engagement radar nearby, and often Pantsir-S1 in the same regiment.

RECOGNISE Launcher Four large missile canisters on an erector, raised vertical to fire. One canister can hold four small 9M96 missiles instead of one 48N6.

Radar 92N6 Grave Stone engagement radar: phased array on MZKT-7930 8x8. 91N6 acquisition radar: back-to-back phased array, MZKT-7930 tractor. 96L6 may sit on a 40V6M tractor.

Chassis Launchers self-propelled on MAZ-543M 8x8 (5P85SM2-01), or on a semitrailer towed by a BAZ-6402-series 6x6 tractor (5P85T2; export 5P85TE2 on BAZ-64022).

Complex 92N6 and 96L6 radars on MZKT-7930 8x8; 55K6 command post on Ural 8x8 truck. A battalion site holds about eight launchers around one 92N6.

NOT TO BE CONFUSED WITH S-300PM1/PM2 (SA-20 Gargoyle) Same 5P85 launcher family, 96L6 radar and a look-alike command post; S-400 has the redesigned 92N6 Grave Stone, not the 30N6 Flap Lid/Tomb Stone.

64N6E Big Bird (S-300PMU family) Same two-faced array layout; the S-400 91N6E is readily told apart by its new-build MZKT-7930 tractor (APA).

SEEN WITH 55K6E command post (30K6E control system) (1 per 30K6E system of up to 6 fire units (export composition)); 91N6E Big Bird acquisition radar (1 per 30K6E system (export composition)); 92N6E Grave Stone engagement radar (domestic 92N6A) (1 per 98Zh6E fire unit); 96L6E all-altitude radar (domestic 96L6-TsP); 5P85SM2-01 / 5P85T2 launcher (APA export list: 5P85SE2 / 5P85TE2) (8 per battalion (Russian regiments, 2012); up to 12 per fire unit (export composition)); BAZ-64022 6x6 tractor (1 per 5P85TE2 semitrailer launcher); 40V6M/MD mast (INA (optional)); 5T58-2 missile transporter and 22T6-2 loading crane; 96K6 Pantsir-S1 (INA; listed in the same regiments)

VARIANTS 5P85SM2-01 Self-propelled launcher on MAZ-543M 8x8; no separate tractor. INA. 5P85T2 Launcher on a semitrailer towed by a BAZ-6402-series 6x6 tractor. INA. 5P90S Self-propelled TEL on BAZ-6909-022; new gantry, folding mast with directional antenna, satnav antennas on a longer baseline. Intended to carry a heavier missile payload than the MAZ-79100-series TELs (APA). 51P6 Launcher on MZKT-7930 chassis (RU Wikipedia only); Oryx lists 51P6 TELs among S-400 losses. INA.

SOURCES APA1 ATP CSIS8 IIS25 [REW02 LENT3 MIL54 MIL55 ORYX1 RUSIAD STRASAM TASS16 WEG65 WP-EN69 WP-RU110 WP-RU111



Vitaly V. Kuzmin, CC BY-SA 4.0



radar.vitaly kuzmin, CC BY-SA 4.0

MOBILITY Wheeled 8x8 vehicles and tractor-semitrailer launchers; WEG estimates 30 km/h on dirt roads for the semitrailer launcher.

Road speed, 5P85TE2 launcher

Cruising range, 5P85TE2 launcher

Time into action

Weight, launcher

Fording

Time out of action

Ground pressure

Length, launcher

Width, launcher

60 km/h road; dirt road 30 km/h, estimated

800 km estimated

5-10 min from travelling position; 3 min to ready from deployed position

INA

INA

INA

INA

INA

INA

Range, 40N6

400 km maximum, aerodynamic targets (STRASAM); APA brochure figure is for export 40N6E, 215 nmi, claim

Altitude, 40N6

30-35 km maximum, 40N6; the 27 km often quoted is reported for the 48N6DM missile specifically, claim

Range, 48N6DM (48N6E3)

250 km maximum, aerodynamic targets; RUSIAD names export 48N6E3 (Army Technology lineage); TASS 2018 state claim: 48N6DM up to 250 km, claim

Altitude, S-400 (missile not stated)

Range, 48N6 (ballistic targets)

Range, 9M96E2

Altitude, 9M96E2

Range, 9M96E

Altitude, 9M96E

Detection range, 91N6E

Detection range, 96L6E

Fire unit distance from command post

Warhead, 48N6

Missiles per launcher, 48N6 family

0.1-27 km reported, reported engagement altitude band

60 km ballistic missile targets, 48N6 series

120 km export 9M96E2; aerodynamic targets; APA gives 64.8 nmi, claim

0.005-30 km range, export 9M96E2; APA gives 15 ft to 100 kft, claim

40 km export 9M96E; aerodynamic targets; APA gives 21.6 nmi, claim

0.005-20 km range, export 9M96E; APA gives 15 ft to 66 kft, claim

580-600 km detection/tracking, target RCS >10 m² (strategic bomber)

300 km max, export 96L6E in S-300PMU family, target not stated; JREW-p1877, PDF p. 1877, claim

up to 100 km maximum, 98Zh6E fire unit from 55K6E, export composition, claim

143 kg HE-frag

4 or up to 16 9M96E/E2, or mixes such as 3 x 48N6 + 4 x 9M96

3

AIR DEF

S-300V4 C-300B4

NATO: SA-23 **TR**

Tracked long-range SAM and theatre anti-ballistic-missile system of Army air defence In service 1988

OPERATED BY air defence missile brigade, air defence missile battalion (up to 4 per brigade, under Polyana-D4M1 brigade CP); battalion CP 9S457M controls up to 4 batteries; WEG: an upper-level 9S52 Polyana-D4 command post may integrate S-300V with Buk into one brigade, S-300V battery: 1 9S32, 4 9A83, 2 9A82, 1 9A84, 2 9A85; S-300V4 battery counts INA [\[D 2017 M\]](#)

air defence missile brigade, INA. IISs lists S-300V and S-300V4 in Army holdings without a count and counts 16 SAM brigades, but does not say which hold them; Izvestia reports S-300V4 brigades under the OSK [\[R 2024 M\]](#)

INDICATES S-300V4 vehicles point to an Army air defence missile brigade: a front-level asset directly under a district (OSK) or front commander, covering main troop groupings. Older S-300V also serves in Aerospace Forces SAM regiments and at the Armenia base (IISs).

RECOGNISE Launcher 9A83 TELAR: four large tubes carried flat, raised vertical to fire; illuminator radar at the front on a folding mast. 9A82: two tubes, radar fixed over the cab.

Launcher Loader-launchers (9A84, 9A85) look like TELARs but carry a hydraulic crane in place of the radar. In S-300V4 heavy missiles ride only on 9A84-2 loader-launchers.

Radar 9S32 guidance station: large phased array raised hydraulically on its own tracked vehicle. S-300VM/V4 9S32M uses a larger High Screen-type space-fed array.

NOT TO BE CONFUSED WITH S-300PS/PM, S-400 [SA-10, SA-20, SA-21] [p. 35](#) Launchers are wheeled: S-300P on 8x8 trucks, S-400 on trailers behind 6x6 tractors. The S-300V family is on tracked vehicles.



TELAR, Alexey Ivanov, CC BY 4.0



radar, Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Heavy tracked vehicles, not amphibious; ford 1.2 m. Deploy from march and fire from unprepared sites in about 5 min; ready to move about 5 min after shutdown.

Weight, 9A83 TELAR **50 t** combat weight
L × W × H **9.30 × 4.50 × 3.40 m** WEG 'hull length'
Road speed **53 km/h** maximum
Road range **250 km** cruising range

Range, 9M83 **75 km** maximum, aircraft targets; S-300V
Minimum range, 9M83 **6 km** aircraft targets; S-300V
Altitude, 9M83 **250 m–25 km** maximum and minimum, aircraft targets; S-300V
Range, 9M82 **100 km** maximum, aircraft and jammer targets; S-300V
Minimum range, 9M82 **13 km** aircraft targets; S-300V, claim
Altitude, 9M82 **1–30 km** maximum and minimum, aircraft targets; S-300V

SEEN WITH 9S457M command post (1 per battalion); 9S15M/9S15MV Obzor-3 all-round surveillance radar (Bill Board) (1 per battalion)

SOURCES APA10 AR48 ATP BMDP7 CSIS7 IIS25 IZV6 JREW02 MISS9 RUSIAD RWW TASS15 VKS WEG4 WEG54 WEG55 WP-EN66 WP-RU107 WP-RU108 WSPOT22 WSPOT23

S-350 Vityaz C-350 «Бумряг»

NATO: SA-28 **WH-AT**

Wheeled medium-range SAM against aircraft, cruise missiles, UAVs and ballistic missiles In service 2019

OPERATED BY air defence missile battalion, Battalion set: 1 command post 50K6A, 2-3 multifunction radars 50N6A, 12 launchers 50P6A, 12 launchers (12 missiles each) per battalion, per Russian MoD statement [\[R 2020 M\]](#)

Unidentified unit, training centre of the Anti-aircraft Missile Forces, Zhukov Military Academy of Aerospace Defence, first S-350 set (crews and instructors training) [\[R 2020-02 M\]](#)

INDICATES An S-350 battalion covers important sites or groups of forces against cruise missiles and aircraft. Russian statements tie deliveries to VKS air defence regiments and brigades and to Navy units; units seen near the front in 2022-25 are unidentified.

RECOGNISE Launcher 50P6 launcher carries 12 missiles in vertical-launch containers; the container pack is raised upright to fire ('cold' vertical launch).

Launcher From photograph: forward-control boxy cab; container pack in two horizontal tiers behind a machinery box. Four axles (8x8 in text sources); two front, gap, two rear.

Radar 50N6A multifunction radar: rotating phased-array antenna, 360° cover, circular and sector modes; unmanned, remote-controlled from the command post.

Chassis Launcher on 8x8 BAZ-6909 (WEG, Almaz-Antey 2023). WEG puts the radar on 6x6 BAZ-69092; Aif puts all modules, launcher included, on BAZ-69092-012.

NOT TO BE CONFUSED WITH S-300PS/PM and S-400 launcher [p. 35](#) Four missile canisters per launcher, against twelve containers on the 50P6.

50P6T launcher-loader Described as very similar to the 50P6 launcher; no distinguishing mark found in text sources.

SOURCES AIF7 AINF3 AR49 ATO GS26 GS27 GS28 IIS25 IZ21 IZ202 RG4 RUSIAD TW212 WEG56 WMC20 WMC21 WP-EN67 WP-EN68 WP-RU109



TELAR, Vitaliy V. Kuzmin, CC BY-SA 4.0



radar, Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Not amphibious; fords about 1.2 m (WEG). Wheeled 8x8 launcher; into and out of action about 5 minutes. Cannot fire on the move.

Weight, launcher vehicle **22 t** BAZ-6909 launcher vehicle
empty
L × W × H **11.3 × 2.75 × 2.85 m** BAZ-6909 launcher vehicle
Road speed **70 km/h** maximum
Road range **1,000 km** maximum

Range, 9M96E2 **120 km** maximum, aerodynamic target; export missile, claim
Minimum range, 9M96E2 **1 km** export missile, claim
Altitude, 9M96E2 **5 m to 20 km** maximum, aerodynamic target; export missile, claim
Range, 9M96E **40 km** maximum, aerodynamic target; export missile, claim
Minimum range, 9M96E **1 km** export missile, claim
Altitude, 9M96E **5 m to 20 km** maximum, aerodynamic target; export missile, claim

SEEN WITH 50K6A command post (1 per battalion)

Buk-M1 / Buk-M2 / Buk-M3 9K37M1; 9K317; 9K317M «Byk-M1»; «Byk-M2»; «Byk-M3»

NATO: SA-11 Gadfly, SA-17 Grizzly, SA-27

TR

Tracked medium-range SAM complex; army-level air defence of ground forces In service 1983

OPERATED BY air defence missile brigade, 4 air defence missile battalions, each 3 batteries; brigade command post Polyana-D4, Buk-M1: 6 TELARs, 3 loader-launchers, 1 CP, 1 search radar per battalion; 24 TELARs per 4-battalion brigade (JIT: 53rd Bde had 3 operational battalions in 2014) [D 2016 H] air defence missile brigade, battalion TELAR, INA per battalion; brigade had 3 operational battalions in 2014 (JIT) [O 2014-07 M] air defence missile brigade, air defence missile battalion (brigade set of Buk-M3), Buk-M3: up to 6 TELARs or 9S36M radars and up to 12 launchers per battalion [R 2019 M] aerospace forces air defence regiment, 4 Aerospace Forces regiments with Buk-M1-2, Buk-M2 and S-300V; about 80 Buk-M1-2/Buk-M2 in Aerospace Forces holdings [R 2024 M]

air defence missile brigade, air defence missile battalion [R 2018 L] **INDICATES** Buk vehicles usually indicate an army-level air defence missile brigade, nominally of four battalions (the 53rd had three in 2014). The army may detach Buk subunits to its divisions, so a battery alone does not fix the brigade's position.

RECOGNISE Launcher M1 and M2 TELAR: four bare missiles on a trainable launcher above the hull, radar at its front. M3 TELAR 9A317M: six missiles in cylindrical containers.

Launcher Loader-launcher has no radar. M1/M2: 8 missiles (4 on rails, 4 on cradles) and a crane. M3 9A316M: 12 containers on two arms. TELAR: 4 missiles (M3: 6).

Radar M1 TELAR carries the 9S35 radar, not a phased array. M2 and M3 TELARs carry a flat phased-array panel. Tell M2 from M3 by missiles: 4 bare vs 6 in tubes.

Radar 9S18M1 search radar: flat array on a tracked hull, 8 m high when raised, turns 360 deg. M2/M3 9S36: array on a telescopic mast up to 21-22 m, no missiles.

Chassis All combat vehicles on similar heavy tracked chassis, about 8-9.6 m long, 3.3 m wide, 30-38 t. M1/M2: GM-567M, 569, 577, 579. M3: GM-5969.

NOT TO BE CONFUSED WITH Tor-M1/M2 (SA-15) p. 38 One self-contained vehicle with its own search radar; missiles in 4-round container modules, not on an arm. No separate radar vehicle; battery has two transloader trucks.

2K12 Kub (SA-6) 2P25 launcher carries three missiles on a lighter 19.5 t chassis and has no radar; radar is on a separate 1S91 vehicle.

S-300V 9A83/9A82 (SA-12) p. 36 Launcher carries four (9A83) or two (9A82) large missiles fired vertically, on an MT-T-based chassis. Buk fires from an inclined arm.

SEEN WITH 9S52 Polyana-D4 brigade command post (1 per brigade); 9S470M1 command post (Buk-M1) (1 per battalion); 9S18M1 Kupol-M1 target acquisition radar (Buk-M1) (1 per battalion); 9A310M1 TELAR (Buk-M1) (2 per battery, 6 per battalion); 9A39M1 loader-launcher (Buk-M1) (1 per battery, 3 per battalion); 9S510 command post (Buk-M2) (1 per battalion; controls up to 6 fire sections); 9S18M1-3 target acquisition radar (Buk-M2) (1 per battalion); 9A317 TELAR (Buk-M2) (Up to 4 per battalion, each with 1 loader-launcher); 9S36 mast-mounted illumination and guidance radar (Buk-M2) (Up to 2 per battalion, each with 2 loader-launchers)

VARIANTS 9K37M1 Buk-M1 TELAR 9A310M1 with 9S35 radar and four bare 9M38M1 missiles; loader-launcher 9A39M1 with 8 missiles. 9K37M1-2 Buk-M1-2 Looks like Buk-M1: TELAR 9A310M1-2 with 9S35M1-2 radar and bare missiles. 9K317 Buk-M2 TELAR 9A317 with flat phased-array radar, four bare missiles; adds 9S36 radar on a 21-22 m mast. 9K317M Buk-M3 Missiles in containers: 6 on TELAR 9A317M, 12 on launcher 9A316M. All vehicles on GM-5969 chassis.



Buk-M2 TELAR. Boevaya mashina, CC BY-SA 3.0



Buk-M3 TELAR. Ministry of Defence of the Russian Federation, CC BY 4.0



9S36 radar on mast. Boevaya mashina, CC BY-SA 4.0

MOBILITY Heavy tracked, not amphibious; fords 1 m. TELAR section ready 5 min from march; 9S36 mast-radar section needs 10-15 min.

Weight, combat	34 t Buk-M1; WEG does not say which vehicle (MISM1 gives 35 t for the 9S18M1 radar)
L × W × H	9.3 × 3.25 × 3.8 m Buk-M1; WEG does not say which vehicle (MISM1 gives 9.59 m for the 9S18M1 radar)
Road speed	65 km/h maximum
Range	500 km maximum
Fording	1 m
Time into action	5 min TELAR/battery from march
Time out of action	5 min
Ground pressure	0.8 kg/cm2
Survivability move	20 s reposition 100-200 m with equipment running

Range, 9M38M1 (Buk-M1)	3.25-34 km maximum, air target
Altitude, 9M38M1 (Buk-M1)	30 m-22 km intercept altitude, air target
Range, 9M317 (Buk-M2)	3-50 km engagement zone, aircraft
Range, 9M317 vs ballistic and cruise missiles	20 km maximum, ALCM-type cruise missile at 30 m (MISM2; ROE gives 100 m, unchecked); Lance-type tactical ballistic missile 15-20 km (MISM2), claim
Altitude, 9M317 (Buk-M2)	15 m-25 km intercept altitude, air target
Range, 9M317M (Buk-M3)	2.5-70 km maximum effective, air target
Altitude, 9M317M (Buk-M3)	35 km maximum, air target
Detection range, 9S18M1 search radar (M1)	100-150 km target size and altitude not stated
Detection range, 9S18M1-3 / 9S18M3 search radar (M2/M3)	160 km RCS 1-2 m2; Buk-M2 9S18M1-3
Detection range, 9S35 TELAR radar (M1)	80 km RCS 2 m2; 100 km vs RCS 3 m2
Detection range, 9A317 / 9A317M TELAR radar (M2/M3)	120 km RCS 1-2 m2 at 3 km altitude and 18-20 km vs same target at 10-15 m (M2); M3: RCS 1-2 m2, altitude not stated
Detection range, 9S36 mast radar (M2)	120 km RCS 1-2 m2 at 3 km altitude; 30-35 km vs same target at 10-15 m
Detection range, 9S36M mast radar (M3)	70 km cruise missile flying at 5 m

SOURCES ATP BMPD3 BMPD4 CNA IIS25 JIT MIL17 MISS1 MISS2 MISS3 MISS4 MISS5 ROE RWW TWI2A WEG25 WEG26 WEG27 WP-EN26 WP-RU46

AIR DEF 3

Tor 9K330 «Top»

NATO: SA-15 Gauntlet

TR

Short-range tracked SAM with own radars; defends troops against aircraft, PGMs and UAVs In service 1986

OPERATION BY motor rifle division, air defence missile regiment, 16 per regiment (4 batteries of 4, bmpd); ATP: two battalions; ru.wikipedia: Tor-M2 'divisional' set of 3 batteries, 12 vehicles [\[D 2024 H\]](#)

motor rifle brigade, air defence missile battalion, three batteries, 12 launcher vehicles per battalion (4 per battery) [\[D 2024 M\]](#)

tank brigade, air defence missile battalion, three batteries, 12 launcher vehicles per battalion (4 per battery) [\[D 2024 M\]](#)

motor rifle division, air defence missile regiment, one regimental set of Tor-M2 (16 combat vehicles) [\[O 2017-11 M\]](#) seen with 933rd Air Defence Missile Regiment, 150th Motor Rifle Division (Millerovo)

tank division, air defence missile regiment, one regimental set of Tor-M2 (16 combat vehicles) [\[O 2017-03 M\]](#) seen with 538th Guards Air Defence Missile Regiment, 4th Guards Tank Division

motor rifle brigade, air defence missile battalion (Tor-M2DT); brigade link by location and retraining report only [\[R 2019-11 L\]](#) seen with Tor-M2DT battalion of Northern Fleet ground and coastal troops, on duty in Pechenga district where a motor rifle brigade is based (number not given)

INDICATES *Four Tor vehicles together suggest a battery of a brigade air defence missile battalion (12 vehicles) or of a division's air defence missile regiment (16). Doctrine sites Tor-M1 5-10 km behind the forward edge.*

RECOGNISE Launcher Missiles stand vertically inside the central unmanned turret. Tor/Tor-M1: two boxes of four (8). Tor-M2: 16 missiles.

Radar Rotating search antenna at turret rear, lowered behind turret when stowed. Phased-array tracking radar on turret front folds down; dome antenna and TV sight nearby.

Chassis Tracked GM-355 chassis (Tor-M2: GM-5955); three crew in front, turret amidships, engine at rear. Not amphibious. Road wheel count INA.

Complex Battery: command post, four combat vehicles, two transloaders on Ural-4320 trucks. Tor-M2DT rides a two-section articulated tracked DT-30 Vityaz carrier.

NOT TO BE CONFUSED WITH **2K22 Tunguska (SA-19 Grison)** p. 40 Tunguska carries two 30 mm cannon on its turret; Tor has no guns and its missiles sit inside the turret.

96K6 Pantsir-S1 (SA-22 Greyhound) p. 39 Pantsir-S1 combines guns with 12 missiles and is usually on a wheeled chassis; Tor (tracked) has no guns and its missiles sit inside the turret.

9K33 Osa (SA-8 Gecko) p. 40 Osa guidance group has two small dishes flanking a central tracker; Tor has one flat phased-array tracking antenna.

SEEN WITH 9S737M Ranzhir-M battery command post (9S737MK with Tor-M2E) (1 per battery); 9T244 transloader (2 per battery); 9T245 transport vehicle; 9V887M2U / 9V887-1M2U maintenance vehicles (9V887M2K / -1M2K in the Tor-M2E export set); GM-355 / GM-5955 tracked chassis (1 per combat vehicle); DT-30 Vityaz articulated tracked carrier (Tor-M2DT; DT-30PM per Army Recognition) (1 per combat vehicle)

VARIANTS 9K331 Tor-M1 Missiles in two four-round containers (9Va281) in the turret; some vehicles on GM-5955 chassis (9A331-1). Second target channel: 2 targets at once; crew 3. 9K331MU Tor-M1-2U INA. Interim upgrade from 2012: 4 targets at once, ceiling 10 km. 9K332 / 9K331M Tor-M2 New surveillance antenna; GM-5955 chassis; 2025 build adds armour and EW fit. 16 9M338K missiles, 4 targets at once, fires on the move; claimed 32 km detection and 16 km range. 9K331MDT Tor-M2DT Two-section articulated tracked DT-30 carrier; weapon module on rear section. Arctic mobility; seen in Ukraine from December 2022.



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Tracked, not amphibious; fords 1.0 m. Tor-M1 halts to fire; Tor-M2 claimed to fire on the move. Tor-M2DT for snow and bog.	
Weight	32 t combat; WEG figure for the 9K330 page, variant not stated
L × W × H	8.0 × 3.2 × 4.0 m
Road speed	65 km/h maximum
Road range	580 km cruising range
Fording	1.0 m
Swimming	not amphibious
Time into action	less than 3 min
Time out of action	less than 5 min
Bridge load class	36 derived from 32 t tracked x 1.10, estimated
Range, 9M330/9M331	12 km maximum, aircraft and PGM targets, Tor/Tor-M1
	1.5 km Tor/Tor-M1
Minimum range, 9M330/9M331	
Altitude, 9M330/9M331	10 m to 6 km maximum, Tor/Tor-M1
Range, 9M338K	16 km maximum, Tor-M2 domestic, air targets, claim
Altitude, 9M338K	10 m to 10 km maximum, Tor-M2 domestic, claim
Range, 9M331 (Tor-M2E)	1 to 12 km maximum, Tor-M2E export; target 300 m/s, claim
Altitude, 9M331 (Tor-M2E)	10 m to 10 km maximum, Tor-M2E export; target 300 m/s (10 m to 6 km at 600 m/s), claim
Detection range, surveillance radar (Tor/Tor-M1)	25-27 km maximum, aircraft; WEG does not state the variant
Detection range, surveillance radar (Tor/Tor-M1), helicopter	12 km maximum, helicopters
Detection range, surveillance radar (Tor/Tor-M1), UAV	9-15 km maximum, UAVs
Detection range, surveillance radar (Tor-M2)	32 km air targets, size not stated, claim
Detection range, tracking radar (Tor/Tor-M1)	15 km maximum, 60-degree sector; target and variant not stated
Detection range, TV sight	20 km maximum, target not stated
Deployment distance, Tor-M1 in defence	5-10 km behind forward edge practical, first defensive position; 5-10 km between vehicles

SOURCES AR58 ATP BMPD9 CNA CSIS9 IHS25 IZV8 JREW02 MISS13 ORYX1 RG25 RG175 RG192 RG232 RG244 RWW WEG67 WP-RU125

Pantsir-S1 96K6 «Панцирь-С1»

NATO: SA-22 Greyhound

WH-AT

Short-range gun and missile air defence of S-400/S-300 battalions and key sites In service 2012

OPERATED BY aerospace forces air defence regiment, Pantsir vehicles deployed with each S-400 battalion (Armada); Lenta: batteries attached to S-400 regiments, 3 per S-400 battalion (Armada, 2024)

D 2024 H

aerospace forces air defence regiment, Pantsir-S battery, 6 Pantsir-S delivered to Eastern MD air defence units on Kamchatka, Dec 2015; the regiment was reported due one battery **O 2015-12 M**

aerospace forces air defence regiment, Pantsir-S1 gun-missile battalion; each battery covers one S-400 battalion **O 2020-02 M** air defence missile regiment, 30 held by the Navy (IISS p. 187)

R 2024 M

aerospace forces air defence regiment, INA per regiment; IISS lists 9 SAM regts with Pantsir-S1 in the 45th, 4 in the 14th, 3 in the 4th and 4 in the 11th AF&AD Army **R 2024 M**

air defence missile regiment, INA (1 SAM regt with S-400 and Pantsir-S1) **R 2024 M**

INDICATES Usually marks a nearby S-400 (or S-300) battalion or a defended strategic site: VKS regiments deploy Pantsir with their long-range SAM battalions (three per S-400 battalion, Armada). IISS lists none in Ground Forces holdings.

RECOGNISE Launcher Turret with two blocks of six missile tubes, one each side, and a twin-barrel 30 mm 2A38M gun beside each block; 12 missiles ready.

Radar Flat rectangular search array, about 1.8 x 0.9 m, rotating on the turret roof; folds forward flat on the roof when stowed.

Radar Tracking radar on the turret front face; small electro-optical post on a pedestal on the turret top, ahead of the search radar.

Chassis Russian systems on an 8x8 KAMAZ-6560 truck; four hydraulic jacks level the vehicle to fire. Some later ones on MZKT-7930 with armoured two-piece-windshield cab.

Complex Battery of up to six combat vehicles linked by data link, with one transloader truck per two combat vehicles.

NOT TO BE CONFUSED WITH 2K22M1 Tunguska-M1 (SA-19 Grison) p. 40 Tracked chassis and 8 missiles; Pantsir-S1 in Russian service is on an 8x8 truck with 12 missiles.

Pantsir-SM-SV Same Pantsir turret family but on a tracked GM-5970-series chassis.

SEEN WITH S-400 (SA-21 Growler) battalion (one Pantsir battery per S-400 battalion (Red Star); 3 Pantsir per battalion (Armada)); Transloader vehicle (designation not confirmed) (1 per 2 combat vehicles); Maintenance, repair, alignment and spares vehicles; KAMAZ-6560 8x8 truck (1 per combat vehicle); 1L122 Garmon radar (INA; reported ringing Pantsir positions in Ukraine)

VARIANTS Pantsir-S2 INA. Modernised Pantsir for Russian forces, in service 2015; IISS lists VKS holdings as Pantsir-S1/S2. Details INA. **Pantsir-SM** New multifunction radar with active phased array; reported on a new 8x8 KAMAZ with armoured cab. KBP claim: engagement range 40 km and detection 75 km, against 20 km and 40 km for Pantsir-S1. **Pantsir-SM-SV** Tracked GM-5970-series chassis instead of a truck. For Ground and Airborne Forces; 1RS2-3 radar; 57E6M and TKB-1055 missiles; imagery with army units reported March 2025 (likely taken 2024). **Pantsir-S1 (MZKT-7930)** MZKT-7930 8x8 chassis with armoured cab and two-piece windshield; one destroyed near Mariupol Feb 2026. Same 72V6 combat module as the KAMAZ version; chassis ordered from Belarus in 2024. **Pantsir-S1M** INA. Export version of Pantsir-SM; new missile with 30 km range (claim). Oryx records one destroyed in Ukraine.



© Mitya K. Avramov
Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY 8x8 truck, fast on roads. KAMAZ chassis has limited off-road mobility and tends to roll over (high centre of gravity, narrow track). Jacked level to fire guns accurately.

Weight	14.85 t	WEG 'curb weight'; basis unclear also up to 30 t depending on chassis WP-RU85
L × W × H	10.3 × 2.5 × 3.27 m	
Road speed	90 km/h	maximum
Road range	1,000 km	maximum also 500 km WP-RU85
Fording	1.8 m	
Time into action	3 min	WEG 'time in action'; WEG also gives deployment time 5 min; which is time into action is not stated also about 15 min WP-RU85
Time out of action	INA	
Ground pressure	INA	

Range, 57E6 missile	20 km	maximum, aircraft-type targets; KBP figures for export 57E6-E, claim also 12 km CSIS6
Minimum range, 57E6 missile	1.2 km	KBP figures for export 57E6-E, claim also 1.5 km CSIS6
Altitude, 57E6 missile	5 m to 15 km	maximum, KBP figures for export 57E6-E; 5 m only in optical mode over water, claim also 10 km CSIS6
Range, 30 mm 2A38M gun	4 km	maximum, air and ground targets, claim
Minimum range, 30 mm 2A38M gun	200 m	air targets, claim
Altitude, 30 mm 2A38M gun	0 to 3 km	maximum, air targets, claim
Detection range, 1RS1 search radar	36 km	maximum, 2 m2 target; S-band; up to 20 targets tracked, claim also 47 km APA8
Detection range, 1RS2-1 tracking radar	28 km	maximum, target size not stated; WEG and Armada, claim also 24 km MISS8 APA8
Detection range, optical post	17-26 km	auto-tracking, F-16 autotracking, 10 km visibility; 13-15 km for HARM missile, claim
Missile speed, 57E6	1,300 m/s	maximum, claim
Target speed	1,000 m/s	maximum, claim
Range, Pantsir-SM missile	40 km	claimed, Pantsir-SM with new missile; KBP chief designer, 2016, claim
Detection range, Pantsir-SM radar	75 km	claimed, target size not stated, claim
Range, Pantsir-S1M missile	30 km	claimed, export Pantsir-S1M; new missile, designation not given by source; High Precision Systems, 2019, claim

SOURCES APA8 AR38 AR39 ARMADA8 CSIS6 IISS25 LENTA2 MIL48 MISS8 ORYX1 RSTAR RUSIAD RWW TASS12 TASS13 WEG44 WP-EN56 WP-RU85

AIR DEF3

Tunguska 2K22 «Тунгуска»

NATO: SA-19 Grison

TR

Tracked short-range gun and missile air defence vehicle, 30 mm guns and SAMs In service 1982

OPERATED BY motor rifle brigade, air defence missile-artillery battalion, Tunguska battery, 6 launcher vehicles per battery [D 2016 M](#)

tank brigade, anti-aircraft missile-artillery battalion, SA-19 battery; ATP 4-161 gives this battalion to tank brigades, but the battery figure is for brigades generally, 6 per battery (generic brigade figure: RWW 'maneuver brigade', ATP fig. 4-19) [D 2024 L](#)

INDICATES A Tunguska battery in a forward position points to a manoeuvre brigade's air defence missile-artillery battalion covering its first-echelon battalions; doctrine places it in brigade air defence, not in area air defence units.

RECOGNISE Launcher 2S6M and 2S6M1 carry eight missile tubes, four each side of the turret; original 2S6 had four, two each side (APA and FM 100-2-3: eight).

Radar Search radar at turret rear, folded over the rear when travelling; tracking radar at turret front; search antenna a cylindrical parabolic section, tracker a dish.

Chassis Tracked, six road wheels per side, three return rollers, rear sprocket; driver front left, engine at rear. GM-352 chassis; GM-5975 on 2S6M1.

NOT TO BE CONFUSED WITH Pantsir-S1 p. 39 Usually an 8x8 truck with 12 missile tubes in two blocks of six, one each side of the turret; the tracked GM-352M1E Pantsir is a rarer offer with 8 missiles. ZSU-23-4 Shilka Gun-only predecessor, four 23 mm barrels, no missile tubes; Tunguska has two twin 30 mm guns with missile tubes outboard and a second radar at the turret front.

SOURCES APA8 ATP FM32 IISS25 JREW02 MISS8 ROE2 RWW WEG70 WP-EN83 WP-RU129



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Fires guns on the move but must be stationary to fire missiles. Not amphibious (WEG); fords 1.0 m (WEG 2S6M asset, RUWIKI).

Weight, combat	34 t
L × W × H	7.93 × 3.24 × 4.02 m radar up; 3.36 m radar down
Speed, maximum road	65 km/h
Range, road	500 km cruising range

Range, 9M311-M1 missile	10 km maximum, Tunguska-M1, air target
Minimum range, 9M311-M1 missile	2.5 km Tunguska-M1, air target; APA gives 2.5 km for 2K22 and 2K22M too
Altitude, 9M311-M1 missile	3.5 km maximum, air target; ceiling for 9M311 family
Range, 30 mm 2A38M guns	4 km maximum, air target
Minimum range, 30 mm 2A38M guns	200 m air target

SEEN WITH 2F77 / 2F77M transporter-loader (6 per battery (WEG; RUWIKI: one per 2S6); APA 2 to 3, WIKI up to 3)

AIR DEF3

Strela-10 and Osa short-range SAM

NATO: SA-13 Gopher

TR

Short-range SAM: tracked Strela-10 (optical/IR) and wheeled Osa (radar command guidance) In service 1971

OPERATED BY motor rifle brigade, air defence missile-artillery battalion, missile battery (SA-13), 6 launchers per battery [D 2024 M](#)

tank brigade, air defence missile-artillery battalion, missile battery (SA-13), 6 launchers per battery [D 2024 L](#)

air defence missile regiment, division air defence regiment (Osa-AKM, being deactivated in 2019) [D 2019 L](#)

INDICATES SA-13 in a brigade fits the missile battery of its air defence missile-artillery battalion, six launchers in doctrine; two platoons cover the artillery group. Osa-AKM fits a division air defence regiment, being withdrawn in 2019; it covers troops on the march.

RECOGNISE Launcher Strela-10: turret with four missile canisters in two pairs, range radar between them. Osa-AK/AKM: six ribbed missile containers; first Osa had four exposed missiles.

Radar Strela-10 9A35 adds Flat Box-B passive antennas on hull and turret; 9A34 has none. Osa: surveillance array folds back 90 degrees at rear, tracking group at front.

Chassis Strela-10: modified MT-LB, tracked, six equally spaced road wheels, no return rollers. Osa: BAZ-5937, 6x6 wheeled, amphibious; 4.20 m high with mast stowed.

Complex Strela-10 battery: one 9A35, three 9A34, PPRU-1M command post with Dog Ear radar. Osa battery: four launch vehicles and two reload vehicles with crane.

NOT TO BE CONFUSED WITH 9K31 Strela-1 (SA-9 Gaskin) Four-wheeled BRDM-2 chassis; Strela-10 is on a tracked MT-LB. 2K22 Tunguska (SA-19 Grison) p. 40 Radar-directed gun and missile hybrid; Strela-10 is visually aimed and has no cannon, only a 7.62 mm MG.

SOURCES APA11 APA12 ATP CNA IISS25 JANESRG JREW02 MISS12 RWW WEG60 WP-EN72 WP-EN73



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Strela-10 is tracked and floats only with fitted buoys (M2). Osa is a wheeled amphibious 6x6 (WH-AT-A class); it fires from short stops, not on the move.

Weight, combat	12.1 t Strela-10
Weight, combat	18.68 t Osa-AKM, APA8 comparative table
L × W × H	6.45 × 2.85 × 2.22 m Strela-10
Road speed	60 km/h maximum, Strela-10

Range, 9M37M/9M333	0.8 to 5 km engagement envelope, Strela-10 engagement zone, all variants; target closing at up to 415 m/s or receding at 310 m/s
Altitude, 9M37M/9M333	25 m to 3.5 km engagement envelope, Strela-10; 9M333 minimum 10 m
Range, 9M33M3	15 km operational range, Osa-AKM missile
Altitude, 9M33M3	4 km maximum flight altitude, Osa-AKM missile; WEG figure
Detection range, Land Roll surveillance radar	30 km acquisition range, Osa; target type not stated in source

4

Radars

IN THIS CHAPTER

Air surveillance radars 41

Counter-battery and ground surveillance radars 42

LOOKING FOR THESE? THEY ARE IN OTHER CHAPTERS

SYSTEM	WHAT IT IS	CHAPTER	PAGE
PRP-4A Argus / PRP-4M Deytery	Mobile artillery reconnaissance post on BMP-1 chassis	7 Command and signals	68
1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)	Tracked command, observation and fire direction vehicles, SP howitzer battalion	7 Command and signals	67

Nebo-M 55K6M «He60-M»

NATO: Tall Rack **WH-AT**

Mobile multiband air surveillance radar complex; paired low-altitude radar Podlet-K1 In service 2012

OPERATED BY Unidentified unit, air defence and radio-technical crews with a division-level command post (MoD release; unit not named) [R 2017-09 L]

NUMBERS Holdings 4 Nebo-M and Nebo-UM combined, stations in Transbaikalia, Khabarovsk and Primorye [2015]

INDICATES Nebo-M or Podlet-K1 indicates an Aerospace Forces radio-technical (radar) unit building the air picture, often near an S-300 or S-400 position that the radars cue. One reported Podlet-K1 site (Kherson, Jul 2022) lay over 100 km behind the front.

RECOGNISE Antenna RLM-M: very large VHF array of short Yagi elements, 24 x 7. RLM-D: flat L-band panel, 38 x 48 elements; outer segments fold and stow flat under clamshell doors.

Antenna Podlet-K1: one large upper phased array, two smaller back-to-back IFF arrays below it, a narrow anti-jam antenna beside it. Antenna lies lowered on the truck in travel.

Mast Nebo-M antennas raised and folded hydraulically on a mast with a compact equipment cabin at its base; small navigation antenna package at the top of each array.

Chassis Nebo-M: each module on a BAZ-6909-015 8x8 truck, the chassis of the S-400 TEL. Podlet-K1: KamAZ-6350 8x8 truck.

NOT TO BE CONFUSED WITH 55Zh6U Nebo-U Transportable, not self-propelled: six transport units and hours to deploy. RLM-M is one 8x8 truck that deploys in about 15 min.

11119 Nebo-SVU Box-shaped T/R module enclosures on the mast and rear backlobe elements; both absent on RLM-M.

59N6E Protivnik-GE Semi-trailer mounted L-band radar; RLM-D uses a smaller antenna on a self-propelled 8x8.



Nebo-M RLM-M. Ministry of Defence of the Russian Federation, CC BY 4.0



RLM-D and KU-RLK control cabin. Ministry of Defence of the Russian Federation, CC BY 4.0

MOBILITY Self-propelled 8x8 trucks; Nebo-M can also move by rail. Each Nebo-M module has its own 100 kW generator. Hydraulic levelling and antenna deployment.

Road speed, KamAZ-6350	95 km/h maximum, base truck, Podlet-K1 carrier
Height, Podlet-K1 antenna deployed	12 m Podlet-K1, radar fully deployed
Time into action, Nebo-M	15 min Nebo-M, manufacturer cited stow and deploy time, claim
Time into action, Podlet-K1	20 min Podlet-K1, deployment or stowage, claim

Detection range, Nebo-M	600 km maximum, complex; target type, size and altitude not stated, claim
Altitude coverage, Nebo-M	600 km maximum, complex, ballistic targets implied; not stated, claim
Detection range, Podlet-K1	200 km maximum, main mode, 10-200 km; target size not stated, claim
Minimum range, Podlet-K1	10 km air targets, lower limit of stated 10-200 km detection zone, claim

SOURCES AIF3 ANNA APA6 APA7 ARMADA7 ESP2 GS21 JREW02 RUSIAD TRUCK TWZ8 TWZ9 U244 VPK6 WEG43 WP-RU80 WP-RU81 ZVEZ

96L6 all-altitude acquisition radar 96Л6 «Всевысотный обнаружитель»

NATO: Cheese Board

WH-AT

3D C-band acquisition and target designation radar for S-300PM/PMU2 and S-400 SAM units In service 1997

OPERATED BY aerospace forces air defence regiment, 3rd SAM battalion, INA (photographed deployed and stowed) [\[O 2010-02 M\]](#)
 aerospace forces air defence regiment, S-400 'battery' in Armada's usage; APA export diagram lists it as a battery component, 1 per S-400 battery (Armada, term undefined; APA lists 96L6E as a battery component without a count); S-300PMU2 count INA [\[R 2014 M\]](#)
INDICATES A 96L6 usually marks a nearby long-range SAM fire unit (S-300PM/PMU2 or S-400) of the Aerospace Forces, which are district-level assets. The maker also offers it to radio-technical units, so on its own it does not confirm a SAM site.

RECOGNISE Antenna Deployed: one flat rectangular array, slightly taller than wide, near-vertical, tilted back a little, on a rotating pedestal just behind the cab; lattice IFF array below.

Antenna, stowed Stowed: array folds forward and lies flat and horizontal over the pedestal and cab roof, giving a long, low, box-like vehicle outline.

Mast Normally no mast. Antenna post can be lifted onto a 40V6M (23.8 m) or 40V6MD lattice mast for low-level cover; the roughly 2 h fitting time is an unverified maker figure.

NOT TO BE CONFUSED WITH 76N6 Clam Shell Split antenna with a large beak-like shroud, usually raised on a 40V6M mast. A mast-mounted 96L6 shows one flat rectangular array.

64N6E / 91N6E Big Bird Janus-faced (back-to-back) space-fed array on a semi-trailer towed by an MZKT-7930 or BAZ-6403 tractor. 96L6 is a single self-propelled vehicle with one array.



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Self-propelled 8x8. Single-vehicle version ready 5 min from march. Two-trailer and mast versions are semi-mobile; maker figures of 30 min to about 2 h not verified. Not amphibious.

L × W × H	11.7 × 3.07 × 2.92 m	WEG figure for MAZ-543 chassis; radar vehicle is MZKT-7930
Road speed	65 km/h	maximum, WEG figure for MAZ-543 chassis
Road range	650 km	cruising, WEG figure for MAZ-543 chassis
Fording	1.3 m	WEG figure for MAZ-543 chassis

Detection range	300 km	instrumented, Range band 5 to 300 km for detected targets; target size and altitude not stated, claim
Maximum target altitude	100 km	All-altitude mode, aerodynamic targets, claim
Band	C-band	claim
Frequency	5.25 to 5.925 GHz	
Elevation cover, all-altitude mode	-1.5° to +20°	360° azimuth

SOURCES APA1 APA2 APA3 ARMA2A1 ATP GS5 JREW02 MIL1 MILRU ORYX1 PVO RS RT RUS11 RUSIAD RWW U241 WEG5 WMC1 WP-RU9

Zoopark-1 / Zoopark-1M 1A219, 1A219M; 1A260 «Зоопарк-1», «Зоопарк-1М»

TR

Tracked radar that locates enemy guns, mortars, rockets and missile launchers In service 1970

OPERATED BY motor rifle brigade, fire control and artillery reconnaissance battery, radar platoon (with SNAR-10M1) [\[D 2016 M\]](#)
 tank brigade, fire control and artillery reconnaissance battery, radar platoon [\[D 2016 M\]](#)
 motor rifle division [\[D 2024 L\]](#)

INDICATES Doctrinally held in the radar platoon of a manoeuvre brigade's fire control battery and by artillery formations. Seeing one suggests brigade-level or higher artillery is nearby and set up for counter-battery fire.

RECOGNISE Antenna One flat rectangular phased-array panel. 1L219/1L219M: raised upright at rear of roof. 1L260: a much taller panel.

Mast No separate mast. Panel is lifted and turned on the roof by a lift and rotation mechanism. For travel it folds down flat along the roof.

Chassis 1L219/1L219M: low MT-LBu tracked carrier, flat sides, raised rear hull. 1L260: larger GM-5955 tracked chassis, 7 road wheels a side, boxy hull with side lockers.

Complex Radar vehicle can work alone. Support: 1130 (1L219M) or 1138 (1L260) maintenance vehicle and a generator; same brigade fire control battery as SNAR-10M1 and PRP-4A.

NOT TO BE CONFUSED WITH SNAR-10 / SNAR-10M1 (1RL232) battlefield surveillance radar [p. 44](#) Same MT-LBu hull but has a turret with a PKT machine gun and a rotating parabolic radar, not a large flat panel.

ARK-1 / ARK-1M Rys (1RL239) counter-battery radar [p. 43](#) Predecessor on the same MT-LBu hull and looks alike. No reliable distance cue found; confirm by markings or close imagery.



Boevaya mashina, CC BY-SA 3.0 de

MOBILITY Tracked, cross-country capable. Emplaces or displaces in about 5 min. Swimming ability not documented for either chassis.

Weight, 1L261 (1L260)	38.1 t
Road speed, 1L219M	62 km/h
Road speed, 1L260	70 km/h
Road range	500 km

Detection range, mortar, 1L219M	20 km	detection of launch position, Mortar firing position; calibre not stated
Detection range, tube artillery, 1L219M	15 km	detection of launch position, Field artillery firing position
Detection range, rocket artillery, 1L219M	30 km	detection of launch position, MLRS firing position
Detection range, tactical missile, 1L219M	40 km	detection of launch position, Tactical missile launch position

SOURCES AR68 AR69 AR70 ARMA2A10 ATP BMPD11 DOGS FISH GWARA2 IZ6 MIL71 MIL72 MIL73 ORYX1 RUMOD RWW SAIDPVO WEG2015 WMC27 WMC28 WP-RU141 not in WEG

ARK-1M Rys 1P/239 «РыбЦб»

NATO: Small Red **TR**

Artillery weapon-locating and fire-control radar In service 1977

OPERATED BY Division reconnaissance battery (Soviet era), division's reconnaissance battery (WEG wording, identical to Wikipedia; Soviet-era) [R 2024 L](#)

INDICATES A single sighting points to an artillery reconnaissance element supporting a formation's guns. Sources give no current Russian subunit or count, so it says little about formation type.

RECOGNISE **Antenna** Parabolic antenna (JREW02); radiator in a compact radio-transparent housing plus a large receiving antenna (TOPWAR). Sizes INA.

Mast No mast described in any read source. Two emitters and a receiving antenna sit outside the hull (MILN); TOPWAR names one emitter housing. Height INA.

Chassis MT-LBu unarmed tracked carrier; electronics mostly inside the armoured hull.

Complex Single self-contained vehicle with own navigation, topographic survey, communications and power supply (ARK-1M generator). Crew 4.

NOT TO BE CONFUSED WITH 1L219 Zoopark-1 p. 42 Also MT-LB based; Zoopark-1 uses a phased array antenna, ARK-1M a parabolic antenna.

1L260 Zoopark-1M p. 42 Built on the GM-5955 tracked chassis with a phased array antenna.

1R1232 SNAR-10 p. 44 Same MT-LBu hull, parabolic antenna and near-identical size; SNAR-10 has a turret with a 762 mm PKT MG. No source arms the ARK-1M.

1K148 Yastreb-AV Four-axle BAZ-6910-025 wheeled chassis, not tracked; antenna post at the rear.



ShinePhantom, CC BY-SA 3.0

MOBILITY Tracked MT-LBu chassis; WEG says it may not be amphibious given radar weight. Forcing depth INA.

Weight	15.7 t
L × W × H	7.61 × 2.85 × 2.72 m Chassis overall
Road speed	61 km/h maximum
Road range	500 km cruising

Detection range, ARK-1M vs artillery (howitzers) **8 km** unspecified, Artillery (WEG), howitzers (JREW02); target and mode not stated by WEG

Detection range, ARK-1M vs mortars **13 km** unspecified, Mortar (WEG); mode not stated

Detection range, ARK-1M vs MRL rockets **25 km** unspecified, Rocket (WEG); mode not stated

Detection range, ARK-1M vs battlefield missiles **30 km** unspecified, Missile (WEG)

Frequency band	NATO H band (6–8 GHz)
Track capacity	3 tracks/min

SEEN WITH MT-LBu tracked chassis (1)

SOURCES ATP JREW02 MIL2 MIL3 ORYX1 PMUL TOPWAR1 TOPWAR2 WEG6 WEG2015 WEG-EN11

1B75 Penicillin 1B75 «Пенициллин»

WH-AT

Passive acoustic-thermal gun locator (1B75); man-portable counter-mortar radar (1L271) In service 2018

OPERATED BY self-propelled artillery battalion, battalion fire control network, conceptual ATP fig. 5-3 (1L271) [D 2024 M](#)

tank division, 1L271; holding subunit not stated [O 2018-01 M](#)

combined arms army, 1L271 in rocket, missile and tube artillery units

[R 2017 L](#)

INDICATES A 1B75 points to an artillery regiment or brigade counter-battery element, as deliveries were planned for those units. A 1L271 was issued down to company and battalion level and marks forward mortar or artillery fire control.

RECOGNISE **Antenna** 1L271: transceiver on hand-turned tripod pedestal; reflector described as single-curvature parabolic or as phased array with mirror. 1B75 has no radar antenna.

Mast 1B75: stabilised head with six TV and six thermal cameras on a raising mast. Per GlobalSecurity only, the mast folds rearward onto the van roof in transit.

Chassis 1B75: KamAZ cross-country truck, 8x8 per GlobalSecurity and Wikipedia; van body and four jacks per GlobalSecurity only. 1L271: man-carried or on a UAZ-2966.

Complex 1B75: two KamAZ vehicles plus remote ground acoustic sensors, up to six vehicles. 1L271: 135 kg set of radar, processor, power unit, console and radio.

NOT TO BE CONFUSED WITH Zoopark-1 / Zoopark-1M p. 43 Radar complex on one armoured amphibious vehicle; 1B75 is a KamAZ truck with a camera mast and no radar.

1B33M AZK-7M Older sound-ranging system, acoustic only; lacks the 1B75 thermal and TV camera mast.



Allcoer, CC BY-SA 3.0; GFDL

MOBILITY 1B75 on KamAZ 8x8 truck; mast raised and ground sensors laid out at site, crew relocates daily. 1L271 man-portable (135 kg in loads) or vehicle-carried.

Weight, 1L271	135 kg complete set: radar, processor, power unit, claim
Road speed, 1B75	90 km/h maximum, KamAZ chassis, claim
Time into action, 1L271	5 min maximum, preparation for work, claim
Weight, 1B75	INA

Detection range, 1B75 **up to 25 km** maximum, radius; artillery, mortar and MLRS firing positions, sound and thermal, claim

Target location time, 1B75 **5 s** maximum, single firing target, claim

Radio range, 1B75 **40 km** maximum, to external subscribers, claim

Detection range, 1L271 **20 km** maximum, tank-type moving ground target, claim

SOURCES AIF4 AR40 ATP BMP05 BMP06 DEFEX GS24 IZV20 IZV22 ORYX1 RADTUT RG152 RG212 RG243 RWW SHEP3 TASS14 WP-EN57 WP-EN58 WP-RU86 WP-RU87 not in WEG

SNAR-10 / SNAR-10M1 Leopard CHAP-10M1 «Aeonapp»

 NATO: Big Fred **TR**

Tracked battlefield surveillance and artillery-adjustment radar In service 1975

OPERATED BY motor rifle brigade, fire control and artillery reconnaissance battery, radar platoon **[D 2016 M]**
 tank brigade, fire control and artillery reconnaissance battery, radar platoon **[D 2016 M]**

military district, artillery units of the tank formation stationed in the South Urals (MoD wording; formation not named) **[R 2019-01 M]** seen with Central Military District

Unidentified unit, captured Kyiv region Apr 2022; Ukraine claimed one destroyed, Kherson left bank, Jan 2024 **[R 2024-01 L]**

INDICATES A SNAR-10M1 points to a maneuver brigade's fire control battery radar platoon, so an artillery-adjustment and ground-surveillance element is working for that formation. Ukraine also holds the SNAR-10, so a sighting alone is not proof of a Russian unit.

RECOGNISE Antenna Rectangular reflector nearly vehicle width, raised electromechanically for use; folds forward when travelling.

Mast No mast described: radar is built into the vehicle's turret, antenna on the roof (SNAR-10); on the M1 the antenna module sits on the rear roof.

Chassis MT-LB family tracked hull; the Big Fred version is a roof-mounted radar on an MT-LB.

Complex Single vehicle with its own diesel generator (M1); sources describe no separate power or command vehicle.

NOT TO BE CONFUSED WITH MT-LBu command vehicle p. 17 MT-LBu has canvas cover extending to rear and extra radios, no folding radar reflector on roof.

PRP-4A Argus p. 68 Built on the BMP-1 infantry fighting vehicle chassis with a mast-mounted radar; SNAR-10 is on the MT-LB tractor hull.



Milru / Yevgeny Kel, CC BY 4.0

MOBILITY Tracked MT-LB derivative; radar deploys from halt, about 5 min into action from prepared position.

Weight, combat **12.7 t** SNAR-10M1, developer figure, claim
 Road speed **60 km/h** maximum, SNAR-10 (WEG); SNAR-10M1 'up to 60 km/h' (developer)
 Road range **500 km** cruising, SNAR-10
 Time into action **5 min** claimed, SNAR-10, prepared position; up to 20 min unprepared, claim

Detection range, SNAR-10 radar **16 km** maximum, moving vehicles, without MTI; 10 km with MTI, claim
 Detection range, SNAR-10 radar **30 km** maximum, moving ships, without MTI, claim
 Detection range, SNAR-10 radar **10 km** maximum, shell impact (100 mm), without MTI, claim
 Detection range, SNAR-10M1 radar **up to 40 km** maximum, tanks and vehicles; Russian MoD figure; instrumented range 0.2–40 km, claim

SOURCES AR51 ARMYINF6 ATP BMPD8 FOCUS6 JANESRG JREW02 MILUA5 PERISC RWW SHEPH UKRIN VPK11 VPK12 WEG48 WEG59 WEG2015 WP-U113

5

Electronic warfare

IN THIS CHAPTER

Radar and strategic jamming	46
Communications jamming	48
GNSS, drone and IED jamming	53

LOOKING FOR THESE? THEY ARE IN OTHER CHAPTERS

SYSTEM	WHAT IT IS	CHAPTER	PAGE
Orlan-10	Catapult-launched recon UAV for artillery spotting and EW	9 Unmanned aircraft	78

Murmansk-BN «Мурманск-БН»

WH-AT

Strategic HF (3–30 MHz) communications intercept and jamming complex, multi-truck In service 2014

OPERATED BY EW brigade, Probably a company of the REB-N (anti-ground) battalion (FOI assessment) **D 2018 H**
Unidentified unit, Radio-jamming node (узел радиономех); its 1st section named at the 471st (KZ), 2 complexes at the Kamchatka centre (KZ, reported Jan 2016; FOI: delivered 2014) **O 2019 H**
EW brigade, At least 1 complex per brigade (Armada); 2 antenna groups observed at 16th EW bde, vehicle count at 19th suggests 2 (PAS imagery, 2019) **O 2019 M**

INDICATES Seeing Murmansk-BN indicates an Army EW brigade (IISS lists 5; 15th under General Staff) or a fleet EW centre, with long-range HF intercept and jamming in place; it takes about 72 hours to deploy, so a raised complex marks a deliberate, fixed position.

RECOGNISE **Antenna** Wire HF antennas and mast guy lines run from the masts to the ground; from afar like a giant spider's web. Antenna field about 800 m a side. Stowed: carried on trucks.

Mast Four hydraulic telescopic masts up to 32 m, guyed. Armada: antenna trucks typically about 35 m apart across a 90-degree arc. Mast lowered for travel.

Chassis Heavy multi-axle KamAZ (or Ural) trucks with unified box bodies (kungs); some masts reported on towed trailers.
Complex Seven trucks: four mast vehicles plus generator, stowed-antenna and command post vehicles. Kola site seen with two groups of four antenna vehicles about 350 m apart.

NOT TO BE CONFUSED WITH Krasukha-2 / Krasukha-4 p. 46 Each is housed on a single truck; Murmansk-BN is spread over several trucks with tall guyed masts.

Samarkand Stationary coastal jamming complex, probably a fixed version of Murmansk-BN; not truck-mounted.

SOURCES 3G ARMADA6 B02 CRFS FOI ICDS IISS25 IZ3 KZ3 MIL45 PAS RG16 RG174 RWW UKRINF2 WEG42



Министерство обороны России, CC BY 4.0

MOBILITY Road-mobile truck column; needs a prepared antenna field about 800 m a side, marking and engineer work; 72 hours to deploy. Trucks up to 35 t need category E drivers.

Weight, vehicle	up to 35 t	Loaded KamAZ trucks of the complex, claim
Time into action	72 h	Normative deployment time, claim
Number of vehicles	7	Complex in travel order, claim
Time out of action	INA	

Jamming range, HF communications	5,000 km	effective (Armada's term), HF (shortwave) radio links, skywave; ship, aircraft and ground emitters, claim
Frequency band	3–30 MHz	(HF)
Power	up to 400 kW	In some operating modes (KZ); generator output (Armada), claim
Mast height	32 m	4 telescopic masts, fully raised, claim
Deployed position area	640,000 m2	Antenna field about 800 x 800 m, claim

Krasukha-2 / Krasukha-4 1Λ269; 1PΛ257 «Kpacыxa-2»; «Kpacыxa-4»

WH-AT

Truck radar jammers: Krasukha-2 vs S-band AEW radar; Krasukha-4 vs X/Ku-band radars In service 2011

OPERATED BY EW brigade, REB-S (counter-air) EW battalion, INA; usually one Krasukha-2 and one Krasukha-4 per brigade (reported 2022) [D 2018 H](#)

EW brigade, part of 1 Krasukha-4 complex (command-post container found near Makariv); Armada estimates 4 complete Krasukha systems in theatre [R 2022-03 M](#)

EW battalion, independent REB-S battalion of an Air and Air Defence Army (VKS); FOI infers, does not name Krasukha there [R 2018 L](#)

Unidentified unit, 1 Krasukha-4 reported hit [R 2023-11 L](#)

Unidentified unit, 1 Krasukha-2 reported [R 2018-08 L](#)

INDICATES Krasukha jammers belong to EW brigades (REB-S battalions); four under the military districts, the 15th (Tombov) under the General Staff. Manoeuvre brigades have EW companies. A sighting points to an EW brigade element or a protected site such as an air base.

RECOGNISE **Antenna** Krasukha-2: one very large solid circular dish, about twice a wheel diameter across, on a turntable at the rear; rotates 360° and tilts. Stowed look not documented.

Antenna Krasukha-4: box-shaped rotating turret on a van-body roof with three dishes, two large on the sides, one small on the front; all far smaller than the Krasukha-2 dish.

Mast Krasukha-4: telescopic mast with a small antenna head behind the turret; the second vehicle also carries a mast antenna. Krasukha-2: no tall mast in photos.

Stowed Krasukha-4: in travel position the turret is stored inside the shelter (AR); in action two hydraulic stabilisers are lowered on each side of the shelter.

Chassis Krasukha-2: BAZ-6910-022 8x8, angular flat-faced cab, axles 2+2 with a wide gap. Krasukha-4: KamAZ-6350 8x8, standard KamAZ cab-over, spare wheel behind cab.

Complex Krasukha-4: two KamAZ-6350 trucks, jammer plus command-post container vehicle. Krasukha-2: three vehicles (one source); probably procured with Moskva-1 (FOI).

NOT TO BE CONFUSED WITH 1Λ269 Krasukha-2 vs 1RL257

Krasukha-4 Krasukha-2: one big dish on a BAZ-6910 with angular cab. Krasukha-4: turret with three smaller dishes on a KamAZ-6350 van body.

Pelena-1 (older S-band counter-AEW jammer) Antenna is a tall curved rectangular reflector on a towed wheeled trailer (APA photos), not a solid circular dish on a self-propelled 8x8 truck.

1Λ267 Moskva-1 ELINT and command complex p. 47 Three KamAZ trucks (FOI: one ELINT, two command; 3G: KamAZ-6350-1335, third with a jamming control station); no Krasukha-4 style dish turret.

GT-01 Murmansk-BN HF jamming complex p. 45 Has KamAZ-6350 support trucks, but its signature is very tall extendable HF antenna masts on KamAZ-53501 trucks, not dishes.

55Zh6M Nebo-M radar modules (BAZ-6909-015) p. 41 Same heavy BAZ 8x8 family, but antennas are a flat rectangular planar array or a Yagi array, not a solid circular dish.

VARIANTS 1Λ269 Krasukha-2 (Krasukha-20) BAZ-6910-022 8x8 with one large solid circular dish at the rear. S-band jammer aimed mainly at AEW radars such as E-3 and E-8. Analogue equipment per ru.wikipedia. 1RL257 Krasukha-4 (Krasukha-S4) KamAZ-6350 8x8 van with a three-dish turret, plus a second KamAZ-6350 carrying the command post. X/Ku-band jammer vs fighter fire-control and ISR radars, UAVs and LEO satellites. Digital equipment per ru.wikipedia.



Krasukha-4. The Ministry of Defence of the Russian Federation, CC BY 4.0



Krasukha-2. The Ministry of Defence of the Russian Federation, CC BY 4.0

MOBILITY Heavy 8x8 trucks. Must halt and emplace to jam: Krasukha-4 needs under 20 min to emplace or displace; Krasukha-2 lowers jacks before operating.

Krasukha-4: weight	39,000 kg also 12,450 kg AR30
Krasukha-2: weight	40,000 kg BAZ-6910-022 chassis full mass; kerb mass 18 t
Krasukha-4: road speed	95 km/h maximum, KamAZ-6350 truck figure (AR)
Krasukha-2: road speed	80 km/h maximum
Krasukha-4: road range	1,000 km KamAZ-6350 truck figure (AR)
Krasukha-2: road range	1,000 km
Krasukha-4: fording	1.75 m KamAZ-6350 truck figure (AR), claim
Krasukha-2: fording	1.4 m BAZ-6910-022 chassis figure (REACH)
Krasukha-4: time into action	less than 20 min

Krasukha-2: jamming range	up to 250 km up to, air target: AEW aircraft radar; target altitude not stated
Krasukha-4: jamming range	150–300 km maximum, air and ground-based radar targets; target size and altitude not stated, claim
Krasukha-2: target band	S-band also 2.3–2.7 GHz CRFS
Krasukha-2: main targets	AEW aircraft radars (E-3 Sentry, E-8 JSTARS) air targets
Krasukha-4: operating band	8–18 GHz (X and Ku band) also 8.5–10.7 and 13.4–17.7 GHz 3G
Krasukha-4: main targets	Fighter and interceptor radars, UAVs, reconnaissance aircraft, LEO satellites air and space targets
Krasukha-4: coverage	360° azimuth; 1° to 60° elevation
Krasukha-4: output power	24 dBW (18 GHz); 23 dBW (13 GHz); 21 dBW (18 GHz) 30 kW
Krasukha-4: power consumption	24 h
Krasukha-4: continuous operation	INA also 7 or 3 REACH
Krasukha-2: crew	INA
Krasukha-4: crew	INA also 3 AR30

SEEN WITH Krasukha-4 command-post vehicle (KamAZ-6350 with container) [1 per Krasukha-4 complex]; 1Λ267 Moskva-1 ELINT and command complex (INA; one Moskva-1 can link up to nine Krasukha-type jammers); KamAZ-6350 8x8 truck [2 per Krasukha-4 complex]; BAZ-6910-022 8x8 chassis [1 per Krasukha-2 jammer vehicle]; RB-109A Bylina EW brigade command system (brigade-level EW C2, not Krasukha-specific)

SOURCES 3G APAs APAG AR30 ARMADA3 ATP CRFS FOI ICDS MIL28 MIL29 MIL30 REACH RWW TWZ6 WEG36 WMC6 WMC7 WMC8 WMC9 WP-EN43 WP-RU69

Moskva-1 1Λ267 «MockBa-1»**WH-AT**

Passive ELINT and automated control of counter-air jammers and air defence cueing In service 2014

OPERATED BY EW brigade, REB-S (counter-air EW) battalion, reported [D 2018 H]

EW brigade, INA (ARM22 lists one Moskva-1 among five brigade systems) [R 2023-09 L]

Unidentified unit, 1 claimed destroyed [R 2022-09 L]

INDICATES Moskva-1 is held by Ground Forces EW brigades, four of which answer to the military districts. Seeing it suggests a brigade counter-air EW element and Krasukha-type jammers at other sites, up to nine of which one complex can task.**RECOGNISE** **Antenna** No automatically raised antenna array, unlike Krasukha. The ELINT module (1L265) was withheld from press; its antenna is not described in text (INA).**Chassis** KamAZ-6350 8x8 trucks with plain box bodies (kung); a Russian reporter wrote the vehicles are hard to tell from other trucks.**Complex** Three KamAZ vehicles per complex: an ELINT module and control vehicles; one account names a power-station truck as the third.**Complex** Deployed: the control module stands on hydraulic legs on the ground while its truck drives 30–100 m away.**NOT TO BE CONFUSED WITH** 1RL257 Krasukha-4 Krasukha deploys an automatically raised antenna block; Moskva-1 vehicles show plain box bodies.

1L269 Krasukha-20 p. 46 Krasukha-20 is on the larger BAZ-6910 chassis, not KamAZ.

Other KamAZ-6350 box-body vehicles p. 75 No reliable external mark in text sources; FOI says Krasukha-20 is probably procured together with Moskva-1, so Krasukha jammers nearby are the main clue.

Italy V. Kuzmin, CC BY-SA 4.0

MOBILITY Road and track mobile truck complex; manufacturer gives 45 min to deploy. Control module can be set down on legs apart from its truck.

Weight	12,450 kg	chassis only
L × W × H	9.3 × 2.5 × 3.25 m	chassis
Road speed	95 km/h	maximum
Road range	1,000 km	cruising

Detection range, airborne emitters
Jamming range, airborne S-band radar

400 km up to, passive; emitting aircraft and cruise missiles; 360 deg sector, claim
250 km up to, air targets; ARM23 says the 1L267 'can engage targets' per unnamed sources; Moskva-1 itself has no jammer (FOI)

Sector of view
Frequency coverage

360 deg claim
UHF, L, S, C, X and Ku bands ELINT module 1L265

Controlled systems

9 EW or air defence complexes tasked simultaneously, claim

SOURCES 3G AEX17 ARMADA4 ARMADA5 ATP CH97 DLG22 FOI ICDS IEEE22 IZ234 RIA15 RWW SDU13 TOPWARS VPKS WEG39 WP-RU66 WP-RU76 WP-RU77**R-330B Mandat-B and R-934BMV** P-3305; P-9345MB «Мангам-Б»**WH-AT**

Ground VHF/UHF radio and air-communication jammers, tactical level In service 1996

OPERATED BY EW company, EW company of a motor rifle or tank brigade: R-330B squad in the VHF-band jamming platoon; R-934B squad (VHF aircraft) in the radio-communications jamming platoon

[D 2016 H]

EW company, At least one Russian EW company deployed to Ukraine in 2014 had two R-934B for airborne radio jamming; company not named, 2 R-934B in one EW company (2014) [R 2014 L]

INDICATES Seen with a brigade or division, it fits the organic EW company. RWW and FOI place these jammers in the EW company of motor rifle and tank brigades. It shows radio and air-link jamming is planned, not the size of the force.**RECOGNISE** **Antenna** R-934UM2: one receiving and direction-finding antenna-feeder system plus seven transmitting antenna-feeder systems. Sizes INA.**Mast** R-330T/B carry a telescoping mast system that is raised when deployed; R-330U stows its antenna in a rectangular housing. Mast height INA.**Chassis** R-934B and R-330B: MT-LBu tracked, or Ural-4320-31 6x6 with K1-4320 body (R-934B); R-330T on KamAZ-43101 or Ural-43203-31 (K2-4320).**Complex** Wheeled versions tow a generator trailer (16 kW for R-934B, ED-30 for R-330); R-934UM2 has a truck with two generators and a shelter.**NOT TO BE CONFUSED WITH** R-378BMV HF jammer (Borisoglebsk-2) p. 48 Same MT-LB carrier in the same nine-vehicle set; no text source gives an external difference. INA.

R-330Zh Zhitel (Diabazol) p. 50 Ural-43203 wheeled carrier, not tracked MT-LB; no external antenna difference given in sources. INA.

SOURCES 3G APAS ATP CRFS DEFBLOG2 FOI ICDS MIL11 MIL12 MIL25 MIL26 RWW TOPWARS UKRINF VRF3 WEG32 WP-RU30

Денис Абрамов (Denis Abramov), CC BY 4.0

MOBILITY Ural-4320 6x6 for R-934B/R-330T/R-934UM2 (truck-class off-road); tracked MT-LBu R-934B/R-330B. Figures below are chassis-class, not station-verified.

Weight, combat	11,000 kg	R-330B on MT-LBu; WEG figure, unverified
Road speed	85 km/h	R-934B on Ural-43203; WEG chassis figure
Road speed	60 km/h	R-330B on MT-LBu; WEG figure, unverified
Road range	1,000 km	R-934B on Ural-43203; WEG chassis figure

Frequency range, R-934B

100–400 MHz Export sheet; aircraft links 100–150 and 220–400 MHz, claim

Transmitter power, R-934B
Jammed links, R-934B

500 W Export sheet, claim
4 at fixed frequencies; 1 with frequency hopping Export sheet, claim

RMS direction-finding error, R-934B
Crew, R-934B

2 deg Export sheet figure copied by WEG; RWW gives not more than 3 deg, claim
3 Export sheet, claim

SEEN WITH R-330KMV command post [1 per Borisoglebsk-2 set]

Borisoglebsk-2 РБ-301Б «Борисоглебск-2»

TR-A

Tactical HF/VHF/UHF radio intercept, direction-finding and jamming complex In service 2013

OPERATED BY motor rifle brigade, EW company (D 2024-02 H)

tank brigade, EW company (D 2024-02 H)

motor rifle division, separate EW battalion (O 2022-01 M) seen with 3rd

Motor Rifle Division (Vislenskaya), separate EW battalion, Belgorod oblast

motor rifle brigade, EW subunit (O 2018 M) seen with 74th Separate Guards

Motor Rifle Brigade

Unidentified unit, EW units (O 2018 M) seen with 201st Military Base, Tajikistan (R-934BMV stations)

Unidentified unit (O 2022-03 L) seen with 69th Separate Covering Brigade

(R-330BMV captured, Kyiv oblast, 2022-03)

INDICATES Usually indicates the EW company of a motor rifle or tank brigade or division within reach; since 2022 also division EW battalions. Vehicles of the complex may deploy singly rather than as a full set.

RECOGNISE Antenna Antennas on a mast, reported for one R-934BMV or R-378BMV station hit by drone. No text source gives antenna shape or size: INA.

Mast Vehicle-mounted antenna mast; height and stowed position INA in text sources.

Chassis MT-LBu tracked amphibious armoured carrier, modified for the EW equipment. One source says one R-325UMV rides on a Ural truck.

Complex Up to 9 vehicles: R-330KMV command post plus pairs of R-378BMV, R-330BMV, R-934BMV and R-325UMV jammers, with diesel generators. May deploy in part.

NOT TO BE CONFUSED WITH R-330Zh Zhitel p. 50 Ural-43203 wheeled truck (Diabazol complex), not tracked MT-LBu.

RB-341V Leer-3 p. 52 Single vehicle working with up to three Orlan-10 UAVs, not a tracked multi-vehicle set.

1L262 Rtut-BM Also on MT-LBu tracked chassis (3G table, tertiary); no text source found giving a visible difference.

SNAR-10 (Big Fred) p. 44 MT-LBu with turret-mounted radar and parabolic antenna; antenna folds forward on the roof when travelling.



Russian Ministry Of Defense, CC BY 4.0

MOBILITY Tracked, amphibious MT-LB chassis. About 15 min to deploy or pack up the complex on arrival.

Weight 11,000 kg WEG figure for base MT-LB chassis, not MT-LBu

also 12.6 t WEG2015

L × W × H 6.45 × 2.86 × 1.86 m WEG figure for base MT-LB chassis

Road speed 60 km/h maximum, base MT-LB chassis

also 61.5 km/h WEG2015

Road range 500 km cruising, base MT-LB chassis

also 500 km WEG2015

Swimming speed 5 km/h maximum

Fording Amphibious

Time into action 15 min or less whole complex, on arrival at site

Time out of action 15 min or less whole complex

Ground pressure 0.46 kg/cm2 standard track; 0.28 kg/cm2 wide track

Employment distance from line of contact 15 to 30 km practical, Donbas 2014-15, with Leer-3, Zhitel, R-934UM, observed

Crew, R-330KMV command post 4 2 of them at automated workstations

Vehicles per complex up to 9
Jammer stations controlled up to 4 pairs per R-330KMV command post

Frequency coverage HF, VHF, UHF intercept and jamming of ground and air radio; also satcom, GPS and cellular terminals

also 3 MHz to 3 GHz 3G

Frequency-hopping signals up to 300 hops/s VHF, detection and jamming, 'per open sources', claim

Performance vs R-330 Mandat 2x bandwidth, up to 100x faster HF and UHF bands; reported, same lineage

Power supply own diesel generators or 220/380 V 50 Hz mains

Jamming range, ground radio INA not published

Detection range, radio emitters INA

also 30 km 3G

Armament INA no source other than WEG names a weapon on the EW vehicles

SEEN WITH R-330KMV command post (1 per complex); R-378BMV HF jamming station (2 per complex); R-330BMV VHF jamming station (2 per complex); R-934BMV airborne VHF/UHF radio jamming station (2 per complex); R-325UMV jamming station (2 per complex); ATO-40 service and repair vehicle; Diesel generators; MT-LBu (up to 9 per complex)

5 EW

Palantin «ПАЛАТИН»

WH-AT

Operational-tactical HF/VHF comms jamming and SIGINT complex; networks other EW systems In service 2019

OPERATED BY combined arms army, EW units of a Western MD combined arms army in Voronezh Oblast (army not named)

R 2019-04 M

combined arms army, EW troops of a Guards combined arms army, Western MD, Luga range, Leningrad Oblast (Palantin-K, Oct 2019); Feb 2022 exercise in Leningrad Oblast names only the Western MD

R 2022-02 M

EW subunits of the Eastern Military District, EW subunits of the Eastern MD in Zabaykalsky Krai (planned deliveries by end 2020)

R 2020-07 L

INDICATES First delivered to EW units of a combined arms army; no source read names the holding subunit or echelon. A Palantin suggests an EW node that may network Moskva, Zhitel or Dzyudoist; in April 2026 one was reported covering air defence assets in Zaporizhzhia.

RECOGNISE Antenna Antenna devices carried on masts raised to 15 m on deployment. From photograph: compact multi-element antenna head at the top of a slender mast.

Mast Antenna masts of 15 m raised at the site. From photograph: masts stand beside the box body, braced by guy lines while being raised.

Chassis Four-axle KamAZ truck. From photograph: large box-body shelter with a side ladder. Generators are started on deployment.

Complex Reported as 22 mobile stations spread along the front; each starts generators, raises masts and joins a data network.

NOT TO BE CONFUSED WITH Murmansk-BN p. 45 Murmansk-BN uses 32 m telescopic masts holding a large HF antenna net between trucks; Palantin masts are about 15 m.

Borisoglebsk-2 p. 48 Borisoglebsk-2 is on MT-LB tracked carriers; Palantin is on four-axle KamAZ trucks.

R-330Zh Zhitel p. 50 Zhitel (Diabazol) is on the Ural-43203 truck; Palantin is on a four-axle KamAZ.



Mil.ru, CC BY 4.0

MOBILITY Truck-mounted on four-axle KamAZ; about 30 min from march to operation including raising 15 m masts.	
Time into action	about 30 min from the march, including raising 15 m antennas, claim
Chassis	four-axle KamAZ truck
Weight	INA
Time out of action	INA
Jamming range, UAV control links	up to 20 km maximum, air target: UAV radio control links, claim also 20 km MIL47
Jamming frontage, radio communications	about 1,000 km wide strip claimed, ground target: radio communications and radio reconnaissance; battalion tactical group exercise near Voronezh, Sep 2019, claim
Jamming frequency range	3–30 MHz (HF) and 30–300 MHz (VHF) ground target: radio communications
Jamming targets	HF/VHF radio incl. SDR-based, cellular and trunked radio, navigation, UAV control links claim
Reconnaissance function	radio-electronic reconnaissance; detects and locates transmitters
Network function	links Moskva, Zhitel and Dzyudoist EW systems into one network claim
Composition	22 mobile stations Palantin-K, claim
Antenna mast height	15 m claim
Crew	INA
SEEN WITH Palantin mobile station (four-axle KamAZ) (22 per complex (reported, Palantin-K)); Moskva EW complex; R-330Zh Zhitel; MKTK-1A Dzyudoist	

5
EW

R-330Zh Zhitel P-330Ж «Жумель»

WH-AT

Truck-and-trailer jammer and direction finder against satcom, GPS and GSM user equipment In service 2008

OPERATED BY EW company, EW company, jamming platoon (radio, GPS, cellular, radio-controlled fuses), jamming squad, 1 (single station on RWW chart; no count stated) [\[D 2016 M\]](#)

EW brigade, REB-S (anti-aircraft communications and navigation) battalion [\[D 2018 M\]](#)

EW company, EW company of motor rifle or tank division (sources confirm the company exists; Zhitel holding not stated) [\[D 2018 L\]](#)

EW company, EW company deployed to Ukraine, 2 in the deployed company [\[R 2014 L\]](#)

motor rifle brigade [\[R 2014 L\]](#) seen with 18th motor rifle brigade (Army Recognition claim, undated; not in WEG; AR article dated 2014 per RUSIAD footnote)

NUMBERS Losses 25 lost in Ukraine Feb 2022 to Sep 2026 (visually confirmed) (2026-09)

INDICATES *Doctrinally one Zhitel sits in the EW company of a manoeuvre brigade, so a sighting usually points to such a formation nearby. It is also held by the REB-S battalions of district EW brigades, so it can indicate an attached higher-level EW element.*

RECOGNISE **Antenna** Four active phased-array transmit antennas, one atop each trailer mast. Truck has a separate DF array: ring of about 8 vertical elements.

Mast Stowed: four telescopic masts upright at trailer corners, just above the box. Deployed: four guyed masts, raised by hand winch.

Chassis, truck Ural-4320-family or KamAZ-43114 6x6 with full-length box body; air-conditioner box at front of body above cab; DF mast at body rear. Houses all operator equipment.

Chassis, trailer Two-axle trailer carrying a tall rectangular box with the antennas and a diesel generator; outrigger jacks at the corners.

Complex In action the trailer is uncoupled and parked well away from the truck, cable-linked. May work in pairs (master and slave) or under an automated EW command post.

NOT TO BE CONFUSED WITH R-934BMV jamming station (Borisoglebsk-2) [p. 48](#) Listed on tracked MT-LB chassis, no trailer; Zhitel is a wheeled 6x6 truck towing a four-mast antenna trailer. RB-301B Borisoglebsk-2 complex (R-330KMV, R-378BMV, R-330BMV, R-934BMV, R-325BMV) [p. 48](#) Group of up to nine tracked MT-LB vehicles; Zhitel is a single wheeled truck plus trailer. R-934UM, R-378UM and other R-330M1P Diabazol stations [p. 47](#) Share the Ural-43203 box-body base. Confirm Zhitel by the separate two-axle trailer with four corner masts; appearance of the others INA. RB-341V Leer-3 (KamAZ box-body truck) [p. 75](#) KamAZ box-body truck working with Orlan-10 UAVs, no antenna trailer; Zhitel has the four-mast trailer and a DF ring mast on the body.

SEEN WITH Antenna trailer with four phased-array antennas and diesel generator (1 per station); Second R-330Zh (paired master/slave for synchronous DF) (1 when paired); R-330KMA EW command post (controls up to 20 stations); R-330M1P Diabazol automated jamming complex (with R-934UM, R-378UM, Altaets-AM); Ural-43203 6x6 (1); KamAZ-43114 6x6 (1)

VARIANTS R-330Zh on Ural-4320-family chassis Bonneted Ural cab with long nose; spare wheel on rear of box body. None reported. FOI and WEG give Ural-43203 as the standard carrier. R-330Zh on KamAZ-43114 chassis Flat-fronted KamAZ cab-over cab. Seen at Army-2015 and in Ukraine 2022. None reported.



Mil.ru (D. Abramov), CC BY 4.0

MOBILITY 6x6 truck, but the towed antenna trailer is fragile; WEG expects it near roads, not cross-country. Masts raised by hand, up to 40 min to deploy.

L × W × H	7.36 × 2.55 × 2.71 m	Truck; trailer not included (unclear in source)
Road speed	85 km/h	maximum, Truck, maximum road
Time into action	40 min or less	Full deployment including masts, claim
Time out of action	40 min or less	Full march order, claim
Payload	4,500 kg	maximum, Truck chassis
Weight, combat	INA	
Road range	INA	
Fording	INA	
Ground pressure	INA	

Jamming range	25 km	maximum effective, Ground user equipment (WEG gives no target type; matches manufacturer's ground-user figure)
Jamming range, airborne users	50 km or more	claimed minimum, User equipment installed on aircraft, claim
Area covered by a pair	About 40 x 20 km	claimed, Two paired stations, jamming ground communications and navigation
Frequency band, intercept and DF	100-2,000 MHz	claim
Jamming bands	1227.6 MHz; 1575.42 MHz; 1500-1900 MHz	GPS L2 and L1, Inmarsat/Iridium and GSM-1800 bands
Targets	Inmarsat and Iridium user terminals; GPS receivers; GSM base stations (WEG: GSM 1900; RWW: GSM-900/1800)	claim
Transmitter output	6,000 W or more	Total energy potential of the four phased arrays, claim
Simultaneous jamming signals	12 or fewer	Quasi-simultaneous, claim
DF accuracy	2 degrees or better	RMS bearing error, claim
Scan speed	800 MHz/s or more (detection); 400 MHz/s or more (DF)	claim
Beam width, azimuth	-90 to 120 deg	WEG labels this azimuth search coverage
Beam width, elevation	-15 to +20 deg	WEG labels this elevation search coverage
Jamming signal types	Spot, spot-barrage and barrage	claim
Power supply	220/380 V 50 Hz	from towed diesel generator or mains; 24 V batteries for emergency claim

SOURCES 3G AR67 ATP CRFS FOI ICDS MIL70 ORYX1 PRO RUSIAD RWW TOPWAR9 TOPWAR10 VRF13 WEG78 WMC10 WMC24 WMC25 WMC26

Leer-2 85Я6 «Леер-2»**WH-AT**

Mobile technical control, radio reconnaissance and jamming vehicle on Tigr-M chassis In service 2012

OPERATED BY airborne division, EW subunit (FOI: VDV EW companies formed from 2016), INA; MoD announced 10 complexes for VDV EW units, Dec 2012 (title only, state claim) [R 2018 M] seen with 106th Guards Airborne Division (Tula), shown Nov 2012, Tula and Ivanovo airborne divisions, trials Sep 2012, Ivanovo and Ulyanovsk VDV formations, in use by 2014

Ground Forces EW units, 'possibly' in FOI; state media: Southern MD column Mar 2018 [R 2018 M] seen with Russian military base in Abkhazia, EW subunit (MoD 2017, article titles only)

Unidentified unit, Position 17 km behind line of contact, Novoluhanske, Donetsk region, Nov 2023, 1 vehicle [R 2023-11 M] seen with No Russian unit named

Navy EW centres [R 2018 L] seen with Navy EW centres, no unit named

INDICATES A Leer-2 most often indicates an Airborne Troops EW subunit; Navy EW centres, the Strategic Rocket Forces and Ground Forces EW units also hold it. It says nothing about the parent unit type.

RECOGNISE Antenna All EW antennas sit on a platform on the vehicle roof. Antenna shapes and sizes: INA, no text source.

Mast No separate mast reported. Two telescopic antennas are raised on the roof when static (3G calls them mast antennas). Height INA.

Chassis GAZ Tigr-M 4x4 armoured car: armoured bonnet, bulletproof hull and door windows, electric winch (4,000 kg, 20 m rope). No armament.

Complex Single vehicle. A forum report (2014) says an earlier army version of the same equipment is on a Kamaz-4350 truck with trailer.

NOT TO BE CONFUSED WITH Leer-3 (RB-341V) p. 52 Leer-3 works through Orlan-10 UAVs from a command vehicle; Leer-2 is a single armoured Tigr-M car with roof antennas.



Ministry of Defence of the Russian Federation, CC BY 4.0

MOBILITY Fords 1.3 m; not amphibious. Chassis figures are Tigr-M, from WEG.

Weight, combat	7,800 kg
L × W × H	5.7 × 2.3 × 2.3 m
Speed, maximum road	140 km/h
Road range	1,000 km cruising

Jamming frequency band 0.02–2.7 GHz Radio emitters incl. mobile communications; 3G table copied from an Airbase.ru forum image, 2014

Signal analysis band 0.1 MHz–18 GHz Radio reconnaissance; direction finding 0.02–18 GHz

Simultaneous suppression 12 emission sources Quasi-simultaneous; VDV EW chief, Nov 2012, via forum, claim

Direction-finding accuracy 3 deg roof antennas, 2 deg mast antennas

Crew 2 (driver, operator) claim

SEEN WITH GAZ Tigr-M (1 per vehicle)

SOURCES 3G AIRBASE AR33 CNA FOI GAGADG ICDS MIL32 DRYX1 RG2018 VPK4 WEG37 WEG2015 WP-RU71

5

EW

Leer-3 ПБ-341В «Aeep-3»

WH-AT

UAV-borne cellular (GSM) jamming, phone location and fake-SMS system using Orlan-10 drones In service 2015

OPERATED BY EW brigade, Leer-3 platoon, platoon operates the complex; number per brigade INA [D 2024 H]

EW battalion, Leer-3 drone/EW company [R 2019 M]

Unidentified unit [R 2015 M] seen with 4th Military Base, South Ossetia military district, Leer-3 crews [R 2022-11 M] seen with Western Military District

motor rifle division, separate EW battalion [R 2022-01 M] seen with 3rd Motor Rifle Division (Vislenskaya), separate EW battalion, Belgorod oblast

Unidentified unit [R 2020-11 M] seen with Russian peacekeeping contingent, Nagorno-Karabakh

INDICATES A Leer-3 control truck points to an EW brigade platoon, an army EW battalion company or a division EW battalion. Expect Orlan-10 flights that jam or imitate cell networks, locate phones for artillery and send SMS within about 6 km of the drone.

RECOGNISE **Antenna** Control truck carries a command-telemetry radio antenna system for the UAVs. From photograph: small drum-shaped radome on a thin vertical pole. Size INA.

Mast From photograph: the antenna pole rises from the rear top corner of the box body; no large telescopic mast seen on the exhibited vehicle. Height INA.

Chassis KamAZ-5350 6x6 truck, closed box body with operator workstations. From photograph: bonneted cab, chamfered roof edges, side door, small windows, spare wheel on rear wall.

Complex Up to three Orlan-10 fixed-wing UAVs, catapult-launched and recovered by parachute; wingspan over 3 m.

Complex From photograph: Orlan-10 has a slim fuselage, straight wing and tail fin.

NOT TO BE CONFUSED WITH Leer-2 p. 51. Leer-2 is built on a Tigr-M 4x4 armoured car; the Leer-3 control vehicle is a KamAZ-5350 6x6 truck with a box body.

Orlan-10 p. 78. Same airframe also flies artillery reconnaissance with UAV companies; the airframe alone does not show the EW role. Look for the Leer-3 control truck or cellular jamming.



Vitaly V. Kuzmin, CC BY-SA 4.0



Orlan-10 UAV, Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Control vehicle is a KamAZ-5350 6x6 truck; complex deploys in about 30 minutes. Leer-3 vehicle weight, dimensions and fording not found.

Road speed	100 km/h maximum, KamAZ-5350 base truck at maximum permitted mass; not a Leer-3 figure, claim also 120 km/h WP-RU72
Road range	1,550 km minimum at reference consumption, KamAZ-5350 base truck, 420 l tanks, at reference fuel consumption; not a Leer-3 figure, claim also 1,000 km WP-RU72
Time into action	30 min whole complex, claim
Weight	INA
Time out of action	INA

Jamming range, cellular terminals (ground)	up to 6 km up to, radius of the zone of blocked GSM terminals around the UAV, claim
Detection range, GSM base stations	6 km within, cellular base stations, GSM and possibly 3G/4G, measured from the UAV
Practical range, Orlan-10 from control vehicle	120 km up to, radius of UAV employment from the control point, claim also over 100 km IZ222 MASH
Endurance, Orlan-10	10 h maximum, maximum flight time of the Orlan-10 in the Leer set, claim also about 12 h (half a day) IZ222
Altitude, Orlan-10	up to 5,000 m up to, operating altitude, claim
Frequency band, jamming and imitation	890-915, 935-960, 1710-1785, 1805-1880 MHz GSM 900 and GSM 1800 networks, claim
Frequency band, UAV control channel	902-922 MHz claim
Networks attacked	GSM 900, GSM 1800 3G and 4G capability claimed after a 2017 upgrade, claim also GSM, 3G, 4G IZ220
Cellular subscribers blocked simultaneously	2,000 claim
Cellular operators blocked simultaneously	3 claim
UAVs controlled simultaneously	2 claim
Crew	5 whole complex, claim also 4 RG15
Speed, Orlan-10	70-130 km/h depends on modification, claim also 90-150 km/h MIL33
Take-off weight, Orlan-10	18 kg claim also 14 kg MIL33

SEEN WITH Orlan-10 (3 per complex); KamAZ-5350 (1 per complex); UAV catapult launcher (1 per complex); Expendable ground jammers

SOURCES 3G ATP CNA CRFS DFR FOI ICDS IZ20 IZ22 IZ222 MASH MIL33 ORYXI RG15 RG223 RWW UNIAN3 UZST VRF8 WMC10 WMC11 WP-RU72 not in WEG

Pole-21 Р-340РП «Поле-21»

FX

Networked GNSS jamming posts protecting fixed sites from GNSS-guided weapons and UAVs In service 2016

OPERATED BY combined arms army, Army EW specialists; first used in an exercise with battalion tactical groups, with R-934BMV and Silok-01 (MoD via AIF) [\[O 2021-04 M\]](#)

INDICATES *A post on a cell tower or fixed mount suggests a protected fixed site such as a strategic facility or command post. A single R-340RP in the field indicates a GNSS-jamming layer near that position.*

RECOGNISE **Antenna** Transmit and receive antennas on cellular communications towers. Small antennas resembling civilian cell antennas, 1 to 3 per post (Ukrainian expert, 2023). Size: INA.

Mast Posts attach to existing cell towers or buildings, or to a stationary ground mount. Height: INA.

Chassis Also shown as a wheeled mobile unit and as a unit deployed in the field. Vehicle type: INA.

NOT TO BE CONFUSED WITH R-330Zh Zhitel p. 50 Zhitel is a vehicle-borne station; Pole-21 posts sit on towers or fixed mounts. FOI notes similarities in function. Focus.ua wrongly calls Pole-21 'Zhitel'.

MOBILITY Posts are fixed on masts or mounts. A wheeled mobile unit is also reported. Weights and dimensions: INA.

Power supply **220 V, 50 Hz; 600 W consumption** WEG (Pole-21E)

Operating temperature **-40 to 50 C** Pole-21E per WEG
Weight **INA**

Jamming range, GNSS receivers (ground or air not stated) **up to 25 km** effective, WEG (Pole-21E); Russian press gives 25 km per jamming module for the domestic Pole-21.

Jamming frequency band **1.176 to 1.602 GHz** (GPS, GLONASS, Galileo, BeiDou)

Coverage area of one complex **150 km2** WEG; number of posts not stated

Output power per module **300 to 1,000 W variable** max radiation pattern, single frequency; Pole-21E

SEEN WITH Jamming posts (up to 100 per complex); jamming modules (1 to 3 per post)

SOURCES 3G AIF6 ATP DPSU FERG FERG2 FOCUS3 FOI IZRU IZV4 KPOST TOPWAR7 UKRINF1 WEG46 WP-UK2

Repellent-1 Репеллент-1

WH-AT

Wheeled counter-UAV EW complex: detects and jams small drone control and GNSS links In service 2016

INDICATES *Sightings so far (Donbas 2018, Kherson 2022) name no holding unit type. Its presence shows a counter-UAV EW asset is deployed in that area, not which formation owns it.*

RECOGNISE **Antenna** Sensor and jamming head on a collapsible telescopic mast; upper section holds ELINT and jamming equipment and a panoramic camera. Size INA.

Mast Long collapsible telescopic mast mounted beside the control cabin on the rear of the truck. Height INA.

Chassis Three-axle truck, reported as MAZ-6317; WEG says Kamaz-6317 8x8 and Armada says Kamaz-6350. Chassis maker varies by customer (Wikipedia).

Complex Single vehicle with rear control cabin (two operator monitors, small-arms and NBC protection). Mobile and stationary versions are offered. Crew 3.

MOBILITY Not amphibious (WEG). Fords 1.0 m; 60% gradient, 0.6 m step (WEG). Chassis type disputed, so figures may differ by chassis.

Weight, combat **23,650 kg** WEG figure, Kamaz-6317 chassis
L × W × H **9.4 × 2.7 × 3.35 m**

Road speed **85 km/h** maximum
Road range **1,000 km** maximum

Detection range, UAV control-link signals **up to 30 km** up to, ground-station command signals; UAV telemetry up to 10 km; target size and altitude not given, claim

Jamming range, small UAV control and telemetry links **up to 30 km** up to, air target: small UAVs; telemetry and satellite-navigation links; control link up to 10 km, claim

Simultaneous UAV targets **up to 12; detect and jam in up to 25 s** claimed, brochure claim, claim

SOURCES 3G ARMADA9 ARMYINF5 BCAT2 IZV5 LIGA OSCE RUSIPL UASV UKRINF22 WEG53 WP-EN65

5
EW

Inf fauna ПБ-5315 «Инфауна»

WH-AT-A

Vehicle-mounted radio jammer and IED jammer, with smoke self-screening In service 2012

OPERATED BY Dedicated EW units and manoeuvre reconnaissance units D 2024 L

EW battalion, EW battalions of the Airborne Forces (VDV) (WEG 2015); RIA 2012 and FOI say VDV EW units R 2015 M

Unidentified unit, Russian military base in Armenia: MoD release title, 27 May 2015 (via FOI); filmed at Alağyaz range, 2019 (state media)

R 2019-04 M

INDICATES Presence suggests a VDV or other EW subunit escorting a column or protecting a battalion-level position against radio-controlled IEDs and guided munitions. WEG 2015 places it in VDV EW battalions; no source gives counts.

RECOGNISE Antenna Main antenna in a domed housing on the roof, raised on a telescopic mast when deployed. Size INA. Auxiliary antenna on starboard side.

Mast Telescopic mast raises the roof antenna. Mast height INA.

Chassis Modified BTR-80 (K1Sh1), 8x8, turret removed. Middle hull larger and higher than BTR-80; four hatches at front, one door at each side centre.

Complex Starboard side carries an optoelectronic sensor and smoke grenade launchers (WEG: 6 x 81 mm in 2 x 3). No crew-served weapon.

NOT TO BE CONFUSED WITH Kushetka-B command vehicle (BTR-80 chassis) p. 20 Small turret, and a long rectangular box where Inf fauna has a camera; a 2018 Horlivka sighting was misidentified as Inf fauna.

Standard BTR-80 p. 20 Inf fauna has no turret; roof carries an antenna dome and a raised middle hull.



Изображение Pygenko, CC BY 4.0

MOBILITY Amphibious 8x8 on BTR-80 derivative; operates on the move and escorts columns.

Weight **13,614 kg** WEG: combat weight, listed under BTR-80 chassis

L × W × H **7.65 × 2.90 × 2.46 m**

Road speed **80 km/h** maximum

Road range **600 km** maximum

Jamming range, target not stated

3 km disruption range, WEG 'Disruption Range'; target and conditions not given

Smoke screen distance

130 m up to, aerosol against laser and TV-guided weapons

Jamming band, communications

20-2020 MHz frequency range as given by WEG, WEG 2015 and RWW; sub-bands not given

Output power
Crew

60 W
3

SEEN WITH K1Sh1 chassis (BTR-80 derivative) (1 per vehicle); Lesoschek portable EW system

5
EW

SOURCES 3G AR26 ATP DFR LAB FOI GS17 ICDS RDP4 RG2 RWW VRF7 WEG33 WEG2015 WP-RU63

6

Engineers

IN THIS CHAPTER

Crossing	56
Obstacle clearing and earthmoving	58
Mine laying	61
Recovery	62
CBRN defence	62

LOOKING FOR THESE? THEY ARE IN OTHER CHAPTERS

SYSTEM	WHAT IT IS	CHAPTER	PAGE
MT-LB	Light armoured amphibious tractor and APC	1 Armoured vehicles	17

TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge TMM-3

WH-AT

Bridging: TMM-3 truck scissors bridge; MTU-72 tank-chassis bridge layer In service 1970

OPERATED BY engineer-sapper battalion, brigade engineer battalion; sets almost always attached to the movement support detachment (OOD), 4 TMM-3 truck-launched bridge sets per brigade engineer battalion, each laying a 10 m span; together a 42 m bridge

2024-02 M

engineer-sapper regiment, engineer and bridge subunit (press-release wording) 2020-11 M

INDICATES TMM-3 sets are held by the brigade engineer battalion and almost always attached to its movement support detachment, which bridges minor gaps ahead of the main body. Four trucks together suggest a route or a crossing of up to 42 m being prepared.

RECOGNISE Chassis TMM-3: KrAZ-255B 6x6 soft-skin truck; spare wheel on a steel frame above the cab. MTU-72: turretless T-72 hull: six road wheels, three return rollers, exhaust left rear.

Working equipment TMM-3: hydraulic slewing mast with forked end and winch lifts the folded span. MTU-72: single bridge launched from the hull, hydraulic front blade. Cantilever type.

Stored TMM-3: each 10.5 m span folds to 5.2 m; spans ride on the truck bed. MTU-72: bridge folded 9.42 m long on a hull 11.64 m long, 3.38 m high.

NOT TO BE CONFUSED WITH MTU-90 T-90 chassis; RWW gives a 25 m bridge, 50 t class, versus 20 m on MTU-72; held in tank brigades per RWW.

MTU-20 and MT-55A T-54/55-family chassis (MT-55A is on T-55) rather than T-72 chassis.



TMM-3. Mil.ru (Ministry of Defence of the Russian Federation), CC BY 4.0



MTU-72. Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY TMM-3 is an unarmoured 6x6 truck (WH-AT). MTU-72 is tracked (TR), armoured. TMM-3 span: gap to 40 m, 3 m deep. Weight, TMM-3 truck with bridge 19,500 kg Weight, MTU-72 40,000 kg Combat weight L × W × H 9.3 × 3.2 × 3.15 m Road speed, TMM-3 71 km/h

Obstacle, TMM-3 bridge set up to 40 m wide, 3 m deep Set of four spans; ATP gives 42 m total length Load class, TMM-3 60 t Wheeled and tracked loads; tracked vehicles at 15 km/h on bridge Bridge geometry, TMM-3 4 spans of 10.5 m; roadway 3.8 m Spans can be connected to 10.5, 21, 31.5 or 42 m (RWW); ATP gives 10 m per set and 42 m together Crew, TMM-3 1 + 1 in cab (driver, passenger) per truck

SOURCES ARMYINF8 ATP IIS25 JANESRG JFRC KOLESA3 MIL59 ORYX1 RWW TRENCH3 WEG66 WEG2015 WP-EN79 WP-EN80 WP-RU123 WP-RU124 WSN5

PMP heavy folding pontoon bridge park ПМП

WH-AT

Truck-carried folding pontoon park for 60-t floating bridges and 40-170 t rafts In service 1961

OPERATED BY motor rifle division, engineer battalion, pontoon bridge company (also in tank divisions), half set: 16 river and 2 shore links and 1 service vehicle on KrAZ-214; 6 BMK-90/150 boats per company [D 1991 M](#)
engineer-sapper brigade, pontoon bridge battalion of the army or front engineer brigade (FM 100-2-3 'Engineer Brigade, Army or Front'), one full set: 32 river and 4 shore links on KrAZ-255, 15 BMK-150 boats [D 1991 M](#)

pontoon-bridge regiment, two pontoon bridge battalions (army or front/district level), three full sets: 96 river and 12 shore links on KrAZ-255 [D 1991 M](#)

motor rifle brigade, engineer battalion, pontoon bridge company, one pontoon park; RWW calls it 'PP-61' but its figures (268 m at 60 t; 90- and 120-t bridges) match PP-91, not PMP. 6 BMK boats, 7 PTS-2 [D 2016 L](#)

Unidentified unit, engineer unit (Воинская часть) of Eastern Military District, INA (new PMP-M park to be received) [R 2017-02 L](#)

NUMBERS Losses 47 PMP(-M) river links and 3 shore links lost (Oryx) (2026-09)

INDICATES A half set (16 river links, 2 shore links, about 6 boats) matches a division engineer battalion's pontoon company. Over 100 link carriers, or three full sets, matches an army or front pontoon-bridge regiment. Either points to a planned bridged river crossing.

RECOGNISE **Chassis** 6x6 truck carriers: KrAZ-214, later KrAZ-255B. Ural-4320 and KamAZ-53501 carriers also seen. Separate trucks carry river links, shore links and boats.

Working equipment Roller bed on the truck; a folding jib and a winch behind the cab recover the link. Truck backs to the water, brakes hard and the link slides off the rollers.

Stowed River link: four hinged steel pontoons folded into one tall block filling the truck bed, 6.75 m long. Shore link is shorter, 5.6 m, with a ramp end.

Stowed, boats BMK-90/150 tug-motor boats ride on their own truck carriers, 6 per half set and 12 per full set; PMP-M sets have 16.

Deployed Link unfolds on hitting the water to a flat deck 6.75 m long, about 8 m wide. Links pin into a continuous steel ribbon, 6.5 m roadway, held in line by boats.

NOT TO BE CONFUSED WITH PP-91 / PP-2005 (PP-91M) pontoon parks Larger links (7.36 x 8.28 m) on KrAZ-260G/Ural-53236 or, for PP-2005, KamAZ-63501 8x8. Truck alone does not settle it: VPK lists Ural-4320-30 for PP-91 too.

TMM-3 truck-launched scissors bridge p. 55 Same KrAZ-255B chassis, but carries a folded 10.5 m scissors span with a launching girder and folding trestle legs, not a pontoon block.

PMM-2 self-propelled ferry p. 57 Tracked, boat-bowed hull with seven road wheels and two box pontoons stacked on top; one vehicle forms a ferry.

VARIANTS PMP-M Same folding river links; shore link has a straightened deck; hydrodynamic shield kit. Carried on KrAZ-255B; KamAZ-53501 carriers also recorded. Bridge usable in currents up to 3 m/s against 2 m/s for PMP; set lists 16 boats instead of 12. **PMP on Ural-4320 carriers** PMP links on Ural-4320 6x6 trucks in place of KrAZ-255B. None known; same links. **PP-91** Bigger river link, 7.36 m long and 8.28 m wide, on KrAZ-260G (Ural-53236) carriers; BMK-225 boats. 60-, 90- and 120-t bridges and double-width roadway; lengths per set differ between sources [see open questions]. **PP-2005 (PP-91M, PP-2005M)** PP-91-type links on KamAZ-63501 8x8 carriers. Boats: BMK-MT with PP-2005M (VPK); BMK-3 seen with PP-91M (AIF, 2025). Bridges to 120 t; rafts 90-360 t (PP-91) or 90/190/380 t (PP-2005M per VPK). In a Group Centre pontoon-bridge company, 2025-02.



Zinnsoled, CC BY-SA 4.0

MOBILITY Carriers must back to the water's edge to launch and recover links. PMP bridge limited to currents of 2 m/s, PMP-M to 3 m/s.

River link weight **6,790 kg** one river link
 Shore link weight **7,252 kg** one shore (ramp) link
 Time into action, bridge **7 m of bridge per minute** ideal conditions

Time out of action, per link **a few minutes** retrieval of one link by winch and jib

River link, unfolded **6.75 m long, 8 m wide**
 Weight, carrier with link **INA**
 Road speed **INA** maximum
 Road range **INA** cruising
 Fording **INA**

Bridge, full set, 60-t **227 m** maximum, 32 river + 4 shore links, 6.5 m roadway

Bridge, full set, 20-t **382 m** maximum, split half-links, 3.29 m roadway

Maximum current **2 m/s** PMP
 Bridge, half set, 60-t **119 m** 16 river + 2 shore links
 Bridge, half set, 20-t **191 m** 16 river + 2 shore links, split
 Load capacity, bridge **60 t** full-width bridge; 20 t on split-link bridge

Roadway width **6.5 m** 60-t bridge
 Rafts **40, 60, 80, 110, 170 t** 2, 3, 4, 5+1 shore, 8+1 shore links

Raft assembly, 40-t **8 min** 2 river links, working party 6
 Tank speed on bridge **up to 30 km/h** tanks on 60-t bridge
 Crew per carrier **3** WEG figure, in the block copied from the PMM-2 page; WEG 2015 KrAZ-255B page gives 3 in cab

Working party, full bridge **approx. 65** full bridge set

SEEN WITH River link on KrAZ-214 or KrAZ-255B (or Ural-4320, KamAZ-53501) carrier (32 per full set, 16 per half set); Shore (ramp) link on carrier truck (4 per full set, 2 per half set); Roadway mats (vystilki) on trucks (2 per full set); BMK-90/150 tug-motor boats on truck carriers (12 per full set (16 for PMP-M), 6 per half set); PTS-2 (or K-61) tracked amphibious transporter (7 in a brigade engineer battalion; 28 in an army engineer brigade (1991)); GSP tracked self-propelled ferry (12 half-ferries per division engineer battalion; 24 in an army engineer brigade (1991)); PMM-2 self-propelled amphibious ferry (replacing GSP); TMM-3 truck-launched bridge (4 per engineer battalion)

SOURCES AIF5 ATP FM23 GS25 MIL49 ORYX1 RWW SPT TI VPK7 VRF9 WEG45 WEG2015 WMC13 WP-EN59 WP-RU88 WP-RU89

PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry ПТС-4

TR-A

Amphibious carrier for cargo and vehicles (PTS-4); self-propelled river ferry (PMM-2) In service 2007

OPERATED BY motor rifle brigade, engineer battalion, 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4) [D 2016 M](#)

tank brigade, engineer battalion, 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4) [D 2016 M](#)

Unidentified unit [R 2017-09 L](#) seen with Russian forces in Syria: PMM-2M sent for the Euphrates crossing near Deir ez-Zor (MoD Road Service operation, TASS)

INDICATES PTS-2 amphibians are a brigade engineer battalion asset in RWW, so PTS-type carriers suggest engineer river-crossing support. ISS25 lists PMM-2 without unit type; no unit is tied to PTS-4 here.

RECOGNISE Chassis PTS-4: tracked amphibious hull with forward armoured cab; T-80 tracks and torsion bars, T-72 engine and gearbox. Road wheel count INA.

Working equipment PTS-4: open rear cargo area, platform 8.28 x 3.3 m, one remote-controlled 12.7 mm machine gun. PMM-2: boat-like bow, PTS-2-style cab.

Stowed PMM-2: MT-T-type chassis, seven road wheels and four return rollers per WEG; two large box pontoons carried one above the other on the hull.

Deployed PMM-2: pontoons hinge out to each side, with wave deflectors and ramps; deck 10 m long, 4.2 m wide; two 5 m loading ramps; two propellers lowered aft.

NOT TO BE CONFUSED WITH PTS-2 / PTS-3 Older, on MT-T chassis with T-64 components; PTS-3 has higher sides. PTS-4 uses T-80 and T-72 units and has a more heavily armoured cab.

PMM-2 vs PTS-2 PMM-2 has similar cab and bow but carries two stacked box pontoons on top of the hull.



Aleksei Kumaev (Aleksy Kitaev), CC BY-SA 4.0

MOBILITY Amphibious tracked. PMM-2 fording 1.2 m, vertical step 2.8 m (WEG15; implausible, see open_questions). PTS-4 vertical step 0.65 m, trench 2.5 m (WEG).

Weight, PTS-4 33.1 t empty, per WEG

Weight, PMM-2 36 t combat weight, per WEG15; RUW gives: ferry mass 36 t

L × W × H 8.3 × 3.3 × 3.2 m WEG; RUW gives 8.28 m for the cargo platform only

Road speed, PTS-4 60 km/h maximum

Payload on land, PTS-4 12 t

Payload on water, PTS-4 18 t

Payload, PMM-2 single ferry 42.5 t WEG; RUWM gives 42.5 t for the PMM-2M

Payload, PMM-2 two joined 85 t WEG gives 85 t for two PMM-2; RUWM gives 85 t for two PMM-2M, assembly 8 min

SOURCES AR42 AR43 ISS25 RWW WEG49 WEG2015 WP-EN62 WP-RU92 WP-RU93 WP-RU94 WWW18 WWW22

IRM Zhuk IPM «Жук»

TR-A

Amphibious engineer reconnaissance of routes, water obstacles and minefields In service 1980

OPERATED BY engineer-sapper battalion [D 1990 M](#)

engineer-sapper battalion, naval engineer battalion [D 1990 L](#)

engineer-sapper brigade [R 2017 L](#) seen with 214th Engineer-Sapper Brigade, 14th Separate Guards Engineer Brigade, 45th Separate Guards Engineer Brigade

engineer-sapper regiment [R 2017 L](#) seen with 56th Engineer-Sapper Regiment, 140th Separate Guards Engineer-Sapper Regiment

NUMBERS Losses 1 destroyed, 1 captured by Ukraine (Oryx) (2026)

INDICATES Sources place the IRM in divisional engineer-sapper battalions and, in an undated list, in engineer brigades and regiments. A sighting suggests engineer reconnaissance of routes, fords or water barriers, for example before a river crossing.

RECOGNISE Chassis BMP-1 based hull with 7 road wheels where the BMP-1 has 6 (WEG). New hull and extra wheel per ENWIKI. Fully tracked, amphibious.

Working equipment Two mine-detector arms on the front of the hull, retractable; PIR-451 periscope at commander's station on right of hull front. Small turret with one 7.62 mm PKT.

Stowed Detector arms retracted. Rocket self-recovery racks on rear deck between propellers (WEG).

Deployed Two three-bladed cowed propellers at the rear sides for swimming. Trim vane extended at the front before entering water. Detector arms forward when scanning.

NOT TO BE CONFUSED WITH BMP-1 ^{p. 11} BMP-1 has 6 road wheels, the IRM 7 (WEG); BMP-1 carries no detector arms or rear-deck rocket racks.

IPR IPR adds a folding snorkel on the rear deck for underwater driving; the IRM does not drive submerged (ARMFORC).



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Amphibious, swims on two rear propellers. Armour is small-arms proof only.

Weight 18,000 kg combat weight

L × W × H 8.30 × 3.15 × 2.42 m hull

Road speed 52 km/h maximum

Road range 500 km on land (ARMFORC 'на зручнє'); WEG gives no condition

Range, 762 mm PKT 1,500 m effective, WEG states no ammunition type and no day or night condition

Echo sounder depth range 0.5 to 20 m (EIR echo sounder) stated range, water depth measured by EIR sounder

Reconnaissance rate land route 8 to 10 km/h; minefield 5 km/h; 100 m water barrier in 5 min typical, WEG attributes these to unnamed Russian sources

Crew 6: commander, driver, 4 dismountable engineers

SOURCES ARMFORC ATP FM32 ISS25 ORYX1 RUSARMY2 WEG34 WEG2015 WP-EN35 WP-RU64

IMR-2 / IMR-2M / IMR-3M combat engineer vehicle IMP-2

TR

Tank-based heavy engineer vehicle for obstacle and minefield clearing In service 1980

OPERATED BY engineer-sapper battalion, mine clearing platoon (brigade engineer battalion), 2 IMR-3M per platoon [D 2016 M](#) engineer-sapper brigade, 2 delivered Nov 2018, 1 more due first half 2019 (Central MD press service, state claim) [R 2018-11 L](#) engineer-sapper brigade, INA; listed with UR-77, BMR-3 and GMZ-3 (English Wikipedia, citing a 2023 Russian regional news item) [R 2023-06 L](#)

INDICATES An IMR-3M in a column suggests the mine clearing platoon of a brigade engineer battalion; its elements may join the movement support detachment (OOD) ahead of the main body to clear routes and minefield lanes. Separate engineer brigades also hold it.

RECOGNISE Chassis Tank chassis: T-72 for IMR-2 (six road wheels a side, torsion bar); T-90 for IMR-3M. No gun; superstructure carries a crane.

Working equipment Wide dozer blade at the front; telescopic boom with grab or bucket on a rotating superstructure where a turret would be.

Stowed Blade raised onto the hull roof; boom drawn back and turned. A KMT track-width mine plough may be fitted.

Deployed Blade works as a dozer, V-blade or grader; boom traverses 360 deg. IMR-3M has a remote 12.7 mm gun mount; IMR-2 a machine gun on the operator's cab.

NOT TO BE CONFUSED WITH BAT-2 p. 58 Also a tracked bulldozer with adjustable blade and boom crane; RWW gives 7.3 m boom reach and 700 hp against about 8.4 m and 840 hp for IMR.

MTU-72 p. 55 Also a T-72 hull with turret removed; carries a bridge-launching system, not a dozer blade and crane.



Kirill Borisenko, CC BY-SA 4.0

MOBILITY Fords 1.2 m unprepared; 5 m with snorkel equipment (Russian Wikipedia). Not amphibious. Heavy: 44 to 50 t.

Weight, IMR-2	44.3 t	combat weight
Weight, IMR-3M	49.5 t	combat weight with KMT-R3 plough
L × W × H	9.55 × 3.735 × 3.68 m	
Road speed, IMR-2	50 km/h	maximum

Crew	2 (commander-operator, driver-mechanic)
Boom lifting capacity	2 t
Boom reach	8.4 m IMR-2M
Dozer capacity	200 to 250 m ³ /h IMR-2 earthmoving
Armament, IMR-3M	12.7 mm NSVT or Kord remote mount
Armament, IMR-2	7.62 mm PKT original IMR-2

SEEN WITH UR-77 mine clearing vehicle (2 per mine clearing platoon); BAT-2 combat engineer vehicle (4 per brigade engineer battalion)

VARIANTS IMR T-54/55 tank chassis; articulating dozer blade and telescoping crane.

SOURCES ATP IIS25 RWW TASS4 TSAMTO WEG19 WEG2015 WP-EN33 WP-EN34 WP-EN61 WP-RU62

BAT-2 BAT-2

TR

Armoured route-clearing tractor and dozer on MT-T chassis In service 1987

OPERATED BY engineer-sapper battalion, route construction and clearing section; usually in the movement support detachment (OOD), 4 per battalion [D 2016 M](#)

engineer-sapper battalion, new-look brigade battalion: engineer-road platoon, CP equipment section, shore section, 7 per battalion (4 road platoon, 2 CP equipment section, 1 shore section) [R INA L](#)

INDICATES Doctrinally a BAT-2 moves with an engineer movement support detachment ahead of a march column, improving the route. A BAT-2 in a column may indicate engineer route preparation for an advance.

RECOGNISE Chassis Tracked, 7 road wheels a side, front idler, rear drive sprocket; MT-T based, engine at rear. Cab at front. Numbers per WeaponSystems.net.

Working equipment Hydraulic dozer blade of variable geometry and a skid in front; a 2 t crane turntable behind the cab; ripper at the rear; storage unit behind the cab.

Stowed Blade carried vertically in front of and over the crew cab when not in use. Fully enclosed armoured cab with bullet-proof windows front, sides and rear.

Deployed Blade at 4.5 m as bulldozer, 4.2 m for road clearing, 4.1 to 4.35 m for grading; wings can be angled to one side. Blade tilts 10 degrees.

NOT TO BE CONFUSED WITH BAT-M Older tractor dozer on AT-T chassis (T-44/T-54 based) with 500 mm tracks and 13-tooth sprocket; BAT-2 is on the MT-T chassis.



Kirill Borisenko, CC BY-SA 4.0

MOBILITY Fords 1.3 m. Ground pressure about 0.8 kg/cm²; one Russian source says it often bogs down.

Weight	39.7 t	combat weight
L × W × H	9.64 × 4.2 × 3.69 m	
Road speed	60 km/h	WEG15 gives INA
Speed, cross-country	28 to 35 km/h	WSNET: 35 km/h on dirt roads

Crew	2	commander and driver; ArmyInform via Army Recognition says up to 3 in Ukrainian service
Troop capacity	8	engineer squad in cab
Blade width	4.5 m	bulldozer position; 4.2 m road clearing; 4.1 to 4.35 m grading
Crane capacity	2 t	boom reach 7.3 m
Winch	25 t	cable length 100 m

SEEN WITH DZ-180A wheeled road grader (2 per engineer battalion)

SOURCES AR7 ATP IIS25 PRIZ RWW WEG8 WEG2015 WP-RU13 WSN3

UR-77 Meteorit YP-77 «Memeopum»

NATO: M1979

TR-A

Tracked mine-clearing vehicle firing rocket-towed explosive line charges In service 1978

OPERATED BY engineer-sapper battalion, mine clearing platoon (organic engineer battalion of a maneuver brigade), 2 per platoon

D 2016 M

engineer-sapper brigade **R 2023 L** seen with 1st Guards Engineer Brigade (Muroom, MUN 11105)

INDICATES Doctrine places two per engineer mine clearing platoon, so one suggests engineers preparing a breach or route clearing. In Ukraine it has also been fired as improvised short-range artillery.

RECOGNISE Chassis Tracked hull derived from the 2S1 Gvozdika, itself built on MT-LB parts; driver and engine compartment front left on the 2S1.

Working equipment Turret-like superstructure with launch ramps at top rear of hull. RWV gives two launch ramps, TWZ two charges; JANESRG says three rockets.

Stowed Charges carried on board: hull holds two launchers and compartments for the line charges (UZP-77 or UZ-67).

Deployed Rocket tows a hose about 7 cm thick and 93 m long (UZP-77) across the minefield, then it is detonated; lane about 6 m wide.

NOT TO BE CONFUSED WITH 2S1 Gvozdika Same chassis; the 2S1 has a fully enclosed turret with a 122 mm howitzer, the UR-77 a launcher superstructure.

UR-67 Older predecessor on a BTR-50PK chassis.

UR-07M BMP-3 chassis, not 2S1; ordered in small numbers, not in series production as of 2024 (GSEC). Named Razgon (GSEC) or Peresortirovka (RUWIKI).

UR-83P Same line-charge role but not mounted on a vehicle chassis: towed or trailer-based.

SOURCES ESPR GS35 IIS25 JANESRG MILUA7 RBTH RWW TWZ14 TWZ15 WEG73 WEG2015 WP-EN34 WP-EN85 WP-RU135 WWW113



Mil.ru / Dmitry Golovko, CC BY 4.0

MOBILITY Tracked, 2S1-derived, amphibious: WEG gives 5 km/h swimming, GlobalSecurity 4.5 km/h. Fording depth and ground pressure INA. Reload after firing takes 30 to 40 minutes.

Weight, combat 15.5 t

L × W × H 8.4 × 2.8 × 3.1 m

Road speed 60 km/h

Road range 500 km cruising

Range, UZP-77 line charge 200 to 500 m not stated, charge launch distance, UZP-77; UZ-67 charge 200 to 350 m (WEG)

Crew 2

Cleared lane length, UZP-77 80 to 90 m anti-tank minefield; UZ-67 charge 75 to 80 m

Cleared lane width up to 6 m one line charge

Line charge length 93 m UZP-77; UZ-67 is 83 m

Cleared lane length, ZRSchh 100 m lane 6 m wide; charge in 6 m sections of 41 kg

SEEN WITH UZP-77 / UZ-67 line charge (2 per vehicle)

BMR-3M BMP-3M «Бенп»

TR

Armoured mine-clearing vehicle on T-90 chassis with roller tawl; clears lanes and routes In service 2000

OPERATED BY Engineer troops units, engineer troops units (not named) **R 2024 M**

Unidentified unit, sapper subunits, Mariupol airport and Volnovakha area **R 2022-09 L**

Unidentified unit, engineer troops demining group on the contact line **R 2024-09 L**

INDICATES Indicates engineer troops clearing routes or lanes through minefields. WEG says it is mainly for road clearance escorting convoys; Russian MoD reports it clearing ground near the contact line, followed by sappers with hand-held detectors.

RECOGNISE Chassis Turretless T-90 tank hull; running gear as T-90: six road wheels, front idler, rear drive sprocket, exhaust on left hull side above last road wheel.

Working equipment Two roller frames at the bow, each with a set of heavy steel wheels, with cutters on the outer side and a chain device in front; electromagnetic attachment above.

Stowed Welded armour wheelhouse with commander's cupola and 12.7 mm MG mount in place of turret; ERA boxes on hull front and wheelhouse; platform at rear to carry tawl section.

Deployed Pushes the roller tawl ahead, leaving two cleared tracks about 0.8 m wide with an uncleared strip between; BMR-3MA TMT-S tawl clears a continuous strip.

NOT TO BE CONFUSED WITH IMR-3M p. 58 Also T-90 based, but has a bulldozer blade and a telescopic boom with manipulator; BMR-3M has a roller tawl and no boom.

BMR-3 Predecessor on T-72A chassis with KMT-7 tawl; lower (2.51 m) and carries a hand-operated jib crane.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Tank-class tracked vehicle, 48 t (WEG), about 50 t with tawl per Russian media. Tawl cuts dirt-road speed to 35-40 km/h; sweeping speed up to 12 km/h. Fords 1.2 m unprepared, 5 m prepared.

Weight, combat 48 t

L × W × H 6.93 × 3.78 × 2.93 m

Road speed 50 km/h maximum

Speed, dirt road with tawl 35-40 km/h tawl fitted

Range, 12.7 mm MG 2,000 m effective, ground targets, 12.7x108 mm

Range, 12.7 mm MG vs air 1,500 m effective, air targets, 12.7x108 mm

Crew 2 (commander, driver) + 3 sappers

Armament 1 x 12.7 mm NSV MG

Swept lane width 2 tracks x 0.8 m KMT-7, against track-width mines; uncleared strip of 1.62 m between

SEEN WITH KMT-7 track-width mine roller tawl (1 per vehicle)

SOURCES AIF AR12 IIS25 JANESRG RWW TECHINS VM WEG15 WP-RU28 WP-RU29 WWW13

Uran-6 Уран-6 «Уран-6»

TR

Remote-controlled tracked demining and obstacle-clearing vehicle In service 2016

OPERATED BY engineer-sapper regiment [R 2021-01 M] seen with separate engineer regiment, Guards Red Banner Combined Arms Army, Western Military District

Engineer units of the Southern and Eastern Military Districts, engineer units [R 2015-11 L] seen with engineer units, Southern Military District, Eastern Military District

Unidentified unit, International Mine Action Center [R 2022-05 L] seen with International Mine Action Center of the Russian Armed Forces (Mariupol coast, per MoD)

INDICATES Seen with an engineer regiment or brigade or the Mine Action Center: points to area clearance of mines and explosive items, reported behind the front line. CNA (2019) saw no Uran robots in combined arms unit structure.

RECOGNISE Chassis Low armoured tracked chassis, about 1.4 m high, unmanned. Reported as outwardly similar to the Croatian DOK-ING MV-4.

Working equipment Tool mounted in front: chain flail (tiller); bulldozer blade 1.8 m wide. Shovel and forklift tools at rear.

Stowed Four cameras: one top centre, two armoured at the sides, one facing rear. Carried on a four-axle KamAZ hook-lift transport truck.

Deployed Moves at a few km/h with the flail turning ahead of the tracks, controlled by radio; a separate control panel and operator stay away from the vehicle.

NOT TO BE CONFUSED WITH DOK-ING MV-4 (Croatia) Foreign design; ATP calls Uran-6 a licence-built copy, Russian Wikipedia says only that it looks similar. No text source for visible differences.

Uran-9 Same manufacturer's armed unmanned vehicle: carries a 30 mm 2A72 gun, 9M120 Ataka missiles and Shmel-M launchers.

SOURCES AIBUS AIF10 AR65 ARTECH ATP CNA [FOR RGNEWS TASS26 WEG76 WP-RU139



Kirill Borisenko, CC BY-SA 4.0

MOBILITY Slow tracked machine, 5 km/h maximum; claimed radio control range 0.3–1 km; operator stays clear of the vehicle.

Weight 6 t approximately; fit not stated, claim
L × W × H 4.56 × 2.01 × 1.47 m with tool, claim
Road speed 5 km/h maximum, tracked, unmanned, claim
Endurance 5 h continuous operation, claim

Remote control distance 1 km claimed maximum, by radio from a control panel, claim

Crew Unmanned; one remote operator
Blast resistance 60 kg TNT equivalent manufacturer figure, claim

Tool set 5 interchangeable tools full set: flail, roller and milling mine trawls, bulldozer blade, mechanical gripper (AIF)

Clearing width 1.6 m flail fitted, claim
Clearing depth 35 cm flail fitted, claim

SEEN WITH Remote control unit (1); Interchangeable tools (5 types) (5 in full set)

Military earthmoving and trenching machines БТМ-3; МДК-3; ПЗМ-2; 30Б-3521; 30Б-3523; 30Б-4421

TR

Machines that dig trenches, pits and shelters for troop and vehicle positions In service 1963

OPERATED BY engineer-sapper battalion, engineer-technical company: engineer-position platoon and command-post equipment squad (brigade battalion), Position squad: 2 MDK-3, 2 BTM-4M, 2 EOY-3521. Command-post squad: 2 EOY-3521 [D 2008 L]

engineer-sapper battalion, engineer-position company, engineer-position platoon (division battalion), Table lists 2 each of BTM, MDK, PZM, EOY (layout unclear) [D 2017 L]

INDICATES Engineer-sapper battalion equipment held for fortifying positions and command-post areas. Its presence points to defensive works being prepared. Ukraine also operates BTM-3 and PZM-2.

RECOGNISE Chassis Tracked: BTM-3 on AT-T, MDK-3 on MT-T tractor. Wheeled: PZM-2 on T-155 tractor; EOY-4421 on Kraz-255B, EOY-3521 on Ural, EOY-3523 on KamAZ-53501 truck.

Working equipment BTM-3: bucket rotor lowered behind hull, star throwers at sides. MDK-3: bladed rotor with thrower, dozer blade, ripper. PZM-2: chain digger, dozer blade, winch.

Working equipment EOY-4421 and EOY-3521: one-bucket backhoe on a rotating platform with its own operator cab, frame with outriggers. EOY-4421 works as backhoe only.

Stowed BTM-3 carries the rotor raised on a lifting linkage that shortens it for travel. MDK-3 is about 10.2–10.3 m long travelling, 11.75 m working, cab at front.

NOT TO BE CONFUSED WITH BAT-2 engineer vehicle p. 58

Tracked bulldozer with adjustable dozer blade and a crane; no rotor or chain digger.

TMK-2 and TMK-3 trench machines Wheeled trenchers on MAZ-538 (TMK-2) or K-703MV (TMK-3) chassis, not T-155.

SOURCES ATP CYBER EXX FOCUS KAT ORYX1 PRIZ RWW WEG30 WEG2015 WP-RU50 WP-RU51 WP-RU52



Anonme, CC0

MOBILITY BTM-3 and MDK-3 tracked, PZM-2 and EOY types wheeled. Mostly unarmoured; BTM-4M has an armoured cab. Slow while digging. Transport speed 36 to 90 km/h by type.

Weight, BTM-3 27.7 t claim
Weight, MDK-3 39.5 t claim
Weight, PZM-2 12.8 t claim
Weight, EOY-4421 20 t claim

Trench depth, BTM-3 up to 1.5 m soils up to category IV, unfrozen, claim

Output, BTM-3, trench 1.1 m deep 270–810 m³/h technical output, depends on soil

Output, BTM-3, trench 1.5 m deep 270–560 m³/h technical output, depends on soil

Pit depth, MDK-3 3 m maximum digging depth
Output, MDK-3, pits 800 m³/h technical output, claim

SEEN WITH Ural-5557 dump truck (2 per engineer-position squad)

ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer ИСДМ «Земледелуе»

WH-AT

Remote minelaying by 122 mm rockets (ISDM) and tracked mechanical minelaying (GMZ-3) In service 2020

OPERATED BY engineer-sapper battalion, obstacle platoon (brigade engineer battalion per RWW; division engineer battalion per WEG), 3 GMZ-3 per obstacle platoon **D 2016 M**
engineer-sapper brigade **D 2024 L**
engineer-sapper regiment **D 2024 L**

INDICATES An ISDM points to an engineer brigade or army engineer regiment remote-mining subunit preparing minefields 5–15 km out. A group of three GMZ-3 suggests a brigade engineer obstacle platoon, likely a mobile obstacle detachment.

RECOGNISE Chassis ISDM: four-axle KamAZ-6560 8x8 truck; cab armoured per Za Rulem (WEG: armour none). The transloader uses the same chassis.

Chassis GMZ-3: tracked, on the SA-4 Ganef chassis; seven road wheels, front drive sprocket, rear idler, four return rollers. Driver front left, engine to his right.

Working equipment ISDM: launcher on rear deck carrying two box containers, each of 25 tubes in five rows of five. Transloader carries a small crane (blog source).

Working equipment GMZ-3: operator compartment at rear; mines leave through ports in the rear hull plate onto a conveyor and plough. PKT MG; three 81 mm smoke launchers each side.

NOT TO BE CONFUSED WITH BM-21 Grad p. 26 Ural 6x6 truck with 40 open tubes; ISDM is an 8x8 KamAZ with two 25-tube box containers.

UMZ ZIL-131 6x6 truck with six firing modules of 30 tubes (180 tubes) scattering mines 30–60 m from the truck.



MIL.ru, CC BY 4.0

MOBILITY ISDM: 8x8 truck, not amphibious, fords 1.8 m per WEG. GMZ-3: tracked, not amphibious, 28.5 t; lays surface mines at up to 16 km/h, buries at 6 km/h.

Weight, GMZ-3 28.5 t combat
L × W × H 10.3 × 2.5 × 3.27 m
Road speed, ISDM 90 km/h maximum, KamAZ-6560 chassis
Road speed, GMZ-3 60 km/h maximum

Range, ISDM mine rocket 15 km maximum, cassette rocket with AP or AT mines, claim
Minimum range, ISDM mine rocket 5 km cassette rocket with AP or AT mines, claim
Laying speed, GMZ-3 16 km/h surface; 6 km/h buried maximum
Range, 762 mm PKT (GMZ-3) 2,000 m max effective, day
Rocket calibre, ISDM 122 mm claim

SOURCES ARMINF2 ATP IIS25 IZV222 JANE3 M360 MPER3 ORYX1 OVC RWW TI2 WEG77 WEG2015 WP-EN90 WP-EN91 WWW114 ZR16 ZR24

UMZ УМЗ Универсальный минный заградитель

WH-AT

Truck-mounted remote scatterable minelayer In service 1970

OPERATED BY Ground Forces (SV); holding unit not named **D 2024 L**

INDICATES Sources say Russian ground forces employ UMZ but name no holding unit or count. It must work close to the area it mines, so it points to an obstacle task near the line. Ukraine also uses UMZ launchers.

RECOGNISE Cab ZIL-131 truck cab and bonnet; the truck is the carrier. Sources write ZIL-131V (Cyrillic ZIL-131B); WEG writes ZIL-131B.

Body Six hexagonal launcher blocks of 30 tubes each in the load body, three each side (ATP: attached to the back of the truck).

Axles Three axles, 6x6 (WEG online gives configuration 6x6).

Details Blocks rotate 360 degrees and tilt, set by hand before laying; control panel is in the cab.

NOT TO BE CONFUSED WITH UMZ-K Six rotary launchers on a newer truck: KamAZ-63501 8x8 (Army-2015) or Asteyes 6x6 (Army-2019); armoured mine body. ZIL-131 bonnet marks the UMZ.

MOBILITY Truck chassis; off-road speed about 30 km/h. Lays mines at 10 to 40 km/h and needs 60 to 90 min to reload.

Weight, with mine load 10,000 kg ZIL-131 carrier
Weight, without mine load 8,300 kg
L × W × H 7.1 × 3.0 × 2.5 m
Road speed 80 km/h

Crew 2
Launch tubes 180 (6 blocks x 30)
Mines per load, PFM-1S / POM-2 / PTM-3 11,520 / 720 / 180 basic load by mine type; one source gives 180 to 11,520 range
Minefield length 1,000 to 1,200 m WEG general figure, mine type not stated; WEG gives one-load maxima of 3,200 m (PFM-1S), 5,000 m (POM-2), 600 m (PTM-3)

SEEN WITH ZIL-131 truck (ZIL-131V) (1 per launcher); Mine cassettes (PFM-1S, POM-2, PTM-3) (180 to 11,520 mines per load)

SOURCES AR61 AR62 ATP DEFUA4 KYIVP3 MILUA6 ORYX1 RTRU RWW WEG72 WEG2015

BREM and BTS armoured recovery vehicles 6P3M

TR

Armoured recovery and repair vehicles for tank and infantry vehicle units In service 1975

OPERATED BY motor rifle battalion, support platoon (evacuation squad), 3 BREM-L (RWW organisation chart) [D 2016 M]
 tank battalion, armoured tracked recovery vehicles with the battalion on the march; type not named [D 2016 L]

INDICATES A BREM seen with a column marks a maintenance or evacuation element of a battalion or brigade. Type of base chassis shows what it supports: T-72 hull for tanks, BMP-3 hull for motor rifle units.

RECOGNISE Chassis Bases: T-72A (BREM-1, turret removed, new superstructure), BMP-1 (BREM-2), BMP-3 (BREM-L), BTR-D (BREM-D), BTR-80 8x8 (BREM-K, keeps a turret).

Working equipment BREM-1: modified T-72A hull, turret removed, new superstructure, front 3.1 m dozer blade that doubles as earth anchor, crane, winch, 12.7 mm NSVT MG.

Stowed Light ARVs: swivel crane and winch; cargo platform on BREM-2 and BREM-L; BREM-K spares platform on turret roof. BTS-4: platform amidships, jib at left front, rear spade.

Deployed Spade or blade lowered as anchor; crane slewed; winch cable run out. BREM-1 crane reach 4.4 m. BREM-L crane reach 4.85 m. BREM-2 crane slews 270 degrees.

NOT TO BE CONFUSED WITH IMR-2 / IMR-3M engineer vehicles p. 58 Also turretless T-72 (IMR-3M: T-90) hulls, but a rotating telescopic crane sits where the turret was and the V-blade folds up; BREM-1 has a fixed superstructure.

T-72 hull ARVs of other makers [WZT-3, VT-72B] p. 24 Slovak (VT-72B) and Polish (WZT-3) ARVs on the T-72 hull, per WEG. No visual distinguishing mark found in text.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY BREM-1 (41 t tank hull) cannot swim; it fords 1.2 m, 5 m with snorkel. BREM-2, -L, -D and wheeled BREM-K swim. BREM-D is air-droppable.

Weight, BREM-1 41 t combat weight

Weight, BREM-2 14 t

Weight, BREM-L 18.7 t

Weight, BREM-D 8 t

Tow load, BREM-1 50 t maximum, semi-rigid tow bars

Crew, BREM-1 3

Crane capacity, BREM-1 19 t at 2 m; 3 t at 4.4 m crane length 4.4 m; WEG and RWW share one lineage

Main winch, BREM-1 25 t; 100 t with snatch blocks; 200 m cable

Dozer blade width, BREM-1 3.1 m

SEEN WITH REM-KL wheeled repair-recovery vehicle (Ural-532362) (1 per motor rifle battalion support platoon)

SOURCES AR15 ATP IIS25 JANESRG MIL13 ORYX1 RWW WEG18 WEG19 WP-RU32 WP-RU33 WP-RU34 WP-RU35 WP-RU36 WP-RU37 WP-RU38

CBRN reconnaissance vehicles (RKhM family) PXM-4, PXM-6, PXM-8, PXM-9

WH-AT-A

Radiation, chemical and biological reconnaissance; marks contaminated ground In service 2000

OPERATED BY RKhBZ brigade, NBC defense company [D 2016 L]

RKhBZ troops, RKhB troops (RKhM-8 and RKhM-9) [R 2021 M]

RKhBZ regiment, RKhB reconnaissance company [R 2025 L] seen with 26th Guards RKhBZ Regiment, 36th Combined Arms Army, Buryatia

INDICATES Seen together, these vehicles suggest an RKhBZ (CBRN defence) reconnaissance subunit. Doctrine holds RKhBZ units at regiment and brigade level and attaches NBC reconnaissance subunits to manoeuvre units.

RECOGNISE Cab RKhM-4 and RKhM-6 keep the BTR-80 layout: driver forward, NBC laboratory amidships, engine aft. Crew 3. RKhM-8 is a 4x4 Tigr armoured vehicle, crew 3.

Body RKhM-4/-6 keep a turret with 14.5 mm KPV and 7.62 mm PKT. RKhM-8 reported unarmed: two banks of two smoke dischargers at front roof, detectors on roof.

Axles Chassis separates the members: BTR-80 8x8 amphibious (RKhM-4, -6); Tigr 4x4 (RKhM-8); three-axle KamAZ-63969 Typhoon (RKhM-9); BRDM-2 4x4 (BRDM-2RKhb).

Details BRDM-2RKhb has two rectangular racks at the rear, one per side, dispensing lane-marking poles with pennants. Base BTR-80 has six smoke dischargers on the turret rear.

NOT TO BE CONFUSED WITH BTR-80 APC p. 20 Same hull and turret outline as RKhM-4/-6. No text source found for an external mark; the NBC equipment is the difference (INA for appearance). BRDM-2 (other variants) p. 22 BRDM-2RKhb carries two rear racks for lane-marking poles with pennants; the base BRDM-2 does not. RKhM-5 (BMD-3 based, airborne) Tracked BMD-3 chassis, not wheeled; used by airborne forces.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY BTR-80 and BRDM-2 based members swim (RKhM-8 at 9 km/h). Swimming for RKhM-8 and RKhM-9 not stated.

Reconnaissance speed is far below road speed.

Weight, RKhM-6 combat 13.6 t

Weight, RKhM-4 full 13.5 t

Weight, RKhM-8 combat 7.5 t claim

Weight, RKhM-9 20 t claim

Detection range, PKhRDD-28 remote chemical detector (RKhM-6) about 3 km claimed, Remote vapour detection; target and conditions not stated, claim

Chemical monitoring radius, RKhM-6 6 km claimed, claim

Detection range, UAV reconnaissance complex (RKhM-9) 25 km claimed, Onboard unmanned aerial complex, terrain survey, claim

Crew, RKhM-4, -6, -8 3

SOURCES AG4 AMAL AR20 IIS25 JANESRG RG21 RG2108 RT21 RWW TOPWAR4 U242 VRF4 VRF5 VRF6 WEG28 WEG2015 WP-RU47 WP-RU48 UNVERIFIED DRAFT

Decontamination vehicles (ARS-14, TMS-65, TZ-74 family)

APC-14KM / TMC-65

WH-AT

Truck-mounted decontamination and smoke vehicles of RKhBZ (NBC defence) troops In service 1969

OPERATED BY motor rifle brigade, RKhBZ company (also supports tank brigade), decontamination platoon, 8 ARS-14KM per RKhBZ company [D 2024 M]

RKhBZ regiment, Eastern Military District NBC protection regiment, Amur Region, received ARS-14KM batch [R 2020 L]

INDICATES An ARS-14KM or TMS-65 means an RKhBZ subunit is present; ATP puts ARS-14KM in a company supporting a motor rifle or tank brigade. The vehicles alone do not show that chemical weapons are expected.

RECOGNISE Chassis ARS-14KM on KamAZ-43114 6x6; ARS-14 on ZIL-131; ARS-15 on Ural-375; TMS-65 on Ural-375E, TMS-65M/U on Ural-4320; TZ-74 on Tatra 148 6x6.

Working equipment ARS: oval-section steel tank on the frame, side equipment boxes, hose rack on top (base ARS-14). KM adds heater, aerosol generator, pumps.

Working equipment TMS-65, TZ-74: gas turbine on the rear bed, operator cab on its left; TMS-65 engine and cab share a turntable. TZ-74 has an obscurant tank on the right.

Stowed TMS-65 engine is locked horizontal for travel; the vehicle tows a PTs-4.2-754V water-tank trailer. TZ-74 carries no trailer in the sources.

NOT TO BE CONFUSED WITH TZ-74 Tatra 148 chassis with turbine at the rear and obscurant tank on the right; TMS-65 is on a Ural with the engine on a turntable.

ARS-14 (ZIL-131) Older base model on ZIL-131 with one pump; ARS-14KM is on KamAZ with heater and aerosol generator.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Wheeled all-terrain trucks. TMS-65 with water trailer is limited to 40 km/h on roads. Work is done at a halt (ARS) or at slow speed (TMS-65).

Weight, ARS-14KM	15,350 kg	WEG figure; other source is equipped station with crew
Weight, TMS-65	8 t	Empty, Ural-375 chassis
Weight, TZ-74	21.9 t	
L × W × H	7.35 × 2.96 × 2.98 m	

Crew, ARS-14KM	1
Crew, TMS-65 and TZ-74	2
Liquid capacity per fill, ARS-14KM	2,500+1,000 L
Throughput, TMS-65, decontamination and disinfection	30–40 vehicles/h
Throughput, TMS-65, degassing	10–15 vehicles/h

Water or aqueous decontamination solution
Gas-droplet stream, one vehicle takes 150–200 L

SOURCES ATP AVIAROS CYCLO LOL54 MOCR ORYX1 SDN STUD1 STUD2 SVRHBZ WEG29 WEG2015 WP-RU49 UNVERIFIED DRAFT

Smoke generator vehicles (TDA-2K, TDA-3, TDM-2K) TDA-2K

WH-AT

Truck-mounted thermal aerosol generators that screen troops and sites from view and IR In service 1982

OPERATED BY motor rifle brigade, RKhBZ company, aerosol platoon (RKhBZ company also supports tank brigades), 5 TDA-2K per aerosol platoon [D 2024 M]

RKhBZ brigade, 5 TDA-3 delivered in one batch [R 2017 L] seen with

RKhBZ large unit of a military district, not named (district press office via TASS)

Unidentified unit, RKhBZ unit of the Central Military District, exercise

[R 2019 L] seen with RKhBZ unit, Central Military District (photo caption)

Unidentified unit, RKhBZ troops (chemists) of Southern Group of Forces, Luhansk axis, about a dozen vehicles in column, types not all stated [R 2024 L] seen with Southern Group of Forces RKhBZ troops (Telegram channel claim)

INDICATES TDA-2K or TDA-3 indicates an RKhBZ subunit, doctrinally an aerosol platoon in the brigade RKhBZ company. A column laying white aerosol suggests a screening task for an advance or a high-value site.

RECOGNISE Chassis TDA-3 on a KamAZ-5350 6x6 truck with the standard cab for its two-man crew. TDA-2K on a KamAZ-4310 (43114) truck. TDM-2K chassis INA.

Working equipment TDA-3: three rectangular equipment modules in line behind the cab (power unit, liquid tanks, powder tank). TDA-2K: insulated tank, gas-thermal generator, two fuel tanks.

Stowed Truck carrying closed rectangular equipment cases and tanks, no weapon. Exact nozzle position INA in the text sources.

Deployed Opaque white aerosol forms in air from a hot gas jet; TDA-2K lays smoke while moving at 10 to 40 km/h (developer sheet).

NOT TO BE CONFUSED WITH TMS-65M decontamination/smoke vehicle p. 63 Ural-4320 6x6 chassis with a movable nozzle (WEG15), not KamAZ; also decontaminates.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY TDA-3: full-time all-wheel drive, central tyre inflation, fords 1.75 m (WEG). Not amphibious. TDA-2K and TDM-2K limits INA.

Weight, combat	15.75 t	TDA-3
Weight	15.1 t	TDA-2K, full weight as stated by Belarus MoD
L × W × H	7.85 × 2.5 × 3.29 m	TDA-3
Road speed	100 km/h	maximum, TDA-3

Smokescreen length	over 1,000 m	claimed, TDA-2K, length of the opaque part of the screen, claim
Smoke duration per fill	4–13 h	claimed, TDA-2K, one tank charge, claim
Smoke duration per fill	110 min–6 h	claimed, TDA-3, depends on flow rate 10–30 l/min, claim
Speed while laying smoke	10–40 km/h	claimed, TDA-2K, claim
Crew	2	TDA-3 and TDA-2K
Smoke mixture tank	4,800 l (4,200 l working)	TDA-2K, claim

SEEN WITH KamAZ-5350 truck (TDA-3 chassis) (1 per vehicle); KamAZ-4310 truck (TDA-2K chassis) (1 per vehicle)

SOURCES AR50 ATP FOCUS5 MILBY ORYX1 TOPWAR8 TWZ13 VRF11 WEG58 WEG2015 UNVERIFIED DRAFT

7

Command and signals

IN THIS CHAPTER

Command and staff vehicles	64
Artillery and air defence command	67
Radio and signals	70

LOOKING FOR THESE? THEY ARE IN OTHER CHAPTERS

SYSTEM	WHAT IT IS	CHAPTER	PAGE
PRP-4A Argus / PRP-4M Deytery	Mobile artillery reconnaissance post on BMP-1 chassis	7 Command and signals	68
1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)	Tracked command, observation and fire direction vehicles, SP howitzer battalion	7 Command and signals	67

R-149 command and staff vehicles P-149

WH-AT-A

Tactical command and staff vehicles with HF/VHF radios and data terminals (ESU TZ) In service 1990

OPERATED BY motor rifle regiment, R-149MA3  2022-09 M
command brigade, R-149AKSh-1  2022-03 L
motor rifle division, R-149AKSh-1  2022-03 L

INDICATES An R-149MA1 or R-149AKSh-1 suggests a brigade, regiment, division or army control-brigade command post nearby. An R-149MA3 is meant for company to squad command and was lost with motor rifle and tank regiments.

RECOGNISE Cab R-149AKSh-1: KamAZ-5350 6x6 truck cab. BTR-based R-149MA1, MA3 and BMR: no cab; commander and driver sit at front of an 8x8 armoured hull.

Body AKSh-1: van body split into electronics compartment (3 workstations) and officer compartment (5). MA1 and BMR: K1Sh1 hull, larger than BTR-80, unarmed turret.

Axles AKSh-1: three axles. BTR-based: four large tyred wheels a side; hatch/door between 2nd and 3rd wheels, upper part opens to right, lower folds down (BTR-80 base).

Details R-149MA3: WEG lists a 14.5 mm KPVT turret (BTR-80 sheet text); one lost MA3 was turretless. R-149BMR carries an ASh-4 telescopic mast, AZI NVIS HF and discone antennas.

NOT TO BE CONFUSED WITH BTR-80 APC p. 20 R-149MA1 and BMR use the K1Sh1 hull, larger than BTR-80 with an unarmed turret; WEG lists the MA3 with the standard armed turret.

BMM-80 ambulance and 1V152 on K1Sh1 p. 67 WEG 2015: the Kushetka-B/R-149BMR 'shares the same configuration as BMM medical vehicles and the 1V152/Kapustnik-B ACRV'.



Ministry of Defence of the Russian Federation, CC BY 4.0

MOBILITY BTR-80-based versions swim (WEG). R-149AKSh-1 on KamAZ-5350 is not amphibious; fords 1.75 m (WEG).

Weight, R-149MA1	15,000-15,050 kg	maximum, claim
Weight, R-149BMR	13.6 t	combat
L × W × H	7.65 × 2.9 × 2.41 m	
Road speed, BTR-based versions	80 km/h	maximum

Range, VHF R-168-100UE-2, R-149MA1	25 km	maximum, on the move, vehicle antenna, claim
Range, VHF R-168-100UE-2 on mast, R-149MA1	55 km	maximum, halted, mast antenna, claim
Range, HF R-168-100KBE, R-149MA1	350 km	maximum, halted, mast antenna; 50 km (20 km moving) on ASh-4 whip, claim
Range, HF air-ground Prima, R-149MA1	450 km	maximum, HF to aircraft, claim
Range, VHF air-ground Prima, R-149MA1	120 km	maximum, VHF to aircraft, claim

SOURCES ACONF2 AR23 AR44 ARMYINF3 ATP ELSIG GAGA JANESRG K15 MIL51 ORYX1 RUSARMY4 UNIAN5 WEG52 WEG2015 WP-EN13 WP-RU102 WP-RU103 WSPOT10 WSPOT11 WSPOT13 WSPOT14 WSPOT15 WSPOT16 WSPOT17 WSPOT18

R-145BM Chayka command and staff vehicle

P-1455M «Чаўка»

WH-AT-A

Turretless radio command and staff vehicle for mobile command posts In service 1960

OPERATED BY anti-tank battalion, ACV listed as 'BTR-60PU/BRDM-2'; battery placement unclear in OCR, 5 per battalion (BTR-60PU or BRDM-2) [D 1991 L]

artillery brigade, 1 per brigade [O 1994 L]

RKhBZ (NBC protection) troops, Wikipedia names the vehicle

BTR-60PU/PUM and equates BTR-60PU with R-145BM [R 2022 L]

NUMBERS Losses 2 R-145BM lost in Ukraine, no unit or date given (Oryx) [2026-09]

INDICATES Built for radio links at mobile command posts, so it points to a command post or communications node. A Soviet-era source places it at tank, motor rifle and artillery regiment and formation HQs; the current echelon is not fixed.

RECOGNISE Cab BTR-60 hull (BTR-60PA base) with no turret; two compartments, command section forward and radio section aft.

Body Collapsible frame antenna on left side of roof; cylindrical box (TZK) on the roof; characteristic 'horns' on left and right upper edges of front armour.

Axles 8x8 wheeled, amphibious, individual torsion-bar suspension.

Details Hinged antenna leg supports on side armour upper edge, mid-length; 10 m telescopic mast and AB-1-P/30 roof generator (Jane's: BTR-60PU has roof generator, 10 m antenna).

NOT TO BE CONFUSED WITH BTR-60PU-12 (9S482) air defence command vehicle p. 69 Telescopic mast on right of front roof, generator on the turret-ring blanking plate, large stowage box on right of hull; turretless like R-145BM.

BTR-60 1V18/1V19 p. 67 Unarmed turret with observation devices; R-145BM has no turret.



VargaA, CC BY-SA 4.0

MOBILITY Amphibious 8x8, light armour; swims (no fording depth given). Petrol engines.

Weight, combat	10.3 t
L × W × H	7.56 × 2.83 × 2.31 m
Road speed	80 km/h
Road range	500 km

Range, R-130M zenith antenna	75 km HF, halted or moving, symmetrical zenith-radiation antenna
Range, R-130M dipole	350 km HF, halted only, symmetrical dipole
Range, R-123MT 4 m whip	20 km VHF, halted or moving
Range, R-111 3.4 m whip	35 km VHF, halted or moving
Crew	2
Radio fit	2 x R-111, R-123MT, R-130M, R-012M call device

VARIANTS BTR-60PUM Not stated in source. Russian late-2000s refit of BTR-60PU with new engines adapted from the BTR-80 series.

SOURCES ACONF ARINF FM32 IISS25 JANESRG ORYX1 RUSARMY3 UNIAN4 WEG51 WEG2015 WP-EN12 WP-EN13 WP-EN63 WP-RU98 WP-RU100 WP-RU101

BMP-1KSh command and staff vehicle

БМП-1КШ «Помок-2»

NATO: M1978

TR

Command and staff vehicle for regimental and battalion commanders In service 1972

OPERATED BY motor rifle regiment, regimental command post, 1 in regimental HQ of a BMP-equipped motor rifle regiment (FM 100-2-3 table, OCR scrambled) [D 1991 M]

tank regiment, regimental command post, INA (FM 100-2-3 table reads 0 for the tank regiment HQ) [D 1990 M]

motor rifle regiment, 501st signal company [O 1986 L]

INDICATES A BMP-1KSh marks a regimental command post; one source also gives one per battalion. Sources describe it for motor rifle and tank regiments. Oryx records 15 Russian losses 2022-26 but names no unit.

RECOGNISE Hull BMP-1 hull: pointed nose, almost horizontal ribbed glacis, driver front left, engine louvers roof right, two bulged rear doors. KSh carries only two roof hatches.

Armament No gun or missile rail; turret fixed; one 7.62 mm PK fired from the rear-door port. Most firing ports blocked. Tube case for mast parts on right rear.

Turret Fixed turret cap carries the telescopic antenna mast (10 m per WEG and VALKA) in place of the gun; only two roof hatches behind the turret.

Roof and rear Box-shaped generator set centre rear of hull roof; four elevatable whip antennas at rear, two each side.

Suspension BMP-1 running gear: six road wheels each side, front sprocket, rear idler, three return rollers, skirts over upper track.

NOT TO BE CONFUSED WITH BMP-1K p. 11 Keeps the 73 mm gun and rotating turret; K1/K2 have a semi-telescopic antenna, K3 a 10 m mast (WEG). KSh has a fixed turret with the mast in place of the gun.

PRP-3 artillery reconnaissance vehicle p. 67 Enlarged two-man turret set to the rear with a folding rectangular radar antenna.

SOURCES FM23 GS11 IISS25 JANESRG ORYX1 VALKA2 VALKACZ WEG12 WEG2015 WP-RU22 WP-RU24



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Amphibious, swims. Radios run on the move from an engine-driven generator; a portable generator set is used at halts (RUWK2).

Weight, combat	13 t
L × W × H	6.74 × 2.94 × 2.15 m
Road speed	65 km/h
Road range	550 km
Swimming speed	7 km/h
Fording	Amphibious, swims

Crew	3 crew plus 4 staff officers
Armament	1 x 7.62 mm PK, 2,000 rounds carried machine gun, fired through the rear-door port (RUWK2); WEG lists it as PKT

Antenna mast height	10 m AMU Hawkeye telescopic mast, raised
Radios	2 x R-111, R-123MT, R-130M

Navigation	TNA-3 gyroscopic navigation device
------------	------------------------------------

VARIANTS BMP-1KShM Extra power sources on top of the troop compartment (VALKA). MP-31 INA.

7

COMMAND

BMD-1KSh-A БМД-1КШ-А «Сопка»

TR-A

Airborne battalion chief-of-staff command and staff vehicle In service 1975

OPERATED BY air assault regiment, 2 to 7 BMD-1KSh per regiment in the 2000 list [R 2000 L]

artillery regiment, 0 to 2 per regiment in the 2000 list [R 2000 L]

signal battalion, 6 to 8 per independent signal battalion in the 2000 list [R 2000 L]

NUMBERS Losses 12 lost: 8 destroyed, 4 captured (Oryx) [2026]

INDICATES The BMD-1KSh is a battalion chief-of-staff vehicle. A 2000 list places it in airborne regiments and in the artillery, air defence, engineer and signal units of airborne divisions.

RECOGNISE **Hull** Turretless BTR-D hull with raised centre compartment; no bow machine guns or firing ports, equipment in their place.

Armament None fitted as standard; bow machine guns replaced by equipment.

Turret No turret. Commander's cupola sits aft and to the left of the base BTR-D position; officers' hatches moved forward to the driver's hatch.

Suspension BTR-D running gear: six road wheels and five return rollers each side, one pair of road wheels more than BMD-1. Adjustable hydropneumatic suspension.

Details Three windows with armoured covers in the superstructure plate; 4 m whip antenna for the HF radio, a mast antenna can be added.

NOT TO BE CONFUSED WITH BTR-D p. 16 Base carrier: has bow machine guns, firing ports and officers' roof hatches over the centre; no armoured windows.

BMD-1R Sinitsa p. 14 HF radio vehicle for regiment-division link; heavy telescopic mast stowed along left of roof, slim mast at rear right.

SOURCES JANESRG ORYX1 WEG2015 WP-EN14 WP-RU14 WP-RU15 WP-RU16 not in WEG



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Amphibious, swims at 10 km/h. Climbs 32 degrees, 0.7 m wall, 2.5 m trench [ru-wiki figures, base BMD-1KSh].

Weight, combat	8.7 t	BMD-1KSh
L × W × H	5.883 × 2.63 × 1.65 to 2.00 m	Hull; same as BTR-D hull (BMD-1 hull 5.40 m)
Speed, road	60 km/h	BMD-1KSh
Range, road	500 km	BMD-1KSh

Range, HF radio R-130	300 km	maximum, Up to, with 4 m whip antenna; BMD-1KSh
Crew	6	BMD-1KSh
Radios	2 R-123M, 2 R-111, 1 R-130	Original BMD-1KSh
Radios, A version	R-168-100U-2, R-168-UN5-2, R-168-100KB	BMD-1KSh-A; replace R-111, R-123, R-130
Power supply	AB-0.5-P/30	petrol generator
Armament	None	as standard

SEEN WITH BTR-D

7

COMMAND

VTS Ladoga BTC «Ладoga»

TR

Armoured command, staff and reconnaissance vehicle for NBC conditions In service 1980

OPERATED BY Unidentified unit, vehicle hit by drone near Lyman, 2024-03 (Oryx: destroyed); no unit named [R 2024-03 L]

INDICATES A very rare Soviet vehicle. One seen at the front in 2024 is a reactivated vehicle of uncertain role (carrier, ambulance or command vehicle are suggested), not evidence of a standard formation.

RECOGNISE **Hull** T-80 tank hull and chassis; a new armoured-steel superstructure holds a front control compartment with the driver's post and a rear passenger compartment.

Armament No gun: the T-80 turret is removed. No weapon fit is named in the sources read.

Turret No turret; in its place a new upper structure with roof hatches and a side door with integrated steps. A mast carries cameras.

Suspension Tracked T-80 running gear and gas-turbine drive; road wheel count INA (no text source read).

NOT TO BE CONFUSED WITH T-80 main battle tank Ladoga has no gun turret; a new armoured upper structure with roof hatches and a side door replaces it.

SOURCES AR32 FOCUS2 NAKED2 ORYX1 TECHIN TWZ7 WWW16 not in WEG

MOBILITY Fords 1.2 m, crosses trenches over 2 m and 1 m obstacles (Army Recognition figures). Heavy T-80 chassis.

Weight	42 t	condition not stated by sources
Road speed	70 km/h	maximum, maximum, highway also 110 km/h NAKED2
Road range	350 km	maximum also 330 km AR32 WWW16
Fording	1.2 m	figure from Army Recognition, original source not named
Bridge load class	47	42 t tracked x 1.10, rounded up, estimated

Crew	2 crew plus 4 in the passenger compartment	also 6 persons inside TECHIN
Endurance, water and food	48 h	stores carried; Rossiyskaya Gazeta via Naked Science, claim

1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) 1B12 «Машина-С»

NATO: M1974

TR-A

Tracked command, observation and fire direction vehicles, SP howitzer battalion In service 1972

OPERATED BY self-propelled artillery battalion, each battery: one 1V13 (battery FDC) and one 1V14 (battery COP); battalion HQ: one 1V15 (battalion COP) and one 1V16 (battalion FDC), 8 per battalion: 3 1V13, 3 1V14, 1 1V15, 1 1V16 [D 2016 H](#)

self-propelled artillery regiment, howitzer battalion with 2S19M1, at least 3 1V13 and 3 1V14 seen with one battalion [R 2021-09 L](#)

INDICATES A 1V13 marks a battery firing position, where it stands behind the guns. A 1V14 or 1V15 marks a battery or battalion command observation post, usually with the supported manoeuvre unit. A full set of eight serves one SP howitzer battalion.

RECOGNISE **Hull** High box-like MT-LBu hull with steep glacis; straight vertical rear with a single door. Three or four antennas on the hull roof.

Armament 1V13 and 1V16: 12.7 mm DShKMT machine gun on the turret. 1V14 and 1V15: 7.62 mm PKMB machine gun on a mount on the turret roof.

Turret Flat, round rotating turret shaped as a truncated cone on the rear half of the hull. On 1V14 and 1V15 it carries a laser rangefinder and optical sights.

Suspension Seven road wheels each side, no return rollers. The MT-LB it derives from has one road wheel fewer.

NOT TO BE CONFUSED WITH MT-LB [p. 17](#) Six road wheels, lower hull and a small machine-gun turret at the front over the driver's compartment instead of a turret at the rear.

ARK-1M Rys [p. 43](#) Same MT-LBu chassis but carries a parabolic radar antenna.



Simon Q from United Kingdom, CC BY 2.0

MOBILITY Amphibious, propelled by its tracks in water.

Weight **14.8 t** 1V13 and 1V14, combat weight

L × W × H **7.475 × 2.85 × 2.425 m** hull

Road speed **62 km/h** maximum, claim

Road range **500 km** maximum, claim

Detection range, 1V14 optical reconnaissance **up to 10,000 m** maximum, day

Detection range, 1V14 optical reconnaissance **up to 2,500 m** maximum, night

Detection range, 1V14 optical reconnaissance (night)

Range, laser target illumination (1V14)

Radio range

up to 7,000 m maximum, for laser-guided (corrected) munitions

20–25 km stated, 1V13 and 1V14, vehicle radios

220–250 km stated, 1V15 and 1V16, vehicle radio

Radio range, R-111

SOURCES AG1 ARMF13 ARMF14 ARMF15 ARMF16 FAS1 JANESRG JREW02 MILINF ORYX1 RWW VRF1 WEG2015 WP-RU1 not in WEG

Artillery command vehicles 1V18/1V19, 1V119, 1V152, 1V1003

1B18, 1B19, 1B119, 1B152, 1B1003 «Клещ-1», «Клещ-2», «Ресомат», «Cnekmp», «Kanycmnyk-B»

WH-AT

Artillery battery and battalion command, observation and fire direction vehicles In service 1979

OPERATED BY artillery battalion, 1V18 battery commander's COP vehicle; 1V19 battalion commander's COP vehicle; towed gun and MRL battalions, 3 x 1V18 and 1 x 1V19 per battalion set [D 2016 M](#)

airborne division, 1V119 in VDV artillery and mortar units (RWW), used with 2S9 Nona-S (WEG15, ru-wiki); formation level not stated

[D 2016 M](#)

INDICATES 1V18/1V19 suggests a towed gun or MRL battalion (three battery and one battalion vehicle per set). 1V119 suggests VDV or naval infantry artillery, often 2S9. 1V152 or 1V1003 marks a gun or MRL battery or battalion command post.

RECOGNISE **Body** Armoured command vehicles on APC hulls: 1V18/1V19 on BTR-60PB, 1V152 and 1V1003 on BTR-80, 1V119 on a tracked airborne hull.

Turret 1V18/1V19: unarmed turret with NNP-21 and DV observation devices and a rangefinder (Darth Vader shape). 1V19 same, plus a stowage box each hull side.

Axles 1V18/1V19 and 1V152: 8x8 wheeled, amphibious. 1V1003 is on a BTR-80 8x8 base; no further axle detail in the sources.

NOT TO BE CONFUSED WITH BTR-60PB / BTR-80 APC [p. 20](#) BTR-60PB APC has a 14.5 mm MG turret; 1V18/1V19 carry an unarmed observer turret. No text source read separates 1V152 or 1V1003 from a BTR-80.

BTR-60PU-12 (9S482) air defence command vehicle [p. 69](#) Turret removed and a telescopic AMU mast fitted at right front roof; 1V18 keeps a turret.



Mike1979 Russia, CC BY-SA 4.0

MOBILITY 1V18/1V19, 1V152, 1V1003: 8x8 wheeled, 1V18 amphibious. 1V119 is tracked (code TR) and can be parachute-landed with airborne troops (WEG15).

Weight, 1V18/1V19 **10.1 t** combat weight, BTR-60PB chassis

Weight, 1V119 **6.7 t** combat weight, BMD-1 chassis

Weight, 1V152 **12.6 t** combat weight, BTR-80 chassis

L × W × H

7.22 × 2.82 × 2.06 m hull

Accuracy, 1V198 Kanonada

up to 10 m claimed, Rostec claim: locating and striking targets from two or more emplacements in succession, claim

Detection range, PSNR-5K Tall Mike (1V119 optional)

3 km people; 12 km vehicles claimed, I-band ground surveillance radar, roof-mounted on 1V119; reference-site figure, claim

Crew, 1V18/1V19

6

Crew, 1V119

6

SOURCES AR5 AR6 JANESRG ORYX1 RUWK152 RWW TASS2 WEG2015 WIKR18 WP-EN12 WP-EN13 WP-RU11 not in WEG

PRP-4A Argus / PRP-4M Deytery ПРП-4А «Арзус»

TR-A

Mobile artillery reconnaissance post on BMP-1 chassis (target location, fire adjustment) In service 1988

OPERATED BY motor rifle brigade, fire control and artillery reconnaissance battery, artillery spotting platoon [D 2016 M](#)
tank brigade, fire control and artillery reconnaissance battery, artillery spotting platoon [D 2016 L](#)
self-propelled artillery battalion, battalion fire control platoon, artillery spotter squad [D 2016 L](#)

INDICATES Doctrinally a brigade- or regiment-level artillery reconnaissance asset feeding a fire direction centre. In defence it sits behind first-echelon companies; in offence it moves with forward reconnaissance detachments up to 20 km ahead of the main body.

RECOGNISE Hull BMP-1 infantry fighting vehicle chassis, amphibious; 4 crew.

Armament One 7.62 mm PKTM machine gun and four smoke dischargers only. PRP-4M is reported with a fake gun mantlet and offset barrel, resembling a BMP-2.

Turret PRP-4/4M/4MU: retractable radar; 4M antenna mount on roof behind radar hatch; 4MU left optical flap hinged at top. Argus: WEG says must-mounted radar.

Turret sides PRP-4 family: extra fairing on right side of turret (GlobalSecurity); protected mountings either side of turret house optical devices (Wikipedia).

NOT TO BE CONFUSED WITH BMP-2 p. 12 PRP-4M with a fake gun mantlet has no real 30 mm gun, only a 7.62 mm machine gun.

PRP-3 Val Earlier post; carries a 90 mm 2P130-1 launcher, which PRP-4 removed.

PRP-4 Nard Earlier base model (1V121, in service 1984); radar 1RL133-1 'Tall Mike' (Wikipedia, WEG 2015) vs WEG's '1L120-1' on Argus. No visible mark in a text source.

SOURCES AR41 ATP GS11 JANESRG JREW02 MIL50 ORYX1 RWW WEG48 WEG2015 WP-EN35 WP-RU91



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Amphibious on BMP-1 chassis; swim speed 7 km/h (WEG). Armour, combat 13.8 t

L × W × H 6.74 × 2.94 × 2.15 m
Road speed 65 km/h maximum, paved road
Road range 550 km paved road

Detection range, radar (moving tank-type target) 12 km up to, moving tank-type target

Detection range, radar (person) 7 km up to, single soldier

Detection range, stationary tank, engine off (sensor not stated) 2 km no less than, stationary tank-type target, engine not running; ARMYR and RWW name no sensor, WEG lists it under radar, claim

Detection range, optical-electronic (tank-type target, day) 8 km up to, tank-type target, daytime

Polyana-D4 9C52 «Поляна-Д4»

WH-AT

Automated command post of an Army air defence missile brigade (S-300V, Buk, Tor) In service 1986

OPERATED BY air defence missile brigade, brigade command post (combat control post MP06 and command-staff vehicle MP02)

[D 2021-12 H](#)

air defence missile brigade, brigade command post; vehicle captioned as 9S52M Polyana-D4M [O 2010-12 M](#)

combined arms army, army air defence and aviation command post (SKP PVO i aviatsii) [R 2021-12 M](#)

INDICATES Indicates the command post of an Army air defence missile brigade (S-300V/V4 or Buk), or an army or district AD command post. Up to four SAM battalions, a radar post and a point-defence CP report to a brigade Polyana.

RECOGNISE Cab MP06 combat control post on BAZ-6950 8x8: forward fibreglass two-door cab, three reverse-raked windscreens, oval vent hatch in centre of flat front wall.

Body Engine compartment behind cab. SKN-6950 'Rodinka' metal-frame van body with wide chamfered sides was built for this chassis for C2 complexes; MP06 body not described.

Axles Four axles in two widely spaced pairs (base 1.5 × 3.8 × 1.5 m); front pair steers. Large single tyres 1300x530-533.

Details Complex adds MP02 command-staff van on Ural-375 or Ural-43203 6x6 with KP4 trailer, MP45 maintenance van and two generator trucks (KamAZ-4310 on D4M).

NOT TO BE CONFUSED WITH 9S457 S-300V battalion command post p. 36 Tracked MT-T chassis shared with other S-300V vehicles, with telescoping datalink mast; Polyana vehicles are wheeled trucks.

Baikal-1 air defence command post Carried on MAZ-7910 8x8 with integrated FL-94 telescoping mast and 5S15 turbine generator; Polyana MP06 is on BAZ-6950 with separate generator trucks.

SOURCES APA4 APA9 ATP IIS25 KZ2019 KZ2021 NASHAVT RDP6 RUSAR1 RUSARMM VPK8 VRF10 WEG47 WP-EN60 WP-EN61 WP-RU90



© Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Wheeled truck complex with generator trucks; needs a road or firm track. Deploys or packs up in about 20 to 35 min. Fording 1.2 m (Ural) and 1.3 m (BAZ) chassis figures.

Weight, gross, Ural-375D chassis (MP02) 13.2 t chassis figure; MP02 as fitted INA

Weight, gross, BAZ-6950 chassis (MP06) 29.4 t chassis figure; MP06 as fitted INA

L × W × H 7.36 × 2.67 × 2.68 m cargo truck
Road speed, Ural-375D 75 km/h maximum

SAM battalions controlled up to 4 9S52 and 9S52M; S-300V (9S457) or Buk (9S470) battalion CPs
up to 250 9S52M1

Air targets tracked simultaneously
Targets received up to 1,000 per 10 s 9S52M1, claim
Linked objects, total up to 14 9S52M1; AEW aircraft, subordinate SAM unit CPs, higher and cooperating CPs, claim
Superior CP interfaces 1 variant not stated

PU-12M7 9C482M7 PY-12M7**WH-AT-A**

Air defence battery command post: fuses radar data and cues SHORAD fire units In service 1977

OPERATED BY motor rifle regiment, air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet), 1 per battery (PU-12 family) D 1995 M

tank regiment, air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet), 1 per battery (PU-12 family) D 1995 M

motor rifle regiment, regiment on BTR, Soviet 1980s establishment, 3 PU-12 per regiment D 1989 L

INDICATES A PU-12-series or Ranzhir vehicle marks the command post of a short-range air defence battery (Strela-10, Tunguska, Tor, Osa, Shilka or Igla teams, up to 4-f fire units) or a regimental air defence chief's post nearby.

RECOGNISE **Body** BTR-80 based K1Sh1 hull: raised centre section and extra roof hatches; no turret (from photograph).

Mast Telescopic mast at hull rear, raised for work; dome-shaped head on top and a box halfway up (from photograph). WEG: passive antennas day/night camera on top.

Axles Four axles, 8x8, front two axles steer; single waterjet at hull rear; amphibious (BTR-80 base).

Details Roof-mounted box sensor with two lenses on a pan-tilt head (from photograph); BTR-80 side door between second and third axles.

NOT TO BE CONFUSED WITH PPRU-1M / PPRU-M1 Also on MT-LBu but carries its own 9S80 Dog Ear radar; 9S737 Ranzhir takes data from external radars such as P-19 or 9S18M1.

BTR-60PU-12 p. 65 BTR-60 hull with large stowage box on right side, generator and telescopic mast; PU-12M6/M7 use the BTR-80 hull.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY PU-12M7 is amphibious on the BTR-80 base. 9S737 and 9S737M are tracked; 9S737M fords 1.0 m.

Weight, PU-12M7

13.6 t

Weight, 9S737 (WEG figure)

25000 kg WEG lists a Tata Lark 2518 truck chassis; see open_questions

Weight, 9S737M

31 t GM-5965.05 tracked chassis; RUSA2 gives 'not more than 31,000 kg'

L × W × H

7.7 × 2.9 × 2.41 m

Coordinates measurement envelope, PU-12M7 (connected radars)

up to 100 km maximum, coordinates measurement envelope from connected radars; PU-12M7 has no radar of its own, claim

Data link range, PU-12M7 by radio

up to 40 km maximum, radio link to higher and subordinate posts, claim

Data link range, PU-12M7 by field cable

up to 15 km maximum, field telephone cable, claim

Data link range, 9S737 to higher post

up to 30 km maximum, radio, claim

SOURCES APA4 ARMF JANESRG ORUZH ORYX1 ROE2 RUSA1 RUSA2 VPK9 WEG50 WMC14 WMC15 WP-RU42 WP-RU95 WP-RU96 WP-RU97 WP-RU98

Barnaul-T «Барнаул-Т»**TR-A**

Automated command and control set for tactical short-range air defence, battery to brigade In service 2015

OPERATED BY motor rifle brigade, brigade AD subunit; BO quotes a 'missile division' (battalion) commander O 2021-12 M

airborne division, air defence missile regiment, one full divisional set O 2018-06 M

airborne division, air defence missile regiment O 2018 L

INDICATES A 9S932-1 marks a short-range air defence battery command post. Strela-10, Tor, Osa, Tunguska or MANPADS fire units it controls are likely nearby; in VDV units the kit belongs to the division's AD regiment.

RECOGNISE **Antenna** 1L122-1E rectangular L-band antenna array, 1.2 × 0.8 m (maker's figure), on the top rear of the hull; folded down in transit.

Mast INA: no text source describes a mast on the 9S932-1.

Chassis Modified MT-LBu tracked amphibious carrier (9S932-1).

Airborne version on BTR-MD/BTR-MDM tracked carrier.

Complex Tracked 9S932-1 at battery, 9S931 on MT-LBu, 9S931-1 on KamAZ 6x6 with trailer at brigade; 9S933 and 9S935 are portable MANPADS kits.

NOT TO BE CONFUSED WITH PPRU-1M / PPRU-M1 Also an MT-LBu AD battery command post, but carries a 9S80-series X-band radar, not the 1.2 × 0.8 m 1L122 L-band array.

1L122-2E Self-propelled Garmon radar on MT-LBu with its antenna on a lifting boom; antenna 1.9 × 1.8 m per TW, but GS gives 1.2 × 0.8 m for both versions.

9S737 Ranzhir p. 69 Older AD battery command post on MT-LBu-derived chassis; check for the 1L122 array on the rear hull.



Doomych, Public domain

MOBILITY Amphibious tracked MT-LBu; swims 5 km/h maximum. The BTR-MDM (VDV) version is parachute-droppable.

Weight, combat

11,000 kg MT-LBu chassis figure; 9S932-1 weight INA

L × W × H

6.45 × 2.86 × 1.86 m MT-LBu chassis

Load speed

60 km/h maximum, maximum

Road range

500 km cruising, cruising

Detection range, 1L122-1E

40 km maximum (1-40 km), aircraft, helicopters, cruise missiles, UAVs at up to 700 m/s; export 1L122-1E brochure figure, claim

Minimum detection range, 1L122-1E

1 km export 1L122-1E, claim

Detection altitude, 1L122-1E

10 km maximum, export 1L122-1E; 'at least 10 km', claim

Data link range, 9S932-1

20 km communication range, stationary; 10 km on the move (9S933: 3 km / 1 km)

Radar band, 1L122-1E

L-band claim

SEEN WITH 1L122-1E Garmon (1 per 9S932-1)

SOURCES APA4 BO DB GS6 KZ MIL4 ORYX1 SDU TOPWAR3 TW22 VALKA WEG7 WMC2 WP-RU12

R-166 Artek P-166 «Арте́к»

WH-AT

Medium-power HF/VHF mobile radio station for command post signal nodes In service 1996

OPERATED BY Unidentified unit, signal and EW subunits

R 2025-08 L

military district, signal units R 2018 L

INDICATES The truck R-166 is built for operational-strategic level radio nets; the armoured R-166-0.5 has been lost with tank and motor rifle brigades. Either suggests a brigade or higher command post signal node nearby.

RECOGNISE **Antenna** Roof antennas: whips, a loop and a loop-slot antenna. Mast antennas: V, T, biconical and log-periodic (listed for the truck R-166).

Mast Mast antenna sets take about 60 min to erect; with roof antennas only the station is ready in 5 min. Mast height 1A.

Chassis R-166-0.5: armoured UNSh-10 or K1Sh1 chassis, both described as based on the BTR-80; or MT-LBu. R-166: KamAZ-4310 or KamAZ-43114 truck.

Complex Works alone or in a field signal node. Used with R-149AKSh on a 2025 exercise; lost beside R-149MA1 at Nova Basan and Novyi Bykiv in 2022.

NOT TO BE CONFUSED WITH R-149MA1 p. 64 Command-staff vehicle on the same K1Sh1 (BTR-80-derived) chassis. No text source gives an external difference; confirm by tactical marks or captions. R-161B p. 70 HF-only automated radio station on MT-LBu; R-166-0.5B on the same chassis is HF and VHF. No text source gives an external difference.

VARIANTS R-166 KamAZ-4310 or KamAZ-43114 truck; 8.3 m long (KamAZ-43114). R-166-0.5 Armoured UNSh-10 or K1Sh1 chassis (BTR-80-based), or MT-LBu, in place of the truck. R-166-0.5B Tracked MT-LBu chassis.

SOURCES AR45 AR46 DBLOG2 MIL52 ORYX1 RIA2 RUSARMY5 VPK10 WMC16 WMC17 WP-EN64 WP-RU79 WP-RU105 WSPOT19 WSPOT20 WSPOT21 WWW19 not in WEG



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Works on the move at up to 30 km/h on dirt roads and 60 km/h on paved roads (R-166 truck). R-166-0.5 armoured chassis figures 1A.

Weight, R-166 (KamAZ-43114)	14,840 kg truck version; not more than (rusarmy.com), claim
L × W × H	8.30 × 2.61 × 3.52 m truck version, claim
Speed while transmitting	30 km/h dirt road; 60 km/h paved truck version, working on the move, claim
Time into action	5 min duplex roof antennas, claim

Communication range, HF, stationary	2,000 km up to, R-166, at the halt, claim
Communication range, HF, on the move	350 km up to, R-166, moving, claim
Communication range, VHF, stationary	250 km up to, R-166, at the halt, claim
Communication range, VHF, on the move	70 km up to, R-166, moving, claim

R-161 Poisk HF radio station P-161 «Поиск»

WH-AT

Medium and high power HF/VHF radio stations on trucks for operational-level links

OPERATED BY Unidentified unit, one Oryx-listed loss; no unit, place or date given R 2022 L

NUMBERS Losses 1 R-161 destroyed (Oryx, photo-confirmed; record undated, variant unknown) (2022)

INDICATES A ZIL-131 radio van with a wide wire-antenna field on telescopic masts suggests the radio centre of a higher-level field command post (operational or operational-tactical). Unit type is not established by the sources read.

RECOGNISE **Body** R-161A2M: one K6-131 van body (kung) on a ZIL-131 chassis; three compartments: operator console, amplifiers, generator at rear.

Details Roof lockers for antennas; heater and mushroom ventilator over the cab; roof whips; telescopic masts carried (12 m per UTIN, 60 ft per ARMYRAD).

Details Generator air intake and exhaust through opening side panels. Deployed: rhombic, V and dipole wire antennas on masts, site about 150 x 100 m.

SOURCES ARMYRAD ORYX1 RSIG UTIN WP-RU104 not in WEG

MOBILITY R-161A2M on ZIL-131 truck. Full antenna set needs a site about 150 x 100 m; station can be carried by An-22 aircraft.

Weight	10430 kg R-161A2M, with fuel
Time into action	1 hour R161-A2M, three-man crew, on site
Length x width x height	7400 x 2500 x 3470 mm R-161A2M

Communication range, HF	2000 km maximum, R-161A2M, stationary; UTIN: V 2x46 antenna, 10-30 MHz
Communication range, moving	75 km, 300 km maximum, R-161A2M moving at 30 km/h: roof whip 75 km, zenith-radiating antenna (1.5-14 MHz) 300 km
Frequency range	1.5 to 59.999 MHz R-161A2 and A2M

SEEN WITH R-160P receiver (1 in R-161A2 and R-161A2M station); R-161-5 Ural-4320 shelter trucks (transmit, receive) (2 per R-161-5 station)

R-142N Deimos-N P-142H «Деѡмос-Н»

WH-AT

Truck-mounted HF/VHF command and staff vehicle for regiment, brigade and division staffs In service 1976

OPERATED BY motor rifle division, regiment (brigade) to division

staffs [D 2020 M]

motor rifle brigade, brigade staff [D 2020 M]

INDICATES A general-purpose command and staff vehicle for regiment (brigade) to division staffs. Seeing one suggests a regiment, brigade or division command post nearby.

RECOGNISE Cab GAZ-66 high-mobility 4x4 truck cab; the cab holds a commander's workstation and the antenna control panel.

Body Heated metal van body K-66N in two compartments, equipment forward and staff rear; rear door to the staff compartment. Vehicle 6.06 m long, 2.4 m wide, 3.3 m high.

Axles Two axles (GAZ-66, 4x4). The R-142NSA version is on a three-axle 6x6 KamAZ-43114 chassis.

Details Roof loop antenna of two spaced vertical frames; 3.4 m and 4 m whips that a motor lowers to horizontal; seven-section 11 m telescopic mast raised at halt.

Details At halt: petrol generator set carried out of the body on its cable, two 25 m wire dipole legs strung on the mast or natural supports.

NOT TO BE CONFUSED WITH R-142NSA Three-axle 6x6 KamAZ-43114 truck, 7.94 m long, against the two-axle GAZ-66 of the R-142N.

R-142D Deimos-D Airborne version on the GAZ-66B with KM-66DS body.

VARIANTS R-142NMR Deimos-NMR INA; GAZ-66 chassis per Russian encyclopedia cited by ruWiki. R-142NSA Larger van on three-axle 6x6 KamAZ-43114 truck.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY GAZ-66 4x4 with metal van body. Fords 0.8 m on firm bottom. Radios work on the move up to 40 km/h; the mast and wire dipole are used only at halt.

Weight **5.97 t** R-142N, complete with crew, not more than

L × W × H **6.06 × 2.40 × 3.30 m** R-142N

Fording **0.8 m** R-142N, firm bottom

Turning radius **9.5 m** R-142N, by front bumper

Communication range, R-130M HF **350 km** R-142N, roof loop (AZI) antenna or wire dipole, day and night, between like stations

Communication range, R-130M HF **50 km day, 20 km night** R-142N, 4 m whip, at halt and on the move

Communication range, R-111 VHF **60 km** R-142N, antenna on 11 m mast, at halt

Communication range, R-123MT VHF **40 km** R-142N, antenna on 11 m mast, at halt

Communication range, R-809M2 air band **6 km** R-142N, discone on mast, to aircraft

SOURCES ARMSNET ATP CRYPTO DUDAK KOLESA2 ORYX1 WEG2015 WP-RU99 not in WEG

P-240I-4 Pereselenets П-240И-4 «Переселенец»

WH-AT

Digital field communications vehicle linking command posts (radio, radio relay, switching) In service 2015

OPERATED BY motor rifle brigade, command company or battalion; CNA says combined arms brigades and regiments, tank units not excluded [D 2019 M]

INDICATES CNA places the vehicle in command companies and battalions of brigades and regiments, linking field command posts. One may indicate a brigade or regiment command post nearby.

RECOGNISE Cab Cab of the KamAZ-5350 Mustang truck: seats three, has a sleeping berth, armour can be added.

Body INA: no text source read describes the communications body. Equipment is a radio, radio relay station, terminal and message processor.

Axles Three axles, six-wheel drive (KamAZ-5350 chassis).

NOT TO BE CONFUSED WITH P-260-U signals vehicle (Redut-2US) [D 74] Redut-2US vehicles carry a 32 m telescopic mast on a KamAZ-63501 eight-wheeler; P-240I-4 is on a three-axle KamAZ-5350.

MOBILITY Truck chassis, all-wheel drive; no figures found for this fit.

Weight **INA**

Road speed **INA**

Road range **INA**

Time into action **INA**

Secure channels **more than 100** claimed, cryptographic channels, claim

Total channel capacity **more than 2,000 kbit/s** claimed, all channels together, claim

Communications range **at least 40 km** claimed, medium and conditions INA, claim

Crew **INA**

SEEN WITH Akveduk-25U radio station (1); Radio relay station and digital radio relay equipment; KamAZ-5350 Mustang (1)

VARIANTS P-240BTZ BTR-80 based switchboard vehicle with Zenit set (per Wikipedia); not a truck.

SOURCES AR37 CNA ORYX1 TASS10 WEG2015 WP-EN13 WP-EN40 WPBTR60 not in WEG

R-416GM radio relay station P-416FM

WH-AT

Truck-mounted digital radio relay station for field command networks In service 2018

OPERATED BY military district, Six sets of R-416GM and R-416GM2 delivered (Russian MoD, reported Dec 2023) [R 2023-12 M] seen with Central Military District

Unidentified unit, reported Jun 2024 [R 2024-06 L] seen with Kursk region, near Kremyane (unit not named; Militarnyi says Kharkiv region)

INDICATES A raised mast marks a working relay link. Ukrainian sources say it links forward positions to command posts; Russian MoD says it builds a field backbone network. Strikes reported in Kursk area 2024 and Luhansk deep rear 2026.

RECOGNISE Cab KamAZ-5350 truck chassis.

Body Container for radio equipment, where the operator works.

Axles INA

Details Tall antenna mast carrying three parabolic antennas, raised vertically to work, reportedly up to 30 m.

NOT TO BE CONFUSED WITH R-419L1 radio relay station INA. Forbes via Focus names R-419L1 as another relay station in use alongside; no visual difference read.

SOURCES ARMYINF4 FOCUS4 KYIVP2 MILUA4 ORYX1 WWW110 not in WEG



Vladislav Belograd, CC BY 4.0

MOBILITY KamAZ-5350 truck chassis; mast must be raised to work. Other limits INA.

Chassis **KamAZ-5350** claim

Mast height, raised **up to 30 m** sometimes reaches, Ukrainian border guard description of R-416G-MS, claim

Antennas **3 parabolic** claim

Crew **INA**

Link range **INA**

Tropospheric scatter radio stations (R-410-7.5, R-412A/AM, R-417)

P-412A; P-412AM; P-417

WH-AT

Beyond-line-of-sight radio relay link between headquarters

OPERATED BY Cold War HQ links, R-412F General Staff-front and front-army; R-412A air defence-air force; R-412B army-grade operational-tactical HQ [D 1983 L]

Unidentified unit, Air base communications: R-417 'Baget' antenna vehicle, 229th Air Base, Rostov-on-Don (ATESH claim) [R 2025 L] seen with 229th Air Base, Rostov-on-Don (4th Air and Air Defence Army)

INDICATES A tropospheric station at a headquarters or air base suggests a long, beyond-horizon link to a higher or neighbouring command post. The sources do not say which unit type holds each model.

RECOGNISE Antenna R-412A: two parabolic antennas, controlled from the I326M control panel in the instrument vehicle. R-417, R-412AM, R-410-7.5: antenna type INA.

Mast R-412F: antennas raised to 20 m on a separate I601 mast vehicle on a ZIL-131 chassis. R-417 has a separate antenna vehicle.

Chassis R-412A: two Ural-375 box-body vehicles (instrument, power). R-412B: MT-LB. R-417: KamAZ-4310 truck. R-412AM and R-410-7.5 chassis INA.

Complex R-412A set: I606 instrument vehicle plus I613M power vehicle with two generators. R-412F adds the I601 mast vehicle.

NOT TO BE CONFUSED WITH R-416G-MS radio relay station p.

72 KamAZ truck with a roof-mounted retractable mast carrying three parabolic antennas; digital link to about 40 km (Ukrainian Border Guard statement).



ShinePhantom, CC BY-SA 3.0

MOBILITY Truck-mounted sets (R-412A, R-412F, R-417); R-412B tracked. Antennas raised by mast vehicle on R-412F. Time into action INA.

Antenna height, R-412F **20 m** antennas lifted by I601 mast, claim

Link range, R-412A **at least 150 km** at least, between two stations of the same type, claim

Frequency band, R-412A **4438–4555 MHz and 4630–4749.98 MHz** two sub-bands, claim

Channels, R-412A **3 to 6 telephone channels, or 1 telegraph channel** frequency modulation, claim

RF power, R-412A **200 W minimum (one transmitter, two antennas); 400 W (two transmitters)** radiated, minimum, claim

Frequency band, R-417 **4435–4750 MHz** Ukrainian R-417MU, a modernised R-417; Russian set not confirmed

SOURCES ORYX1 SLOVO U246 VALKA3 VNTU WP-RU127 WP-RU128 not in WEG

R-439 Legenda-2 P-439 «Легенда-2»

WH-AT

Military satellite communication stations: portable, container and BTR-80-based

OPERATED BY Unidentified unit [R 2022-08 M] seen with Unit not stated; 'Legenda-2' complex reported destroyed near Blahodatne, Mykolaiv region, 2022-08 [Operational Command South claim]

INDICATES An R-439-MD2 or -BK on a BTR-80-based K1Sh1 hull suggests a tactical-level command element: the MD2 is built for satellite links of mobile tactical-level command posts, and the BK is described as regiment level.

RECOGNISE Antenna Dish antennas: R-439-MD2 has 1.2 m (4/6 GHz) and 0.8 m (7/8 GHz) dishes plus an array for 0.2/0.4 GHz; R-439-BK has one 1.5 m dish.

Antenna Portable R-439-P: 1.2 m dish in five segments; antenna module about 1.16 x 0.88 x 1.20 m. R-439-MD1 uses a 0.5 m phased array for use on the move.

Chassis BK and MD2 ride the K1Sh1 command hull of the BTR-80 8x8, described as an enlarged hull with no turret armament.

Complex R-439-KUL hub: container on a KK-2.02 base with a 2.5 m five-segment dish that can stand up to 25 m from the container. R-439-MD3 packs into three aluminium cases.

NOT TO BE CONFUSED WITH R-149BMR and other K1Sh1 command vehicles p. 24 Same K1Sh1 (BTR-80) hull; the R-439 versions carry satellite dishes (MD2: 1.2 m and 0.8 m; BK: 1.5 m).

R-448 portable satellite station Ku- and C-band VSAT with an antenna assembled from six elements; the R-439-P dish is in five segments.

VARIANTS R-439-P Legenda-2P Transportable set, 1.2 m dish in five segments; carried by any transport. C band (4/6 GHz), one link, 1.2 to 9.6 kbit/s; company or battalion level; deploys in 15 min. R-439-BK Legenda-2BK BTR-80 (K1Sh1) hull with one 1.5 m dish. Jam-resistant C-band links via ESSS-2 relays, 4 phone and 1 telegraph channel; regiment level; deploys in 20 min. R-439-BK1 INA. Improved R-439-BK; details INA. R-439-MD1 0.5 m phased-array antenna on a command vehicle. 4/6 and 7/8 GHz with automatic tracking; works on the move; up to 4 digital phone channels. R-439-MD2 K1Sh1 hull with 1.2 m and 0.8 m dishes and an antenna array. Three bands: 0.2/0.4 GHz on the move, 7/8 GHz at short halts, 4/6 and 7/8 GHz parked.



Mil.ru, CC BY 4.0

MOBILITY BK and MD2 on K1Sh1 (BTR-80) hull; MD2 links only on 0.2/0.4 GHz while moving, 7/8 GHz at short halts. P, MD3 and KUL are carried sets that must be set up.

Weight, R-439-P set **132.5 kg** Transportable station; MoD catalogue figure as quoted by RUWIKI, claim also 90 kg WP-RU106
90 kg or less claim

Weight, R-439-MD3 main set
Time into action, R-439-P **15 min** russianarms.ru figure as quoted by RUWIKI, claim

Time into action, R-439-BK **20 min** claim

Time into action, R-439-MD3 **15 min or less** Crew of 1 to 2, claim also 15 to 20 min RAMD3

Time into action, R-439-KUL **1 h** Plus up to 20 min warm-up after power-on, claim

Weight, vehicle (K1Sh1)

Time out of action

Crew, R-439-P **1 to 2** russianarms.ru figure as quoted by RUWIKI, claim

Crew, R-439-KUL **4** claim
Frequency bands, R-439-MD2 **Rx 3400-3900, 7490-7540, 280-290 MHz; Tx 5725-6225, 7900-8400, 345-385 MHz** C, X and UHF satellite bands, claim
Frequency bands, R-439-BK **Rx 3400-3900 MHz; Tx 5725-6225 MHz** claim

Transmitter power, R-439-MD2 **10 / 10 / 40 W** 4/6 GHz / 7/8 GHz / 0.2/0.4 GHz, minimum, claim

Transmitter power, R-439-BK **130 W** claim

Data rate, R-439-P and MD3 **1.2, 2.4, 4.8, 9.6 kbit/s** P also 48 kbit/s per RUWIKI, claim

Links, R-439-KUL **10 directions** claim

Anti-jam method **Frequency hopping uplink; spread-spectrum PSK downlink** MD1 and MD3, claim

Power use, R-439-MD1 **0.8 kW at 27 V DC** claim

Crew, R-439-BK and MD2 **INA**

SEEN WITH K1Sh1 (BTR-80-based command vehicle hull) [1 per R-439-BK or R-439-MD2]; KK-2.02 container base (1 per R-439-KUL); R-439-KUL hub station (Up to 10 links to R-439-P and other stations); Globus-1, Yamal, Express relay satellites

Redut-2US «Pegym-2YC»

WH-AT

Mobile digital radio-relay and broadband wireless communications complex for command posts In service 2011

OPERATED BY command brigade [O 2017 M] seen with 175th command brigade

command brigade, communications company [R 2019 M]

Combined-arms formations of the Western Military District

[R 2015 L] seen with Western Military District combined-arms formations

NUMBERS Losses 4 vehicles of the complex recorded lost (Oryx) (2026-10)

INDICATES Marks an operational-level command post or communications node, typically of a command brigade, rather than a battalion tactical post. A complex comprises an R-431AM antenna module and P-260 equipment vehicles.

RECOGNISE Cab KamAZ 8x8 (KamAZ-63501); forward-control cab; P-260T seen with spare wheel carried behind cab (from photograph).

Body Closed equipment shelter (from photograph); the complex carries data transmitters on a 32 m telescopic mast.

Axles Four axles, 8x8 all-terrain chassis (KamAZ-63501).

Details Antenna module: mast assembly behind cab with cable drums and railed roof platform; levelling jacks lowered when deployed (from photograph).

NOT TO BE CONFUSED WITH P-2401-4 Pereselenets p. 71

Tactical system of combined-arms brigades and regiments on KamAZ-5350, not an eight-wheeler with 32 m mast.

R-416G-MS radio-relay station p. 72 KamAZ with mast of up to 30 m carrying three parabolic antennas (Ukrainian report).

SEEN WITH R-431AM digital radio-relay station (antenna module);

P-260-O terminal communications vehicle; P-260-U node communications vehicle; P-260-T transit communications vehicle; P-261 communications control vehicle

SOURCES AR37 AR47 BORG CNA MIL53 OLLUA ORYX1 RWW UKRINF4 WMC18 WMC19 not in WEG



MILRU, CC BY 4.0

MOBILITY Eight-wheel all-terrain truck (KamAZ-63501).

Antenna module lowers levelling jacks when deployed (from photograph). No cross-country figures found.

Weight	INA
Road speed	INA
Road range	INA
Time into action	INA

Radio-relay data rate	5-155 Mbit/s claimed, multi-hop radio-relay lines, 4 directions simultaneously; official documents quoted, claim
Radio-relay range	up to 55 km maximum, radio-relay link at up to 155 Mbit/s
Broadband wireless access	up to 37 Mbit/s, up to 200 subscribers claimed, broadband wireless network; official documents quoted, claim
Wireless access range	up to 30 km maximum, broadband wireless access at up to 37 Mbit/s

Strelets КРРК «Смрелеу»

MP+AD

Soldier-worn reconnaissance, command and communications set; fire direction link

In service 2007

OPERATED BY Squad to company level, Squad to company level; platoon and section leaders wear the full set [D 2024 H]

TOS-1A batteries allocated to manoeuvre brigades, TOS-1A battery allocated to a manoeuvre brigade gets Orlan-10 data over the Strelets network [D 2024 L]

reconnaissance battalion, Ground forces reconnaissance subunits, first recipients; echelon not stated [R 2021 M]

Spetsnaz brigade, 'Commandos' (TASS reprint); Spetsnaz brigade is an inference [R 2021 L]

INDICATES Soldiers wearing tablets and a vest-mounted radio set suggest a squad to company level group able to pass target coordinates to artillery or aircraft. Reconnaissance units were first to receive it.

RECOGNISE Terminal Portable computer (tablet) with peripherals spread over a load-bearing vest. Commander tablet type TT; rank-and-file tablet type AK.

Vest Commander set is a vest loaded with radios, antennas, tablet and batteries. A squad member carries a small pager-like terminal.

Components Operating console, vital-signs recorder, R-853-V2M radio, headset, PDU-4 rangefinder on a tripod, SBR-5M Fara-VR radar.

Complex Worn by dismounts. Lighter kit reportedly seen in Syria: tablet, distance-and-coordinates calculator and a communications module.

NOT TO BE CONFUSED WITH Andromeda-D Airborne troops automated C2 complex with similar functions; Strelets gives individual soldiers worn terminals.



Vitaly V. Kuzmin, CC BY-SA 4.0

MOBILITY Worn by one soldier; battery life 12-14 h (ATP). Scouts can use it after a parachute jump (ARMYREC).

Weight	2.4 kg	Modernised set, claim
Battery life	12-14 h	VHF link to company network
Water resistance	2 m briefly (tablets, above IP68)	TT and AK tablets; FOCUS confirms above IP68 only, claim
Time to fix position	3 min, 10 m error	claimed, After parachute jump; TASS reprint, error

Range, squad radio network	about 1.5 km	claimed, Line of sight; each set acts as relay, so reach grows, claim
Data rate, internal network	11 Mbit/s	claimed, Direct visibility, up to 1.5 km (ARMYREC); VPK gives 11 Mbit/s for a backup Wi-Fi channel, claim
Range, R-853-V2M to aircraft	up to 8 km	claimed, Target designation data to aircraft, no relay, claim
Range, PDU-4 laser rangefinder	3-5 km	operating range, Operating range ('дальность действия'); target type not stated, claim

SOURCES AR52 ATP CNA FOCUS7 RG6 VPK13 WP-RU114 not in WEG

8

Logistics

IN THIS CHAPTER

Trucks 75

LOOKING FOR THESE? THEY ARE IN OTHER CHAPTERS

SYSTEM	WHAT IT IS	CHAPTER	PAGE
BREM and BTS armoured recovery vehicles	Armoured recovery and repair vehicles for tank and infantry vehicle units	6 Engineers	62
PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry	Amphibious carrier for cargo and vehicles	6 Engineers	57

KamAZ military trucks (4310 and Mustang family)

KAMA3-4310, KAMA3-5350, KAMA3-6350 «Мусманг»

WH-AT

General-purpose 6x6 and 8x8 cargo and tractor truck chassis for supply and special bodies In service 1980

OPERATED BY motor rifle battalion, logistical support platoon

D 2016 M

command brigade, communications company (Redut-2US on

KamAZ-63501) **D 2019 M**

matériel support brigade, transport units (heavy lift) **D 2016 L**

INDICATES Mustang trucks are the main army lift and are found in supply platoons and MTO units. A 6350 or 63501 in an artillery, signal or air defence unit usually means a special body or tractor role.

RECOGNISE Cab Cab over engine, no bonnet. Mustang family shares one three-man cab that takes an add-on armour kit. Sleeper berth: sources disagree.

Body Standard cargo body is a drop-side platform with drop tailgate and removable bows and tarpaulin. Cranes, shelters and containers are common alternatives.

Axles Count axles: 4310 and 5350/53501 have three (6x6), 6350/63501 four (8x8). The 4350 (4x4) has two.

Axles Length: 5350 is 7.96 m, 53501 is 8.5 m (ruwiki). Exterior difference between 5350 and 53501 is length. The 53501 looks like the 43114 and 43118.

NOT TO BE CONFUSED WITH Ural-4320 / Ural-5323 (Motovoz) Ural-5323-20 is fitted with a KamAZ cab, so the cab alone does not separate it from the 63501. Check the chassis.

KamAZ-43114 / 43118 The 53501 resembles them in outline; 4310 is the older design with the 43114 as successor.

KamAZ-63968 Typhoon-K p. 23 A separate MRAP-type vehicle.

MM-501 on a 5350 is a box slid onto a truck bed, not a monocoque hull.



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Wheeled all-wheel drive with central tyre inflation, not amphibious. Mustang fords 1.75 m. RWW: best for long road hauls, Ural better off-road.

Weight, kerb, 5350 **9.2 t** KamAZ-5350
Gross weight, 5350 **15.85 t** KamAZ-5350
Weight, 6350 **11.4 t** KamAZ-6350
Weight, kerb, 4310 **7 t** KamAZ-4310

Payload, 5350 **6 t** KamAZ-5350
Payload, 53501 **10 t** KamAZ-53501; ruwiki only
Payload, 6350 **10 t** KamAZ-6350
Payload, 63501 **15 t** KamAZ-63501; RWW only
Payload, 4310 **6 t** KamAZ-4310 basic
Towed load, 6350 **12 t** KamAZ-6350

SEEN WITH Pantsir-S1 (96K6) [1 per launcher; chassis stated differently, see open_questions]; 9A52-4 Tornado MRL

VARIANTS KamAZ-5350 Mustang 6x6, 7.96 m long, 6 t payload; standard platform.

SOURCES ATP CNA D243 KYIV ORYX1 RWW WEG35 WP-EN40 WP-EN41 WP-RU65 WP-RU66 WP-RU67 ZR3 UNVERIFIED DRAFT

Ural-6370 / 63706 / 532301 trucks (newer Ural family) Урал-63706

WH-AT

«Торнадо-У»

Heavy 6x6 and 8x8 cargo trucks and tractors; Tornado-U has armoured cab

In service 2016

OPERATED BY materiel support battalion, MTO (supply and transport) subunits [R 2020 L] seen with Southern Military District, unnamed MTO subunits

Unidentified unit [R 2021 L] seen with Russian forces, Syria (Khmeimim parade towing M-46 guns, 2021)

NUMBERS Losses 47 Ural-63706-0122 and 3 Ural-63706-0011 in Oryx loss list (2026-10)

INDICATES Per Izvestia's MoD sources, Tornado-U goes first to materiel support (MTO) subunits and is meant for hauling ammunition, medicine and food to forward units. Sightings point to supply traffic, not to a fighting unit.

RECOGNISE Cab Ural-63706 Tornado-U: bonneted, frame-panel cab with its own bonnet, three seats, hung steps and handrails at the door; armour fitted under the panels.

Body Flat cargo body of large size; the same chassis is built as a bolster or semi-trailer tractor. Military 63706 trucks go to units with a roof mount fitted.

Axles Ural-63706 and 6370: three axles, 6x6, solid axles on leaf springs, large tyres. Ural-532301: four axles, 8x8.

Details Armoured roof turret with shield for one gunner, PKM 7.62 mm shown in state media; turret can take a 30 mm grenade launcher or 12.7 mm MG.

Details Ural-6370 civil base and Ural-532301 are cab-over (cab above the engine, no bonnet); Ural-4320 and Tornado-U are bonneted.

NOT TO BE CONFUSED WITH Ural-4320 Older 6x6 with a 6 t payload; Tornado-U carries 16 t and has a new frame-panel cab, bonnet and roof gun mount.

Ural Typhoon-U p. 23 Armour capsule for troops behind the cab instead of a cargo platform, independent suspension.

KamAZ Tornado-K p. 75 Same role (armoured supply truck) but built on a KamAZ chassis.

Ural-5323 (unarmoured 8x8) Same cab-over 8x8 as Ural-532301; Oryx entry is the one with armoured cab.



Ministry of Defence of the Russian Federation, CC BY 4.0

MOBILITY Heavy road train: payload 16 t and gross 30 t to 33 t for Tornado-U. Large tyres and axle locks for rough ground; no figure for ground pressure.

Gross weight, Ural-63706 Tornado-U **33,000 kg** Full laden mass, series vehicle, claim also 30,000 kg AR64

L × W × H **9,550 m × 2,550 m × 3,330 m** Ural-63706 Tornado-U, claim

Road speed **100 km/h** maximum, Ural-63706 Tornado-U, maximum, claim

Road range **1,000 km** Ural-63706 Tornado-U, one fill; 2 × 210 l tanks per RG21, 400 l per STKT, claim

Fording depth **1.8 m** Ural-63706 / 63704 Tornado-U; source does not say if prepared

Payload, Ural-63706 Tornado-U **16,000 kg** Cargo truck, claim

Towed load, Ural-63706 Tornado-U **12,000 kg** Semi-trailer, claim

Ground clearance **400 mm** Ural-63706 Tornado-U

Cab protection **7.62 mm SVD and AK bullets** claimed, Ural-63706 Tornado-U; windows armoured glass; engine armoured

Engine power, Ural-63706 Tornado-U **440 hp** YaMZ-652 diesel also 412 hp WP-RU138

Crew **3 seats in cab** Driver plus two passengers; gunner stands in roof turret

Roof weapon **7.62 mm PKM in protected mount** Ural-63706 Tornado-U, claim

Payload, Ural-532301 **10,000 kg** 8x8 cargo truck; road train up to 34 t

Gross weight and payload, Ural-6370 civil base **34,500 kg / 19,500 kg** Civil commercial truck, not the military Tornado-U figures

SEEN WITH Ural-63704 bolster-body truck (armoured vehicle transporter); Semi-trailers up to 12 t (1 per tractor)

VARIANTS Ural-63706-0010 Tornado-U Base military 6x6 truck with armoured bonneted cab. Difference to -0011 and -0122: INA. 16 t payload, 12 t trailer (state media figures for Tornado-U).

Ural-63706-0011 INA. INA. Ural-63706-0122 INA. INA.

Ural-6370-0121-30 INA; listed by Oryx as 6x6 cargo truck, one loss. INA. Ural-532301 Cab-over 8x8, four axles, armoured cab. 10 t payload; older Ural-5323 line, not the Tornado-U line.

9

Unmanned aircraft

IN THIS CHAPTER

Small reconnaissance drones	78
Larger reconnaissance and strike drones	82
Loitering munitions and one-way attack	85
FPV drones	89

LOOKING FOR THESE? THEY ARE IN OTHER CHAPTERS

SYSTEM	WHAT IT IS	CHAPTER	PAGE
Leer-3	UAV-borne cellular	5 Electronic warfare	52
Repellent-1	Wheeled counter-UAV EW complex	5 Electronic warfare	53
Pole-21	Networked GNSS jamming posts protecting fixed sites from GNSS-guided weapons...	5 Electronic warfare	53

ZALA 421-16E ZALA 421-16E

AIR-FW

Short-range electric reconnaissance and fire-correction UAV; spots for Lancet In service 2011

OPERATED BY combined arms army, Z-16 reconnaissance UAV crews [O 2026-01 L](#)

artillery brigade, UAV reconnaissance-strike subunit (Z-16 with Lancet) [R 2024-12 L](#)

artillery regiment, UAV reconnaissance-strike subunit (Z-16 with Lancet) [R 2024-12 L](#)

INDICATES Z-16 flights suggest a reconnaissance-strike chain at work: the maker, relaying Russian media, says Z-16 crews correct artillery, MLRS and air strikes, spot for Lancet, and serve in artillery regiments and brigades.

RECOGNISE Wings Flying wing, span 2,815 mm (E; one source gives the same for E2) or 2,800 mm (E2); two removable outer wing panels.

Engines One electric motor driving a pusher propeller; low noise claimed by maker and state media.

Fuselage Central fuselage, 1,020 mm long without payload, carrying a gyro-stabilised camera or thermal-imager payload.

Tail No separate tail unit: flying-wing layout.

Launch and recovery Launched from a pneumatic, elastic or mechanical catapult (E2 also by hand); lands by parachute with an air shock absorber, or into a net.

NOT TO BE CONFUSED WITH ZALA 421-16E5 Much larger: wingspan 5,000 mm and take-off weight 29.5 kg.

SuperCam S350 p. 79 Larger: wingspan 3.5 m.

VARIANTS ZALA 421-16EM INA. ZALA 421-16E2 Span 2,800 mm (one source: 2,815 mm); lighter, 7.5 kg take-off weight; can be hand-launched. ZALA 421-16E HD Light camouflage finish and composite materials (state media, 2022).

SOURCES AR66 ATP GS37 IIS25 INT K11 LENTA4 MIL66 MIL67 MIL68 MIL69 RG8 ROBOT RUSI2 TAD TMD8 VPK14 ZAIF ZALA1 ZALA2 ZALA3 ZALA4 ZALA6 ZALA7 ZALA8 ZALAP not in WEG



Mike1979 Russia, CC BY-SA 4.0

MOBILITY Catapult launch from unprepared sites, parachute recovery. **Maker limits:** wind up to 15 m/s, -30 to +40 C (E).

Take-off weight, maximum	8-10.5 kg 421-16E, claim
Speed	65-110 km/h 421-16E, claim
Range, flight distance	200 km claimed, 421-16E per 2011 article text, in more than 3.5 h; the article's table gives 210 km, claim
Wingspan	2,815 mm 421-16E, claim

Control range, video/radio link	50 km radius of action, 421-16E; average radius of video/radio link, claim
Altitude, maximum	3,600 m 421-16E, claim
Altitude, operating	500-1,500 m operating altitude range, current Z-16E, claim
Employment depth	operational depth practical, RUSI on Orian and Zala UAVs, 2024-25; Zala model not given, observed
Payload mass	up to 1.5 kg 421-16E, claim

9
DRONES

Orlan-10 «Орлан-10»

AIR-FW

Catapult-launched recon UAV for artillery spotting and EW; Orlan-30 adds laser designator In service 2010

OPERATED BY motor rifle brigade, UAV company, short-range platoon [D 2019 H](#)

EW brigade, Leer-3 platoon, 3 per Leer-3 complex [D 2024 H](#)

motor rifle division, UAV company [D 2019 M](#)

artillery brigade, UAV company, UAS platoon (BKAR artillery reconnaissance version) [D 2019 M](#)

EW battalion, Leer-3 crews (battalion or company of a combined arms army) [D 2019 M](#)

Spetsnaz brigade, Leer-3 complex (with Orlan-10); FOI says 'Spetsnaz units' [R 2017-12 L](#)

INDICATES *Doctrinally the Orlan-10 sits in brigade and division UAV companies and EW Leer-3 platoons, and is used to adjust artillery. A catapult or GCS truck suggests such a unit and nearby fire support; Orlan-30 adds laser-guided Krasnopol.*

RECOGNISE Wings High-mounted wing, span 3.1 m on Orlan-10; Orlan-30 is larger with a 3.9 m span.

Engines One small petrol engine in the nose driving a pulling (tractor) propeller; two-blade propeller stated for Orlan-30.

Fuselage Overall length 1.8 m, of mixed metal and plastic construction. Orlan-30 has a spherical stabilised optical ball in the rear fuselage.

Tail Conventional tail; the fin is much larger than the narrow tailplane.

Launch Launched from a folding catapult and recovered by parachute; needs no runway. Ground station in a truck nearby.

NOT TO BE CONFUSED WITH Orlan-30 Larger, 3.9 m span against 3.1 m, with a spherical optical ball in the rear fuselage.

SEEN WITH Ground control station, truck-mounted (1 per Leer-3 complex; controls up to 4 UAVs); Folding launch catapult; RB-341V Leer-3 (3 Orlan-10 per complex (ATP, FOI); 2 in a 2015 Eastern MD description (GS)); Strelets; 2S19 Msta-S with Krasnopol

VARIANTS Orlan-30 Larger airframe, 3.9 m span; spherical gyro-stabilised optical ball in the rear fuselage. Laser designator-rangefinder for guided shells such as Krasnopol; about 8 h endurance. BKAR INA. Orlan-10 with electronic reconnaissance pods that locate artillery and other radars and cue fire.



Mike1979 Russia, CC BY-SA 4.0

MOBILITY Catapult launch and parachute recovery, no runway. Launch wind limit 15 m/s (Orlan-10M). Operates -30 to +40 C. Ground station truck must stay within link range.

Max take-off weight, Orlan-10	18 kg Orlan-10M figure (STC); War & Sanctions figure via TV24, claim also 15 kg RWW
Max take-off weight, Orlan-30	30 kg export Orlan-30, Rosoboronexport at ADEX-2024, claim also not more than 40 kg W1030
Flight range, Orlan-10 under radio control	150 km maximum, maximum link range with ground antenna, Orlan-10M, claim also 150 km ATP
Flight range, Orlan-10 autonomous	600 km maximum, pre-programmed flight without radio link also up to 300 km from ground station ATP
Flight range, Orlan-30 under radio control	120 km maximum, data-link range, export Orlan-30, claim also 500 km ATP
Flight range, Orlan-30 autonomous	300 km maximum, autonomous mode, export Orlan-30, claim
Time into action	under 15 min crew deploys equipment and launches UAV, claim
Wingspan, Orlan-10	3.1 m also just over 1 m KI1
Wingspan, Orlan-30	3.9 m export Orlan-30, claim

Payload, Orlan-10	5 kg maximum also 3 kg STC15
Payload, Orlan-30	5 kg maximum, export Orlan-30, claim also 6-8 kg normal, 12 kg maximum WP-RU83
Detection range, Leer-3 payload	6 km GSM, possibly 3G and 4G, cellular base stations; distance from the UAV
Practical range, artillery spotting	40-50 km distance mentioned in Russian press articles on artillery spotting c. 2015, per RWW; RWW suggests a video-link limit
Practical range, Leer-3 EW tasks	over 100 km distance from unit deployment site, Russian MoD statement, after Feb 2022, claim
Laser range, Orlan-30 designator-rangefinder	200-8000 m range of measured distances, measured distance, 1064 nm laser, export Orlan-30, claim
Crew	6 crew chief, 2 UAV pilots, 2 payload operators, 1 technician, claim
UAVs controlled per ground station	up to 4 simultaneous control
Bomb load, Orlan-10 strike version	4 HE-frag bomblets converted VOG-25 grenades in underwing blocks; Russian MoD video, May 2022, claim

SOURCES ATP CNA EDR24 FOI GS23 IISS25 KI1 KP262 MASH MIL33 MIL92 RUSIPL RWW STC15 TV24 W1030 W171 WP-EN55 WP-RU83 not in WEG

Supercam S350 «Суперкам»

AIR-FW

Small electric fixed-wing reconnaissance and artillery fire-adjustment UAV In service 2014

OPERATED BY airborne division, separate unmanned systems battalion [O 2026-03 L] seen with Ivanovo Guards VDV formation (separate UAV battalion), Dnestr Group of Forces

Unidentified unit, UAV crews [R 2024-01 L] seen with Vostok Group of Forces, South Donetsk direction

INDICATES A Supercam S350 overhead suggests a Russian reconnaissance UAV crew working with artillery: MoD reports describe crews passing target coordinates straight to a battery or howitzer battalion and adjusting its fire.

RECOGNISE Wings Flying-wing layout; span 3.2 m per maker (other reports 3.5 m).

Engines Single electric motor driving a propeller; flies almost silently.

Fuselage Modular composite airframe; wing panels detach on landing to avoid damage.

Launch and recovery Launched from an elastic or pneumatic catapult (MoD: two rubber cords suffice); lands under a parachute.

NOT TO BE CONFUSED WITH SKAT 350M (Kalashnikov) Derived from Supercam 350; differs by a new wing design, battery controller and cameras.

Supercam S180 Smaller and faster; reinforced wing centre section with permanently attached wing panels.

Supercam SX350 Hybrid VTOL tiltrotor (convertiplane); takes off vertically, no catapult.

Orlan-10 p. 78 Similar span (Orlan-10 3.1 m), but petrol engine; the S350 has an electric motor and is near silent.



Supercamacro, CC BY-SA 4.0

MOBILITY Catapult launch, parachute recovery; deploys in 15 min (maker). Wind up to 15 m/s, -40 to +45 C, moderate rain and snow.

Maximum take-off weight	15.5 kg claim
Speed	65–120 km/h claim
Flight range	at least 240 km maximum, total flight distance per maker, not radius of action, claim
Time into action	15 min complex deployment, claim

Range, command and telemetry radio link	100 km maximum, radio line of sight to ground control station, claim
Range, video link	70 km maximum, standard video channel to ground control station, claim
Practical range, video link	about 70 km good picture and link, operator's account 2024-08, observed
Payload, day camera	photo camera 20/24/42/60 MP claim

SOURCES AIF9 DSA IISS25 IZ5 I11 LIGA2 MEDUZA MFR MFRAB NEWSRU2 REFORM RWW TASS17 TASS18 TASS19 W1A2 W1B2 W1C2 W1D2 W1E W1F W1G W1H W1I W1J WP-RU115 WSPOT24 ZAIF not in WEG

Granat-1 / Granat-4 Гранат-1 / Гранат-4 «Гранат»

AIR-FW

Small reconnaissance UAVs for artillery target acquisition and spotting In service 2010

OPERATED BY motor rifle brigade, UAV company, short-range platoon (Granat-4 with Orlan-10) [D 2016 H]

motor rifle brigade, UAV company, mini-platoon (hand-launched Granat-1) [D 2016 M]

motor rifle regiment, Attached Granat or Eleron platoon from the brigade or division UAV company; also battalions, at times companies [D 2019 M]

artillery battalion, Granat-1 operator station, probably at battery command and observation post [D 2016 L]

INDICATES A Granat launch site points to a UAV platoon of a brigade or division UAV company, often attached to a regiment or battalion, or to the Navodchik-2 complex of a tube or rocket artillery battalion; Granat-1 likely near the battery command post.

RECOGNISE Wings Granat-4: aeroplane type, wingspan 3.2 m. Granat-1: flying wing of composite, span 0.82 m.

Engines Granat-4: internal combustion engine, pusher propeller. Granat-1: electric motor, propeller at the front.

Fuselage Granat-4 length given as 2.6 m. Granat-1 length 0.52 m, height 0.24 m. Granat-4 system carried in a KamAZ van.

Tail INA. No text source describes the tail of either type; see image candidate.

NOT TO BE CONFUSED WITH RB-341V Leer-3 (ground station) p. 52 OSCE SMM first reported a Navodchik-2 control station as an RB-341V Leer-3 (Nov 2020), then corrected it. No text source gives a distinguishing feature.



Vitaliy V. Kuzmin, CC BY-SA 4.0

MOBILITY Granat-4 catapult-launched, parachute recovery; system moved in a KamAZ van. Granat-1 hand or elastic catapult launch, carried in a backpack.

Weight, Granat-4 takeoff	30 kg manufacturer figure via Kalashnikov, claim
Weight, Granat-1 takeoff	2.4 kg claim
Time into action, Granat-4	15 minutes not more than, claim
Time into action, Granat-1	5 minutes not more than, claim

Data link range, Granat-4	70 km claimed, photo, video and thermal data transmitted near real time, claim
Range of action, Granat-4	100 km claimed, flight range per Kalashnikov, claim
Endurance, Granat-4	6 h maximum, maximum, claim
Ceiling, Granat-4	4,000 m maximum, above sea level; working band 300–2,000 m, claim
Range, laser designation, Granat-4	3 km claimed, laser illumination of targets for guided munitions (Krasnopol), claim

SOURCES ATP AV21 CHARTER CNA DEFNDR IISS25 MIL24 ROBO1 ROBO4 RWW TADV WMC5 WP-RU59 WP-RU60 ZAXID not in WEG

Eleron-3SV Элерон-3СВ «Элерон»

AIR-FW

Man-portable short-range electro-optical reconnaissance and artillery-spotting UAV In service 2013

OPERATED BY motor rifle brigade, UAS company, Eleron platoon; platoons attached to battalions or companies [\[D 2019 M\]](#) motor rifle division, UAS company (division asset), Eleron platoon attached to regiments [\[D 2019 M\]](#) artillery brigade, UAS company of an army artillery brigade; each platoon flies Orlan-10, Eleron-SV or Granat-4 [\[D 2019 L\]](#) Unidentified unit, Eleron-3SV captured in Ukraine (Ukrainian capture reports 2022-2024); no unit named [\[R 2024-04 M\]](#)

Russian forces in Syria, Russian group in Syria; artillery spotting [\[R 2016 M\]](#) seen with Russian forces in Syria

motor rifle division [\[R 2018 L\]](#) seen with 150th Motor Rifle Division

INDICATES Before 2022 Eleron platoons sat in brigade or division UAS companies and were attached to regiments and battalions to spot for artillery; an Eleron may mean such a unit within about 25 km. FSB, MVD and Strategic Missile Forces also use it.

RECOGNISE Wings Eleron-3SV: small flying wing (blended wing body), span 1.47 m; original Eleron-3 has delta wings on a conventional fuselage.

Engines One electric motor driving a pusher propeller at the rear; acoustically near silent.

Fuselage About 5 kg; launched from a bungee (T23P) or pneumatic (T27P) catapult, recovered by parachute; carried in two backpacks.

Tail Eleron-3SV has no tailplane; original Eleron-3 has a tailplane.

Variant T-16 Eleron T-16: larger (about 20 kg) canard; swept wings with leading-edge extensions and downturned tips, two fins, underwing pylons.

NOT TO BE CONFUSED WITH Takhion [p. 81](#) Also an electric, catapult-launched, parachute-recovered flying wing, but larger: sized between Eleron and Orlan.

Orlan-10 [p. 78](#) Heavier (16–18 kg) with a petrol engine, so audible; Eleron is about 5 kg and electric.

Eleron-10 Same family but larger: T10 air vehicle 2206 x 883 x 384 mm, 15.5 kg, 2.5 h endurance.

VARIANTS Eleron-3 Conventional fuselage with delta wings and tailplane. Analogue link, 20 km link range, 90 min, 4.9 kg take-off weight. Eleron-3SV Tailless flying wing, span 1.47 m. Ground Forces version: digital closed link to 25 km, about 100 min, 5.3 kg, 1 kg payload. **Eleron T-16** Much larger canard with two fins and underwing pylons. About 20 kg, up to 6 kg payload including underwing containers; flew 150 km in 2018 tests.



Nickel nitride, CDD

MOBILITY Carried by the crew in two backpacks; needs a catapult to launch and lands by parachute; battery limits flight to about 1.5–2 h.	
Take-off weight, Eleron-3SV	5.3 kg maximum, claim also 4.9 kg ATP
Take-off weight, Eleron T-16	about 20 kg
Control range, real-time link	25 km maximum, real-time video over data link, claim also 20 km ATP
Flight range, autonomous	50 km maximum, no link, imagery recorded on board, claim
Time into action	10 min maximum, deploy or pack up complex, claim also 15 min BG137
Wingspan, Eleron-3SV	1.47 m claim
Length, Eleron-3SV	0.6 m
Transport containers	2 backpacks, each 830 x 560 x 230 mm claim
Flight speed, Eleron-3SV	70–130 km/h level flight, airspeed band, claim also 50–130 km/h AR22
Operating altitude, Eleron-3SV	3,000 m maximum, above launch point, claim also 4,000 m SHEP2
Payload, Eleron-3SV	1 kg maximum, claim also 0.5 kg BG137
Payload, Eleron T-16	up to 6 kg maximum, incl. underwing containers also 5–6 kg DBLOG
Sensors	photo camera, gyro-stabilised TV camera, or thermal imager with low-light colour camera claim
Other payloads	radio relay or jammer module claim
Navigation	GLONASS/GPS; autonomous, radio-command and orbit modes; automatic return claim
UAVs per complex	2 claim
Simultaneous UAVs per GCS	2
Crew	2 claim
SEEN WITH Eleron-3SV air vehicle (2 per complex); Ground control station (1 per complex; controls 2 UAVs at once); T23P bungee catapult or T27P pneumatic catapult; 15Ts6M Typhoon-M anti-sabotage vehicle	

Takhion Тахюх «Тахюх»

AIR-FW

Small electric reconnaissance and fire-correction UAV In service 2015

OPERATED BY combined arms army [O 2022-11 M] seen with 49th Combined Arms Army, Southern Military District

Unidentified unit, UAV unit (Ukrainian state media; delivery date not given) [O 2022-05 L] seen with Russian military base in South Ossetia, 58th CAA, Southern MD

Unidentified unit, Anti-underwater-saboteur unit, trials (mil.ru 2016 via Wikipedia) [O 2016 L] seen with Northern Fleet anti-saboteur unit

Unidentified unit, reconnaissance unit of a motor rifle formation [R 2020-12 M] seen with Motor rifle formation (soyedineniye), 41st Combined Arms Army, Central MD, Altai Krai (number not given)

INDICATES A Takhion points to a Russian ground-force reconnaissance subunit or a UAV unit, possibly spotting for artillery. Reported holders are a motor rifle formation, military bases in Armenia and South Ossetia, a combined arms army and Navy units.

RECOGNISE **Wings** Flying wing, span about 2 m, with winglets at the wingtips.

Engines One electric motor with a two-bladed propeller; sources disagree whether it is at the nose (Military Factory) or a pusher (Focus 2022).

Fuselage Flying wing layout; removable payload modules (TV, infrared or photo camera).

Tail Tailless flying wing; winglets at the wingtips.

Launch Catapult launch from ground station; lands by parachute.

SEEN WITH Takhion UAV (2 per set); Ground control station; Launch catapult; Interchangeable payload modules (TV, IR, photo)

VARIANTS Fuel-cell modification INA.



Nickel nitride, CCO

MOBILITY Catapult launch, parachute recovery; runs from a ground control station. Operates in wind gusts to 15 m/s and -30 to +40 C.

Weight, take-off 25 kg Wikipedia lineage (WRU, GSEC, RUARM copy one table)
Wingspan 2,000 mm TASS: about 2 m
Length 610 mm Wikipedia lineage only
Radius of action 40 km claimed, kalashnikov.club: within direct radio line of sight, claim

Range, radius of action 40 km claimed, Radius of action; kalashnikov.club: within direct radio line of sight, claim

Design employment distance from line of contact 5 to 40 km Design figure, not observed: 'designed to operate at a distance of 5 to 40 km from the line of contact'

Endurance 2 h claimed, claim
Speed, maximum 120 km/h Wikipedia table; cruise 65 km/h

SOURCES AR54 ARMYINF7 FOCUS8 GS32 IIS25 KALCLUB MF2 RIASV RWW TASS20 VRF12 WP-RU121 ZVEZDA not in WEG

Zastava Застава «Застава»

MP

Short-range reconnaissance and fire-adjustment mini-UAV In service 2012

OPERATED BY motor rifle battalion, several complexes

[R 2014-01 L] seen with separate motor rifle battalion, Central Military District, Orenburg Oblast

At least 12 Russian Army units, not named, INA; held by at least 12 units per InformNapalm (via Ukrinform, Ukrainian state media)

[R 2020 L]

Unidentified unit, present in Ukraine (CNA, March 2023); no unit named

[R 2023-03 L]

INDICATES RWW lists Zastava among Ground Forces UAVs fielded as artillery spotters; CNA lists it as a 10 km ISR UAV seen in Ukraine. Seeing one suggests short-range reconnaissance or artillery fire adjustment.

RECOGNISE **Wings** Fixed-wing aircraft, wingspan 2.2 m. Wing position and planform: INA, no text source read.

Engines One electric motor on a swappable battery. Propeller position: INA, no text source read.

Fuselage Length 0.8 m, height 0.24 m. Rubber (spring) catapult launch; parachute recovery with a 180 degree flip before landing (Bird-Eye 400 figures).

Tail INA. No text source read describes the tail.



Mil.ru, CC BY 4.0

MOBILITY Packs into two backpacks for a two-person crew; 2012 trials found it unsuited to cold climates; operating range -20 to +49 C. Carried by truck or car.

Weight, maximum take-off 5.5 kg
Speed, maximum 100 km/h relative to air
Time into action 20 min
Time out of action 10 min
Length 0.8 m
Wingspan 2.2 m
Height 0.24 m

Range of action 10 km up to, Flight range from the ground station; ROBOT 'up to 10 km', CNA 'range', WPRU (Bird-Eye 400) 'radius'

Payload 1.2 kg, TV and IR
Crew 3 servicing crew
Launch and recovery Catapult launch, parachute recovery
Positioning GPS

SOURCES AVIA CNA23 IIS25 ROBOT2 RWW UASV2 UKRIN2 WP-RU140 not in WEG

Merlin-VR Мерлин-ВР

AIR-FW

Long-endurance reconnaissance and artillery-spotting UAV In service 2021

INDICATES An experimental long-endurance reconnaissance UAV, rarely reported. IISS lists it among Russian Army ISR UAVs; no source names the holding unit type, so a sighting does not by itself identify an echelon.

RECOGNISE Wings Fixed-wing aeroplane type. Span about 5 m (4.9 m reported); large for a Russian tactical UAV.

Engines Piston engine with starter-generator reported from wreckage; announced as a hybrid with electric mode. Propeller layout INA.

Fuselage Length 2.02 m reported. Electro-optical payload module with day and thermal camera. Catapult launch, parachute recovery. **Tail** INA. No text source describes the tail layout.

NOT TO BE CONFUSED WITH Orlan-10 p. 78. Smaller: span about 3 m against about 5 m for Merlin-VR (single low-reliability comparison table).



Mil.ru, CC BY 4.0

MOBILITY Catapult launch with rubber cord needs a prepared site; recovery by parachute with air cushion.

Weight, payload 6.5 kg maximum, Maximum payload; claim as repeated by Ukrainian state media, claim

Flight range 600 km claimed, Kyiv Post, basis and source not given

Length 2.02 m

Wingspan about 5 m 4.9 m in PLANDIY, 5 m in W1RU25

Detection range, optical payload 30 km claimed, 90x optical, 200x digital zoom; Russian serviceman's statement; day and thermal camera

Datalink range 150 km claimed, Flight distance without loss of link; Russian serviceman's statement

Payload sensor Electro-optical and thermal camera, 90x optical zoom Russian online press (www1.ru)

Crew INA

SOURCES ARMY22 DEPOST IISS25 KYIVP PLANDIY W1RU25 W1RU26 not in WEG

Orion «Орион»

AIR-FW

Medium-altitude long-endurance reconnaissance and strike UAV In service 2019

OPERATED BY Reconnaissance assets of the reconnaissance-fire system, medium-range reconnaissance UAV among reconnaissance assets of the reconnaissance-fire system (ATP fig. 5-8, OCR garbled)

D 2024-02 L

aviation regiment, Naval Aviation UAV regiment (also holds Forpost)

R 2024 M

Aerospace Forces (VKS), Aerospace Forces (VKS) UAV holdings; IISS lists 'some Inokhodets' **R 2024 M**

INDICATES A large V-tailed pusher UAV indicates a Navy or Aerospace Forces MALE UAV unit (IISS). Export Orion-E radius is 250 km, 300 km with relay; domestic Inokhodets also has satcom. It can find and strike single vehicles.

RECOGNISE Wings Long, straight, mid-mounted wing of high aspect ratio with slight taper; span about 16 m.

Engines Single piston engine (115 hp) in the rear fuselage driving a two-blade pusher propeller.

Fuselage Long slender fuselage, asymmetric cross-section; EO/IR turret in forward payload bay; three pylons: one centreline, one under each wing.

Tail V-tail.

Details MoD video in 2022 showed a launch container under the fuselage; RT's experts took it for the Kornet-derived Kh-BPLA missile.

NOT TO BE CONFUSED WITH Bayraktar TB2 Smaller: wingspan 12 m against 16.3 m for Orion-E, max take-off weight 650 kg against 1,000 kg.

Sirius (Inokhodets-RU) Twin engines and much larger span (20 m in ru-Wikipedia, 30 m in en-Wikipedia); Orion has one pusher engine and about 16 m span.

SOURCES AMR ATP D242 DPOST EDR GS22 IISS25 KP3 MEZHA MIL ORYX1 RT4 RUSIPL TWZ10 TWZ11 UPDIU WP-EN54 WP-RU82 not in WEG



Boevaya mashina, CC BY-SA 3.0

MOBILITY Conventional wheeled take-off and landing from an airfield, with automatic take-off and landing equipment; operating radius limited by line-of-sight datalink unless relay or satcom is fitted.

Max take-off weight 1,000 kg Orion-E export, developer figure, claim

Length 8 m Orion-E, claim

Wingspan 16.3 m Orion-E, claim

Height 3.2 m Orion-E, claim

Range, Kh-BPLA 10 km maximum, air-launched Kornet derivative; ground or low slow air target; TWZ gives 6.2 miles from RIA source, claim

Payload 200 kg Orion-E, maximum payload; normal 60 kg, claim

Hardpoints 3 (1 centreline, 1 under each wing) Orion-E, claim

Weapon loads 1x100 kg bomb, or 6x 20 kg bombs (KAB-20 per EDR), or 3x 50 kg guided bombs, or 4 ATGMs, or 2x 50 kg missiles Orion-E brochure, claim

Forpost «Фопост»

AIR-FW

Medium-range reconnaissance UAV; Forpost-R adds light guided bombs and missiles In service 2013

OPERATED BY Higher-echelon or Aerospace Forces reconnaissance assets, higher-echelon or Aerospace Forces reconnaissance assets (ATP fig. 5-8) [D 2024-02 L]

Aerospace Forces UAV units, Aerospace Forces UAV units

[R 2024-11 H]

unmanned systems unit, UAV detachment of a naval aviation base, 6 UAV (Kamchatka, 2014) [R 2014-01 M]

unmanned systems unit, fleet UAV subunit [R 2026-05 M]

INDICATES IISS lists Forpost only with Aerospace Forces and Naval Aviation, not with Army brigade UAV units. A Forpost overhead points to a fleet or Aerospace Forces UAV unit working for a higher echelon, possibly with light guided bombs.

RECOGNISE Wings Straight high-mounted wing with unswept tips, set toward the rear of the fuselage; span 8.55 m. Forpost-R has two hardpoints for light bombs.

Engines Single piston engine in the fuselage pod driving a pusher propeller; Searcher II has a three-blade pusher.

Fuselage Rectangular-section fuselage pod, 5.85 m long, carrying sensors and engine.

Tail Twin tail booms, each with a vertical fin, joined by a horizontal stabiliser mounted on top.

NOT TO BE CONFUSED WITH IAI Searcher Mk II Same outline: MoD says Forpost-R kept only the external outline. No source gives an external difference.

Orlan-10 p. 78 Far smaller: 3.1 m wingspan against 8.55 m for Forpost. IAI Searcher (Mk I) Originally a two-blade pusher propeller; Searcher II has a three-blade propeller.

SOURCES ARMINF ATP BCAT1 CALIB D242 EAN EDR IFAX IISS25 IZV2 JANES MAILRU MIL18 MK MPER ORYX1 RT2 RWW TASSNN VIKNA WP-EN28 WP-RU55 WWW1A2 WWW1B2 not in WEG



Mike1979 Russia, CC BY-SA 4.0

MOBILITY Needs a runway for take-off and recovery.

Maximum take-off weight, 500 kg claim

Forpost-R

Maximum speed, Forpost-R 200 km/h claim

Operating range 250 km range of employment, directional antenna; 150 km with omnidirectional antenna, claim 8.55 m claim

Wingspan

Range, KAB-20 8 km maximum stand-off, released from high altitude; KAB-20L laser or KAB-20S satellite guidance

UAVs per complex 3 UAV and 1 ground control station claim

Payload, Forpost-R 100 kg claim

Sensors Day TV and thermal camera; side-looking radar or ELINT payload as option claim

Korsar «Корсар»

AIR-FW

Medium reconnaissance UAV, fixed wing; strike and EW payloads planned In service 2011

OPERATED BY Aerospace Forces, Aerospace Forces holding; unit not stated [R 2024 L]

unmanned systems unit, platoon in brigade or division UAS company (CNA 2019 conjecture) [R 2019 L]

INDICATES IISS lists Korsar only among Aerospace Forces ISR UAVs (late 2024); holder unit unknown. CNA (2019) expected the Army might field it in platoons of UAS companies.

RECOGNISE Wings or rotors Fixed wing, wingspan 6.5 m (manufacturer figure).

Engines Single piston engine driving a pusher propeller.

Fuselage Length 4.2 m; conventional layout (normal aerodynamic scheme). Other fuselage detail INA.

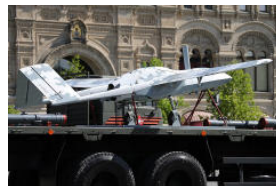
Tail Twin-boom tail.

NOT TO BE CONFUSED WITH Forpost p. 83 Larger: 8.55 m span against Korsar's 6.5 m; fin on each tail boom, joined by a top-mounted tailplane. Both twin-boom pushers in IISS Aerospace Forces list.

SEEN WITH Ground control post and several UAVs per system; UAV transport truck, shown with ATGM containers at 2018 parade

VARIANTS Korsar-M INA. Seaborne variant for the Navy (Navy Recognition, Oct 2019, via English Wikipedia); service status unknown.

Korsar (export) INA. Export figures, shown 2019: 100 to 180 km/h, range up to 100 km, ceiling 100 to 4,000 m, endurance 7.5 h. Not the domestic figures.



MIL.ru, CC BY 4.0

MOBILITY Range 120 km from control post (manufacturer claim); 250 km upgrade planned.

Maximum take-off weight 200 kg WPRU gives MTOW; ARMREC and ARMADA say only weight 200 kg, claim also 400 kg WP-EN42

Maximum speed 150 km/h claim

Flight range, radius from control post 120 km claimed, terrain reconnaissance mission, claim also up to 200 km WP-RU68

Length 4.2 m claim

Wingspan 6.5 m claim

Height 1.2 m claim

Cruise speed 120 km/h claim

Planned armament ATGM Ataka or rocket grenades claimed, not confirmed in service, claim

Engine power 50 hp piston engine; 70 hp planned, claim

Crew

INA

SOURCES AR29 ARMADA2 ARMINF CNA HOTNEWS IISS25 ORYX1 WP-EN42 WP-RU68 not in WEG

Mohajer-6 Moxagkep-6

AIR-FW

Iranian-built medium reconnaissance and light strike UAV used by Russia In service 2018

OPERATED BY Aerospace Forces, Aerospace Forces (IISS list, p.38); subunit INA [R 2024-12 M]

Unidentified unit, One reported on Dzhankoy airfield, Crimea, 2026-03 (Ukrainian SSU claim); holding unit INA [R 2026-03 L]

Russian Navy, Reported delivered to the Russian Navy (Guardian via Iran International); operator INA [R 2023-02 L]

NUMBERS Holdings 6 reportedly delivered to the Russian Navy (Guardian via Iran International) (2023-02)

INDICATES A Mohajer-6 points to a runway-based Aerospace Forces ISR or light strike drone; reported use is mainly over southern and coastal Ukraine and Crimea. Holding unit is unknown.

RECOGNISE **Wings** Straight wings mounted high and to the rear of the body, uncanted wingtips. Hardpoints under each wing: two on variant A, four on variant B.

Engines One piston engine driving a three-bladed pusher propeller (twin-boom pusher layout); air intakes on top and bottom of the engine.

Fuselage Rectangular fuselage with upward-sloping nose. Chin gimbal for sensors, fixed forward camera, fixed tricycle landing gear.

Tail Twin tailbooms with a top-mounted horizontal stabilizer. Tail numbers of Russian-held examples seen with ER- prefix (ER-858).

NOT TO BE CONFUSED WITH **Shahed 129** Larger Iranian UAV; Wikipedia calls it the larger type that the Mohajer-6 complements. **Bayraktar TB2** TB2 has an inverted V-tail on its booms and a two-blade pusher propeller; Mohajer-6 has a flat top-mounted tailplane and three blades. Confused in 2024.



Hamid Vakil / Fars News Agency, CC BY 4.0

MOBILITY Runway takeoff and landing. Reported to be launched and recovered from runways.

Max takeoff weight	600 kg maximum, Open-data figure, Iranian-built type also 600 to 670 kg WP-EN48
Maximum speed	200 km/h
Flight range	2,000 km claimed, "Operational range (radius of action)" 2,000 to 2,400 km per Wikipedia, from Iranian press, claim also 200 km MIL35 MIL37 IRINTL
Length	5.67 m Iran Ministry of Defense sheet, as cited by Wikipedia, claim also 7.5 m WP-EN48
Wingspan	10 m Iran MoD sheet per Wikipedia; Army Recognition gives the same, claim
Endurance	12 h maximum
Maximum altitude	5,400 m "Open data" figure in Ukrainian press also 16,000 to 18,000 ft WP-EN48

Payload	100 kg maximum also 100 to 150 kg WP-EN48
Data link range, line of sight	200 km claimed, Ground control station data link, direct line of sight, claim also 500 km WP-EN48
Range, Qaem-series guided bomb	12 to 40 km claimed, Glide release from 1,219 to 3,048 m; series-wide figure stated by Iranian minister Hatami, 2019, claim also 12 to 20 km AR36
Release altitude, Qaem-series guided bomb	1,219 to 3,048 m claimed, Release height 4,000 to 10,000 ft, same Iranian statement, claim also 5 km or higher AR36
Payload munitions	up to 4 maximum, Guided bombs or missiles
Crew	2 operators per ground control station
EO turret, Oghab-4	36x optical zoom, laser rangefinder, thermal camera Oghab-4H or 4D; which fit Russian examples carry is INA, claim

SEEN WITH Ground control station (INA (2 operators per station)); Qaem-series guided bombs (2 to 4 per aircraft); Almas anti-tank missile (INA (four hardpoints))

VARIANTS **Mohajer-6 A** Two hardpoints, one under each wing. Carries two munitions. Which variant Russia uses is INA.

Mohajer-6 B Four hardpoints, two under each wing. Carries four munitions.

Lancet Изделие 51 «Ланцет»

AIR-FW

Loitering munition for precision strikes on artillery, air defence and armour In service 2019

OPERATED BY high-power artillery brigade, unmanned systems troops Lancet reconnaissance-strike crew (O 2026-06 L) seen with 45th

Svir High-Power Artillery Brigade (Zapad group)

combined arms army, 56th separate special-purpose battalion

(O 2024 L) seen with 56th separate special-purpose battalion, 51st Guards

Combined Arms Army

air assault division (O 2023-07 L) seen with 76th Guards Air Assault Division

airborne division (O 2023-07 L) seen with 98th Guards Airborne Division

Special Operations Forces detachments (first users), Special

Operations Forces detachments (first users) (O 2023-07 L) seen with

Senezh SOF detachment, Kubinka-2 SOF detachment

army corps, Kaskad unit (O 2023-07 L) seen with Kaskad unit, 1st Army

Corps

INDICATES A Lancet strike usually means a crew working with a scout UAV (ZALA or Orlan-10) within tens of km. A main counter-battery tool; reported in artillery brigade and regiment recce-strike subunits (2024) and unmanned systems troops (2026).

RECOGNISE Wings Two X-shaped wing sets in tandem. Izdeliye 52: both sets of equal span. Izdeliye 51: larger-span front X-wing with a smaller rear set.

Engine Electric motor driving a pusher propeller in the tail; near silent in flight.

Fuselage Slender, rocket-like composite body about 1.65 m long; optical sensor in the nose, warhead and battery mid-body.

Launch Launched from a pneumatic catapult (domestic crews and Lancet-E); the control station looks like a case. Izdeliye 53 fires from a four-round container.

Complex Usually works with a ZALA scout UAV (Z-16 or 421-16E2) circling above the target to find it and record the strike.

NOT TO BE CONFUSED WITH KUB-BLA p. 86 Shorter and wider:

0.95 m long with 1.21 m span; the Lancet was developed from it.

Scalpel Vostok design bureau munition, also X-winged; HUR calls it the Lancet's younger brother. No reliable visual discriminator found.

Geran-2 (Shahed-136) p. 87 Wing about 3 by 4 m, used against targets deep in the rear; the Lancet is a small X-winged battlefield munition.

SEEN WITH ZALA Z-16 reconnaissance UAV; ZALA 421-16E2

reconnaissance UAV; Orlan-10 reconnaissance UAV; Pneumatic catapult launcher; Ground control station

VARIANTS Izdeliye 52 Two X-shaped wing sets of equal span. Izdeliye 51

Larger-span front X-wing, smaller rear wing set. Izdeliye 53 Four large front

wings at 45 degrees, folding; launched from a four-round container, no

catapult. Lancet-1 Smaller airframe of the same tandem X-wing layout.

Lancet-E Export set. 51-E has a larger-span front X-wing; 52-E has two X-wings of equal span; pneumatic launcher.



Nickel nitride, COO

MOBILITY Catapult-launched by a small team; launch team of two plus two operators. Flies 80-110 km/h on battery power; usually needs a scout UAV for targets.

Weight, maximum take-off, 12 kg Lancet-3 (Izdeliye 52 class)

Lancet-3

Speed, level flight

80-110 km/h speed range, Lancet-3 and Lancet-1; developer figure per UAHB

appendix; GS gives Lancet-3 cruise 80,

maximum 110, claim

300 km/h Lancet-3

under 1 min Izdeliye 51E from KPU 51E

single-use launcher, export, one-person

crew, claim

1.65 m Lancet-3, approximate

Speed, terminal dive

Time into action

Length

Range, Lancet-3 (Izdeliye 52)

40 km flight range with 3 kg warhead

Range, Izdeliye 51

50 km communication range, control link range, 5 kg warhead; ZALA website figure reported by RT, claim

Range, Lancet

about 70 km practical, MiG-29 struck at Dolgintsevo airfield 2023-09, shown on video (KYIV); about 70 km from the front per Russian claims reported by Wikipedia, observed

Warhead, Izdeliye 52

3 kg HE-frag or shaped charge (RT); Izvestia describes one 3 kg head combining both effects; wreckage showed a KZ-6 shaped charge (MILN)

Warhead, Izdeliye 51

5 kg 5 kg per RT (Izdeliye 51); RUSI gives 5 kg shaped charge for Lancet-3M; RG gives 3 or 5 kg payloads of HE-frag, shaped-charge or thermobaric type

Guidance

coordinate, electro-optical and combined TV channel to operator; terminal homing on designated object (RUSIMP); thermal imager on newer lots (RT)

Crew

4 launch team of two (munition and pneumatic catapult) and two operators (scout UAV and strike), one Lancet team, 2024, claim

Fuze

contact or non-contact contact or non-contact (UAHB table, 2023); non-contact detonation reported on newer drones (MILN, 2023)

Strike record claim

about 2,000 verified strikes, Jan to early Jul 2024 Kalashnikov statistics: 309 guns and mortars, 280 SP guns, over 200 tanks, 85 SAM systems, 51 radars, claim

SOURCES ATP ED2R EDRL GS18 GS19 HUR IZV3 KYIV2 MIL31 ORYX1 RAMB RG3 RT3 RUSI5 RUSI6 TASS7 UAHB VM2 WP-EN44 WP-EN45 WWW17 not in WEG

KUB-BLA КУБ-БЛА «КУБ»

AIR-FW

Catapult-launched electric loitering munition, 3 kg warhead, tactical strike In service 2021

INDICATES A KUB strike or wreck suggests a reconnaissance UAV is working with the launch team, within about 25 km [export KUB-E radius]; one trade report says KUB is mostly used by special forces and VDV units.

RECOGNISE Wings Wide triangular (delta) wing, span 1.21 m; overall 1.21 x 0.95 x 0.165 m.

Engines One electric motor driving a pusher propeller at the tail; battery at rear; claimed near-silent.

Fuselage Very flat airframe, 165 mm high, apparently fibreglass. Original version thought to carry no camera; a 2024 version has an onboard camera.

Launch Catapult-launched from ground (or boat); munitions carried in large shock-resistant cases.

NOT TO BE CONFUSED WITH Lancet-3 [Izdeliye 51/52] p. 85 Lancet has two X-shaped (cruciform) wing sets, front and rear; KUB has one flat delta wing.

KUB-2 KUB-2 has a square-section body, straight trapezoidal low wing and high tailplane, like Switchblade; KUB-BLA is a flat delta.

SEEN WITH Ground control station (1 per complex); Unified launch equipment (catapult) (1 per complex); KUB guided carrier with HE-frag warhead and battery (25 per complex); Battery charging station (1 per complex); ZALA 421-16E2 or other reconnaissance UAV

VARIANTS KUB-2 Square-section body, straight trapezoidal wing, high tailplane; nose optical-electronic sensor (KUB-2-E export description). KUB-10E INA.



Mztourist, CC BY-SA 4.0

MOBILITY Munition and launcher carried in cases; catapult launch. Carrier vehicle not stated in sources.

Weight, maximum take-off	8 kg
Speed	80–130 km/h claim
Time into action	20 min complex deployment, claim
Time out of action	10 min or less maximum, complex strike-down after use, claim

Range, KUB HE-frag warhead	25 km radius, radius of action, KUB-E export figure (Kalashnikov via TASS), claim
Altitude, KUB HE-frag warhead	150–1,000 m operating, flight altitude band, KUB-E export figure (Kalashnikov via TASS), claim
Altitude, KUB-10E	100–2,500 m operating, flight altitude band; export designation, claim
Warhead mass	3 kg original HE-frag warhead; heavier 'increased-power' warhead delivered from Dec 2023, mass not stated, claim

SOURCES AR31 EDR23 EDR25 EDRL F45 IIS25 NAKED ORYX3 RAMBLER RG242 SNOB TASS6 WP-EN44 WP-RU70 WWW1A3 WWW1B3 WWW1C WWW1D WWW1I not in WEG

Italmas Италмас

AIR-FW

One-way attack drone (loitering munition), piston engine In service 2023

OPERATED BY Unidentified unit, used against Ukraine, reported Oct 2023 [R 2023-10 L]

INDICATES Reported only as a drone used against Ukraine; no source names the operating unit, so sighting does not indicate a particular unit type.

RECOGNISE Wings Flying-wing layout like the Geran-2; span 2.5 m as claimed. Fuel reportedly carried in the wings.

Engines Piston engine in the nose with tractor propeller; an off-the-shelf model-aircraft engine (DLE-60 per Wikipedia). Ukrainians call it 'lawnmower' from its sound.

Fuselage Plywood construction, 'assembled from sticks'; a plastic bottle reported as fuel tank; length 2.5 m as claimed. Captured examples carried a radar reflector.

NOT TO BE CONFUSED WITH Geran-2 (Shahed 136) p. 87 Same flying-wing concept but larger; Italmas is described as linearly smaller and lighter, with a lighter payload.

MOBILITY Launched from a special catapult (2025 footage, Russian press). Road transport INA.

Flight range	200 km claimed, claimed, over 200 km per Rossiya 1 TV; Ukrainian sources call the claim possible propaganda, claim
Length	2.5 m as claimed in a Russian news report, claim
Wingspan	2.5 m as claimed in a Russian news report, claim
Launch method	catapult seen in footage before a combat launch, 2025-02

Range, one-way attack	200 km claimed, claimed (Rossiya 1 TV: over 200 km); warhead not specified, claim
Warhead mass	50 kg claimed, as claimed in a Russian news report (2025-01), claim
Warhead type	Shaped-charge (cumulative) warhead claimed, claimed against armoured vehicles incl. HIMARS; artillery and personnel also named as targets; Rossiya 1 TV, claim

SOURCES AR28 UNIAN2 WP-EN39 WWW2 WWW3 WWWJ5 not in WEG

Geran-2 Герань-2 «Герань»

AIR-FW

Long-range one-way attack drone (Iranian Shahed design), used against fixed deep targets In service 2022

OPERATED BY unmanned systems unit, strike crews; regiment has one UAV strike battalion, 2 Geran-2 crews per drone regiment

O 2025-05 H seen with 7th Separate Regiment of Unmanned Systems Troops unmanned systems unit, launch teams, combat groups

O 2026-09 M seen with GROM Kaskad Separate Unmanned Aviation Brigade (Aerospace Forces)

unmanned systems unit, strike UAV detachment, Aerospace Forces, 5 detachments (2025) **R 2025-03 M** seen with detachments at Millerovo, Primorsko-Akhtarsk, Shatalovo, Kursk and Oktyabrskoye airfields

INDICATES Reported Geran operators are Aerospace Forces strike UAV detachments at airfields and the GROM Kaskad brigade at droneports; a drone regiment holds only two Geran-2 crews. A launch site points to these strike units.

RECOGNISE **Wings** Cropped delta wing blended into the fuselage, span about 2.5 m (Geran-2). Vertical fins at both wingtips; no horizontal tailplane.

Engines Geran-1/-2: piston engine in the rear fuselage driving a two-bladed pusher propeller; loud moped-like sound heard kilometres away. Geran-3: turbojet, tail reshaped.

Fuselage Tube-like fuselage with short nose cone, about 3.3–3.5 m long; warhead in nose. Some Russian and Iranian examples painted matte black.

Tail Tailless: wingtip fins serve as rudders. Geran-2 fins extend above and below the wing; Geran-1 fins only above. Geran-3 tail reshaped for jet.

NOT TO BE CONFUSED WITH Gerbera p. 88 Cheap decoy and recce copy: smaller (about 2 m long), plywood frame under foam skin; same cropped-delta pusher layout.

Parodiya Shahed-mimicking decoy named with Gerbera by Ukrainian intelligence (Nov 2024); no visual discriminator found.

Geran-5 Jet with cylindrical body up to 6 m long and 5.5 m span: cruise-missile shape, not a delta flying wing.

Lancet p. 85 Much smaller tactical loitering munition (12 kg) with two sets of X-shaped wings, electric motor.

SEEN WITH Truck-mounted launch container (rocket booster) (up to 5 drones per container [Iranian-shown launcher]); Gerbera decoy and reconnaissance drone

VARIANTS Geran-1 Smaller: about 2.6 m long, 2.2 m span; wingtip fins point upward only. Range 700–1,000 km, warhead 10–20 kg, 135 kg launch weight. Geran-2 Baseline: 3.3–3.5 m long, 2.5 m span, piston pusher. Later series add cameras, antennas, black paint. 50 kg (BCh-50) or 90 kg (BCh-90) warhead; jam-resistant satnav; some series camera-guided via mesh modem or Starlink. Geran-2 E series R-60 missile on a rail above the forward fuselage, nose camera and second camera behind rail; or a Verba MANPADS launcher on the back (WPEN, early 2026). Operator-cued launch of R-60 against helicopters and aircraft hunting drones (state claim, Ukraine, Dec 2025). Geran-2 b series Camera on a gyrostabilised gimbal under the forward fuselage, not in the nose cone. Onboard camera; reported in use from September 2025. Geran-2 MS series INA: infrared camera fitted; no text source read describes its position. Infrared camera, Nvidia Jetson computer, video modem; scouts routes for mobile air defence before strikes (Ukrainian intelligence, Jun 2025, via WPEN).



S. Hossein Alaei / Fars News, CC BY 4.0

MOBILITY Rail-launched with rocket booster from trucks or fixed sites; pre-programmed one-way flight, no recovery.

Launch weight, Geran-2	200 kg
Launch weight, Geran-1	135 kg
Launch weight, Geran-3	250–370 kg
Speed, Geran-3	Shahed 238 figure 300–370 km/h maximum, telemetry T80 turbojet; maximum reached in defended zones and terminal dive, claim
Flight range, Geran-2	1,350 km reported, standard 50 kg warhead; figure from leaked Alabuga (manufacturer) documents, claim
Flight range, Geran-2 with BCh-90	650 km reported, 90 kg warhead, smaller fuel tank; leaked Alabuga documents, claim
Flight range, Geran-1	700–1,000 km estimated, estimate; DIA estimate 900 km, estimated
Flight range, Geran-3	up to 1,000 km operational, U-series, telemetry T80 turbojet; Ukrainian intelligence, claim
Length, Geran-2	3.3–3.5 m

Range, Geran-2	1,350 km reported, 50 kg warhead; leaked Alabuga documents; see mobility data for alternatives, claim
Flight altitude, Geran-2	60–4,000 m operating, Ukrainian General Staff handbook, claim
Flight altitude, Geran-2 observed	700–2,000 m en route; down to 200 m near target practical, Ukrainian General Staff handbook, observed 2022–23; as low as 20 m along rivers and roads, claim
Ceiling, Geran-3	up to 7 km maximum
Flight time to target, observed	15–30 min average practical, Ukrainian General Staff handbook, Shahed-136 use 2022–23, claim
Warhead, Geran-2	40–50 kg (BCh-50) HE-fragmentation; leaked Alabuga slide; Ukrainian General Staff handbook (state claim) gives 40 kg, claim
Warhead, Geran-2 heavy	90 kg (BCh-90) reduces range; combat-tested Nov 2023 per leaked Alabuga documents; reported fielded autumn 2025, claim
Warhead, Geran-1	10–20 kg
Warhead, Geran-3	50 kg expert view; some Russian variants reported at 90 kg
Guidance	Inertial with GNSS (GPS/GLONASS) correction jam-resistant Kometa CRPA antenna, 4 to 16 elements by 2025; some series add camera and mesh, LTE or Starlink link

SOURCES AINF2 CSIS1 CSIS2 IIS25 ISIS JAMES LBUA MIL20 MIL21 MIL22 PROF3 RUSI3 UAHB WP-EN30 WP-EN31 WP-RU56 not in WEG

Gerbera «Гербера»

AIR-FW

Cheap foam one-way drone: Shahed decoy, reconnaissance, small-warhead strike, relay In service 2024

OPERATED BY Unidentified unit, launch team with car; car struck on the ground, Dec 2025 R 2025-12 L

INDICATES Gerberas are flown with Shahed/Geran-2 in mass raids to draw fire and locate air defences. A vehicle carrying a ready Gerbera marks a launch point.

RECOGNISE **Wings** Tailless delta with double-swept (cranked) leading edge; elevons on trailing edge. Span about 2 to 2.5 m (sources differ); a little smaller than Shahed-136.

Engines One small piston engine at the rear driving a two-blade fixed-pitch pusher propeller.

Fuselage Foam body over plywood frame, about 2 m long; white or black. Camera in nose on recon and strike versions.

Tail Single central vertical fin just ahead of the propeller; no fins on the wingtips.

NOT TO BE CONFUSED WITH Geran-2 (Shahed-136) p. 87

Geran-2 has stabilisers at the wingtips; Gerbera has one central fin just ahead of the pusher propeller.

Parodiya Parodiya is a high-wing monoplane, much smaller than a Shahed; Gerbera is a tailless delta.

SEEN WITH Geran-2 (Shahed-136); Launch vehicle (car); FPV drone carried as payload

VARIANTS **Gerbera (decoy)** No warhead; outwardly the basic airframe. **Gerbera (reconnaissance)** Gimbal camera (Topotek KHY10S90) in nose; 4G or mesh-network modem and antennas. **Gerbera (strike)** Warhead in the nose; camera for operator terminal guidance on some.



dpsu.gov.ua (State Border Guard Service of Ukraine), CC BY 4.0

MOBILITY Carried by car to a launch point; catapult launch, no runway (reported). Slow (max about 160 km/h); range shrinks with a warhead.

Weight	about 18 kg
Speed, maximum	dry (CAPSS); TSN gives no condition
Flight range	160 km/h
	600 km maximum, without payload; shorter with warhead; some have a second fuel tank in the nose for extended range
Wingspan	2.5 m

Operating altitude	up to 1 km maximum, Espresso report via Kyiv Post, 2025-02
Warhead	up to 5 kg strike version; 2.5 kg HE-frag seen in nose 2025-02, observed
Guidance	pre-programmed route (presumed); CRPA anti-jam satnav; operator camera guidance on strike version
Datalink	4G modem with SIM; mesh-network modem (Xingkai XK-F358) on some observed

SOURCES ARMYINF2 CAPSS FAKTY ISIS2 KP24 KP25 KP26 MIL23 MILUA3 TSN TWZ5 WP-EN31 WP-RU57 not in WEG

Molniya «Молния»

AIR-FW

Cheap fixed-wing FPV one-way attack drone; also recon, fire spotting, FPV carrier, courier In service 2024

INDICATES Points to a drone crew of an unmanned systems, airborne or Rubikon unit within the reported 30–60 km range of the target, more if Starlink-fitted. Crews move launch point after each launch.

RECOGNISE **Wings** Wing about 1.5 m span, built of plywood; airframe also uses foam and plastic.

Engines Electric. Molniya-1: one tractor motor in the nose. Molniya-2: two motors under the wings. Export Lightning 13 has four.

Fuselage Fuselage of two longitudinal aluminium tubes with a plywood centre section; electronics and batteries inside; camera in the nose.

Tail Tail section can be taken apart for transport (MoD).

NOT TO BE CONFUSED WITH Lancet loitering munition p. 85

Lancet has two X-shaped wing sets on a slender body. Molniya is visibly improvised: plywood, foam and exposed aluminium tubes; catapult or hand launch.

Ukrainian copies of Molniya Ukraine builds copies from captured examples (Oct 2024); no visual difference reported.

SEEN WITH Pneumatic catapult with compressor; FPV goggles and hand controller; FPV quadcopter carried as payload (1 to 2 per Molniya (carrier role)); Starlink terminal (on some airframes)

VARIANTS **Molniya-1** Single electric motor in the nose. **Molniya-2** Two electric motors under the wings. **Molniya-2R** INA.



Russian Ministry of Defence (mil.ru), CC BY 4.0

MOBILITY Man-portable; breaks down into about 10 pieces carried by three soldiers in backpacks. Launched by pneumatic catapult or by hand; one-way, not recovered.

Weight	10 kg
Speed	Variant not stated
	200 km/h Molniya-2, serviceman in MoD video, claim
Wingspan	1.5 m
Endurance	Variant not stated
	up to 40 min maximum, Variant not stated

Range, Molniya-2 with warhead	up to 50 km maximum, Molniya-2; MoD figure, control link not stated, claim
Range, Molniya-1 with warhead	about 40 km maximum, Kharkiv regional prosecutor's office; manual FPV control, no GPS, claim
Warhead mass, Molniya-2	up to 6 kg explosive claim
Warhead types	shaped-charge, fragmentation, thermobaric; also improvised: grenades, RPG rounds, anti-tank mines, thermite

SOURCES BI FN GWARA ISW2 K14 KP2 MIL31 MIL38 MIL39 MIL40 MODMR U243 VIKNA2 W1A W1B W1C W1D WP-EN50 not in WEG

FPV drones

AIR-RW

Pilot-flown one-way attack quadcopters, radio- or fibre-optic-controlled In service 2023

OPERATED BY Rubicon detachment, FPV drone operators

O 2025-12 M seen with Rubicon Center for Advanced Unmanned Technologies
motor rifle brigade, FPV drone crews **O 2026-03 L** seen with 4th Motor Rifle Brigade (Southern grouping)

artillery brigade, FPV drone crews; a subunit of the anti-tank battalion works with UAVs **O 2026-05 L** seen with 238th Artillery Brigade, 8th Guards Combined Arms Army

Unidentified unit, FPV drone operators **O 2025-12 L** seen with 58th Separate Spetsnaz Battalion (Okhotnik detachment), 51st CAA

Unidentified unit **O 2025-05 L** seen with Sudny Den

unmanned systems unit, FPV crews of the unmanned systems troops
R 2026-02 M

INDICATES An FPV strike means a drone crew in flight range: 7-10 km claimed for the radio VT-40, 10 to over 40 km for fibre-optic types.

Russian reports name crews in motor rifle and artillery brigades, a Spetsnaz battalion and drone units; the type fixes no echelon.

RECOGNISE Rotors Quadcopter on a carbon fibre, aluminium or polymer frame; fibre-optic types need a larger frame, motors and battery. A Ukrainian commander reports them at 20-50 m up.

Frame Carbon fibre, aluminium or polymer frame with electronics in the middle and the battery strapped on top. Captured VT-40 had the frame braced with extra carbon rods.

Payload Munition strapped under the frame: RPG-7 grenades, KZ-6 shaped charges, or improvised charges such as a plastic tube of explosive wrapped in steel balls.

Link Radio types carry standard analogue or digital antennas. Fibre-optic types carry a large cylindrical spool between frame and munition and trail reflective fibre.

NOT TO BE CONFUSED WITH Ukrainian FPV drones Same layout, largely the same parts as a captured VT-40; fibre-optic types are used by both sides. The VT-40 differed by a stamped serial number and carbon-rod bracing.

Bomber drones Hover over the target and drop a heavier payload, mostly at night; an FPV dives into the target.

Reconnaissance quadcopters Quadcopter spotter drones cue FPV strikes; an attack FPV carries its munition strapped under the frame.



National Police of Ukraine, CC BY 4.0

MOBILITY Fibre-optic types are slower and less agile: the spool adds weight and limits sharp turns; the fibre can snap or be cut.

Flight altitude, fibre-optic types **20-50 m** practical, reported by a Ukrainian drone commander, observed
 Weight **INA**

Range, KVN fibre-optic	20 km maximum, fibre-optic spool; figure 'according to Russian resources', Jan 2025, claim also 50-65 km MIL19
Range, Provod fibre-optic	over 40 km claimed, spool length; Ovod centre statement via TASS, Aug 2026, claim also 25-30 km W1PROV
Range, VT-40	7-10 km claimed, munitions of 1.3-3 kg, claim also 7-8 km TEXTY
Range, fibre-optic types	over 10 km practical, Russian fibre-optic drones striking Kostiantynivka, just over 10 km from the contact line, Apr 2025, observed
Payload, KVN fibre-optic	3.5 kg according to Russian resources, Jan 2025, claim
Payload, VT-40	1.3-3 kg claim also about 2 kg TEXTY
Payload, Ovod	up to 4 kg claim
Radio links, VT-40	video 1.3 GHz; control 800-900 MHz captured example, Sep 2023, observed
Fibre break rate, KVN	1-2 per 10 launches maker's figure from nearly 1,000 uses, quoted Jan 2025, claim
Crew	INA

SEEN WITH Fibre-optic spool (1 per fibre-optic drone); Reconnaissance UAV crews; Signal booster (relay) drones

VARIANTS KVN Knyaz Vandal Novgorodsky Fibre-optic spool under the frame; resembles the Chinese Skywalker commercial fibre-optic quadcopter. Fibre-optic link, immune to jamming. Russian sources claim 20 km, then 50-65 km with new spools; contact fuze. Thermal imager version in series from Jan 2025. Provod Fibre-optic spool; spool length differs between versions. Maker claims use at over 40 km (Aug 2026); the designer gave 25-30 km with up to 4.5 kg in Feb 2026. Boomerang Carries its fibre-optic spool on the frame, like KVN. Fibre-optic; a Russian officer says it proved itself at the line alongside KVN (2025).

Piranha-5 Pocket size: 12 cm diameter, 0.3 kg. Fibre-optic; Russian claim of up to 1 kg payload, 220 km/h and 1-3 km radius; mainly for operator training. Ovod INA. Radio-controlled; developer claims lift of up to 4 kg and Ploshchad onboard homing after loss of link.

SOURCES AIF2 AR24 AR25 CYCLO2 ESP ISW1 IZ IZ2 KI2 KP LENTA MIL19 NGE PRIME RG RIA RUSI2 TASS3 TEXTY W1KVN W1PROV W1RU2 WP-EN29 not in WEG

Table 1. Master list of designations

Every system, vehicle, component and munition in this guide's lexicon, 875 in all, sorted by designation (digits sort as numbers). NATO reporting names and codes and US designations are set in **BLUE CAPITALS**, for example **STONE**, **SS-26** or **M1974**. NATO designations and reporting names can also be looked up in Table 1b. **(UA)** marks a type reported in Russian use in Ukraine since 2014. A **bold** page number leads to a full entry; a *grey italic* one to the entry of the system it belongs to. Look up other names and Cyrillic forms in Tables 1b and 1c.

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
1B44		1B44	meteorological station - of 9K58 Smerch	28
1B75 1B75E	Penicillin	«Пенициллин»	acoustic-thermal locating system (UA)	43
1K113 3F5, 3VF4	Smelchak	1K113 «Смелчак», 3Ф5	guided mortar bomb - of 2S4 Tyulpan (UA)	31
1K148	Yastreb-AV	1K148 «Ястреб-АВ»	counter-battery radar (UA)	
1L13-3 1L13	Nebo-SV BOX SPRING	1Л13, «Небо-СВ»	air surveillance radar	
1L119	Nebo-SVU	«Небо-СВУ» 1Л119	air surveillance radar (UA)	
1L122	Garmon, Garmony	1Л122 «Гармонь»	portable air surveillance radar (UA)	
1L125	Niobium-SV		multifunction radar (UA)	41
1L219 Zoopark-1	Zoopark-1	1Л219 «Зоопарк-1», 1Л219М	counter-battery radar (UA)	42
1L222	Avtobaza-M	«Автомобаз»	ELINT system	
1L259		1Л259	counter-battery radar (UA)	42
1L260 Zoopark-1M	Zoopark-1M	1Л260, «Зоопарк-1М»	counter-battery radar (UA)	42
1L261			counter-battery radar (UA)	42
1L262E SPR-2M	Rtut-BM, Mercury-BM	1Л262З «Ртуть-БМ»	proximity-fuze jammer (UA)	
1L265		1Л265	ELINT system	47
1L266		1Л266	EW command post	47
1L267	Moskva-1	«Москва-1»	EW command post (UA)	47
1L269 Krasukha-20	Krasukha-2	1Л269 «Красуха-2»	radar jammer (UA)	46
1L271	Aistenok, Aistyonok	1Л271 «Аистенок»	portable counter-battery radar (UA)	43
1L277	Sobolyatnik		portable ground surveillance radar (UA)	
1RL144M 1RL144M1, 1RL144, 1RL138	HOT SHOT	1РЛ144М1	fire-control radar (UA)	40
1RL239 ARK-1, ARK-1M	Rys SMALL FRED (disputed)	1РЛ239 / АРК-1 / «Рысь»	counter-battery radar (UA)	43
1RL257	Krasukha-4, Krasukha-4S	1РЛ257 «Красуха-4»	radar jammer (UA)	46
1T12-2M		1Т12-2М	topographic survey vehicle - of 9K58 Smerch	28
1V13 1V13M, M1974/1		1В13	artillery battery fire control centre (UA)	67
1V14 1V15, 1V15M, 1V16			artillery command vehicle (UA)	67
1V18 1V19, M1979-2B	Klyon-1, Klen-1, Klen-2		artillery command vehicle (UA)	
1V110			artillery command vehicle - of 9K51 Grad <i>legacy</i> (UA)	26
1V119	Spektr	1В119	artillery command vehicle (UA)	
1V119 1V152, 1V1003, M1979-2B	Klyon-1, Klen-1	1В18, 1В19	artillery battery and battalion command, observation and fir	67
1V126 1V126M, 1V152, 1V153, 1V152M	Kapustnik-B, Kapustnik-BM	1В126 «Капустник-Б»	artillery fire control system - of 9K59 Prima	
1V152		1В152	artillery command vehicle	
1V1003			artillery command vehicle (UA)	
2A36 M1976	Giatsint-B	2А36 «Гиацинт-Б»	152 mm towed gun (UA)	32
2A45M 2A45MR	Sprut-B		125 mm towed anti-tank gun <i>legacy</i>	
2A65 M1987	Msta-B	2А65 «Мста-Б»	152 mm towed howitzer (UA)	32
2B9 2B9M	Vasilek	2В9 «Василёк»	82 mm automatic gun-mortar (UA)	
2B11		2В11	120 mm mortar - of 2S12 Sani (UA)	
2B14	Podnos	2В14 «Поднос»	82 mm mortar	
2B16	Nona-K	2В16 «Нона-К»	120 mm towed gun-mortar (UA)	
2B17M	Tornado-G	2В17М	122 mm multiple rocket launcher (UA)	26
2B26	Grad-K	2В26	122 mm multiple rocket launcher - of 9K51 Grad (UA)	26
2F77 2F77M		2Ф77, 2Ф77М	transloader - of 2K22 Tunguska (UA)	40
2K22 2K22M, 2K22M1, SA-19	Tunguska, Tunguska-M1 GRISON	2К22, 2К22М «Тунгуска-М»	gun-missile air defence system (UA)	40
2S1 M1974	Gvozdika	2С1 «Гвоздика»	122 mm self-propelled howitzer (UA)	
2S3 Object 303, M1973	Akatsiya	2С3 «Акация»	152 mm self-propelled howitzer (UA)	
2S3M 2S3M1, 2S3M2	Akatsiya	2С3М	152 mm self-propelled howitzer - of 2S3 Akatsiya (UA)	
2S4 Object 305, M1975	Tyulpan	2С4 «Тюльпан»	240 mm self-propelled mortar (UA)	31

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
2S5 Object 307, M1981	Giatsint-S	2С5 «Гиацинт-С»	152 mm self-propelled gun UA	29
2S5M		2С5М	152 mm self-propelled gun - of 2S5 Giatsint-S <i>legacy</i>	29
2S6 2S6M, 2S6M1		2С6, 2С6М	gun-missile air defence vehicle UA	40
2S7 Object 216, M1975	Pion	2С7 «Пион»	203 mm self-propelled gun UA	31
2S7M	Malka	2С7М «Малка»	203 mm self-propelled gun UA	31
2S9 2S9-1	Nona-S	2С9 «Нона-С»	120 mm self-propelled gun-mortar UA	
2S12 2S12A	Sani	2С12 «Саня», 2С12А	120 mm mortar UA	
2S19 Object 316, M1990	Msta-S FARM	2С19 «Мста-С»	152 mm self-propelled howitzer UA	30
2S19M1	Msta-S	2С19М1	152 mm self-propelled howitzer UA	30
2S19M2 2S33	Msta-SM	2С19М2, 2С33 «Мста-СМ»	152 mm self-propelled howitzer UA	30
2S23	Nona-SVK	2С23 «Нона-СВК»	120 mm self-propelled gun-mortar UA	
2S25	Sprut-SD	2С25	125 mm self-propelled anti-tank gun	
2S31	Vena	2С31 «Вена»	120 mm self-propelled gun-mortar UA	
2S34 2SIM	Khosta, Chosta	2С34 «Хоста»	120 mm self-propelled gun-mortar UA	
2S35	Koalitsiya-SV	2С35 «Коалиция-СВ»	152 mm self-propelled howitzer UA	30
2S38 ZAK-57	Derivatsiya-PVO	2С38, ЗАК-57	57 mm self-propelled anti-aircraft gun <i>trial</i> UA	
2S40	Floks	2С40, «Флокс»	120 mm self-propelled gun-mortar UA	
2S43	Malva	2С43 «Мальва»	152 mm wheeled self-propelled howitzer UA	30
3BK13		3БК13	artillery round - of 2S1 Gvozdika <i>legacy</i>	
3BK19		3БК19	artillery round	
3F2 3VF2		3Ф2	mortar bomb - of 2S4 Tyulpan	31
3K55 K-300P, K300P, SSC-5	Bastion, Bastion-P STOOGEE	3К55, Бастуион	coastal missile system UA	
3K60 SSC-6	Bal, Bal-E SENNIGHT	3К60, Бал	coastal missile system UA	
3M14 3M14T, 3M14K, SS-N-30A, SS-N-27	Kalibr SAGARIS	Калибр, 3М-14	cruise missile UA	
3M16		3М16	122 mm cluster rocket - of 9K51 Grad <i>legacy</i>	26
3-O-8 3VO11		3-О-8	mortar bomb - of 2S4 Tyulpan UA	31
3-O-13 3VO13, 3VO14		3-О-13	artillery round UA	
3-O-23 3VO28, 3VO29, 3VO30		3-О-23	artillery round	
3OF24		3ОФ24	artillery round - of 2S1 Gvozdika	
3OF25 3VOF32, 3VOF33		3ОФ25	artillery round <i>legacy</i>	
3OF29 3VOF39, 3VOF40		3ОФ29	artillery round - of 2S5 Giatsint-S	29
3OF38 3VOF63, 3VOF66	Santimetr	3ОФ38	guided artillery round	
3OF39 3VOF64, 3VOF93, 2K25	Krasnopol	3ОФ39, «Краснополь»	guided artillery round UA	
3OF39M 3VOF64M, 2K25M	Krasnopol-M1	3ОФ39М	guided artillery round - of 3OF39 Krasnopol UA	
3OF43 3VOF34, 3VOF42		3ОФ43	artillery round - of 2S7 Pion	31
3OF44 3VOF35		3ОФ44	artillery round - of 2S7 Pion	31
3OF45 3OF45-1, 3VOF58, 3VOF72, 3VOF73		3ОФ45	artillery round - of 2S19 Msta-S	30
3OF49 3VOF49		3ОФ49	artillery round	
3OF50		3ОФ50	artillery round	
3OF56		3ОФ56	artillery round - of 2S1 Gvozdika	
3OF59 3VOF86, 3VOF87		3ОФ59	artillery round - of 2S5 Giatsint-S	29
3OF61 3VOF91		3ОФ61	artillery round	
3OF64 3VOF96, 3VOF97, 3VOF98		3ОФ64	artillery round	
3OF69 Kitolov-2	Kitolov-2	«Кумолов-2»	guided mortar bomb UA	
3OF69M Kitolov-2M	Kitolov-2M	«Кумолов-2М»	guided mortar bomb - of 3OF69 Kitolov-2 UA	
3OF75	Santimetr-M	3ОФ75	guided artillery round	
3RB30 3VRB36, 3VRB37, 3VRB38		3РБ30	communications jammer shell	
3S55 K-340P		К-340П	missile transporter erector launcher - of 3K55 Bastion UA	
5N63S 3ON6	FLAP LID B	5Н63С	fire-control radar - of S-300PS S-300PS <i>legacy</i> UA	
5N84A Oborona-14	Oborona-14 TALL KING C	5Н84А «Оборона»	early warning radar UA	
5P85SD 5P85SM, 5P85S, 5P85D		5П85С, 5П85Д	SAM launcher - of S-300PS S-300PS <i>legacy</i> UA	
5P85SM2-01		5П85СМ2-01	SAM launcher UA	35
5P85T2		5П85Т2	SAM launcher UA	35
5V55R 5V55K, 5V55RM, V-500R		5В55Р, 5В55К	surface-to-air missile - of S-300PS S-300PS <i>legacy</i> UA	

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
6P50 6P49, 6P57, 6P58, 6P59	Kord	Kopp	12.7 mm heavy machine gun UA	
9A33B 9A33BM3		9A335	SAM launcher - of 9K33 Osa UA	40
9A35		9A35	SAM launcher - of 9K35 Strela-10 UA	40
9A39 9A39M1		9A39, 9A39M1	SAM loader-launcher - of 9K37 Buk stored UA	37
9A39M1-2			SAM loader-launcher - of 9K37M1-2 Buk-M1-2 UA	37
9A51	Prima	9A51	122 mm multiple rocket launcher - of 9K59 Prima legacy	
9A52 BM-30	Smerch	9A52	300 mm multiple rocket launcher legacy UA	28
9A52-2 BM-30	Smerch-M	9A52-2	300 mm multiple rocket launcher UA	28
9A52-2T		9A52-2T	300 mm multiple rocket launcher - of 9K58 Smerch export	28
9A52-4	Kama	9A52-4	300 mm multiple rocket launcher trial	28
9A53	Uragan-1M	9A53	220/300 mm multiple rocket launcher UA	27
9A54 9A53-S	Tornado-S	9A54	300 mm multiple rocket launcher UA	28
9A82		9A82	SAM launcher - of S-300V S-300V	36
9A83 9A83-1, 9A83M2		9A83	SAM launcher - of S-300V S-300V UA	36
9A83M2			SAM launcher - of S-300V4 S-300V4 UA	36
9A84		9A84	SAM loader-launcher - of S-300V S-300V UA	36
9A84-2			SAM loader-launcher - of S-300V4 S-300V4 UA	36
9A85 9A85-2		9A85	SAM loader-launcher - of S-300V S-300V UA	36
9A85-2			transloader - of S-300V4 S-300V4 UA	36
9A310 9A310M1, 9A38		9A310, 9A38	SAM launcher - of 9K37 Buk stored UA	37
9A310M1-2		9A310M1-2	SAM launcher - of 9K37M1-2 Buk-M1-2 UA	37
9A316		9A316	SAM loader-launcher - of 9K317 Buk-M2 UA	37
9A317		9A317	SAM launcher - of 9K317 Buk-M2 UA	37
9A317M		9A317M	SAM launcher - of 9K317M Buk-M3 UA	37
9A330		9A330	SAM launcher legacy UA	38
9A331 9A331-1		9A331, 9A331-1	SAM launcher UA	38
9A331M 9A331M2		9A331M2	SAM launcher UA	38
9A4172 9K121, AT-16	Vikhr SCALLION		air-launched anti-tank guided missile UA	
9K11 9M14, AT-3	Malyutka, Malyutka-2 SAGGER	9K11, 9M14	anti-tank guided missile system legacy	
9K32 9K32M, SA-7	Strela-2 GRAIL	9K32, «Стрела-2»	MANPADS legacy	
9K33 9K33M2, 9K33M3, SA-8	Osa, Osa-AK, Osa-AKM GECKO	9K33, 9K33M2	short-range SAM system UA	40
9K34 SA-14	Strela-3 GREMLIN	9K34, «Стрела-3»	MANPADS	
9K35 9K35M3, SA-13	Strela-10, Strela-10M3 GOPHER	9K35 «Стрела-10», 9K35M3 «Стрела-10M3»	short-range SAM system UA	40
9K37 9K37M1, SA-11	Buk, Buk-M1 GADFLY	9K37, 9K37M1 «Бук-M1»	medium-range SAM system stored UA	37
9K37M1-2 SA-11	Buk-M1-2 GADFLY	9K37M1-2 «Бук-M1-2»	medium-range SAM system UA	37
9K38 SA-18	Igla GROUSE	9K38, «Игла»	MANPADS UA	
9K51 M-21, M1964	Grad	9K51 «Град»	122 mm multiple rocket launcher system UA	26
9K51M	Tornado-G	9K51M «Торнадо-Г»	122 mm multiple rocket launcher system UA	26
9K51M 2B17M, 9A53-G	Tornado-G	9K51M, Торнадо-Г	122 mm 40-tube truck mri; automated grad upgrade for brigade	26
9K52 FROG-7	Luna-M	9K52, Луна-M	unguided rocket system legacy	
9K54 BM-21V, M-21V, M1975	Grad-V	9K54 «Град-В»	122 mm multiple rocket launcher system legacy	26
9K55 BM-21b, M1976	Grad-1	9K55 «Град-1»	122 mm multiple rocket launcher system stored UA	26
9K55-1	Grad-1	9K55-1 «Град-1»	122 mm multiple rocket launcher system legacy UA	26
9K57 BM-27, M1977	Uragan	9K57 «Ураган»	220 mm multiple rocket launcher system UA	27
9K57-1	Uragan-1		220 mm multiple rocket launcher legacy UA	27
9K58 BM-30	Smerch	9K58 «Смерч»	300 mm multiple rocket launcher system UA	28
9K59	Prima	9K59 «Прима»	122 mm multiple rocket launcher system legacy	
9K72 8K14, R-17, SS-1c, R-300	Elbrus SCUD B	9K72, P-17	ballistic missile system stored	

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
9K79 9K79-1, OTR-21, SS-21	Tochka, Tochka-U SCARAB	9K79, Точка-У	ballistic missile system (UA)	
9K111 9K111M, 9P135, AT-4	Fagot SPIGOT	9K111, Фазот	anti-tank guided missile system (UA)	
9K111-1 9K111-1M, 9K113, 9P135M, AT-5	Konkurs, Konkurs-M SPANDREL	9K111-1, Конкурс	anti-tank guided missile system (UA)	
9K112 9M112, AT-8	Kobra SONGSTER		gun-launched anti-tank missile system <i>legacy</i>	
9K114 AT-6	Shturm SPIRAL	9K114, Штурм	anti-tank guided missile system (UA)	
9K115 9P151, AT-7	Metis SAXHORN	9K115, Метис	anti-tank guided missile system (UA)	
9K115-1 AT-13	Metis-M SAXHORN-2	9K115-1, Метис-М	anti-tank guided missile system	
9K115-2 AT-13	Metis-M1 SAXHORN-2	9K115-2, Метис-М1	anti-tank guided missile system · of 9K115-1 Metis-M	
9K116 9K116-1, 9K116-2, 9K116-3, 3UBK10, AT-12	Kastet, Bastion, Sheksna SWINGER	9K116, Кастет	gun-launched anti-tank missile system	
9K119 9K119M, 9K120, AT-11	Svir, Refleks, Refleks-M SNIPER	9K119M, Рефлекс-М	gun-launched anti-tank missile system (UA)	
9K123 9K123-1, AT-15	Khrizantema, Khrizantema-S SPRINGER	9K123, Хризантема	anti-tank guided missile system (UA)	
9K132 9P149M, AT-9	Shturm-SM SPIRAL-2	9K132, 9П149М	anti-tank guided missile system · of 9K114 Shturm (UA)	
9K135 9K129, AT-14	Kornet, Kornet-E, Kornet-EM SPRIGGAN	9K135, Корнет	anti-tank guided missile system (UA)	
9K310 SA-16	Igla-1 GIMLET	9K310, «Игла-1»	MANPADS	
9K317 SA-17	Buk-M2 GRIZZLY	9K317 «Бук-М2»	medium-range SAM system (UA)	37
9K317M SA-27	Buk-M3 GOLLUM	9K317M «Бук-М3»	medium-range SAM system (UA)	37
9K330 SA-15	Tor GAUNTLET	9K330	short-range SAM system <i>legacy</i> (UA)	38
9K331 9K331M, SA-15	Tor-M1 GAUNTLET	9K331 «Тор-М1»	short-range SAM system (UA)	38
9K331MDT SA-15	Tor-M2DT GAUNTLET	«Тор-М2ДТ»	short-range SAM system (UA)	38
9K331MU SA-15	Tor-M1-2U, Tor-M2U GAUNTLET	«Тор-М1-2У»	short-range SAM system (UA)	38
9K332 SA-15	Tor-M2 GAUNTLET	9K332 3PK «Тор-М2»	short-range SAM system (UA)	38
9K333 SA-29	Verba GIZMO	9K333, «Верб»	MANPADS (UA)	
9K334	Gibka-S	«Гибка-С»	MANPADS launcher vehicle (UA)	
9K338 SA-24	Igla-S GRINCH	9K338, «Игла-С»	MANPADS	
9K512	Uragan-1M	9K512 «Ураган-1М»	220/300 mm multiple rocket launcher system (UA)	27
9K515	Tornado-S	9K515	300 mm multiple rocket launcher system (UA)	28
9K720 SS-26, SSC-7, SSC-8	Iskander-M, Iskander, Iskander-K STONE, SOUTHPAW	9K720, Искангеп	ballistic missile system (UA)	33
9M22		9M22	122 mm HE-fragmentation rocket · of 9K51 Grad <i>legacy</i>	26
9M22M	Malysh	9M22M	122 mm rocket · of 9P132 Grad-P <i>legacy</i>	
9M22S		9M22C	122 mm incendiary rocket · of 9K51 Grad <i>legacy</i> (UA)	26
9M22U 9M22U-1, M-210F		9M22Y	122 mm HE-fragmentation rocket · of 9K51 Grad (UA)	26
9M27D		9M27A	220 mm leaflet rocket · of 9K57 Uragan <i>legacy</i>	27
9M27F		9M27D	220 mm HE rocket · of 9K57 Uragan	27
9M27K		9M27K	220 mm cluster rocket · of 9K57 Uragan (UA)	27
9M27K1		9M27K1	220 mm cluster rocket · of 9K57 Uragan (UA)	27
9M27K2		9M27K2	220 mm scatterable-mine rocket · of 9K57 Uragan	27
9M27K3		9M27K3	220 mm scatterable-mine rocket · of 9K57 Uragan	27
9M27S		9M27C	220 mm incendiary rocket · of 9K57 Uragan <i>legacy</i>	27
9M28D		9M28A	122 mm leaflet rocket · of 9K51 Grad <i>legacy</i>	26

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
9M28F		9M28D	122 mm HE-fragmentation rocket - of 9K51 Grad legacy UA	26
9M28K		9M28K	122 mm cluster rocket - of 9K51 Grad legacy	26
9M28S		9M28C	122 mm incendiary rocket - of 9K51 Grad legacy UA	26
9M33 9M33M2, 9M33M3, 9M33M		9M33, 9M33M3	surface-to-air missile	40
9M37 9M37M, 9M37MD		9M37	surface-to-air missile UA	40
9M38		9M38	surface-to-air missile - of 9K37 Buk UA	37
9M39		9M39	surface-to-air missile - of 9K38 Igla UA	
9M42		9M42	122 mm illumination rocket - of 9K51 Grad legacy	26
9M43		9M43	122 mm cluster rocket - of 9K51 Grad legacy	26
9M51		9M51	220 mm thermobaric rocket - of 9K57 Uragan legacy	27
9M53F		9M53D	122 mm HE rocket - of 9K59 Prima legacy	
9M53K		9M53K	122 mm cluster rocket - of 9K59 Prima legacy	
9M53S		9M53C	122 mm incendiary rocket - of 9K59 Prima legacy	
9M55F		9M55D	300 mm HE-fragmentation rocket - of 9K58 Smerch	28
9M55K		9M55K	300 mm cluster rocket - of 9K58 Smerch UA	28
9M55K1		9M55K1	300 mm cluster rocket - of 9K58 Smerch UA	28
9M55K4		9M55K4	300 mm scatterable-mine rocket - of 9K58 Smerch	28
9M55K5		9M55K5	300 mm cluster rocket - of 9K58 Smerch	28
9M55S		9M55C	300 mm thermobaric rocket - of 9K58 Smerch	28
9M59		9M59	220 mm scatterable-mine rocket - of 9K57 Uragan	27
9M79 9M79-1, SS-21	SCARAB	9M79	ballistic missile - of 9K79 Tochka UA	
9M82 9M82M, 9M82MD, SA-12B	GIANT	9M82, 9M82M	surface-to-air missile - of S-300V S-300V UA	36
9M83 9M83M, SA-12A	GLADIATOR	9M83	surface-to-air missile - of S-300V S-300V UA	36
9M96 9M96E, 9M96E2, 9M96M		9M96E2, 9M96E	surface-to-air missile - of S-400 S-400 UA	35
9M100		9M100	surface-to-air missile - of S-350 S-350	36
9M111 9M111-2, 9M111M, AT-4	Fagot, Faktoriya SPIGOT	9M111	anti-tank guided missile - of 9K111 Fagot UA	
9M113 AT-5	Konkurs SPANDREL	9M113	anti-tank guided missile - of 9K111-1 Konkurs UA	
9M113M AT-5B	Konkurs-M SPANDREL-B	9M113M	anti-tank guided missile - of 9K111-1 Konkurs	
9M114 AT-6	Kokon SPIRAL	9M114	anti-tank guided missile - of 9K114 Shturm UA	
9M115 AT-7	Metis SAXHORN	9M115	anti-tank guided missile - of 9K115 Metis UA	
9M117 9M117M, 9M117M1, AT-10	Bastion, Kan, Arkan STABBER	9M117	gun-launched anti-tank guided missile - of 9K116 Kastet UA	
9M119 9M119M, 9M119M1, AT-11B, AT-11C, AT-11	Svir, Refleks, Invar SNIPER	9M119M, 9M119M1	gun-launched anti-tank guided missile - of 9K119 Svir UA	
9M120 AT-9	Ataka SPIRAL-2	9M120	anti-tank guided missile - of 9K132 Shturm-SM	
9M120-1	Ataka-1	9M120-1	anti-tank guided missile - of 9K132 Shturm-SM	
9M123 9M123F, AT-15	Khrizantema SPRINGER	9M123, 9M123D	anti-tank guided missile - of 9K123 Khrizantema UA	
9M131 9M131M, AT-13	Metis-M SAXHORN-2	9M131	anti-tank guided missile - of 9K115-1 Metis-M	
9M133 9M133F-1, AT-14	Kornet SPRIGGAN	9M133	anti-tank guided missile - of 9K135 Kornet UA	
9M217		9M217	122 mm cluster rocket - of 9K51M Tornado-G	26
9M218 9M541		9M218	122 mm cluster rocket - of 9K51M Tornado-G	26
9M311 9M311-1M		9M311	surface-to-air missile UA	40
9M313		9M313	surface-to-air missile - of 9K310 Igla-1	
9M317		9M317	surface-to-air missile - of 9K317 Buk-M2 UA	37
9M317M			surface-to-air missile - of 9K317M Buk-M3 UA	37
9M330 9M330-2		9M330	surface-to-air missile legacy UA	38
9M331		9M331	surface-to-air missile UA	38
9M333		9M333	surface-to-air missile UA	40
9M335		9M335	surface-to-air missile - of 9GK6 Pantsir-S1 UA	39
9M336		9M336	surface-to-air missile - of 9K333 Verba UA	
9M338 9M338K		9M338	surface-to-air missile UA	38

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
9M340		9M340	surface-to-air missile · of 9M340 Sosna <i>trial</i>	
9M340	Sosna, Sosna-R	«Сосна»	short-range SAM system <i>trial</i> (UA)	
9M342		9M342	surface-to-air missile · of 9K338 Igla-S	
9M519		9M519	122 mm radio jammer rocket · of 9K51 Grad <i>legacy</i>	26
9M521		9M521	122 mm extended-range HE rocket · of 9K51M Tornado-G	26
9M522		9M522	122 mm extended-range HE rocket	
9M525		9M525	300 mm cluster rocket · of 9K58 Smerch <i>trial</i>	28
9M528		9M528	300 mm HE-fragmentation rocket · of 9K58 Smerch	28
9M529		9M529	300 mm thermobaric rocket · of 9K58 Smerch <i>trial</i>	28
9M531		9M531	300 mm cluster rocket · of 9K58 Smerch <i>trial</i>	28
9M533		9M533	300 mm cluster rocket · of 9K58 Smerch <i>trial</i>	28
9M534		9M534	300 mm rocket with reconnaissance UAV · of 9K58 Smerch <i>trial</i>	28
9M542		9M542	300 mm guided rocket · of 9K515 Tornado-S <i>trial</i>	28
9M544		9M544	300 mm guided cluster rocket · of 9K515 Tornado-S (UA)	28
9M549		9M549	300 mm guided cluster rocket · of 9K515 Tornado-S (UA)	28
9M723 9M723K1, SS-26	STONE	9M723	ballistic missile (UA)	33
9M728 R-500, SSC-7	SOUTHPAW	9M728, P-500	cruise missile (UA)	33
9M729 SSC-8	SCREWDRIIVER	9M729	cruise missile (UA)	33
9P78-1 9P78		9П78-1	missile transporter erector launcher (UA)	33
9P122	Malyutka-M	9П122	ATGM launcher vehicle · of 9K11 Malyutka <i>legacy</i>	
9P125 BM-21V, M1975		9П125, 6M-21B	122 mm multiple rocket launcher · of 9K54 Grad-V <i>legacy</i>	26
9P129 9P129-1M		9П129	missile transporter erector launcher · of 9K79 Tochka (UA)	
9P132 DKZ-66, BM-21P, Grad-1P, 9K510	Grad-P, Partizan	9П132	man-portable 122 mm rocket launcher · of 9K51 Grad <i>legacy</i> (UA)	26
9P133	Malyutka-P	9П133	ATGM launcher vehicle · of 9K11 Malyutka <i>legacy</i>	
9P137			ATGM launcher vehicle · of 9K111-1 Konkurs <i>legacy</i>	
9P138 BM-21b, M1976	Grad-1	9П138	122 mm multiple rocket launcher <i>stored</i> (UA)	26
9P139		9П139	122 mm multiple rocket launcher · of 9K55-1 Grad-1 <i>legacy</i> (UA)	26
9P140 BM-27, BM-22, M1977	Uragan	9П140	220 mm multiple rocket launcher (UA)	27
9P148	Konkurs	9П148	ATGM launcher vehicle · of 9K111-1 Konkurs <i>legacy</i> (UA)	
9P149	Shturm-S	9П149	ATGM launcher vehicle · of 9K114 Shturm (UA)	
9P157-2 9P157	Khrizantema-S	9П157, 9П157-2	ATGM launcher vehicle · of 9K123 Khrizantema (UA)	
9P162 9K128-1, 9P163M-1	Kornet-T, Kornet-LR	9П162, 9K128-1	ATGM launcher vehicle · of 9K135 Kornet (UA)	
9P163-3 Kornet-D1	Kornet-D	9П163-3	ATGM launcher vehicle · of 9K135 Kornet	
9P163M-1	Kornet-P	9П163M-1	man-portable ATGM launcher · of 9K135 Kornet (UA)	
9P163M-2 9P163-2	Kvartet	9П163M-2	vehicle-mounted ATGM launcher module · of 9K135 Kornet	
9S15M 9S15MT2E, 9S15	Obzor-3 BILL BOARD	9C15M	SAM surveillance radar · of S-300V S-300V	36
9S18 9S18M1, 9S18M1-1, 9S18M1-2, 9S18M1-3	Kupol SNOW DRIFT	9C18, 9C18M1	SAM surveillance radar · of 9K37M1-2 Buk-M1-2 (UA)	37
9S19M2 9S19	Imbir HIGH SCREEN	9C19M2	SAM surveillance radar · of S-300V S-300V (UA)	36
9S32 9S32M1	GRILL PAN	9C32	fire-control radar · of S-300V S-300V (UA)	36
9S36 9S36M		9C36M	fire-control radar · of 9K317 Buk-M2 (UA)	37
9S52 9S52M, 9S52M1	Polyana-D4, Polyana-D4M1	Поляна-Д4, 9C52	air defence command post	68
9S80 9S80-1, PPRU-1, PPRU-1M	Sborka, Ovod		air defence command post · of 9K35 Strela-10 (UA)	40
9S457		9C457	SAM command post · of S-300V S-300V	36
9S470 9S470M1, 9S470M1-2		9C470	SAM command post · of 9K37M1-2 Buk-M1-2 (UA)	37
9S510 9S510M		9C510M	SAM command post · of 9K317 Buk-M2	37
9S552		9C552	command and staff vehicle · of 9K720 Iskander-M	33

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
9S729M1	Slepok-1	9C729M1 «Слепок-1»	artillery fire-control system - of 9K58 Smerch UA	28
9S737	Ranzhir		air defence command post	69
9S920		9C920	missile data preparation vehicle - of 9K720 Iskander-M	33
9S931 MP		9C931	planning module vehicle	69
9S931-1 MP-K		МП-K	planning module vehicle	69
9S931-2 9S931-3	Rakushka, Rakushka-M	Pakyuka, Pakyuuka-M	airborne tracked amphibious apc and carrier chassis for vdv	15
9S932-1 9S932-2, MRU-B	Barnaul-T	9C932-1	air defence command post UA	69
9S933		ПМЮ	portable fire control module - of Barnaul-T	69
9S935		KCAC	squad automation complex - of Barnaul-T UA	69
9S936			command and staff vehicle - of 9K515 Tornado-S UA	28
9T217 9T217B		9T217Б	transloader - of 9K33 Osa UA	40
9T218 9T218-1		9T218	transloader - of 9K79 Tochka	
9T232M		9T232M	transloader - of 9K59 Prima legacy	
9T234		9T234	transloader - of 9K58 Smerch legacy UA	28
9T234-2		9T234-2	transloader - of 9K58 Smerch UA	28
9T234-2T		9T234-2T	transloader on Tatra chassis - of 9K58 Smerch export	28
9T234-4		9T234-4	transloader on KamAZ chassis - of 9A52-4 Kama trial	28
9T243M			transloader - of 9K317M Buk-M3 UA	37
9T244		9T244	transloader - of 9K331 Tor-M1 UA	38
9T248			life support vehicle - of 9K720 Iskander-M UA	33
9T249		9T249	transloader - of 9K512 Uragan-1M	27
9T250 9T250E		9T250	transloader - of 9K720 Iskander-M UA	33
9T254			ammunition carrier - of 9K51 Grad legacy	26
9T255		9T255	transloader - of 9K515 Tornado-S	28
9T450		9T450	transloader - of 9K55 Grad-1 stored	26
9T451		9T451	transloader on MT-LBu chassis - of 9K55-1 Grad-1 legacy	26
9T452		9T452	transloader - of 9K57 Uragan UA	27
35N6	Kasta	35Н6 «Каста»	air surveillance radar UA	
39N6 39N6E, Kasta-2E2	Kasta-2-2 SQUAT EYE	39Н6 «Каста-2-2», 39Н6Е «Каста-2Е2»	low-altitude detection radar UA	
40N6 40N6E		40Н6Е	surface-to-air missile - of S-400 S-400 UA	35
48N6 48N6E, 48N6E2, 48N6E3, 48N6DM		48Н6Е, 48Н6Е2	surface-to-air missile - of S-400 S-400 UA	35
48Ya6-K1 Podlet-K1	Podlet-K1	48Я6-K1 «Подъём»	low-altitude detection radar UA	41
50K6A 50K6E		50К6А	SAM command post	36
50N6A 50N6E		50Н6А	fire-control radar UA	36
50P6A 50P6E		50П6А	SAM launcher UA	36
51P6		51П6	SAM launcher UA	35
55K6 55K6A		55К6, 55К6Е	SAM command post - of S-400 S-400 UA	35
55Zh6M 55Zh6ME	Nebo-M	55Ж6М «Небо-М», 55Ж6МЕ	multi-band air surveillance radar UA	41
55Zh6U	Nebo-U TALL RACK		air surveillance radar UA	
55Zh6UM	Nebo-UM		air surveillance radar	
57E6 57E6-E, 57E6M		5736Е	surface-to-air missile - of 96K6 Pantsir-S1 UA	39
59N6 59N6-E	Protivnik-G, Protivnik-GE	Противник-ГЕ	3-D air surveillance radar	
64N6 64N6E	BIG BIRD		target acquisition radar - of S-300PM S-300PM	
67N6 67N6E	Gamma-DE	Гамма-ДЕ	UHF 3-D air surveillance radar export	
77Ya6-DM	Voronezh-DM	«Воронеж-ДМ»	early warning radar UA	
85Ya6	Leer-2	85Я6 «Леер-2»	communications jammer UA	51
91N6A 91N6E		91Н6Е	air surveillance radar - of S-400 S-400 UA	35
92N6A 92N6E		92Н6Е	fire-control radar - of S-400 S-400 UA	35
96K6 SA-22	Pantsir-S1 GREYHOUND	96К6, Панцирь-С1	gun-missile air defence system UA	39
96L6 96L6E, 96L6-TsP	CHEESE BOARD	96Л6Е	air surveillance radar UA	42
AGS-17	Plamya	АГС-17, Пламя	30 mm automatic grenade launcher UA	
AGS-30		АГС-30	30 mm automatic grenade launcher	
AGS-40	Balkan	АГС-40, Балкан	40 mm automatic grenade launcher	

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
Akatsiya-E	Akatsiya-E		air defence command and control complex	
Akatsiya-M	Akatsiya-M		command and control automation system	
Altaets	Altaets, Altayets, Altaets-AM	«Алтаец»	communications jammer · of R-330M1P Diabazol	
Alurgit	Alurgit		GNSS jammer	
AMN-233114 GAZ-233114	Tigr-M	Тигр-М, AMH-233114	armoured car (UA)	51
AMN-590951	VPK Ural	AMH-590951, ВПК-Урал	armoured car (UA)	
Andromeda-D	Andromeda-D		command and control system	
APA-80			engine starting vehicle (UA)	
Argus-Antidron	Argus-Antidron		UAV jammer	
Armata	Armata	Армата	tracked armoured platform <i>trial</i> (UA)	
ARS-14 ARS-14KM		APC-14KM	decontamination vehicle (UA)	
ARS-14 ARS-14KM, ARS-15		APC-14KM, TMC-65	truck-mounted decontamination and smoke vehicles of rkhbz (n	63
AS-4350.1			ambulance (UA)	
AShT-4310			topographic command and staff vehicle (UA)	
ASN-233115	Tigr-M SpN	ACH-233115	armoured car · of AMN-233114 Tigr-M (UA)	51
Asteis-7020	Patrul	Патруль	armoured car (UA)	
ASUNO-S 1V168			artillery fire control system	
ATO-40			repair vehicle · of RB-301B Borisoglebsk-2	48
ATS-59G		ATC-59Г	tracked artillery tractor <i>legacy</i> (UA)	
Auriga-1.2V			satellite communications terminal (UA)	
AZK-7	AZK-7		acoustic artillery reconnaissance system	
B-4 M1931		Б-4	203 mm towed howitzer <i>legacy</i>	
Barnaul-T	Barnaul-T	Барнаул-Т	air defence command and control system (UA)	69
BAT-2 Object 454		БАТ-2	route-clearance vehicle (UA)	
BAT-2 Object 454	Izdeliye	БАТ-2, Изделие	armoured route-clearing tractor and dozer on mt-t chassis	58
BAT-M			route-clearance vehicle <i>legacy</i>	
BAZ-6909 BAZ-69092			heavy truck chassis (UA)	
BAZ-69092			transloader · of S-400 S-400 (UA)	35
BKAR	Orlan-10, Orlan-30	Орлан-10, Орлан-30	catapult-launched recon uav for artillery spotting and ew, o	78
BLA-07			reconnaissance UAV	
BM-1		БМ-1	220 mm heavy flamethrower launcher (UA)	29
BM-13	Katyusha	БМ-13	132 mm multiple rocket launcher <i>stored</i>	
BM-21 2B5, M1964		БМ-21, 2Б5	122 mm multiple rocket launcher (UA)	26
BM-21-1 2B17, 2B17-1		БМ-21-1, 2Б17	122 mm multiple rocket launcher (UA)	26
BM-35	Italmas	Италмас	one-way attack drone (UA)	
BM-35	Italmas, Product	Италмас	one-way attack drone (loitering munition), piston engine	86
BMD-1		БМД-1	airborne fighting vehicle <i>stored</i> (UA)	
BMD-1KSh-A			armoured command vehicle (UA)	
BMD-1KSh-A BMD-1KSh	Obj	БМД-1КШ-А, Сопка	airborne battalion chief-of-staff command and staff vehicle	66
BMD-2		БМД-2	airborne fighting vehicle (UA)	
BMD-2 BMD-1P, Object 916, M1981/1, M1970		БМД-2, БМД-1П	air-droppable amphibious tracked ifv of the vdv; bmd-1 is th	14
BMD-3 Object 950		БМД-3, Обьект 950	airborne fighting vehicle <i>legacy</i>	
BMD-4		БМД-4	airborne fighting vehicle <i>legacy</i> (UA)	
BMD-4M		БМД-4М	airborne fighting vehicle · of BMD-4 (UA)	
BMD-4M BMD-4, BMD-3M	Sadovnitsa, Kornet-D1	БМД-4М, БМД-4	air-droppable amphibious tracked ifv of the airborne forces	13
BMK-130M BMK-150			towing and motor boat (UA)	
BMK-460		BMK-460	towing and motor boat (UA)	
BMK-MO			towing and motor boat (UA)	
BMK-MT			towing and motor boat (UA)	
BMO-T Object 564		БМО-Т, Обьект 564	armoured personnel carrier (UA)	
BMP-1 Object 765, BMP Model 1970		БМП-1, Обьект 765	infantry fighting vehicle (UA)	
BMP-1 Object 765, BMP-76PB, M-1967	Model, Basurmanin	БМП-1, БМП-1П	amphibious tracked infantry fighting vehicle, 73 mm gun and	11
BMP-1AM	Basurmanin	БМП-1AM, Басурманин	infantry fighting vehicle · of BMP-1 (UA)	
BMP-1KSh BMP-1KShM			command and staff vehicle (UA)	65

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
BMP-1P		БМП-1П	infantry fighting vehicle - of BMP-1 (UA)	
BMP-1U	Shkval	БМП-1У, Шквал	infantry fighting vehicle - of BMP-1 <i>legacy</i> (UA)	
BMP-2 Object 675		БМП-2, Объект 675	infantry fighting vehicle (UA)	
BMP-2 Object 675, BMP-2M, M1981	Berezhok	БМП-2, БМП-2К	tracked amphibious ifv: carries a 7-man squad; 30 mm cannon	12
BMP-2D		БМП-2Д	infantry fighting vehicle - of BMP-2 (UA)	
BMP-2M	Berezhok	БМП-2М, Бережок	infantry fighting vehicle - of BMP-2 (UA)	
BMP-3 Object 688		БМП-3, Объект 688	infantry fighting vehicle (UA)	
BMP-3F		БМП-3Ф	infantry fighting vehicle - of BMP-3	
BMP-3M		БМП-3М	infantry fighting vehicle - of BMP-3	
BMPT	Terminator	БМПТ, Терминатор	tank support fighting vehicle (UA)	
BMPT Object 199, BMPT-72	Terminator, Spiridon	БМПТ, Терминатор	tank support vehicle: twin 30 mm guns and atgms on a tank hu	11
BMPT-72 Object 183	Terminator-2, Terminator	БМПТ-72, Терминатор-2	tank support fighting vehicle - of BMPT Terminator (UA)	
BMR-3M BMR-3MA, Object 197	Vepr	БМР-3М, Венры	mine-clearing vehicle (UA)	
BMR-3M Object 197, BMR-3MA	Vepr, Prokhod-1	БМР-3М, Венры	armoured mine-clearing vehicle on t-90 chassis with roller t	59
BPM-97 KamAZ-43269, AMN-590951	Vystrel, Spartak	БПМ-97, КамАЗ-43269	light armoured cars and mrap-type utility vehicles for secur	24
BRDM-2 GAZ-41-08		БРДМ-2, ГАЗ-41-08	armoured reconnaissance vehicle (UA)	
BRDM-2 BTR-40PB, BTR-40P-2	Bardak	БРДМ-2, БРДМ-2А	amphibious 4x4 armoured scout car for reconnaissance and pat	22
BRDM-2A		БРДМ-2А	armoured reconnaissance vehicle - of BRDM-2 (UA)	
BRDM-2M		БРДМ-2М	armoured reconnaissance vehicle - of BRDM-2 (UA)	
BRDM-2MS		БРДМ-2МС	armoured reconnaissance vehicle - of BRDM-2 (UA)	
BRDM-2RKhb BRDM-2RKh			CBRN reconnaissance vehicle (UA)	
BREM BREM-1, BREM-1M	Beglyanka	БРЭМ, БРЭМ-1	armoured recovery and repair vehicles for tank and infantry	62
BREM-1 Object 608		БРЭМ-1	armoured recovery vehicle (UA)	
BREM-2		БРЭМ-2	armoured recovery vehicle (UA)	
BREM-4	BREM-Ch		armoured recovery vehicle (UA)	
BREM-64			armoured recovery vehicle	
BREM-D	BREhM-D	БРЭМ-Д	armoured recovery vehicle (UA)	
BREM-K		БРЭМ-К	armoured recovery vehicle (UA)	
BREM-L	Beglyanka, Beglianka	БРЭМ-Л	armoured recovery vehicle (UA)	
BRM-1K BRM-1KM, Object 676, BMP M1976/2		БРМ-1К, БРМ-1	armoured reconnaissance vehicle (UA)	
BRM-1K Object 676, BRM-1, M1976/2		БРМ-1К, БРМ-1	tracked amphibious combat reconnaissance vehicle on bmp-1 ch	13
BRM-3K Object 501	Rys	БРМ-3К, Рысь	armoured reconnaissance vehicle	43
BS-3 M1944		БС-3	100 mm field gun <i>stored</i> (UA)	
BTM-3		БТМ-3	trench digger (UA)	
BTM-3 MDK-3, PZM-2		БТМ-3, МДК-3	machines that dig trenches, pits and shelters for troop and	60
BTR-50		БТР-50	tracked armoured personnel carrier <i>legacy</i> (UA)	
BTR-50PK		БТР-50ПК	tracked armoured personnel carrier - of BTR-50 <i>legacy</i> (UA)	
BTR-60PB BTR-60		БТР-60ПБ, БТР-60	armoured personnel carrier <i>legacy</i> (UA)	65
BTR-70		БТР-70	armoured personnel carrier (UA)	
BTR-70M		БТР-70М	armoured personnel carrier - of BTR-70 (UA)	
BTR-80 GAZ-5903		БТР-80, ГАЗ-5903	armoured personnel carrier (UA)	
BTR-80 BTR-82, BTR-82A, M1978 , BTR-70		БТР-80, БТР-70	wheeled 8x8 amphibious apc; btr-80a and btr-82a carry a 30 m	20
BTR-80A		БТР-80А	armoured personnel carrier - of BTR-80	
BTR-82		БТР-82	armoured personnel carrier - of BTR-80 (UA)	
BTR-82A BTR-82AM		БТР-82А, БТР-82АМ	armoured personnel carrier - of BTR-80 (UA)	
BTR-82AT		БТР-82АТ	armoured personnel carrier - of BTR-80 (UA)	
BTR-90 GAZ-5923	Rostok	БТР-90, ГАЗ-5923	armoured personnel carrier (UA)	
BTR-90 GAZ-5923	Rostok	БТР-90, Росмок	8x8 amphibious armoured personnel carrier with 30 mm turret	21
BTR-D		БТР-Д	armoured personnel carrier (UA)	

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
BTR-MDM	Rakushka	БТР-МДМ, Пакушка	armoured personnel carrier UA	
BTR-T			heavy armoured personnel carrier <i>legacy</i>	
BTS-4A		БТС-4А	armoured recovery vehicle UA	
BTS-4V		БТС-4В	armoured recovery vehicle UA	
Buran	Buran, RIDA Buran, Buran ASV		armoured utility vehicle UA	
CBRN RKhM-4, RKhM-4-01	Povozka	РХМ-4, РХМ-6	radiation, chemical and biological reconnaissance; marks con	62
Chistyulya	Chistyulya	Чистюля	UAV jammer UA	
D-1 M1943		Д-1	152 mm towed howitzer <i>stored</i> UA	
D-20		Д-20	152 mm towed gun-howitzer UA	
D-30 2A18, D-30A, 2A18M		Д-30, Д-30А	122 mm towed howitzer UA	
D-44		Д-44	85 mm towed gun <i>legacy</i> UA	
DIM			vehicle-mounted mine detector <i>legacy</i>	
DShK 38/46	DShK, DShKM	ДШК	12.7 mm heavy machine gun <i>legacy</i> UA	
DT-10PM	Vityaz		articulated tracked carrier UA	36
DT-30 DT-30PM, DT-10PM	Vityaz	ДТ-30, ДТ-30ПМ	articulated tracked carrier UA	36
Eleron-3 Eleron-3SV		Элерон-3, Элерон-3СВ	reconnaissance UAV UA	
Eleron T-16			reconnaissance UAV UA	
EOV-3521			military excavator UA	
EOV-3523			military excavator UA	
EOV-4421			military excavator UA	
ESU TZ	YeSU TZ	ЕСУ ТЗ	tactical-level command and control system	
Fateh-360	Fath-360		short-range ballistic missile	
Forpost	Forpost, Searcher II	Форпост	reconnaissance UAV UA	
Forpost	Forpost, Forpost-R	Форпост, Форпост-Р	medium-range reconnaissance uav; forpost-r adds light guided	83
Forpost-R	Forpost-R		reconnaissance and strike UAV UA	
FPV KVN, VT-40	Knyaz Vandal Novgorodsky, Provod	Князь Вандал Новгородский, Провод	pilot-flown one-way attack quadcopters, radio- or fibre-opti	89
FPV drones	FPV drones, FPV kamikaze drones	FPV-дроны	first-person-view attack drones (family) UA	
Garpun-3	Garpun-3, Harpoon-3	Гарпун-3	UAV jammer UA	
GAZ-51		ГАЗ-51	cargo truck 4x2 <i>legacy</i> UA	
GAZ-66		ГАЗ-66	cargo truck 4x4 <i>legacy</i> UA	
GAZ-2330 GAZ-233014, GAZ-233114	Tigr, Tigr-M	ГАЗ-233014, Туз	light 4x4 armoured cars for escort, reconnaissance, patrol a	22
GAZ-3307		ГАЗ-3307	cargo truck 4x2 UA	
GAZ-3308 GAZ-33081	Sadko, Eger-II	ГАЗ-3308 «Сажко»	cargo truck 4x4 UA	
GAZ-3344-20	Aleut		articulated tracked carrier UA	
GAZ-3344-20	Aleut, Plastun-SN	ГАЗ-3344-20, Алеут	two-unit unarmoured (aleut) and light-armoured open-top (pla	19
GAZ-39371	Vodnik		armoured utility vehicle UA	
GAZ-233014 GAZ-2330, VPK-233136, GAZ-233036	Tigr	ГАЗ-233014, ГАЗ-2330	armoured car UA	
GAZ-233036 SPM-2	Tigr SPM-2	ГАЗ-233036, СПМ-2	armoured car · of GAZ-233014 Tigr UA	
Geran-1 Shaded-131, Shaded 131	Geran-1	Герань-1	one-way attack drone UA	
Geran-2 Shaded-136, Shaded 136	Geran-2	Герань-2	one-way attack drone UA	
Geran-2	Geran-2, Geran-1	Герань-2, Герань	long-range one-way attack drone (iranian shaded design), use	87
Geran-3	Geran-3		one-way attack drone <i>trial</i> UA	
Gerbera	Gerbera	Гербера	one-way attack drone UA	
Gerbera	Gerbera, Gerber	Гербера	cheap foam one-way drone: shaded decoy, reconnaissance, smal	88
GMZ-3 Object 318		ГМЗ-3	minelayer UA	61
Granat-1		Гранат-1	reconnaissance UAV UA	
Granat-1	Granat-1, Granat-4	Гранат-1, Гранат-4	small reconnaissance uavs for artillery target acquisition a	79
Granat-4		Гранат-4	reconnaissance UAV UA	
Groza.04.K	Groza-04K	Гроза.04.К	UAV jammer UA	
GSP Object 55		ГСП	ferry <i>legacy</i>	
GT-01	Murmansk-BN	Мурманск-БН	HF communications jammer UA	45
Harpya	Harpya		UAV jammer	
IMR IMR-1			combat engineer vehicle <i>legacy</i> UA	58

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
IMR-2 IMR-2M, IMR-2M1, Object 637		ИМР-2, ИМР-2М	combat engineer vehicle (UA)	58
IMR-3 IMR-3M		ИМР-3М	combat engineer vehicle (UA)	58
Irbis	Irbis		counter-battery radar	
IRM	Zhuk		engineer reconnaissance vehicle (UA)	
IRM	Zhuk, Beetle	ИРМ, Жук	amphibious engineer reconnaissance of routes, water obstacle	57
ISDM	Zemledeliye	ИСДМ, Земледелие	truck-mounted remote mining system (UA)	61
Izdeliye 51 Izdeliye 52	Lancet, Lancet-3	Ланцет, Изделие 51/52	loitering munition (UA)	
Izdeliye 53			loitering munition - of Izdeliye 51 Lancet (UA)	
K1Sh1 GAZ-59032	UNSh	К1Ш1	command vehicle chassis (BTR-80)	
K155M	Krasnopol-M2		guided artillery round - of 30F39 Krasnopol (UA)	
K-53949 K-63968, K-4386	Typhoon-K, Typhoon-U	КамаЗ-63968, КамаЗ-53949	mine-protected 4x4 and 6x6 armoured trucks: troop carrier, r	23
KamAZ-4310		КАМАЗ-4310	cargo truck 6x6 <i>legacy</i>	
KamAZ-4310 KamAZ-5350, KamAZ-53501	Mustang	КАМАЗ-4310, КАМАЗ-5350	general-purpose 6x6 and 8x8 cargo and tractor truck chassis	75
KamAZ-4386	Typhoon-VDV		mine-resistant ambush-protected vehicle (UA)	
KamAZ-5350	Mustang	КАМАЗ-5350	cargo truck 6x6 (UA)	
KamAZ-6350		КАМАЗ-6350	cargo truck 8x8 (UA)	
KamAZ-6560		КАМАЗ-6560	heavy truck chassis 8x8	
KamAZ-43269 BPM-97	Vystrel	КамаЗ-43269, БПМ-97	armoured car (UA)	
KamAZ-53501		КАМАЗ-53501	cargo truck 6x6 (UA)	
KamAZ-53949 Typhoon-K 4x4	Typhoon-K	КамаЗ-53949, Тайфун-К	mine-resistant ambush-protected vehicle (UA)	
KamAZ-53949	Linza	КамаЗ-53949, Линза	armoured ambulance (UA)	
KamAZ-63501		КАМАЗ-63501	cargo truck 8x8 (UA)	
KamAZ-63968 Typhoon-K 6x6	Typhoon-K	КамаЗ-63968, Тайфун-К	mine-resistant ambush-protected vehicle (UA)	
KamAZ-65225		КАМАЗ-65225	tractor truck 6x6	
Kapustnik-BM	Kapustnik-BM	Капустник-БМ	artillery fire-control system - of 9K51M Tornado-G	26
Kh-22 AS-4	Burya KITCHEN	X-22 «Буря»	air-launched anti-ship missile (UA)	
Kh-32		X-32	air-launched cruise missile (UA)	
Kh-47M2 AS-24	Kinzhalt KILLJOY	X-47M2, Кинжал	air-launched ballistic missile (UA)	
Kh-55 Kh-55SM, AS-15	KENT	X-55	air-launched cruise missile (UA)	
Kh-59 Kh-59M, AS-13, AS-18	Ovod KINGBOLT, KAZOO	X-59 «Обод»	air-launched cruise missile (UA)	
Kh-69		X-69	air-launched cruise missile (UA)	
Kh-101 AS-23A	KODIAK	X-101	air-launched cruise missile (UA)	
KM-8	Gran	КМ-8, «Грань»	guided mortar bomb (UA)	
KMM			bridgelaye <i>legacy</i>	
KMT-5			mine plough and roller set	
KMT-7			mine roller	
KMT-8			mine plough	
KMT-10			mine roller	
KN-23 Hwasong-IIA			short-range ballistic missile (UA)	
Korsar	Korsar	Корсар	reconnaissance UAV (UA)	
Korsar	Korsar, Korsar-M	Корсар, Корсар-М	medium reconnaissance uav, fixed wing; strike and ew payload	83
KPV KPVТ		КПВ	14.5 mm heavy machine gun (UA)	
KrAZ-255 KrAZ-255B		КрАЗ-255Б	cargo truck 6x6 <i>legacy</i> (UA)	
KrAZ-260		КрАЗ-260	cargo truck 6x6 <i>legacy</i> (UA)	
Kredo-1E Credo-1E, RP-200, 1RL133-3, PSNR-8	Kredo-1, Credo-1	1РЛ133-3 «Крего-1»	ground surveillance radar	
Kredo-M1 PSNR-8M, PSNR-9	Kredo-M1		portable ground surveillance radar (UA)	
KRUS	Strelets, Strelets-M	КРУС, Стрелец	soldier-worn reconnaissance, command and communications set;	74
KRUS-VR	Strelets, Strelets-M	Стрелец, Стрелец-М	soldier command and control terminal (UA)	
KRVD			temporary road deployment kit (UA)	
KT-L			recovery vehicle (UA)	
KUB KUB-2, KUB-10E		КУБ, КУБ-2	catapult-launched electric loitering munition, 3 kg warhead,	86
KUB-BLA	Kub-BLA	Куб-БЛА	loitering munition (UA)	

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
Kurganets-25	Kurganets-25	Курганец-25	infantry fighting vehicle <i>trial</i>	
KZKT-7428	Rusich		tank transporter <i>legacy</i>	
Ladoga	Ladoga, VTS Ladoga		heavy command and reconnaissance vehicle (UA)	
Lancet	Lancet, Izdeliye	Изделие, Ланцет	loitering munition for precision strikes on artillery, air d	85
Lancet launcher	Lancet launcher		UAV control station · of Izdeliye 51 Lancet (UA)	
LPD-801			UAV jammer (UA)	
M-30 M-1938		M-30	122 mm towed howitzer <i>stored</i> (UA)	
M-46		M-46	130 mm towed gun (UA)	
M-160 M1949, 52-M-853		M-160	160 mm mortar <i>stored</i>	
M-1938 PM-38		ПМ-38	120 mm mortar <i>stored</i>	
M1977			armoured recovery vehicle	27
MAZ-537		MA3-537	tank transporter <i>legacy</i> (UA)	
MAZ-543 MAZ-7310		MA3-543/MA3-7310	heavy truck chassis 8x8 <i>legacy</i>	
MAZ-63172			tractor truck (UA)	
MAZ-74101	Oplot-T		tractor truck (UA)	
MAZ-531605			cargo truck (UA)	
MDK-3		МДК-3	trench digger (UA)	
Merlin-VR	Merlin-VR		reconnaissance UAV (UA)	
Merlin-VR	Merlin-VR	Мерлин-ВР	long-endurance reconnaissance and artillery-spotting uav	82
MKTK-1	Plavsk	«Плавок»	radio monitoring station	
MKTK-1A	Dzyudoist, Judoist	«Дзюдюист»	SIGINT system (UA)	
ML-20 M-1937		МЛ-20	152 mm towed gun-howitzer <i>stored</i> (UA)	
MLZh-VF-VT	MLZh-VF-VT		railway pontoon bridge (UA)	
MO.1.01.04			220 mm thermobaric rocket · of TOS-1A Sointsepek	29
MO.1.01.04M			220 mm thermobaric rocket, extended range · of TOS-1A Sointsepek (UA)	29
MO.1.01.04M2			220 mm thermobaric rocket, heavier warhead · of TOS-2 Tosochka	29
Mohajer-6			reconnaissance and strike UAV (UA)	
Mohajer-6	Mohajer-6, Qods Mohajer-6	Мохажер-6	iranian-built medium reconnaissance and light strike uav use	84
Molniya	Molniya, Molniya-1	Молния, Молния-1	cheap fixed-wing fpv one-way attack drone; also recon, fire	88
Molniya-1 Molniya-2	Molniya		FPV drone (UA)	
MON-50		МОН-50	directional anti-personnel mine (UA)	
MON-90		МОН-90	directional anti-personnel mine (UA)	
MON-100			directional anti-personnel mine (UA)	
MON-200			directional anti-personnel mine (UA)	
MP-32M1 MP32M1			command post vehicle · of 9S729M1 Slepok-1 (UA)	
MRO-A	Borodach		72.5 mm disposable thermobaric launcher (UA)	
MRT0	MRT0	МРТ0	maintenance vehicle · of 9K720 Iskander-M	33
MT-12 2A29, T-12A	Rapira	MT-12, 2A29	100 mm anti-tank gun (UA)	
MT-12R 2A29R	Ruta		100 mm anti-tank gun · of MT-12 Rapira <i>legacy</i>	
MT-55A			bridgelaye	
MT-LB		MT-ЛБ	tracked armoured transporter (UA)	
MT-LBM 6MA MT-LBM-6MA, Object 6MA		Объект 6MA	tracked armoured transporter · of MT-LB (UA)	
MT-LBM 6MB MT-LBM-6MB, Object 6MB		Объект 6MB	tracked armoured transporter · of MT-LB (UA)	
MT-LBV		MT-ЛБВ	tracked armoured transporter · of MT-LB <i>legacy</i> (UA)	
MT-LBVM MT-LBVMK, MT-LB VM1K, MT-LBVM1K		MT-ЛБВМ, MT-ЛБВМК	tracked armoured transporter · of MT-LB (UA)	
MTK-2 MTK			mine-clearing vehicle (UA)	
MTO-AT			repair vehicle (UA)	
MTO-UB2 MTO-UB			repair vehicle (UA)	
MTP-A2.1 MTP-A2			recovery vehicle (UA)	
MTP-LB		МТП-ЛБ	repair vehicle	
MTU-20 Object 602		МТУ-20	bridgelaye (UA)	
MTU-72 Object 632		МТУ-72	bridgelaye (UA)	55

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
MTU-90		MTY-90	bridgelaye	
MZKT-7930	Astrolog	МЗКТ-7930, Астролог	heavy truck chassis 8x8	39
Nebo-T	Nebo-T		air surveillance radar	
NSV NSVT, NSVS	Utyos	НСВ, Умёс	12.7 mm heavy machine gun (UA)	
Object 6 MT-LBVM1K, 6MA, M1970		MT-ЛБВМ1К, Обьект 6МА	light armed amphibious tractor and apc; base for many arm	17
Object 10	MT-LBu	MT-Лбу	tracked armoured transporter - of MT-LB (UA)	
Object 564	Firebug	Обьект 564	heavy flamethrower-crew carrier on t-72 chassis, carries rpo	18
Object 925 BMD-1KSh, BMD-1R, M1979	Robot, Skrezhet	БМД-1КШ, БМД-1Р	airborne tracked apc on a lengthened bmd-1 hull; base of a v	16
OF-843B 53-OF-843B, 30F843B		53-ОФ-843Б	mortar bomb	
Oreshnik RS-SS-28 MOD	Oreshnik	Орешник	intermediate-range ballistic missile (UA)	
Orion	Orion, Inokhodets	Орион, Иноходец	reconnaissance and strike UAV (UA)	
Orion	Orion, Inokhodets	Орион, Иноходец	medium-altitude long-endurance reconnaissance and strike uav	82
Orlan-10		Орлан-10	reconnaissance UAV (UA)	
Orlan-30		Орлан-30	reconnaissance UAV (UA)	
OZM-72	Lyagushka, Vedma	ОЗМ-72	bounding anti-personnel mine (UA)	
P-18 1RL131, P-18M	Terek SPOON REST D	П-18 «Терек», 1РЛ131	early warning radar (UA)	
P-19 1RL134	Dunai FLAT FACE B	П-19 «Дунай», 1РЛ134	air surveillance radar	
P-35 P-37, 1RL110, 1RL139	Saturn, Mech BAR LOCK	П-35 «Сатурн»	early warning radar <i>legacy</i> (UA)	
P-240 P-240I-4	Pereselenets, Pereselenetz		digital communications vehicle (UA)	
P-240I-4 P-240, P-240BTZ	Pereselenets, Pereselents	П-240И-4, Переселенец	digital field communications vehicle linking command posts (	71
P-260-O P-260-U, P-260-T	Redut-ZUS	Регут-ЗУС	mobile digital radio-relay and broadband wireless communicat	74
P-260-U P-260			signals vehicle - of Redut-ZUS (UA)	
P-800 3M55, SS-N-26	Oniks STROBILE	П-800 «Оникс»	anti-ship cruise missile - of 3K55 Bastion (UA)	
PAK-200 PAK-200M			field kitchen (UA)	
Palantin	Palantin, Palantin-K	«Палантин», «Палантин-К»	communications jammer (UA)	49
Pantsir-S1M	Pantsir-SM, Pantsir-S1M	Панцирь-СМ	gun-missile air defence system (UA)	39
Pantsir-S2 SA-22	Pantsir-S2 GREYHOUND	Панцирь-С2	gun-missile air defence system (UA)	39
Pantsir-SM-SV	Pantsir-SM-SV	Панцирь-СМ-СВ	gun-missile air defence system	39
Pantsir-SMD-E	Pantsir-SMD-E	Панцирь-СМД-Е	short-range SAM system	
Parodist	Parodist		radar and communications jammer <i>trial</i>	
PAZ-32053			bus (UA)	
PDM-1			anti-landing mine <i>legacy</i>	
PFM-1 PFM-1S	Lepestok	ПФМ-1, ПФМ-1С	scatterable anti-personnel mine (UA)	
PKM			man-portable minelaying kit	
Plastun-SN	Plastun-SN		light tracked all-terrain vehicle (UA)	
PMM-2 PMM-2M		ПММ-2, ПММ-2М	amphibious ferry	57
PMN PMN-1, PMN-2, PMN-4		ПМН	anti-personnel blast mine (UA)	
PMP PMP-M		ПМП, ПМП-М	heavy folding pontoon bridge (UA)	56
PMR-3			towed minelayer <i>legacy</i>	
PMZ-4			towed minelayer	
POM-2 POM-2S		ПОМ-2	scatterable anti-personnel mine (UA)	
POM-2R		ПОМ-2Р	scatterable anti-personnel mine - of POM-2 (UA)	
POM-3	Medalyon		scatterable anti-personnel mine (UA)	
PP-91		ПП-91	pontoon park <i>legacy</i> (UA)	56
PP-91M PP-2005		ПП-91М, ПП-2005	pontoon park (UA)	56
PPS-84		ППС-84	pontoon park <i>legacy</i>	
Predel-E	Predel-E		coastal defence radar (UA)	
Prokhod-1	Prokhod-1		remote-controlled mine-clearing system (UA)	
PRP-3 1Zh3, Object 767	Val SMALL FRED (disputed)	ПРП-3 «Вал»	artillery reconnaissance vehicle (UA)	
PRP-4 1V121, 1V125, Object 779	Nard	ПРП-4 Нард	artillery reconnaissance vehicle (UA)	68
PRP-4A	Argus	ПРП-4А «Аргус»	artillery reconnaissance vehicle (UA)	68

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
PRP-4M PRP-4MU, 1V145	Deyteriy	ПРП-4М «Дейтерий»	artillery reconnaissance vehicle UA	68
PSNR-5 PSNR-5M, 1RL133	Kredo-M TALL MIKE		ground surveillance radar · of PRP-4M Deyteriy UA	68
PT-76		ПТ-76	light amphibious tank <i>legacy</i>	
PTM-1 PTM-1S, PTM-1G		ПТМ-1	scatterable anti-tank mine UA	
PTM-3 KPTM-3		ПТМ-3	scatterable anti-tank mine UA	
PTM-4	Nokaut-2	ПТМ-4, Hokaym-2	scatterable anti-tank mine UA	
PTS-2 PTS, PTS-M		ПТС-2, ПТС-M	amphibious transporter UA	57
PTS-3		ПТС-3	amphibious transporter <i>legacy</i>	57
PTS-4		ПТС-4	amphibious transporter	57
PU-12 BTR-60PU-12, PU-12M, PU-12M6, PU-12M7		ПУ-12	air defence command post UA	69
PZM-2		ПЗМ-2	trench digger UA	
R-142N R-142NM, R-142NMR, R-142NSA	Deimos-N	P-142H, Деѳмос-Н	command and staff vehicle UA	
R-142N R-142, R-142NM	Deimos-N, Deimos-NMR	P-142H, Деѳмос-Н	truck-mounted hf/vhf command and staff vehicle for regiment,	71
R-145BM R-145BM1, R-145BMA			command and staff vehicle UA	65
R-149AKSh-1			command and staff vehicle UA	64
R-149BMR R-149BMRg	Kushetka-B	P-1495MP, Кушетка-Б	command and staff vehicle UA	64
R-149MA3 R-149MA1			command and staff vehicle UA	64
R-161 R-161A2, R-161A2M	Poisk	P-161, Пoиск	radio station UA	
R-161 R-161A2, R-161A2M	Poisk, Ekvator-3M	P-161, Пoиск	medium and high power hf/vhf radio stations on trucks for op	70
R-166 R-166-0.5, R-166S	Artek	P-166, Артек	radio station UA	
R-166 R-166-0.5, R-166-05	Artek, Artek-1A	P-166, Артек	medium-power hf/vhf mobile radio station for command post si	70
R-168 R-168E	Akveduk	P-168, Аквeгyк	radio set family UA	
R-187 R-187-P1	Azart		radio set family UA	
R-325UMV R-325BMV		P-325YMB	communications jammer · of RB-301B Borisoglebsk-2 UA	48
R-330	Mandat	P-330 «Мангат»	communications jammer complex <i>legacy</i>	
R-330B	Mandat-B	P-330B «Мангат-Б»	communications jammer <i>legacy</i>	47
R-330BMV		P-330BMB	communications jammer UA	47
R-330KMA		P-330KMA	EW command post · of R-330M1P Diabazol	
R-330KMV		P-330KMB	EW command post · of RB-301B Borisoglebsk-2	48
R-330M1P	Diabazol	P-330M1П «Дѳабазол»	communications jammer complex UA	
R-330T		P-330T	communications jammer <i>export</i>	47
R-330Zh	Zhitel	P-330Ж «Житeль»	GNSS and cellular jammer UA	50
R-340RP Pole-21M, Pole-21E	Pole-21	P-340PП «Пoлe-21»	GNSS jammer UA	53
R-378A R-378B		P-378A(B)	communications jammer · of R-330 Mandat <i>legacy</i> UA	
R-378BMV		P-3785MB	communications jammer · of RB-301B Borisoglebsk-2	48
R-378UM			communications jammer · of R-330M1P Diabazol	
R-381T R-381TM, R-381T2M	Taran, Taran-M	P-381T, «Таран»	SIGINT system UA	
R-410-7.5			tropospheric scatter station UA	
R-410-7.5 R-412A, R-412AM	Torf, Rezoner	P-412A, P-412AM	beyond-line-of-sight radio relay link between headquarters	72
R-412A R-412AM	Torf, Rezoner	P-412AM, Резонер	tropospheric scatter station UA	
R-416GM			radio relay station UA	
R-416GM R-416G-MS, R-416GM2		P-416ГМ	truck-mounted digital radio relay station for field command	72
R-417 R-417S	Baget		tropospheric scatter station UA	
R-431AM			radio relay antenna module · of Redut-2US UA	
R-438			satellite communications terminal	
R-439 R-439-P, R-439-BK, R-439-KUL	Legenda-2	P-439, Легенда-2	satellite communications terminal UA	
R-439 R-439-P, R-439-BK	Legenda-2, Legenda-2P	P-439, Легенда-2	military satellite communication stations: portable, contain	73
R-934B	Sinitsa	P-934B «Синѳца»	communications jammer UA	47
R-934BMV		P-9345MB	communications jammer UA	47
R-934UM	Sinitsa		communications jammer UA	47
RB-105B	Less		SIGINT system	
RB-108S			EW command post	

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
RB-109A	Bylina	РБ-109А «Былина»	EW command post (UA)	
RB-301B	Borisoglebsk-2	РБ-301Б «Борисоглебск-2»	communications jammer complex (UA)	48
RB-341V	Leer-3	РБ-341В «Леер-3»	cellular jammer (UA)	52
RB-531B	Infaua	РБ-531Б «Инфауна»	counter-IED jammer (UA)	54
RB-636AM2 RB-636	Svet-KU	РБ636 «СВет-КУ»	SIGINT system (UA)	
RB-636 Svet-VSG	Svet-VSG		SIGINT system - of RB-636AM2 Svet-KU	
REBovets Valley	REBovets Valley		UAV jammer <i>trial</i>	
Redut-2US			signal and communications system (UA)	
REM-KL			recovery vehicle (UA)	
Repellent-1	Repellent-1	Репеллент-1	UAV jammer (UA)	53
Repeynik	Repeynik		portable counter-UAS radar (UA)	
Rezonans-N	Rezonans-N, Resonans-N, Rezonans-NE		early warning radar	
RKhM-4		РХМ-4	CBRN reconnaissance vehicle (UA)	
RKhM-6	Povozka	РХМ-6	CBRN reconnaissance vehicle (UA)	
RKhM-8		РХМ-8	CBRN reconnaissance vehicle	
RKhM-9		РХМ-9	CBRN reconnaissance vehicle	
RLM-D		РЛМ-Д	UHF radar module - of 55Zh6M Nebo-M (UA)	41
RLM-M		РЛМ-М	VHF radar module - of 55Zh6M Nebo-M (UA)	41
RLM-S		РЛМ-С	S-band radar module - of 55Zh6M Nebo-M (UA)	41
RM-G			armoured recovery vehicle	
RP-377L RP-377LA	Lorandit, Lorandit-M		communications jammer (UA)	
RP-377UVM1 RP-377UVM2, RP-377UM2	Lesochek		counter-IED jammer (UA)	
RPG-7 RPG-7V, RPG-7V2, 663		РПГ-7	anti-tank rocket launcher (UA)	
RPG-22	Netto	РПГ-22, Hemmo	72.5 mm disposable anti-tank launcher <i>legacy</i> (UA)	
RPG-26	Aglen	РПГ-26, Аглень	disposable anti-tank rocket launcher (UA)	
RPG-27	Tavolga	Таволга	105 mm disposable anti-tank launcher (UA)	
RPG-28	Klyukva	РПГ-28, Клюкба	125 mm disposable anti-tank launcher (UA)	
RPG-29 TKB-0175	Vampir	РПГ-29 «Вампир», ТKB-0175	105 mm anti-tank rocket launcher	
RPG-30	Kryuk	РПГ-30, Kplok	105 mm disposable anti-tank launcher	
RPG-32	Barkas, Hashim, Nashshab	РПГ-32	105 mm reusable anti-tank launcher	
RPMK-1 1B44, 1B44-1	Ulybka, Ulibka-M		meteorological station (UA)	
RPO-A RPO-M, RPO	Shmel	РПО-А «Шмель»	thermobaric rocket launcher (UA)	
Rys	Rys, Iveco LMV	Рысь	armoured utility vehicle (UA)	43
S-60 AZP-57		C-60, АЗП-57	57 mm anti-aircraft gun (UA)	
S-300PM S-300PM1, S-300PM2, S-300PMU1, S-300PMU2, SA-20	S-300PM, Favorit GARGOYLE	C-300ПМ, C-300ПМУ1	long-range SAM system (UA)	
S-300PS 75R6, S-300PMU, SA-10	S-300PS GRUMBLE	C-300ПС, 75Р6	long-range SAM system <i>legacy</i> (UA)	
S-300V 9K81, SA-12A, SA-12B	S-300V, Antey-300 GLADIATOR, GIANT	C-300В, 9К81	long-range SAM system (UA)	36
S-300V4 S-300VMD, SA-23	S-300V4	C-300В4	long-range SAM system (UA)	36
S-350 50R6A	S-350, Vityaz	C-350 «Витязь», 50Р6А	medium-range SAM system (UA)	36
S350 S350M	Supercam	Суперкам	small electric fixed-wing reconnaissance and artillery fire-	79
S-400 40R6, SA-21	S-400, Triumph GROWLER	C-400 «Триумф», 40Р6	long-range SAM system (UA)	35
S-500 55R6M	S-500, Prometey, Triumphator-M	C-500 «Прометей», 55Р6М	long-range SAM and missile defence system <i>trial</i> (UA)	
SA-14RM/SM			encryption command post (UA)	
Saniya	Saniya, Sania	«Саняя»	UAV jammer (UA)	
Sapfir	Sapfir		UAV jammer (UA)	
Sarmat-2 Sarmat-3, LSTC-1944	Sarmat, STS Sarmat		armoured utility vehicle (UA)	
SBA-60K2	Bulat	«Булат»	armoured personnel carrier	
SBR-5M	Fara-VR		portable battlefield surveillance radar (UA)	
SERP-VS6D	Serp	СЕРП-BC6Д	UAV jammer (UA)	
Shipovnik-Aero	Shipovnik-Aero	«Шиповник-АЭРО»	UAV jammer (UA)	
Silok-01	Silok	«Силок-01»	UAV jammer (UA)	
SNAR-10 IRL232, IRL-127-1	Leopard BIG FRED	СНАР-10 «Леопард», 1РЛ232 «Леопард»	battlefield surveillance radar (UA)	44

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
SNAR-10M SNAR-10M1, 1RL232-1M	Pantera	СНАР-10М, 1РА232-1М «Пантера»	battlefield surveillance radar (UA)	44
Sopka-2 12A6	Sopka-2		air surveillance radar	
Sozvezdie	Sozvezdie		tactical command and control system <i>legacy</i>	
SPG-9 SPG-9M, SPG-9D	Kopye	СПГ-9, Копье	73 mm recoilless gun (UA)	
SPN-4 SPN-4M, 1RL248-4	HEART ACHE		radar jammer	
Strizh-3	Strizh-3		UAV jammer (UA)	
Stupor	Stupor		UAV jammer (UA)	
Supercam S350			reconnaissance UAV (UA)	
Syuzhet-MF	Syuzhet-MF		meteorological station <i>trial</i>	
T-12 2A19		T-12, 2A19	100 mm anti-tank gun <i>stored</i>	
T-14 Object 148	Armata	T-14, Объект 148	main battle tank · of Armata <i>trial</i>	
T-15	Armata	T-15	infantry fighting vehicle · of Armata <i>trial</i>	
T-16	Eleron-3SV, Eleron-3	Элерон-3СВ, Элерон	man-portable short-range electro-optical reconnaissance and	80
T-54 Object 137		T-54, Объект 137	main battle tank <i>legacy</i> (UA)	
T-54 T-55, Object 155		T-54, T-55	medium tank; reserve stock, reactivated since 2023	10
T-54-T			armoured recovery vehicle <i>legacy</i>	
T-55		T-55	main battle tank <i>stored</i> (UA)	
T-55A		T-55A	main battle tank · of T-55 <i>stored</i> (UA)	
T-55AM T-55AMV		T-55AM, T-55AMB	main battle tank · of T-55 <i>legacy</i>	
T-62 Object 166		T-62, Объект 166	main battle tank <i>legacy</i> (UA)	
T-62 Object 166, T-62M		T-62, T-62 обр. 1967	medium tank with 115 mm smoothbore gun; cold war reserve typ	9
T-62M		T-62M	main battle tank · of T-62 (UA)	
T-62MV		T-62MB	main battle tank · of T-62 (UA)	
T-64A Object 434		T-64A, Объект 434	main battle tank (UA)	
T-64A/B/BV Object 434, Object 447A		T-64Б, T-64A	main battle tank, 3-man crew with autoloader (older kharkiv	9
T-64B Object 447, T-64BV		T-64Б, Объект 447	main battle tank (UA)	
T-64BV			main battle tank · of T-64B (UA)	
T-64BVK			armoured command vehicle · of T-64B <i>legacy</i> (UA)	
T-72 T-72 Ural, Object 172M	Ural	T-72, T-72 «Урал»	main battle tank (UA)	
T-72A		T-72A	main battle tank · of T-72 Ural (UA)	
T-72AV		T-72AB	main battle tank · of T-72A (UA)	
T-72B Object 184		T-72Б, Объект 184	main battle tank · of T-72 Ural (UA)	
T-72B1 T-72B1K		T-72Б1, T-72Б1К	main battle tank · of T-72B (UA)	
T-72B3		T-72Б3	main battle tank · of T-72B (UA)	
T-72B3A			main battle tank · of T-72B3 <i>legacy</i> (UA)	
T-72B3M		T-72Б3М	main battle tank · of T-72B3 (UA)	
T-72BA		T-72БA	main battle tank · of T-72B (UA)	
T-72BM		T-72БМ	main battle tank · of T-72B <i>legacy</i> (UA)	
T-72M1 T-72M		T-72М1, T-72М	main battle tank · of T-72 Ural <i>legacy</i>	
T-80B Object 219R		T-80Б, Объект 219Р	main battle tank <i>stored</i> (UA)	
T-80B/BV/BVM T-80U, T-80B, M1989		T-80, T-80Б	main battle tank with gas turbine; 125 mm gun that also fire	8
T-80BV		T-80БВ	main battle tank · of T-80B (UA)	
T-80BVK		T-80БВК	armoured command vehicle · of T-80BV <i>legacy</i> (UA)	
T-80BVM		T-80БВМ	main battle tank · of T-80BV (UA)	
T-80U Object 219AS		T-80У, Объект 219АС	main battle tank (UA)	
T-80UE-1		T-80УЕ-1	main battle tank · of T-80U <i>legacy</i> (UA)	
T-80UK		T-80УК	armoured command vehicle · of T-80U <i>legacy</i> (UA)	
T-80UM2		T-80УМ2	main battle tank · of T-80U <i>legacy</i> (UA)	
T-90 Object 188		T-90, Объект 188	main battle tank (UA)	
T-90 Object 188, T-72BU	Vladimir, Proryv-3	T-90, Владимир	main battle tank; 125 mm gun-launcher with autoloader	7
T-90A		T-90A	main battle tank · of T-90 (UA)	
T-90AK		T-90АК	armoured command vehicle · of T-90A (UA)	
T-90M Proryv-3	Proryv-3	T-90М, Прорыв-3	main battle tank · of T-90A	

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
T-90S T-90SK		T-90C, T-90CK	main battle tank - of T-90 UA	
Tachyon	Tachyon, Takhion	Тахуон	reconnaissance UAV UA	
Takhion	Takhion, Tachyon	Тахуон	small electric reconnaissance and fire-correction uav	81
TBS-M3		ТБС-М3	220 mm thermobaric rocket, long range - of TOS-2 Tosochka UA	29
TDA-2K TDA-2M		ТДА-2К, ТДА-2М	smoke generator vehicle UA	
TDA-3			smoke generator vehicle UA	
TDA-3 TDM-2K, TDA-2M		ТДА-2К, ТДА-3	truck-mounted thermal aerosol generators that screen troops	63
TDM-2K			smoke generator vehicle UA	
Tetrazdr	Tetrazdr		UAV jammer <i>trial</i>	
Tirada-2S Tirada-2.3, RB-371A	Tirada-2	«Tupaga-2C», «Tupaga 2.3»	satellite communications jammer UA	
TM-46		TM-46	anti-tank blast mine <i>legacy</i>	
TM-57		TM-57	anti-tank blast mine <i>legacy</i>	
TM-62 TM-62M		TM-62	anti-tank blast mine UA	
TM-83		TM-83	side-attack anti-tank mine	
TMM-3 TMM		TMM-3	truck-mounted bridgelay UA	55
TMM-6		TMM-6	8x8 heavy bridgelay UA	55
TMS-65 TMS-65M, TMS-65U		TMC-65	decontamination vehicle	
Tobol	Tobol	«Тобол»	satellite communications jammer UA	
Torn	Torn, Torn-MDM		SIGINT system UA	
Toros	Toros		armoured utility vehicle	
TOS-1 Object 634	Buratino	ТОС-1 «Буратино»	220 mm heavy flamethrower system <i>legacy</i> UA	29
TOS-1A Object 634B	Solntsepek	ТОС-1А «Солнце́пек»	220 mm heavy flamethrower system UA	29
TOS-2	Tosochka	ТОС-2 «Тосочка»	220 mm heavy flamethrower system UA	29
TOS-3	Drakon	ТОС-3 «Дракон»	heavy flamethrower system, tracked <i>trial</i> UA	29
TS-3132	Gusar		armoured utility vehicle UA	
TZ-74			decontamination and smoke vehicle <i>legacy</i>	
TZM-T		ТЗМ-Т	rocket transloader - of TOS-1A Solntsepek UA	29
UAZ-452	Bukhanka	УАЗ-452, Буханка	utility van UA	
UAZ-469 UAZ-3151	Kozlik	УАЗ-469	utility vehicle UA	
UAZ-469RKh UAZ-469RKhB			CBRN reconnaissance and decontamination vehicle UA	
UAZ-39094	Fermer		utility pickup UA	
UMPB D-30SN			jet-powered glide bomb UA	
UMPK		УМПК	glide bomb guidance kit UA	
UMZ		УМЗ	truck-mounted minelayer UA	
UMZ UMZ-3		УМЗ	truck-mounted remote scatterable minelayer	61
UMZ-G UMZ-K, UMZ-T	Kleshch-G, Kleshch-G		universal tracked remote minelayer <i>trial</i>	
UR-67		УР-67	mine-clearing line charge <i>legacy</i> UA	
UR-77 Object 256, M1979	Meteorit	УР-77	mine-clearing line charge vehicle UA	59
UR-83P		УР-83П	portable mine-clearing line charge	
Ural-375D Ural-375		Урал-375Д	cargo truck 6x6 <i>legacy</i> UA	
Ural-4320 Ural-43203		Урал-4320	cargo truck 6x6 UA	
Ural-4320-7810	Motovoz-M		cargo truck UA	
Ural-6370 Ural-6370-0121-30			cargo truck 6x6 UA	
Ural-6370	Ural-6370, Ural-63706-0010	Урал-63706, Торнадо-У	heavy 6x6 and 8x8 cargo trucks and tractors; tornado-u has a cargo truck 4x4 UA	76
Ural-43206		Урал-43206	mine-resistant ambush-protected vehicle	
Ural-63095	Typhoon-U	Урал-63095, Тайфун-У	cargo truck 6x6 UA	
Ural-63706-0122 Ural-63706-0011	Tornado-U		armoured truck UA	
Ural-432009	Ural-VV	Урал-432009, Урал-BB	cargo truck with armoured cab UA	
Ural-532301			tank transporter UA	
Ural-542301			remote-controlled mine-clearing robot UA	
Uran-6		Уран-6	remote-controlled tracked demining and obstacle-clearing veh	60
Uran-6	Uran-6	Уран-6	satellite communications jammer UA	
Volna Kupol Garant	Volna Kupol Garant		UAV jammer UA	
Volnerez	Volnerez	«Волнорез»	armoured personnel carrier <i>trial</i>	
VPK-7829 K-16, K-17	Bumerang, Boomerang	Бумеранг		

DESIGNATION	NAME	CYRILLIC	WHAT IT IS	PAGE
VPK-233136 SBM	Tigr SBM	ВПК-233136	armoured car · of GAZ-233014 Tigr UA	
VSM-1		BCM-1	helicopter remote mining system	
VT-72A			armoured recovery vehicle	
VTs	Ladoga, Debut	BTC, Aagoza	armoured command, staff and reconnaissance vehicle for nbc c	66
Z-STs	Akhmat, Remdiesel Z-STs Akhmat		armoured utility vehicle UA	
ZALA 421-16E, Z-16		421-16E, 421-16E2	short-range electric reconnaissance and fire-correction uav; reconnaissance UAV UA	77
ZALA 421-16 Z-16				
Zaslon-REB	Zaslon-REB		cellular jammer	
Zastava	Zastava	Застава	reconnaissance UAV UA	
Zastava	Zastava, Bird-Eye	Застава	short-range reconnaissance and fire-adjustment mini-uav	81
Zavarukha	Zavarukha		portable acoustic reconnaissance system UA	
ZIL-131		ЗИЛ-131	cargo truck 6x6 <i>legacy</i> UA	
ZIL-157		ЗИЛ-157	cargo truck 6x6 <i>legacy</i>	
ZSU-23-4 ZA6	Shilka	ЗСУ-23-4, ZA6	23 mm self-propelled anti-aircraft gun <i>stored</i> UA	
ZU-23-2 ZU-23, 2A13		ЗУ-23, ЗУ-23-2	23 mm anti-aircraft gun UA	

Table 1b. Other names and designations

Names, NATO names and codes (in **BLUE CAPITALS**), and other designations, each pointing (→) to its row in Table 1. Page numbers as in Table 1. English translations of Russian names are in *italics*.

NAME OR DESIGNATION	PAGE	NAME OR DESIGNATION	PAGE	NAME OR DESIGNATION	PAGE
1B44 → RPMK-1		2A19 → T-12		3VOF32 → 3OF25	
1B44-1 → RPMK-1		2A29 → MT-12		3VOF33 → 3OF25	
1B75E → 1B75	43	2A29R → MT-12R		3VOF34 → 3OF43	31
1L13 → 1L13-3		2A45MR → 2A45M		3VOF35 → 3OF44	31
1RL110 → P-35		2B5 → BM-21	26	3VOF39 → 3OF29	29
1RL131 → P-18		2B9M → 2B9		3VOF40 → 3OF29	29
1RL133 → PSNR-5	68	2B17 → BM-21-1	26	3VOF42 → 3OF43	31
1RL133-3 → Kredo-1E		2B17-1 → BM-21-1	26	3VOF49 → 3OF49	
1RL134 → P-19		2B17M → 9K51M	26	3VOF58 → 3OF45	30
1RL138 → 1RL144M	40	2F77M → 2F77	40	3VOF63 → 3OF38	
1RL139 → P-35		2K22M → 2K22	40	3VOF64 → 3OF39	
1RL144 → 1RL144M	40	2K22M1 → 2K22	40	3VOF64M → 3OF39M	
1RL144M1 → 1RL144M	40	2K25 → 3OF39		3VOF66 → 3OF38	
1RL232 → SNAR-10	44	2K25M → 3OF39M		3VOF72 → 3OF45	30
1RL232-1M → SNAR-10M	44	2S1M → 2S34		3VOF73 → 3OF45	30
1RL248-4 → SPN-4		2S3M1 → 2S3M		3VOF86 → 3OF59	29
1V13M → 1V13	67	2S3M2 → 2S3M		3VOF87 → 3OF59	29
1V15 → 1V14	67	2S6M → 2S6	40	3VOF91 → 3OF61	
1V15M → 1V14	67	2S6M1 → 2S6	40	3VOF93 → 3OF39	
1V16 → 1V14	67	2S9-1 → 2S9		3VOF96 → 3OF64	
1V19 → 1V18		2S12A → 2S12		3VOF97 → 3OF64	
1V121 → PRP-4	68	2S33 → 2S19M2	30	3VOF98 → 3OF64	
1V125 → PRP-4	68	3F5 → 1K113	31	3VRB36 → 3RB30	
1V126M → 1V126		3M14K → 3M14		3VRB37 → 3RB30	
1V145 → PRP-4M	68	3M14T → 3M14		3VRB38 → 3RB30	
1V152 → 1V126		3M55 → P-800		5P85D → 5P85SD	
1V152 → 1V119	67	3OF45-1 → 3OF45	30	5P85S → 5P85SD	
1V152M → 1V126		3OF843B → OF-843B		5P85SM → 5P85SD	
1V153 → 1V126		3UBK10 → 9K116		5V55K → 5V55R	
1V153M → 1V126		3VF2 → 3F2	31	5V55RM → 5V55R	
1V168 → ASUNO-S		3VF4 → 1K113	31	6G3 → RPG-7	
1V1003 → 1V119	67	3VO11 → 3-O-8	31	6MA → Object 6	17
1Zh3 → PRP-3		3VO13 → 3-O-13		6P49 → 6P50	
2A6 → ZSU-23-4		3VO14 → 3-O-13		6P57 → 6P50	
2A13 → ZU-23-2		3VO28 → 3-O-23		6P58 → 6P50	
2A18 → D-30		3VO29 → 3-O-23		6P59 → 6P50	
2A18M → D-30		3VO30 → 3-O-23		8K14 → 9K72	

NAME OR DESIGNATION	PAGE
9A33BM3 → 9A33B	40
9A38 → 9A310	37
9A39M1 → 9A39	37
9A53-G → 9K51M	26
9A53-S → 9A54	28
9A83-1 → 9A83	36
9A83M2 → 9A83	36
9A85-2 → 9A85	36
9A310M1 → 9A310	37
9A331-1 → 9A331	38
9A331M2 → 9A331M	38
9K32M → 9K32	
9K33M2 → 9K33	40
9K33M3 → 9K33	40
9K35M3 → 9K35	40
9K37M1 → 9K37	37
9K79-1 → 9K79	
9K81 → S-300V	36
9K111-1M → 9K111-1	
9K111M → 9K111	
9K113 → 9K111-1	
9K116-1 → 9K116	
9K116-2 → 9K116	
9K116-3 → 9K116	
9K119M → 9K119	
9K120 → 9K119	
9K121 → 9A4172	
9K123-1 → 9K123	
9K128-1 → 9P162	
9K129 → 9K135	
9K331M → 9K331	38
9K510 → 9P132	26
9M14 → 9K11	
9M22U-1 → 9M22U	26
9M33M → 9M33	40
9M33M2 → 9M33	40
9M33M3 → 9M33	40
9M37M → 9M37	40
9M37MD → 9M37	40
9M79-1 → 9M79	
9M82M → 9M82	36
9M82MD → 9M82	36
9M83M → 9M83	36
9M96E → 9M96	35
9M96E2 → 9M96	35
9M96M → 9M96	35
9M111-2 → 9M111	
9M111M → 9M111	
9M112 → 9K112	
9M117M → 9M117	
9M117M1 → 9M117	
9M119M → 9M119	
9M119M1 → 9M119	
9M123F → 9M123	
9M131M → 9M131	
9M133F-1 → 9M133	
9M311-1M → 9M311	40
9M330-2 → 9M330	38
9M338K → 9M338	38
9M541 → 9M218	26
9M723K1 → 9M723	33
9P78 → 9P78-1	33

NAME OR DESIGNATION	PAGE
9P129-1M → 9P129	
9P135 → 9K111	
9P135M → 9K111-1	
9P149M → 9K132	
9P151 → 9K115	
9P157 → 9P157-2	
9P163-2 → 9P163M-2	
9P163M-1 → 9P162	
9S15 → 9S15M	36
9S15MTZE → 9S15M	36
9S18M1 → 9S18	37
9S18M1-1 → 9S18	37
9S18M1-2 → 9S18	37
9S18M1-3 → 9S18	37
9S19 → 9S19M2	36
9S32M1 → 9S32	36
9S36M → 9S36	37
9S52M → 9S52	68
9S52M1 → 9S52	68
9S80-1 → 9S80	40
9S470M1 → 9S470	37
9S470M1-2 → 9S470	37
9S482 → PU-12	69
9S482M6 → PU-12	69
9S482MU → PU-12	69
9S510M → 9S510	37
9S931-3 → 9S931-2	15
9S932-2 → 9S932-1	69
9T128-1 → 9T128	
9T127B → 9T127	40
9T250E → 9T250	33
12A6 → Sopka-2	
30N6 → 5N63S	
39N6E → 39N6	
40N6E → 40N6	35
40R6 → S-400	35
48N6DM → 48N6	35
48N6E → 48N6	35
48N6E2 → 48N6	35
48N6E3 → 48N6	35
50K6E → 50K6A	36
50N6E → 50N6A	36
50P6E → 50P6A	36
50R6A → S-350	36
52-M-853 → M-160	
53-OF-843B → OF-843B	
55K6A → 55K6	35
55R6M → S-500	
55Zh6ME → 55Zh6M	41
57E6-E → 57E6	39
57E6M → 57E6	39
59N6-E → 59N6	
64N6E → 64N6	
67N6E → 67N6	
75R6 → S-300PS	
91N6E → 91N6A	35
92N6E → 92N6A	35
96L6E → 96L6	42
96L6-TsP → 96L6	42
421-16E → ZALA	77
Aglen → RPG-26	
Agriculture [English] → ISDM	61

NAME OR DESIGNATION	PAGE
Aistenok → 1L271	43
Aistyonok → 1L271	43
Akatsiya → 2S3	
Akatsiya → 2S3M	
Akhmat → Z-STS	
Akveduk → R-168	
Aleut → GAZ-3344-20	
Altaets-AM → Altaets	
Altayets → Altaets	
AMN-590951 → BPM-97	24
Antey-300 → S-300V	36
Argus → PRP-4A	68
ARK-1 → 1RL239	43
ARK-1M → 1RL239	43
Arkan → 9M117	
Armata → T-14	
Armata → T-15	
ARS-14KM → ARS-14	
ARS-15 → ARS-14	63
Artek → R-166	
Artek-1A → R-166	70
AS-4 → Kh-22	
AS-13 → Kh-59	
AS-15 → Kh-55	
AS-18 → Kh-59	
AS-23A → Kh-101	
AS-24 → Kh-47M2	
Astrolog → MZKT-7930	39
AT-3 → 9K11	
AT-4 → 9K111	
AT-4 → 9M111	
AT-5 → 9K111-1	
AT-5 → 9M113	
AT-5B → 9M113M	
AT-6 → 9K114	
AT-6 → 9M114	
AT-7 → 9K115	
AT-7 → 9M115	
AT-8 → 9K112	
AT-9 → 9K132	
AT-9 → 9M120	
AT-10 → 9M117	
AT-11 → 9K119	
AT-11 → 9M119	
AT-11B → 9M119	
AT-11C → 9M119	
AT-12 → 9K116	
AT-13 → 9K115-1	
AT-13 → 9K115-2	
AT-13 → 9M131	
AT-14 → 9K135	
AT-14 → 9M133	
AT-15 → 9K123	
AT-15 → 9M123	
AT-16 → 9A4172	
Ataka → 9M120	
Ataka-1 → 9M120-1	
Avtobaza-M → 1L222	
Azart → R-187	
AZP-57 → S-60	
Baget → R-417	
Bal → 3K60	

NAME OR DESIGNATION	PAGE
Bal-E → 3K60	
Balkan → AGS-40	
BAR LOCK → P-35	
Bardak → BRDM-2	22
Barkas → RPG-32	
Barnaul-T → 9S932-1	69
Bastion → 3K55	
Bastion → 9K116	
Bastion → 9M117	
Bastion-P → 3K55	
Basurmanin → BMP-1AM	
Basurmanin → BMP-1	11
BAZ-69092 → BAZ-6909	
Beetle (English) → IRM	
Beglianka → BREM-L	
Beglyanka → BREM-L	
Beglyanka → BREM	62
Berezhok → BMP-2M	
Berezhok → BMP-2	12
BIG BIRD → 64N6	
BIG FRED → SNAR-10	44
BILL BOARD → 9S15M	36
Bird-Eye → Zastava	81
BM-21b → 9K55	26
BM-21b → 9P138	26
BM-21P → 9P132	26
BM-21V → 9K54	26
BM-21V → 9P125	26
BM-22 → 9P140	27
BM-27 → 9K57	27
BM-27 → 9P140	27
BM-30 → 9A52	28
BM-30 → 9A52-2	28
BM-30 → 9K58	28
BMD-1 → BMD-2	14
BMD-1KSh → BMD-1KSh-A	66
BMD-1KSh → Object 925	16
BMD-1P → BMD-2	14
BMD-1R → Object 925	16
BMD-3M → BMD-4M	13
BMD-4 → BMD-4M	13
BMK-150 → BMK-130M	
BMP-1KShM → BMP-1KSh	65
BMP-2M → BMP-2	12
BMP-76PB → BMP-1	11
BMP M1976/2 → BRM-1K	
BMP Model 1970 → BMP-1	
BMPT-72 → BMPT	11
BMR-3MA → BMR-3M	
Boar (English) → BMR-3M	
Boomerang → VPK-7829	
Borderline (English) → KM-8	
Borisoglebsk-2 → RB-301B	48
Borodach → MRO-A	
BOX SPRING → 1L13-3	
BPM-97 → KamAZ-43269	
BRDM-2RKh → BRDM-2RKhb	
Breakwater (English) → Volnorez	
BREhM-D → BREM-D	
BREM-1 → BREM	62
BREM-1M → BREM	62
BREM-Ch → BREM-4	

NAME OR DESIGNATION	PAGE
BRM-1 → BRM-1K	13
BRM-1KM → BRM-1K	
BTR-40P-2 → BRDM-2	22
BTR-40PB → BRDM-2	22
BTR-60 → BTR-60PB	65
BTR-60PU-12 → PU-12	69
BTR-70 → BTR-80	20
BTR-82 → BTR-80	20
BTR-82A → BTR-80	20
BTR-82AM → BTR-82A	
Buk → 9K37	37
Buk-M1 → 9K37	37
Buk-M1-2 → 9K37M1-2	37
Buk-M2 → 9K317	37
Buk-M3 → 9K317M	37
Bukhanka → UAZ-452	
Bulat → SBA-60K2	
bulldozer on artillery tractor (English) → BAT-2	
Bumerang → VPK-7829	
Buran ASV → Buran	
Burattino → TOS-1	29
Burya → Kh-22	
Bylina → RB-109A	
Carnation (English) → 2S1	
CHEESE BOARD → 96L6	42
Chosta → 2S34	
Coalition-SV (English) → 2S35	30
Cornflower (English) → 2B9	
Credo-1 → Kredo-1E	
Credo-1E → Kredo-1E	
D-30A → D-30	40
Dagger (English) → Kh-47M2	
Danube (English) → P-19	
Debut → VTS	66
Deimos-N → R-142N	
Deimos-NMR → R-142N	71
Derivatsiya-PVO → 2S38	
Deuterium (English) → PRP-4M	68
Deyteriy → PRP-4M	68
Diabazol → R-330M1P	
DKZ-66 → 9P132	26
Drakon → TOS-3	29
DShK → DShK 38/46	
DShKM → DShK 38/46	
DT-10PM → DT-30	36
DT-30PM → DT-30	36
Dunai → P-19	
Dzyudoist → MKTK-1A	
Eger-II → GAZ-3308	
Ekvator-3M → R-161	70
Elbrus → 9K72	
Eleron-3 → T-16	80
Eleron-3SV → Eleron-3	
Eleron-3SV → T-16	80
Fagot → 9K111	
Fagot → 9M111	
Faktoriya → 9M111	
Fara-VR → SBR-5M	
FARM → 2S19	30
fast trench machine (English) → BTM-3	
Fath-360 → Fateh-360	

NAME OR DESIGNATION	PAGE
Favorit → S-300PM	
Fermer → UAZ-39094	
Firebug → Object 564	18
FLAP LID B → 5N63S	
FLAT FACE B → P-19	
Floks → 2S40	
Forpost-R → Forpost	83
FPV kamikaze drones → FPV drones	
Frog (English) → OZM-72	
FROG-7 → 9K52	
GADFLY → 9K37	37
GADFLY → 9K37M1-2	37
Gamma-DE → 67N6	
GARGOYLE → S-300PM	
Garmon → 1L122	
Garmony → 1L122	
GAUNTLET → 9K330	38
GAUNTLET → 9K331	38
GAUNTLET → 9K331MDT	38
GAUNTLET → 9K331MU	38
GAUNTLET → 9K332	38
GAZ-41-08 → BRDM-2	
GAZ-2330 → GAZ-233014	
GAZ-5903 → BTR-80	
GAZ-5923 → BTR-90	
GAZ-33081 → GAZ-3308	
GAZ-59032 → K1Sh1	
GAZ-233014 → GAZ-2330	22
GAZ-233036 → GAZ-233014	
GAZ-233114 → AMN-233114	51
GAZ-233114 → GAZ-2330	22
GECKO → 9K33	40
Geran-1 → Geran-2	87
Gerber → Gerbera	88
GIANT → 9M82	36
GIANT → S-300V	36
Giatsint-B → 2A36	32
Giatsint-S → 2S5	29
Gibka-S → 9K334	
GIMLET → 9K310	
GIZMO → 9K333	
GLADIATOR → 9M83	36
GLADIATOR → S-300V	36
Goat (English) → UAZ-469	
GOLLUM → 9K317M	37
GOPHER → 9K35	40
Grad → 9K51	26
Grad-1 → 9K55	26
Grad-1 → 9K55-1	26
Grad-1 → 9P138	26
Grad-1P → 9P132	26
Grad-K → 2B26	26
Grad-P → 9P132	26
Grad-V → 9K54	26
GRAIL → 9K32	
Gran → KM-8	
Granat-4 → Granat-1	79
GREMLIN → 9K34	
GREYHOUND → 96K6	39
GREYHOUND → Pantsir-S2	39
GRILL PAN → 9S32	36
GRINCH → 9K338	

NAME OR DESIGNATION	PAGE
GRISON → 2K22	40
GRIZZLY → 9K317	37
GROUSE → 9K38	
GROWLER → S-400	35
Groza-04K → Groza.04.K	
GRUMBLE → S-300PS	
Gusar → TS-3132	
Gvozдика → 2S1	
Hail (English) → 9K51	26
Harpoon-3 → Garpun-3	
Hashim → RPG-32	
HEARTACHE → SPN-4	
HIGH SCREEN → 9S19M2	36
HOT SHOT → 1RL144M	40
Hurricane (English) → 9K57	27
Hurricane-1M (English) → 9K512	27
Hwasong-11A → KN-23	
Hyacinth (English) → 2A36	32
Hyacinth (English) → 2S5	29
Igla → 9K38	
Igla-1 → 9K310	
Igla-S → 9K338	
Imbir → 9S19M2	36
IMR-1 → IMR	58
IMR-2M → IMR-2	58
IMR-2M1 → IMR-2	58
IMR-3M → IMR-3	58
Infauna → RB-531B	54
Inokhodets → Orion	
Invar → 9M119	
Invar-M → 9M119	
IRL-127-1 → SNAR-10	44
Iskander → 9K720	33
Iskander-E → 9K720	33
Iskander-K → 9K720	33
Iskander-M → 9K720	33
Italmas → BM-35	
Iveco LMV → Rys	43
Izdeliye → BAT-2	58
Izdeliye → Lancet	85
Izdeliye 52 → Izdeliye 51	
Judoist → MKTK-1A	
K-16 → VPK-7829	
K-17 → VPK-7829	
K-300P → 3K55	
K300P → 3K55	
K-340P → 3S55	
K-4386 → K-53949	23
K-63968 → K-53949	23
Kalibr → 3M14	
Kama → 9A52-4	28
KamAZ-5350 → KamAZ-4310	75
KamAZ-43269 → BPM-97	24
KamAZ-53501 → KamAZ-4310	75
Kan → 9M117	
Kapustnik-B → 1V126	
Kapustnik-BM → 1V126	
Kasta → 35N6	
Kasta-2-2 → 39N6	
Kasta-2E2 → 39N6	
Kastet → 9K116	
Katyusha → BM-13	

NAME OR DESIGNATION	PAGE
KAZOO → Kh-59	
KENT → Kh-55	
Kh-55SM → Kh-55	
Kh-59M → Kh-59	
Khosta → 2S34	
Khrizantema → 9K123	
Khrizantema → 9M123	
Khrizantema-S → 9K123	
Khrizantema-S → 9P157-2	
KILLJOY → Kh-47M2	
KINGBOLT → Kh-59	
Kinzhalt → Kh-47M2	
KITCHEN → Kh-22	
Kitolov-2 → 30F69	
Kitolov-2M → 30F69M	
Klen-1 → 1V18	
Klen-1 → 1V119	67
Klen-2 → 1V18	
Klesch-G → UMZ-G	
Kleshch-G → UMZ-G	
Klyon-1 → 1V18	
Klyon-1 → 1V119	67
Klyukva → RPG-28	
Knyaz Vandal Novgorodsky → FPV	89
Koalitsiya-SV → 2S35	30
Kobra → 9K112	
KODIAK → Kh-101	
Kokon → 9M114	
Konkurs → 9K111-1	
Konkurs → 9M113	
Konkurs → 9P148	
Konkurs-M → 9K111-1	
Konkurs-M → 9M113M	
Kopie → SPG-9	
Kord → 6P50	
Kornet → 9K135	
Kornet → 9M133	
Kornet-D → 9K135	
Kornet-D → 9P163-3	
Kornet-D1 → 9P163-3	
Kornet-D1 → BMD-4M	13
Kornet-E → 9K135	
Kornet-EM → 9K135	
Kornet-LR → 9P162	
Kornet-P → 9P163M-1	
Kornet-T → 9P162	
Korsar-M → Korsar	83
Kozlik → UAZ-469	
KPTM-3 → PTM-3	
KPVT → KPVT	
Krasnopol → 30F39	
Krasnopol-M1 → 30F39M	
Krasnopol-M2 → K155M	
Krasukha-2 → 1L269	46
Krasukha-20 → 1L269	46
Krasukha-4 → 1RL257	46
Krasukha-4S → 1RL257	46
KrAZ-255B → KrAZ-255	
Kredo-1 → Kredo-1E	
Kredo-M → PSNR-5	68
Kryuk → RPG-30	
KUB-2 → KUB	86

NAME OR DESIGNATION	PAGE
KUB-10E → KUB	86
Kub-BLA → KUB-BLA	
Kupol → 9S18	37
Kushetka-B → R-149BMR	64
Kvartet → 9P163M-2	
KVN → FPV	89
Ladoga → VTS	66
Lancet → Izdeliye 51	
Lancet-3 → Izdeliye 51	
Leer-2 → 85Ya6	51
Leer-3 → RB-341V	52
Legenda-2 → R-439	
Legenda-2P → R-439	73
Leopard → SNAR-10	44
Lepeshtok → PFM-1	
Lesoshek → RP-377UVM1	
Less → RB-105B	
Linza → KamAZ-53949	
Loaf (English) → UAZ-452	
Lorandit → RP-377L	
Lorandit-M → RP-377L	
LSTC-1944 → Sarmat-2	
Luna-M → 9K52	
Lyagushka → OZM-72	
Lynx (English) → 1RL239	43
M-21 → 9K51	26
M-210F → 9M22U	26
M-21V → 9K54	26
M1931 → B-4	
M-1937 → ML-20	
M-1938 → M-30	
M1943 → D-1	
M1944 → BS-3	
M1949 → M-160	
M1964 → 9K51	26
M1964 → BM-21	26
M-1967 → BMP-1	11
M1970 → BMD-2	14
M1970 → BTR-80	20
M1970 → Object 6	17
M1973 → 2S3	
M1974 → 2S1	
M1974/1 → 1V13	67
M1975 → 2S4	31
M1975 → 2S7	31
M1975 → 9K54	26
M1975 → 9P125	26
M1976 → 2A36	32
M1976 → 9K55	26
M1976 → 9P138	26
M1976/2 → BRM-1K	13
M1977 → 9K57	27
M1977 → 9P140	27
M1978 → BTR-80	20
M1979 → UR-77	59
M1979 → Object 925	16
M1979-2B → 1V18	
M1979-2B → 1V119	67
M1981 → 2S5	29
M1981 → BMP-2	12
M1981/1 → BMD-2	14
M1987 → 2A65	32

NAME OR DESIGNATION	PAGE
M1989 → T-80B/BV/BVM	8
M1990 → ZS19	30
Malka → ZS7M	31
Malva → ZS43	30
Malysh → 9M22M	
Malyutka → 9K11	
Malyutka-2 → 9K11	
Malyutka-M → 9P122	
Malyutka-P → 9P133	
Mandat → R-330	
Mandat-B → R-330B	47
MAZ-7310 → MAZ-543	
MDK-3 → BTM-3	60
Mech → P-35	
Medallion (English) → POM-3	
Medalyon → POM-3	
Mercury-BM → 1L262E	
Meteorit → UR-77	59
Metis → 9K115	
Metis → 9M115	
Metis-M → 9K115-1	
Metis-M → 9M131	
Metis-M1 → 9K115-2	
Model → BMP-1	11
Molniya → Molniya-1	
Molniya-1 → Molniya	88
Molniya-2 → Molniya-1	
Moskva-1 → 1L267	47
Motovoz-M → Ural-4320-7810	
MP → 9S931	69
MP32M1 → MP-32M1	
MP-K → 9S931-1	69
MRU-B → 9S932-1	69
Msta-B → 2A65	32
Msta-S → 2S19	30
Msta-S → 2S19M1	30
Msta-SM → 2S19M2	30
MT-LB VMIK → MT-LBVM	
MT-LBM-6MA → MT-LBM 6MA	
MT-LBM-6MB → MT-LBM 6MB	
MT-Lbu → Object 10	
MT-LBVM1K → MT-LBVM	
MT-LBVM1K → Object 6	17
MT-LBVMK → MT-LBVM	
MTK → MTK-2	
MTO-UB → MTO-UB2	
MTP-A2 → MTP-A2.1	
Murmansk-BN → GT-01	45
Mustang → KamAZ-5350	
Mustang → KamAZ-4310	75
Nard → PRP-4	68
Nashshab → RPG-32	
Nebo-M → 5S2h6M	41
Nebo-SV → 1L13-3	
Nebo-SVU → 1L119	
Nebo-U → 5S2h6U	
Nebo-UM → 5S2h6UM	
Netto → RPG-22	
Niobium-SV → 1L125	41
Nokaut-2 → PTM-4	
Nona-K → ZB16	
Nona-S → ZS9	

NAME OR DESIGNATION	PAGE
Nona-SVK → ZS23	
NSVS → NSV	
NSVT → NSV	
Obj → BMD-1KSh-A	66
Object 6MA → MT-LBM 6MA	
Object 6MB → MT-LBM 6MB	
Object 55 → GSP	
Object 137 → T-54	
Object 148 → T-14	
Object 155 → T-54	10
Object 166 → T-62	
Object 172M → T-72	
Object 183 → BMPT-72	
Object 184 → T-72B	
Object 188 → T-90	
Object 197 → BMR-3M	
Object 199 → BMPT	11
Object 216 → ZS7	31
Object 219AS → T-80U	
Object 219R → T-80B	
Object 256 → UR-77	59
Object 303 → ZS3	
Object 305 → ZS4	31
Object 307 → ZS5	29
Object 316 → ZS19	30
Object 318 → GMZ-3	61
Object 434 → T-64A	
Object 434 → T-64A/B/BV	9
Object 447 → T-64B	
Object 447A → T-64A/B/BV	9
Object 454 → BAT-2	
Object 501 → BRM-3K	43
Object 564 → BMO-T	
Object 602 → MTU-20	
Object 608 → BREM-1	
Object 632 → MTU-72	55
Object 634 → TOS-1	29
Object 634B → TOS-1A	29
Object 637 → IMR-2	58
Object 675 → BMP-2	
Object 676 → BRM-1K	
Object 688 → BMP-3	
Object 765 → BMP-1	
Object 767 → PRP-3	
Object 779 → PRP-4	68
Object 916 → BMD-2	14
Object 950 → BMD-3	
Oborona-14 → 5N8AA	
Obzor-3 → 9S15M	36
Oniks → P-800	
Onyx (English) → P-800	
Oplot-T → MAZ-74101	
Orlan-10 → BKAR	78
Orlan-30 → BKAR	78
Osa → 9K33	40
Osa-AK → 9K33	40
Osa-AKM → 9K33	40
OTR-21 → 9K79	
Ovod → 9S80	40
Ovod → Kh-59	
P-18M → P-18	
P-37 → P-35	

NAME OR DESIGNATION	PAGE
P-240 → P-240I-4	71
P-240BTZ → P-240I-4	71
P-240I-4 → P-240	
P-260 → P-260-U	
P-260-T → P-260-O	74
P-260-U → P-260-O	74
PAK-200M → PAK-200	
Palantin-K → Palantin	49
Pantera → SNAR-10M	44
Pantsir-S1 → 96K6	39
Pantsir-SM → Pantsir-S1M	39
Partizan → 9P132	26
Patrul → Asteis-7020	
Penicillin → 1B75	43
Peony (English) → ZS7	31
Pereselenets → P-240	
Pereselenets → P-240I-4	71
Pereselenetz → P-240	
Pereselents → P-240I-4	71
Petal (English) → PFM-1	
PFM-1S → PFM-1	
Phlox (English) → ZS40	
Pion → ZS7	31
Plamya → AGS-17	
Plastun-SN → GAZ-3344-20	19
Plavsk → MKTK-1	
PM-38 → M-1938	
PMN-2M → PMN-2	57
PMN-1 → PMN	
PMN-2 → PMN	
PMN-4 → PMN	
PMP-M → PMP	56
Podlet-K1 → 48Ya6-K1	41
Podnos → ZB14	
Polisk → R-161	
Pole-21 → R-340RP	53
Pole-21E → R-340RP	53
Pole-21M → R-340RP	53
Polyana-D4 → 9S52	68
Polyana-D4M1 → 9S52	68
POM-2S → POM-2	
Povozka → RKHM-6	
Povozka → CBRN	62
PP-2005 → PP-91M	56
PPRU-1 → 9S80	40
PPRU-1M → 9S80	40
Prima → 9A51	
Prima → 9K59	
Product → BM-35	86
Prokhod-1 → BMR-3M	59
Prometey → S-500	
Prometheus (English) → S-500	
Proryv-3 → T-90M	
Proryv-3 → T-90	7
Protivnik-G → 59N6	
Protivnik-GE → 59N6	
Provod → FPV	89
PRP-4MU → PRP-4M	68
PSNR-5M → PSNR-5	68
PSNR-8 → Kreda-1E	
PSNR-8M → Kreda-M1	
PSNR-9 → Kreda-M1	

NAME OR DESIGNATION	PAGE
PTM-1G → PTM-1	
PTM-1S → PTM-1	
PTS → PTS-2	57
PTS-M → PTS-2	57
PU-12M → PU-12	69
PU-12M6 → PU-12	69
PU-12M7 → PU-12	69
PZM-2 → BTM-3	60
Qods Mohajer-6 → Mohajer-6	84
R-17 → 9K72	
R-142 → R-142N	71
R-142NM → R-142N	
R-142NMR → R-142N	
R-142NSA → R-142N	
R-145BM1 → R-145BM	65
R-145BM → R-145BM	65
R-149BMRg → R-149BMR	64
R-149MA1 → R-149MA3	64
R-161A2 → R-161	
R-161A2M → R-161	
R-166-0.5 → R-166	
R-166-05 → R-166	70
R-166S → R-166	
R-168E → R-168	
R-187-P1 → R-187	
R-300 → 9K72	
R-325BMV → R-325UMV	48
R-378B → R-378A	
R-381T2M → R-381T	
R-381TM → R-381T	
R-412A → R-410-7.5	72
R-412AM → R-412A	
R-412AM → R-410-7.5	72
R-416G-MS → R-416GM	72
R-416GM2 → R-416GM	72
R-417S → R-417	
R-439-BK → R-439	
R-439-KUL → R-439	
R-439-P → R-439	
R-500 → 9M728	33
Rakushka → BTR-MDM	
Rakushka → 9S931-2	15
Rakushka-M → 9S931-2	15
Ranzhir → 9S737	69
Rapira → MT-12	
RB-371A → Tirada-2S	
RB-636 → RB-636AM2	
Redut-2US → P-260-0	74
Refleks → 9K119	
Refleks → 9M119	
Refleks-M → 9K119	
regimental digging machine (English) → PZM-2	
Remdiesel Z-STs Akhmat → Z-STs	
Resonance-N (English) → Rezonans-N	
Rezonans-N → Rezonans-N	
Rezonans-NE → Rezonans-N	
Rezonir → R-412A	
Rezonir → R-410-7.5	72
ribbon bridge (English) → PMP	56
RIDA Buran → Buran	
RKhM-4 → CBRN	62

NAME OR DESIGNATION	PAGE
RKhM-4-01 → CBRN	62
Robot → Object 925	16
Rostok → BTR-90	
RP-200 → Kreda-1E	
RP-377LA → RP-377L	
RP-377UM2 → RP-377UVM1	
RP-377UVM2 → RP-377UVM1	
RPG-7V → RPG-7	
RPG-7V2 → RPG-7	
RPO → RPO-A	
RPO-M → RPO-A	
RS-SS-28 MOD → Oreshnik	
Rtut-BM → 1L262E	
Rusich → KZKT-7428	
Ruta → MT-12R	
Rys → 1RL239	43
Rys → BRM-3K	43
S-300PM1 → S-300PM	
S-300PM2 → S-300PM	
S-300PMU → S-300PS	
S-300PMU1 → S-300PM	
S-300PMU2 → S-300PM	
S-300VMD → S-300V4	36
S350M → S350	79
SA-7 → 9K32	
SA-8 → 9K33	40
SA-10 → S-300PS	
SA-11 → 9K37	37
SA-11 → 9K37M1-2	37
SA-12A → 9M83	36
SA-12A → S-300V	36
SA-12B → 9M82	36
SA-12B → S-300V	36
SA-13 → 9K35	40
SA-14 → 9K34	
SA-15 → 9K330	38
SA-15 → 9K331	38
SA-15 → 9K331MDT	38
SA-15 → 9K331MU	38
SA-15 → 9K332	38
SA-16 → 9K310	
SA-17 → 9K317	37
SA-18 → 9K38	
SA-19 → 2K22	40
SA-20 → S-300PM	
SA-21 → S-400	35
SA-22 → 9K6K6	39
SA-22 → Pantsir-S2	39
SA-23 → S-300V4	36
SA-24 → 9K338	
SA-27 → 9K317M	37
SA-29 → 9K333	
Sadko → GAZ-3308	
Sadovnitsa → BMD-4M	13
SAGARIS → 3M14	
SAGGER → 9K11	
Sani → 2S12	
Sania → Saniya	
Santimetr → 30F38	
Santimetr-M → 30F75	
Sarmat → Sarmat-2	
Sarmat-3 → Sarmat-2	

NAME OR DESIGNATION	PAGE
Saturn → P-35	
SAXHORN → 9K115	
SAXHORN → 9M115	
SAXHORN-2 → 9K115-1	
SAXHORN-2 → 9K115-2	
SAXHORN-2 → 9M131	
SBM → VPK-233136	
Sborka → 9S80	40
SCALLION → 9A4172	
SCARAB → 9K79	
SCARAB → 9M79	
SCREWDRIIVER → 9K720	33
SCREWDRIIVER → 9M729	33
SCUD B → 9K72	
Searcher II → Forpost	
SENNIGHT → 3K60	
Serp → SERP-VS6D	
Shahed-131 → Geran-1	
Shahed 131 → Geran-1	
Shahed-136 → Geran-2	
Shahed 136 → Geran-2	
Sheksna → 9K116	
Shilka → ZSU-23-4	
Shkval → BMP-1U	
Shmel → RPO-A	
Shturm → 9K114	
Shturm-S → 9P149	
Shturm-SM → 9K132	
Silok → Silok-01	
Sinitsa → R-934B	47
Sinitsa → R-934UM	47
Skrezhet → Object 925	16
Sled (English) → 2S12	
Slepok-1 → 9S729M1	28
SMALL FRED (disputed) → 1RL239	43
SMALL FRED (disputed) → PRP-3	
Smelchak → 1K113	31
Smerch → 9A52	28
Smerch → 9K58	28
Smerch-M → 9A52-2	28
SNAR-10M1 → SNAR-10M	44
SNIPER → 9K119	
SNIPER → 9M119	
SNOW DRIFT → 9S18	37
Sobolyatnik → 1L277	
Solntsepek → TOS-1A	29
SONGSTER → 9K112	
Sosna → 9M340	
Sosna-R → 9M340	
SOUTHPAW → 9K720	33
SOUTHPAW → 9M728	33
SPANDREL → 9K111-1	
SPANDREL → 9M113	
SPANDREL-B → 9M113M	
Spartak → BPM-97	24
Spektr → IV119	
SPG-9D → SPG-9	
SPG-9M → SPG-9	
SPIGOT → 9K111	
SPIGOT → 9M111	
SPIRAL → 9K114	
SPIRAL → 9M114	

NAME OR DESIGNATION	PAGE
SPIRAL-2 → 9K132	
SPIRAL-2 → 9M120	
Spiridon → BMPT	11
SPM-2 → GAZ-233036	
SPN-4M → SPN-4	
SPOON REST D → P-18	
SPR-2M → 1L262E	
SPRIGGAN → 9K135	
SPRIGGAN → 9M133	
SPRINGER → 9K123	
SPRINGER → 9M123	
Sprut-B → 2A45M	
Sprut-SD → 2S25	
SQUAT EYE → 39N6	
SS-1c → 9K72	
SS-21 → 9K79	
SS-21 → 9M79	
SS-26 → 9K720	33
SS-26 → 9M723	33
SS-N-26 → P-800	
SS-N-27 → 3M14	
SS-N-30A → 3M14	
SSC-5 → 3K55	
SSC-6 → 3K60	
SSC-7 → 9K720	33
SSC-7 → 9M728	33
SSC-8 → 9K720	33
SSC-8 → 9M729	33
STABBER → 9K117	
STONE → 9K720	33
STONE → 9M723	33
STOOGEE → 3K55	
Strela-2 → 9K32	
Strela-3 → 9K34	
Strela-10 → 9K35	40
Strela-10M3 → 9K35	40
Strelets → KRUS-VR	
Strelets → KRUS	74
Strelets-M → KRUS-VR	
Strelets-M → KRUS	74
STROBILE → P-800	
STS Sarmat → Sarmat-2	
Supercam → S350	79
Svet-KU → RB-636AM2	
Svet-VSG → RB-636 Svet-VSG	
Svir → 9K119	
Svir → 9M119	
SWINGER → 9K116	
T-12A → MT-12	
T-55 → T-54	10
T-55AMV → T-55AM	
T-62M → T-62	9
T-64BV → T-64B	
T-72BIK → T-72B1	
T-72BU → T-90	7
T-72M → T-72M1	
T-72 Ural → T-72	
T-80B → T-80B/BV/BVM	8
T-80U → T-80B/BV/BVM	8
T-90SK → T-90S	
Tachyon → Takhion	81
Takhion → Tachyon	

NAME OR DESIGNATION	PAGE
TALL KING C → 5N84A	
TALL MIKE → PSNR-5	68
TALL RACK → 5S2h6U	
Taran → R-381T	
Taran-M → R-381T	
Tavolga → RPG-27	
TDA-2M → TDA-2K	
TDA-2M → TDA-3	63
TDM-2K → TDA-3	63
Terek → P-18	
Terminator → BMPT	
Terminator → BMPT-72	
Terminator-2 → BMPT-72	
Tigr → GAZ-233014	
Tigr → GAZ-2330	22
Tigr-M → AMN-233114	51
Tigr-M → GAZ-2330	22
Tigr-M SpN → ASN-233115	51
Tigr SBM → VPK-233136	
Tigr SPM-2 → GAZ-233036	
Tirada-2 → Tirada-2S	
Tirada-2.3 → Tirada-2S	
TKB-0175 → RPG-29	
TM-62M → TM-62	
TMM → TMM-3	55
TMS-65M → TMS-65	
TMS-65U → TMS-65	
Tochka → 9K79	
Tochka-U → 9K79	
Tor → 9K330	38
Tor-M1 → 9K331	38
Tor-M1-ZU → 9K331MU	38
Tor-M2 → 9K332	38
Tor-M2DT → 9K331MDT	38
Tor-M2U → 9K331MU	38
Torf → R-412A	
Torf → R-410-75	72
Torn-MDM → Torn	
Tornado [English] → 9K58	28
Tornado-G → 2B17M	26
Tornado-G → 9K51M	26
Tornado-S → 9A54	28
Tornado-S → 9K515	28
Tornado-U → Ural-63706-0122	
Tosochka → TOS-2	29
Tray [English] → 2B14	
Triumf → S-400	35
Triumfator-M → S-500	
Tulip [English] → 2S4	31
Tunguska → 2K22	40
Tunguska-M1 → 2K22	40
Typhoon-K → KamAZ-53949	
Typhoon-K → KamAZ-63968	
Typhoon-K → K-53949	23
Typhoon-K 4x4 → KamAZ-53949	
Typhoon-K 6x6 → KamAZ-63968	
Typhoon-U → Ural-63095	
Typhoon-U → K-53949	23
Typhoon-VDV → KamAZ-4386	
Tyulpan → 2S4	31
UAZ-469RKhB → UAZ-469RKh	
UAZ-3151 → UAZ-469	

NAME OR DESIGNATION	PAGE
Ulibka-M → RPMK-1	
Ulybka → RPMK-1	
UMZ-3 → UMZ	61
UMZ-K → UMZ-G	
UMZ-T → UMZ-G	
UNSh → K1Sh1	
Uragan → 9K57	27
Uragan → 9P140	27
Uragan-1 → 9K57-1	27
Uragan-1M → 9A53	27
Uragan-1M → 9K512	27
Ural → T-72	
Ural-375 → Ural-375D	
Ural-6370-0121-30 → Ural-6370	
Ural-43203 → Ural-4320	
Ural-63706-0010 → Ural-6370	76
Ural-63706-0011 → Ural-63706-0122	
Ural-VV → Ural-432009	
Utyos → NSV	
V-500R → 5V55R	
Val → PRP-3	
Vampir → RPG-29	
Vasilek → 2B9	
Vedma → OZM-72	
Vena → 2S31	
Vepr → BMR-3M	
Verba → 9K333	
Vikhr → 9A4172	
Vityaz → DT-10PM	36
Vityaz → DT-30	36
Vityaz → S-350	36
Vladimir → T-90	7
Vodnik → GAZ-39371	
Voronezh-DM → 77Va6-DM	
VPK-233136 → GAZ-233014	
VPK Ural → AMN-590951	
VT-40 → FPV	89
VTS Ladoga → Ladoga	
Vystrel → KamAZ-43269	
Vystrel → BPM-97	24
Witch [English] → OZM-72	
Yastreb-AV → 1K148	
YeSU TZ → ESU TZ	
Z-16 → ZALA 421-16	
Z-16 → ZALA	77
ZAK-57 → 2S38	
Zemledeliye → ISDM	61
Zhitel → R-330Zh	50
Zhuk → IRM	
Zoopark-1 → 1L219	42
Zoopark-1M → 1L260	42
ZU-23 → ZU-23-2	

Table 1c. Cyrillic designations and names

Sorted in Russian alphabetical order, each pointing (→) to its row in Table 1.			
ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE	ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
1Б44 → 1В44	28	2С34 «Хоста» → 2S34	
1В119 → 1V119		2С35 «Ковалиция-СВ» → 2S35	30
1В119 → Object 925	16	2С38 → 2S38	
1В126 «Капустник-Б» → 1V126		2С3 «Акация» → 2S3	
1В13 → 1V13	67	2С3М → 2S3М	
1В152 → 1V152		2С40 → 2S40	
1В18 → 1V119	67	2С43 «Мальва» → 2S43	30
1В19 → 1V119	67	2С4 «Тюльпан» → 2S4	31
1К113 «Смелычак» → 1K113	31	2С5 «Гуацинт-С» → 2S5	29
1К148 «Ястреб-АВ» → 1K148		2С5М → 2S5М	29
1Л122 «Гармония» → 1L122		2С6 → 2S6	40
1Л13 → 1L13-3		2С6М → 2S6	40
1Л219 «Зоопарк-1» → 1L219	42	2С6М1 → 2S6	40
1Л219М → 1L219	42	2С7М «Малка» → 2S7М	31
1Л259 → 1L259	42	2С7 «Пион» → 2S7	31
1Л260 → 1L260	42	2С9 «Нона-С» → 2S9	
1Л2623 «Ртуть-БМ» → 1L262E		2Ф77 → 2F77	40
1Л265 → 1L265	47	2Ф77М → 2F77	40
1Л266 → 1L266	47	35Н6 «Касма» → 3SН6	
1Л269 «Красуха-2» → 1L269	46	39Н6Е «Касма-2Е2» → 39N6	
1Л271 «Аистенок» → 1L271	43	39Н6 «Касма-2-2» → 39N6	
1Р131 → P-18		3К13 → 3BK13	
1Р133-3 «Крего-1» → Kredo-1E		3К19 → 3BK19	
1Р134 → P-19		3ВФ4 → 1K113	31
1Р144М1 → 1RL144М	40	3К55 → 3K55	
1Р132-1М «Пантера» → SNAR-10М	44	3К60 → 3K60	
1Р132 «Леопард» → SNAR-10	44	3М-14 → 3M14	
1Р13239 / АРК-1 / «Рысь» → 1RL239	43	3М16 → 3M16	26
1Р13257 «Красуха-4» → 1RL257	46	3-0-13 → 3-0-13	
1Т12-2М → 1T12-2М	28	3-0-23 → 3-0-23	
2А13 → ZU-23-2		3-0-8 → 3-0-8	
2А18М → D-30		30Ф24 → 30F24	
2А19 → T-12		30Ф25 → 30F25	
2А29 → MT-12		30Ф29 → 30F29	29
2А36 «Гуацинт-Б» → 2A36	32	30Ф38 → 30F38	
2А6 → ZSU-23-4		30Ф39 → 30F39	
2А65 «Мста-Б» → 2A65	32	30Ф39М → 30F39М	
2В11 → 2В11		30Ф43 → 30F43	31
2В14 «Погнос» → 2В14		30Ф44 → 30F44	31
2В16 «Нона-К» → 2В16		30Ф45 → 30F45	30
2В17 → BM-21-1	26	30Ф49 → 30F49	
2В17М → 2В17М	26	30Ф50 → 30F50	
2В26 → 2В26	26	30Ф56 → 30F56	
2В5 → BM-21	26	30Ф59 → 30F59	29
2В9 «Василёк» → 2В9		30Ф61 → 30F61	
2К22 → 2K22	40	30Ф64 → 30F64	
2К22М1 «Тунгуска-М1» → 2K22	40	30Ф75 → 30F75	
2К22М «Тунгуска-М» → 2K22	40	3РБ30 → 3RB30	
2С12А → 2S12		3Ф2 → 3F2	31
2С12 «Сани» → 2S12		3Ф5 → 1K113	31
2С19М1 → 2S19М1	30	40Н6Е → 40N6	35
2С19М2 → 2S19М2	30	40Р6 → S-400	35
2С19 «Мста-С» → 2S19	30	421-16Е → ZALA	77
2С1 «Гвоздика» → 2S1		421-16Е2 → ZALA	77
2С23 «Нона-СВК» → 2S23		48Н6ДМ → 48N6	35
2С25 → 2S25		48Н6Е → 48N6	35
2С31 «Вена» → 2S31		48Н6Е2 → 48N6	35
2С33 «Мста-СМ» → 2S19М2	30	48Я6-К1 «Погнёт» → 48Ya6-K1	41
		50К6А → 50K6A	36
		50Н6А → 50N6A	36
		50П6А → 50P6A	36
		50Р6А → S-350	36
		51П6 → 51P6	35
		53-0Ф-В43Б → OF-B43B	
		55Ж6МЕ → 55Zh6M	41
		55Ж6М «Небо-М» → 55Zh6M	41
		55К6 → 55K6	35
		55К6Е → 55K6	35
		55Р6М → S-500	
		5736Е → 57E6	39
		5В55К → 5V55R	
		5В55Р → 5V55R	
		5Н63С → 5N63S	
		5Н84А «Оборона» → 5N84A	
		5Н85Д → 5P85SD	
		5Н85С → 5P85SD	
		5Н85СМ2-01 → 5P85SM2-01	35
		5Н85Т2 → 5P85T2	35
		75Р6 → S-300PS	
		83м215-ВР → KRUS	74
		85Я6 «Леер-2» → 85Ya6	51
		91Н6Е → 91N6A	35
		92Н6Е → 92N6A	35
		96К6 → 96K6	39
		96А6Е → 96L6	42
		9А310 → 9A310	37
		9А310М1-2 → 9A310M1-2	37
		9А316 → 9A316	37
		9А317 → 9A317	37
		9А317М → 9A317M	37
		9А330 → 9A330	38
		9А331 → 9A331	38
		9А331-1 → 9A331	38
		9А331М2 → 9A331M	38
		9А335 → 9A33B	40
		9А35 → 9A35	40
		9А38 → 9A310	37
		9А39 → 9A39	37
		9А39М1 → 9A39	37
		9А51 → 9A51	
		9А52 → 9A52	28
		9А52-2 → 9A52-2	28
		9А52-2Т → 9A52-2T	28
		9А52-4 → 9A52-4	28
		9А53 → 9A53	27
		9А54 → 9A54	28
		9А82 → 9A82	36
		9А83 → 9A83	36
		9А84 → 9A84	36
		9А85 → 9A85	36
		9К11 → 9K11	
		9К111 → 9K111	
		9К111-1 → 9K111-1	
		9К114 → 9K114	
		9К115 → 9K115	
		9К115-1 → 9K115-1	

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE	ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE	ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
9K115-2 → 9K115-2		9M22 → 9M22	26	9M83 → 9M83	36
9K116 → 9K116		9M22M → 9M22M		9M96E → 9M96	35
9K119M → 9K119		9M22C → 9M22S	26	9M96E2 → 9M96	35
9K123 → 9K123		9M22Y → 9M22U	26	9P122 → 9P122	
9K123-1 → 9K123		9M27A → 9M27D	27	9P125 → 9P125	26
9K128-1 → 9P162		9M27K → 9M27K	27	9P129 → 9P129	
9K132 → 9K132		9M27K1 → 9M27K1	27	9P132 → 9P132	26
9K135 → 9K135		9M27K2 → 9M27K2	27	9P133 → 9P133	
9K310 → 9K310		9M27K3 → 9M27K3	27	9P138 → 9P138	26
9K317 «Бук-М2» → 9K317	37	9M27C → 9M27S	27	9P139 → 9P139	26
9K317M «Бук-М3» → 9K317M	37	9M27D → 9M27F	27	9P140 → 9P140	27
9K32 → 9K32		9M28A → 9M28D	26	9P148 → 9P148	
9K33 → 9K33	40	9M28K → 9M28K	26	9P149 → 9P149	
9K330 → 9K330	38	9M28C → 9M28S	26	9P149M → 9K132	
9K331 «Тор-М1» → 9K331	38	9M28D → 9M28F	26	9P151 → 9K115	
9K332 3PK «Тор-М2» → 9K332	38	9M311 → 9M311	40	9P157 → 9P157-2	
9K333 → 9K333		9M313 → 9M313		9P157-2 → 9P157-2	
9K338 → 9K338		9M317 → 9M317	37	9P162 → 9P162	
9K33M2 → 9K33	40	9M33 → 9M33	40	9P163-3 → 9P163-3	
9K33M3 → 9K33	40	9M330 → 9M330	38	9P163M-1 → 9P163M-1	
9K34 → 9K34		9M331 → 9M331	38	9P163M-2 → 9P163M-2	
9K35M3 «Стрела-10M3» → 9K35	40	9M333 → 9M333	40	9P78-1 → 9P78-1	33
9K35 «Стрела-10» → 9K35	40	9M335 → 9M335	39	9C15M → 9S15M	36
9K37 → 9K37	37	9M336 → 9M336	37	9C18 → 9S18	37
9K37M1-2 «Бук-М1-2» → 9K37M1-2	37	9M338 → 9M338	38	9C18M1 → 9S18	37
9K37M1 «Бук-М1» → 9K37	37	9M33M3 → 9M33	40	9C19M2 → 9S19M2	36
9K38 → 9K38		9M340 → 9M340		9C32 → 9S32	36
9K512 «Ураган-1М» → 9K512	27	9M342 → 9M342		9C36M → 9S36	37
9K515 → 9K515	28	9M37 → 9M37	40	9C457 → 9S457	36
9K51 «Град» → 9K51	26	9M38 → 9M38	37	9C470 → 9S470	37
9K51M → 9K51M	26	9M39 → 9M39		9C510M → 9S510	37
9K51M «Торнадо-Г» → 9K51M	26	9M42 → 9M42	26	9C52 → 9S52	68
9K52 → 9K52		9M43 → 9M43	26	9C552 → 9S552	33
9K54 «Град-В» → 9K54	26	9M51 → 9M51	27	9C729M1 «Сенек-1» → 9S729M1	28
9K55-1 «Град-1» → 9K55-1	26	9M519 → 9M519	26	9C920 → 9S920	33
9K55 «Град-1» → 9K55	26	9M521 → 9M521	26	9C931 → 9S931	69
9K57 «Ураган» → 9K57	27	9M522 → 9M522		9C931-2 → 9S931-2	15
9K58 «Смерч» → 9K58	28	9M525 → 9M525	28	9C932-1 → 9S932-1	69
9K59 «Прима» → 9K59		9M528 → 9M528	28	9T2175 → 9T217	40
9K72 → 9K72		9M529 → 9M529	28	9T218 → 9T218	
9K720 → 9K720	33	9M531 → 9M531	28	9T232M → 9T232M	
9K79 → 9K79		9M533 → 9M533	28	9T234 → 9T234	28
9K79-1 → 9K79		9M534 → 9M534	28	9T234-2 → 9T234-2	28
9K81 → S-300V	36	9M53K → 9M53K		9T234-2T → 9T234-2T	28
9M100 → 9M100	36	9M53C → 9M53S		9T234-4 → 9T234-4	28
9M111 → 9M111		9M53D → 9M53F		9T244 → 9T244	38
9M113 → 9M113		9M542 → 9M542	28	9T249 → 9T249	27
9M113M → 9M113M		9M544 → 9M544	28	9T250 → 9T250	33
9M114 → 9M114		9M549 → 9M549	28	9T255 → 9T255	28
9M115 → 9M115		9M55K → 9M55K	28	9T450 → 9T450	26
9M117 → 9M117		9M55K1 → 9M55K1	28	9T451 → 9T451	26
9M119M → 9M119		9M55K4 → 9M55K4	28	9T452 → 9T452	27
9M119M1 → 9M119		9M55K5 → 9M55K5	28	«Абмобага» → 1L222	
9M120 → 9M120		9M55C → 9M55S	28	Аглень → RPG-26	
9M120-1 → 9M120-1		9M55D → 9M55F	28	AGC-17 → AGS-17	
9M123 → 9M123		9M59 → 9M59	27	AGC-30 → AGS-30	
9M123D → 9M123		9M723 → 9M723	33	AGC-40 → AGS-40	
9M131 → 9M131		9M728 → 9M728	33	A3П-57 → S-60	
9M133 → 9M133		9M729 → 9M729	33	Ak8egyik → R-168	
9M14 → 9K11		9M79 → 9M79		Алеуи → GAZ-3344-20	19
9M217 → 9M217	26	9M82 → 9M82	36	«Алмаец» → Altaets	
9M218 → 9M218	26	9M82M → 9M82	36	AMH-233114 → AMN-233114	51

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
АМН-590951 → АМН-590951	
Армама → Armata	
АРС-14 → ARS-14	63
АРС-14КМ → ARS-14	
Артек → R-166	
АСН-233115 → ASN-233115	51
Астролог → MZKT-7930	39
АТС-59Г → ATS-59G	
Б-4 → B-4	
Бал → 3К60	
Балкан → AGS-40	
Бал-3 → 3К60	
Барнаул-Т → Barnaul-T	69
Бастион → 3К55	
Басурманин → BMP-1AM	
БАТ-2 → BAT-2	
Берёжок → BMP-2M	
БМ-1 → BM-1	29
БМ-13 → BM-13	
БМ-21 → BM-21	26
БМ-21-1 → BM-21-1	26
БМ-21В → 9P125	26
БМД-1 → BMD-1	
БМД-1КШ → BMD-1KSh-A	66
БМД-1КШ → Object 925	16
БМД-1КШ-А → BMD-1KSh-A	66
БМД-1П → BMD-2	14
БМД-1Р → Object 925	16
БМД-2 → BMD-2	
БМД-2К → BMD-2	14
БМД-3 → BMD-3	
БМД-4 → BMD-4	
БМД-4 → BMD-4M	13
БМД-4М → BMD-4M	
БМК-460 → BMK-460	
БМО-Т → BMO-T	
БМП-1 → BMP-1	
БМП-1АМ → BMP-1AM	
БМП-1АМ → BMP-1	11
БМП-1П → BMP-1P	
БМП-1П → BMP-1	11
БМП-1У → BMP-1U	
БМП-2 → BMP-2	
БМП-2Д → BMP-2D	
БМП-2Д → BMP-2	12
БМП-2К → BMP-2	12
БМП-2М → BMP-2M	
БМП-3 → BMP-3	
БМП-3М → BMP-3M	
БМП-3Ф → BMP-3F	
БМПТ → BMPT	
БМПТ-72 → BMPT-72	
БМПТ-72 → BMPT	11
БМР-3М → BMR-3M	
БМР-3МА → BMR-3M	59
БПМ-97 → KamAZ-43269	
БПМ-97 → BPM-97	24
БРДМ-2 → BRDM-2	
БРДМ-2А → BRDM-2A	
БРДМ-2А → BRDM-2	22
БРДМ-2М → BRDM-2M	
БРДМ-2М → BRDM-2	22

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
БРДМ-2МС → BRDM-2MS	
БРМ-1 → BRM-1K	
БРМ-1К → BRM-1K	
БРМ-1КМ → BRM-1K	13
БРМ-3К → BRM-3K	43
БРЭМ → BREM	62
БРЭМ-1 → BREM-1	
БРЭМ-1 → BREM	62
БРЭМ-1М → BREM	62
БРЭМ-2 → BREM-2	
БРЭМ-Д → BREM-D	
БРЭМ-К → BREM-K	
БРЭМ-Л → BREM-L	
БС-3 → BS-3	
БТМ-3 → BTM-3	
БТР-50 → BTR-50	
БТР-50ПК → BTR-50PK	
БТР-60 → BTR-60PB	65
БТР-60ПБ → BTR-60PB	65
БТР-70 → BTR-70	
БТР-70 → BTR-80	20
БТР-70М → BTR-70M	
БТР-70М → BTR-80	20
БТР-80 → BTR-80	
БТР-80А → BTR-80A	
БТР-82 → BTR-82	
БТР-82А → BTR-82A	
БТР-82АМ → BTR-82A	
БТР-82АТ → BTR-82AT	
БТР-90 → BTR-90	
БТР-Д → BTR-D	
БТР-МДМ → BTR-MDM	
БТС-4А → BTS-4A	
БТС-4В → BTS-4V	
«Булат» → SBA-60K2	
Бумеранг → VPK-7829	
Бумеранг → FPV	89
Буханка → UAZ-452	
Вепрь → BMR-3M	
«Верба» → 9K333	
Владимир → T-90	7
Водник → GAZ-2330	22
«Волкорез» → Volnorez	
«Воронеж-ДМ» → 77Ys6-DM	
ВПК-233136 → VPK-233136	
ВПК-Урал → АМН-590951	
ВСМ-1 → VSM-1	
ВТС → VTS	66
Выстрел → KamAZ-43269	
Выстрел → BPM-97	24
ГАЗ-2330 → GAZ-233014	
ГАЗ-233014 → GAZ-233014	
ГАЗ-233014 → GAZ-2330	22
ГАЗ-233036 → GAZ-233036	
ГАЗ-3307 → GAZ-3307	
ГАЗ-3308 «Саян» → GAZ-3308	
ГАЗ-3344-20 → GAZ-3344-20	19
ГАЗ-41-08 → BRDM-2	
ГАЗ-51 → GAZ-51	
ГАЗ-5903 → BTR-80	
ГАЗ-5923 → BTR-90	
ГАЗ-66 → GAZ-66	

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
Гамма-ДЕ → 67N6	
Гарпун-3 → Garpun-3	
Герань → Geran-2	87
Герань-1 → Geran-1	
Герань-1 → Geran-2	87
Герань-2 → Geran-2	
Гербера → Gerbera	
«Губка-С» → 9K334	
ГМЗ-3 → GMZ-3	61
Гранат → Granat-1	79
Гранат-1 → Granat-1	
Гранат-4 → Granat-4	
Гранат-4 → Granat-1	79
«Грань» → KM-8	
Гроза.04.К → Groza.04.K	
ГСП → GSP	
Д-1 → D-1	
Д-20 → D-20	
Д-30 → D-30	
Д-30А → D-30	
Д-44 → D-44	
Деймос-Н → R-142N	
«Деривация-ПВО» → ZS38	
«Дзюгоист» → MKTK-1A	
ДТ-30 → DT-30	36
ДТ-30ПМ → DT-30	36
ДШК → DShK 38/46	
ЕСУ ТЗ → ESU TZ	
Жук → IRM	57
ЗАК-57 → ZS38	
Застава → Zastava	
Земледелец → ISDM	61
ЗИЛ-131 → ZIL-131	
ЗИЛ-157 → ZIL-157	
«Зоопарк-1М» → 1L260	42
ЗСУ-23-4 → ZSU-23-4	
ЗУ-23 → ZU-23-2	
ЗУ-23-2 → ZU-23-2	
«Игла» → 9K38	
«Игла-1» → 9K310	
«Игла-С» → 9K338	
Изделие → BAT-2	58
Изделие → Lancet	85
Изделие 51/52 → Izdeliye 51	
ИМР-2 → IMR-2	58
ИМР-2М → IMR-2	58
ИМР-3М → IMR-3	58
Инвар → 9M119	
Иноходец → Orion	
ИРМ → IRM	57
ИСДМ → ISDM	61
Искангер → 9K720	33
Искангер-К → 9K720	33
Искангер-М → 9K720	33
Италамас → BM-35	
К1Ш1 → K1Sh1	
КЗООП → 3K55	
К-340П → 3S55	
Калибр → 3M14	
КАМАЗ-4310 → KamAZ-4310	
КАМАЗ-43269 → KamAZ-43269	
КАМАЗ-43269 → BPM-97	24

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
КамаЗ-4386 → K-53949	23
КАМАЗ-5350 → KamAZ-5350	
КАМАЗ-5350 → KamAZ-4310	75
КАМАЗ-53501 → KamAZ-53501	
КамаЗ-53949 → KamAZ-53949	
КамаЗ-53949 → K-53949	23
КАМАЗ-6350 → KamAZ-6350	
КАМАЗ-6350 → KamAZ-4310	75
КАМАЗ-63501 → KamAZ-63501	
КамаЗ-63968 → KamAZ-63968	
КАМАЗ-63968 → K-53949	23
КАМАЗ-65225 → KamAZ-65225	
КАМАЗ-6560 → KamAZ-6560	
Капустник-БМ → Kapustnik-BM	26
Кастет → 9K116	
Кишкэл → Ki-47M2	
«Китолоб-2» → 30F69	
«Китолоб-2М» → 30F69M	
Клокова → RPG-28	
КМ-8 → KM-8	
Князь Вангал Новгородский → FPV	89
Конкурс → 9K111-1	
Конкурс-М → 9K111-1	
Копьё → SPG-9	
Корг → 6P50	
Корнет → 9K135	
Корнет-Д → 9K135	
Корнет-З → 9K135	
Корнет-ЗМ → 9K135	
Корсар → Korsar	
Корсар-М → Korsar	83
КПВ → KPV	
КраЗ-2556 → KrAZ-255	
КраЗ-260 → KrAZ-260	
«Краснополь» → 30F39	
КРУС → KRUS-VR	
КРУС → KRUS	74
Крёлк → RPG-30	
КСАС → 9S935	69
КУБ → KUB	86
КУБ-103 → KUB	86
КУБ-2 → KUB	86
КУБ-БЛА → KUB-BLA	
Кураанец-25 → Kurjanets-25	
Кушетка-Б → R-149BMR	64
Лагога → VTS	66
Ланцет → Izdeliye 51	
Ланцет → Lancet	85
Легенда-2 → R-439	
Линза → KamAZ-53949	
Луна-М → 9K52	
М-160 → M-160	
М-30 → M-30	
М-46 → M-46	
МАЗ-537 → MAZ-537	
МАЗ-543/МАЗ-7310 → MAZ-543	
Manlomka → 9K11	
МДК-3 → MDK-3	
МДК-3 → BTM-3	60
Мерлин-ВР → Merlin-VR	82
Метус → 9K115	
Метус-М → 9K115-1	

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
Метус-М1 → 9K115-2	
МЭКТ-7930 → MZKT-7930	39
МА-20 → ML-20	
Молния → Molniya	88
Молния-1 → Molniya	88
Молния-2 → Molniya	88
МОН-50 → MON-50	
МОН-90 → MON-90	
«Москва-1» → 1L267	47
Мохаджер-6 → Mohajer-6	64
МП-К → 9S931-1	69
МРТО → MRT0	33
MT-12 → MT-12	
MT-ЛБ → MT-LB	
MT-ЛБВ → MT-LBV	
MT-ЛБВМ → MT-LBVM	
MT-ЛБВМК → MT-LBVM	
MT-ЛБВМК → Object 6	17
MT-ЛБВМК → MT-LBVM	
MT-Лбу → Object 10	
МТП-ЛБ → MTP-LB	
MTY-20 → MTU-20	
MTY-72 → MTU-72	55
MTY-90 → MTU-90	
Мурманск-БН → GT-01	45
«Небо-СВ» → 1L13-3	
«Небо-СВУ» 1L119 → 1L119	
Нетмо → RPG-22	
Nokaym-2 → PTM-4	
НСВ → NSV	
Объект 137 → T-54	
Объект 148 → T-14	
Объект 166 → T-62	
Объект 172M → T-72	
Объект 184 → T-72B	
Объект 188 → T-90	
Объект 219AC → T-80U	
Объект 219P → T-80B	
Объект 434 → T-64A	
Объект 447 → T-64B	
Объект 501 → BRM-3K	43
Объект 564 → BMO-T	
Объект 564 → Object 564	18
Объект 675 → BMP-2	
Объект 676 → BRM-1K	
Объект 688 → BMP-3	
Объект 6MA → MT-LBM 6MA	
Объект 6MA → Object 6	17
Объект 6MB → MT-LBM 6MB	
Объект 6MB → Object 6	17
Объект 765 → BMP-1	
Объект 950 → BMD-3	
ОЗМ-72 → OZM-72	
Орешник → Oreshnik	
Орион → Orion	
Орлан-10 → Orlan-10	
Орлан-10 → BKAR	78
Орлан-30 → Orlan-30	
Орлан-30 → BKAR	78
П-18 «Терек» → P-18	
П-19 «Дунай» → P-19	
П-2406T3 → P-2401-4	71

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
П-2401-4 → P-2401-4	71
П-35 «Самурн» → P-35	
П-800 «Оникс» → P-800	
«Палантин» → Palantin	49
«Палантин-К» → Palantin	49
Панцирь-С1 → 96K6	39
Панцирь-С2 → Pantisir-S2	39
Панцирь-СМ → Pantisir-SLM	39
Панцирь-СМД-Е → Pantisir-SMD-E	
Панцирь-СМ-СВ → Pantisir-SM-SV	39
Патруль → Asteis-7020	
«Пенциллиш» → 1B75	43
Пересвенец → P-2401-4	71
ПЗМ-2 → PZM-2	
ПЗМ-2 → BTM-3	60
«Плавок» → MTKK-1	
Пламя → AGS-17	
Плассун-CH → GAZ-3344-20	19
ПМ-38 → M-1938	
ПММ-2 → PMM-2	57
ПММ-2M → PMM-2	57
ПМН → PMN	
ПМП → PMP	56
ПМП-М → PMP	56
ПМУО → 9S933	69
Поиск → R-161	
Поляна-Д4 → 9S52	68
ПОМ-2 → POM-2	
ПОМ-2P → POM-2R	
ПП-2005 → PP-91M	56
ПП-91 → PP-91	56
ПП-91M → PP-91M	56
ППС-84 → PPS-84	
Провод → FPV	89
Прорыв-3 → T-90M	
Противник-ГЕ → 59N6	
ПРП-3 «Вал» → PRP-3	
ПРП-4А «Арзус» → PRP-4A	68
ПРП-4М «Дейтерий» → PRP-4M	68
ПРП-4 Harp → PRP-4	68
ПТ-76 → PT-76	
ПТМ-1 → PTM-1	
ПТМ-3 → PTM-3	
ПТМ-4 → PTM-4	
ПТС-2 → PTS-2	57
ПТС-3 → PTS-3	57
ПТС-4 → PTS-4	57
ПТС-М → PTS-2	57
ПУ-12 → PU-12	69
ПФМ-1 → PFM-1	
ПФМ-1C → PFM-1	
Р-142N → R-142N	
Р-142HMP → R-142N	71
Р-149БМР → R-149BMR	64
Р-161 → R-161	
Р-161A2M → R-161	70
Р-166 → R-166	
Р-166-0 → R-166	70
Р-168 → R-168	
Р-17 → 9K72	
Р-325УМВ → R-325UMV	48
Р-330Б «Манган-Б» → R-330B	47

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
P-3305MB → R-330BMV	47
P-330Ж «Жульель» → R-330Zh	50
P-330KMA → R-330KMA	
P-330KMB → R-330KMV	48
P-330M1P «Диаболо» → R-330M1P	
P-330 «Манган» → R-330	
P-330T → R-330T	47
P-340RP «Поле-21» → R-340RP	53
P-378A(B) → R-378A	
P-3785MB → R-378BMV	48
P-381T → R-381T	
P-412A → R-410-75	72
P-412AM → R-412A	
P-412AM → R-410-75	72
P-416GM → R-416GM	72
P-417 → R-410-75	72
P-439 → R-439	
P-439-П → R-439	73
P-500 → 9M728	33
P-9345MB → R-934BMV	47
P-9345 «Синица» → R-934B	47
Paкyшka → BTR-MDM	
Paкyшka → 9S931-2	15
Paкyшka-M → 9S931-2	15
«Папира» → MT-12	
PБ-109A «Былина» → RB-109A	
PБ-301Б «Борисоглебск-2» → RB-301B	48
PБ-341B «Лерер-3» → RB-341V	52
PБ-531Б «Инфауна» → RB-531B	54
PБ636 «Свет-КУ» → RB-636AM2	
Peгyм-2YC → P-260-0	74
Peзoнep → R-412A	
Peпeлeнт-1 → Repellent-1	53
Peфлeкс-M → 9K119	
PAМ-A → RLM-D	41
PAМ-M → RLM-M	41
PAМ-C → RLM-S	41
Pocтoк → BTR-90	
PPГ-22 → RPG-22	
PPГ-26 → RPG-26	
PPГ-28 → RPG-28	
PPГ-29 «Вампир» → RPG-29	
PPГ-30 → RPG-30	
PPГ-32 → RPG-32	
PPГ-7 → RPG-7	
PПO-A «Шмель» → PPO-A	
PXM-4 → RKhM-4	
PXM-4 → CBRN	62
PXM-6 → RKhM-6	
PXM-6 → CBRN	62
PXM-8 → RKhM-8	
PXM-8 → CBRN	62
PXM-9 → RKhM-9	
Pыcь → BRM-3K	43
Pыcь → Rys	43
C-300B → S-300V	36
C-300B4 → S-300V4	36
C-300PM → S-300PM	
C-300PM2 → S-300PM	
C-300PMY → S-300PS	
C-300PMY1 → S-300PM	

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
C-300ПС → S-300PS	
C-350 «Вулкэз» → S-350	36
C-400 «Триумф» → S-400	35
C-500 «Прометей» → S-500	
C-60 → S-60	
«Cаншя» → Saniya	
CEPП-BC6A → SERP-VS6D	
«Cилoк-01» → Silok-01	
Cиpуyc → Orion	82
CHAP-10 «Леонард» → SNAR-10	44
CHAP-10M → SNAR-10M	44
Copoкa → BMD-1KSh-A	66
«Cocнa» → 9M340	
CПГ-9 → SPG-9	
CПM-2 → GAZ-233036	
«Cтpeлa-2» → 9K32	
«Cтpeлa-3» → 9K34	
Cтpeлeц → KRUS-VR	
Cтpeлeц → KRUS	74
Cтpeлeц-M → KRUS-VR	
Cyпepкaм → S350	79
T-12 → T-12	
T-12A → MT-12	
T-14 → T-14	
T-15 → T-15	
T-54 → T-54	
T-54Б → T-54	10
T-55 → T-55	
T-55 → T-54	10
T-55A → T-55A	
T-55AM → T-55AM	
T-55AMB → T-55AM	
T-62 → T-62	
T-62M → T-62M	
T-62MB → T-62MV	
T-62 oбp. 1967 → T-62	9
T-62 oбp. 1975 → T-62	9
T-64A → T-64A	
T-64A → T-64A/B/BV	9
T-64Б → T-64B	
T-64Б → T-64A/B/BV	9
T-64Б1 → T-64A/B/BV	9
T-72 → T-72	
T-72A → T-72A	
T-72AB → T-72AV	
T-72Б → T-72B	
T-72Б1 → T-72Б1	
T-72Б1K → T-72Б1	
T-72Б3 → T-72Б3	
T-72Б3M → T-72Б3M	
T-72БA → T-72БA	
T-72BM → T-72BM	
T-72M → T-72M1	
T-72M1 → T-72M1	
T-72 «Урал» → T-72	
T-80 → T-80B/BV/BVM	8
T-80Б → T-80B	
T-80Б → T-80B/BV/BVM	8
T-80БB → T-80BВ	
T-80БB → T-80B/BV/BVM	8
T-80БBK → T-80BVK	
T-80БBM → T-80BVM	

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
T-80Y → T-80U	
T-80YE-1 → T-80UE-1	
T-80YK → T-80YK	
T-80YМ2 → T-80YМ2	
T-90 → T-90	
T-90A → T-90A	
T-90AK → T-90AK	
T-90K → T-90	7
T-90M → T-90M	
T-90C → T-90S	
T-90CK → T-90S	
Taбoлa → RPG-27	
Taйфyн-K → KamAZ-53949	
Taйфyн-K → KamAZ-63968	
Taйфyн-Y → Ural-63095	
«Tapaн» → R-381T	
Taxиoн → Tachyon	
Taxиoн → Takhion	81
TБC-M3 → TBS-M3	29
TДA-2K → TDA-2K	
TДA-2K → TDA-3	63
TДA-2M → TDA-2K	
TДA-3 → TDA-3	63
TДM-2K → TDA-3	63
Tepминaтop → BMPT	
Tepминaтop-2 → BMPT-72	
TЗM-T → TZM-T	29
Tизp → GAZ-233014	
Tизp → GAZ-2330	22
Tизp-M → AMN-233114	51
Tиpara 2.3» → Tirada-2S	
Tиpara-2C» → Tirada-2S	
TКБ-0175 → RPG-29	
TM-46 → TM-46	
TM-57 → TM-57	
TM-62 → TM-62	
TM-83 → TM-83	
TMM-3 → TMM-3	55
TMM-6 → TMM-6	55
TMC-65 → TMS-65	
TMC-65 → ARS-14	63
«Toбoл» → Tobol	
«Toп-M1-2Y» → 9K331MU	38
«Toп-M2AД» → 9K331MDT	38
Topнaгo-Г → 9K51M	26
Topнaгo-Y → Ural-6370	76
TOC-1A «Coлнцeпeк» → TOS-1A	29
TOC-1 «Bypaтинo» → TOS-1	29
TOC-2 «Tocoчka» → TOS-2	29
TOC-3 «Дpaкoн» → TOS-3	29
Toчka-Y → 9K79	
YAZ-452 → UAZ-452	
YAZ-469 → UAZ-469	
YМЗ → UMZ	
YMPK → UMPK	
YP-67 → UR-67	
YP-77 → UR-77	59
YP-83П → UR-83P	
Ypaл-375A → Ural-375D	
Ypaл-4320 → Ural-4320	
Ypaл-432009 → Ural-432009	
Ypaл-43206 → Ural-43206	

ОБОЗНАЧЕНИЕ, НАЗВАНИЕ	PAGE
Урал-63095 → Ural-63095	
Урал-6370 → Ural-6370	76
Урал-63706 → Ural-6370	76
Урал-ВВ → Ural-432009	
Уран-6 → Uran-6	
Утёс → NSV	
Фагот → 9K111	
«Флокс» → 2S40	
Форпост → Forpost	
Форпост-Р → Forpost	83
Форпост-РУ → Forpost	83
Х-101 → Kh-101	
Х-22 «Буря» → Kh-22	
Х-32 → Kh-32	
Х-47М2 → Kh-47М2	
Х-55 → Kh-55	
Х-59 «Овод» → Kh-59	
Х-69 → Kh-69	
Хризантема → 9K123	
Чистюля → Chistyulya	
«Шиповник-АЭРО» → Shipovnik-Aero	
Шквал → BMP-1U	
Штурм → 9K114	
Элерон → T-16	80
Элерон-3 → Eleron-3	
Элерон-3 → T-16	80
Элерон-ЗСВ → Eleron-3	
Элерон-ЗСВ → T-16	80
Эльбрус → 9K72	
FPV-гроны → FPV drones	

Table 2. Unit type to equipment

Tag gives basis, date and confidence. See page 5 for explanations.

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
Aerospace Forces	Korsar	Aerospace Forces holding; unit not stated	R 2024 L	83
	Mohajer-6	Aerospace Forces (IISS list, p.38); subunit INA	R 2024-12 M	84
Aerospace Forces (VKS)	Orion	Aerospace Forces (VKS) UAV holdings; IISS lists 'some Inokhodets'	R 2024 M	82
aerospace forces air defence regiment	96L6 all-altitude acquisition radar	3rd SAM battalion. INA (photographed deployed and stowed)	O 2010-02 M	42
	96L6 all-altitude acquisition radar	S-400 'battery' in Armada's usage; APA export diagram lists it as a battery component. 1 per S-400 battery (Armada, term undefined; APA lists 96L6E as a battery component without a count); S-300PMU2 count INA	R 2014 M	42
	Buk-M1 / Buk-M2 / Buk-M3	4 Aerospace Forces regiments with Buk-M1-2, Buk-M2 and S-300V; about 80 Buk-M1-2/Buk-M2 in Aerospace Forces holdings	R 2024 M	37
	Pantsir-S1	Pantsir vehicles deployed with each S-400 battalion (Armada); Lenta: batteries attached to S-400 regiments. 3 per S-400 battalion (Armada, 2024)	D 2024 H	39
	Pantsir-S1	Pantsir-S battery. 6 Pantsir-S delivered to Eastern MD air defence units on Kamchatka, Dec 2015; the regiment was reported due one battery	O 2015-12 M	39
	Pantsir-S1	Pantsir-S1 gun-missile battalion; each battery covers one S-400 battalion	O 2020-02 M	39
	Pantsir-S1	INA per regiment; IISS lists 9 SAM regts with Pantsir-S1 in the 45th, 4 in the 14th, 3 in the 4th and 4 in the 11th AF&AD Army	R 2024 M	39
	S-400 Triumf	IISS: 13 Aerospace Forces regts with S-400 and Pantsir-S1, plus 2 mixed with S-300	R 2024 M	35
Aerospace Forces UAV units	Forpost	Aerospace Forces UAV units	R 2024-11 H	83
air assault brigade	BTR-D	57 in 2000 listing	R 2000 L	16
air assault division	BMD-4M	air assault battalion of an air assault regiment (battalion set). 31 per battalion (nominal battalion set)	R 2020-06 M	13
	Lancet		O 2023-07 L	85
air assault regiment	BMD-1KSh-A	2 to 7 BMD-1KSh per regiment in the 2000 list	R 2000 L	66
	BTR-D	42 in 247th regiment (2017 table); 26-38 per regiment in a 2000 listing	R 2017 L	16
	BTR-MDM Rakushka	Air assault or parachute battalion, issued with BMD-4M in a battalion set. 8 per battalion set of 31 BMD-4M (June 2020); 6 in a 2021 set; some earlier sets 16	O 2021-07 M	15
air defence missile battalion	S-350 Vityaz	Battalion set: 1 command post 50K6A, 2-3 multifunction radars 50N6A, 12 launchers 50P6A, 12 launchers (12 missiles each) per battalion, per Russian MoD statement	R 2020 M	36
	S-400 Triumf	IISS: 2 SAM bn in Belarus; 1 SAM bty in Syria	R 2024 M	35
air defence missile brigade	Buk-M1 / Buk-M2 / Buk-M3	4 air defence missile battalions, each 3 batteries; brigade command post Polyana-D4, Buk-M1: 6 TELARs, 3 loader-launchers, 1 CP, 1 search radar per battalion; 24 TELARs per 4-battalion brigade (JIT: 53rd Bde had 3 operational battalions in 2014)	D 2016 H	37
	Buk-M1 / Buk-M2 / Buk-M3	battalion TELAR. INA per battalion; brigade had 3 operational battalions in 2014 (JIT)	O 2014-07 M	37
	Buk-M1 / Buk-M2 / Buk-M3	air defence missile battalion (brigade set of Buk-M3). Buk-M3: up to 6 TELARs or 9S36M radars and up to 12 launchers per battalion	R 2019 M	37
	Buk-M1 / Buk-M2 / Buk-M3	air defence missile battalion	R 2018 L	37
	Polyana-D4	brigade command post (combat control post MP06 and command-staff vehicle MP02)	D 2021-12 H	68
	Polyana-D4	brigade command post; vehicle captioned as 9S52M Polyana-D4M	O 2010-12 M	68
	S-300V4	air defence missile battalion (up to 4 per brigade, under Polyana-D4M1 brigade CP); battalion CP 9S457M controls up to 4 batteries; WEG: an upper-level 9S52 Polyana-D4 command post may integrate S-300V with Buk into one brigade. S-300V battery: 1 9S32, 4 9A83, 2 9A82, 1 9A84, 2 9A85; S-300V4 battery counts INA	D 2017 M	36
	S-300V4	INA. IISS lists S-300V and S-300V4 in Army holdings without a count and counts 16 SAM brigades, but does not say which hold them; Izvestia reports S-300V4 brigades under the OSK	R 2024 M	36
air defence missile regiment	Pantsir-S1	30 held by the Navy (IISS p. 187)	R 2024 M	39
	Pantsir-S1	INA (1 SAM regt with S-400 and Pantsir-S1)	R 2024 M	39
	S-400 Triumf	about 12 launchers reported at the unit site; 4 launchers and 96L6 and 92N6 radars reported destroyed	O 2025-11 M	35

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
	S-400 Triumf	air defence missile battalion (division). usually 2 battalions of 8 launchers each (16 per regiment); 2012 press report relaying MoD press service	R 2012 M	35
	S-400 Triumf	Navy: regiments under fleet air defence divisions. IISS: 64 S-400 in Navy; 3 SAM regts: 2 Baltic Fleet, 1 Pacific Fleet (naval aviation)	R 2024 M	35
	Strela-10 and Osa short-range SAM	division air defence regiment (Osa-AKM, being deactivated in 2019)	D 2019 L	40
airborne division	Artillery command vehicles 1V18/1V19, 1V119, 1V152, 1V1003	1V119 in VDV artillery and mortar units (RWW), used with 2S9 Nona-S (WEG15, ru-wiki); formation level not stated	D 2016 M	67
	Barnaul-T	air defence missile regiment. one full divisional set	O 2018-06 M	69
	Barnaul-T	air defence missile regiment	O 2018 L	69
	BMD-2	airborne regiment, battalion, company: 3 BMD per platoon, one per squad (BMD-1 in source), over 100 BMD per regiment, BMD-1 as squad vehicle (Soviet structure)	D 1991 M	14
	BMD-2	parachute regiment (Kostroma); MoD claim via RIA	O 2021 M	14
	BMD-2	parachute regiment	O 2011-09 L	14
	BMD-4M	lost at Hostomel, 3 Mar 2022, as the 31st Guards Air Assault Brigade	O 2022-03 M	13
	BMD-4M	parachute battalion of a parachute regiment (battalion set). 31 per battalion (nominal battalion set)	R 2021-07 M	13
	BTR-D	150 BTR-D in all Airborne Forces (IISS estimate); per-division count INA	R 2024-12 M	16
	BTR-MDM Rakushka	Procurement plan, replacing BTR-D; not a holding. Plan (2015) of 2,500 by 2025; not met, IISS estimate 150 in 2024	D 2015 L	15
	BTR-MDM Rakushka	Battalion sets in regiments; some units mix BMD-4M and BTR-MDM companies	O 2021-07 M	15
	BTR-MDM Rakushka	150 in Airborne Forces (IISS estimate)	R 2024 M	15
	Lancet		O 2023-07 L	85
	Leer-2	EW subunit (FOI: VDV EW companies formed from 2016). INA; MoD announced 10 complexes for VDV EW units, Dec 2012 (title only, state claim)	R 2018 M	51
	Supercam S350	separate unmanned systems battalion	O 2026-03 L	79
	T-90	INA; 7th, 76th and 106th Airborne Divisions each have one tank battalion (IISS). 30 T-90M across the Airborne Forces (IISS estimate); IISS does not say which formation holds them	R 2024 L	7
Airborne Forces	2A36 Giatsint-B 152 mm towed gun	INA; IISS lists the type with Airborne Forces without a count	R 2024-12 L	32
	Other armoured cars and utility vehicles (family)	Toros: on Airborne Forces AUV list; no unit named	R 2024-12 M	24
Airborne Forces (VDV)	2S5 Giatsint-S	Airborne Forces artillery; unit not named	R 2024 M	29
Airborne Forces (VDV) units	BRDM-2	Airborne Forces (VDV) units; no formation named	R 2023 L	22
anti-tank battalion	MT-LB	anti-tank gun battery (prime mover for 100 mm MT-12)	D 2024 M	17
	R-145BM Chayka command and staff vehicle	ACV listed as 'BTR-60PU/BRDM-2'; battery placement unclear in OCR. 5 per battalion (BTR-60PU or BRDM-2)	D 1991 L	65
Army	Other armoured cars and utility vehicles (family)	Vystrel: on Army AUV list; no unit or subunit given	R 2024-12 M	24
army corps	Iskander-M	corps missile brigade. 1 missile brigade (12 launchers)	O 2021 M	33
	Lancet	Kaskad unit	O 2023-07 L	85
artillery battalion	Artillery command vehicles 1V18/1V19, 1V119, 1V152, 1V1003	1V18 battery commander's COP vehicle; 1V19 battalion commander's COP vehicle; towed gun and MRL battalions. 3 x 1V18 and 1 x 1V19 per battalion set	D 2016 M	67
	Granat-1 / Granat-4	Granat-1 operator station, probably at battery command and observation post	D 2016 L	79
artillery brigade	2A36 Giatsint-B 152 mm towed gun	gun battalion (152 mm) of the army artillery brigade; 2A36 or 2S5 or M-46. 36 per brigade (two 18-tube gun battalions) or 48 (four 24-tube battalions)	D 1991 M	32
	2A36 Giatsint-B 152 mm towed gun	gun brigade of the front artillery division; 2A36 or 2S7. 72 per brigade	D 1991 M	32
	2A36 Giatsint-B 152 mm towed gun		R 2026 L	32
	2A65 Msta-B	artillery battalions (several per brigade; 2S19 or 2A65)	D 2019 M	32
	2S35 Koalitsiya-SV	self-propelled artillery battalions. INA; planned replacement of the brigade's 2S19 by 2S35, reported 2017, not carried out by late 2024	R 2019 L	30
	2S5 Giatsint-S	gun battalion (152 mm). 18 per battalion (24 in forward armies); 2S5 or 2A36 or M-46	D 1991 M	29

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
	2S7 Pion / 2S7M Malka	2S7 battalion (IZ17, 2017); 2S7M Malka (mil.ru via RDP18), 1 battalion; 12 x 2S7 (RUWIKI, 2019)	O 2018-04 M	31
	2S7 Pion / 2S7M Malka	12 x 2S7 each (RUWIKI, 2019)	O 2019 L	31
	Eleron-3SV	UAS company of an army artillery brigade; each platoon flies Orlan-10, Eleron-SV or Granat-4	D 2019 L	80
	FPV drones	FPV drone crews; a subunit of the anti-tank battalion works with UAVs	O 2026-05 L	89
	Orlan-10	UAV company, UAS platoon (BKAR artillery reconnaissance version)	D 2019 M	78
	R-145BM Chayka command and staff vehicle	1 per brigade	O 1994 L	65
	Uragan-1M	MRL battalion (Uragan 'being succeeded' by Uragan-M1; no holding confirmed)	R 2019-10 L	27
	ZALA 421-16E	UAV reconnaissance-strike subunit (Z-16 with Lancet)	R 2024-12 L	77
artillery regiment	BM-21 Grad 122 mm MRL	one Grad MRL battalion in division field artillery regiment, being replaced by Tornado-G	D 2019 M	26
	BMD-1KSh-A	0 to 2 per regiment in the 2000 list	R 2000 L	66
	ZALA 421-16E	UAV reconnaissance-strike subunit (Z-16 with Lancet)	R 2024-12 L	77
At least 12 Russian Army units, not named	Zastava	INA; held by at least 12 units per InformNapalm (via Ukrinform, Ukrainian state media)	R 2020 L	81
aviation regiment	Orion	Naval Aviation UAV regiment (also holds Forpost)	R 2024 M	82
Cold War HQ links	Tropospheric scatter radio stations (R-410-7.5, R-412A/AM, R-417)	R-412F General Staff-front and front-army; R-412A air defence-air force; R-412B army-grade operational-tactical HQ	D 1983 L	72
combined arms army	1B75 Penicillin	1L271 in rocket, missile and tube artillery units	R 2017 L	43
	Iskander-M	army missile brigade. 1 missile brigade (12 launchers) per army	D 2024 H	33
	Iskander-M	army missile brigade. 12 launchers per brigade	O 2023 M	33
	Lancet	56th separate special-purpose battalion	O 2024 L	85
	Palantin	EW units of a Western MD combined arms army in Voronezh Oblast (army not named)	R 2019-04 M	49
	Palantin	EW troops of a Guards combined arms army, Western MD, Luga range, Leningrad Oblast (Palantin-K, Oct 2019); Feb 2022 exercise in Leningrad Oblast names only the Western MD	R 2022-02 M	49
	Pole-21	Army EW specialists; first used in an exercise with battalion tactical groups, with R-934BMV and Silok-01 (MoD via Aif)	O 2021-04 M	53
	Polyana-D4	army air defence and aviation command post (SKP PVO i aviatsii)	R 2021-12 M	68
	T-62	army units with T-62MV obr. 2022 (mine ploughs, cages, RP-377), per Forbes as retold by UNIAN	R 2024-02 L	9
	Takhion		O 2022-11 M	81
	Typhoon family (Typhoon-K, Typhoon-U, Typhoon-VDV, Linza)	separate special-purpose (deep reconnaissance) company	R 2021 L	23
	ZALA 421-16E	Z-16 reconnaissance UAV crews	O 2026-01 L	77
Combined-arms formations of the Western Military District	Redut-2US		R 2015 L	74
command brigade	KamAZ military trucks (4310 and Mustang family)	communications company (Redut-2US on KamAZ-63501)	D 2019 M	75
	R-149 command and staff vehicles	R-149AKSh-1	O 2022-03 L	64
	Redut-2US		O 2017 M	74
	Redut-2US	communications company	R 2019 M	74
Dedicated EW units and manoeuvre reconnaissance units	Infaua		D 2024 L	54
Division reconnaissance battery (Soviet era)	ARK-1M Rys	division's reconnaissance battery (WEG wording, identical to Wikipedia; Soviet-era)	R 2024 L	43
Engineer troops units	BMR-3M	engineer troops units (not named)	R 2024 M	59
Engineer units of the Southern and Eastern Military Districts	Uran-6	engineer units	R 2015-11 L	60
engineer-sapper battalion	BAT-2	route construction and clearing section; usually in the movement support detachment (OOD). 4 per battalion	D 2016 M	58
	BAT-2	new-look brigade battalion: engineer-road platoon, CP equipment section, shore section. 7 per battalion (4 road platoon, 2 CP equipment section, 1 shore section)	R INA L	58

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
	IMR-2 / IMR-2M / IMR-3M combat engineer vehicle	mine clearing platoon (brigade engineer battalion). 2 IMR-3M per platoon	D 2016 M	58
	IRM Zhuk		D 1990 M	57
	IRM Zhuk	naval engineer battalion	D 1990 L	57
	ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer	obstacle platoon (brigade engineer battalion per RWW; division engineer battalion per WEG). 3 GMZ-3 per obstacle platoon	D 2016 M	61
	Military earthmoving and trenching machines	engineer-technical company; engineer-position platoon and command-post equipment squad (brigade battalion). Position squad: 2 MDK-3, 2 BTM-4M, 2 EOY-3521. Command-post squad: 2 EOY-3521	D 2008 L	60
	Military earthmoving and trenching machines	engineer-position company, engineer-position platoon (division battalion). Table lists 2 each of BTM, MDK, PZM, EOY (layout unclear)	D 2017 L	60
	TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge	brigade engineer battalion; sets almost always attached to the movement support detachment (OOD). 4 TMM-3 truck-launched bridge sets per brigade engineer battalion, each laying a 10 m span; together a 42 m bridge	D 2024-02 M	55
	UR-77 Meteorit	mine clearing platoon (organic engineer battalion of a maneuver brigade). 2 per platoon	D 2016 M	59
engineer-sapper brigade	IMR-2 / IMR-2M / IMR-3M combat engineer vehicle	2 delivered Nov 2018, 1 more due first half 2019 (Central MD press service, state claim)	R 2018-11 L	58
	IMR-2 / IMR-2M / IMR-3M combat engineer vehicle	INA; listed with UR-77, BMR-3 and GMZ-3 (English Wikipedia, citing a 2023 Russian regional news item)	R 2023-06 L	58
	IRM Zhuk		R 2017 L	57
	ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer		D 2024 L	61
	PMP heavy folding pontoon bridge park	pontoon bridge battalion of the army or front engineer brigade (FM 100-2-3 'Engineer Brigade, Army or Front'). one full set: 32 river and 4 shore links on KrAZ-255, 15 BMK-150 boats	D 1991 M	56
	Typhoon family (Typhoon-K, Typhoon-U, Typhoon-VDV, Linza)	assault and obstacle-clearing battalion	O 2017 L	23
	UR-77 Meteorit		R 2023 L	59
engineer-sapper regiment	IRM Zhuk		R 2017 L	57
	ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer		D 2024 L	61
	TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge	engineer and bridge subunit (press-release wording)	O 2020-11 M	55
	Uran-6		R 2021-01 M	60
EW battalion	Infaua	EW battalions of the Airborne Forces (VDV) (WEG 2015); RIA 2012 and FOI say VDV EW units	R 2015 M	54
	Krasukha-2 / Krasukha-4	independent REB-S battalion of an Air and Air Defence Army (VKS); FOI infers, does not name Krasukha there	R 2018 L	46
	Leer-3	Leer-3 drone/EW company	R 2019 M	52
	Orlan-10	Leer-3 crews (battalion or company of a combined arms army)	D 2019 M	78
EW brigade	Krasukha-2 / Krasukha-4	REB-S (counter-air) EW battalion. INA; usually one Krasukha-2 and one Krasukha-4 per brigade (reported 2022)	D 2018 H	46
	Krasukha-2 / Krasukha-4	part of 1 Krasukha-4 complex (command-post container found near Makariv); Armada estimates 4 complete Krasukha systems in theatre	R 2022-03 M	46
	Leer-3	Leer-3 platoon. platoon operates the complex; number per brigade INA	D 2024 H	52
	Moskva-1	REB-S (counter-air EW) battalion, reported	D 2018 H	47
	Moskva-1	INA (ARM22 lists one Moskva-1 among five brigade systems)	R 2023-09 L	47
	Murmansk-BN	Probably a company of the REB-N (anti-ground) battalion (FOI assessment)	D 2018 H	45
	Murmansk-BN	At least 1 complex per brigade (Armada); 2 antenna groups observed at 16th EW bde, vehicle count at 19th suggests 2 (PAS imagery, 2019)	O 2019 M	45
	Orlan-10	Leer-3 platoon. 3 per Leer-3 complex	D 2024 H	78
	R-330Zh Zhitel	REB-S (anti-aircraft communications and navigation) battalion	D 2018 M	50
EW company	R-330B Mandat-B and R-934BMV	EW company of a motor rifle or tank brigade: R-330B squad in the VHF-band jamming platoon; R-934B squad (VHF aircraft) in the radio-communications jamming platoon	D 2016 H	47

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
EW subunits of the Eastern Military District Ground Forces (SV); holding unit not named Ground Forces EW units high-power artillery brigade	R-330B Mandat-B and R-934BMV	At least one Russian EW company deployed to Ukraine in 2014 had two R-934B for airborne radio jamming; company not named. 2 R-934B in one EW company (2014)	R 2014 L	47
	R-330Zh Zhitel	EW company, jamming platoon (radio, GPS, cellular, radio-controlled fuses), jamming squad. 1 (single station on RWW chart; no count stated)	D 2016 M	50
	R-330Zh Zhitel	EW company of motor rifle or tank division (sources confirm the company exists; Zhitel holding not stated)	D 2018 L	50
	R-330Zh Zhitel	EW company deployed to Ukraine. 2 in the deployed company	R 2014 L	50
	Palantin	EW subunits of the Eastern MD in Zabaykalsky Krai (planned deliveries by end 2020)	R 2020-07 L	49
	UMZ		D 2024 L	61
	Leer-2	'possibly' in FOI; state media: Southern MD column Mar 2018	R 2018 M	51
	2S4 Tyulpan	2S4 battalion (ATP figure 5-18 shows two 2S4 groups; subunit name not given). ATP fig. 5-18: groups of 6 and 4 2S4 (OCR uncertain); 8 reported in 45th brigade (2017)	D 2024 H	31
	2S4 Tyulpan	heavy mortar battalion (240-mm); brigade allocated to a front (Soviet RVGK)	D 1991 M	31
	2S7 Pion / 2S7M Malka	2S7 battalions (45th bde: two battalions of eight, 2017). ATP fig. 5-18: 24 x 2S7 per brigade, 8 x 2S7 battalion block (OCR uncertain); 16 reported in 45th bde, 2017	D 2024-02 H	31
Higher-echelon or Aerospace Forces reconnaissance assets Manoeuvre units materiel support battalion materiel support brigade military district	2S7 Pion / 2S7M Malka	16 x 2S7 Pion (with 8 x 2S4 Tyulpan)	O 2017-12 M	31
	Lancet	unmanned systems troops Lancet reconnaissance-strike crew	O 2026-06 L	85
	Forpost	higher-echelon or Aerospace Forces reconnaissance assets (ATP fig. 5-8)	D 2024-02 L	83
	BMP-2	combat reconnaissance patrol, platoon-sized, task-organised from a manoeuvre unit (ATP example). 2 per patrol	D 2024 M	12
	Ural-6370 / 63706 / 532301 trucks (newer Ural family)	MTO (supply and transport) subunits	R 2020 L	76
	KamAZ military trucks (4310 and Mustang family)	transport units (heavy lift)	D 2016 L	75
	BMPT Terminator	Central Grouping of Forces, Sievierodonetsk axis; UK MoD assessment (state claim), at most 10 deployed; the only formation fielding BMPT	R 2022-05 L	11
	Leer-3	Leer-3 crews	R 2022-11 M	52
	R-166 Artek	signal units	R 2018 L	70
	R-416GM radio relay station	Six sets of R-416GM and R-416GM2 delivered (Russian MoD, reported Dec 2023)	R 2023-12 M	72
Military police units missile battalion missile brigade motor rifle battalion	S-400 Triumf	SAM regiments of the district's Air Force and Air Defence Army. IISS late 2024, SAM regiments listed with S-400 in each district's Air Force and Air Defence Army: 45th (Leningrad MD) 5, plus 4 listed with S-300PS, S-300PM1 and S-400; 14th (Central MD) 4; 4th (Southern MD) 3; 11th (Eastern MD) 4 listed with S-300PS and S-400, plus 1 with S-300V and S-400; 6th (Moscow MD) none (Moscow area: 5 regiments under Aerospace Defence Command). Launchers per district INA	D 2024 H	35
	SNAR-10 / SNAR-10M1 Leopard	artillery units of the tank formation stationed in the South Urals (MoD wording; formation not named)	R 2019-01 M	44
	Typhoon family (Typhoon-K, Typhoon-U, Typhoon-VDV, Linza)		R 2021 L	23
	Typhoon family (Typhoon-K, Typhoon-U, Typhoon-VDV, Linza)	sent to Syria with KamAZ-63968, early 2017	R 2017 L	23
	Iskander-M	4 launchers	O 2008 M	33
	Iskander-M	3 missile battalions of 2 batteries each; 2 launchers per battery. 12 launchers and 12 transloaders per brigade; 4 of each per battalion	D 2021 H	33
	BMP-1	motor rifle companies: 3 per platoon; battalion HQ also has 1 BMP-1K. 42 BMP-1/BMP-2 per BMP-equipped battalion (Soviet table)	D 1991 M	11
	BMP-2	motor rifle companies, one vehicle per squad. 37 per BMP battalion (BMP-2 or BMP-3); 3 per platoon	D 2016 M	12
	BREM and BTS armoured recovery vehicles	support platoon (evacuation squad). 3 BREM-L (RWW organisation chart)	D 2016 M	62
	BTR-80	motor rifle companies. about 44 BTR-80/BTR-82/MT-LB per BTR-mounted battalion	D 2016 M	20

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
	KamAZ military trucks (4310 and Mustang family)	logistical support platoon	D 2016 M	75
	MT-LB	motor rifle companies (squad carriers in BTR/MT-LB-based battalions), about 44 BTR-80/BTR-82/MT-LB per BTR- or MT-LB-based battalion	D 2016 M	17
motor rifle brigade	Zastava	several complexes	R 2014-01 L	81
	2S19 Msta-S	two self-propelled artillery battalions, each of three batteries of 6 howitzers; ATP notes 2S3 may be held instead. 18 per battalion (6 per battery), 36 per brigade; ATP: some battalions up to 24 systems	D 2024 M	30
	Barnaul-T	brigade AD subunit; BO quotes a 'missile division' (battalion) commander	O 2021-12 M	69
	BM-21 Grad 122 mm MRL	MRL battalion; 6 launchers per battery. 18 per battalion	D 2016 M	26
	BM-21 Grad 122 mm MRL	BTG, attached BM-21 battery. 1 battery	O 2022-04 M	26
	BMO-T	flamethrower platoon, RKHBZ company. 3 per RKHBZ company flame platoon (ATP); BMP-2-based BMO-1 may be used instead (ATP, RWW)	D 2024 M	18
	BMP-2	motor rifle battalions	O 2023-06 H	12
	Borisoglebsk-2	EW company	D 2024-02 H	48
	Borisoglebsk-2	EW subunit	O 2018 M	48
	BTR-80	motor rifle battalions	O 2022-09 H	20
	BTR-80		O 2022-03 M	20
	Decontamination vehicles (ARS-14, TMS-65, TZ-74 family)	RKHBZ company (also supports tank brigade), decontamination platoon. 8 ARS-14KM per RKHBZ company	D 2024 M	63
	Eleron-3SV	UAS company, Eleron platoon; platoons attached to battalions or companies	D 2019 M	80
	FPV drones	FPV drone crews	O 2026-03 L	89
	Granat-1 / Granat-4	UAV company, short-range platoon (Granat-4 with Orlan-10)	D 2016 H	79
	Granat-1 / Granat-4	UAV company, mini-platoon (hand-launched Granat-1)	D 2016 M	79
	MT-LB	motor rifle battalions, especially in northern (Arctic) brigades	D 2024 M	17
	MT-LB	MANPADS battery of the brigade's anti-aircraft missile-artillery battalion (ATP fig. 4-19). 9 per MANPADS battery (27 launchers)	D 2024 M	17
	MT-LB	battalion tactical group sent to Ukraine; brigade under Navy Coastal Troops (MILU). 36 MT-LB left the brigade's base in May 2022	O 2022-05 L	17
	Orlan-10	UAV company, short-range platoon	D 2019 H	78
	P-240I-4 Pereselenets	command company or battalion; CNA says combined arms brigades and regiments, tank units not excluded	D 2019 M	71
	PMP heavy folding pontoon bridge park	engineer battalion, pontoon bridge company. one pontoon park; RWW calls it 'PP-61' but its figures (268 m at 60 t; 90- and 120-t bridges) match PP-91, not PMP. 6 BMK boats, 7 PTS-2	D 2016 L	56
	PRP-4A Argus / PRP-4M Deytery	fire control and artillery reconnaissance battery, artillery spotting platoon	D 2016 M	68
	PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry	engineer battalion. 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4)	D 2016 M	57
	R-142N Deimos-N	brigade staff	D 2020 M	71
	R-330Zh Zhitel		R 2014 L	50
	Smoke generator vehicles (TDA-2K, TDA-3, TDM-2K)	RKHBZ company, aerosol platoon (RKHBZ company also supports tank brigades). 5 TDA-2K per aerosol platoon	D 2024 M	63
	SNAR-10 / SNAR-10M1 Leopard	fire control and artillery reconnaissance battery, radar platoon	D 2016 M	44
	Strela-10 and Osa short-range SAM	air defence missile-artillery battalion, missile battery (SA-13). 6 launchers per battery	D 2024 M	40
	T-62	platoon; alternative attribution for the T-62MV obr. 2022 platoon filed under the 5th Tank Brigade	R 2024-02 L	9
	T-80B/BV/BVM, T-80U	tank battalion. 26 T-80BVM in the brigade (2022)	O 2022-03 H	8
	T-80B/BV/BVM, T-80U	38th Bde: 40 (2019); others INA	R 2019 L	8
	T-90	INA; T-90A elements geolocated in Luhansk oblast	O 2014 L	7
	Tor	air defence missile battalion, three batteries. 12 launcher vehicles per battalion (4 per battery)	D 2024 M	38
	Tor	air defence missile battalion (Tor-M2DT); brigade link by location and retraining report only	R 2019-11 L	38
	Tornado-G	brigade artillery, MRL battalion. INA for Tornado-G; it replaces the BM-21, 18 per brigade MRL battalion (RWW p.100); 18 delivered to one brigade 2012 [RBASE]	D 2019 M	26
	Tornado-G	18	O 2012-04 M	26

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
	Tunguska	air defence missile-artillery battalion, Tunguska battery, 6 launcher vehicles per battery	D 2016 M	40
	Typhoon family (Typhoon-K, Typhoon-U, Typhoon-VDV, Linza)	INA; first K-53949 built went to this brigade for trials	R 2021 L	23
	Zoopark-1 / Zoopark-1M	fire control and artillery reconnaissance battery, radar platoon (with SNAR-10M1)	D 2016 M	42
motor rifle division	Borisoglebsk-2	separate EW battalion	O 2022-01 M	48
	BRM-1K	reconnaissance battalion; two of its three troops operate BRM-1K. 3 per reconnaissance battalion (ARMYREC)	D 2019 M	13
	BTR-80		O 2024-10 M	20
	Eleron-3SV	UAS company (division asset), Eleron platoon attached to regiments	D 2019 M	80
	Eleron-3SV		R 2018 L	80
	Leer-3	separate EW battalion	R 2022-01 M	52
	Orlan-10	UAV company	D 2019 M	78
	PMP heavy folding pontoon bridge park	engineer battalion, pontoon bridge company (also in tank divisions). half set: 16 river and 2 shore links and 1 service vehicle on KrAZ-214; 6 BMK-90/150 boats per company	D 1991 M	56
	R-142N Deimos-N	regiment (brigade) to division staffs	D 2020 M	71
	R-149 command and staff vehicles	R-149AKSh-1	O 2022-03 L	64
	T-54 / T-55	one subunit, presumably of the 42nd Guards MRD, near Robotyne with unmodernised T-55s	O 2024-02 L	10
	T-90		R 2020-04 M	7
	Tor	air defence missile regiment. 16 per regiment (4 batteries of 4, bmpd); ATP: two battalions; ru.wikipedia: Tor-M2 'divisional' set of 3 batteries, 12 vehicles	D 2024 H	38
	Tor	air defence missile regiment. one regimental set of Tor-M2 (16 combat vehicles)	O 2017-11 M	38
	Tornado-G	artillery regiment, MRL battalion (replacing Grad battalion)	D 2019 H	26
	Zoopark-1 / Zoopark-1M		D 2024 L	42
motor rifle regiment	BM-21 Grad 122 mm MRL	BTG, attached MRL element. 3 in one BTG	O 2022-04 M	26
	BMP-1	reconnaissance company. 3 BMP-1/BMP-2 with 1 BRM-1 (Soviet table)	D 1991 M	11
	BMP-1KSh command and staff vehicle	regimental command post. 1 in regimental HQ of a BMP-equipped motor rifle regiment (FM 100-2-3 table, OCR scrambled)	D 1991 M	65
	BMP-1KSh command and staff vehicle	501st signal company	O 1986 L	65
	BMP-2	reconnaissance company, tracked platoon (Soviet TOE). 3 BMP-1/ BMP-2	D 1991 L	12
	BMP-2	air defence battery (Soviet TOE), some regiments. 6 BMP-2, each carrying three SA-16 launchers	D 1991 L	12
	BRM-1K	reconnaissance company. 1 per company (Soviet era, WPEN)	D 1991 L	13
	BTR-80		O 2022-09 M	20
	Granat-1 / Granat-4	Attached Granat or Eleron platoon from the brigade or division UAV company; also battalions, at times companies	D 2019 M	79
	PU-12M7	air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet). 1 per battery (PU-12 family)	D 1995 M	69
	PU-12M7	regiment on BTR, Soviet 1980s establishment. 3 PU-12 per regiment	D 1989 L	69
	R-149 command and staff vehicles	R-149MA3	O 2022-09 M	64
Naval Infantry	Tigr and Vodnik armoured cars	branch; no unit named	R 2026 L	22
Naval Infantry (Coastal Troops of the Navy)	2A65 Msta-B	Naval Infantry artillery; brigade not named	R 2024 M	32
naval infantry brigade	BTR-80	Naval Infantry branch total 200 BTR-82A, 100 BTR-80 (IISS estimate); IISS lists 6 naval infantry brigades and 1 regiment	R 2024 H	20
	BTR-MDM Rakushka	INA; ATP says only 'also used by naval infantry'	D 2024 L	15
	MT-LB	IISS estimate 250 MT-LB in Naval Infantry (6 brigades, 1 regiment)	R 2024 M	17
	Other armoured cars and utility vehicles (family)	Vystrel on Naval Infantry equipment list; no unit named	R 2024-12 M	24
	T-54 / T-55	30 T-55A across Naval Infantry (IISS estimate); not allocated by brigade	R 2024 M	10
	T-80B/BV/BVM, T-80U	IISS estimate 30 T-80B/BVM in all Naval Infantry (2024)	O 2024-10 H	8

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
	T-80B/BV/BVM, T-80U		R 2019 L	8
Navy EW centres	Leer-2		R 2018 L	51
pontoon-bridge regiment	PMP heavy folding pontoon bridge park	two pontoon bridge battalions (army or front/district level), three full sets: 96 river and 12 shore links on KrAZ-255	D 1991 M	56
Reconnaissance and special operations units	Tigr and Vodnik armoured cars	force protection for forward air controller detachments, Syria	R 2019 L	22
Reconnaissance assets of the reconnaissance-fire system	Orion	medium-range reconnaissance UAV among reconnaissance assets of the reconnaissance-fire system (ATP fig. 5-8, OCR garbled)	D 2024-02 L	82
reconnaissance battalion	BRDM-2	reconnaissance companies of division or brigade	D 2016 M	22
	Strelets	Ground forces reconnaissance subunits, first recipients; echelon not stated	R 2021 M	74
	Tigr and Vodnik armoured cars		D 2016 M	22
RKhBZ (NBC protection) troops	R-145BM Chayka command and staff vehicle	Wikipedia names the vehicle BTR-60PU/PUM and equates BTR-60PU with R-145BM	R 2022 L	65
RKhBZ battalion	BRDM-2	BRDM-2RKh: divisional chemical defence battalion or company, regimental chemical defence company or platoon, reconnaissance battalion (Soviet structure)	D 1999 L	22
RKhBZ brigade	CBRN reconnaissance vehicles (RKhM family)	NBC defense company	D 2016 L	62
	Smoke generator vehicles (TDA-2K, TDA-3, TDM-2K)	5 TDA-3 delivered in one batch	R 2017 L	63
	TOS-1A Solntsepek	flamethrower battalion of 3 batteries, each 6 BM-1 and 6 TZM-T; RWW places these battalions in NBC defence brigades and regiments and does not state the echelon. 18 per battalion (6 per battery)	D 2016 M	29
	TOS-1A Solntsepek	TOS battalion of 3 companies of 6 launchers (ATP para 5-146). 18 per battalion	D 2024 M	29
RKhBZ regiment	CBRN reconnaissance vehicles (RKhM family)	RKhB reconnaissance company	R 2025 L	62
	Decontamination vehicles (ARS-14, TMS-65, TZ-74 family)	Eastern Military District NBC protection regiment, Amur Region, received ARS-14KM batch	R 2020 L	63
RKhBZ troops	CBRN reconnaissance vehicles (RKhM family)	RKhBZ troops (RKhM-8 and RKhM-9)	R 2021 M	62
rocket artillery brigade	BM-27 Uragan	rocket launcher battalions. 54 per brigade (3 battalions of 18)	D 1991 M	27
	BM-30 Smerch	three groups of 4 x 9A52-2 in ATP fig. 5-16; whether each group is a battalion or a battery is not readable from the text. 12 x 9A52-2 shown in ATP fig. 5-16 (3 groups of 4)	D 2024 M	28
	BM-30 Smerch	7 delivered April 2017, 5 more announced	O 2017-04 M	28
	BM-30 Smerch		O 2026-09 M	28
rocket artillery regiment	BM-27 Uragan	three rocket launcher battalions. 54 per regiment (3 battalions of 18)	D 1991 M	27
Rosgvardia unit	Other armoured cars and utility vehicles (family)	Z-STs Akhmat: Chechen forces; vehicle also on National Guard list. INA (first 20 delivered July 2022)	R 2024-10 M	24
	Tigr and Vodnik armoured cars	OMON; police SPM-1 and SPM-2 versions	R 2024 M	22
Rubicon detachment	FPV drones	FPV drone operators	O 2025-12 M	89
Russian forces in Syria	2A65 Msta-B	Russian deployment in Syria; unit not named	R 2024 M	32
	Eleron-3SV	Russian group in Syria; artillery spotting	R 2016 M	80
Russian Navy	Mohajer-6	Reportedly delivered to the Russian Navy (Guardian via Iran International); operator INA	R 2023-02 L	84
Second- and third-line units	T-54 / T-55	used as mobile artillery	R 2024 M	10
self-propelled artillery battalion	1B75 Penicillin	battalion fire control network, conceptual ATP fig. 5-3 (1L271)	D 2024 M	43
	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)	each battery: one 1V13 (battery FDC) and one 1V14 (battery COP); battalion HQ: one 1V15 (battalion COP) and one 1V16 (battalion FDC). 8 per battalion: 3 1V13, 3 1V14, 1 1V15, 1 1V16	D 2016 H	67
	2S35 Koalitsiya-SV	self-propelled howitzer battery. 6 per battery, 18 per battalion	D 2016 M	30
	PRP-4A Argus / PRP-4M Deytery	battalion fire control platoon, artillery spotter squad	D 2016 L	68
self-propelled artillery regiment	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)	howitzer battalion with 2S19M1. at least 3 1V13 and 3 1V14 seen with one battalion	R 2021-09 L	67
	2S35 Koalitsiya-SV	8 2S35 on troop trials	R 2020-05 M	30

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
	BM-27 Uragan	Uragan rocket artillery battalion. 8 per battalion	R 2017-12 M	27
	Uragan-1M	rocket artillery battalion (planned replacement for 9K57 Uragan, MoD statement; no holding confirmed)	R 2017-12 L	27
signal battalion	BMD-1KSh-A	6 to 8 per independent signal battalion in the 2000 list	R 2000 L	66
Special Operations Forces detachments (first users)	Lancet	Special Operations Forces detachments (first users)	O 2023-07 L	85
Spetsnaz brigade	Orlan-10	Leer-3 complex (with Orlan-10); FOI says 'Spetsnaz units'	R 2017-12 L	78
	Strelets	'Commandos' (TASS reprint); Spetsnaz brigade is an inference	R 2021 L	74
	Typhoon family (Typhoon-K, Typhoon-U, Typhoon-VDV, Linza)		O 2018 L	23
Squad to company level	Strelets	Squad to company level; platoon and section leaders wear the full set	D 2024 H	74
tank army	Iskander-M	army missile brigade. 1 missile brigade (12 launchers)	O 2021 M	33
	T-80B/BV/BVM, T-80U	INA; 'majority' of the army's tanks were T-80U (2022)	R 2022-03 M	8
tank battalion	BREM and BTS armoured recovery vehicles	armoured tracked recovery vehicles with the battalion on the march; type not named	D 2016 L	62
	T-90	tank companies; tank type not fixed by doctrine. 31 tanks per battalion of a motor rifle brigade; 41 with four companies (strength for tanks of any type; sources do not tie the T-90 to it)	D 2024 L	7
tank brigade	2S19 Msta-S	two howitzer (self-propelled artillery) battalions, three batteries of 6.18 per battalion, 36 per brigade	D 2016 M	30
	BMO-T	flamethrower platoon, RKhBZ company. 3 per RKhBZ company flame platoon (ATP); BMP-2-based BMO-1 may be used instead	D 2024 L	18
	Borisoglebsk-2	EW company	D 2024-02 H	48
	PRP-4A Argus / PRP-4M Deytery	fire control and artillery reconnaissance battery, artillery spotting platoon	D 2016 L	68
	PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry	engineer battalion. 7 PTS-2 per engineer battalion (RWW names PTS-2, not PTS-4)	D 2016 M	57
	SNAR-10 / SNAR-10M1 Leopard	fire control and artillery reconnaissance battery, radar platoon	D 2016 M	44
	Strela-10 and Osa short-range SAM	air defence missile-artillery battalion, missile battery (SA-13). 6 launchers per battery	D 2024 L	40
	T-62	platoon	R 2024-02 L	9
	Tor	air defence missile battalion, three batteries. 12 launcher vehicles per battalion (4 per battery)	D 2024 M	38
	Tunguska	anti-aircraft missile-artillery battalion, SA-19 battery; ATP 4-161 gives this battalion to tank brigades, but the battery figure is for brigades generally. 6 per battery (generic brigade figure: RWW 'maneuver brigade', ATP fig. 4-19)	D 2024 L	40
	Zoopark-1 / Zoopark-1M	fire control and artillery reconnaissance battery, radar platoon	D 2016 M	42
tank division	1B75 Penicillin	1L271; holding subunit not stated	O 2018-01 M	43
	BMPT Terminator	BMPT company (first formed Dec 2021); a further subunit formed 2024. 9 per company	O 2024-12 H	11
	T-80B/BV/BVM, T-80U	about 200 T-80 in the division (2019, Russian Wikipedia)	O 2022-03 H	8
	T-90	tank regiments (80th, 6th, 239th named as its regiments, not tied to T-90M). on paper about 300 tanks, mostly T-90M (analyst estimate quoted by TRENCH); IISS lists the division with 2 tank regiments	R 2025 L	7
	Tor	air defence missile regiment. one regimental set of Tor-M2 (16 combat vehicles)	O 2017-03 M	38
	Tornado-G	artillery regiment, MRL battalion	D 2019 H	26
tank regiment	BMP-1KSh command and staff vehicle	regimental command post. INA (FM 100-2-3 table reads 0 for the tank regiment HQ)	D 1990 M	65
	BMP-2	INA (BMP-2M Berezhek)	O 2023-02 M	12
	BMPT Terminator	separate BMPT company. 9 per company	O 2021-12 M	11
	PU-12M7	air defence missile-artillery battery; also regimental air defence chief's post (ARMF: PU-12 family 1977-95; ORUZH: undated PU-12M7 sheet). 1 per battery (PU-12 family)	D 1995 M	69
TOS-1A batteries allocated to manoeuvre brigades	Strelets	TOS-1A battery allocated to a manoeuvre brigade gets Orlan-10 data over the Strelets network	D 2024 L	74
unmanned systems unit	Forpost	UAV detachment of a naval aviation base. 6 UAV (Kamchatka, 2014)	R 2014-01 M	83
	Forpost	fleet UAV subunit	R 2026-05 M	83

UNIT TYPE	EQUIPMENT	HOLDING SUBUNIT AND NUMBER	TAG	PAGE
	FPV drones	FPV crews of the unmanned systems troops	R 2026-02 M	89
	Geran-2	strike crews; regiment has one UAV strike battalion. 2 Geran-2 crews per drone regiment	O 2025-05 H	87
	Geran-2	launch teams, combat groups	O 2026-09 M	87
	Geran-2	strike UAV detachment, Aerospace Forces. 5 detachments (2025)	R 2025-03 M	87
	Korsar	platoon in brigade or division UAS company (CNA 2019 conjecture)	R 2019 L	83

Table 3a. Reach of indirect fire

Guns, howitzers, mortars and rocket launchers, including mine-laying rockets. Practical rows give observed employment ranges.				
SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	TERM	PAGE
BM-30 Smerch 9M544 and 9M549 guided Tornado-S guided cluster rockets, inertial and GLONASS; Tornado-S guided rockets; crew statement CEP 5-10 m (inertial and GLONASS; official claim)	120 km	20-40 km	maximum (claim), minimum (claim)	28
Uragan-1M 9M544 guided 300 mm 9M544 guided rocket; claim in Russian press (ZR gives 120 km as the system maximum without naming the rocket)	120 km		claimed	27
BM-30 Smerch Employment range, Tornado-S guided rockets against a command post (2026-08) and a fortified strongpoint (2026-09); MoD figures via state media	about 100 km		practical (claim)	28
BM-30 Smerch 9M528 HE-frag unguided extended-range rocket, separable HE-frag warhead	90 km		maximum (claim)	28
2S35 Koalitsiya-SV 2S35 guided or extended-range shell precision-guided or rocket-assisted projectile; munition not named	80 km		maximum (claim)	30
BM-30 Smerch 9M55K cluster unguided, range-corrected rocket with fragmentation submunitions CEP 150 m (inertial range and attitude correction, no terminal guidance; also given for 9M528)	70 km	20 km	maximum, minimum	28
BM-30 Smerch 9M55F HE-frag unguided, separable HE-frag warhead	70 km	20 km	maximum, minimum	28
BM-30 Smerch Employment range, Smerch unguided rockets, Ukraine 2022, used at or near maximum range	35-70 km		practical	28
Uragan-1M 9M55F HE-frag 300 mm 9M55F HE-frag rocket with trajectory correction	70 km	20 km	maximum, minimum	27
BM-27 Uragan Employment range, observed 9K57 and 9K58 Smerch together, Donbas, May-June 2022; RUSI does not separate the two	35-70 km		practical	27
2S7 Pion / 2S7M Malka HE rocket-assisted RAP projectile, 103 kg	47.5 km		maximum	31
2S35 Koalitsiya-SV 2S35 standard HE-frag standard (unassisted) rounds	40 km		maximum	30
BM-21 Grad 122 mm MRL 9M521 HE-frag 9M521 HE-frag, export round for 9K51M Tornado-G; in 9K51 only with a manual electronic-fuze setter (RUWP); not adopted by the Russian Army (MISSIL)	40 km	15 km	maximum (claim), minimum (claim)	26
Tornado-G 9M521 HE-frag 9M521 extended-range HE-frag rocket; attribution to 9M521 from RUWIKI (Karpenko), WEG, ZR and WUW1 give 40 km as system maximum	40 km		maximum	26
2S7 Pion / 2S7M Malka HE-frag Conventional HE-frag, 110 kg projectile	37.5 km		maximum	31
Tornado-G 9M522 HE-frag 9M522 rocket, separable HE-frag warhead	37.5 km		maximum	26
2S19 Msta-S rocket-assisted HE rocket-assisted projectile (HERA/RAP), type not named	36 km		range	30
Uragan-1M 9M27F HE-frag 220 mm 9M27F unguided HE-frag rocket	35 km	10 km	maximum effective, minimum effective	27
Uragan-1M 9M27K cluster 220 mm 9M27K rocket, 30 9N210 fragmentation submunitions	35 km		maximum effective	27
BM-27 Uragan 9M27F HE-frag 9M27F HE-frag rocket	35 km	10 km	maximum, minimum	27
BM-27 Uragan 9M27K cluster 9M27K rocket, 30 9N210 HE-frag submunitions	35 km	10 km	maximum, minimum	27
BM-27 Uragan 9M27K2 AT mines 9M27K2 rocket, 24 PTM-1 anti-tank mines	35 km	10 km	maximum, minimum	27
BM-27 Uragan 9M27K3 AP mines 9M27K3 rocket, PFM-1 anti-personnel mines	35 km	10 km	maximum, minimum	27
2A36 Giatsint-B 152 mm towed gun rocket-assisted 3VOF41 active-rocket projectile 3OF30	33.5 km		maximum	32
2S5 Giatsint-S rocket-assisted HE-frag 3OF30 rocket-assisted HE-frag	33 km		maximum	29
2S5 Giatsint-S HE-frag 3OF59 HE-frag 3OF59, improved aerodynamic shape, full charge	30.5 km		maximum	29
2A65 Msta-B OF61 HE-frag base bleed OF61 (3OF61) HE-frag with base-bleed gas generator, round 3VOF91	29 km		maximum (claim)	32
2S19 Msta-S HE-frag base bleed 3OF61 HE-frag base bleed (round 3VOF91)	29 km		maximum	30

SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	TERM	PAGE
2A36 Giatsint-B 152 mm towed gun HE-frag 3VOF39 HE-frag 30F29, full charge; ammunition not stated; listed with HE-frag maximum	28.5 km	8.6 km	maximum, minimum	32
2S5 Giatsint-S HE-frag 30F29 HE-frag 30F29, full charge; HE-frag, indirect fire	28.4 km	8,600 m	maximum, minimum	29
2A65 Msta-B OF45 HE-frag OF45 (30F45) HE-frag projectile; OF45 HE-frag projectile	24.7 km	6.5 km	maximum (claim), minimum (claim)	32
2S19 Msta-S HE-frag 30F45 (OF-45) HE-frag, full charge	24.7 km		maximum	30
2S35 Koalitsiya-SV 2S43 HE-frag 2A64 gun; munition not stated by WEG or WPMALRU (Rostec: HE-frag more than 24 km)	24.5 km		maximum	30
2S5 Giatsint-S HE-frag 30F29 reduced charge HE-frag 30F29 with reduced charge 4Zh48	21.5 km		maximum	29
BM-21 Grad 122 mm MRL 9M22U HE-frag 9M22U (M-210F) HE-fragmentation rocket	20.4 km	5 km	maximum, minimum	26
Tornado-G 9M22U HE-frag legacy Grad 9M22U HE-frag rocket	20.4 km	5 km	maximum effective, minimum effective	26
2S5 Giatsint-S Krasnopol guided 30F39 laser-guided HE-frag Krasnopol	20 km		maximum	29
2S19 Msta-S guided Krasnopol laser-guided projectile; laser designation by an observer required	20 km		maximum	30
Tornado-G 9M538 HE-frag 9M538 rocket, enhanced monobloc HE-frag warhead	20 km		maximum (claim)	26
Tornado-G 9M539 HE-frag 9M539 rocket, separable HE-frag warhead; Tecmach statement at IDEX 2019; 9M539 rocket; Tecmach statement at IDEX 2019	20 km	5 km	maximum (claim), minimum (claim)	26
Tornado-G 9M541 cluster 9M541 rocket, cluster warhead with HEAT-frag submunitions	20 km		maximum	26
2S4 Tyulpan 308 rocket-assisted cluster mine cluster mine, 14 submunitions	19.3 km	7.1 km	maximum, minimum	31
2S4 Tyulpan Sayda incendiary rocket-assisted mine incendiary (napalm) rocket-assisted mine; incendiary rocket-assisted mine	19 km	0.7 km	maximum, minimum	31
2S4 Tyulpan 3F2 rocket-assisted HE mine 228 kg rocket-assisted HE mine; WEG and GS do not name the round, 3F2 per RUWIKI	18 km	0.8 km	maximum, minimum	31
BM-21 Grad 122 mm MRL 9M28F HE-frag 9M28F HE-fragmentation rocket	15 km	1.5 km	maximum, minimum effective	26
Tornado-G 9M28F HE-frag legacy Grad 9M28F HE-frag rocket	15 km	1.5 km	maximum effective, minimum effective	26
ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer ISDM mine rocket cassette rocket with AP or AT mines	15 km	5 km	maximum (claim), minimum (claim)	61
BM-27 Uragan 9M51 thermobaric 9M51 thermobaric rocket	13 km	5 km	maximum, minimum	27
2S4 Tyulpan 53-F-864 HE mine HE mine 130.7 kg, full charge; HE mine, smallest charge	9.65 km	0.8 km	maximum, minimum	31
2S4 Tyulpan Smel'chak-M laser-guided mine laser-guided mine, 134 kg	9.5 km	0.9 km	maximum, minimum	31
2S4 Tyulpan 3F5 Smel'chak laser-guided mine laser-guided mine, target designated by 1D15 from 0.2-5 km CEP 80-90% within a 2-3 m circle (semi-active laser homing, target lit in final 400-800 m of flight)	9.2 km	3.6 km	maximum, minimum	31
CBRN reconnaissance vehicles (RKhM family) Chemical monitoring radius, RKhM-6	6 km		claimed (claim)	62
TOS-1A Soltsepek thermobaric rocket MO.1.01.04M thermobaric rocket, 3.7 m, 217 kg	6,000 m		maximum	29
TOS-1A Soltsepek incendiary rocket MO.1.01.04M.OP incendiary rocket, 3.7 m, 204 kg	6,000 m		maximum effective	29
TOS-1A Soltsepek original short rocket MO.1.01.04 thermobaric rocket, 3.3 m, 173 kg, fired from TOS-1A	2,700 m		maximum	29
TOS-1A Soltsepek minimum range MO.1.01.04M rocket		600 m	minimum	29
IRM Zhuk Echo sounder depth range water depth measured by EIR sounder	0.5 to 20 m (EIR echo sounder)		stated range	57

Table 3b. Reach of long-range missiles and one-way attack drones

Ballistic and cruise missiles and one-way attack drones. CEP, where known, is printed under the munition.				
SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	TERM	PAGE
Iskander-M 9M723 ballistic missile 9M723, domestic Iskander-M; From the MIS Iskander-E (export) data table, max 280 km; domestic 9M723 minimum not confirmed CEP 10 to 30 m (With electro-optical terminal seeker)	up to 500 km	50 km	maximum, minimum	33
Iskander-M 9M728 (R-500) cruise missile 9M728 cruise missile, Iskander-K	500 km		official maximum (claim)	33

Table 3c. Reach of direct fire

Guns, cannons, machine guns and grenade launchers, smallest calibre first, with the vehicles that carry them. The headline range follows the manufacturer, then WEG, then other sources; 'Sources' gives the spread where they disagree. Line charges are in the engineering table.			
WEAPON, TARGET AND AMMUNITION	TERM	RANGE	SOURCES
PKT 7.62 mm vehicle machine gun Carried by, e.g.: T-62M, T-62MV, T-72A, T-72B, T-72B3, T-72B3M, T-80BV, T-80U, T-80BVM, T-90, T-90A, T-90M, T-14, BMPT, BMP-1, BMP-2, BMP-3, BMD-1, BMD-2, BMD-3, BMD-4, BMD-4M, BTR-4, BTR-60PB, BTR-70, BTR-80, BTR-80A, BTR-82, BTR-82A, BTR-82AM, BRDM-2, BRM-1K, BRM-3K, MT-LB, GAZ-3937 Vodnik, 2S3, 2S3M Akatsiya, 2S5 Giatint-S [29], 2S34 Khosta, 2S25 Sprut-SD, 9K35 Strela-10 [40], SNAR-10 [44], PRP-4M [68], GMZ-3			
ground. ball and API (57-N-323S, 7BZ-3)	effective	1,500 m (disputed)	1,000 to 2,000 m
ground. ball and API (57-N-323S, 7BZ-3)	maximum	4,000 m	3,800 to 4,000 m
DSHKM 12.7 mm heavy machine gun; anti-aircraft gun on older tanks Carried by, e.g.: T-62M, T-62MV, T-55A			
ground. API (B-32), API-T (BZT-44)	effective	2,000 m	2,000 to 3,500 m
air. API (B-32), API-T (BZT-44)	effective	1,000 m	
ground. API (B-32)	maximum	7,850 m	2,500 to 7,850 m
Kord 6P49 12.7 mm heavy machine gun, vehicle version of the Kord Carried by, e.g.: T-72B3 [2016 standard], T-90 [late production], T-90M, T-14, 2S35 Koalitsiya-SV [30], MT-LBVMK, GAZ-3937 Vodnik, Tigr, Tigr-M			
ground, personnel, light armour. B-32 API, BZT-44 API-T	effective	2,000 m	
air. B-32 API, BZT-44 API-T	effective	1,000 m (disputed)	1,000 to 1,500 m
ground. B-32 API	maximum	7,850 m	6,000 to 7,850 m
NSVT 12.7 mm heavy machine gun, vehicle version of the NSV Carried by, e.g.: T-72A, T-72B, T-72B3 (early), T-80BV, T-80U, T-90 (early production), T-62M (some), 2S19 Msta-S [30], BREM-1, MT-LBVM			
ground. API (B-32), API-T (BZT-44)	effective	2,000 m	
air. API (B-32), API-T (BZT-44)	effective	1,000 m (disputed)	1,000 to 1,500 m
ground. API (B-32)	maximum	7,850 m	2,000 to 7,850 m
KPVT 14.5 mm heavy machine gun, vehicle version of the KPV Carried by, e.g.: BTR-60PB, BTR-70, BTR-80, BTR-82, BRDM-2, MT-LBM 6MA (Object 6MA), GAZ-3937 Vodnik			
ground, light armour. API, API-T (B-32, BZT)	effective	2,000 m	
air. API, API-T (B-32, BZT)	effective	1,500 m	1,000 to 1,500 m
ground. 14.5 mm (not specified)	maximum	9,000 m	
2A14 23 mm automatic cannon, twin mount ZU-23-2 Carried by, e.g.: Ural-4320, Ural-43206 (truck-mounted ZU-23-2), KamAZ trucks (truck-mounted ZU-23-2), MT-LB, MT-LBM 6MA4, BTR-ZD Skrezhet			
ground. BZT, OFZ	effective	2,500 m	2,000 to 2,500 m
air. BZT, OFZ	effective	2,500 m	
2A38M 30 mm twin-barrel anti-aircraft automatic cannon Carried by: 2S6M, 2S6M1 Tunguska-M, -M1 [40], Pantsir-S1, Pantsir-SM [96K6] [39]			
air. HE-I, AP-T	effective	4,000 m	2,000 to 4,000 m
ground	maximum	2,000 m	
2A42 30 mm automatic cannon Carried by, e.g.: BMP-2, BMP-2D, BMP-2M Berezhok, BMD-2, BMD-3, BTR-90, BTR-82AT, BMPT, BMPT-72, Ka-52, Mi-28N, Mi-28NM, Ka-29, T-15, Kurganets-25, VPK-7829 Bumerang			
light armour. armour-piercing (BT, APDS)	effective	1,500 m	1,500 to 2,000 m
personnel. HE (OFZ, OT)	effective	4,000 m	
air. HE	effective	2,000 m	2,000 to 4,000 m
2A72 30 mm automatic cannon Carried by, e.g.: BMP-3, BMP-3M, BMD-4, BMD-4M, BTR-80A, BTR-82A, BTR-82AM, BRM-3K Rys, BTR-90M, MT-LBM 6MB, BMP-1AM Basurmanin, GAZ Vodnik with BPPU, Uran-9			
light armour	effective	1,500 m	1,500 to 2,000 m
personnel. HE	effective	4,000 m	2,000 to 4,000 m
air. HE	effective	4,000 m (disputed)	1,500 to 4,000 m
AGS-17 30 mm automatic grenade launcher Carried by, e.g.: BMPT (Terminator), BMD-3, Mi-24 (Hind), Mi-8T, LuAZ-967M			
personnel, ground. Frag-HE VOG-17M	effective	1,200 m	800 to 1,730 m
personnel, ground. Frag-HE VOG-17M	maximum	1,730 m	
personnel, ground. Frag-HE GPD-30 (extended range)	maximum	2,100 m	

TABLES

WEAPON, TARGET AND AMMUNITION	TERM	RANGE	SOURCES
AGS-30 30 mm automatic grenade launcher Carried by, e.g.: BMP-2M Berezhek			
personnel, ground. Frag-HE VOG-17M, VOG-30	maximum	1,700 m	
personnel, ground. Frag-HE GPD-30 [extended range]	maximum	2,100 m	2,100 to 2,200 m
57 mm gun of AU-220M Baikal (2A91 or 2A90, disputed) 57 mm automatic gun in AU-220M remote weapon station Carried by, e.g.: BMP-3 'Derivatsiya' (with AU-220M), BRM-3K with AU-220M, 2S38 Derivatsiya-PVO			
ground	effective	3,000 m	
air	effective	7,500 m	6,000 to 7,500 m
ground	maximum	more than 14,000 m	12,000 to 14,500 m
2A28 73 mm low-pressure smoothbore gun Carried by: BMP-1, BMP-1P, BRM-1K, BMD-1, BMD-1P, Improvised towed mount			
armour. HEAT-FS PG-15V	effective	1,300 m	500 to 1,300 m
armour. HEAT-FS PG-15V	maximum	1,300 m	
ground. Frag-HE OG-15V	maximum	4,400 m	4,400 to 4,500 m
SPG-9 73 mm recoilless gun Carried by, e.g.: UAZ-469			
armour. HEAT PG-9VS (PG-9V)	effective	1,300 m	800 to 1,300 m
armour. tandem HEAT PG-9VNT	effective	1,000 m	700 to 1,000 m
personnel, ground. Frag-HE OG-9VMI	maximum	4,500 m	
2A70 100 mm rifled gun-launcher Carried by, e.g.: BMP-3, BMP-3M, BMP-3F, BMD-4M, BMD-4			
ground. HE-Frag 3UOF17 (30F32)	effective	4,000 m	
air. HE-Frag	effective	4,000 m	
ground. HE-Frag 3UOF19 (30F70)	maximum	6,500 m	6,500 to 7,000 m
D-10T2S 100 mm rifled tank gun of the T-54B/T-55, manually loaded Carried by, e.g.: T-55, T-55A, T-55AM, T-55AMV			
armour. APFSDS (BM-412M, BM-25 type)	effective	2,000-2,500 m	
armour. HEAT (BK-17)	effective	1,000 m	
ground. HE-Frag (OF-32)	effective	2,500-4,000 m	2,500 to 6,000 m
ground. HE-Frag	maximum	15,000 m	14,600 to 16,000 m
MT-12 100 mm smoothbore towed anti-tank gun Carried by, e.g.: MT-LB, Ural-4320, Ural-375D			
armour. APFSDS (BM-412M)	effective	2,500 m	1,880 to 2,500 m
armour. direct fire sight OP4M-40U	sighting	3,000 m	
ground. HE-Frag (OF-15)	maximum	8,200 m	
2A20 115 mm smoothbore tank gun of the T-62, manually loaded Carried by: T-62M, T-62MV			
armour. APFSDS (BM-6, BD/36-2)	effective	1,800 m	1,800 to 2,100 m
armour. HEAT (3BK4)	effective	1,500 m	1,200 to 1,500 m
ground. HE-Frag (OF-27)	effective	4,000 m	4,000 to 6,000 m
ground. HE-Frag	maximum	9,000 m	
2A45M 125 mm smoothbore towed anti-tank gun with auxiliary power unit Carried by, e.g.: MT-LB			
armour. APFSDS (BM-42M)	effective	3,000 m (disputed)	2,000 to 3,000 m
ground. HE-Frag OF-26	direct fire	5,000 m	
ground. HE-Frag	maximum	12,200 m	12,000 to 12,200 m
2A46M 125 mm smoothbore tank gun and ATGM launcher, autoloading Carried by, e.g.: T-72B, T-72BM, T-90, T-80BV, T-80U, T-80BVM, T-64BV			
armour. APFSDS (3BM42M type)	effective	3,000 m	2,500 to 4,000 m
armour. HEAT (3BK29M, 3BK27 type)	effective	3,000 m	2,000 to 4,000 m
ground. HE-Frag (30F26)	effective	5,000 m	4,000 to 5,000 m
ground. HE-Frag	maximum	10,000 m	
2A46M-5 125 mm smoothbore tank gun and ATGM launcher, autoloading; improved-accuracy 2A46M Carried by, e.g.: T-72B3, T-72B3M, T-90A, T-90M			
armour. APFSDS (3BM42M type)	effective	3,000-4,000 m	3,000 to 4,000 m
armour. HEAT-MP (3BK29M)	effective	4,000 m	
ground. HE-Frag (30F26)	effective	5,000 m	

WEAPON, TARGET AND AMMUNITION	TERM	RANGE	SOURCES
air. HE-shrapnel, time-fuzed (Ainet)	effective	4,000-5,000 m	
ground. HE-Frag	maximum	10,000 m	
2A75 125 mm smoothbore gun with autoloader, a version of the 2A46M tank gun Carried by: 2S25 Sprut-SD, 2S25M Sprut-SDM1			
armour. APFSDS (BM-42M / 3BM42)	effective	3,000 m	
armour. APFSDS, HEAT (BK-29M) and guided rounds with 1A40M-1 sight	sighting	4,000 m	
ground. HE-Frag OF-26 (3OF26)	effective	5,000 m	
2A82-1M 125 mm smoothbore tank gun of the T-14, in an unmanned turret Carried by: T-14 Armata			
armour. APFSDS	effective	3,500+ m	3,500 to 5,000 m
ground. HEAT, HE-Frag	effective	4,700 m	
ground	maximum	12,000 m (disputed)	7,000 to 12,000 m

Other direct-fire figures in the entries

SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	TERM	PAGE
T-54 / T-55 100 mm HE Indirect fire with side level and azimuth pointer; round type not named for this figure	14,600 m		maximum	10
T-64A/B/BV 125 mm HE-Frag OF-26 day; night 800-1,300 m; maximum aimed range 5,000 m	5,000 m		max effective	9
T-80B/BV/BVM, T-80U HE-Frag direct fire, day; RUWIKI's 5,000 m is 'greatest sighting range'	5,000 m		maximum effective	8
BRM-1K 73 mm (munition not stated) ammunition type not stated by WEG	4,500 m		maximum firing range	13
BMD-2 73 mm OG-15V HE-Frag BMD-1, HE-fragmentation	4,400 m		maximum	14
BMD-2 30 mm (type not stated) ammunition not stated by WEG	4,000 m		effective, maximum	14
BMD-2 30 mm HE (BMD-2M) high-explosive, BMD-2M fire control	4,000 m		effective (claim)	14
BMD-4M HE-Frag 100 mm, day; ammunition not stated by WEG	4,000 m		maximum effective	13
BMP-2 30 mm HE ground targets, Frag-HE; day (1,300 m night, WEG15)	4,000 m		effective	12
T-54 / T-55 9M117 Bastion Only T-55M/AM/MV with Volna; day; laser beam-riding round fired through the gun	4,000 m		effective	10
T-62 HE-Frag OF-27 HE-Frag, day, direct fire	4,000 m		maximum aimed	9
T-64A/B/BV 9M112 Kobra (AT-8 Songster) gun-launched, radio command guidance; day sight only unless target illuminated	4,000 m		range	9
T-90 HEAT 3BK12M day/night not stated by WEG	4,000 m		effective	7
T-90 HE-Frag 3OF19 day/night not stated by WEG	4,000 m		effective	7
T-64A/B/BV 125 mm APFSDS-T BM-42M day; night 800-1,300 m; maximum aimed range 3,000-4,000 m	2,000-3,000 m		max effective	9
T-64A/B/BV 125 mm HEAT-MP BK-29M day; night 800-1,300 m; maximum aimed range 4,000 m	3,000 m		max effective	9
T-80B/BV/BVM, T-80U APFSDS day	2,000-3,000 m		effective	8
T-80B/BV/BVM, T-80U HEAT 3BK31 Start, day	3,000 m		maximum effective	8
T-90 APFSDS 3BM42M day; night	2,000-2,600 m		effective	7
BMP-2 30 mm, air targets helicopters and slow aircraft; ammunition not stated	2,500 m		effective	12
BTR-MDM Rakushka 23 mm ZU-23-2 ammunition Ground targets; ZU-23-2 fit on BTR-MDM, 2021 exercise; Air targets; ZU-23-2 fit on BTR-MDM, 2021 exercise	2,500 m		claimed (claim)	15
2A65 Msta-B Direct fire range ATP Table E-2 and RWW table; projectile not stated; the same table's maximum-range row is misaligned	2,130 m		direct fire	32
BMD-2 30 mm AP (BMD-2M) armour-piercing, BMD-2M fire control	2,000 m		effective (claim)	14

SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	TERM	PAGE
BMD-2 9M111 Fagot 9M111-2; 9M111M reaches 2,500 m	2,000 m		maximum	14
BMP-2 30 mm APDS day (1,300 m night)	2,000 m		maximum effective	12
BMR-3M 12.7 mm MG ground targets, 12.7x108 mm	2,000 m		effective	59
2S19 Msta-S secondary MG 12.7x108 mm, ground and air targets	2,000 m ground; 1,500 m air		effective	30
T-62 APFSDS APFSDS, 2 m high target, day; 2,260 m against 3 m high target	1,870 m		effective	9
2A36 Giatsint-B 152 mm towed gun direct fire ammunition and target not stated in source	1,530 m		direct fire	32
BMP-2 30 mm AP-T armour, day (1,300 m night, WEG15)	1,500 m		effective	12
BMR-3M 12.7 mm MG vs air air targets, 12.7x108 mm	1,500 m		effective	59
T-62 HEAT BK-4 HEAT, day	1,500 m		maximum effective	9
BMD-2 73 mm PG-15V HEAT BMD-1, anti-armour	1,300 m		maximum	14
T-54 / T-55 100 mm HE-frag, direct fire Grazing shot at 2 m high target	1,100 m		grazing shot	10
T-54 / T-55 100 mm AP, direct fire Grazing shot (point-blank range) at 2 m high target	1,080 m		grazing shot	10
2S5 Giatsint-S HEAT BP-540 HEAT BP-540, direct fire	1,000 m		direct fire	29
BRM-1K HEAT-FS day; firing on the move; night	800-1,000 m		maximum effective	13
BTR-D 7.62 mm API (bow machine gun) Day; 400-500 m on the move	1,000 m		maximum effective	16
T-62 APFSDS (night) night, with IR night vision equipment; WEG does not name the ammunition	800 m		effective	9
BM0-T RPO-A rocket 93 mm RPO-A thermobaric rocket	400 m		effective	18

Table 3d. Reach of anti-tank weapons

Anti-tank guided missiles.				
SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	TERM	PAGE
BMP-T Terminator 9M120-1/9M120-1F Ataka-T tandem HEAT (9M120-1) or thermobaric (9M120-1F); laser beam riding	6,000 m		maximum (claim)	11
BMD-4M 9M117M1 Arkan ATGM laser beam-riding, fired through 100 mm gun; tanks; 3UBK23-3 round	5,500 m	100 m	maximum (claim), minimum	13
BMP-2 9M133 Kornet (BMP-2M) Kornet-LR; day (3,500 m night)	5,500 m		maximum effective	12
T-80B/BV/BVM, T-80U 9M119 Refleks ATGM T-80U, T-80BVM; laser beam-riding; ATP: 4,000 m, 5 km against helicopters	4,000-5,000 m		operational	8
T-90 ATGM 9M119M Refleks (AT-11 Sniper-B) day	5,000 m		maximum effective	7
BMD-2 9M113 Konkurs SACLOS wire-guided ATGM, BMD-2 and BMD-1P	4,000 m	70 m	maximum, minimum	14
BMP-2 9M113 Konkurs SACLOS, wire-guided ATGM	4,000 m	70 m	maximum, minimum	12
BTR-90 Rostok 9M113 Konkurs 9M113 Konkurs (WEG15: also AT-5B Konkurs-M, 4,000 m), wire-guided SACLOS, four missiles; WEG15 term; WEG online calls 70 m - 4 km 'operational range'; 9M113 Konkurs, lower bound of WEG 'operational range'	4 km	70 m	maximum effective, minimum	21
T-62 9M117 ATGM 9K116-2 Sheksna laser beam-riding missile, T-62M and T-62MV only, day only	4,000 m		effective	9
T-80B/BV/BVM, T-80U 9M112 Kobra ATGM T-80B; radio command guidance; RUWIKI: 4,000 m greatest sighting range for guided round	4,000 m	100 m	effective	8
BMD-2 9M14 Malyutka MCLOS wire-guided ATGM, BMD-1	3,000 m	500 m	maximum, minimum	14
BMP-1 9M14M Malyutka (AT-3 Sagger) MCLOS wire-guided ATGM, day only; BMP-1; MCLOS wire-guided ATGM; BMP-1	3,000 m	500 m	effective, minimum	11
BMP-1 9M111 Fagot (AT-4 Spigot) SACLOS ATGM from roof launcher; BMP-1P	2,000 m		maximum effective	11
BMP-2 9M111 Fagot SACLOS, wire-guided ATGM	2,000 m		maximum effective	12

Table 3e. Reach of air defence weapons

Missiles and guns. Range is horizontal; altitude is the vertical reach. Figures are against the target named in the condition.					
SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	ALTITUDE	TERM	PAGE
S-300V4 9M82MD air targets incl. AEW aircraft; S-300V4 new long-range missile	400 km			claimed maximum (claim)	36
S-400 Triumf 40N6 aerodynamic targets (STRASAM); APA brochure figure is for export 40N6E, 215 nmi; 40N6; the 27 km often quoted is reported for the 48N6DM missile specifically	400 km		30-35 km	maximum (claim)	35
S-400 Triumf 48N6DM (48N6E3) aerodynamic targets; RUSIAD names export 48N6E3 (Army Technology lineage); TASS 2018 state_claim: 48N6DM up to 250 km	250 km			maximum (claim)	35
S-300V4 9M82M aircraft targets; S-300VM	200 km		25 m-30 km	maximum (claim), maximum and minimum (claim)	36
S-350 Vityaz 9M96D extended-range domestic 9M96D; target type not stated by TWZ	about 121 km (75 miles)			range	36
S-350 Vityaz 9M96E2 aerodynamic target; export missile	120 km	1 km	5 m to 30 km	maximum (claim), minimum (claim)	36
S-400 Triumf 9M96E2 export 9M96E2; aerodynamic targets; APA gives 64.8 nmi; export 9M96E2; APA gives 15 ft to 100 kft	120 km		0.005-30 km	range (claim)	35
S-300V4 9M82 aircraft and jammer targets; S-300V; aircraft targets; S-300V	100 km	13 km	1-30 km	maximum, minimum (claim), maximum and minimum	36
S-300V4 9M83 aircraft targets; S-300V	75 km	6 km	250 m-25 km	maximum, minimum, maximum and minimum	36
Buk-M1 / Buk-M2 / Buk-M3 9M317M (Buk-M3) air target	2.5-70 km		35 km	maximum effective, maximum	37
S-350 Vityaz S-350E system aerodynamic targets; export exhibit figure, also given by Russian MoD for the domestic S-350 (state claim); aerodynamic targets; export system brochure	1.5 to 60 km		10 m to 30 km	maximum (claim)	36
S-400 Triumf 48N6 (ballistic targets) ballistic missile targets, 48N6 series	60 km			range	35
Buk-M1 / Buk-M2 / Buk-M3 9M317 (Buk-M2) aircraft; air target	3-50 km		15 m-25 km	engagement zone, intercept altitude	37
Pantsir-S1 Pantsir-SM missile Pantsir-SM with new missile; KBP chief designer, 2016	40 km			claimed (claim)	39
S-300V4 9M83/9M82 vs ballistic missile ballistic targets; S-300V and S-300VM; ballistic targets; S-300V (S-300VM: 1-30 km)	40 km		1-25 km	maximum, maximum and minimum (claim)	36
S-350 Vityaz 9M96E aerodynamic target; export missile	40 km	1 km	5 m to 20 km	maximum (claim), minimum (claim)	36
S-400 Triumf 9M96E export 9M96E; aerodynamic targets; APA gives 21.6 nmi; export 9M96E; APA gives 15 ft to 66 kft	40 km		0.005-20 km	range (claim)	35
Buk-M1 / Buk-M2 / Buk-M3 9M38M1 (Buk-M1) air target	3.25-34 km		30 m-22 km	maximum, intercept altitude	37
Pantsir-S1 Pantsir-S1M missile export Pantsir-S1M; new missile, designation not given by source; High Precision Systems, 2019	30 km			claimed (claim)	39
S-350 Vityaz S-350E system vs ballistic ballistic targets	1.5 to 30 km		2 to 25 km	maximum (claim)	36
S-350 Vityaz 9M96 basic domestic 9M96; target type not stated by TWZ	about 29 km (18 miles)			range	36
S-400 Triumf S-400 (missile not stated) reported engagement altitude band			0.1-27 km	reported	35
Buk-M1 / Buk-M2 / Buk-M3 9M317 vs ballistic and cruise missiles ALCM-type cruise missile at 30 m (MISM2; ROE gives 100 m, unchecked); Lance-type tactical ballistic missile 15-20 km (MISM2)	20 km			maximum (claim)	37
Pantsir-S1 57E6 missile aircraft-type targets; KBP figures for export 57E6-E; KBP figures for export 57E6-E; 5 m only in optical mode over water	20 km	1.2 km	5 m to 15 km	maximum (claim), minimum (claim)	39
Tor 9M338K Tor-M2 domestic, air targets	16 km		10 m to 10 km	maximum (claim)	38

SYSTEM, MUNITION AND CONDITION	RANGE	MINIMUM	ALTITUDE	TERM	PAGE
Strela-10 and Osa short-range SAM 9M33M3 Osa-AKM missile; Osa-AKM missile; WEG figure	15 km		4 km	operational range, maximum flight altitude	40
Tor 9M330/9M331 aircraft and PGM targets, Tor/Tor-M1	12 km	1.5 km	10 m to 6 km	maximum, minimum	38
Tor 9M331 (Tor-M2E) Tor-M2E export; target 300 m/s; Tor-M2E export; target 300 m/s (10 m to 6 km at 600 m/s)	1 to 12 km		10 m to 10 km	maximum (claim)	38
S-350 Vityaz 9M100 IR-homing short-range missile; fielding unconfirmed	10 km			maximum (claim)	36
Tunguska 9M311-M1 missile Tunguska-M1, air target; Tunguska-M1, air target; APA gives 2.5 km for 2K22 and 2K22M too; air target; ceiling for 9M311 family	10 km	2.5 km	3.5 km	maximum, minimum	40
Pantsir-S1 TKB-1055 small missile anti-drone round, up to 48 per vehicle in cannon-less versions	7 km			claimed (claim)	39
Strela-10 and Osa short-range SAM 9M37M/9M333 Strela-10 engagement zone, all variants; target closing at up to 415 m/s or receding at 310 m/s; Strela-10; 9M333 minimum 10 m	0.8 to 5 km		25 m to 3.5 km	engagement envelope	40
Pantsir-S1 30 mm 2A38M gun air and ground targets; air targets	4 km	200 m	0 to 3 km	maximum (claim), minimum (claim)	39
Tunguska 30 mm 2A38M guns air target	4 km	200 m	3 km	maximum, minimum	40

Table 3f. Reach of sensors

Detection, tracking and location ranges, grouped by what the sensor is for. Target type and size are part of the figure: a radar that sees a bomber at 600 km sees a small drone much closer.

REACH	SYSTEM, DATUM AND CONDITION	TARGET	TERM	PAGE
Ground surveillance				
up to 40 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10M1 radar. tanks and vehicles; Russian MoD figure; instrumented range 0.2-40 km	tank, vehicle	maximum (claim)	44
30 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10 radar. moving ships, without MTI	ship	maximum (claim)	44
20 km	1B75 Penicillin Detection range, 1L271. tank-type moving ground target	tank	maximum (claim)	43
16 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10 radar. moving vehicles, without MTI; 10 km with MTI	vehicle	maximum (claim)	44
15 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10M1 radar. moving person; developer figure	person	claimed (claim)	44
12 km	PRP-4A Argus / PRP-4M Deytery Detection range, radar (moving tank-type target). moving tank-type target	tank	up to	68
up to 10,000 m	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) Detection range, 1V14 optical reconnaissance. day	not stated	maximum	67
10 km	PRP-4A Argus / PRP-4M Deytery Recognition range, laser rangefinder. up to three tank-type targets; large targets incl. buildings 25 km; WEG names the 1D14-1 periscopic rangefinder	tank	up to (claim)	68
10 km	PRP-4A Argus / PRP-4M Deytery Detection range, PSNR-5M radar on PRP-4M. moving tank, PRP-4M application; Jane's 2001 export data, confidence reported	tank	range (claim)	68
8 km	PRP-4A Argus / PRP-4M Deytery Detection range, optical-electronic (tank-type target, day). tank-type target, daytime	tank	up to	68
7 km	PRP-4A Argus / PRP-4M Deytery Detection range, radar (person). single soldier	person	up to	68
over 3,300 m	T-90 Detection range, T-90MS thermal sight. tank target; T-90MS export variant	tank	detection	7
3 km	BTR-80 Detection range, TKN-AI commander's sight (BTR-82A). ground targets, with laser rangefinder	not stated	detection (claim)	20
3 km	PRP-4A Argus / PRP-4M Deytery Detection range, optical-electronic (tank-type target, night). tank-type target, night	tank	up to	68
up to 2,500 m	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) Detection range, 1V14 optical reconnaissance (night). night	not stated	maximum	67
2 km	PRP-4A Argus / PRP-4M Deytery Detection range, stationary tank, engine off (sensor not stated). stationary tank-type target, engine not running; ARMYR and RWW name no sensor, WEG lists it under radar	tank	no less than (claim)	68
100-1,500 m	BRDM-2 Range, 762 mm ball (PKT). ground target, day; FAS calls 1,500 m maximum aimed range	not stated	effective (with sight adjustments)	22
750 m	1B75 Penicillin Minimum range, 1L271. mortar firing positions; 200 m for moving ground targets	mortar	minimum (claim)	43
Counter-battery				
65 km	Zoopark-1 / Zoopark-1M Detection range, tactical missile, 1L260. Launch positions of Lance / ATACMS type missiles	tactical missile	up to	42
45 km	Zoopark-1 / Zoopark-1M Detection range, rocket artillery, 1L260. MLRS firing position (DOGS, unverified, specifies M270 type)	rocket launcher, artillery	up to	42
40 km	Zoopark-1 / Zoopark-1M Detection range, tactical missile, 1L219M. Tactical missile launch position	tactical missile	detection of launch position	42
mortar 22, artillery 20, MLRS 35, tactical missile 40 km	Zoopark-1 / Zoopark-1M Fire-correction range, 1L219M. Adjusting friendly fire, by weapon type	not stated	fire correction range	42
30 km	ARK-1M Rys Detection range, ARK-1M vs battlefield missiles. Missile (WEG)	not stated	unspecified	43
30 km	Zoopark-1 / Zoopark-1M Detection range, rocket artillery, 1L219M. MLRS firing position	rocket launcher, artillery	detection of launch position	42
23-27 km	Zoopark-1 / Zoopark-1M Detection range, tube artillery, 1L260. 155-203 mm gun firing positions (23 km for 155 mm, 27 km for 203 mm)	artillery	up to	42

REACH	SYSTEM, DATUM AND CONDITION	TARGET	TERM	PAGE
Iskander 80, Smerch 70, Uragan 50, Grad 40, 203 mm 32, 152 mm 27 km	Zoopark-1 / Zoopark-1M Own-fire tracking range, 1L260. Tracking friendly fire, by weapon	not stated	up to	42
25 km	ARK-1M Rys Detection range, ARK-1M vs MRL rockets. Rocket (WEG); mode not stated	rocket launcher	unspecified	43
up to 25 km	1B75 Penicillin Detection range, 1B75. radius; artillery, mortar and MLRS firing positions, sound and thermal	rocket launcher, mortar, artillery	maximum (claim)	43
20 km	Zoopark-1 / Zoopark-1M Detection range, mortar, 1L219M. Mortar firing position; calibre not stated	mortar	detection of launch position	42
20 km	Zoopark-1 / Zoopark-1M Detection range, mortar, 1L260. Mortar firing position; calibre not stated. ATP lists this figure under '1L219M Zoopark-1M'	mortar	range as listed by ATP (no qualifier)	42
15 km	1B75 Penicillin Detection range, 1L271 (fire adjustment). by shell bursts; by trajectory 5 km	shell bursts	maximum (claim)	43
15 km	Zoopark-1 / Zoopark-1M Detection range, tube artillery, 1L219M. Field artillery firing position	artillery	detection of launch position	42
13 km	ARK-1M Rys Detection range, ARK-1M vs mortars. Mortar (WEG); mode not stated	mortar	unspecified	43
10 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10 radar. shell impact (100 mm), without MTI	shell bursts	maximum (claim)	44
10 km	SNAR-10 / SNAR-10M1 Leopard Detection range, SNAR-10M1 radar. shell burst on land; 20 km on water; developer figure	shell bursts	claimed (claim)	44
8 km	ARK-1M Rys Detection range, ARK-1M vs artillery (howitzers). Artillery (WEG), howitzers (IREW02); target and mode not stated by WEG	artillery	unspecified	43
5 km	1B75 Penicillin Detection range, 1L271 (mortar positions). mortar firing positions, 81-120 mm	mortar	maximum (claim)	43
SIGINT and DF				
400 km	Moskva-1 Detection range, airborne emitters. passive; emitting aircraft and cruise missiles; 360 deg sector	cruise missile, emitters	up to (claim)	47
up to 30 km	Repellent-1 Detection range, UAV control-link signals. ground-station command signals; UAV telemetry up to 10 km; target size and altitude not given	UAV, emitters	up to (claim)	53
6 km	Leer-3 Detection range, GSM base stations. cellular base stations, GSM and possibly 3G/4G, measured from the UAV	UAV, cell base station	within	52
1,300 m	T-80B/BV/BVM, T-80U Detection range, TPN-3-49 night sight. T-80B; active IR; 850 m passive	not stated	sighting	8
650 m	BMP-2 Detection range, BPK-1-42 sight. night, passive starlight channel; 350 m with IR searchlight	not stated	vision range	12
about 400 m	BMD-2 Detection range, 1PN22M1 night channel. BMD-1 gunner sight, passive image intensifier, night	not stated	vision range	14
400 m	BMP-1 Detection range, 1PN22M1 sight. night, passive image intensifier; 900 m with IR searchlight	not stated	maximum	11
Laser designation				
200-8000 m	Orlan-10 Laser range, Orlan-30 designator-rangefinder. measured distance, 1064 nm laser, export Orlan-30	not stated	range of measured distances (claim)	78
up to 7,000 m	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) Range, laser target illumination (1V14). for laser-guided (corrected) munitions	not stated	maximum	67
3 km	Granat-1 / Granat-4 Range, laser designation, Granat-4. laser illumination of targets for guided munitions (Krasnopol)	not stated	claimed (claim)	79
250-300 m	BRDM-2 Detection range, TKN-1S night sight. night, with OU-3 infrared illuminator; target not stated	not stated	observation range	22
Air tracking and fire control				
120 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9A317 / 9A317M TELAR radar (M2/M3). RCS 1-2 m ² at 3 km altitude and 18-20 km vs same target at 10-15 m (M2); M3: RCS 1-2 m ² , altitude not stated	RCS 1-2 m ²	detection range	37
80 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9S35 TELAR radar (M1). RCS 2 m ² ; 100 km vs RCS 3 m ²	RCS 2 m ²	detection range	37

REACH	SYSTEM, DATUM AND CONDITION	TARGET	TERM	PAGE
80 km	Strela-10 and Osa short-range SAM Detection range, Dog Ear (PPRU-1M) battery radar. Strela-10 battery acquisition radar, F/G-band; target type not stated; tracking 35 km (JREW02)	not stated	acquisition range	40
36 km	Pantsir-S1 Detection range, 1RS1 search radar. 2 m2 target; S-band; up to 20 targets tracked	RCS 2 m²	maximum (claim)	39
28 km	Pantsir-S1 Detection range, 1RS2-1 tracking radar. target size not stated; WEG and Armada	not stated	maximum (claim)	39
17-26 km	Pantsir-S1 Detection range, optical post. F-16 autotracking, 10 km visibility; 13-15 km for HARM missile	fighter, anti-radiation missile	auto-tracking (claim)	39
20 km	Strela-10 and Osa short-range SAM Detection range, Land Roll tracking radar. Osa; target type not stated in source	not stated	tracking range	40
20 km	Tor Detection range, TV sight. target not stated	not stated	maximum	38
15 km	Tor Detection range, tracking radar (Tor/Tor-M1). 60-degree sector; target and variant not stated	not stated	maximum	38
13 km	Tunguska Detection range, 1RL144M tracking radar. J-band tracker	not stated	maximum	40
Air surveillance				
600 km	Nebo-M Detection range, Nebo-M. complex; target type, size and altitude not stated	not stated	maximum (claim)	41
600 km	Nebo-M Altitude coverage, Nebo-M. complex, ballistic targets implied; not stated	ballistic missile	maximum (claim)	41
580-600 km	S-400 Triumf Detection range, 91N6E. target RCS >10 m2 (strategic bomber)	RCS 10 m², bomber	detection/ tracking	35
330 km	S-300V4 Detection range, 9S15 Bill Board. target size not stated by WEG	not stated	maximum	36
300 km	96L6 all-altitude acquisition radar Detection range. Range band 5 to 300 km for detected targets; target size and altitude not stated	not stated	instrumented (claim)	42
300 km	S-400 Triumf Detection range, 96L6E. export 96L6E in S-300PMU family, target not stated; JREW-p1877, PDF p. 1877	not stated	max (claim)	35
200 km	Nebo-M Detection range, Podlet-K1. main mode, 10-200 km; target size not stated	not stated	maximum (claim)	41
175 km	S-300V4 Detection range, 9S19 High Screen. ballistic missile search; target size not stated	ballistic missile	maximum	36
160 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9S18M1-3 / 9S18M3 search radar (M2/M3). RCS 1-2 m2; Buk-M2 9S18M1-3	RCS 1-2 m²	detection range	37
100-150 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9S18M1 search radar (M1). target size and altitude not stated	not stated	detection range	37
150 km	S-300V4 Detection range, 9S32 Grill Pan. 2 m2 RCS target, manual mode (140 km automatic)	RCS 2 m²	maximum acquisition	36
120 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9S36 mast radar (M2). RCS 1-2 m2 at 3 km altitude; 30-35 km vs same target at 10-15 m	RCS 1-2 m²	detection range	37
100 km	96L6 all-altitude acquisition radar Maximum target altitude. All-altitude mode, aerodynamic targets	aircraft	maximum (claim)	42
75 km	Pantsir-S1 Detection range, Pantsir-SM radar. target size not stated	not stated	claimed (claim)	39
70 km	Buk-M1 / Buk-M2 / Buk-M3 Detection range, 9S36M mast radar (M3). cruise missile flying at 5 m	cruise missile	detection range	37
40 km	Barnaul-T Detection range, 1L122-1E. aircraft, helicopters, cruise missiles, UAVs at up to 700 m/s; export 1L122-1E brochure figure	cruise missile, helicopter, UAV	maximum (1-40 km) (claim)	69
32 km	Tor Detection range, surveillance radar (Tor-M2). air targets, size not stated	aircraft	detection (claim)	38
30 km	Strela-10 and Osa short-range SAM Detection range, Land Roll surveillance radar. Osa; target type not stated in source	not stated	acquisition range	40
25-27 km	Tor Detection range, surveillance radar (Tor/Tor-M1). aircraft; WEG does not state the variant	aircraft	maximum	38
25 km	CBRN reconnaissance vehicles (RKhM family) Detection range, UAV reconnaissance complex (RKhM-9). Onboard unmanned aerial complex, terrain survey	UAV	claimed (claim)	62
18 km	Tunguska Detection range, 1RL144 surveillance radar. E-band; low-flying search mode for targets at about 15 m (JREW02)	not stated	maximum	40

REACH	SYSTEM, DATUM AND CONDITION	TARGET	TERM	PAGE
9–15 km	Tor Detection range, surveillance radar (Tor/Tor-M1), UAV, UAVs	UAV	maximum	38
12 km	Tor Detection range, surveillance radar (Tor/Tor-M1), helicopter, helicopters	helicopter	maximum	38
10 km	Barnaul-T Detection altitude, 1L122-1E, export 1L122-1E; 'at least 10 km'	not stated	maximum (claim)	69
10,000 m	BRM-1K Detection range, PSNR-5K (1RL133-1), target type not stated by WEG	not stated	maximum	13
10 km	Nebo-M Minimum range, Podlet-K1, air targets, lower limit of stated 10–200 km detection zone	aircraft	minimum (claim)	41
10 km	Nebo-M Altitude coverage, Podlet-K1, maximum detection altitude	not stated	maximum (claim)	41
0.45 to 10 km	Strela-10 and Osa short-range SAM Detection range, 9S86 Snap Shot, Strela-10 range-only radar; target type not stated; minimum 0.45 km	not stated	maximum	40
7,000 m	BMPT Terminator Detection range, thermal sight, thermal channel	not stated	detection (claim)	11
5,000 m	BMPT Terminator Detection range, gunner's sight, optical channel, day	not stated	detection (claim)	11
5 km day; 3.5 km night	BRDM-2 Detection range, BRDM-2MS thermal sight, BRDM-2MS thermal camera in optronic head at turret rear; target not stated	not stated	claimed (claim)	22
3 km people; 12 km vehicles	Artillery command vehicles 1V18/1V19, 1V119, 1V152, 1V1003 Detection range, PSNR-5K Tall Mike (1V119 optional), I-band ground surveillance radar, roof-mounted on 1V119; reference-site figure	not stated	claimed (claim)	67
3,000 m	BRDM-2 Detection range, TPKU-2B commander's sight, day; 5x periscope, target not stated	not stated	observation range	22
about 3 km	CBRN reconnaissance vehicles (RKhM family) Detection range, PKhRDD-2B remote chemical detector (RKhM-6), Remote vapour detection; target and conditions not stated	not stated	claimed (claim)	62
2–3 km	T-62 Detection range, 1PN96MT-02 thermal sight, T-62M/MV obr. 2022, uncooled thermal; target type not stated	not stated	approximate	9
2,600 m	T-80B/BV/BVM, T-80U Detection range, Agava thermal sight, T-80UK, night	not stated	acquisition	8
2.5 km	T-90 Detection range, Agava-2 thermal sight, target identification, night	not stated	identification	7
2.5 km	Tigr and Vodnik armoured cars Detection range, Arbalet-DM TV camera, target identification; thermal imager 1.5 km	not stated	claimed (claim)	22
1,750 m	T-80B/BV/BVM, T-80U Detection range, T-80U gunner night sight, night; classification 1,500 m	not stated	detection	8
1.2–1.5 km	T-90 Detection range, Buran-PA night sight, target identification, night	not stated	identification	7
1 km	Barnaul-T Minimum detection range, 1L122-1E, export 1L122-1E	not stated	minimum (claim)	69
1,000 m	BTR-80 Range, KPVT 14.5 mm, air target, air target; BTR-80 1PZ-2 sight; round not named	aircraft	sight range	20
400 m	T-62 Detection range, commander's night sight, IR night channel with OU-3GK searchlight (specs match TKN-3)	not stated	range	9
about 40 m	MT-LB Detection range, driver's TVN-2 IR periscope, night, with OU-3GK IR searchlight	not stated	approximate	17
Other				
200 km	Mohajer-6 Data link range, line of sight, Ground control station data link, direct line of sight	not stated	claimed (claim)	84
up to 100 km	PU-12M7 Coordinates measurement envelope, PU-12M7 (connected radars), coordinates measurement envelope from connected radars; PU-12M7 has no radar of its own	not stated	maximum (claim)	69
50–80 km	BRM-1K Reconnaissance depth, BRM-1KM, Front-line reconnaissance, distance from own troops' positions	not stated	intended (claim)	13
land route 8 to 10 km/h; minefield 5 km/h; 100 m water barrier in 5 min	IRM Zhuk Reconnaissance rate, WEG attributes these to unnamed Russian sources	not stated	typical	57

REACH	SYSTEM, DATUM AND CONDITION	TARGET	TERM	PAGE
1,500 m	BMD-2 Range, 762 mm PKT. ball, coaxial	not stated	maximum effective (with sight adjustments)	14

Table 3g. Reach of communications

Radio and data-link ranges. Ranges depend on antenna, terrain and whether the vehicle is halted.				
REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE	
2000 km	R-161 Poisk HF radio station Communication range, HF. R-161A2M, stationary; UTIN: V 2x46 antenna, 10-30 MHz	maximum	70	
2,000 km	R-166 Artek Communication range, HF, stationary. R-166, at the halt	up to (claim)	70	
450 km	R-149 command and staff vehicles Range, HF air-ground Prima, R-149MA1. HF to aircraft	maximum (claim)	64	
350 km	R-142N Deimos-N Communication range, R-130M HF. R-142N, roof loop (AZI) antenna or wire dipole, day and night, between like stations	communication range	71	
350 km	R-142N Deimos-N Communication range, R-142NMR, R-142NMR, HF 1.5-30 MHz and VHF 30-108 MHz FH, voice and data	maximum range	71	
350 km	R-145BM Chayka command and staff vehicle Range, R-130M dipole. HF, halted only, symmetrical dipole	range	65	
350 km	R-149 command and staff vehicles Range, HF R-168-100KBE, R-149MA1, halted, mast antenna; 50 km (20 km moving) on ASH-4 whip	maximum (claim)	64	
350 km	R-149 command and staff vehicles Range, HF, R-149BMR. HF band 1.5-30 MHz	maximum (claim)	64	
350 km	R-166 Artek Communication range, HF, on the move. R-166, moving	up to (claim)	70	
300 km	BMD-1KSh-A Range, HF radio R-130. Up to, with 4 m whip antenna; BMD-1KSh	maximum	66	
15-300 km	R-149 command and staff vehicles Range, HF R-168-5KV, R-149AKSh-1 1.5-30 MHz, 2 sets	maximum	64	
220-250 km	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) Radio range, R-111. 1V15 and 1V16, vehicle radio	stated	67	
250 km	R-166 Artek Communication range, VHF, stationary. R-166, at the halt	up to (claim)	70	
150 km	Merlin-VR Data link range. Flight distance without loss of link; Russian serviceman's statement	claimed	82	
at least 150 km	Tropospheric scatter radio stations (R-410-7.5, R-412A/AM, R-417) Link range, R-412A, between two stations of the same type	at least (claim)	72	
130 km	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) Radio range, R-130MT. 1V15 and 1V16, HF vehicle radio	stated	67	
120 km	R-149 command and staff vehicles Range, VHF air-ground Prima, R-149MA1. VHF to aircraft	maximum (claim)	64	
100 km	Supercam S350 Range, command and telemetry radio link. radio line of sight to ground control station	maximum (claim)	79	
75 km	R-145BM Chayka command and staff vehicle Range, R-130M zenith antenna. HF, halted or moving, symmetrical zenith-radiation antenna	range	65	
75 km, 300 km	R-161 Poisk HF radio station Communication range, moving. R-161A2M moving at 30 km/h: roof whip 75 km, zenith-radiating antenna (1.5-14 MHz) 300 km	maximum	70	
70 km	Granat-1 / Granat-4 Data link range, Granat-4, photo, video and thermal data transmitted near real time	claimed (claim)	79	
70 km	R-166 Artek Communication range, VHF, on the move. R-166, moving	up to (claim)	70	
60 km	R-142N Deimos-N Communication range, R-111 VHF. R-142N, antenna on 11 m mast, at halt	communication range	71	
60 km	R-149 command and staff vehicles Range, VHF, R-149BMR. VHF band 30-108 MHz	maximum (claim)	64	
25-60 km	R-149 command and staff vehicles Range, VHF R-168-100U-2, R-149AKSh-1 30-108 MHz, 2 sets	maximum	64	
55 km	R-149 command and staff vehicles Range, VHF R-168-100UE-2 on mast, R-149MA1, halted, mast antenna	maximum (claim)	64	
up to 55 km	Redut-2US Radio-relay range, radio-relay link at up to 155 Mbit/s	maximum	74	
50 km	Lancet Range, Izdeliye 51, control link range, 5 kg warhead; ZALA website figure reported by RT	communication range (claim)	85	
50 km day, 20 km night	R-142N Deimos-N Communication range, R-130M HF. R-142N, 4 m whip, at halt and on the move	communication range	71	
50 km	ZALA 421-16E Control range, video/radio link. 421-16E; average radius of video/radio link	radius of action (claim)	77	

REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE
at least 40 km	P-240I-4 Pereselenets Communications range, medium and conditions INA	claimed (claim)	71
40 km	1B75 Penicillin Radio range, 1B75, to external subscribers	maximum (claim)	43
up to 40 km	PU-12M7 Data link range, PU-12M7 by radio, radio link to higher and subordinate posts	maximum (claim)	69
40 km	R-142N Deimos-N Communication range, R-123MT VHF, R-142N, antenna on 11 m mast, at halt	communication range	71
25-40 km	R-149 command and staff vehicles Range, VHF R-168-25U-2, R-149MA3, 30-108 MHz, 2 sets	maximum	64
35 km	R-145BM Chayka command and staff vehicle Range, R-111 3.4 m whip, VHF, halted or moving	range	65
up to 30 km	PU-12M7 Data link range, 9S737 to higher post, radio	maximum (claim)	69
up to 30 km	Redut-2US Wireless access range, broadband wireless access at up to 37 Mbit/s	maximum	74
20-25 km	1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16) Radio range, 1V13 and 1V14, vehicle radios	stated	67
25 km	R-149 command and staff vehicles Range, VHF R-168-100UE-2, R-149MA1, on the move, vehicle antenna	maximum (claim)	64
20 km	Barnaul-T Data link range, 9S932-1, stationary; 10 km on the move (9S933: 3 km / 1 km)	communication range	69
20 km	R-145BM Chayka command and staff vehicle Range, R-123MT 4 m whip, VHF, halted or moving	range	65
6-20 km	R-149 command and staff vehicles Range, UHF R-168-MRA, R-149MA3, 1.5-1.75 GHz data radio	maximum	64
up to 15 km	PU-12M7 Data link range, PU-12M7 by field cable, field telephone cable	maximum (claim)	69
up to 8 km	Strelets Range, R-853-V2M to aircraft, Target designation data to aircraft, no relay	claimed (claim)	74
6 km	PRP-4A Argus / PRP-4M Deytery Dismounted observation post, distance from vehicle, dismounted observation post	maximum	68
6 km	R-142N Deimos-N Communication range, R-809M2 air band, R-142N, discone on mast, to aircraft	communication range	71
up to 5 km	PU-12M7 Data link range, 9S737 to fire units, radio	maximum (claim)	69
3-5 km	Strelets Range, PDU-4 laser rangefinder, Operating range ('гальность геуцмбуя'); target type not stated	operating range (claim)	74
about 1.5 km	Strelets Range, squad radio network, Line of sight; each set acts as relay, so reach grows	claimed (claim)	74
up to 30 m	R-416GM radio relay station Mast height, raised, Ukrainian border guard description of R-416G-MS	sometimes reaches (claim)	72

Table 3h. Reach of electronic attack

Jamming ranges. Most are manufacturer claims.			
REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE
5,000 km	Murmansk-BN Jamming range, HF communications. HF (shortwave) radio links, skywave; ship, aircraft and ground emitters	effective (Armada's term) (claim)	45
about 1,000 km wide strip	Palantín Jamming frontage, radio communications. ground target: radio communications and radio reconnaissance; battalion tactical group exercise near Voronezh, Sep 2019	claimed (claim)	49
150–300 km	Krasukha-2 / Krasukha-4 Krasukha-4: jamming range. air and ground-based radar targets; target size and altitude not stated	maximum (claim)	46
up to 250 km	Krasukha-2 / Krasukha-4 Krasukha-2: jamming range. air target: AEW aircraft radar; target altitude not stated	up to	46
250 km	Moskva-1 Jamming range, airborne S-band radar. air targets; ARM23 says the 1L267 'can engage targets' per unnamed sources; Moskva-1 itself has no jammer (FOI)	up to	47
50 km or more	R-330Zh Zhitel Jamming range, airborne users. User equipment installed on aircraft	claimed minimum (claim)	50
up to 30 km	Repellent-1 Jamming range, small UAV control and telemetry links. air target: small UAVs; telemetry and satellite-navigation links; control link up to 10 km	up to (claim)	53
up to 25 km	Pole-21 Jamming range, GNSS receivers (ground or air not stated). WEG (Pole-21E); Russian press gives 25 km per jamming module for the domestic Pole-21	effective	53
25 km	R-330Zh Zhitel Jamming range. Ground user equipment (WEG gives no target type; matches manufacturer's ground-user figure)	maximum effective	50
22 km	2S19 Msta-S Range, jammer round. 3RB30 radio jamming projectile; jamming radius 700 m	range (claim)	30
up to 20 km	Palantín Jamming range, UAV control links. air target: UAV radio control links	maximum (claim)	49
up to 6 km	Leer-3 Jamming range, cellular terminals (ground). radius of the zone of blocked GSM terminals around the UAV	up to (claim)	52
3 km	Infaua Jamming range, target not stated. WEG 'Disruption Range'; target and conditions not given	disruption range	54

Table 3i. Reach of UAVs

How far from its control point a UAV operates, and at what altitude.			
REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE
2,000 km	Mohajer-6 Flight range. 'Operational range (radius of action)' 2,000 to 2,400 km per Wikipedia, from Iranian press	claimed	84
1,350 km	Geran-2 Range, Geran-2. 50 kg warhead; leaked Alabuga documents; see mobility data for alternatives	reported (claim)	87
1,350 km	Geran-2 Flight range, Geran-2. standard 50 kg warhead; figure from leaked Alabuga (manufacturer) documents	reported	87
700–1,000 km	Geran-2 Flight range, Geran-1. estimate; DIA estimate 900 km	estimated	87
up to 1,000 km	Geran-2 Flight range, Geran-3. U-series, Telefly JT80 turbojet; Ukrainian intelligence	operational	87
650 km	Geran-2 Flight range, Geran-2 with BCh-90. 90 kg warhead, smaller fuel tank; leaked Alabuga documents	reported	87
600 km	Gerbera Flight range. without payload; shorter with warhead; some have a second fuel tank in the nose for extended range	maximum	88
600 km	Merlin-VR Flight range. Kyiv Post, basis and source not given	claimed	82
600 km	Orlan-10 Flight range, Orlan-10 autonomous. pre-programmed flight without radio link	maximum	78
300 km	Orlan-10 Flight range, Orlan-30 autonomous. autonomous mode, export Orlan-30	maximum	78
250 km	Forpost Operating range. directional antenna; 150 km with omnidirectional antenna	range of employment	83
250 km	Orion Radius of action. without relay UAV; Orion-E (WIKIRU); HUR claim (UPDIU)	up to	82
at least 240 km	Supercam S350 Flight range. total flight distance per maker, not radius of action	maximum	79
200 km	Italmas Range, one-way attack. claimed (Rossiya 1 TV: over 200 km); warhead not specified	claimed (claim)	86
200 km	Italmas Flight range. claimed, over 200 km per Rossiya 1 TV; Ukrainian sources call the claim possible propaganda	claimed	86
200 km	ZALA 421-16E Range, flight distance. 421-16E per 2011 article text, in more than 3.5 h; the article's table gives 210 km	claimed	77
150 km	Orlan-10 Flight range, Orlan-10 under radio control. maximum link range with ground antenna, Orlan-10M	maximum	78
120 km	Korsar Flight range, radius from control post. terrain reconnaissance mission	claimed	83
120 km	Leer-3 Practical range, Orlan-10 from control vehicle. radius of UAV employment from the control point	up to (claim)	52
120 km	Orlan-10 Flight range, Orlan-30 under radio control. data-link range, export Orlan-30	maximum	78
100 km	Granat-1 / Granat-4 Range of action, Granat-4. flight range per Kalashnikov	claimed (claim)	79
over 100 km	Orlan-10 Practical range, Leer-3 EW tasks. distance from unit deployment site, Russian MoD statement, after Feb 2022	practical (claim)	78
about 70 km	Lancet Range, Lancet. MiG-29 struck at Dolgintsevo airfield 2023-09, shown on video (KYIV); about 70 km from the front per Russian claims reported by Wikipedia	practical	85
70 km	Supercam S350 Range, video link. standard video channel to ground control station	maximum (claim)	79
about 70 km	Supercam S350 Practical range, video link. good picture and link, operator's account 2024-08	practical	79
50 km	Eleron-3SV Flight range, autonomous. no link, imagery recorded on board	maximum	80
up to 50 km	Molniya Range, Molniya-2 with warhead. Molniya-2; MoD figure, control link not stated	maximum (claim)	88
40–50 km	Orlan-10 Practical range, artillery spotting. distance mentioned in Russian press articles on artillery spotting c. 2015, per RWW; RWW suggests a video-link limit	practical	78
over 40 km	FPV drones Range, Provod fibre-optic. spool length; Ovod centre statement via TASS, Aug 2026	claimed (claim)	89
40 km	Lancet Range, Lancet-3 (Izdelye 52). flight range with 3 kg warhead	range	85

REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE
12 to 40 km	Mohajer-6 Range, Qaem-series guided bomb. Glide release from 1,219 to 3,048 m; series-wide figure stated by Iranian minister Hatami, 2019	claimed (claim)	84
about 40 km	Molniya Range, Molniya-1 with warhead. Kharkiv regional prosecutor's office; manual FPV control, no GPS	maximum (claim)	88
40 km	Takhion Range, radius of action. Radius of action; kalashnikov.club: within direct radio line of sight	claimed (claim)	81
40 km	Takhion Radius of action. kalashnikov.club: within direct radio line of sight	claimed	81
25 km	Eleron-3SV Control range, real-time link. real-time video over data link	maximum	80
25 km	KUB-BLA Range, KUB HE-frag warhead. radius of action, KUB-E export figure (Kalashnikov via TASS)	radius (claim)	86
20 km	FPV drones Range, KVN fibre-optic. fibre-optic spool; figure 'according to Russian resources', Jan 2025	maximum (claim)	89
15 km	Granat-1 / Granat-4 Range, Granat-1 photo. line of sight, still camera	maximum (claim)	79
7-10 km	FPV drones Range, VT-40. munitions of 1.3-3 kg	claimed (claim)	89
over 10 km	FPV drones Range, fibre-optic types. Russian fibre-optic drones striking Kostiantynivka, just over 10 km from the contact line, Apr 2025	practical	89
10 km	Granat-1 / Granat-4 Range, Granat-1 video. line of sight, video camera	claimed (claim)	79
10 km	Orion Range, Kh-BPLA. air-launched Kornet derivative; ground or low slow air target; TWZ gives 6.2 miles from RIA source	maximum (claim)	82
10 km	Zastava Range of action. Flight range from the ground station; ROBOT 'up to 10 km', CNA 'range', WPRU (Bird-Eye 400) 'radius'	up to	81
8 km	Forpost Range, KAB-20. released from high altitude; KAB-20L laser or KAB-20S satellite guidance	maximum stand-off	83
7,500 m	Orion Ceiling. Orion-E developer figure (WIKIRU); Inokhodets per RT, Russian state media	maximum	82
up to 7 km	Geran-2 Ceiling, Geran-3	maximum	87
6,000 m	Forpost Service ceiling, Forpost-R		83
6000 m	Orlan-10 Ceiling, Orlan-10. above sea level	maximum	78
5,400 m	Mohajer-6 Maximum altitude. 'Open data' figure in Ukrainian press	maximum	84
5100 m	Korsar Ceiling	maximum	83
up to 5,000 m	Leer-3 Altitude, Orlan-10. operating altitude	up to (claim)	52
5 km	Merlin-VR Service ceiling. Claimed; operating altitude 450-2500 m per PLANDIY	maximum	82
150-5,000 m	Supercam S350 Operating altitude		79
60-4,000 m	Geran-2 Flight altitude, Geran-2. Ukrainian General Staff handbook	operating (claim)	87
4,000 m	Granat-1 / Granat-4 Ceiling, Granat-4. above sea level; working band 300-2,000 m	maximum (claim)	79
4,000 m	Takhion Ceiling. Minimum flight height 50 m (WRUJ)	maximum	81
3,600 m	ZALA 421-16E Altitude, maximum. 421-16E	maximum (claim)	77
3,500 m	Granat-1 / Granat-4 Ceiling, Granat-1. working band 40-1,500 m above ground	maximum (claim)	79
3500 m	Orlan-10 Ceiling, Orlan-30. above sea level, export Orlan-30	maximum	78
1,219 to 3,048 m	Mohajer-6 Release altitude, Qaem-series guided bomb. Release height 4,000 to 10,000 ft, same Iranian statement	claimed (claim)	84
3,000 m	Eleron-3SV Operating altitude, Eleron-3SV. above launch point	maximum (claim)	80

REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE
60-3,000 m	Lancet Altitude, flight. Lancet type, Ukrainian General Staff figure		85
100-2,500 m	KUB-BLA Altitude, KUB-10E. flight altitude band; export designation	operating (claim)	86
2200 m	Zastava Ceiling, above sea level	maximum	81
700-2,000 m en route; down to 200 m near target	Geran-2 Flight altitude, Geran-2 observed. Ukrainian General Staff handbook, observed 2022-23; as low as 20 m along rivers and roads	practical (claim)	87
500-1,500 m	ZALA 421-16E Altitude, operating. current Z-16E	operating altitude range (claim)	77
up to 1 km	Gerbera Operating altitude. Espresso report via Kyiv Post, 2025-02	maximum	88
150-1,000 m	KUB-BLA Altitude, KUB HE-frag warhead. flight altitude band, KUB-E export figure (Kalashnikov via TASS)	operating (claim)	86
up to 300-400 m	Molniya Operating altitude. Molniya-2, serviceman in MoD video		88
20-50 m	FPV drones Flight altitude, fibre-optic types. reported by a Ukrainian drone commander	practical	89

Table 3j. Deployment distances

Where the system has been seen deployed or employed relative to the line of contact, and how far its elements spread. These are observations and reports, not rules; they indicate what may lie at a given depth.

REACH	SYSTEM, DATUM AND CONDITION	TERM	PAGE
up to 100 km	S-400 Triumf Fire unit distance from command post. 98Zh6E fire unit from 55K6E, export composition	maximum (claim)	35
5 to 40 km	Takhion Design employment distance from line of contact. Design figure, not observed: 'designed to operate at a distance of 5 to 40 km from the line of contact'	design	81
15 to 30 km	Borisoglebsk-2 Employment distance from line of contact. Donbas 2014-15, with Leer-3, Zhitel, R-934UM	practical	48
25 km	BM-27 Uragan Distance from front line, observed. one launcher hit by FPV drone, reported c. 2024-07	practical	27
about 15 km from enemy positions	2S4 Tyulpan Employment distance. reported MoD video, Ukraine, Nov 2022	practical (claim)	31
5-10 km behind forward edge	Tor Deployment distance, Tor-M1 in defence. first defensive position; 5-10 km between vehicles	practical	38
1.5 to 2.0 km	BMO-T Dismount distance from enemy line. Doctrinal scenario: platoon attached to motor rifle battalion, then advances on foot to about 600 m	practical	18
1 km	Uran-6 Remote control distance. by radio from a control panel	claimed maximum (claim)	60

Table 3k. Engineering equipment

What engineer vehicles can do, grouped by task: gap crossing, breaching, mine laying, earthmoving and transport across water. Mine-laying rockets also appear under indirect fire.			
SYSTEM, DATUM AND CONDITION	CAPABILITY	PAGE	
Gap crossing			
PMP heavy folding pontoon bridge park	227 m	56	
Bridge, full set, 60-t. 32 river + 4 shore links, 6.5 m roadway			
Bridge, full set, 20-t. split half-links, 3.29 m roadway	382 m		
Bridge, half set, 60-t. 16 river + 2 shore links	119 m		
Bridge, half set, 20-t. 16 river + 2 shore links, split	191 m		
Load capacity, bridge, full-width bridge; 20 t on split-link bridge	60 t		
Roadway width, 60-t bridge	6.5 m		
Rafts, 2, 3, 4, 5+1 shore, 8+1 shore links	40, 60, 80, 110, 170 t		
Raft assembly, 40-t. 2 river links, working party 6	8 min		
Maximum current, PMP	2 m/s		
Tank speed on bridge, tanks on 60-t bridge	up to 30 km/h	55	
Working party, full bridge, full bridge set	approx. 65		
TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge	up to 40 m wide, 3 m deep		
Obstacle, TMM-3 bridge set. Set of four spans; ATP gives 42 m total length			
Load class, TMM-3. Wheeled and tracked loads; tracked vehicles at 15 km/h on bridge	60 t		
Bridge geometry, TMM-3. Spans can be connected to 10.5, 21, 31.5 or 42 m (RWW); ATP gives 10 m per set and 42 m together	4 spans of 10.5 m; roadway 3.8 m		
Bridge, MTU-72. Single-span bridge; ENT72 and RUWM support only the 18 m span and 50 t	20 m long, 3.3 m wide, spans 18 m; 6,400 kg		
Load class, MTU-72	50 t		
Breaching and mine clearing			
BMR-3M	2 tracks x 0.8 m		59
Swept lane width, KMT-7, against track-width mines; uncleared strip of 1.62 m between			
Swept lane width, continuous, belly mines with contact fuze; 6-7 m vs magnetic fuzes	3.2 m		
Swept lane width, Vepr. Vepr on T-90A, 2024	almost 4 m	58	
IMR-2 / IMR-2M / IMR-3M combat engineer vehicle	6 km/h		
Mine plough lane clearing, IMR-2M. AT mines, lane clearing rate (WEG term)			
UR-77 Meteorit	200 to 500 m	59	
Range, UZP-77 line charge. charge launch distance, UZP-77; UZ-67 charge 200 to 350 m (WEG)			
Cleared lane length, UZP-77. anti-tank minefield; UZ-67 charge 75 to 80 m	80 to 90 m		
Cleared lane width, one line charge	up to 6 m		
Line charge length, UZP-77; UZ-67 is 83 m	93 m		
Cleared lane length, ZRShch. lane 6 m wide; charge in 6 m sections of 41 kg	100 m		
Time to clear a path, one full demining cycle	3 to 5 min	60	
Uran-6	1.6 m		
Clearing width, flail fitted			
Clearing depth, flail fitted	35 cm	59	
Mine laying			
BMR-3M	KMT-7 (or KMT-8)		
Mine roller trawl, BMR-3MA; TMT-S			
IRM Zhuk	RShM-2 wide-span detector; ferrous mines to 0.3 m depth in a 3.6 m lane		
Mine detection, antitank mines with ferromagnetic parts			
ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer	15 km		
Range, ISDM mine rocket, cassette rocket with AP or AT mines			
Minimum range, ISDM mine rocket, cassette rocket with AP or AT mines	5 km		
Reload, ISDM, container swap by transloader at a technical position	several minutes		
Mine load, GMZ-3	208 AT mines		
Minefield length per load, GMZ-3, one vehicle, one load	1 km (pressure mines); 2 km (influence mines)	61	
Platoon minefield, GMZ-3, 3 GMZ-3 of obstacle platoon	1,200 m, three rows, 624 mines in 26 min		
Laying speed, GMZ-3	16 km/h surface; 6 km/h buried		
Laying rate, GMZ-3	8 mines/min surface; 4 mines/min buried		
Mine spacing, GMZ-3	5 or 10 m		
Smoke generator vehicles (TDA-2K, TDA-3, TDM-2K)	10-40 km/h		
Speed while laying smoke, TDA-2K		63	

SYSTEM, DATUM AND CONDITION	CAPABILITY	PAGE
UMZ	11,520 / 720 / 180	61
Mines per load, PFM-1S / POM-2 / PTM-3. basic load by mine type; one source gives 180 to 11,520 range		
Minefield length. WEG general figure, mine type not stated; WEG gives one-load maxima of 3,200 m (PFM-1S), 5,000 m (POM-2), 600 m (PTM-3)	1,000 to 1,200 m	
Laying distance from vehicle. vehicle moving 10 to 40 km/h	30 to 60 m	
Laying speed	10 to 40 km/h	
Earthmoving		
BAT-2	350 to 450 m ³ /h	58
Earthwork capacity, filling ditches and craters. ditch digging 200 to 250 m ³ /h		
BREM and BTS armoured recovery vehicles	3.1 m	62
Dozer blade width, BREM-1		
IMR-2 / IMR-2M / IMR-3M combat engineer vehicle	2 t	58
Boom lifting capacity		
Boom reach, IMR-2M	8.4 m	
Dozer capacity, IMR-2 earthmoving	200 to 250 m ³ /h	
Military earthmoving and trenching machines	up to 1.5 m	60
Trench depth, BTM-3. soils up to category IV, unfrozen		
Output, BTM-3, trench 1.1 m deep. technical output, depends on soil	270 - 810 m/h	
Output, BTM-3, trench 1.5 m deep. technical output, depends on soil	270 - 560 m/h	
Trench depth, PZM-2. unfrozen and frozen ground	1.2 m	
Digging depth, EOY-4421. backhoe, bucket 0.65 m ³	3.25 m	
Digging depth, EOY-3521M. backhoe, bucket 0.65 m ³ , Ural-4320 chassis	4.0 m	
Digging depth, EOY-3523. bucket 0.8 m ³ , KamAZ-53501 chassis	5 m	
Uran-6	1.8 m	60
Bulldozer blade width		
Transport across water		
PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry	12 t	57
Payload on land, PTS-4		
Payload on water, PTS-4	18 t	
Payload, PMM-2 single ferry. WEG; RUWM gives 42.5 t for the PMM-2M	42.5 t	
Payload, PMM-2 two joined. WEG gives 85 t for two PMM-2; RUWM gives 85 t for two PMM-2M, assembly 8 min	85 t	

Table 4. Mobility and time into action

SYSTEM	CODE	WEIGHT	FORDING	INTO ACTION	OUT OF ACTION	PAGE
1B75 Penicillin	WH-AT	INA	INA	5 min	INA	43
1V12 artillery fire control complex (1V13, 1V14, 1V15, 1V16)	TR-A	14.8 t	Amphibious	INA	INA	67
2A36 Giatsint-B 152 mm towed gun	TW	9,760 kg	INA	4 min	4 min	32
2A65 Msta-B	TW	7,000 kg	INA	2 to 2.5 min	2 min	32
2S19 Msta-S	TR	42 t	1.2 m unprepared; 5 m prepared	1.5 min	1.5 min	30
2S35 Koalitsiya-SV	TR	48-55 t	2.1 m	1.5 min	under 1 min	30
2S4 Tyulpan	TR	27.5 t	1.0 m	25 min	25 min	31
2S5 Giatsint-S	TR	28.2 t	1.05 m	less than 2 min	less than 2 min	29
2S7 Pion / 2S7M Malka	TR	46,500 kg	1.2 m	5-6 min	3-5 min	31
96L6 all-altitude acquisition radar	WH-AT	INA	1.3 m	5 min	5 min	42
Aleut and Plastun articulated and light tracked carriers	TR	11 t	INA	INA	INA	19
ARK-1M Rys	TR	15.7 t	6 km/h	5 min	5 min	43
Artillery command vehicles 1V18/1V19, 1V119, 1V152, 1V1003	WH-AT	10.1 t	10 km/h	10 min	INA	67
Barnaul-T	TR-A	11,000 kg	5 km/h	5 min	7 min	69
BAT-2	TR	39.7 t	1.3 m	INA	INA	58
BM-21 Grad 122 mm MRL	WH-AT	13.7 t	1.5 m	3 min	2 min	26
BM-27 Uragan	WH-AT	20 t	1.2 m	3 min	3 min	27
BM-30 Smerch	WH-AT	43.7 t	1.1 m	3 min	3 min	28
BMD-1KSh-A	TR-A	8.7 t	10 km/h	INA	INA	66
BMD-2	TR-A+AD	8,225 kg	10 km/h	INA	INA	14
BMD-4M	TR-A+AD	13.5 t	10 km/h	INA	INA	13

SYSTEM	CODE	WEIGHT	FORDING	INTO ACTION	OUT OF ACTION	PAGE
BMO-T	TR	43.9 t	1.2 m	INA	INA	18
BMP-1	TR-A	13,500 kg	7 km/h	INA	INA	11
BMP-1KSh command and staff vehicle	TR	13 t	7 km/h	INA	INA	65
BMP-2	TR-A	14,300 kg	amphibious	INA	INA	12
BMPT Terminator	TR	48 t	1.2 m unprepared; 1.8 m prepared; 5 m with snorkel	INA	INA	11
BMR-3M	TR	48 t	1.2 m unprepared; 5 m prepared	INA	INA	59
Borisoglebsk-2	TR-A	11,000 kg	5 km/h	15 min or less	15 min or less	48
BRDM-2	WH-AT-A	7.0 t	10 km/h	INA	INA	22
BREM and BTS armoured recovery vehicles	TR	41 t	1.2 m	INA	INA	62
BRM-1K	TR-A	13.3 t	7 km/h	INA	INA	13
BTR-80	WH-AT-A	13,600 kg	9 km/h	INA	INA	20
BTR-90 Rostok	WH-AT	22,000 kg	amphibious, no preparation; 9 km/h afloat	INA	INA	21
BTR-D	TR+AD	8,200 kg	10 km/h	INA	INA	16
BTR-MDM Rakushka	TR-A+AD	13.2 t	10 km/h	INA	INA	15
Buk-M1 / Buk-M2 / Buk-M3	TR	34 t	1 m	5 min	5 min	37
CBRN reconnaissance vehicles (RKhM family)	WH-AT-A	13.6 t	9 km/h	INA	INA	62
Decontamination vehicles (ARS-14, TMS-65, TZ-74 family)	WH-AT	15,350 kg	1.75 m	6-8 min	10-12 min	63
Eleron-3SV	AIR-FW	5.3 kg	INA	10 min	INA	80
Forpost	AIR-FW	500 kg	INA	INA	INA	83
FPV drones	AIR-RW	INA	INA	INA	INA	89
Geran-2	AIR-FW	200 kg	INA	INA	INA	87
Gerbera	AIR-FW	about 18 kg	INA	INA	INA	88
Granat-1 / Granat-4	AIR-FW	30 kg	INA	15 minutes	INA	79
IMR-2 / IMR-2M / IMR-3M combat engineer vehicle	TR	44.3 t	1.2 m unprepared; 5.0 m with equipment	6 min	INA	58
Infaua	WH-AT-A	13,614 kg	9 km/h	INA	INA	54
IRM Zhuk	TR-A	18,000 kg	10 km/h	INA	INA	57
ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer	WH-AT	INA	1.8 m	INA	INA	61
Iskander-M	WH-AT	42.85 t	1.4 m	16 min to first launch from the march; 4 min from highest readiness	INA	33
Italmas	AIR-FW	50 kg	INA	INA	INA	86
KamAZ military trucks (4310 and Mustang family)	WH-AT	9.2 t	1.75 m	INA	INA	75
Korsar	AIR-FW	200 kg	INA	INA	INA	83
Krasukha-2 / Krasukha-4	WH-AT	39,000 kg	1.75 m	less than 20 min	less than 20 min	46
KUB-BLA	AIR-FW	8 kg	INA	20 min	10 min or less	86
Lancet	AIR-FW	12 kg	INA	under 1 min	INA	85
Leer-2	WH-AT	7,800 kg	1.3 m	INA	INA	51

SYSTEM	CODE	WEIGHT	FORDING	INTO ACTION	OUT OF ACTION	PAGE
Leer-3	WH-AT	INA	INA	30 min	INA	52
Merlin-VR	AIR-FW	6.5 kg	INA	INA	INA	82
Military earthmoving and trenching machines	TR	277 t	1.5 m	5-7 min	INA	60
Mohajer-6	AIR-FW	600 kg	INA	INA	INA	84
Molniya	AIR-FW	10 kg	INA	INA	INA	88
Moskva-1	WH-AT	12,450 kg	1.75 m	45 min	INA	47
MT-LB	TR-A	11,900 kg	5-6 km/h	INA	INA	17
Murmansk-BN	WH-AT	up to 35 t	INA	72 h	INA	45
Nebo-M	WH-AT	INA	INA	12 m	15 min	41
Orion	AIR-FW	1,000 kg	INA	INA	INA	82
Orlan-10	AIR-FW	18 kg	INA	under 15 min	INA	78
Other armoured cars and utility vehicles (family)	WH-AT	10.5 t	1.75 m	INA	INA	24
P-240I-4 Pereselenets	WH-AT	INA	INA	INA	INA	71
Palantin	WH-AT	INA	INA	about 30 min	INA	49
Pantsir-S1	WH-AT	14.85 t	1.8 m	3 min	INA	39
PMP heavy folding pontoon bridge park	WH-AT	INA	INA	7 m of bridge per minute	a few minutes	56
Pole-21	FX	INA	INA	INA	INA	53
Polyana-D4	WH-AT	13.2 t	1.2 m	35 min	35 min	68
PRP-4A Argus / PRP-4M Deytery	TR-A	13.8 t	7 km/h	2 min	INA	68
PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry	TR-A	33.1 t	INA	5 min	INA	57
PU-12M7	WH-AT-A	13.6 t	10 km/h	3 min	INA	69
R-142N Deimos-N	WH-AT	5.97 t	0.8 m	INA	INA	71
R-145BM Chayka command and staff vehicle	WH-AT-A	10.3 t	10 km/h	INA	INA	65
R-149 command and staff vehicles	WH-AT-A	15,000-15,050 kg	10 km/h	10 min on the move / 30 min halted	INA	64
R-161 Poisk HF radio station	WH-AT	10430 kg	INA	1 hour	INA	70
R-166 Artek	WH-AT	14,840 kg	INA	5 min	INA	70
R-330B Mandat-B and R-934BMV	WH-AT	11,000 kg	INA	30 min	INA	47
R-330Zh Zhitel	WH-AT	INA	INA	40 min or less	40 min or less	50
R-416GM radio relay station	WH-AT	INA	INA	INA	INA	72
R-439 Legenda-2	WH-AT	132.5 kg	INA	15 min	INA	73
Redut-2US	WH-AT	INA	INA	INA	INA	74
Repellent-1	WH-AT	23,650 kg	1.0 m	15 min	INA	53
S-300V4	TR	50 t	1.2 m	5 min	5 min	36
S-350 Vityaz	WH-AT	22 t	1.2 m	5 min	5 min	36
S-400 Triumf	WH-AT	INA	INA	5-10 min	INA	35
Smoke generator vehicles (TDA-2K, TDA-3, TDM-2K)	WH-AT	15.75 t	1.75 m	5.5 min	INA	63
SNAR-10 / SNAR-10M1 Leopard	TR	12.7 t	INA	5 min	INA	44
Strela-10 and Osa short-range SAM	TR	12.1 t	1.2 m	3 to 5 min	3 to 5 min	40
Strelets	MP+AD	2.4 kg	INA	INA	INA	74
Supercam S350	AIR-FW	15.5 kg	INA	15 min	INA	79
T-54 / T-55	TR	41,000 kg	1.4 m unprepared; 5.5 m with snorkel	INA	INA	10
T-62	TR	40,000 kg	1.40 m	INA	INA	9
T-64A/B/BV	TR	39.5 t	1.8 m unprepared; 5.0 m with snorkel	INA	INA	9
T-80B/BV/BVM, T-80U	TR	43.7 t	1.8 m unprepared; 5.0 m with snorkel	N/A	INA	8

SYSTEM	CODE	WEIGHT	FORDING	INTO ACTION	OUT OF ACTION	PAGE
T-90	TR	46 t	1.2 m	INA	INA	7
Takhion	AIR-FW	25 kg	INA	INA	INA	81
Tigr and Vodnik armoured cars	WH-AT	6 t	1.2 m	INA	INA	22
TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge	WH-AT	19,500 kg	1.0 m	72 min	8 min	55
Tor	TR	32 t	1.0 m	less than 3 min	less than 5 min	38
Tornado-G	WH-AT	14 t	1.75 m	2 min	3 min	26
TOS-1A Soltsepek	TR	44.3 t	1.2 m	90 s from halt to fire	INA	29
Tropospheric scatter radio stations (R-410-7.5, R-412A/AM, R-417)	WH-AT	INA	INA	INA	INA	72
Tunguska	TR	34 t	1.0 m	not more than 5 min	not more than 5 min	40
Typhoon family (Typhoon-K, Typhoon-U, Typhoon-VDV, Linza)	WH-AT	13,700kg	1.5 m	INA	INA	23
UMZ	WH-AT	10,000 kg	1.4 m	INA	INA	61
UR-77 Meteorit	TR-A	15.5 t	5 km/h	INA	30 to 40 min	59
Uragan-1M	WH-AT	40 t	1.4 m	3 min	3 min	27
Ural-6370 / 63706 / 532301 trucks (newer Ural family)	WH-AT	33,000 kg	1.8 m	INA	INA	76
Uran-6	TR	6 t	INA	INA	INA	60
VTS Ladoga	TR	42 t	1.2 m	INA	INA	66
ZALA 421-16E	AIR-FW	8-10.5 kg	INA	INA	INA	77
Zastava	MP	5.5 kg	INA	20 min	10 min	81
Zoopark-1 / Zoopark-1M	TR	38.1 t	INA	5 min	5 min	42

Table 5. Source register

Codes are printed at the foot of each entry. Web sources are coded by publisher and numbered: APA Air Power Australia, AR Army Recognition, ARMADA Armada International, BCAT Bellingcat, BMPD bmpd blog, CSIS CSIS Missile Threat, FAS Federation of American Scientists, GS GlobalSecurity, MIL Militaryny, MISS missilery.info, ORYX Oryx, RDP Russian Defense Policy, RUSI RUSI web, TASS TASS, TOPWAR Topwar, TWZ The War Zone, VPK VPK, VRF Vooruzhenie Rossii, WEG WEG online (ODIN), WMC Wikimedia Commons, WP Wikipedia (language after the hyphen), WSPOT WarSpotting. Other codes name a single document.

3G Russian Electronic Warfare Systems, Analytic Insight Report, pp. 42–44, 3GIMBALS, 2023–06–07.
https://sprtyv7.com.ua/wp-content/uploads/2023/11/COGINT_Analytic_Insight_Report_Russian_EW_Systems_231119_114942.pdf

ACONF R-145BM [armedconflicts.com]. Armed Conflicts, 2005.
<https://www.armedconflicts.com/topic/view/28534>

ACONF2 Russian Command Staff Vehicle R-149MA1. armedconflicts.com.
<https://www.armedconflicts.com/Russian-Command-Staff-Vehicle-R-149MA1-t252192>

ADU Aleut snow and swamp-going vehicle will be demonstrated for the first time at the Army 2021 forum in Ulan-Ude. Aviation & Defence Universe, 2021–08–19.
<https://aviation-defence-universe.com/aleut-snow-and-swamp-going-vehicle-will-be-demonstrated-for-the-first-time-at-the-army-2021-forum-in-ulan-ude>

AEX17 V podrazdeleniya REB postupili novye komplekсы Krasukha-20 i Moskva-1. AEX.RU [relaying Russian MoD], 2017–08–31.
<https://www.aex.ru/news/2017/8/31/17451/>

AG1 1B12, Командная машина. Army Guide.
<https://www.army-guide.com/rus/product/4755.html>

AG2 2C43 Мальва, Самоходная гаубица. Army Guide.
<https://www.army-guide.com/rus/product.php?prodID=6099&printmode=1>

AG3 2S4 Tyulpan [Samokhodnyy minomët]. Army Guide.
<http://www.army-guide.com/rus/product31805.html>

AG4 BRDM-2 [Reconnaissance Vehicle]. army-guide.com.
<http://www.army-guide.com/eng/product3952.html>

AG5 Kapustnik-B, Command vehicle. Army Guide.
<https://www.army-guide.com/eng/product.php?prodID=2383&printmode=1>

AIBUS Russia deploys mine-clearing robots in eastern Ukraine. AI Business, 2022–05.
<https://aibusiness.com/verticals/russia-deploys-mine-clearing-robots-in-eastern-ukraine>

AIF Что за новейшую машину разминирования БМР-3М «Вепрь» применили в Мариуполе? Argumenty i Fakty, 2022–09–07.
https://aif.ru/politics/world/chto_za_noveshuyu_mashinu_razminirovaniya_bmr-3m_vepr_primenili_v_mariupole

AIF2 «Смерть на поводке». Жуткое оружие устроило кровавое побоище под Курском. Argumenty i Fakty, 2025.
<https://aif.ru/politics/russia/smer-t-na-povodka-zhutkoe-oruzhie-ustroilo-krovavoie-poboishche-pod-kurskom>

AIF3 Что за новье nizkovysotnye RLS Podlet-K1 postupili v VKS RF? Argumenty i Fakty, 2022–07–06.
https://aif.ru/society/army/chto_za_novye_nizkovysotnye_rls_podlet-k1_postupili_v_vks_rf

AIF4 Что за «Аустёнок» ведёт контрбатареиную борьбу на Украине? Argumenty i Fakty, 2022.
https://aif.ru/society/army/chto_za_aistyonok_vedyot_kontrbatareynuyu_borbu_na_ukraine

AIF5 Вода — не предрада. Бойцы «Центра» отработали наведение pontonnoy переправы. Аргументы и факты, 2025–02–26.
<https://aif.ru/politics/russia/voda-ne-pregrada-boycy-centra-otrabotali-navedenie-pontonnoy-pereravry>

AIF6 Что izvestno o nepronitsaemom kupole zashchity ot dronov Pole-21? Argumenty i Fakty, 2021–04–17.
https://aif.ru/society/army/chto_izvestno_o_nepronitsaemom_kupole_zashchity_ot_dronov_pole_21

AIF7 Что представляет собой зенитный ракетный комплекс С-350 «Вульф»? Argumenty i Fakty.
https://aif.ru/dontkows/files/chto_predstavlyaet_soboy_zenitnyy_raketnyy_kompleks_s-350_vityaz

AIF8 В Тверскую область поступили семь новых РСЗО «Смерч». Aif Tver, 2017–04–18.
https://tver.aif.ru/society/army/v_tverskuyu_oblast_postupili_sem_novyh_rszo_smerch

AIF9 Rabota bespilotnikov Superkam na yuzhnodonetskom napravlenii. Argumenty i Fakty, 2024–01–24.
https://aif.ru/politics/gallery/rabota_bespilotnikov_superkam_na_yuzhnodonetskom_napravlenii

AIF10 Что за роботы-саперы «Уран-6» проводят разминирование на Украине? Argumenty i Fakty, 2022–05.
https://aif.ru/society/army/chto_za_roboty-sapery_uran-6_provodyat_razminirovaniye_na_ukraine

AIF22 Чмо за комплекс РЗБ «Палантин» применяют на Украине? Argumenty i Fakty, 2022–06–07.
https://aif.ru/society/army/chto_za_kompleks_reb_palantin_primeniyayut_na_ukraine

AINF «Борисоглебск-2»: про ворожий комплекс радиоэлектронной борьбы. ArmyInform (Ukrainian MoD), 2022–04–26.
<https://armyinform.com.ua/2022/04/26/borysoglebsk-2-pro-vorozhij-kompleks-radioelektronnoyi-borotby/>

AINF2 R-60 missile on a Geran: DIU unveils a new modification of a Russian strike UAV. ArmyInform (Ukrainian MoD), 2025–12–15.
<https://armyinform.com.ua/en/2025/12/15/r-60-missile-on-a-geran-diu-unveils-a-new-modification-of-a-russian-strike-uav>

AINF3 ЗПК С-350 «Вульф»: що про нього відома. ArmyInform (Ukrainian MoD), 2022–08–17.
<https://armyinform.com.ua/2022/08/17/zrk-s-350-vytyaz-shho-pro-nogo-vidomo/>

AINF22 ZSU destroyed Aleut in artillery strike (Ukr.). ArmyInform (Ukrainian MoD), 2022–04–01.
<https://armyinform.com.ua/2022/04/01/zsu-pid-chas-tochnogo-artudaru-po-technicz-okupantiv-znyshchyl-eksklyuzyv-bojoviy-arktychnyj-vsyudyhid/>

AINF23 Another Russian tracked novelty burned in Zaporizhzhia (Ukr.). ArmyInform (Ukrainian MoD), 2023–11–23.
<https://armyinform.com.ua/2023/11/23/shhe-odnu-rojskiju-gusenychnu-novynku-spalyl-na-zaporizhzh/>

AIRBASE Sredstva REB i RTR podrazdelenii VDV [forum thread, MoD press quotes, 2012–14]. forums.airbase.ru, 2014–05–23.
<https://forums.airbase.ru/2014/05/t89725--sredstva-reb-i-rtr-podrazdelenii-vdv.html>

AMAL 26-й полк ПХБЗ: участие В спецоперации, структура и вооружение. amalantra.ru, 2025.
<https://amalantra.ru/26-polk-rhbz/>

AMR Russia unveils its Orion-E MALE drone in attack configuration. Asian Military Review, 2021–03.
<https://www.asianmilitaryreview.com/2021/03/russia-unveils-its-orion-e-male-drone-in-attack-configuration/>

ANNA RLS Nebo-M razvernuli pod gorodom Naryan-Mar [citing Izvestia, 17 Nov 2020]. ANNA News, 2020–11–17.
<https://anna-news.info/rls-nebo-m-razvernuli-pod-gorodom-naryan-mar/>

AP1 Almaz-Antey 40R6 / S-400 Triumf Self Propelled Air Defence System / SA-21 (Kopp). Air Power Australia, 2009–05, updated 2012–04.
<https://www.ausairpower.net/APA-S-400-Triumf.html>

AP2 Almaz S-300P/PT/PS/PMU/PMU2 Self Propelled Air Defence System / SA-10/20 (Kopp). Air Power Australia, 2012–04.
<https://www.ausairpower.net/APA-Grumble-Gargoyle.html>

APA3 Soviet/Russian Acquisition and GCI Radars (Kopp). Air Power Australia.
<https://www.ausairpower.net/APA-Acquisition-GCI.html>

APA4 Warsaw Pact / Russian Air Defence Command Posts. Air Power Australia.
<https://www.ausairpower.net/APA-Rus-ADCP-CP.html>

APA5 Russian EW systems: R-330B/T, R-934B/BM. Air Power Australia, 2014.
<https://www.ausairpower.net/APA-REB-Systems.html>

APA6 NNIRIT Nebo SVU / RLM-M Nebo M annex (Kopp). Air Power Australia, 2012–10–03.
<https://www.ausairpower.net/APA-Nebo-M-Annex.html>

AP7 NNIRIT 1119 Nebo SVU / RLM-M Nebo M, APA-TR-2008-0402 (Kopp). Air Power Australia, 2014.
<https://ausairpower.net/APA-Nebo-SVU-Analysis.html>

APA8 KBP 2K22 Tunguska / 96K6 Pantsir S1 SPAAAGM, APA-TR-2009-0703 (Kopp). Air Power Australia, 2012 [page updated 2014–01–27].
<https://www.ausairpower.net/APA-96K6-Pantsir-2K22-Tunguska.html>

APA9 S-300V: 202nd Air Defence Brigade, Naro-Fominsk, AP-PE-2011-0501. Air Power Australia, 2011.
<https://www.ausairpower.net/APA-202nd-AD-Brigade.html>

APATO S-300V 9K81/9K81-1/9K81M Self Propelled Air Defence System / SA-12/SA-23 Giant/Gladator (Kopp), APA-TR-2006-1202. Air Power Australia, 2012–04 [page updated 2014–01–01].
<https://www.ausairpower.net/APA-Giant-Gladator.html>

- APA11** KBTochmas 9K35 Strela 10 Self Propelled Air Defence System / SA-13 Gopher, APA-TR-2009-0801. Air Power Australia (Kopp), 2012.
<https://www.airsuppower.net/APA-9K35-Strela-10.html>
- APA12** NIEMI 9K33 Osa/Romb Self Propelled Air Defence System / SA-8 Gecko, APA-TR-2009-0704. Air Power Australia (Kopp), 2012.
<https://www.airsuppower.net/APA-9K33-Osa.html>
- AR1** 2S35 Koalitsiya-SV. Army Recognition.
- AR2** Russian All-Terrain Tracked Vehicle GAZ-3344-20 at BIDEC 2017. Army Recognition, 2017.
<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2017/russian-all-terrain-tracked-vehicle-gaz-3344-20-at-bidec-2017>
- AR3** Russian tank unit in Buryatia receive Aleut articulated amphibious vehicles. Army Recognition, 2021.
<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2021/russian-tank-unit-in-buryatia-receive-aleut-articulated-amphibious-vehicles>
- AR4** Eastern district tests GAZ-3344-20 Aleut in low temperatures. Army Recognition, 2021.
<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2021/russian-eastern-district-tests-gaz-3344-20-aleut-snow-and-swamp-going-vehicles-in-low-temperatures>
- AR5** Ukraine forces capture Russian 1V119 Reostat artillery fire direction vehicle. Army Recognition, 2022.
<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/ukraine-forces-capture-russian-1v119-reostat-artillery-fire-direction-vehicle>
- AR6** Ukrainian army captured Russian 1V1003 artillery command and observation vehicle. Army Recognition, 2022-09.
<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/ukrainian-army-captured-russian-1v1003-artillery-command-and-observation-vehicle>
- AR7** The Soviet-era BAT-2 Engineering Bulldozer still defends Ukraine's frontline. Army Recognition (From Armyinform), 2023-08.
<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/the-soviet-era-bat-2-engineering-bulldozer-still-defends-ukraine-s-frontline>
- AR8** Russian paratroopers receive refurbished BMD-2 and BTR-RD armored vehicles. Army Recognition, 2023-10.
<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/russian-paratroopers-receive-refurbished-bmd-2-and-btr-rd-armored-vehicles>
- AR9** BMD-4M airborne tracked armored infantry fighting vehicle. Army Recognition.
<https://armyrecognition.com/military-products/army/infantry-fighting-vehicles/airborne-vehicles/bmd-4m-russia-uk>
- AR10** Russian Bakhcha-U combat turret from KBP will increase firepower of IFV (TASS text). Army Recognition / TASS, 2017.
<https://armyrecognition.com/focus-analysis-conflicts/army/defence-security-industry-technology/russian-bakhcha-u-combat-turret-from-kbp-will-increase-firepower-of-ifv-infantry-fighting-vehicle-tass-10404171>
- AR11** Russian army in Siberia received first BMP-2M IFV armed with Berezhok combat module. Army Recognition, 2018.
<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2018/russian-army-in-siberia-received-first-bmp-2m-ifv-armed-with-berezhok-combat-module>
- AR12** Deployment of new Russian BMR-3MA Vepr armored mine-clearing vehicles in Ukraine. Army Recognition, 2024-04.
<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2024/deployment-of-new-russian-bmr-3ma-vepr-armored-mine-clearing-vehicles-in-ukraine>
- AR13** Borisoglebsk-2 EW system delivered to a Russian military base in Tajikistan. Army Recognition, 2018.
<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2018/borisoglebsk-2-ew-system-delivered-to-a-russian-military-base-in-tajikistan>
- AR14** Ukrainian army recuperates abandoned Russian MTLB-mounted Borisoglebsk-2 jamming system. Army Recognition, 2022-09-16.
<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/ukrainian-army-recuperates-abandoned-russian-mtlb-mounted-borisoglebsk-2-jamming-system>
- AR15** Russia receives new batch of BREM-1M armored recovery vehicles. Army Recognition, 2023-09.
<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2023/russia-receives-new-batch-of-brem-1m-armored-recovery-vehicles>
- AR16** BRM-1K Model 2021 BRM-1KM reconnaissance tracked armored vehicle. Army Recognition.
<https://www.armyrecognition.com/military-products/army/armored-vehicles/reconnaissance-vehicles/brm-1k-model-2021-brm-1km-reconnaissance-tracked-armored-vehicle-data>
- AR17** Russia transfers experimental BTR-90 'Rostok' IFV from museum to Ukraine. Army Recognition, 2023-10-16.
<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/russia-transfer-experimental-btr-90-rostok-ifv-from-museum-to-ukraine>
- AR18** Russia Delivers New Batch of BTR-MDM Armored Personnel Carriers. Army Recognition, 2024-10.
<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/ukraine-russia-conflict/russia-delivers-new-batch-of-btr-mdm-armored-personnel-carriers-to-its-armed-forces-in-ukraine>
- AR19** BTR-82A / BTR-82AM APC. Army Recognition.
<https://www.armyrecognition.com/military-products/army/armoured-personnel-carriers/wheeled-vehicles/btr-82a-russia-uk>
- AR20** Army-2019: New RKhM-8 wheeled CBRN reconnaissance armored vehicle. Army Recognition, 2019.
<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2019/army-2019-new-rkhm-8-wheeled-cbrn-reconnaissance-armored-vehicle>
- AR21** Ukrainian Forces Captures first Latest-Generation Russian Eleron T-16 UAV. Army Recognition, 2023-08.
<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/ukrainian-forces-captures-first-latest-generation-russian-eleron-t-16-uav>
- AR22** Western Military District of Russian army to receive Granat and Eleron-3 UAVs. Army Recognition (TASS text), 2016.
<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2016/western-military-district-of-russian-army-to-receive-granat-and-eleron-3-unmanned-aerial-vehicles-11112184>
- AR23** Russia: R-149MA1 command and staff vehicle adopted for service in Southern Military District. Army Recognition (SMD press service), 2018.
<https://www.armyrecognition.com/archives/archives-land-defense/2018/russia-r-149ma1-command-and-staff-vehicle-adopted-for-service-in-southern-military-district>
- AR24** Russia intensifies use of armed FPV drones to strike Ukrainian positions. Army Recognition, 2024-03-18.
<https://armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/russia-intensifies-use-of-armed-fpv-first-person-view-drones-to-strike-ukrainian-positions>
- AR25** Russian Company Develops AI-Powered Combat Drone Capable of Autonomous Target Engagement. Army Recognition, 2024-04.
<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2024/russian-company-develops-ai-powered-combat-drone-capable-of-autonomous-target-engagement>
- AR26** Infauza KISH1 UNSH-12. Army Recognition.
<https://armyrecognition.com/military-products/army/electronic-warfare/infauza-russia-uk>
- AR27** Iskander-K 9K728. Army Recognition.
<https://www.armyrecognition.com/military-products/army/missiles/ballistic-missiles/iskander-k-mobile-short-range-cruise-missile-system-russia-data>
- AR28** Russian company Aeroscan unveils Product 54 Italmas long-range loitering munition. Army Recognition, 2023-09.
<https://armyrecognition.com/archives/archives-land-defense/land-defense-2023/russian-company-aeroscan-unveils-product-54-italmas-long-range-loitering-munition>
- AR29** Eleron-3 and upgraded Korsar UAVs for Russian army. Army Recognition, 2018.
<https://armyrecognition.com/news/army-news/2018/eleron-3-and-upgraded-korsar-uavs-for-russian-army>
- AR30** 1RL257 or Krasukha-4 Jamming Station. Army Recognition, 2026-01-02.
<https://www.armyrecognition.com/military-products/army/electronic-warfare/1rl257-or-krasukha-4-jamming-station>
- AR31** Russia confirms deployment in Ukraine of new KUB loitering munition able to reach target at 50 km. Army Recognition, 2024-04.
<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/russia-confirms-deployment-in-ukraine-of-new-kub-loitering-munition-able-to-reach-target-at-50-km>
- AR32** Russian army uses a very rare Ladoga autonomous armored vehicle for combat operations in Ukraine. Army Recognition, 2024-03.
<https://armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/russian-army-uses-a-very-rare-ladoga-autonomous-armored-vehicle-for-combat-operations-in-ukraine>

AR33 Ukraine destroys Russian Leer-2 electronic warfare system. Army Recognition, 2023-11.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2023/ukraine-destroy-a-russian-leer-2-electronic-warfare-system>

AR34 Ukrainian Forces Destroy Two Chechen Z-STs Akhmat Armored Vehicles in Kursk region. Army Recognition, 2024-10.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/ukrainian-forces-destroy-two-chechen-z-sts-akhmat-armored-vehicles-in-kursk-region>

AR35 Buran 4x4 armoured vehicle Russia Russian defense industry MAKs 2017. Army Recognition, 2017-07.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/defence-security-industry-technology/buran-4x4-armoured-vehicle-russia-russian-defense-industry-maks-2017-12007172>

AR36 First armed Iranian-made Mohajer-6 UAV used by Russia crashes near Kursk. Army Recognition, 2024-05.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/first-armed-iranian-made-mohajer-6-uav-used-by-russia-in-ukraine-crashes-near-kursk>

AR37 Russian Central Military District take delivery of new Communications System. Army Recognition [reproducing district press release], 2015.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2015/russian-central-military-district-take-delivery-of-new-communications-system>

AR38 Russian Army deploys first Pantsir SM SV air defense system. Army Recognition, 2025-03.

<https://www.armyrecognition.com/news/army-news/2025/russian-army-deploys-first-pantsir-sm-sv-air-defense-system-to-improve-coverage-in-off-road-conditions>

AR39 Russia's use of Belarusian-made Pantsir-S1 on MZKT chassis in Ukraine confirmed. Army Recognition, 2026-04.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/russias-use-of-belarusian-made-pantsir-s1-on-mzkt-chassis-in-ukraine-confirmed-after-new-analysis>

AR40 Russian artillery learn to use Aistyonok radar [TASS text]. Army Recognition, reprinting TASS, 2017.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2017/russian-artillery-learn-to-use-aistyonok-radar-81708173>

AR41 PRP-4A Argus artillery fire control posts for Russian army. Army Recognition, 2018-06.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2018/prp-4a-argus-artillery-fire-control-posts-for-russian-army>

AR42 Russian Defense Ministry plans to buy new amphibious tracked armoured vehicle PTS-4 in 2014. Army Recognition, 2013-07-24.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2013/russia-24-july-2013-news-uk>

AR43 Russia has deployed PMM-2M amphibious bridging ferry in Syria. Army Recognition, 2017-09.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2017/russia-has-deployed-pmm-2m-amphibious-bridging-ferry-in-syria>

AR44 Russian Central Military District signal units receive advanced hardware. Army Recognition, 2019-12.

<https://www.armyrecognition.com/news/army-news/army-news-2019/russian-central-military-district-signal-units-receive-advanced-hardware>

AR45 Russian communications units to receive R-149MA1 unified command and staff vehicle. Army Recognition [TASS text], 2017.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2017/russia-communication-units-receive-r-149ma1-vehicle-80806171>

AR46 Russian Army to get new radio stations. Army Recognition [Zvezda weekly text], 2019.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2019/russian-army-to-get-new-radio-stations>

AR47 6,000 new military communication equipment for Western Military District of Russian army. Army Recognition, 2015.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/defence-security-industry-technology/6-000-new-military-communication-equipment-for-western-military-district-of-russian-army>

AR48 S-300V 9K81 Antey-300 SA-12 Gladiator. Army Recognition.

<https://www.armyrecognition.com/military-products/army/air-defense-systems/air-defense-vehicles/s-300v-russia-uk>

AR49 Russian Southern district to receive S-350 Vityaz air defense system in 2021. Army Recognition [TASS text], 2020.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2020/russian-southern-district-to-receive-s-350-vityaz-air-defense-system-in-2021>

AR50 Russian army CBRN unit receives TDA-3 mobile smokescreen generators (quoting TASS). Army Recognition, 2017.

<https://armyrecognition.com/news/army-news/2017/russian-army-southern-military-district-cbrn-unit-receives-tda-3-mobile-smokescreen-generators-tass-82103173>

AR51 SNAR-10 M1 self-propelled antitank radar enters service in Russian Central Military District. Army Recognition, 2019-01.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2019/snar-10-m1-self-propelled-antitank-radar-enters-service-in-russian-central-military-district>

AR52 Russia upgrades Strelets reconnaissance system - Part 1. Army Recognition (from TASS / Nezavisimoye Voennoye Obozreniye), 2021.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2021/russia-upgrades-strelets-reconnaissance-system-part-1>

AR53 T-90M Model 2017 Proryv-3 MBT technical data sheet. Army Recognition, 2026-01.

<https://www.armyrecognition.com/military-products/army/main-battle-tanks/main-battle-tanks/t-90m-model-2017-mbt-main-battle-tank-technical-data-sheet>

AR54 Russian army integrates Orlan-10 and Takhion UAVs with Msta-B howitzers. Army Recognition (from TASS/Mod), 2021.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2021/russian-army-integrates-orlan-10-and-takhion-uavs-with-msta-b-howitzers>

AR55 Ukrainian soldiers seize extremely rare Russian GAZ-3937 Vodnik armored vehicle. Army Recognition, 2023-03.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/ukrainian-soldiers-seize-extremely-rare-russian-gaz-3937-vodnik-armored-vehicle>

AR56 Russia up-armored 'Mad Max' Tigr-M SpN spotted in Ukraine. Army Recognition, 2023-10.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/russia-up-armored-mad-max-tigr-m-spn-spotted-in-ukraine>

AR57 Analysis: Russian airborne troops armored vehicles used in combat in Ukraine. Army Recognition, 2022.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/analysis-russian-airborne-troops-armored-vehicles-used-in-combat-in-ukraine>

AR58 Ukrainian soldiers destroy Russian TOR-M2DT new air defense missile system. Army Recognition, 2023-02.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/ukrainian-soldiers-destroy-russian-tor-m2dt-new-air-defense-missile-system>

AR59 More heavy flamethrower units to appear in Russian armies. Army Recognition (citing Izvestia, 27 Feb 2018), 2018-02-27.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2018/more-heavy-flamethrower-units-to-appear-in-russian-armies>

AR60 Each Russian military district to have TOS-1A flamethrower battalion. Army Recognition (citing Izvestia), 2019-10-22.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/analysis-defense-and-security-industry/each-russian-military-district-to-have-tos-1a-flamethrower-battalion>

AR61 Russia unveils new UMZ-K multipurpose mine-laying vehicle at Army-2015. Army Recognition, 2015.

<https://armyrecognition.com/focus-analysis-conflicts/army/defence-security-industry-technology/russia-defence-industry-unveils-new-umz-k-multipurpose-mine-laying-vehicle-at-army-2015-12706155>

AR62 Russian engineering units receive robotized UMZ-K and UMZ-T minelayers. Army Recognition, 2019.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2019/russian-engineering-units-receive-robotized-umz-k-and-umz-t-minelayers>

AR63 Is Russia secretly deploying Uragan-1M bi-caliber MLRS in Ukraine? Army Recognition, 2024-01.

<https://armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/is-russia-secretly-deploying-uragan-1m-bi-caliber-mlrs-in-ukraine>

AR64 Development of Russian Ural Tornado-U truck is at final stage. Army Recognition [TASS text], 2017.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/defence-security-industry-technology/development-of-russian-ural-tornado-u-truck-is-at-final-stage>

AR65 Russian Uran-6 robotic demining UGV undergoing modernization following Ukraine war experience. Army Recognition, 2023-06.

<https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2023/russian-uran-6-robotic-demining-ugv-undergoing-modernization-following-ukraine-war-experience>

AR66 Ukrainian sniper downs Russian ZALA 421-16Ev2 surveillance drone. Army Recognition, 2024-03.

<https://www.armyrecognition.com/focus-analysis-conflicts/army/conflicts-in-the-world/russia-ukraine-war-2022/ukrainian-sniper-downs-russian-zala-421-16ev2-surveillance-drone>

AR67 R-330ZH Zhitel. Army Recognition, 2025–11–23.

<https://www.armyrecognition.com/military-products/army/electronic-warfare/r-330zh-jamming-station-russia-uk>

AR68 Zoopark-1M 1L260 artillery counter-battery radar system data. Army Recognition.

<https://www.armyrecognition.com/military-products/army/radars/counter-battery-radars/zoopark-1m-1l260-artillery-counter-battery-radar-system-data>

AR69 Zoopark-1L219 counter-battery radar system data. Army Recognition.

<https://www.armyrecognition.com/military-products/army/radars/counter-battery-radars/zoopark-1-l-219-counter-battery-radar-system-russia-data>

AR70 Russian Central Military District induces Zoopark-1M counterbattery radars. Army Recognition (TASS text), 2018–03–01. <https://www.armyrecognition.com/archives/archives-land-defense/land-defense-2018/russian-central-military-district-induces-zoopark-1m-counterbattery-radars>

ARCH36 BMD-4m Tanker, 5th Air Assault Company, 2nd Air Assault Battalion, 31st Air Assault Brigade. The 36th Division Archive, 2023.

<https://www.36thdivisionarchive.com/bmd4m-tanker/pa-ru-bmd4m-tanker>

ARINF Ukrainian defenders captured rare command-staff Tigr. Armyinform (Ukrainian MoD), 2022–09–30.

<https://armyinform.com.ua/2022/09/30/ukrayinski-zahysnyky-zahopyly-ridkisyj-komandno-shtabnyj-tygr/>

ARMADA1 Russian IADS Redux Part-7: The Effectors (Withington). Armada International, 2023–09–07.

<https://www.armadainternational.com/2023/09/09/russian-air-force-sam-systems/>

ARMADA2 Russia's Rapid UAV Expansion. Armada International, 2019–03.

<https://www.armadainternational.com/2019/03/russias-rapid-uav-expansion/>

ARMADA3 All Yours. Krasukha, Krasukha, Krasukha, Ja, Ja! Armada International (Thomas Withington), 2022–03–24.

<https://www.armadainternational.com/2022/03/ukraine-forces-capture-russian-electronic-warfare-system/>

ARMADA4 Russian Radiation (Withington). Armada International, 2022–01.

<https://www.armadainternational.com/2022/01/ukraine-russia-electronic-warfare/>

ARMADA5 Russian IADS Redux Part-8: Shocking Behaviour (Withington). Armada International, 2023–09.

<https://www.armadainternational.com/2023/09/russian-air-defence-electronic-warfare/>

ARMADA6 Murmurs of Murmansk (Withington). Armada International, 2024–09.

<https://www.armadainternational.com/2024/09/murmansk-electronic-warfare-system-iran/>

ARMADA7 Russian IADS Redux Part 3: Strategic Skywatchers. Armada International, 2023–05.

<https://www.armadainternational.com/2023/05/russian-air-defence-analysis-part-3/>

ARMADA8 To Kill a Pantsir (Withington). Armada International, 2024–07.

<https://www.armadainternational.com/2024/07/to-kill-a-pantsir-electronic-warfare/>

ARMADA9 Exaggerated Russian electronic warfare capabilities. Armada International, 2022–05.

<https://www.armadainternational.com/2022/05/exaggerated-russian-electronic-warfare-capabilities/>

ARMADA10 The Art of Countering Artillery Fires in Ukraine (Withington). Armada International, 2022–12–30.

<https://www.armadainternational.com/2022/12/the-art-of-countering-artillery-fires-in-ukraine/>

ARMF PU-12 (9S482) mobile control post. armforc.su.

https://armforc.su/srk_pu12.htm

ARMF13 Машина старшего офицера батареи 1813; Машина командира батареи 1814; Машина начальника штаба дивизиона 1816. armforc.su.

https://armforc.su/rva_1v3.htm; https://armforc.su/rva_1v4.htm; https://armforc.su/rva_1v16.htm

ARMF14 Машина командира батареи 1814. armforc.su.

https://armforc.su/rva_1v14.htm

ARMF15 Машина командира дивизиона 1815. armforc.su.

https://armforc.su/rva_1v15.htm

ARMF16 Машина начальника штаба дивизиона 1816. armforc.su.

https://armforc.su/rva_1v16.htm

ARMFORC Inzhenernaya razvedyvatelnaya mashina IMR Zhuk. armforc.su.

https://armforc.su/ism_ivm.htm

ARMINF Сім мільйонів доларів у морі: що відома про збитий російською БПЛА «Форпост-Р». Armyinform (Ukrainian MoD news agency), 2024–04–05.

<https://armyinform.com.ua/2024/04/05/sim-milijoniv-dolariv-u-more-shho-vidomo-pro-zbytij-rosijskij-bpla-forpost-r/>

ARMINF2 ZSU vidzhaly v rashystiv minny zahorodzhuvach GMZ-3. Armyinform (Ukrainian MoD), 2022–10–10.

<https://armyinform.com.ua/2022/10/10/zsu-vidzhaly-v-rashystiv-minny-zagorodzhuvach-gmz-3/>

ARMSNET Deimos-NSA Russian 6x6 Command and Control Vehicle, R-142 NSA. armsnet (mirror of US Army ODIN/WEG data format).

<https://armsnet.duckdns.org/equipment/fc51a371-0c5e-4687-ee78-08de6e3c1543>

ARMY22 Ukrainian troops hunted down a Russian Merlin (Armyinform, 2022–06–17). Armyinform (Ukrainian MoD), 2022–06–17.

<https://armyinform.com.ua/2022/06/17/ukrayinski-vijskovi-vpolyuvaly-rosijskogo-merlyna/>

ARMYIN Бойова машина вознеметників БМО-Т: що відома. Armyinform (Ukrainian MoD), 2022–04–17.

<https://armyinform.com.ua/2022/04/17/bojova-mashyna-vognemetnykiv-bmo-t-shho-vidomo-pro-ridkisyj-ekzemplar-vazhkoyi-tehniky-agresora/>

ARMYINF Ови ворожа: що відома про безлічотнику комплексы «Ексерн-3СВ». Armyinform (Ukrainian MoD), 2022–04–21.

<https://armyinform.com.ua/2022/04/21/ochi-voroga-shho-vidomo-pro-bezplyotnyky-kompleksu-elon-3sv/>

ARMYINF2 Destroyed a Russian Gerbera on wheels: struck the car carrying it. Armyinform (Ukrainian MoD news agency), 2025–12–25.

<https://armyinform.com.ua/2025/12/25/znishchyly-z-kolesamy-rosijsku-gerberu-vrazlyly-z-avto-yake-yiji-perevezlyo/>

ARMYINF3 Komandno-shtabna mashyna R-149MA1 vdruge . Armyinform (Ukrainian MoD news agency), 2022–09–12.

<https://armyinform.com.ua/2022/09/12/komandno-shtabna-mashyna-r-149ma1/>

ARMYINF4 Rare catch for the Defence Forces: station to replace Starlink hit. Armyinform (Ukrainian MoD), 2026–03–26.

<https://armyinform.com.ua/2026/03/26/vkray-ridkisyj-dlya-syl-oborony-ulov-unikalnu-stanciyu-dlya-zaminy-starlink-urazlyly-novitrim-midstarkom/>

ARMYINF5 Repellent-1: shcho vidomo pro znyshchenyi rosiyskiy kompleks REB. Armyinform (Ukrainian MoD), 2022–06–08.

<https://armyinform.com.ua/2022/06/08/repellent-1-shho-vidomo-pro-znyshchenyi-rosijskij-kompleks-reb/>

ARMYINF6 Українські війни захопили станцію радіолокаційної розвідки російської армії CHAP-10M1. Armyinform, 2022–04–05.

<https://armyinform.com.ua/2022/04/05/ukrayinski-voiny-zahopyly-stanciyu-radiolokaczij-noyi-rozvidky-rosijskoyi-armiyi-snar-10m1/>

ARMYINF7 Російською БПЛА «Такіон»: що відома про чергову збитий українськими бойові. Armyinform (Ukrainian MoD news agency), 2022–03–03.

<https://armyinform.com.ua/2022/03/03/rosijskij-bpla-takion-shho-vidomo-pro-chergovozdobych-ukrayinskyh-voiniv/>

ARMYINF8 Wanted to build a crossing, but MTU-72 hit earlier (Army Inform). Army Inform (Ukrainian MoD), 2024–10–31.

<https://armyinform.com.ua/2024/10/31/hotily-zvesty-perepravu-ale-rosijskij-mostokladach-nashi-urazlyly-ranish/>

ARMYRAD Zil Radio Truck (R161-A2M article by Joe Bell, translated Russian manual). Army Radio Sales Co. / VMARS, 2003–01–03.

https://armyradio.com/Zil_Radio_Truck.html

ARMYT MSTA-S 2S19 152mm Self-Propelled Howitzer. Army Technology, 2000–10–03.

<https://www.army-technology.com/projects/mstas/>

ARTECH URAN 6: El robot para desminado de Rusia [copy of Army Technology page]. FIE UNDEF.

<https://www.fie.undef.edu.ar/ceptm/?p=1370>

ATO S-350 out in the open at MAKs. ATO.ru (Russia Aviation Insider), 2013–08.

<https://www.ato.ru/content/s-350-out-open-maks>

ATP ATP 7-100.1 Russian Tactics, table E-2 and para A-6. US Army, 2024–02.

<https://archive.org/details/atp-7-100-1-russian-tactics-2024>

AV21 БПЛА Гранат-4, как новый стандарт тактической эффективности. aviation21.ru, 2025.

<https://aviation21.ru/bpla-granat-4-kak-noviy-standart-takticheskoy-effektivnosti/>

AVIA Застава. Технические характеристики. Фото. avia.pro.

<https://avia.pro/blog/zastava-tehnicheskie-harakteristiki-foto>

AVIAROS Теплового машина TMC-65. aviaros.narod.ru.

<https://aviaros.narod.ru/tms-65.htm>

BABEL CIT: Russia has started de-canning old Soviet T-55 tanks. Babel, 2023-03.

<https://babel.ua/en/news/91852-cit-russia-has-started-de-canning-old-soviet-t-55-tanks-and-sending-them-to-ukraine>

BART15 Russian Arctic brigades put off to 2015. Barents Observer, updated [about 2011-2012].

<https://www.thebarentsobserver.com/security/russian-arctic-brigades-put-off-to-2015/>

BARENTS Arctic brigade severely weakened. The Barents Observer, 2022-12.

<https://thebarentsobserver.com/en/security/2022/12/arctic-brigade-severely-weakened>

BCAT1 Exclusive Access to the Russian Forpost Drone Shot Down in Ukraine. Bellingcat, 2015-06-13.

<https://www.bellingcat.com/news/uk-and-europe/2015/06/13/the-russian-forpost-drone-shot-down-in-ukraine>

BCAT2 Novye rossiyskie sistemy radioelektronnoy borby na vostoке Ukrainy. Bellingcat (Russian edition of DFLab), 2018-09-10.

<https://ru.bellingcat.com/novosti/russia/2018/09/10/new-russian-ew-donbas/>

BG69 И полярники, и военные, и спецслужбы (ENICS Eleron). Business Online (business-gazeta.ru), c. 2011.

<https://www.business-gazeta.ru/article/69375>

BG137 «...и попасть в него очень трудно» [Eleron-3SV downed in Syria]. Business Online (business-gazeta.ru), 2015.

<https://m.business-gazeta.ru/article/137067/>

BI Ukrainian soldiers say Russian strike drones are flying smaller drones into battle like an 'aircraft carrier'. Business Insider (via AOL), 2015-11-28.

<https://www.aol.com/articles/ukrainian-soldiers-russian-strike-drones-122011876.html>

BMPD1 Вооруженные Силы России получили первые серийные самоходные гаубицы 2С35 «Коалиция-СВ». bmpd (CAST blog), 2024-01-05.

<https://bmpd.livejournal.com/4792565.html>

BMPD2 Самоходное оружие 2С44 «Гуашин-К». bmpd (CAST blog), 2025-02.

<https://bmpd.livejournal.com/4895454.html>

BMPD3 53-я зенитная ракетная бригада в Курске перевооружается на ЗРК «Бук-М3». bmpd blog (CAST), 2017-12-04.

<https://bmpd.livejournal.com/2985151.html>

BMPD4 61-я зенитная ракетная бригада в Центральном военном округе получает ЗРК «Бук-М3». bmpd blog (CAST), 2019-11-22.

<https://bmpd.livejournal.com/3847762.html>

BMPD5 Комплекс звукотепловой разведки 1Б75 «Пенициллин» на форуме «Армия-2021». bmpd (CAST blog), quoting Rossiyskaya Gazeta, 2021-08.

<https://bmpd.livejournal.com/4383315.html>

BMPD6 РЛС «Аистенок» в Донецке. bmpd (CAST blog), 2015-01.

<https://bmpd.livejournal.com/1136931.html>

BMPD7 Zhitnaya raketnaya sistema S-300V4 v Krymu. bmpd (CAST blog), 2016-11.

<https://bmpd.livejournal.com/2296725.html>

BMPD8 Деятельность Концерна ПВО «Алмаз-Антей» в 2013 году: НПО «Стрела». bmpd (CAST blog), 2014.

<https://bmpd.livejournal.com/905167.html>

BMPD9 Зенитный ракетный полк 150-й мотострелковой дивизии вооружен зенитными ракетными системами Тор-М2. bmpd (CAST blog), 2017-11-10.

<https://bmpd.livejournal.com/2946915.html>

BMPD10 Защищенный автомобиль К-53949 «Таифун-К» в бригаде спецназа ЮЮВ. bmpd (LiveJournal), CAST, 2018.

<https://bmpd.livejournal.com/3207846.html>

BMPD11 Радиолокационный комплекс «Зоопарк-1М» под Пальмирой. bmpd blog (CAST), 2016-04-04.

<https://bmpd.livejournal.com/1827333.html>

BO New Barnaul-T air defense system comes to motorized rifle brigade in Pechenga. The Barents Observer, 2021-12-17.

<https://thebarentsobserver.com/en/security/2021/12/new-barnaul-t-air-defense-system-comes-motorized-rifle-brigade-pechenga>

BO2 Russia exercises long-range radio-electronic warfare on Kola. The Barents Observer, undated [after spring 2019].

<https://www.thebarentsobserver.com/security/russia-exercises-longrange-radioelectronic-warfare-on-kola/>

<https://www.thebarentsobserver.com/security/russia-exercises-longrange-radioelectronic-warfare-on-kola/>

BORG ЗСУ знищили комунікаційний вузол окупантів полкового рівня (Bigeo). Borg Expert (citing InformNapalm).

<https://borgexpert.com/news/zsu-znyshchyly-komunikatsijny-vuzol-okupantiv-polkovoho-rivnia-video>

BTVT Боевая машина огнеметчиков-тяжелая [БМО-Т]. btvt.narod.ru. <http://btvt.narod.ru/4/bmot/bmot.htm>

CALIB Forpost-R enters service with Russian navy. Calibre Defence, 2025-06-09.

<https://www.calibredefence.co.uk/forpost-r-enters-service-with-russian-navy/>

CAPSS Low-Cost Gerbera Drones [CAPSS InFocus]. Centre for Air Power Studies, India, 2025-09-16.

https://capsindia.org/wp-content/uploads/2025/09/CAPSS_InFocus_DPK_16_9_25.pdf

CH97 VSU unichtozhili REB Moskva-1 stoimostiu 57 millionov. Charter97 (citing Censor.net), 2022-09-09.

<https://charter97.org/ru/news/2022/9/9/514866/>

CHARTER Russian Drone Granat-4 Was Shot Down In Ukraine. Charter97, 2025-11-27.

<https://charter97.org/en/news/2025/11/27/664600/>

CNA The Russian Army: Organization and Modernization, p. 13. CNA, 2019-10.

<https://www.cna.org/reports/2019/10/10/P-2019-U-021801-Final.pdf>

CNA The Russian Army: Organization and Modernization (Ramm), p. 45. CNA, 2019-10.

<https://www.cna.org/>

CNA21 Russian Forces in the Western Military District, p. 31. CNA, 2021-06.

<https://www.cna.org/reports/2021/06/Russian-Forces-in-the-Western-Military-District.pdf>

CNA23 Russia's Use of Uncrewed Systems in Ukraine [DRM-2022-U-034223]. CNA, 2023-03.

<https://www.cna.org/reports/2023/03/Russian-Uncrewed-Systems-Ukraine.pdf>

CRFS Hertz and Minds (Withington), p. 6 and note 12. CRFS, 2024.

https://www.crfs.com/hubs/whitepapers/Hertz_and_Minds_Report.pdf

CRYPTO R-142 Mobile HF/VHF radio station. Crypto Museum, 2024-01-04.

<https://cryptomuseum.com/radio/ussr/r142>

CSIS1 Shahed-131 and 136. CSIS Missile Defense Project, 2026.

<https://missilethreat.csis.org/missile/shahed-131-and-136/>

CSIS2 Shahed-238. CSIS Missile Defense Project, 2026.

<https://missilethreat.csis.org/missile/shahed-238/>

CSIS3 9K720 Iskander (SS-26). CSIS Missile Threat, 2024-04-23.

<https://missilethreat.csis.org/missile/ss-26-2/>

CSIS4 9M729 [SSC-8]. CSIS Missile Threat, 2024-04-23.

<https://missilethreat.csis.org/missile/ssc-8-novator-9m729/>

CSIS5 OTR-21 Tochka (SS-21). CSIS Missile Threat, 2024-04-23.

<https://missilethreat.csis.org/missile/ss-21/>

CSIS6 Pantsir S-1. CSIS Missile Threat, undated [after 2021].

<https://missilethreat.csis.org/defsys/pantsir-s-1/>

CSIS7 S-300 (Missile Defense Project). CSIS.

<https://missilethreat.csis.org/defsys/s-300/>

CSIS8 S-400 Triumph. CSIS Missile Threat, undated [c. 2017-2022].

<https://missilethreat.csis.org/defsys/s-400-triumf/>

CSIS9 Tor (SA-15 Gauntlet). CSIS Missile Defense Project, undated (cites to 2020).

<https://missilethreat.csis.org/defsys/tor/>

CULT Samokhodnyy minomiyot 2S4 Tyulpan. ar.culture.ru (Russian Ministry of Culture portal).

<https://ar.culture.ru/ru/subject/samokhodnyy-minomiyot-2s4-tyulpan>

CYBER Предназначение, организация, вооружение отдельного инженерно-саперного батальона. cyberpediasu.com (copied course text), 2017-06-19.

<https://cyberpediasu.com/5x393c.html>

CYCLO APC-14KM. Циклонегия, 2024.

<https://cyclowiki.org/wiki/%D0%90%D0%A0%D0%A1-14%D0%9A%D0%9C>

CYCLO2 BT-40. Cyclopedia (cyclowiki.org), 2026.

<https://cyclowiki.org/wiki/BT-40>

D24 Rosjanie dopanczerzyli dopanczerzy BMP-2. Defenc24, 2023-05-05.

<https://defenc24.pl/wojna-na-ukrairie-raport-specjalny-defenc24/rosjanie-dopanczerzyli-dopanczerzy-bmp-2>

D24A Rosjanie prezentują nową wersję BRDM-2. Defenc24, undated, after Sep 2022.

<https://defenc24.pl/przemysl/rosjanie-prezentuja-nowa-wersje-brdm-2>

D24A2 Rosja: Rusza seryjna produkcja nastepcy wyrzutni Grad, Uragan i Smiercz. Defenc24, 2021-08.

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

<https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz>

[https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-sm](https://defenc24.pl/rosja-rusza-seryjna-produkcja-nastepcy-wyrzutni-grad-uragan-i-smiercz)

D24B Rosjanie wysyłają zmodernizowane BRDM-2 na front. Defence24, 2023 (embeds post of 27 Mar 2023).
<https://defence24.pl/wojna-na-ukraينه-raport-specjalny-defence24/rosjanie-wysylaja-zmodernizowane-brdm-2-na-front>

D24B2 Rosjanie wysłali na Ukrainę ulepszone artylerie rakietowa. Defence24, 2024.
<https://defence24.pl/wojna-na-ukraينه-raport-specjalny-defence24/rosjanie-wyslali-na-ukraينه-ulepszone-artylerie-rakietowa>

D242 Rosyjski predator w cieniu Bayraktara. Bezzałogowiec Orion uderza w Ukraińców. Defence24, 2026.
<https://defence24.pl/wojna-na-ukraينه-raport-specjalny-defence24/rosyjski-predator-w-cieniu-bayraktara-bezzałogowiec-orion-uderza-w-ukraينcow>

D243 Rosjanie wprowadzają opancerzone Mustangi. Defence24.
<https://defence24.pl/sily-zbrojne/rosjanie-wprowadzaja-opancerzone-mustangi>

D244 Wyrzutnie Tornado-G dostarczane rosyjskiej armii. Defence24, 2019.
<https://defence24.pl/przemysl/wyrzutnie-tornado-g-dostarczane-rosyjskiej-armii>

D245 Rosyjski ulepszony Uragan trafił do WDW. Defence24, 2025-03-16.
<https://defence24.pl/sily-zbrojne/rosyjski-ulepszony-uragan-trafil-do-wdw>

DB Russian Airborne Troops received modern Barnaul-T air defence control systems. Defence Blog, 2018-06.
<https://defence-blog.com/russian-airborne-troops-received-modern-barnaul-t-air-defence-control-systems/>

DB17 Russia modernises batch of 2s4 Tyulpan self-propelled nuclear mortars. Defence Blog, 2017-11-13.
<https://defence-blog.com/russia-modernises-batch-of-2s4-tyulpan-self-propelled-nuclear-mortars/>

DBLOG Ukraine claims it shot down Russia's newest drone. Defence Blog, 2023-08.
<https://defence-blog.com/ukraine-claims-it-shot-down-russias-newest-drone/>

DBLOG2 A Syrian Journalist Has Posted Photos of Russian Radio station R-166-05. Defence Blog, 2015-09.
<https://defence-blog.com/a-syrian-journalist-has-posted-photos-of-russian-radio-station-r-166-05/>

DBSAR Russian new armored vehicle spotted on Ukraine battlefield. Defence Blog, 2024-10-21.
<https://defence-blog.com/russian-new-armored-vehicle-spotted-on-ukraine-battlefield/>

DEF24 Bóg wojny waży ciężkie - 2s7 w działaniach na Ukrainie. Defence24, 2022-03.
<https://defence24.pl/polityka-obrona/bog-wojny-wazy-ciezkie-2s7-w-dzialaniach-na-ukraينه>

DEFBLOG Russia's rare BTR-90 troop carrier spotted in Ukraine. Defence Blog, 2023-10.
<https://defence-blog.com/russias-rare-btr-90-troop-carrier-spotted-in-ukraine/>

DEFBLOG2 Belarus receives first Groza R-934UM2 jamming stations. Defence Blog, 2017-12-18.
<https://defence-blog.com/belarus-receives-first-groza-r-934um2-jamming-stations/>

DEFBLOG3 Russian Army's new heavy trucks fell into the hands of Ukrainian Soldiers. Defence Blog, 2022.
<https://defence-blog.com/russian-armys-new-heavy-trucks-fell-into-the-hands-of-ukrainian-soldiers/>

DEFEX The russians could lose the extremely rare Penicillin system in Ukraine. Defence Express, 2023-05.
https://en.defence.ua.com/weapon_and_tech/the_russians_could_lose_the_extremely_rare_the_penicillin_artillery_reconnaissance_system_in_ukraine-6952.html

DEFNDR Sting interceptor by Wild Hornets downs costly Russian Granat-4. Defender Media, 2026-03.
<https://thedefender.media/en/2026/03/sting-intercepted-rare-target/>

DEFOST Russian Electronic Warfare Drone Shot Down in Ukraine: Report. The Defense Post, 2022-06-21.
<https://www.thedefensepost.com/2022/06/21/russia-electronic-warfare-drone-ukraine>

DEFUA Ukraine's Military Destroyed Another Z-STS Akhmat. Defence-UA, 2022-11.
https://en.defence.ua.com/weapon_and_tech/ukraines_military_destroyed_another_z_sts_akhmat_what_manufactured_in_25_days_for_kadyrov_mercenaries-4843.html

DEFUA1 Ukrainian Military Destroy a Rare russian BTR-90 Rostok APC Near Advivka. Defence Express, 2023-12.
https://en.defence.ua.com/weapon_and_tech/ukrainian_military_destroy_a_rare_russian_btr_90_rostok_apc_near_advivka-8838.html

DEFUA2 For New Assaults in the East, the russians are Even Using Rare BTR-90. Defence Express, 2023-10.
https://en.defence.ua.com/news/for_new_assaults_in_the_east_the_russians_are_even_using_rare_btr_90-8242.html

DEFUA3 Russia's tank stocks are running out [Defence Express]. Defence Express, 2024-03.
https://en.defence.ua.com/analysis/russias_tank_stocks_are_running_out_more_and_more_outdated_equipment_is_being_used_by_invaders_against_ukraine-9619.html

DEFUA4 Ukrainian landmine finds a rare Russian vehicle (UMZ on BTR-80). Defence Express, 2023.
https://en.defence.ua.com/news/ukrainian_landmine_finds_a_rare_russian_vehicle_that_hindered_counteroffensive_video-7330.html

DFR Electronic Warfare by Drone and SMS. DFRLab (Atlantic Council), 2017-05-18.
<https://dfriab.org/2017/05/18/electronic-warfare-by-drone-and-sms/>

DFRLab #MinskMonitor: New Russian Electronic Warfare in Horlivka? DFRLab, 2018-04-27.
<https://dfriab.org/2018/04/27/minskmonitor-new-russian-electronic-warfare-in-horlivka/>

DIND BTR-82A Armoured Personnel Carrier (Military Industrial Company LLC product page). defence-industries.com.
<https://www.defence-industries.com/products/military-industrial-company-llc/btr-82a-armoured-personnel-carrier>

DLG22 Novelishii rossiiskii kompleks REB Moskva-1 otpravilsia vsled za otdnoimennym kreiserom. Dialog.ua [citing Censor.net], 2022-09-09.
https://www.dialog.ua/war/258422_1662714543

DOGS РАС контрбатареиной борьбы 1А260 «Зoopарк-1М». dogswar.ru, 2022-09-22.
<http://www.dogswar.ru/specsredstva/58-specsredstva/8955-rs-kontrbareinoi-1m.html>

DPOST Russia Offers Orion-E Combat Drone for Export. The Defense Post, 2021-06-29.
<https://www.thedefensepost.com/2021/06/29/russia-orion-e-drone-export/>

DPSU Border guards in Kharkiv region destroyed the latest Russian EW system. State Border Guard Service of Ukraine, 2024-04-12.
<https://dpsu.gov.ua/en/news/%20video%20border-guards-in-kharkiv-region-destroyed-the-latest-russian-electronic-warfare-system>

DSA Russia's Supercam S350 UAV Showcased First Time in Caspian Region During ADEX-2024. Defence Security Asia, 2024-09.
<https://defencesecurityasia.com/en/russias-supercam-s350-uav-showcased-in-caspian-region-during-adex-2024/>

DUA BMPT Terminator Reappears on Ukrainian Frontline for Midnight Missions. Defence Express, 2023-07.
https://en.defence.ua.com/weapon_and_tech/bmpt_terminator_reappears_on_ukrainian_frontline_for_midnight_missions-7262.html

DUA2 Photo of destroyed russian modern satellite communication vehicle appeared. Defence Express [en.defence.ua.com], 2022.
https://en.defence.ua.com/weapon_and_tech/photo_of_destroyed_russian_modern_satellite_communication_vehicle_appeared-3792.html

DUDAK Ustroistvo i ekspluatatsiya sredstv svyazi (Dudak, Fedorenko, Gorovenko), ch. 5 KSHM R-142N, pp. 34-37, 64-84. Belarusian State University of Informatics and Radioelectronics, Military Faculty, 2020.
https://libelcod.bsuir.by/bitstream/123456789/40729/1/Dudak_2020.pdf

EAN Екатеринбургский авиазавод представил новый ударный БПЛА. EAN (Yekaterinburg), 2025-05-27.
<https://eaneews.ru/yekaterinburg/20250527150057/yekaterinburgskiy-aviazavod-predstavil-novyy-udarnyy-bpla>

EDR Russia exhibits the latest version of its Orion-E MALE UAV. EDR On-Line, 2021-11.
<https://www.edrmagazine.eu/russia-exhibits-the-latest-version-of-its-orion-e-male-uav>

EDR2 UMEX 2026 - Zala unveils new KPU 51E disposable launcher (Novichkov). EDR Magazine, 2026-01-20.
<https://www.edrmagazine.eu/umex-2026-zala-unveils-new-kpu-51e-disposable-launcher>

EDR3 IDEX 2019: Tornado-G MRL system makes international debut. European Defence Review, 2019-02.
<https://www.edrmagazine.eu/idx-2019-tornado-g-mrl-system-makes-international-debut>

EDR23 Russia receives improved KUB-BLA loitering munition. EDR Magazine, 2023-12.
<https://www.edrmagazine.eu/russia-receives-improved-kub-bla-loitering-munition>

EDR24 ADEX 2024 - Rosoboronexport showcases the Orlan-30 UAV for the first time in the Caspian region. EDR Magazine, 2024-09.
<https://www.edrmagazine.eu/rosoboronexport-showcases-the-orlan-30-uav-for-the-first-time-in-the-caspian-region>

EDR25 IDEX 2025: Kalashnikov presents its latest attack UAVs. EDR Magazine, 2025-02.
<https://www.edrmagazine.eu/?p=42207>

EDRL Army-2024: Lancet loitering munitions offered for export. EDR Magazine, 2024-08.
<https://www.edrmagazine.eu/?p=38441>

ELSIG Unified command and staff vehicle R-149MA1 (catalogue sheet, pp. 20–21). Elektrosignal (Sovzvezdie group), 2018.

<http://elektrosignal.ru/wp-content/uploads/2018/06/R-149MA1.pdf>

EMP Russia's thermobaric launchers used to just drive up and fire (new TOS-1As). Euromaidan Press, 2026–08–23.

<https://euromaidanpress.com/2026/08/23/new-tos-1as/>

ESD New Russian ISKANDER Missile Brigade. European Security & Defence, 2022–02–08.

<https://euro-sd.com/2022/02/articles/exclusive/25236/iskander-missile/>

ESP Fiber-optic drones: can they change Russian-Ukrainian war? *Espresso Global*, 2025–01–02.

<https://global.espresso.tv/military-news-fiber-optic-drones-can-they-change-russian-ukrainian-war>

ESP2 What is Podlet radar. *Espresso TV*, 2024–12.

<https://global.espresso.tv/russia-ukraine-war-what-is-podlet-radar-latest-russian-radar-ukrainians-keep-destroying-but-cant-capture>

ESPR Ukrainian Armed Forces destroy Russian Meteorit demining system in Kharkiv region. *Espresso*, 2024.

<https://global.espresso.tv/ukrainian-armed-forces-destroy-russian-meteorit-demining-system-in-kharkiv-region-the-10th-since-war-started>

ESPRESSO T-62MV with mineplow, drone protection: media show latest version of Russian tank. *Espresso Global*, citing Forbes and analyst Naalsio26, 2024–02.

<https://global.espresso.tv/russia-ukraine-war-t-62mv-with-mineplow-and-drone-protection-media-show-video-of-latest-version-of-russian-tank>

EXK UMG 30B-3523: технические характеристики. *exkavator.ru*.

https://exkavator.ru/exkapedia/technika/eksmash_rov-3523

F45 Russia attacked U.S. supplied howitzers in Ukraine with kamikaze drones and rockets. *19FortyFive*, 2022–05–19.

<https://www.19fortyfive.com/2022/05/russia-attacked-u-s-supplied-howitzers-in-ukraine-with-kamikaze-drones-and-rockets/>

FAKTY Gerbery z fanery: GUR on what Russians make Shahed imitations from. *Fakty (fakty.com.ua)*, 2024–11–18.

<https://fakty.com.ua/uk/ukraine/20241118-gerbery-z-fanery-u-gur-rozpoznavly-z-chogo-rosiyani-vygotovlyayut-imitaciyu-shahedivamp/>

FA51 MT-LBu (1V12 series) ACRV M1974 Artillery Command and Reconnaissance Vehicle. Federation of American Scientists, 1999–08–21.

<https://man.fas.org/dod-101/sys/land/row/mt-lbu.htm>

FA52 2S5 Giatsint-S 152-mm Self-Propelled Gun. Federation of American Scientists, 1999–06–19.

<https://man.fas.org/dod-101/sys/land/row/2s5.htm>

FA53 2S7 203-mm Self-Propelled Gun. Federation of American Scientists, 1999–10–16.

<https://man.fas.org/dod-101/sys/land/row/2s7.htm>

FA54 BRDM-2 Armored Reconnaissance Vehicle. Federation of American Scientists (Military Analysis Network), c. 1999.

<https://man.fas.org/dod-101/sys/land/row/brdm-2.htm>

FA55 BTR-D BMD M1979. Federation of American Scientists, 1999–06–19.

<https://man.fas.org/dod-101/sys/land/row/btr-d.htm>

FA56 MT-LB (DoD 101 Systems). Federation of American Scientists, about 2000.

<https://man.fas.org/dod-101/sys/land/row/mt-lb.htm>

FA57 AT-12 SWINGER (DoD 101 Systems). Federation of American Scientists, 1999–06–19.

<https://man.fas.org/dod-101/sys/land/row/at-12.htm>

FA58 T-80 Main Battle Tank (FAS Military Analysis Network). Federation of American Scientists, 2000–01–16.

<https://man.fas.org/dod-101/sys/land/row/t80tank.htm>

FA59 220-mm Multiple Rocket Launcher (9P140). Federation of American Scientists, 1999–06–19.

<https://man.fas.org/dod-101/sys/land/row/9p140.htm>

FERG Russian base in Tajikistan to be fitted with Pole-21 electronic countermeasures system. *Fergana News*, 2019–11–26.

<https://en.fergana.agency/news/112784/>

FERG2 Russian Electronic Warfare System Is Capable Of Neutralizing GPS, Galileo and BeiDou Systems (Sputnik, 1 Feb 2021, reposted). Resilient Navigation and Timing Foundation (rntfnd.org), 2021–02–01.

<https://rntfnd.org/?p=35483>

FISH 1A260 3oonapk-1M. *fisshi.net* (repost of militaryrussia.ru material).

<https://fisshi.net/anti/2340561-1i260-3oonapk-1m.html>

FM3 FM 100–2–3 The Soviet Army: Troops, Organization and Equipment

(pp. 4–120, 4–135, 5–69). US Army, 1991.

https://archive.org/details/FM_100_2_3_T_S_A_T_O_F_1991

FM23 FM 100–2–3 The Soviet Army: Troops, Organization, and Equipment, pp. 4–11 (regimental HQ table, PDF p. 43), 5–34 to 5–35 (BMP variants, PDF p. 223), battalion HQ table (PDF p. 58). US Army, 1991.

[https://www.bits.de/NRANEU/ethers/amd-us-archive/fm100-2-3\[91\].pdf](https://www.bits.de/NRANEU/ethers/amd-us-archive/fm100-2-3[91].pdf)

FM32 FM 100–2–3 The Soviet Army: Troops, Organization and Equipment. US Army, 1991.

https://archive.org/download/FM_100_2_3_T_S_A_T_O_F_1991/FM%20100-2-3%20The%20Soviet%20Army-Troops%2C%20Organization%20and%20Equipment%20%281991%29.djvu.txt

FN Molniya: The wooden FPV drone that's made in Russia and copied in Ukraine. *Forces News* (BFBS), 2024–10–29.

<https://www.forcesnews.com/ukraine/molniya-wooden-fpv-drone-thats-made-russia-and-copied-ukraine>

FOCUS 10 машин на всю Украину: В ВСУ заметили пегкий инженерный БТМ–3. *Focus.ua* (relaying Forbes), 2023–08.

<https://focus.ua/voennoye-novosti/585223-10-mashin-na-vssyu-ukrainu-v-vsu-zametil-redkiy-inzhenernyy-btm-3-chto-izvestno>

FOCUS2 Zashchita ot yadernogo dozhdyha NATO: VСУ podorvali eshche odnu mashinu RF (Forbes). *Focus.ua*, 2024–07–01.

<https://focus.ua/voennoye-novosti/655273-zashchita-ot-yadernogo-dozhdyha-nato-vsu-podorvali-esheche-odnu-mashinu-rf-dlya-yadernoy-vovny-forbes>

FOCUS3 VS RF ustanavliwayut moduli REB Pole-21 na domakh i vyshkakh svyazi. *Focus.ua*, 2023–03–28.

<https://focus.ua/digital/557429-vs-rf-ustanavliwayut-moduli-reb-pole-21-na-domakh-i-vyshkah-svyazi-zachem-eto-im-nuzhno>

FOCUS4 Forbes via Focus: Ukraine destroyed R-416GM. *Focus.ua* (citing Forbes), 2024–06.

<https://focus.ua/voennoye-novosti/652373-v-rf-ogranichen-dostup-k-sputnikam-vsu-unichtozhili-r-416gm-otslyavshuyu-sigaly-v-troposferu-forbes>

FOCUS5 VS RF primeniyayut v boyakh dymovyye mashiny TDM-2K. *Focus.ua*, 2024–05.

<https://focus.ua/voennoye-novosti/644074-vs-rf-primeniyayut-v-boyah-dymovyye-mashiny-tdm-2k-kak-s-nimi-borotsya-analiz-ekspertov>

FOCUS6 ВСУ уничтожили станцию разведки ЧАП-10 "Леопард", которая работает чеоловека за 15 км. *Focus.ua*, 2023–08.

<https://focus.ua/voennoye-novosti/583806-vsu-unichtozhili-stanciyu-razvedki-snar-10-leopard-kotoryaya-razlachaet-cheloveka-za-15-km-foto-video>

FOCUS7 В Украине затрофеили российский комплекс Стрелец. *Focus.ua*, 2024 (undated on page).

<https://focus.ua/digital/674075-v-ukraine-zatrofeili-rossiyskiy-kompleks-strelec-zachem-on-nuzhen-razvedchikom-vs-rf-foto>

FOCUS8 3СУ захопили пікірський російський дрон «Тяхіон» під Харковом. *Focus.ua*, 2022.

<https://focus.ua/uk/digital/532547-vsu-zahvatili-redkiy-rossiyskiy-dron-tahion-pod-harkovom-chem-on-opasen-foto>

FOI Russian Electronic Warfare (Kjellen), FOI–R–4625--SE, pp. 36, 43–45. FOI, 2018–09.

<https://www.foi.se/rest-api/report/FOI-R--4625--SE>

GAGA The Ukrainian armed forces seized a Russian command and control vehicle R-149AKSh-1. *gagadget.com*, 2022–09–09.

<https://gagadget.com/en/165185-the-ukrainian-armed-forces-seized-a-russian-command-and-control-vehicle-r-149aksh-1-to-ensure-communication-via-closed-in/>

GAGADG The Ukrainian Defense Forces destroyed the Russian Leer-2 ... for the first time. *Gagadget*, 2022–08–01.

<https://gagadget.com/en/152749-the-ukrainian-defense-forces-destroyed-the-russian-leer-2-electronic-warfare-station-based-on-the-tiger-m-armed-vehicle/>

GS1 2A36 Giatsint-B (M1976) 152-mm Towed Gun. *GlobalSecurity.org*.

<https://www.globalsecurity.org/military/world/russia/2a36-specs.htm>

GS2 2A65 (M1987) 152-mm Towed Gun-Howitzer. *GlobalSecurity.org*.

<https://www.globalsecurity.org/military/world/russia/2a65-specs.htm>

GS3 2S4 M-1975 240-mm Tulpan self-propelled mortar.

GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/2s4.htm>

GS4 2S5 Giatsint-S 152-mm Self-Propelled Gun. *GlobalSecurity.org*.

<https://www.globalsecurity.org/military/world/russia/2s5.htm>

GS5 96L6 / 96L6E – CHEESE BOARD Radar. *GlobalSecurity.org*.

<https://www.globalsecurity.org/military/world/russia/96l6.htm>

GS6 11L22 Garmon / Harmonium. *GlobalSecurity.org*.

<https://www.globalsecurity.org/military/world/russia/garmon.htm>

GS7 Airborne Assault Troops (VDV) – Organization. *GlobalSecurity.org*, undated (Soviet-era text, later additions).

<https://www.globalsecurity.org/military/world/russia/vdv.org.htm>

GS8 Russian Defense Ministry to receive 250 BMD–4M, Rakushka vehicles (TASS, reproduced by GlobalSecurity). TASS, 2015–05–13.

<https://www.globalsecurity.org/wmd/library/news/russia/2015/russia-150513-tass04.htm>

GS9 Modernized BMD-4M (Boyevaya Mashina Desanta). GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/bmd-4m.htm>

GS10 BMO-T (Object 564) Firebug. GlobalSecurity.org, undated [c. 2012]. <https://www.globalsecurity.org/military/world/russia/bmo-t.htm>

GS11 BMP-1 Fighting Vehicle - Variants. GlobalSecurity.org. <https://www.globalsecurity.org/military/world/russia/bmp-1-var.htm>

GS12 Airborne Assault Troops [VDV] - Order of Battle. GlobalSecurity, 2020.

<https://www.globalsecurity.org/military/world/russia/vdv-orbat.htm>

GS13 BTR-MD Rakushka (Ob'yekt 955). GlobalSecurity, 2015. <https://www.globalsecurity.org/military/world/russia/btr-md.htm>

GS14 BTR-82 specs. GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/btr-82-specs.htm>

GS15 BTR-82A 8x8 infantry fighting vehicle - Program. GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/btr-82-program.htm>

GS16 BTR-70 Armored Personnel Carrier. GlobalSecurity.org. <https://www.globalsecurity.org/military/world/russia/btr-70.htm>

GS17 Russian Airborne Gets New ECM System (RIA Novosti, 6 Jul 2012). GlobalSecurity.org, 2012-07-06.

<https://www.globalsecurity.org/wmd/library/news/russia/2012/russia-120706-rianovosti01.htm>

GS18 Lancet specs. GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/lancet-specs.htm>

GS19 Lancet. GlobalSecurity.org, undated [quotes TASS of 2023-09-22]. <https://www.globalsecurity.org/military/world/russia/lancet.htm>

GS20 2S19 MST-A 152-mm Self-Propelled Howitzer. GlobalSecurity.org. <https://www.globalsecurity.org/military/world/russia/2s19.htm>

GS21 55ZH6M Nebo-M TALL RACK Mobile multi-band radar complex. GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/nebo-m.htm>

GS22 Orion UAV / Inokhodets [Pacer]. GlobalSecurity. <https://www.globalsecurity.org/military/world/russia/orion.htm>

GS23 Orlan-10. GlobalSecurity.org, undated, after 2023-08. <https://www.globalsecurity.org/military/world/russia/orlan-10.htm>

GS24 1B75 Penicillin Automated Sound-Thermal Artillery Reconnaissance Complex. GlobalSecurity.

<https://www.globalsecurity.org/military/world/russia/penicillin.htm>

GS25 Engineers - River Crossings. GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/engineer-river.htm>

GS26 S-350E Vityaz / SA-28 - Specifications. GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/s-350-specs.htm>

GS27 S-350 Vityaz [Knight / Hero] (S0R6) / SA-28. GlobalSecurity.org. <https://www.globalsecurity.org/military/world/russia/s-350.htm>

GS28 S-350E Vityaz / SA-28 - Design. GlobalSecurity.org. <https://www.globalsecurity.org/military/world/russia/s-350-design.htm>

GS29 Ten T-80 Tanks Deployed to Russian Pacific Fleet to Bolster Marines (Sputnik, reposted by GlobalSecurity). Sputnik / GlobalSecurity.org, 2019-09-10.

<https://www.globalsecurity.org/wmd/library/news/russia/2019/russia-190910-sputnik01.htm>

GS30 Contract on Delivering T-80BMV Tanks to Russian Army Fulfilled - Rostec (Sputnik, reposted by GlobalSecurity). Sputnik / GlobalSecurity.org, 2019-12-12.

<https://www.globalsecurity.org/wmd/library/news/russia/2019/russia-191212-sputnik04.htm>

GS31 T-90 specifications. GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/t-90-specs.htm>

GS32 Tachyon, GlobalSecurity. GlobalSecurity, 2022.

<https://www.globalsecurity.org/military/world/russia/tachyon.htm>

GS33 Russia's Western Military District gets newest Tornado-G multiple rocket launchers. TASS [reproduced by GlobalSecurity.org], 2015-07-22. <https://www.globalsecurity.org/wmd/library/news/russia/2015/russia-150722-tass02.htm>

GS34 TOS-1A Soltisepek (Sun) 220mm Multiple Rocket Launcher. GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/tos-1a.htm>

GS35 UR-77 Meteorit. GlobalSecurity.

<https://www.globalsecurity.org/military/world/russia/ur-77.htm>

GS36 9P140 Uragan (Hurricane) 220-mm Multiple Rocket Launcher. GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/9p140.htm>

GS37 ZALA421-16E5. GlobalSecurity.org.

<https://www.globalsecurity.org/military/world/russia/zala421-16e5.htm>

GWARA Russia equips Molniya drones with Starlink terminals, ISW says. Gwara Media, 2026-01-14.

<https://gwaramedia.com/en/russia-equips-molniya-drones-with-starlink-terminals-isw-says/>

GWARA2 Kharkiv Oblast in ruins: identification of Russian servicemen, part 2. Gwara Media, 2022-12.

<https://gwaramedia.com/en/kharkiv-oblast-in-ruins-identification-of-russian-servicemen-part-2/>

HOTNEWS FOTO Ruşii au aruncat in luptă o dronă extrem de rădă. HotNews.ro, 2022-11.

<https://hotnews.ro/?p=98013>

HUR Ukraine's Military Intelligence Releases Detailed Information on Russian Loitering Munitions Lancet and Scalpel. ArmyInform (Ukrainian MoD), 2026-03-23.

<https://armyinform.com.ua/en/2026/03/23/ukraines-military-intelligence-releases-detailed-information-on-russian-loitering-munitions-lancet-and-scalpel/>

ICDS Russia's Electronic Warfare Capabilities to 2025 (McDermott), pp. 15, 17, 23, 24, A-2. ICDS, 2017-09.

https://icds.ee/wp-content/uploads/2018/ICDS_Report_Russias_Electronic_Warfare_to_2025.pdf

IEEE22 The Fall and Rise of Russian Electronic Warfare (Clark). IEEE Spectrum, 2022.

<https://spectrum.ieee.org/the-fall-and-rise-of-russian-electronic-warfare>

IFAX Serial delivery of Forpost-R reconnaissance drone to begin in 2020. Interfax, 2019-08-24.

<https://interfax.com/newsroom/top-stories/19316/>

IISS25 The Military Balance 2025, chapter 4, extract pp. 31, 36, 38 (printed pp. 182, 187, 189). IISS, 2025-02.

<https://www.iiiss.org/publications/the-military-balance/>

INT Беспилотный самолёт ZALA 421-16E: технологическая новинка Российской армии. Integral-Russia, 2017-01-16.

<https://integral-russia.ru/2017/01/16/bespilotnyj-samolyot-zala-421-16e-tehnologicheskay-a-novinka-rossijskoj-armii/>

INTFX В российской армии появилось первое подразделение боевых машин "Терминатор". Interfax, 2021-12-01.

<https://www.interfax.ru/russia/806008>

IRINTL Iran Smuggles Drones To Russia By Boat, Airliners: Guardian. Iran International, 2023-02-13.

<https://iranintl.com/en/202302131788>

ISIS Alabuga's Shahed 136 (Gen 2) Warheads: A Dangerous Escalation. Institute for Science and International Security, 2024-05-09.

<https://isis-online.org/isis-reports/alabugas-shahed-136-geran-2-warheads-a-dangerous-escalation>

ISIS2 Russian Decoy Drones that Depend on Western Parts Pose a Great Challenge to Ukrainian Defenses (Anokhin, Faragasso). Institute for Science and International Security, 2024-12-18.

<https://isis-online.org/isis-reports/russian-decoy-drones-that-depend-on-western-parts-pose-a-great-challenge/>

ISW1 Russian Offensive Campaign Assessment, December 26, 2025. Institute for the Study of War, 2025-12-26.

<https://www.understandingwar.org/backgrounder/russian-offensive-campaign-assessment-december-26-2025>

ISW2 Russian Offensive Campaign Assessment, January 13, 2026. Institute for the Study of War, 2026-01-13.

<https://understandingwar.org/research/russia-ukraine/russian-offensive-campaign-assessment-january-13-2026/>

IZ Расчеты FPV-дронов уничтожили гаубицы «Богдана-Б» и Д-30 ВСУ в зоне СВО. Izvestia [citing Russian MoD], 2026-05-19.

<https://iz.ru/2099591/2026-05-19/raschety-fpv-dronov-unichtozhili-gaubitsy-bogdana-b-i-d-30-vs-u-v-zone-svo>

I22 В НПЦ «Ушкунік» назвали «Князя Вандал» результативным массовым дроном на СВО. Izvestia [citing TASS], 2026-08-30.

<https://iz.ru/2157676/v-nptc-ushkunik-nazvali-kniaz-vandal-rezultativnym-massovym-dronom-na-svo-iz>

I23 Glushitelnii uspekhi: kompleks REB nakroet Evropu iz-pod Kaliningrada. Izvestia, 2019-04-26.

<https://iz.ru/871450/aleksei-ramm-aleksei-kozachenko-bogdan-stepovoi/glushitelnii-uspekhi-kompleks-reb-nakroet-evropu-iz-pod-kaliningrada>

I24 Удар РСЗО «Смерч» уничтожил артракетом ВСУ на границе с Белгородской областью. Izvestia, 2024-03-21.

<https://iz.ru/2168950/2024-03-21/vs-rf-s-pomoshchyu-rszo-smersch-razgromili-artrachet-vs-u-na-granitse-belgorodskoi-oblasti>

I25 Raschet Supercam S350 vskryl punkt upravleniya BPLA VSU v Khersonskoi oblasti. Izvestia, 2026-03-01.
<https://iz.ru/2051080/2026-03-01/raschet-supercam-s350-vskryl-punkt-upravleniya-bpla-vsu-v-khersonskoi-oblasti>

I26 Артиллеристы рассказали об особенностях комплекса разведки «Зоопарк-1М». Izvestia (relaying Russian MoD video), 2023-07-27.
<https://iz.ru/1550170/2023-07-27/artilleryisty-rasskazali-ob-osobennostiakh-kompleksa-razvedki-zoopark-1m>

I217 Artilleriya narashchivaet moshch. Izvestia, 2017-12-14.
<https://iz.ru/675176/evgeniy-andreev-bogdan-stepovoi-aleksei-ramm/artilleriya-narashchivaet-moshch>

I220 В Нагорном Карабахе появились российские комплексы «Леер-3». Izvestia, 2020-11-25.
<https://iz.ru/1092202/2020-11-25/v-nagornom-karabakhe-poiavilis-rossiiskie-kompleksy-leer-3>

I221 Командующий армией ЮВО сообщил о скором поступлении С-350 «Витязь». Izvestia, 2021-08-12.
<https://iz.ru/1206053/2021-08-12/komanduiushchii-armii-yuvo-soobshchil-o-skornom-postuplenii-s-350-vitiaz>

I222 В Западном округе создадут батальоны радиоэлектронной борьбы. Izvestia, 2022-02-02.
<https://iz.ru/1285489/2022-02-02/v-zapadnom-okruge-sozdaiut-batalony-radioelektronnoi-borby>

I223 Правящая «Коалиция» в зоне спецоперации появилась самая дальнобоинная САУ. Izvestia, 2023-12.
<https://iz.ru/1624356/roman-kretcul-iulija-leonova-bogdan-stepovoi/privashchaila-koaliticia-v-zone-spetcoptacitii-poiavilas-samaia-dalnoboinaia-sau>

I223B Dali nam samye krutye pushki iz tekhn, chto est v mire. Izvestia, 2023-02.
<https://iz.ru/1618242/dmitrii-astakhan/dali-nam-samye-krutye-pushki-iz-tekhn-chto-est-v-mire>

I226 Расчет САУ «Гиацинт-С» уничтожил пункт управления вражеских беспилотников. Izvestia (citing Russian MoD), 2026-09-23.
<https://iz.ru/2171815/2026-09-23/raschet-sau-giacint-s-unichtozhil-punkt-upravleniya-vrazheskikh-bespiilotnikov>

I2202 Минобороны РФ назвало дислокацию первой ЗРС С-350 «Витязь». Izvestia, 2020-01-14.
<https://iz.ru/964166/2020-01-14/minobrony-rf-nazvalo-diskolacitui-pervoi-zrs-s-350-vitiaz>

I2222 Появились кадры работы комплекса «Леер-3» по подавлению сотовой связи в зоне СВО. Izvestia, 2022-11-30.
<https://iz.ru/1433400/2022-11-30/poiavilis-kadry-raboty-kompleksa-leer-3-po-podavleniiu-sotovoii-svazi-v-zone-svo>

I2232 Бригады подряд: VS RF usiliat artilleriyskie soedineniya bolshoy moshchnosti. Izvestia, 2023.
<https://iz.ru/1612479/aleksei-ramm-bogdan-stepovoi/brigady-podriadi-vs-rf-usiliat-artilleriyskie-soedineniya-bolshoi-moshchnosti>

I2233 Гаубицы «Мста-С» и пушки «Гиацинт-С» поразили живую силу и технику ВСУ. Izvestia (citing Russian MoD), 2023-02-28.
<https://iz.ru/1476303/2023-02-28/gaubitsy-msta-s-i-pushki-giacint-s-porazili-zhivuuiu-silu-i-tekhniku-vsu>

I2234 МО показало боевую работу специалистов РЕБ ЗВО на купинском направлении СВО. Izvestia, 2023-12-17.
<https://iz.ru/1621743/2023-12-17/mo-pokazalo-boevuiu-rabotu-spetcialistov-reb-zvo-na-kupinskomp-napravlenii-svo>

I2MAL Новейшая гаубица «Мальва» выдержала попадание из HIMARS в зоне СВО. Izvestia, 2024-07-16.
<https://iz.ru/1728408/2024-07-16/noveishaia-gaubitca-malva-vyderzhala-popadanie-iz-himars-v-zone-svo>

I2RU Минобороны заглушит GPS с выске сотовой связи. Izvestiya, 2016-08-25.
<https://iz.ru/news/628766>

I2RU2 В энциптрее «Торнадо»: новые бронегрузовики поступают на Северный Кавказ. Izvestia, date not shown in text.
<https://iz.ru/1024825/aleksei-ramm-bogdan-stepovoi/v-epitcentre-tornado-novye-bronegruzoviki-postupiat-na-severnyi-kavkaz>

I2V Уралвагонзавод переименовал БМТП «Терминатор» в «Спиридон». Izvestia, 2026-06-02.
<https://iz.ru/2107613/na-uralvagonzavode-bmtp-terminator-pereimenovali-v-spiridona-zi>

I2V2 «Форпост» принял: ВМФ начал осваивать гроны дальнего радиуса действия. Izvestia, 2025-06-05.
<https://iz.ru/export/google/amp/1898170>

I2V3 S ostrotoy Lantseta: kakie novshestva primenyayutsya v kontrbateraynoy borbe. Izvestia, 2024-12 (inferred from a 2024-12-06 derivative).
<https://iz.ru/1802349/bogdan-stepovoi-vladimir-matveev/s-ostrotoi-lanceta-kakie-novshestva-priimenautsa-v-kontrbateraynoi-borbe>

I2V4 Russia Integrates Military Capabilities into Civil Communications Infrastructure [OE Watch, quoting Izvestiya 25 Aug 2016]. US Army FMSO / OE Watch, 2016-10.
<https://mftrnd.org/wp-content/uploads/Civil-Tower-Jammers-Russia.pdf>

I2V5 Vrazheskie bespiilotniki ubyt 'Repellentom'. Izvestia, 2016-12-29.
<https://iz.ru/news/635954>

I2V6 Kozachenko A., Ramm A., 'Severnoe vliyanie: v Arktike razvernuto sverkhdalnoboinnye S-300V4'. Izvestia, 2019-08-19.
<https://iz.ru/904692/aleksei-kozachenko-aleksei-ramm/severnoe-vliyanie-v-arktike-razvernuto-sverkhdalnoboinnye-s-300v4>

I2V7 Танкобый «Прорыв»: войска в Центральной России уккомплектуют Т-90М. Izvestia, 2020-03.
<https://iz.ru/export/google/amp/989082>

I2V8 Зенитчики рассказали об особенностях работы ЗРК Тор-М2 в зоне СВО. Izvestia, 2024-05-02.
<https://iz.ru/1690614/2024-05-02/zenitchiki-rasskazali-ob-osobennostiakh-raboty-zrk-tor-m2-v-zone-svo>

I2V9 Колесить по тылам: армейский спецназ получит новый броневтомобиль. Izvestia, 2021.
<https://iz.ru/1164819/roman-kretcul-anna-cherepanova/kolesit-po-tylam-armeiskii-spetcnaz-poluchit-novyy-broneavtomobil>

I2V20 «Пенициллин» угём в разведку. Izvestia, 2020-04.
<https://iz.ru/1007743/bogdan-stepovoi-anton-lavrov/penicillin-idet-v-razvedku-artilleriiu-navedut-na-tsel-elektronnye-kompleksy>

I2V22 ВС РФ получили комплексы разведки «Пенициллин» для применения в зоне СВО. Izvestia, 2022-12-17.
<https://iz.ru/1442261/2022-12-17/vs-rf-poluchili-kompleksy-razvedki-penicillin-dlia-primeneniia-v-zone-svo>

I2V222 Kurs «Zemledeliya»: yug Rossii usilyat «reaktivnymi» minnymi polyami. Izvestia, 2022-02-16.
<https://iz.ru/1291330/roman-kretcul-bogdan-stepovoi/kurs-zemledeliia-ug-rossii-usiliat-reaktivnymi-minnymi-poliami>

JAMES Russia's Unmanned Systems Forces Become Wildcard in Moscow's Military Modernization. Jamestown Foundation, Eurasia Daily Monitor, 2026.
<https://jamestown.org/russias-unmanned-systems-forces-become-wildcard-in-moscows-military-modernization/>

JANES Ukraine conflict: Russia employs Forpost-R UAV. Janes, 2022-03-15.
<https://www.janes.com/defence-news/air-platforms/latest/ukraine-conflict-russia-employs-forpost-r-uav>

JANES2 Ukraine conflict: Russian forces employ guided rockets. Janes, 2022-03-11.
<https://www.janes.com/osint-insights/defence-news/weapons/ukraine-conflict-russian-forces-employ-guided-rockets>

JANES3 Ukraine conflict: Russian mine-laying system makes combat debut. Janes, 2022-03-30.
<https://www.janes.com/defence-news/news-detail/ukraine-conflict-russian-mine-laying-system-makes-combat-debut>

JANES19 Russia's Palantin electronic warfare system makes training debut. Janes, 2019-09-19.
<https://www.janes.com/osint-insights/defence-news/russias-palantin-electronic-warfare-system-makes-training-debut>

JANES25 Analysis: Russia regenerates its tank force. Janes, 2025-09.
<https://www.janes.com/osint-insights/defence-and-national-security-analysis/russia-regenerates-tank-force>

JANESRG Jane's Tanks and Combat Vehicles Recognition Guide, 2nd ed. (Foss), 251 entry, variants, p. 439; MT-LB entry, key recognition features. HarperCollins / Jane's, 2000.
 ISBN 0-00-472452-6

JANESRG Jane's Tanks and Combat Vehicles Recognition Guide, 2nd ed. (Foss): BMD-1, 2S9 and BTR-60 variant lists. HarperCollins / Jane's, 2000.
 ISBN 0-00-472452-6 (printed book; no public URL)

JANESRG Jane's Tanks and Combat Vehicles Recognition Guide, 2nd ed. HarperCollins / Jane's, 2000.

JANESRG Jane's Tanks and Combat Vehicles Recognition Guide, BMP-1 entry, pp. 175-176. HarperCollins / Jane's, 2000.
 ISBN 0-00-472452-6 (printed book)

- JANESRG** Jane's Tanks and Combat Vehicles Recognition Guide, BMD-1 entry variants, p. 179. HarperCollins / Jane's, 2000.
- JANESRG** Jane's Tanks and Combat Vehicles Recognition Guide, 2nd ed., T-54/T-55 and T-62 entries (pp. 71-75). HarperCollins / Jane's, 2000.
sources_cache/janes_foss.txt
- JANESRG** Jane's Tanks and Combat Vehicles Recognition Guide, 2nd ed., T-72 key recognition features. HarperCollins / Jane's, 2000.
ISBN 0-00-472452-6 (printed book; local text sources_cache/janes_foss.txt)
- JF18** Russian CMD receives Tornado-G MLRS. Joint Forces News (reproducing Russian MoD release of 25 Oct 2018), 2018-10.
<https://www.joint-forces.com/world-news/18989-russian-cmd-receives-tornado-g-mlrs>
- JFOR** Engineering troops of the CAA of the Western MD received URAN-6 (MoD release 24 Jan 2021). joint-forces.com, 2021-01-24.
<https://www.joint-forces.com/?p=39843>
- JFRC** Russian Northern Fleet Temporary River Bridging Exercise. Joint Forces News (Northern Fleet press release), 2020-11-02.
<https://www.joint-forces.com/exercise-news/37305-russian-northern-fleet-temporary-river-bridging-exercise>
- JIT** Update investigation JIT MH17, speaker's text press meeting 24 May 2018. Netherlands Public Prosecution Service (JIT), 2018-05-24.
<https://www.prosecutionservice.nl/topics/mh17-plane-crash/criminal-investigation-jit-mh17/speakers-text-jit-mh17-press-meeting-24-5-2018>
- JREW02** Jane's Radar and EW Systems 2001-2002 extract: ARK-1M Rys (JREW-p1781, PDF pp. 1781-1782); Zoopark-1 (JREW-p1766, PDF pp. 1766-1767). Jane's Information Group, 2001.
Not public
- JREW02** Jane's Radar and EW Systems 2002-2003: TOR-M1 surface-to-air missile radar system. Jane's Information Group, 2001-11-09.
Not public; JREW02, JREW-p1795, PDF pp. 1795-1796 (also JREW-u1992 Rombl)
- JREW02** Jane's Radar and EW Systems 2001-2003, entry JREW-p2011 (1RL144M), PDF pp. 2011-2012. Jane's Information Group, 2001-07-07.
- JT21** Russia's Military Boosts Electromagnetic Spectrum Capability (McDermott). Jamestown Foundation, 2021-09-22.
<https://jamestown.org/program/russias-military-boosts-electromagnetic-spectrum-capability/>
- KALCLUB** БПЛА «Тахюн» – круглосуточный разведчик. kalashnikov.club, 2022-12.
<https://kalashnikov.club/а/bpla-takhion-kruglosutochnyy-razvedchik->
- KAT** Sredstva inzhenernogo vooruzheniya, Katalog, Kniga 1, 2nd ed. (Russian engineer troops, for official use). Engineer Troops of the Russian Armed Forces (copy hosted by sprotvyg7.com.ua), after 2008, undated.
https://sprotvyg7.com.ua/wp-content/uploads/2023/12/Katalog_Sredstva_inzhenernogo_vooruzheniya_Kniga_1.pdf
- K11** Opinion: A look at the drone arsenal Russia uses against Ukraine (Kharkiv). Kyiv Independent, 2024-04-16.
<https://kyivindependent.com/opinion-a-look-at-the-drone-arsenal-russia-uses-against-ukraine/>
- K12** As Russia's fiber optic drones flood the battlefield, Ukraine is racing to catch up (F. Farrell). The Kyiv Independent, 2025-05-20.
<https://kyivindependent.com/as-russias-fiber-optic-drones-flood-the-battlefield-ukraine-is-racing-to-catch-up/>
- K13** Ukraine's intelligence reveals Russian brigades responsible for Sumy attack. Kyiv Independent, 2025-04-13.
<https://kyivindependent.com/ukraines-intelligence-reveals-russian-brigades-responsible-for-sumy-attack/>
- K14** Exclusive: Majority of companies involved in Russian Molniya drone production unsanctioned. Kyiv Independent, 2026-09-25.
<https://kyivindependent.com/exclusive-majority-of-companies-involved-in-russian-molniya-drone-production-unsanctioned-research-shows/>
- K15** Hundreds of Russian vehicles, weapon systems seized during Kharkiv counterattack. Kyiv Independent, 2022-09.
<https://kyivindependent.com/hundreds-of-russian-vehicles-weapon-systems-seized-during-kharkiv-counterattack/>
- KOLESA** Перекинутом ли БТР-82А американского «Страйкера»? Kolesa.ru, 2015-06-18.
<https://www.kolesa.ru/article/perezhitv-lt-btr82a-amerikanskogo-strajkera-2015-06-18>
- KOLESA2** Радиотехнические, десантируемые, для космической связи и не только: спецверсии ГАЗ-66. Kolesa.ru.
<https://www.kolesa.ru/article/radiotekhnicheskie-desantiруемые-dlya-kosmicheskoy-svyazi-i-ne-tolko-spetsverzii-gaz-66>
- KOLESA3** Military trades of the KrAZ-255 (kolesa.ru). Kolesa.ru.
<https://www.kolesa.ru/article/mostoukladchik-svaezibatel-i-dazhe-gigantskiy-pylososyvoennyy-professii-kraz-255>
- KP** Russia's 'Vandal' Fiber Optic FPV Drone Is Probably Another Chinese Import. Kyiv Post, 2024, after August.
<https://www.kyivpost.com/post/38831>
- KP2** Ukrainian Drone Hunters Turn Their Attention to Russian Operators. Kyiv Post, 2025-04-03.
<https://www.kyivpost.com/amp/post/50064>
- KP3** Ukraine Strikes Russian Fuel Trains, Destroys Rare 'Inokhodets' Drones in Crimea. Kyiv Post, 2026-04-04.
<https://www.kyivpost.com/amp/post/73228>
- KP24** Russia Releases Video of Shahed Drones 'Little Sister'. Kyiv Post, 2024-07.
<https://www.kyivpost.com/post/36577>
- KP25** Russia Turns Cheap Decoy Drones Into Kamikaze Weapons. Kyiv Post, 2025-02-06.
<https://www.kyivpost.com/post/46692>
- KP26** Russia Repurposes Grabber Drone as FPV Carrier for First Time. Kyiv Post, 2026-02.
<https://www.kyivpost.com/post/69364>
- KP262** Operatory Orlan-30 ... porazhayut tseli vysokotochnymi snaryadami. Komsomolskaya Pravda, 2026-07-09.
<https://www.kp.ru/daily/277796/5274528/>
- KPEN** Ukrainian Army Destroys Huge Russian Complex in Mykolaiv Region. Kyiv Post, 2022-08-18.
<https://www.kyivpost.com/russias-war/ukrainian-army-destroys-huge-russian-complex-in-mykolaiv-region.html>
- KPOST** Prykordonnky pokazaly, yak znyshchyly novitini rosiyskiy kompleks REB R-340RP Pole-21. Kyiv Post [Ukrainian edition], 2024-04.
<https://www.kyivpost.com/uk/amp/post/73228>
- KYIV** Russian armored vehicles crash on Crimean Bridge. Kyiv Post, 2023.
<https://kyivpost.com/post/15010>
- KYIV2** What is the Lancet? Kyiv Post, 2023-11.
<https://www.kyivpost.com/analysis/23923>
- KYIVP** \$600K in 30 Minutes: Ukraine Downs Two of Russia's Most Expensive Recon Drones. Kyiv Post, 2025.
<https://www.kyivpost.com/post/51783>
- KYIVP2** Ukrainian Special-Ops Pick Off Russia's Cutting-Edge Radio Relay Station. Kyiv Post, 2024-06-13.
<https://www.kyivpost.com/amp/post/34226>
- KYIVP3** Ukraine Uses Hummers as Improvised Remote Anti-Tank Mine Layers. Kyiv Post, c. 2024 (undated page).
<https://www.kyivpost.com/post/43317>
- KZ** Оборона под управлением системы «Барнаул-Т». Krasnaya Zvezda [Russian MoD], 2018-06-18.
<http://redstar.ru/oborona-pod-upravleniem-sistemy-barnaul-t/>
- KZ1** Воюющие на САУ «Мальва» бойцы называют ее снайперской винтовкой. Mail.ru V fokuse, reprinting Krasnaya Zvezda [MoD newspaper], 2026-06-05.
<https://vfokus.mail.ru/news/69624868-voyuyushchie-na-sau-malva-bojtsy-nazyvayut-ee-snaiperskoj-vintovkoj/>
- KZ2** Самоходки «Мальва» ведут непрерывную огневую работу по позициям ВСУ. Mail.ru V fokuse, reprinting Krasnaya Zvezda [MoD newspaper], 2026-08-14.
<https://vfokus.mail.ru/news/69636296-samohodki-malva-vedut-neprepryvnuyu-ognevuyu-rabotu-po-pozitsiyam-vsuv/>
- KZ3** Na Kamchatke uspešno osvoini kompleks Murmansk-BN (Krasnaya Zvezda, reposted on LiveJournal). Krasnaya Zvezda (repost by starcom68), 2016-01.
<https://starcom68.livejournal.com/1959794.html>
- KZ2019** Войска надёжно прикрыты от угаров с воздуха (interview, Lt Gen A. Leonov). Krasnaya Zvezda, 2019-12.
<http://redstar.ru/vojska-nadyazhno-prikryty-ot-udarov-s-vozduha/>
- KZ2021** Небо над войсками надёжно защищено (interview, Lt Gen A. Leonov, chief of troop air defence). Krasnaya Zvezda, 2021-12.
<http://redstar.ru/nebo-nad-vojskami-nadyazhno-zashishhenno/>
- LBUA** What is known about jet-powered Shaheds. LB.ua, 2025-08-01.
https://en.lb.ua/news/2025/08/01/36522_reactive_shaheds_when.html
- LENTA** «Князь Вангал» разбил колонну ВСУ под Курском. Lenta.ru, 2024-10-30.
<https://lenta.ru/news/2024/10/30/knyaz-vandal-razbil-kolonnu-vsuv-pod-kurskom-chto-iz-sebya-predstavlyayet-revolutsionnoe-oruzhie-rossii/>
- LENTA2** Polk smistie S-400 na Kamchatke poluchil Pantsiri. Lenta.ru (citing Interfax, Eastern MD press service), 2015-12-02.
<https://lenta.ru/news/2015/12/02/pantsir/>

LENTA3 Vostochnyy voyennyy okrug poluchil polkovoy kompleks S-400. Lenta.ru [citing RIA Novosti and MoD press service], 2012-06-09.

<https://lenta.ru/news/2012/06/09/s400/>

LENTA4 В России рассказали о применении незаметных для украинских войск беспилотников. Lenta.ru, 2022-04-27.

<https://lenta.ru/news/2022/04/27/bespiotniki/>

LENTA16 Lenta.ru: first Malka firing in an Eastern MD artillery brigade, Amur oblast. Lenta.ru, 2016-04-04.

<https://lenta.ru/news/2016/04/04/malka/>

LENTA22 В районе проведения спецоперации заметили БМП «Басурманин». Lenta.ru, 2022-04-27.

<https://lenta.ru/news/2022/04/27/bmp1am/>

LENTA23 Artilleristy TsVo ... Zoopark-1M. Lenta.ru, 2023-07-27.

<https://lenta.ru/news/2023/07/27/zoopark/>

LEXCN Lexicon records vystrel, vpk-ural, patrol-7020, sarmat, z-sts-akhmat, gusar, buran-asv, toros. Project lexicon [pilot/lexicon/lexicon_final.json], 2026-09-30.

LIGA Rossiya ispytyvaet v Donbasse noveyshe komplekсы REB (Bellingcat/DFRLab summary). Liga.net, 2018-09.

<https://news.liga.net/politics/news/rossiya-ispytyvaet-v-donbasse-noveyshe-kompleksy-reb---bellingcat>

LIGA2 Interesting trophy: Ukrainian forces dismantled Russian SuperCam drone. LIGA.net, 2023-09.

<https://news.liga.net/politics/news/interesnyy-trofej-vsu-razobrali-rossiyskiy-dron-supercam-foto>

LOL54 TMC-65Y - армейская дымовуха. lol54.ru.

<https://lol54.ru/army/137248-tms-65u-armejskaja-dymovukha.html>

M360 «Умные» минные поля установили на Авдеевском направлении. 360.ru citing RIA Novosti and Russian MoD, 2024-09-16.

<https://360.ru/news/ukrainian-crisis/umnye-minnye-polja-ustanovili-na-avdeevskom-napravlenii/>

MAILRU Беспилотник «Форпост-РЗ» покажут на Dubai Airshow 2025. Hi-Tech Mail.ru, 2025-11.

<https://hi-tech.mail.ru/news/137458-bespiotnik-forpost-re-pokazhut-na-dubai-airshow-2025/>

MAILRU2 Расчет «Торнадо-С» уничтожил подземные укрытия (232nd Gds Rocket Artillery Brigade). Mail.ru News, citing Russian MoD, 2026-09-09.

<https://news.mail.ru/politics/72289828/>

MAILRU3 Российский реактивный снаряд для «Торнадо-Г» впервые показали за рубежом. Hi-Tech Mail.ru, 2018-11.

<https://hi-tech.mail.ru/news/40544-tornado-g-za-rubezhom/>

MASH Специалисты РЗБ успешно применяют комплексы «Леер-3» и БПА «Орлан-10» в зоне СВО. Mashnews, undated [c. 2022].

<https://mashnews.ru/specialisty-reb-uspeshno-primenyayut-kompleksy-leer-3-i-bpa-orlan-10-v-zone-svo.html>

MDB Iskander the Great (Barabanov), Moscow Defense Brief 4/2008. CAST, 2008.

https://www.files.ethz.ch/isn/94687/2008_4.pdf

MEDUZA Supercam S350, one of the drones the Russian army uses in Ukraine (photos). Meduza, 2025-02-18.

<https://meduza.io/feature/2025/02/18/supercam-s350-odin-iz-bespiotnikov-kotorye-rossiyskaya-armiya-ispolzuet-v-ukraine-fotografii>

MEZHA Ukrainian forces struck fuel trains in Luhansk and shot down Russian Inokhodets drone in Crimea. Mezha, 2026-04.

https://mezha.net/eng/bukvy/ukrainian_forces_struck-23/

MEZHA2 Lasar's Group destroyed Russian Palantine EW system on Zaporizhzhia front. Mezha, 2026-04.

https://mezha.net/eng/bukvy/lasar-s_group_destroyed-2/

MF1 2S5 Giatint-S (Hyacinth) [M1981] Self-Propelled Artillery. MilitaryFactory.com, 2014.

https://www.militaryfactory.com/armour/detail.php?armor_id=704

MF2 Kalashnikov Takhion (Tachyon) Mini-UAV. Military Factory, 2022-04.

https://www.militaryfactory.com/aircraft/detail.php?aircraft_id=2466

MFR Supercam S350 (product page). GK Bespiotnyye sistemy (Unmanned Systems Group), undated, read 2026-10.

<https://supercam.aero/catalog/supercam-s350>

MFRAB About the company. GK Bespiotnyye sistemy, undated, read 2026-10.

<https://supercam.aero/about>

MIL Ukrainian paratroopers down Russian Orion UAV over Kursk oblast. Military, 2024-10.

<https://mil.in.ua/en/news/ukrainian-paratroopers-down-russian-orion-uav-over-kursk-oblast>

MIL1 Defence Intelligence of Ukraine Strikes S-400 Launcher and Radar in Crimea. Military, 2026-07-27.

<https://military.com/en/news/defence-intelligence-of-ukraine-strikes-s-400-launcher-and-radar-in-crimea/>

MIL2 Russian ARK-1 Rys' radar neutralized in the Kherson region. Military.

<https://military.com/en/news/russian-ark-1-rys-radar-neutralized-in-the-kherson-region/>

MIL3 Counter-battery radars of Russia under Ukrainian fire (Ukrainian text). Military, 2025.

<https://military.com/uk/articles/konturbatarejini-radii-rosiyi-pid-vogmem-zsu-shho-znyshcheno-i-chomu-tse-vazhlyvo/>

MIL4 Ukrainian Drones Hit Module of Russia's Rare Barnaul-T System. Military, 2025-08 (inferred from image path).

<https://military.com/en/news/ukrainian-drones-hit-module-of-russia-s-rare-barnaul-t-system/>

MIL5 БМД-2 (reference article). Military.

<https://military.com/uk/articles/bmd-2/>

MIL6 Russian Army receives new BMP-3 and BMD-4M. Military, 2024.

<https://military.com/en/news/russian-army-receives-new-bmp-3-and-bmd-4m/>

MIL7 Modernized Russian BMP-1AM Falls into Hands of Ukrainian 58th Brigade. Military, 2023-11.

<https://military.com/en/news/modernized-russian-bmp-1am-falls-into-hands-of-ukrainian-58th-brigade/>

MIL8 Армія Росії отримала нові БМПТ «Терминатор». Military, 2025-10.

<https://military.com/uk/news/armiya-rosiyi-otrymala-novi-bmpt-terminator/>

MIL9 У Росії офіційно перейменували БМПТ «Терминатор» на «Спирюгон». Military, 2026-06.

<https://military.com/uk/news/perejimenuvali-bmpt-terminator-na-spyrydon/>

MIL10 Ukrainian Forces Destroy Third Borisoglebsk-2 Electronic Warfare System in a Week. Military, April, year not shown on fetched text.

<https://military.com/en/news/ukrainian-forces-destroy-third-borisoglebsk-2-electronic-warfare-system-in-a-week/>

MIL11 Ukrainian military destroy the Russian R-934BMV Borisoglebsk-2 EW system. Military, 2023-10-16.

<https://military.com/en/news/ukrainian-military-destroy-the-russian-r-934bm-borisoglebsk-2-ew-system/>

MIL12 414th Strike Drone Regiment destroyed a Russian R-934BMV electronic warfare station of the Borisoglebsk-2 system. Military, 2024-11.

<https://military.com/en/news/414th-strike-drone-regiment-destroyed-a-russian-r-934bm-electronic-warfare-station-of-the-borisoglebsk-2-system/>

MIL13 Russian BREM-L destroyed in the Zaporizhzhia region. Military, 2023.

<https://military.com/en/news/russian-brem-l-destroyed-in-the-zaporizhzhia-region/>

MIL14 Ukrainian Air Assault Forces demonstrated an unknown modernized version of BRM-1K. Military.

<https://military.com/en/news/ukrainian-air-assault-forces-demonstrated-an-unknown-modernized-version-of-brm-1k/>

MIL15 A rare BTR-90 was spotted among the defeated column of Russians near Avdiivka. Military, 2023-12.

<https://military.com/en/news/a-rare-btr-90-was-spotted-among-the-defeated-column-of-russians-near-avdiivka/>

MIL16 Volunteers restored the captured BTR-MD Rakushka for Ukrainian paratroopers. Military, 2023-06.

<https://military.com/en/news/volunteers-restored-the-captured-btr-md-rakushka-for-ukrainian-paratroopers/>

MIL17 Ukrainian Forces Destroy Russian Buk-M2 Air Defense Systems. Military, 2025-04-21.

<https://military.com/en/news/ukrainian-forces-destroy-russian-buk-m2-air-defense-systems/>

MIL18 Unmanned Systems Forces Shoot Down Russian Forpost-R Drone Over Kursk Region at 4.3 km Altitude. Military, 2026-07-27.

<https://military.com/en/news/unmanned-systems-forces-shoot-down-russian-forpost-r-drone-over-kursk-region-at-4-3-km-altitude/>

MIL19 Range of 50-65 km: Russians Modernized Fiber Optic FPV Drone Prince Vandal. Military, 2025-12-27.

<https://military.com/en/news/range-of-50-65-km-russians-modernized-fiber-optic-fpv-drone-prince-vandal/>

MIL20 Russians complete formation of first UAV regiment, present it at May 9 parade. Military, 2025-05.

<https://military.com/en/news/russians-completes-formation-of-first-uav-regiment-presents-it-at-may-9-parade/>

- MIL21** Ukraine's Unmanned Systems Forces destroy training base of Russian GROM Kaskad drone brigade. *Militarnyi*, 2026-09.
<https://militarnyi.com/en/news/ukraine-s-unmanned-systems-forces-destroy-training-base-of-russian-grom-kaskad-drone-brigade/>
- MIL22** Ukrainian intelligence releases detailed analysis of Russian Geran-3 jet-powered UAV (series UJ). *Militarnyi*, 2025-09.
<https://militarnyi.com/en/news/ukrainian-intelligence-releases-detailed-analysis-of-russian-geran-3-jet-powered-uav-series-uj>
- MIL23** Caught on the Ground: NEMESIS Unmanned Systems Brigade Strikes Russian Gerbera UAV on Launch Vehicle. *Militarnyi*, 2025-12.
<https://militarnyi.com/en/news/caught-on-the-ground-nemesis-unmanned-systems-brigade-strikes-russian-gerbera-uav-on-launch-vehicle/>
- MIL24** Drones and Artillery Strike Russian Navodchik-2 UAV. *Militarnyi*.
<https://militarnyi.com/en/news/drones-and-artillery-strike-russian-navodchik-2-uav/>
- MIL25** Ukrainian military destroys Russian R-934B Sinitsa electronic warfare station. *Militarnyi*, 2024-10 (inferred from image upload path; no date in text).
<https://militarnyi.com/en/news/ukrainian-military-destroys-russian-r-934b-sinitsa-electronic-warfare-station/>
- MIL26** Ukrainian bomber drones destroy two stations of Borisoglebsk-2 EW system. *Militarnyi*, 2025-04 (inferred from image upload path; no date in text).
<https://militarnyi.com/en/news/ukrainian-bomber-drones-destroy-two-stations-of-borisoglebsk-2-ew-system/>
- MIL27** Sumy Missile Attack Traced to Russia's 112th and 448th Brigades. *Militarnyi*, 2025-04-13.
<https://militarnyi.com/en/news/sumy-missile-attack-traced-to-russia-s-112th-and-448th-brigades/>
- MIL28** In the Kyiv region, Russian servicemen has lost part of the Krasukha-4 EW system. *Militarnyi*, 2022-03-23.
<https://militarnyi.com/en/news/in-the-kyiv-region-russian-servicemen-has-lost-part-of-the-krasukha-4-ew-system/>
- MIL29** Ukrainian Forces eliminate Russian Krasukha-4 EW System in Zaporizhzhia region. *Militarnyi*, 2023-11-01.
<https://militarnyi.com/en/news/ukrainian-forces-eliminate-russian-krasukha-4-ew-system-in-zaporizhzhia-region/>
- MIL30** Hackers Leak Detailed Photos of Russian EW System 1RL257 Krasukha-4. *Militarnyi*, 2025-10-09.
<https://militarnyi.com/en/news/hackers-leak-detailed-photos-of-russian-ew-system-1rl257-krasukha-4/>
- MIL31** ZALA Lancet: engagement and countereffort (Yan). *Militarnyi*, 2023-07-18.
<https://militarnyi.com/en/articles/zala-lancet-engagement-and-countereffort/>
- MIL32** Ukrainian Defence Forces destroyed the Russian Leer-2 EW system. *Militarnyi*, 2023-11.
<https://militarnyi.com/en/news/ukrainian-defence-forces-destroyed-the-russian-leer-2-ew-system/>
- MIL33** Mykolaiv paratroopers shot down enemy drone capable of jamming GSM connection. *Militarnyi*.
<https://militarnyi.com/en/news/mykolaiv-paratroopers-shot-down-enemy-drone-capable-of-jamming-gsm-connection/>
- MIL34** Armed Forces of Ukraine seize new Russian Buran armored vehicle in Kursk oblast. *Militarnyi*, 2024.
<https://militarnyi.com/en/news/armed-forces-of-ukraine-seize-new-russian-buran-armored-vehicle-in-kursk-oblast/>
- MIL35** Ukrainian military for the first time shot down Iranian Mohajer-6. *Militarnyi*, 2022-09-23.
<https://militarnyi.com/en/news/ukrainian-military-for-the-first-time-shot-down-iranian-mohajer-6-drone-used-by-russians/>
- MIL36** Russia's own Mohajer-6 drone spotted crashed in Kursk region. *Militarnyi*, 2024-05.
<https://militarnyi.com/en/news/russia-s-own-mohajer-6-drone-spotted-crashed-in-kursk-region/>
- MIL37** In Crimea, SSU struck Iranian Mohajer-6 drone and Pantsir-S2 systems. *Militarnyi*, 2026-03-06.
<https://militarnyi.com/en/news/in-crimea-ssu-struck-mohajer-6-pantsir-s2/>
- MIL38** Ukrainian Forces Shoot Down AI-Equipped Molniya UAV for the First Time. *Militarnyi*, 2026-06.
<https://militarnyi.com/en/news/ukraine-shoot-down-ai-molniya-uav-first/>
- MIL39** General Cherry Air Drones Destroyed Over 1000 Russian Molniya UAVs in December. *Militarnyi*, 2026-01.
<https://militarnyi.com/en/news/general-cherry-air-drones-downed-over-1000-russian-molniya-uavs-in-december/>
- MIL40** Lightning 13: РФ розробила нову версію ударного БпЛА «Молнія». *Militarnyi* (Ukrainian edition), 2026-06.
<https://militarnyi.com/uk/news/lightning-13-versiya-udarnogo-bpla-molnija/>
- MIL41** HIMARS destroyed a battery of Russian Msta-S. *Militarnyi*, 2023-06-14.
<https://militarnyi.com/en/news/himars-destroyed-a-battalion-of-russian-msta-s/>
- MIL42** MT-LB: combat chariot of the Russo-Ukrainian war. *Militarnyi*, 2023.
<https://militarnyi.com/en/articles/mt-lb-combat-chariot-of-the-russo-ukrainian-war>
- MIL43** Russian invaders lost a rare MT-LB 6MB. *Militarnyi*, undated (2022-2023).
<https://militarnyi.com/en/news/russian-invaders-lost-a-rare-mt-lb-6mb/>
- MIL44** РФ вігнравляє в Україні 80-ту арктичну бригаду з фінського кордону. *Militarnyi*, 2022.
<https://militarnyi.com/uk/news/rf-vidpravlyaye-v-ukrayinu-80-tu-arktychnu-bryadgu-z-finskogo-kordonu-popyr-vstup-krayiny-v-nato/>
- MIL45** Defense Intelligence of Ukraine Struck Russian Murmansk-BN Electronic Warfare station in Crimea. *Militarnyi*, 2026-07.
<https://militarnyi.com/en/news/diu-struck-murmansk-bn-ew-station-in-crimea/>
- MIL46** Ukrainian Armed Force destroy the Russian Palantin-K EW system for the first time. *Militarnyi*, 2023-02.
<https://militarnyi.com/en/news/ukrainian-armed-force-destroy-the-russian-palantin-k-ew-system-for-the-first-time/>
- MIL47** Ukrainian Special Forces Eliminate Russian Palantin EW System. *Militarnyi*, 2024-03-11.
<https://militarnyi.com/en/news/ukrainian-special-forces-eliminate-russian-palantin-ew-system/>
- MIL48** Belarus Supplying Russia With MZKT Armored Chassis for New Pantsir Version Since 2024. *Militarnyi*, 2025-11.
<https://militarnyi.com/en/news/belarus-supplying-russia-with-mzkt-armored-chassis-for-new-pantsir-version-since-2024/>
- MIL49** Ukrainian Special Operation Forces destroyed a pontoon unit in the Kursk region. *Militarnyi*, 2024-08.
<https://militarnyi.com/en/news/ukrainian-special-operation-forces-destroyed-a-pontoon-unit-in-the-kursk-region/>
- MIL50** Russians Lost Rare PRP-4 Vehicles in Combat. *Militarnyi*, 2025-11-07.
<https://militarnyi.com/en/news/russians-lost-rare-prp-4-vehicles-in-combat/>
- MIL51** The Ukrainian military seized a Russian R-149MA1 command post carrier. *Militarnyi*, 2022.
<https://militarnyi.com/en/news/the-ukrainian-military-seized-a-russian-r-149ma1-command-post-carrier/>
- MIL52** The Armed Forces had seized the russian R-166-0.5 radio station in the Sumy region. *Militarnyi*, 2022.
<https://militarnyi.com/en/news/the-armed-forces-had-seized-the-russian-r-166-0-5-radio-station-in-the-sumy-region/>
- MIL53** Ukrainian Drones Strike Tor, Redut-2US, and Kaira Operator Post. *Militarnyi*, 2026-05-14.
<https://militarnyi.com/en/news/ukraine-drones-strike-tor-redut-2us-kaira/>
- MIL54** Strike on Novorossiysk: Shescharis oil terminal and air-defense base hit. *Militarnyi*, 2025-11-14.
<https://militarnyi.com/en/news/strike-on-novorossiysk-shescharis-oil-terminal-and-air-defense-base-hit/>
- MIL55** Satellite images confirm destruction of S-400 launchers and radars in Novorossiysk. *Militarnyi*, 2025-11.
<https://militarnyi.com/en/news/satellite-images-confirm-destruction-of-s-400-launchers-and-radars-in-novorossiysk/>
- MIL56** T-54 and T-55: What are those tanks, and how Russia uses them against Ukraine. *Militarnyi*, 2023-07.
<https://militarnyi.com/en/articles/t-54-and-t-55-what-are-those-tanks-and-how-russia-uses-them-against-ukraine/>
- MIL57** Russians Show T-62M and T-62MV Tanks Upgrades: Self-Entrenching Blades, Tank Sweeps and EW. *Militarnyi*.
<https://militarnyi.com/en/news/russians-show-t-62m-and-t-62mv-tanks-upgrades-self-entrenching-blades-tank-sweeps-and-ew/>
- MIL58** Details of the elimination of the first T-90M Proryv in the Kharkiv region have appeared. *Militarnyi*, 2022-05.
<https://militarnyi.com/en/news/details-of-the-elimination-of-the-first-t-90m-proryv-in-the-kharkiv-region-have-appeared/>
- MIL59** Drones of National Guard disrupted MTU-72 bridge layer in Donetsk (Mylitskiy). *Militarnyi*, 2025-12-03.
<https://militarnyi.com/uk/news/drony-natsgardiyi-zirvaly-robotu-mostoukladacha-mtu-72-na-donechchyni/>

MIL60 Solntsepyok, Buratino, and Tosochka: The Threat of Russian Thermobaric Flamethrowers in Ukraine (Kozatskyi). Militarynyi, 2024-10-14. <https://militarynyi.com/en/articles/solntsepyok-buratino-and-tosochka-the-threat-of-russian-thermobaric-flamethrowers-in-ukraine/>

MIL61 Russians lost the state-of-the-art TOS-2 Tosochka flamethrower system. Militarynyi, 2025-02-16. <https://militarynyi.com/en/news/russians-lost-the-state-of-the-art-tos-2-tosochka-flamethrower-system/>

MIL62 Russia presents Drakon TOS-3 MRL for the first time. Militarynyi, 2024-06-04. <https://militarynyi.com/en/news/russia-presents-drakon-tos-3-mrl-for-the-first-time/>

MIL63 Russia modernizes BM-27 Uragan rocket launchers with BAZ-69092 chassis upgrade. Militarynyi, 2023-12-20. <https://militarynyi.com/en/news/russia-modernizes-bm-27-uragan-rocket-launchers-with-baz-69092-chassis-upgrade/>

MIL64 Силы обороны зничили пікірку РСЗВ «Ураган» на шасі вантажівки «БАЗ». Militarynyi, 2024-08-25. <https://militarynyi.com/uk/news/sily-oborony-znyshlyu-rdkisnu-rszv-uragan-na-shasi-vantazhivky-baz/>

MIL65 Tornado-U trophy armored vehicle restored for Kraken Regiment. Militarynyi, 2023-09. <https://militarynyi.com/en/news/tornado-u-trophy-armored-vehicle-restored-for-kraken-regiment/>

MIL66 The Ukrainian servicemen shot down an enemy ZALA 421-16E2 drone. Militarynyi, 2022-05 (approx.). <https://militarynyi.com/en/news/the-ukrainian-servicemen-shot-down-an-enemy-zala-421-16e2-drone/>

MIL67 Modern Russian ZALA 421-16E2 UAV neutralized near Izium. Militarynyi, 2022-07 (approx.). <https://militarynyi.com/en/news/modern-russian-zala-421-16e2-uav-neutralized-near-izium/>

MIL68 Ukrainian Bukovel-AD EW system landed Russian ZALA 421-16E2 UAV. Militarynyi, 2023-03 (approx.). <https://militarynyi.com/en/news/ukrainian-bukovel-ad-ew-system-landed-russian-zala-421-16e2-uav/>

MIL69 The Russians lost two Zala UAVs that were adjusting strikes in Ukraine. Militarynyi, 2024-07 (approx.). <https://militarynyi.com/en/news/the-russians-lost-two-zala-uavs-that-were-adjusting-strikes-in-ukraine/>

MIL70 Russian R-330Zh Zhitel EW system destroyed in the Kharkiv Region. Militarynyi, 2022-09. <https://militarynyi.com/en/news/russian-r-330zh-zhitel-ew-system-destroyed-in-the-kharkiv-region/>

MIL71 Ukrainian Forces Target Russian Counter-Battery Radars: What Was Hit and Why It Matters. Militarynyi, 2025-08-18. <https://militarynyi.com/en/articles/ukrainian-forces-target-russian-counter-battery-radars-what-was-hit-and-why-it-matters/>

MIL72 Ukrainian military hit 4 Zoopark-1 radar stations. Militarynyi, 2023-11-04. <https://militarynyi.com/en/news/ukrainian-military-hit-4-zoopark-1-radar-stations/>

MIL73 Destroyed Zoopark-1M radar found in Kherson region. Militarynyi, 2023-02. <https://militarynyi.com/en/news/destroyed-zoopark-1m-radar-found-in-kherson-region/>

MIL92 The 92nd Mechanized Brigade destroyed the Russian Orlan-10 UAV mobile control post. Militarynyi, 2023. <https://mil.in.ua/en/news/the-92nd-mechanized-brigade-of-the-armed-forces-of-ukraine-destroyed-the-russian-orlan-10-uav-mobile-control-post/>

MIL110 Ukrainian troops destroyed Russian Uragan MLRS in the Kharkiv sector. Militarynyi, 2024-07. <https://mil.in.ua/en/news/ukrainian-troops-destroyed-russian-uragan-mlrs-in-the-kharkiv-sector>

MILBY TDA-2K, Voyska RKNB zashchity, equipment page. Ministry of Defence of Belarus. <https://www.mil.by/ru/forces/special/rhzb/equipment/542/8369/>

MILIN The Russian army received a batch of modernized T-80BVM tanks. Militarynyi (mil.in.ua), undated, c. 2024. <https://mil.in.ua/en/news/the-russian-army-received-a-batch-of-modernized-t-80bvm-tanks>

MILINF Военный Осведомитель, Telegram post 70948. Voennyy Osvedomitel (Telegram channel), 2021-09-04. <https://t.me/s/milinfolive/70948>

MILIN Russian 203mm Pion self-propelled guns were destroyed in the Kherson region. Militarynyi, 2022. <https://mil.in.ua/en/news/russian-203mm-pion-self-propelled-guns-were-destroyed-in-the-kherson-region>

MILRU 96A6-1 / 96A6E Всевысотный обнаружкитель. MilitaryRussia.ru (DIMMI), 2014-05-07. <https://web.archive.org/web/2026/http://militaryrussia.ru/blog/topic-805.html>

MILUA Ukrainian defenders destroy Russian Aleut swamp buggy. Militarynyi, undated (Aleut loss 2022 referred to as earlier). <https://mil.in.ua/en/news/ukrainian-defenders-destroy-russian-aleut-swamp-buggy/>

MILUA2 Ukrainian border guards seized newest Russian T-16 UAV. Militarynyi (mil.in.ua), 2023-08. <https://mil.in.ua/en/news/ukrainian-border-guards-seized-newest-russian-t-16-uav>

MILUA3 The drone found in Kyiv region turned out to be a new Russian Gerbera UAV. Ukrainian Military Center (mil.in.ua), 2024-07-29. <https://mil.in.ua/en/news/the-drone-found-in-kyiv-region-turned-out-to-be-a-new-russian-gerbera-uav/>

MILUA4 Ukrainian troops seize Russian satellite communications station in Kursk Oblast. Militarynyi (mil.in.ua), 2024. <https://mil.in.ua/en/news/ukrainian-troops-seize-russian-satellite-communications-station-in-kursk-oblast>

MILUA5 SNAR-10M1 radar reconnaissance station seized in Ukraine. Militarynyi, 2022-04. <https://mil.in.ua/en/news/the-snar-10m1-radar-reconnaissance-station-of-the-russian-army-was-seized-in-ukraine/>

MILUA6 Ukrainian FPV-drones hit Russian minelayers. Militarynyi (mil.in.ua), 2023. <https://mil.in.ua/en/news/ukrainian-fpv-drones-hit-russian-minelayers/>

MILUA7 Luhansk region: Ukrainian troops used UR-77 Meteorit mine clearing vehicle. Militarynyi, 2024. <https://mil.in.ua/en/news/luhansk-region-ukrainian-troops-used-ur-77-meteorit-mine-clearing-vehicle-against-russian-invaders>

MILUA22 Armed Forces captured upgraded Russian BMP-2M in Kherson Region. Militarynyi (mil.in.ua), 2022. <https://mil.in.ua/en/news/armed-forces-captured-upgraded-russian-bmp-2m-in-kherson-region/>

MILUA23 Special Operations Forces operators seized russian Eleron UAV. Militarynyi (mil.in.ua), 2022. <https://mil.in.ua/en/news/special-operations-forces-operators-seized-russian-eleron-uav/>

MILUA24 Ukrainian National Guard destroys Russian armored group with rarely protected BMP-2. Militarynyi (mil.in.ua), 2024. <https://mil.in.ua/en/news/ukrainian-national-guard-destroys-russian-armored-group-with-rarely-protected-bmp-2>

MISS1 Anti-aircraft missile system 9K37 Buk-M1 (EN and RU pages). Missilery.info. <https://en.missilery.info/missile/bukm1>

MISS2 Зенитный ракетный комплекс средней дальности 9K317 «Бук-М2». Missilery.info. <https://missilery.info/missile/buk-2m>

MISS3 Зенитный ракетный комплекс средней дальности 9K317M «Бук-М3». Missilery.info. <https://missilery.info/missile/bukm3>

MISS4 Зенитный ракетный комплекс 2K12 «Куб». Missilery.info. <https://missilery.info/missile/kub>

MISS5 Зенитный ракетный комплекс «Тор-М2». Missilery.info. <https://missilery.info/missile/tor-m2>

MISS6 PC30 9K51 «Граб». missilery.info. <https://missilery.info/missile/grab>

MISS7 9K720 Iskander operational-tactical missile system. missilery.info. <https://en.missilery.info/missile/iscander>

MISS8 Universal anti-aircraft missile and gun system Pantsir-S1. missilery.info (Raketnaya tekhnika). <https://en.missilery.info/missile/panz>

MISS9 Anti-aircraft missile system S-300V / S-300VM Antey-2500. missilery.info. <https://en.missilery.info/missile/c300v>

MISS10 Multiple launch rocket system 9K58 Smerch. missilery.info (S. V. Gurov). <https://en.missilery.info/missile/smerch>

MISS11 Реактивная система залпового огня 9K515 «Торнадо-С». missilery.info (S. V. Gurov), 2023. <https://missilery.info/missile/tornado-s>

MISS12 Anti-aircraft missile system 9K333 Wasp. missilery.info. <https://en.missilery.info/missile/loasa>

MISS13 Anti-aircraft missile system Tor-M2E. missilery.info. <https://en.missilery.info/missile/tor-m2>

MISS14 Heavy flamethrower system TOS-1A. missilery.info.

<https://en.missilery.info/missile/tos-1a>

МК Беспилотники «Форпост» поставлены на Тихоокеанский флот. Moskovskiy Komsomolets, 2014–01–16.

<https://www.mk.ru/politics/army/news/2014/01/16/971083-bespilotniki-quotforpostquot-postavleni-na-tihookeanskiy-flot.html>

МК2 «Длинная рука» артиллерии: расчет «Торнадо-С» уничтожил скопление ВСУ. Moskovskiy Komsomolets, 2025–04–09.

<https://www.mk.ru/politics/2025/04/09/dlinnaya-ruka-artillerii-raschet-tornado-s-unichtozhil-skoplenie-vsu.html>

МК26 Tyulpany VS RF razbili podzemnye komandnye punkty VSU pod Kupyanskom. Moskovskiy Komsomolets, 2026–02–09.

<https://www.mk.ru/politics/2026/02/09/tyulpany-vs-rf-razbili-podzemnye-komandnye-punkty-vsu-pod-kupyanskom.html>

МОСР Decontamination System TZ-74. Czech Ministry of Defence.

<https://mocr.moc.gov.cz/scripts/detail.php?id=6066>

МОДМР Минобороны показало работу новейшего БПЛА «Молния-2» в Курской области. News.mail.ru, quoting Russian MoD, 2024–11–19.

<https://news.mail.ru/svo/63702601/>

МРЕР Searcher (overview). Military Periscope.

<https://www.militaryperiscope.com/weapons/unmanned/unmanned-aerial-vehicles/searcher/overview/>

МРЕР2 220-mm BM-22 Uragan MRL, overview. Military Periscope.

<https://www.militaryperiscope.com/weapons/munitions/multiple-rocket-launchers/220-mm-bm-22-uragan-mrl/overview/>

МРЕР3 GMZ-3 tracked minelayer, overview. Military Periscope.

<https://www.militaryperiscope.com/weapons/ground-combat-vehicles/combat-support/gmz-3-tracked-minelayer/overview/>

NAKED Rossiya ispytala morskoy udarnyy BPLA. Naked Science (citing TASS/Kalashnikov), 2021–08.

<https://naked-science.ru/article/tech/rossiya-ispytala-morskoy-udarnyy-bpla>

NAKED2 V Rossii vosstanovili bronomashinu Sudnogo dnya. Naked Science.

<https://naked-science.ru/article/tech/v-rossii-vosstanovili-bronomashinu-sudnogo-dnya>

NASHAYT БА3-6950 семейство «Основа». nashi-avto.ru.

<http://nashi-avto.ru/ru/bzkt/s/%D0%B1%D0%B0%D0%B7-6950.html>

NEWSRU Терминатор выпустил 400 снарядов по ВСУ. News.ru, 2023–07–11.

<https://news.ru/world/terminator-vypustil-400-snyadov-po-vsu-novosti-svo-na-utro-11-iyulya/>

NEWSRU2 Vskryli punkt upravleniya BPLA VSU: uspekhi VS RF k utru 1 marta. News.ru, 2026–03–01.

<https://news.ru/russia/vskryli-punkt-upravleniya-bpla-vsu-uspehi-vs-rf-k-utru-1-marta>

NGE Страшное оружие на тонкой леске. Novaya Gazeta Europe, 2025–09–25.

<https://novyagazeta.eu/articles/2025/09/25/strashnoe-oruzhie-na-tonkoi-leske>

NI The Russian Army just Received Its First New T-90M Proryv Tanks (Suci, The National Interest). Warrior Maven / The National Interest, 2020–04.

<https://warriormaven.com/land/the-russian-army-just-received-its-first-new-t-90m-proryv-tanks>

NIKIT БТР-МДМ: десантируемая Пакушка. nikitinskiy LiveJournal (repost), 2014.

<https://nikitinskiy.livejournal.com/240438.html>

OLUA Watch: border guards destroyed a rare radar station of the occupiers in Luhansk region. online.ua, 2026.

<https://news.online.ua/en/watch-border-guards-destroyed-a-rare-radar-station-of-the-occupiers-in-luhansk-region-90262/>

ORUZH 9S482M7 (PU-12M7) battery mobile control post. oruzhie.info.

<http://oruzhie.info/vojska-pvo/243-9s482m7-pu-12m7>

ORYX1 Attack On Europe: Documenting Russian Equipment Losses During The Russian Invasion Of Ukraine, section Artillery and Missile Support Vehicles. Oryx, continuously updated.

<https://www.oryxspioenkop.com/2022/02/attack-on-europe-documenting-equipment.html>

ORYX2 Destination Disaster: Russia's Failure At Hostomel Airport. Oryx, 2022–04.

<https://www.oryxspioenkop.com/2022/04/destination-disaster-russias-failure-at.html>

ORYX3 Hit or miss: Russian loitering munition strikes (Kub and Lancet-3[M]). Oryx, 2023–03.

<https://www.oryxspioenkop.com/2022/11/hit-or-miss-russian-loitering-munition.html>

OSCE Latest from the OSCE SMM to Ukraine, based on information received as of 19:30, 10 August 2018. OSCE, 2018–08–11.

<https://www.osce.org/special-monitoring-mission-to-ukraine/390236>

OSONE Многоножка из Муассы (Урал-532301). os1.ru, about 2003.

<http://www.os1.ru/article/7612-mnogonogka-iz-miassa-ural-532301>

OVC Preimushchestva sistemy minirovaniya «Zemledeliye» i ispolzovaniya eyo na SVO. Overclockers.ru (user blog).

<https://overclockers.ru/blog/Blog9/JACKa/show/101493/Preimushchestva-sistemy-minirovaniya-Zemledeliye-i-ispol-zovanie-eyo-na-SVO>

OVD Elements of Russia's Elite 1st Guards Tank Army Withdrawn From Ukraine. Overt Defense, 2022–03–29.

<https://overtdefense.com/2022/03/29/elements-of-russias-elite-1st-guards-tank-army-withdrawn-from-ukraine>

PAS Mursmank-BN HF EW Complex. Planesandstuff (blog), 2019–09–09.

<https://planesandstuff.wordpress.com/2019/09/09/mursmank-bn-hf-ew-complex/>

PERISC Central Military District Gets New Surveillance Radars. Military Periscope, 2019–01–10.

<https://www.militaryperiscope.com/news/central-military-district-gets-new-surveillance-radars>

PLANDIY Merlin-VR UAV: experimental Russian scout (plandiyy.com.ua). plandiyy.com.ua, 2026.

<https://plandiyy.com.ua/ru/merlin-bpla-eksperimentalnyi-rozvidnyk-rosiskoi-armii-ikakiy-stav-doroholu-zdobychchiliu-dlia-zsu/>

PMUL Russian artillery support vehicles (PRP-3 section). pmulchahy.com (hobby site).

http://www.pmulchahy.com/artillery_support_vehicles/russian_asv.htm

PRIME Новый ударный дрон "Провод" начал массово поступать в войска. Prime (RIA Novosti group), 2026–02–05.

<https://prime.ru/20260205/dron-867201482.html>

PRIME26 Raschety minometov уничтожили punkt dislokatsii VSU pod Kharkovom. PRIME (Rossiya Segodnya), 2026–01–17.

<https://prime.ru/20260117/vsu-866619242.html>

PRIZ Engineer-sapper battalion of the new-look brigade. prizyvnik.info (blog).

<https://www.prizyvnik.info/blogs/entry/3239-inzhenerno-sapernyy-batalyon-brigady-novogo-oblika/>

PRO Автоматизированная станция помех Р-330Ж. ОАО NVP Protek (manufacturer), via Wayback Machine, 2015–10–05.

<https://web.archive.org/web/20160413003527/http://www.protek-vrn.ru/production/avtomata-tizirovannaya-stantsiya-pomeh-r-330zh/>

PROF Уралвагонзавод отправил в войска партию БМПТ. Profile.ru, 2025–10.

<https://profile.ru/news/military/uralvagonzavod-otpravil-v-vojska-partiyu-bmpt-ranshe-ih-nazyvali-terminatorami-1762052/amp/>

PROF2 Какие беспилотники есть на вооружении у российской армии. Profile.ru, undated, c. 2019.

<https://profile.ru/military/kakie-bespilotniki-est-na-vooruzhenii-u-rossijskoj-armii-275319/amp/>

PROF3 Длинная рука Москвы: какие БПЛА использует Россия для ударов по тылам Украины. Profi (Russia), 2026–03–02.

<https://profile.ru/military/dlinnaya-ruka-moskvy-kakie-bpla-ispolzuet-rossiya-dlya-udarov-po-tylam-ukrainy-1827608/amp/>

PROFILE В трёх регионах России начались масштабные учения ВДВ. Profile.ru (citing RIA Novosti and MoD), 2021–02.

<https://profile.ru/news/protection/army/v-trex-regionax-rossii-nachalis-masshtabnye-ucheniya-vdv-644509/>

PVO Радиолокационная станция 96А6Е. pvo.guns.ru (Nevsky Bastion material), 2008–11–30.

<https://pvo.guns.ru/rtrv/966.htm>

RADTUT 1L271 (Radartutorial, saved mirror). Radartutorial.eu via I.xiffr mirror.

<https://i.xiffr/files/elec/radar/radartutorial-eu.save/19.karte/04.battle/karte036.en.html>

RAKUL Станция спутниковой связи Р-439-КУА. RusArmy.com, 2004–2010.

<http://www.rusarmy.com/svyaz/sputnik/r-439-kul.html>

RAMB Минобороны РФ сообщило об успешном применении дронов Lancet. Rambler News (quoting Russian MoD and RT), 2026–06–10.

<https://news.rambler.ru/army/56580923-minoborony-rf-soobshchilo-ob-ushpnom-primenenii-dronov-lantset/>

RAMBLER Вооружение КUB dosrochno postavili na SVO. Rambler (relaying TASS), c. 2025, undated page.

<https://finance.rambler.ru/army/55553441-boopriasy-kub-dosrochno-postavili-na-svo/>

RAMD1 Мобильная станция спутниковой связи Р-439-МД1.

RusArmy.com, 2004–2010.

<http://www.rusarmy.com/svyaz/sputnik/r-439-md1.html>

RAMD2 Станция спутниковой связи P-439-МД2. RusArmy.com, 2004-2010.

<http://www.rusarmy.com/svyaz/sputnik/r-439-md2.html>

RAMD3 Станция спутниковой связи P-439-МД3. RusArmy.com, 2004-2010.

<http://www.rusarmy.com/svyaz/sputnik/r-439-md3.html>

RAND The Future of the Russian Military (RR-3099-A), PDF pp. 73 and 76. RAND Corporation, 2019.

https://www.rand.org/content/dam/rand/pubs/research_reports/RR3000/RR3099/RAND_RR3099.pdf

RBASE Реактивная система залпового огня 9K51M Тornado-Г. Ракетная техника [rbase.new-factoria.ru / missilery.info], undated, after 2018.

<http://rbase.new-factoria.ru/missile/wobb/tornado-g/tornado-g.shtml>

RBC Украинские СМИ сообщили об участии псковского десанта в боях за Луганск. RBC, 2014-08.

<https://www.rbc.ru/news/570420f79a794760d3d40d9c>

RBTH UR-77: Clearing one landmine at a time. Russia Beyond, 2017-03-29.

https://www.rbth.com/multimedia/video/2017/03/29/ur-77-clearing-one-landmine-at-a-time_729866

RDPI Upgunning Artillery. Russian Defense Policy (blog), 2018-04-29.

<https://russiandefencepolicy.com/2018/04/29/upgunning-artillery/>

RDPI2 New airborne armor deliveries. Russian Defense Policy (blog), 2020-06-25.

<https://russiandefencepolicy.com/2020/06/25/new-airborne-armor-deliveries/>

RDPI3 Russian Defense Policy, 26 July 2021 post. Russian Defense Policy (blog), 2021-07-26.

<https://russiandefencepolicy.com/2021/07/26/>

RDPI4 Electronic warfare: myths and truth (translation of Aleksey Ramm, VPK, Oct 2015). russiandefencepolicy.com, 2015.

<https://russiandefencepolicy.com/tag/r-330zh/>

RDPI5 Iskander-M (tag archive). Russian Defense Policy blog, 2014 to 2017 posts.

<https://russiandefencepolicy.com/tag/iskander-m/>

RDPI6 Polyana-D4M1 (tag page; Ground Troops procurement 2011). Russian Defense Policy blog, 2011.

<https://russiandefencepolicy.com/tag/polyana-d4m1/>

RDPI7 Tag archive: Tornado-G (Motovilikhа 2019 GQZ report). Russian Defense Policy blog, 2019-12.

<https://russiandefencepolicy.com/tag/tornado-g/>

REACH Красуха-2 [IA269]. reachaim, LiveJournal, 2014-03-17.

<https://reachaim.livejournal.com/5097.html>

REDSTAR «Гиацинт-К» в жарком боксу. Krasnaya Zvezda (MoD newspaper), undated (2025 or later).

<http://redstar.ru/giatsint-k-v-zharkom-boksu/>

REFORM Russian Company Begins Supplying Belarus With Upgraded Supercam S350 Drones. Reform.news, 2026-06-17.

<https://reform.news/en/russian-company-begins-supplying-belarus-with-upgraded-supercam-s350-drones>

RG IT-специалист развернул в Туле производство FPV-дронов «Обод». Rossiyskaya Gazeta, 2025-04-01.

<https://rg.ru/2025/04/01/reg-cfo/tulskiy-privanik-kak-indikator-to-go-chto-dron-ne-ustarel.html>

RG2 Ognenniy schit: effektivnui rabotu kompleksa "Infauна" sniali na video. Rossiyskaya Gazeta, 2019-04-08.

<https://rg.ru/2019/04/08/ognenniy-schit-effektivnui-rabotu-kompleksa-infauна-sniali-na-video.html>

RG3 Chto nuzhno znat o боеприпасах Lantset. Rossiyskaya Gazeta, 2023-04-06.

<https://rg.ru/2023/04/06/ot-lanceta-gaubica-ne-spriachetsia.html>

RG4 C-350 в четыре раза повышает эффективность борьбы с крылатыми ракетами. Rossiyskaya Gazeta, 2020-04-10.

<https://rg.ru/2020/04/10/s-350-v-chetyre-raza-povyshayet-effektivnost-borby-s-krilatymi-raketami.html>

RG5 PC30 «Торнадо-С» уничтожила заглубленный командный пункт ВСУ под Доброполем. Rossiyskaya Gazeta, 2026-08-05.

<https://rg.ru/2026/08/05/reg-dnr/rzso-tornado-s-unichtozhila-zaglublenniy-komandnyj-punkt-vsu-pod-dobropolem.html>

RG6 Комплекс Стрелец-М помогает ВКС РФ поражать сложные цели ВСУ. Rossiyskaya Gazeta (from RIA Novosti), 2022-06-07.

<https://rg.ru/2022/06/07/kompleks-strelec-m-pomogaet-vks-rf-porazhat-slozhnye-celi-vsu.html>

RG7 Характеристики «летающего танка» Т-80Б8М рассекретили на «Армии-2020». Rossiyskaya Gazeta, 2020-08-23.

<https://rg.ru/2020/08/23/harakteristiki-letaiushchego-tank-a-t-80bvm-rassekretili-na-armii-2020.html>

RG8 Росгвардия применила незаметные для украинских войск беспилотники. Rossiyskaya Gazeta, 2022-04-27.

<https://rg.ru/2022/04/27/rosgvardiya-primenila-nezametnye-dlia-ukrainskikh-vojsk-bespiilotniki.html>

RG15 Подразделения Р36 на Дальнем Востоке получают беспилотники «Алеп-3» (Gavrilov). Rossiyskaya Gazeta, 2015-06-28.

<https://rg.ru/2015/06/28/leer-site.html>

RG16 Zglushat za gorizontom (Kompleks REB Murmansk-BN obedinili v ediniui sistem). Rossiyskaya Gazeta, 2016-11-14.

<https://rg.ru/2016/11/14/kompleksy-radioelektronnoy-borby-murmansk-bn-obedinili-v-ediniui-sistemu.html>

RG17 Novyy divizion na tyazhelykh minometakh poyavitsya v TsVO. Rossiyskaya Gazeta, 2017-07-26.

<https://rg.ru/2017/07/26/reg-urf/divizion-na-tyazhelykh-minometah.html>

RG19 Применение комплекса "Палантин-К" впервые показали на вуге. Rossiyskaya Gazeta, 2019-10-08.

<https://rg.ru/2019/10/08/primenenie-kompleksa-palantin-k-vpervye-pokazali-na-video.html>

RG20 Вторая жизнь: чем примечательна БМП-1АМ «Басурманин». Rossiyskaya Gazeta, 2020-08-28.

<https://rg.ru/2020/08/28/vtoraia-zhizn-chem-primechatelna-bmp-1am-basurmanin.html>

RG21 В 2021 году военные получат новые машины химвразбегки РХМ-8 и РХМ-9. Rossiyskaya Gazeta, 2021-01-05.

<https://rg.ru/2021/01/05/r-2021-godu-voennoye-poluchat-novye-mashiny-himrazvedki-rhm-8-i-rhm-9.html>

RG22 Razrushitel bunkerov: kak samokhodnye minometry Tyulpan unichtozhayut ukrepaiony VSU. Rossiyskaya Gazeta, 2022-11-18.

<https://rg.ru/2022/11/18/razrushitel-bunkerov-kak-samokhodnye-minometry-tyulpan-unichtozhaiut-ukrepaiony-vsu.html>

RG22B На учениях в ЗВО применили комплекс Р36 "Палантин-К". Rossiyskaya Gazeta, 2022-02-08.

<https://rg.ru/2022/02/08/reg-szfo/na-ucheniiah-v-zvo-primenili-kompleks-reb-palantin-k.html>

RG23 Бронированный «Басурманин» и БРМ-1К показали преимущество перед обычной БМП-1. Rossiyskaya Gazeta, 2023-09-14.

<https://rg.ru/2023/09/14/bronirovannyi-basurmanin-i-brm-1k-pokazali-preimushchestvo-pered-obychnoj-bmp-1.html>

RG24 «Коалиция-СВ» принята на вооружение и вышла на переобу. Rossiyskaya Gazeta, 2024-01-09.

<https://rg.ru/2024/01/09/koaliciya-prinita-na-vooruzhenie-i-vyshla-na-pereobuiu.html>

RG25 ЗПК Тор-М2 получил дополнительную броню и средства Р36. Rossiyskaya Gazeta, 2025-08-13.

<https://rg.ru/2025/08/13/zrk-tor-m2-poluchil-dopolnitelnuu-broniui-i-sredstva-reb.html>

RG152 Артиллеристы ЮВО получили новые комплексы радиоаолокационной разведки. Rossiyskaya Gazeta, 2015-03-11.

<https://www.rg.ru/2015/03/11/aistenok.html>

RG172 К 2020 году ВДВ получат более трехсот новых бронемашин. Rossiyskaya Gazeta, 2017-04-28.

<https://rg.ru/2017/04/28/reg-pfo/r-2020-godu-vdv-poluchat-bolee-trehstot-novykh-bronemashin.html>

RG173 На вооружение ЗВО поступил уникальный комплекс «Борисозлебск-2». Rossiyskaya Gazeta, 2017-04-01.

<https://rg.ru/2017/04/01/reg-cfo/borisozlebsk-2.html>

RG174 В Крыму военные развернули сверхмощный комплекс REB Murmansk. Rossiyskaya Gazeta, 2017-03-10.

<https://rg.ru/2017/03/10/reg-ufu/v-krymu-voennoye-ravzernuli-sverhmoshchnyy-kompleks-reb-murmansk.html>

RG175 Первый полк Тор-М2 пришел в войска. Rossiyskaya Gazeta, 2017-03-25.

<https://rg.ru/2017/03/25/pervyy-polk-tor-m2-prishel-v-vojska.html>

RG192 Арктический комплекс Тор-М2Дт заступил на боевое дежурство. Rossiyskaya Gazeta, 2019-11-30.

<https://rg.ru/2019/11/30/reg-szfo/arkicheskiy-kompleks-tor.html>

RG202 Новейший комплекс Р36 "Палантин" поступит на вооружение ВВО. Rossiyskaya Gazeta, 2020-07-27.

<https://rg.ru/2020/07/27/reg-dfo/noveishij-kompleks-reb-palantin-postupit-na-vooruzhenie-vvo.html>

RG212 В войска передан новейший комплекс разведки «Пенициллин». Rossiyskaya Gazeta, 2021-01-22.

<https://rg.ru/2021/01/22/r-vojska-peredan-noveishij-kompleks-razvedki-penicillin.html>

- RG213** Броневомобиль «Тайфун-ВДВ» приняли на снабжение ВДВ. Rossiyskaya Gazeta, 2021-08-04.
<https://rg.ru/2021/08/04/broneavtomobil-tajfun-adv-priniali-na-snabzhenie-adv.html>
- RG214** Новейшие «Торнадо-У» приняли участие в параде на авиабазе Хмеймим. Rossiyskaya Gazeta, 2021-05-12.
<https://rg.ru/2021/05/12/novejschie-tornado-u-priniali-uchastie-v-parade-na-aviabaze-hmeimim.html>
- RG222** ВВВ получат боевые машины «Басурманин». Rossiyskaya Gazeta, 2022-02-02.
<https://rg.ru/2022/02/02/reg-dfo/vvo-poluchit-boevye-mashiny-basurmanin.html>
- RG223** В Белгородской области будет развернут новый батальон радиоэлектронной борьбы. Rossiyskaya Gazeta, 2022-01-25.
<https://rg.ru/2022/01/25/na-granice-s-ukrainoj-ravnernut-noviy-batalon-radioelektronnoj-borby.html>
- RG224** Комплекс РЗБ «Палантин» подает средства связи штабам ВСУ. Rossiyskaya Gazeta, 2022-07-27.
<https://rg.ru/2022/07/27/ kompleks-reb-palantin-podavliaet-sredstva-svazi-shtabov-vsua.html>
- RG232** Командир комплекса Тор-М2 рассказал, как сбил с начала СВО 50 воздушных целей. Rossiyskaya Gazeta, 2023-11-24.
<https://rg.ru/2023/11/24/komandir-kompleksa-tor-m2-rasskazal-kak-sbil-s-nachala-svo-50-vozdushnyh-celej.html>
- RG242** На SVO применили новейшую версию дрона-камикадзе KUB. Rossiyskaya Gazeta, 2024-12-27.
<https://rg.ru/2024/12/27/novij-udarnyj-dron-primenen-na-fronte.html>
- RG243** Как комплекс артразведки «Генциаллин» вскрывает позиции артиллерии ВСУ. Rossiyskaya Gazeta, 2024-02-01.
<https://rg.ru/2024/02/01/otvetnyj-udar-po-zvuku-vystrela.html>
- RG244** Боицы расчета рассказали, в чем успех ракетного комплекса Тор-М2. Rossiyskaya Gazeta, 2024-09-13.
<https://rg.ru/2024/09/13/reg-cfo/bojcy-rascheta-rasskazali-v-chem-uspeh-raketnogo-kompleksa-tor-m2.html>
- RG245** Боицы ВС РФ показали преимущества бронированной кабины вездехода «Торнадо-У». Rossiyskaya Gazeta, 2024-12-06.
<https://rg.ru/2024/12/06/bojcy-vs-ff-pokazali-preimushchestva-bronirovannoj-kabiny-vezdehoda-tornado-u.html>
- RG2018** 'Leer-2' prikrыл tanki T-72B3 i bronetransportery BTR-82A. Rossiyskaya Gazeta, 2018-03-26.
<https://rg.ru/2018/03/26/leer-2-prikrыл-tanki-t-72b3-i-bronetransportery-btr-82a.html>
- RG2108** Специальный колесный разведчик РХМ-9 можно увидеть на Армии-2021. Rossiyskaya Gazeta, 2021-08-24.
<https://rg.ru/2021/08/24/specialnyj-kolesnyj-razvedchik-rhm-9-mozhno-videt-na-armii-2021.html>
- RGMAL** «Ростех» отправил в войска партию самоходных артиллерийских орудий 2С43 «Мальва». Rossiyskaya Gazeta, 2024-08-04.
<https://rg.ru/2024/08/04/gaibica-s-nezhnym-imenem-no-ochen-groznaia-i-podvizhnaja.html>
- RGMAL2** Новейшая САУ 2С43 «Мальва» замечена в зоне СВО. Rossiyskaya Gazeta, 2024-06-02.
<https://rg.ru/2024/06/02/novejschaja-sau-2s43-malva-zamechena-v-zone-svo.html>
- RGNEWS** Роботов-саперов «Уран-6» модернизировали по результатам применения в зоне СВО. Rossiyskaya Gazeta, 2025-03-07.
<https://rg.ru/2025/03/07/robotov-saperov-uran-6-modernizirovali-po-razultatam-primenenija-v-zone-svo.html>
- RGPAR** Новейшие российские САУ «Мальва» и «Гиацинт-К» впервые приняли участие в параде. Rossiyskaya Gazeta, 2025-05-09.
<https://rg.ru/2025/05/09/novejschie-rossijskie-sau-malva-i-giacint-k-vpervye-priniali-uchastie-v-parade.html>
- RGRU** Разведка на броне: в зоне СВО появились новые боевые машины БРМ-1К. Rossiyskaya Gazeta (state), 2023-02-14.
<https://rg.ru/2023/02/14/razvedka-na-brone-v-zone-svo-poyavilis-novye-boevye-mashiny-brm-1k.html>
- RGRU1** Ивановская дивизия ВДВ получила боевые машины БМД-4М и БТР-МДМ «Пакушка». Rossiyskaya Gazeta (state), 2021-07-26.
<https://rg.ru/2021/07/26/reg-cfo/ivanovskaja-divizia-adv-poluchila-boevye-mashiny-bmd-4m-i-btr-mdm-rakushka.html>
- RGRU2** Десантные БТР-МДМ «Пакушка» вооружили скорострельными «рогатками». Rossiyskaya Gazeta (state), 2021-09-28.
<https://rg.ru/2021/09/28/desantnye-btr-mdm-rakushka-vooruzhili-skorostrelnymi-rogakami.html>
- RIA** Расчет БПЛА уничтожил технику ВСУ почти на три миллиона долларов. RIA Novosti, 2023-06-04.
<https://ria.ru/20260304/spetsoperatsiya-2078346762.html>
- RIA2** Новейшие радиостанции P-166 впервые задействованы на учениях в ЦВО. RIA Novosti, 2012-09-18.
<https://ria.ru/20120918/753121813.html>
- RIA15** KRET в 2015 году перedal Vooruzhennym silam 9 kompleksov REB Moskva-1. RIA Novosti, 2015-12-25.
<https://ria.ru/20151225/1348750286.html>
- RIA22** Самоходные орудия «Гиацинт-С» уничтожили подразделение гаубиц М777. RIA Novosti, 2022-05-23.
<https://ria.ru/20220523/gaibitsy-1790076832.html>
- RIA23** В зону спецоперации прибыли модернизированные машины-разведчики БРМ-1К. RIA Novosti (state media), 2023-02-14.
<https://ria.ru/20230214/brm-1k-1851839705.html>
- RIA222** Боевые машины пехоты «Басурманин» поступят в ВВВ 8 февраля. RIA Novosti, 2022-02-02.
<https://ria.ru/20220202/vooruzheniya-1770605498.html>
- RIASV** Беспилотник «Таксион» — что это? Инфографика. RIA SV, 2022-11-08.
<https://riassv.ru/obshchestvo/2022/11/08/besplotnik-tahion-cto-eto-infografika/>
- RIAVRN** Первый в вооруженных силах комплекс радиоэлектронной борьбы поступил в Воронежскую область. RIA Voronezh, 2019-04-11.
<https://riavrn.ru/news/pervyy-v-vooruzhennykh-silakh-kompleks-radioelektronnoy-borby-postupil-v-voronezhskuyu-oblast/>
- ROBO1** Гранам-1 (Robotrends catalogue). Robotrends.
<https://robotrends.ru/roboedia/granat-1>
- ROBO4** Гранам-4 (Robotrends catalogue). Robotrends.
<https://robotrends.ru/roboedia/granat-4>
- ROBOT** ZALA 421-16E. Robotrends.ru, 2019 (news items to 2019-05).
<https://robotrends.ru/roboedia/zala-421-16e>
- ROBOT2** Застава (Robopedia). robotrends.ru, 2015.
<https://robotrends.ru/roboedia/zastava>
- ROE** Buk-M2E (9K317E). Rosoboronexport.
<https://web.archive.org/web/20251028012324/https://roe.ru/en/production/protivovozdushnaya-oborona/zentnye-raketnye-kompleksy-i-ustanovki/zentnye-raketnye-kompleksy-sredney-dальности/buk-m2e/>
- ROE2** Air Defence Systems Export Catalogue (PU-12M6, PU-12M7, 9S737M, Tor-M1 pages). Rosoboronexport, 2003.
<https://ebin.pub/air-defence-systems-export-catalogue.html>
- ROSTEC23** Ростех поставил вооруженным силам новую партию БМП-2М «Бережок». Ouzhie Rossii (www1.ru), Rostec / High-Precision Systems release, 2023-09-06.
<https://www1.ru/news/2023/09/06/rostec-postavil-vooruzhennym-silam-novuyu-partiyu-bmp-2m-berezok-usovershenstvovannye-boevye-mashiny.html>
- ROSTEC23** In demand in the Middle East: Rostec assessed the prospects of the Berezhok module with Kornets missiles. Ouzhie Rossii (www1.ru), Rostec statement, 2025-02-13.
<https://www1.ru/en/news/2025/02/13/vostrebovan-na-bliznem-vostoke-rostec-ocenil-perspektivy-modulya-berezok-s-raketami-kornet.html>
- RS** 96L6 всевысотный радиолокатор целеуказания. Russkaya Sila (русская-сила.рф).
<https://xn--7bb5ahjiaqd2mxn--plai/guide/army/r/96l6.shtml>
- RS19** Первый в Российской армии комплекс РЗБ «Палантин» гебютировал на учении под Воронежем. Krasnaya Zvezda, 2019-09.
<http://redstar.ru/perviy-v-rossijskoj-armii-kompleks-reb-palantin-debyutiroval-na-ucheniipod-voronezhem/>
- RSIG** The Wireless-Set-No19 Group, Russian equipment index (R161 entries). Royal Signals Amateur Radio Society.
<https://www.royalsignals.org.uk/Russian/index.html>
- RSTAR** Pod prikrытием Pantsirey. Krasnaya Zvezda (Russian MoD), 2020-02-10.
<http://redstar.ru/pod-prikrытием-pantsirei/>
- RSUPPLY** Russia transports tanks of the 1960s to the Ukrainian front. Railway Supply, citing CFTS, 2022-06-01.
<https://www.railway.supply/en/russia-transports-tanks-of-the-1960s-to-the-ukrainian-front/>
- RT** 96L6E Cheese Board. radartutorial.eu (Wolff), mirror copy, 2015.
<https://xlif.fr/files/lehc/radar/radartutorial-eu.save/19.kartei/02.surv/kartei044.en.html>
- RT2** Новые возможности: какими будут российские ударные и разведывательные БПЛА «Альтиус» и «Форпост». RT (Russian state media), 2020.
<https://russian.rt.com/russia/article/715411-rossiya-besplotniki-altius-forpost>
- RT3** Sistemnaya modernizatsiya: kak sovershenstvuet'sya semeystvo barzahiruyushchikh boepriglasov Lantset. RT (Russian state media), 2024 (after Army-2024, Aug 2024; exact date not on page).
<https://russian.rt.com/russia/article/1354523-lantset-modernizatsiya-zala-svo>

RT4 Inokhodets: combat capabilities of the Russian UAV. RT (Russian state media), 2022-03.
<https://russian.rt.com/russia/article/976629-inokhodets-orion-ukraina-specoperaciya/amp>

RT5 Чем уникален новый авиадесантируемый броневтомобиль «Таифун-ВДВ». RT (Russian state media), 2020.
<https://russian.rt.com/russia/article/858725-taifun-vidv-mashina-pushka-bronya>

RT18 «Тахон», «Орлан» и «Элерон»: зачем в ВДВ России создают подразделение беспилотной авиации. RT (Russian), 2018-05-29.
<https://russian.rt.com/russia/article/513759-vidv-bespesantniki-besplotniki-podrazdeleniya-sozdanijeliotniki>

RT21 Готовы к самым разным вызовам: новейшие машины РКБ3. RT (Russian), 2021.
<https://russian.rt.com/russia/article/844614-rhbz-mashina-rhm-tigr-taifun>

RTRU Podpolkovnik VS RF o minnykh polyakh i inzhenernoy tekhnike v zone SVO (RT interview with Lt Col O. Klyukov, relayed). Rambler / RT, 2023.
<https://news.rambler.ua/army/51313592-podpolkovnik-vs-rf-o-minnykh-polyah-i-inzhenernoy-tehnike-v-zone-svo/>

RUBR Russia deploys its only Terminator company to Sievierodonetsk - British intelligence. Rubryka, 2022-05-22.
<https://rubryka.com/en/2022/05/22/ru-brytanskij-razviditsi-povidomyly-pro-perekydannya-rosiyeyu-pid-syevyeronodetsk-svoeyi-yedynoi-grupy-z-terminatoramy>

RUMOD Модернизированный разведывательный комплекс «Зонарок-1М» (MoD news item). Russian Ministry of Defence (mil.ru).
<https://mil.ru/et/news/more.htm?id=12191174@egNews>

RUSA1 Mobile control posts PU-12M7, PU-12M6. RusArmy.com, 2004-2010.
https://www.rusarmy.com/pvo/pvo_vsk/su_pu-12m7_pu-12m6.html

RUSA2 Unified battery command post 9S737M; 9S737 Ranzhir. RusArmy.com, 2004-2010.
https://www.rusarmy.com/pvo/pvo_vsk/su_9s737m.html

RUSAR1 Автоматизированная система управления смешанной группировкой ЗРК 9C52M1 «Поляна-Д4М1». RusArmy.com, 2004-2010.
https://rusarmy.com/pvo/pvo_vsk/su_polyana-d4m1.html

RUSARMM Комплекс 9C52M «Поляна-Д4М». RusArmy.com, 2004-2010.
https://rusarmy.com/pvo/pvo_vsk/su_polyana-d4m.html

RUSARMY 152-мм пушка 2А36 Гауцинт-Б. RusArmy.com, 2004-2010.
https://rusarmy.com/brone/bau_2a36_giatsint-b.htm

RUSARMY2 Inzhenerno-razvedyatel'naya mashina IRM Zhuk. RusArmy.com.
https://www.rusarmy.com/brone/rzm_irm_zhuk.htm

RUSARMY3 Командно штабная машина P-1456M «Чаюка». RusArmy.com, 2004-2010.
https://www.rusarmy.com/svyaz/kshm/kshm_r-1456m.html

RUSARMY4 Komandno-shtabnaya mashina R-149BMR. RusArmy.com, 2010.
https://www.rusarmy.com/svyaz/kshm/kshm_r-149bmr.html

RUSARMY5 Мобильный пункт связи P-166. RusArmy.com, 2004-2010.
<http://www.rusarmy.com/svyaz/radio/r-166.html>

RUSI1 Defending Mother Russia's Skies (Withington). RUSI Commentary, 2022-07-13.
<https://www.rusi.org/explore-our-research/publications/commentary/defending-mother-russias-skies>

RUSI2 Tactical Developments During the Third Year of the Russo-Ukrainian War (Watling, Reynolds). RUSI, 2025-02.
<https://static.rusi.org/tactical-developments-third-year-russo-ukrainian-war-february-2025.pdf>

RUSI3 Russia's Iranian-Made UAVs: A Technical Profile (Uzi Rubin). RUSI, 2023-01-13.
<https://www.rusi.org/explore-our-research/publications/commentary/russias-iranian-made-uavs-technical-profile>

RUSI4 The Iskander-M and Iskander-K: A Technical Profile (Cranny-Evans, Kaushal). RUSI, 2022-08-08.

RUSI5 Mass Precision Strike: Designing UAV Complexes for Land Forces (Bronk, Watling). RUSI, 2024-04-11.
<https://static.rusi.org/mass-precision-strike-final.pdf>

RUSI6 Meatgrinder: Russian Tactics in the Second Year of Its Invasion of Ukraine (Watling, Reynolds). RUSI, 2023-05-19.
<https://static.rusi.org/403-SR-Russian-Tactics-web-final.pdf>

RUSIAD Modern Russian and Chinese Integrated Air Defence Systems (Bronk). RUSI, 2020.
<https://rusi.org>

RUSIPL Preliminary Lessons in Conventional Warfighting, Feb to Jul 2022, p. 15. RUSI, 2022-11.
<https://static.rusi.org/359-SR-Ukraine-Preliminary-Lessons-Feb-July-2022-web-final.pdf>

RUSTEAM Комплекс радиоэлектронной борьбы Борисоглебск-2. rus.team, n.d. (latest event 2019-01-10).
<https://rus.team/articles/kompleks-radioelektronnoy-borby-borisoglebsk-2>

RUWK152 1V152 (Russian Wikipedia mirror, dic.academic.ru). Wikipedia mirror, 2010.
<https://dic.academic.ru/dic.nsf/ruwiki/1415735>

RWW The Russian Way of War: Force Structure, Tactics, and Modernization of the Russian Ground Forces (Grau, Bartles), pp. 234, 251-252. FMSQ, 2016.
https://archive.org/details/The_Russian_Way_of_War_Force_Structure_Tactics_and_Moderniza

SAIDPVO ПАК «Зонарок-1М» в Алабино. saidpvo LiveJournal blog, reposting a militaryrussia.ru table, 2015.
<https://saidpvo.livejournal.com/418815.html>

SDN В полк РКБ3 ВВ0 поступили APC-14KM. Сделано у нас (blog), undated, about 2020.
<https://sdelanounas.ru/blogs/86040/>

SDU Комплект средств автоматизации «Барнаул-Т». Sdelano u nas (user blog), 2011-10-16.
<https://sdelanounas.ru/blogs/8245/>

SDU13 Kompleks 1L267 Moskva-1 s moduliami 1L265 / 1L266 voshel v serijnoe proizvodstvo. Sdelano u nas (user post), 2013-12-14.
<https://sdelanounas.ru/blogs/44869/>

SHEP Russia deploys 2S43 Malva to Kharkiv offensive. Shephard Media, 2024-06.
<https://www.shephardmedia.com/news/landwarfareint/russia-deploys-2s43-malva-to-kharkiv-offensive/>

SHEP2 Russian MoD to receive Eleron-3 UAS. Shephard Media, 2018-05.
<https://shephardmedia.com/news/uv-online/russian-mod-receive-eleron-3-uas/>

SHEP3 Central MD unit receives Aistyonok radar complex. Shephard Media, 2018-01-19.
<https://www.shephardmedia.com/news/landwarfareint/central-md-unit-receives-aistyonok-radar-complex/>

SHEPH Russian Army receives SNAR-10 M1 radar. Shephard Media, 2019-01.
<https://shephardmedia.com/news/digital-battlespace/russian-cmd-receives-portable-snar-10-m1-radar>

SLOVO Border Guard Phoenix unit destroys R-416G-MS in Luhansk region. Slovo i Dilo, 2026-03-26.
<https://www.slovoidilo.ua/2026/03/26/novyna/bezpeka/unikalna-czil-prykordonnky-obrizaly-zvyazok-okupantam-luhanshyny>

SNOB Kalashnikov predstavil samovrivyayushchiysya bespilotnik. Snob (quoting Rostec), 2019-02.
<https://snob.ru/news/172572/>

SPT Понтонно-мостовой парк ПМП. spstechnika.ru (surplus equipment dealer).
<https://spstechnika.ru/voennaya-tehnika/pontonno-mostovoj-park-pmp/>

STC15 Primeneniye BLA semeystva Orlan dlya aerasymochnykh rabot (conference slides). Special Technology Centre (STC), via robotrends.ru, 2015-06.
https://robotrends.ru/images/1540/467822/2015_STC_BPLA_ORLAN.pdf

STKT «Торнадо-У». Новый военный «Урал». st-kt.ru (Za Rulem reprint), 2015-2016.
<https://st-kt.ru/articles/%C2%ABtornado-%C2%BB-novyi-voennoy-%C2%ABural/%C2%BB>

STOL Пакетная гузль «Торнадо-С» и HIMARS (79th Gds Rocket Artillery Brigade). StolicaMedia, 2026-09-17.
<https://stolica.media.ru/news/2624426/>

STRASAM S-400 Triumf Air Defence System (Part 1). STRASAM, 2026-04.
<https://strasam.org/en/defense/aerospace-industry/s-400-triumf-air-defence-system-part-1-4079>

STUD1 Тактико-технические данные машины TMC-65. studfile.net (training manual).
<https://studfile.net/preview/553111/page/2/>

STUD2 APC-14, устройство и работа станци. studfile.net (training manual).
<https://studfile.net/preview/553066/page/2/>

TOPWAR7 Kompleksy REB Pole-21 v rossijskoj armii (English mirror). Voennoe obozrenie (topwar.ru), 2021-04-21.

<https://en.topwar.ru/182196-kompleksy-rieb-pole-21-v-rossijskoj-armii.html>

TOPWAR8 Smoke machine TDA-3. topwar.ru (English edition), 2021.

<https://en.topwar.ru/207753-dymovaja-mashina-tda-3.html>

TOPWAR9 Станция постановки помех P-330Ж «Житель». Военное обозрение (topwar.ru), R. Skomorokhov, 2016-07-26.

<https://web.archive.org/web/2017/https://topwar.ru/98467-stanciya-postanovki-pomeh-r-330zh-zhitel.html>

TOPWAR10 P-330Ж «Житель». Возвращаясь к написанному.

Военное обозрение (topwar.ru), R. Skomorokhov, 2017-11-01.

<https://web.archive.org/web/2019/https://topwar.ru/128607-r-330zh-zhitel-vozvraschayae-k-napisannomu.html>

TRENCH Russia is desperate for tanks. Is it desperate enough to use T-64s? (Axe/Jompy lineage). The Trench Art, Substack, 2025-10.

<https://www.trenchart.us/p/russia-is-desperate-for-tanks-is>

TRENCH2 The ground shakes in eastern Ukraine (Trench Art, Axe). Trench Art (Substack), 2025.

<https://www.trenchart.us/p/the-ground-shakes-in-eastern-ukraine>

TRENCH3 Desperate to cross the Vovcha (Trench Art excerpt). Trench Art / Euromaidan Press (Axe), 2026-06.

<https://www.trenchart.us/p/desperate-to-cross-the-vovcha-the>

TRUCK Podlet 1K (2015). truck-encyclopedia.com.

<https://truck-encyclopedia.com/modern/russia/podlet-1k.php>

TSAMTO В отбельную инженерную бригаду в Башкортостане поступили две машины разграждения IMP-3М на базе танка Т-90. ЦАМТО (Tsamto), citing the Central Military District press service, 2018-11-16.

<http://www.armstrade.org/includes/periodics/news/2018/11/6/100049656/detail.shtml>

TSN RF could strike Europe with Gerberas from sea containers: details of the threat. TSN (tsn.ua), 2026 (page updated, after 2025-09).

<https://tsn.ua/ru/zbroiya/chto-takoe-gerbera-rossiyskiy-dron-koryuyet-mozhet-zapusksya-s-korabley-i-ugrozhat-nato-i-evrope-3174659.html>

TV24 Orlan-10: kharakteristiki drona, tsena, dalnost. 24 Kanal (24tv.ua), 2026.

https://24tv.ua/military/ru/orlan-10-kharakteristiki-drona-cena-dalnost-skorost-i-jeffektivnost_n3044152

TWZ1 Runddown of Russia's arsenal of artillery. The War Zone, 2022-02.

<https://www.twz.com/44579/this-is-russias-much-feared-artillery-arsenal-that-could-wreak-havoc-on-ukraine>

TWZ2 One Of Russia's Newest Air Defense Systems Has Been Captured By Ukraine. The War Zone (Trevithick), 2022-03.

<https://www.twz.com/44981/ukraine-captured-one-of-russias-newest-air-defense-systems>

TWZ3 Hasta La Vista, Baby! Ukraine Kills Its First Russian Terminator Combat Vehicle. The War Zone, 2023-02-09.

<https://www.twz.com/hasta-la-vista-baby-ukraine-kills-its-first-russian-terminator-combat-vehicle>

TWZ4 Russia's New Buk-M3 Air Defense Missile System Now Appears To Be In Ukraine (Newdick). The War Zone, 2022-03-22.

<https://www.twz.com/44866/russias-new-buk-m3-air-defense-missile-system-now-appears-to-be-in-ukraine>

TWZ5 New Russian Long-Range Drone Appears In Ukraine (Newdick). The War Zone, 2024-07-29.

<https://www.twz.com/news-features/new-russian-long-range-drone-appears-in-ukraine>

TWZ6 Ukraine Just Captured Part Of One Of Russia's Most Capable Electronic Warfare Systems. The War Zone (Joseph Trevithick), 2023-03-22.

<https://www.twz.com/44879/ukraine-just-captured-part-of-one-of-russias-most-capable-electronic-warfare-systems>

TWZ7 Ultra-Rare Russian Armored VIP Escape Vehicle Spotted In Ukraine. The War Zone, 2024-03.

<https://www.twz.com/news-features/ultra-rare-russian-armored-vip-escape-vehicle-spotted-in-ukraine>

TWZ8 Prized Russian Long Range Radar System Attacked In Eastern Ukraine. The War Zone, 2024-05-28.

<https://www.twz.com/news-features/prized-russian-long-range-radar-system-attacked-in-eastern-ukraine>

TWZ9 Russia's Advanced Podlet-1K Radar Captured By Anti-Assad Forces In Syria. The War Zone, 2024-12.

<https://www.twz.com/land/russias-advanced-podlet-1k-radar-captured-by-anti-assad-forces-in-syria>

TWZ10 Russia's Predator-Style Drone With Big Export Potential Has Launched Its First Missiles. The War Zone, 2020-12-28.

<https://twz.com/38446/russias-predator-style-drone-with-big-export-potential-has-launched-its-first-missiles>

TWZ11 Russia's Predator-Like Drone Is Now Shooting Down Other Drones. The War Zone, 2021-12.

<https://www.twz.com/43585/russias-predator-like-drone-is-now-shooting-down-other-drones>

TWZ12 Ukraine Appears To Have Destroyed Its First Russian S-350 Air Defense System. The War Zone, updated (reports the Feb 2024 loss as earlier).

<https://www.twz.com/land/ukraine-appears-to-have-destroyed-its-first-russian-s-350-air-defense-system>

TWZ13 Russian Troops Deploy Huge Smokescreen To Hide From Marauding Drones. The War Zone, 2024.

<https://www.twz.com/news-features/russian-troops-deploy-huge-smokescreen-to-hide-from-marauding-drones>

TWZ14 Russian mine clearing vehicle detonates in massive fireball from drone bomblet. The War Zone, 2023.

<https://www.twz.com/russian-mine-clearing-vehicle-detonates-in-massive-fireball-from-drone-bomblet>

TWZ15 Mine-clearing system seen being used as devastating urban artillery in Ukraine. The War Zone, 2023.

<https://www.twz.com/mine-clearing-system-seen-being-used-as-devastating-urban-artillery-in-ukraine>

U241 Ukraine blinds Russia's S-400 in Crimea, then sets its launcher ablaze. United24 Media, 2026-07-27.

<https://united24media.com/war-in-ukraine/ukraine-blinds-russias-s-400-in-crimea-then-sets-its-launcher-ablaze-21083>

U242 Ukrainian Drone Strike Destroys Rare RKhM-4 Chemical Recon Vehicle. UNITED24 Media, updated (1 April, year not stated).

<https://united24media.com/latest-news/ukrainian-drone-strike-destroys-rare-rkhm-4-chemical-recon-vehicle-video-7239>

U243 Russia Tests Molniya Strike Drones as Flying Couriers. UNITED24 Media, 2026-09-24.

<https://united24media.com/war-in-ukraine/russia-tests-molniya-strike-drones-as-flying-couriers-for-carrying-weapons-and-food-to-frontline-22837>

U244 Ukraine Blinds Russia's Air Defenses in Crimea: Podlet and Nebo-M Radars Destroyed. United24 Media, 2025-09-09.

<https://united24media.com/latest-news/ukraine-blinds-russias-air-defenses-in-crimea-podlet-and-nebo-m-radars-destroyed-11491>

U245 Ukraine Destroys \$20M Russian "Palantin" EW System, Opening Skies for Drone Operations. United24 Media, 2026-04.

<https://united24media.com/latest-news/ukraine-destroys-20m-russian-palantin-ew-system-opening-skies-for-drone-operations-17562>

U246 Partisans blow up Russian military communications station in Rostov (Spanish edition). United24 Media, 2025 (year not shown on page).

<https://united24media.com/es/latest-news/partisanos-dinamitan-una-estacion-de-comunicaciones-militares-rusa-en-rostov-en-un-sabotaje-de-ataque-a-largo-alcance-13085>

U247 Fuerzas especiales ucranianas destruyen un sistema Buk-M3 y un raro MLRS Uragan. United24 Media, 2025-05.

<https://united24media.com/es/latest-news/fuerzas-especiales-ucranianas-destruyen-un-sistema-ruso-de-defensa-antiaerea-buk-m3-de-45-millones-de-dolares-y-un-raro-mirs-uragan-8334>

UAHB Guidelines: combined arms units to combat Iranian-made strike UAVs Shahed-136 (Geran-2) and Lancet, VP 3-00(116)120. General Staff of the Armed Forces of Ukraine, 2023-03.

https://mil.ee/wp-content/uploads/2024/10/2023_Ukrainian-handbook-C-UAS-ON-IRANIAN-MADE-SHAHEED-136-GERAN-2-AND-LANCET-2-ASSOCIATION-AIRCRAFT.pdf

UASV New Russian Mobile Anti-Drone System. UAS Vision (from Sputnik/ Izvestia), 2017-01-06.

<https://www.uasvision.com/2017/01/06/new-russian-mobile-anti-drone-system/>

UASV2 Russian-Built Israeli Drone on Display. UAS Vision, 2018-05-29.

<https://www.uasvision.com/2018/05/29/russian-built-israeli-drone-on-display/>

UKDI Подразделения РФ вынуждены пересаживаться на бронетранспортеры 1950-х годов – британская разведка. Yevropeiska Pravda, 2022-07-09.

<https://www.eurointegration.com.ua/rus/news/2022/07/9/7142882/>

UKMOD Russia replenishing armored vehicle losses with old T-62 tanks – UK Intel. Ukrinform, quoting UK Ministry of Defence intelligence update, 2023-03.

<https://ukrinform.net/rubic-ato/3678721-ukraine-replenishing-armored-vehicle-losses-with-old-t62-tanks-uk-intel.html>

UKRIN Ukrainian forces destroy Russia's Leopard radar on left bank of Dnipro River. Ukrinform, 2024-01-17.

<https://ukrinform.net/rubic-ato/3814762-ukrainian-forces-destroy-russias-leopard-radar-on-left-bank-of-dnipro-river.html>

UKRIN2 InformNapalm: Drone downed in Donbas in service with twelve Russian army units. Ukrinform, 2020.
<https://www.ukrinform.net/rubric-defense/3000100-informnapalm-drone-downed-in-donbas-in-service-with-twelve-russian-army-units.html>

UKRINF Ukraine's special forces destroy two Russian EW systems near Avdiivka. Ukrinform, 2023-11-11.
<https://ukrinform.net/rubric-ato/3785708-ukraines-special-forces-destroy-two-russian-ew-systems-near-avdiivka.html>

UKRINF1 Ukrainian defenders destroy Russia's Pole-21 EW system. Ukrinform, 2023-11.
<https://www.ukrinform.net/rubric-ato/3782899-ukrainian-defenders-destroy-russias-pole-21-ew-system.html>

UKRINF2 Ukrainian drones strike Russian Murmansk-BN EW system in Crimea. Ukrinform, 2026-07-31.
<https://www.ukrinform.net/rubric-ato/4149676-ukrainian-drones-strike-russian-murmanskbn-ew-system-in-crimea.html>

UKRINF3 Ukraine forces destroy Russia's Palantin e-warfare system. Ukrinform, 2023-02.
<https://www.ukrinform.net/rubric-ato/3670026-ukraine-forces-destroy-russias-palantin-ewarfare-system.html>

UKRINF4 Ukrainian USFs hit Russia's air defense system, communications complex, drone command post. Ukrinform, 2026-05-14.
<https://ukrinform.net/rubric-ato/4123339-ukrainian-usfs-hit-russias-air-defense-system-communications-complex-drone-command-post.html>

UKRINF5 Army destroys Russian EW system, two drones, over 40 invaders in southern Ukraine. Ukrinform, 2022-05-01.
<https://ukrinform.net/rubric-ato/3471882-army-destroys-russian-ew-system-two-drones-over-40-invaders-in-southern-ukraine.html>

UKRINF22 Ukrainian forces destroy enemy electronic warfare system near Kakhovka HPP. Ukrinform, 2022-07-20.
<https://www.ukrinform.net/rubric-ato/353272-ukrainian-forces-destroy-enemy-electronic-warfare-system-near-kakhovka-hpp.html>

UNIAN В Україні могли знищити майже всі БМО-Т окупантів. UNIAN, с. 2025.
<https://www.unian.ua/weapons/v-ukrajini-mogli-znishchiti-mayzhe-vsi-bmo-t-okupantiv-ekspert-13273467.html>

UNIAN2 Drony Italmas: chto izvestno o novykh bespilotnikakh RF. UNIAN, 2023-10.
<https://www.unian.net/weapons/drony-italmas-chto-izvestno-o-novykh-bespilotnikakh-rf-12436263.html>

UNIAN3 РБ-341В Леер-3 - ЗСУ затрофеїли пігкисний російський комплекс РЕБ. UNIAN, 2023-02-04.
<https://www.unian.ua/war/rb-341v-leer-3-zsu-zatrofeili-pigkisniy-rosiyskiy-kompleks-reb-foto-12132975.html>

UNIAN4 ZSU destroyed Chayka command vehicle and Groza EW station. UNIAN.
<https://www.unian.ua/war/viynna-v-ukrajini-zsu-prodovzhuyut-znishchuvati-tehniku-okupantiv-12660045.html>

UNIAN5 Ukrainian troops captured R-149AKSh-1 command and staff vehicle. UNIAN, 2022-09-09.
<https://www.unian.net/war/ukrainskie-voennye-zahvatili-cennyi-trofej-komandno-shtabnuvu-mashinu-r-149aksh-1-okupantov-11972223.html>

UNIAN6 Танк Т-62МВ образца 2022 - лучший вариант Т-62, но есть важный нюанс. UNIAN, 2024-02-12.
<https://www.unian.net/weapons/tank-t-62mv-obrazca-2022-luchshiy-variant-t-62-no-est-vazhnyy-nyuans-12541386.html>

UPDIU Defence Intelligence of Ukraine reveals data on Russian Orion drone production. Ukrainska Pravda (via Yahoo News), 2025-11-05.
<https://www.yahoo.com/news/articles/ukraines-defence-intelligence-reveals-data-074800301.html>

UTIN Радиостанции полевых узлов связи (Utin, Makaterchik, Masseichik), учебно-методическое пособие. BSUIR Military Faculty, Minsk, 2018.
https://libelcod.bsuir.by/bitstream/123456789/33800/1/Utin_radio.pdf

UZST Технические характеристики автомобилей КамАЗ 5350. UZST (KamAZ dealer, operator-manual data).
<https://uzst.ru/poleznaya-informaciya/rukovodstvo-po-ekspluatsii-avtomobilej-kamaz/ekspluatatsionnyy-parametry/tehnicheskie-harakteristiki-avtomobilej-kamaz-5350>

VALKA Barnaul-T (RUS) automated air defence command system. valka.cz / armedconflicts.com.
<https://www.armedconflicts.com/Barnaul-T-1217337>

VALKA2 SOV - BMP-1K (velitel'ske vozidlo) / BMP-1KS. valka.cz.
<https://en.valka.cz/topic/view/64809>

VALKA3 SOV - R-412 (troposfericka radiova stanice). valka.cz (armedconflicts.com mirror), 2007 source book; 1983 manual.
<https://www.armedconflicts.com/-/t93909>

VALKACZ SOV - BMP-1KŠ / BMP-1KŠM (velitel'ske vozidlo), datasheet. valka.cz.
<https://www.valka.cz/DOV-BMP-1KS-BMP-1KSM-velitel'ske-vozidlo-t64809>

VIKNA БПЛА Форпост стоимостью \$7 млн: характеристика. Vikna, 2024-04.
<https://vikna.tv/ru/dlia-tebe/bpla-forpost-a-z-7 mln-unichtozhen-vsu-chto-izvestno-o-tth/>

VIKNA2 БПЛА Молния: что известно о новой российской разведке. Vikna (STB), 2025-05.
<https://vikna.tv/ru/dlia-tebe/bpla-molnija-chto-izvestno-o-novoi-rossijskoj-razrabotke/>

VKS Luzan A, 'Novoe kachestvo ... nekotorye predlozheniya po modernizatsii ZRS S-300V4', *Vozdushno-kosmicheskaya sfera* No. 3(92), pp. 56-61. *Vozdushno-kosmicheskaya sfera* (journal), 2017-09.
<https://www.socium-a.ru/public/flippingbook/vks20173/20/assets/basic-html/page60.html>

VM Железные саперы: первыми на минные поля пойдут бронированные машины. Vechernyaya Moskva (quoting Russian MoD information department), 2018-11-28.
<https://vm.ru/news/561956.html>

VM2 Lantset primeril teln'yashku. Vechernyaya Moskva, 2025-06.
<https://vm.ru/opinion/1241338-lantset-primeril-teln'yashku>

VMUS 152-мм пушка «Гуацшм-Б» 2А36. СССР. Victory Museum (Moscow).
<https://victorymuseum.ru/encyclopedia/technic/artilleriya/152-mm-pushka-giatsint-b-2a-36-sssr/>

VNTU Osoblyvosti vykorystannya radioreleynoho ta troposferneho zvyazku s'ohodni (Kakovkin). VNTU conference APOZBT, 2021.
<https://conferences.vntu.edu.ua/index.php/apozbt/apozbt2021/paper/download/13982/11846>

VPK1 БМП-1АМ «Басурманин». Как легендарной советской «копейке» подарили вторую жизнь (TASS, reposted by VPK.name). TASS, 2022-02.
https://vpk.name/news/581211_bmp-1am_basurmanin_kak_legendarnoi_sovetskoi_kopeike_podarili_vtoruyu_zhizn.html

VPK2 БТР-МДМ Пакушка-М. vpk.name library.
<https://vpk.name/library/f/btr-mdm.html>

VPK3 Self-propelled launcher 9P78-1. vpk.name.
<https://vpk.name/library/f/spu-9p78-1.html>

VPK4 Mobilnyy avtomatizirovannyi kompleks REB 'Leer-2' (copy of arms-expo.ru catalogue). vpk.name, 2021.
<https://vpk.name/library/f/leer-2.html>

VPK5 Moskva-1 (library entry). VPK.name.
<https://vpk.name/library/f/moscow-1.html>

VPK6 Novaya RLS Nebo-M postupila na voruzhenie voisk v Povolzhe (TASS via VPK.name). TASS, 2018-04-25.
https://vpk.name/news/1213609_novaya_radiolokatsionnaya_stanciya_nebo-m_postupila_na_voruzhenie_voisk_v_povolzhe.html

VPK7 Понтонный парк ПП-91 (ПП-2005). ВПК.name.
<https://vpk.name/library/f/pp-91.html>

VPK8 Поляна-Д4. VPK.name library.
<https://vpk.name/library/f/polyana-d.html>

VPK9 Battery command post 9S482M7 PU-12M7. VPK.name.
<https://vpk.name/library/f/pu-12m7.html>

VPK10 Автомобильная КВ/УКВ радиостанция Р-166 «Артек». VPK.name, undated (archived 2019).
<https://vpk.name/library/f/r-166.html>

VPK11 НПО «Стрела» показало машину радиолакационной разведки СНАР-10М1. VPK.name (Torwar lineage), 2014-08.
https://vpk.name/news/116377_npo_strela_pokazalo_mashinu_radiolokatsionnoi_razvedki_snar-10m1.html

VPK12 В Войска ЗВО в ближайшее время поступят станции наземной артиллерийской разведки «Пантера». VPK.name relaying TsAMTO, 2016-03-11.
https://vpk.name/news/151126_v_vojska_zvo_v_blizhaishhe_vremya_postupyat_stancii_nazemnoi_artilleriiskoi_razvedki_pantera.html

VPK13 Комплекс разведки, управления и связи Стрелец и Стрелец-М дая Ратник. vpk.name (Novosti VPK), undated, c2016-2018.
<https://vpk.name/library/f/strelec.html>

VPK14 Беспилотный летательный аппарат ZALA 421-16(E). VPK.name.
<https://vpk.name/library/f/421-16.html>

VRF1 Комплекс управления огнем артиллерии 1812М. Вооружение.ф.
<https://xn--b1agac3aeas4a.xn--p1ai/armament/1v12m-mashina/>

VRF2 Комплекс радиоэлектронной борьбы Р5-3015 Борисозлебск-2. вооружение.рф.
<https://xn--b1agac13aeas4a.xn--p1ai/armament/rb-3015-borisoglebsk-2/>

VRF3 Станция постановки помех Р-9346МВ. вооружение.рф.
<https://xn--b1agac13aeas4a.xn--p1ai/armament/rb-9346mv/>

VRF4 РХМ-4 (Army Guide). army-guide.com Russian mirror.
<https://xn--b1agac13aeas4a.xn--p1ai/armament/rhm-4/>

VRF5 РХМ-6 (Army Guide). army-guide.com Russian mirror.
<https://xn--b1agac13aeas4a.xn--p1ai/armament/rhm-6/>

VRF6 РХМ-8 (Army Guide). army-guide.com Russian mirror.
<https://xn--b1agac13aeas4a.xn--p1ai/armament/rhm-8/>

VRF7 Р5-5315 «Инфаун». vooruzhenie.rf.
<https://xn--b1agac13aeas4a.xn--p1ai/armament/rb-5315-infauna/>

VRF8 Р5-341В «Леер-3» (catalogue entry). вооружение.рф.
<https://xn--b1agac13aeas4a.xn--p1ai/armament/rb-341v-leer-3/>

VRF9 Понтонно-мостовой парк ПМП: обзор. Вооружение.рф.
<https://xn--b1agac13aeas4a.xn--p1ai/armament/pmp/>

VRF10 Автоматизированная система управления Поляна-Д4. вооружение.рф.
<https://xn--b1agac13aeas4a.xn--p1ai/armament/polyana-d4/>

VRF11 Дымовая машина ТДА-2К: обзор характеристик. armyguide (xn--b1agac13aeas4a.xn--p1ai).
<https://xn--b1agac13aeas4a.xn--p1ai/armament/tda-2k/>

VRF12 Электрический беспилотный летательный аппарат Тахюн. xn--b1agac13aeas4a.xn--p1ai (armament catalogue).
<https://xn--b1agac13aeas4a.xn--p1ai/armament/tahyon/>

VRF13 Р-330Ж «Житель». vooruzhenie.rf (вооружение.рф), Russian reference site.
<https://xn--b1agac13aeas4a.xn--p1ai/armament/r-330zh-zhite/>

W1A Раскрыты характеристики новейшего ударного БПЛА «Молиния-2». www1.ru, quoting Russian MoD, 2024-12-05.
<https://www1.ru/news/2024/12/05/raskrytye-kharakteristiki-noveisogo-udarnogo-bpla-molniiia-2-okazalsia-proshhe-lanceta.html>

W1A2 Glaza i ushi voennykh: v Minoborony RF nazvali preimushchestva dronov Supercam S350. www1.ru, 2024-07-14.
<https://www1.ru/news/2024/07/14/glaza-i-usi-voennykh-v-minoborony-rf-nazvali-preimushchestva-dronov-razvedchikov-supercam-s350.html>

W1A3 Ustroit kontrbatareinyu vzbukhu: chem khorosha perspektivnaya RSZO Uragan-1M. www1.ru, 2024-10-08.
<https://www1.ru/news/2024/10/08/ustroit-kontrbatareinyu-vzbukhu-chem-khoroша-perspektivnaia-rszo-uragan-1m.html>

W1B Крылатая смерть: раскрыта дальность полёта дрона-камикадзе «Молиния-2». www1.ru, quoting Russian MoD press service, 2025-01-28.
<https://www1.ru/news/2025/01/28/krylataya-smert-raskryta-sumassedsaia-dalnost-poleta-noveisogo-drona-kanikadze-molniiia-2.html>

W1B2 Новое поколение воздушной разведки: Supercam S350 zadeistvovali v otslezhivanii manevrov. www1.ru, 2024-08-26.
<https://www1.ru/news/2024/08/26/novoe-pokolenie-vozdusnoi-razvedki-supercam-s350-zadeistvovali-v-otslezhivanii-manevrov-protivnika.amp.html>

W1C Военные показали работу дрона-камикадзе «Молиния-2». www1.ru, quoting Russian MoD, 2025-02-03.
<https://www1.ru/news/2025/02/03/voennye-pokazali-rabotu-preodolevavshhego-pomekhi-sredstv-reb-drona-kanikadze-molniiia-2.html>

W1C2 Sovsem strakh poterai: armiya Rossii svela poteri dronov Supercam pochti k nulyu. www1.ru, 2024-10-25.
<https://www1.ru/news/2024/10/25/sovsem-strax-poterai-armiiia-rossii-svela-poteri-dronov-razvedchikov-supercam-pochti-k-nuliu.amp.html>

W1C3 Brosit vyzov amerikanskoj HIMARS: noveishaya RSZO Uragan-1M. www1.ru, 2024-10-28.
<https://www1.ru/news/2024/10/28/brosit-vyzov-amerikanskoi-himars-noveisaia-rszo-uragan-1m-ustroit-kontrbatareinyu-vzbukhu.html>

W1D Раскрыты официальные характеристики дрона «Молиния-2». www1.ru, quoting Russian MoD, 2025-05-02.
<https://www1.ru/news/2025/05/02/raskrytye-ofitsialnye-kharakteristiki-drona-molniiia-2-sostoit-iz-aluminiyevykh-trub-i-plastika.html>

W1D2 Ustoichiv k REB, rabotaet kruglye sutki: chto mozhet novyi bespilotnik Superkam. www1.ru, 2024-01-27.
<https://www1.ru/news/2024/01/27/ustoiichiv-k-reb-rabotaet-kruglye-sutki-cto-mozhet-novyi-rossiiskii-bespilotnik-supercam.amp.html>

W1E Gaubitsu 2S3 Akatsiya nachali ispolzovat v svyazke s razvedronom Supercam. www1.ru, 2024-08-15.
<https://www1.ru/news/2024/08/15/cvetok-kalibra-152-mm-gaubitsu-2s3-akacia-nachali-ispolzovat-v-sviazke-s-razvedronom-supercam.html>

W1E2 Poyavilis pervye kadry boevogo primeneniya noveishei RSZO Uragan-1M (English edition). www1.ru, 2025-02-06.
<https://www1.ru/en/news/2025/02/06/poyavilis-pervye-kadry-boevogo-primeneniya-noveisei-rszo-uragan-1m.amp.html>

W1F Supercam S350 corrected the destruction of a drone control point. www1.ru, 2026-03-18.
<https://www1.ru/en/news/2026/03/18/bespilotnik-supercam-s350-skorrektiroval-unichtozhenie-punkta-upravleniia-dronami.amp.html>

W1G Rossiiskie voennye poluchat prodvinutuyu versiyu Supercam S350. www1.ru, 2025-05-22.
<https://www1.ru/en/news/2025/05/22/rossiiskie-voennye-poluchat-prodvinutuyu-versiyu-bespilotnika-razvedchika-supercam-s350.amp.html>

W1H Supercam S350 records hundreds of targets at the front. www1.ru, 2026-03-22.
<https://www1.ru/en/news/2026/03/22/hpla-supercam-s350-fiksiрует-sotni-celej-na-pereodovoi-operator-nazval-kameru-odnoi-iz-lucsic-sredi-analogov.html>

W1I Nazvana glavnaya modifikatsiya drona Supercam S350. www1.ru, 2025-01-20.
<https://www1.ru/news/2025/01/20/nazvana-glavnaia-modifikatsiia-drona-supercam-s350-novoe-pokolenie-vozdusnoi-razvedki.html>

W1J Protivniku ne ostavlyаем shansov: rossiiskie voennye ispolzuyut bespilotniki Superkam. www1.ru, 2024-04-29.
<https://www1.ru/news/2024/04/29/protivniku-ne-ostavlyаем-sansov-rossiiskie-voennye-ispolzuyut-bespilotniki-supercam-dlia-tocnoi-ramp.html>

W1KVN Передовые дроны «Князь Вандал Новгородский» с тепловизором вышли в серийное производство. www1.ru, 2025-01-14.
<https://www1.ru/news/2025/01/14/pereodovye-drony-kniaz-vandal-novgorodskii-s-teplovizorom-vysli-v-seriinoe-proizvodstvo.html>

W1O30 Lazernaya ukazka Orlana navodit artilleriyu i vysokotochnye snaryady. www1.ru, 2024-11-20.
<https://www1.ru/news/2024/11/20/lazernaia-ukazka-orlana-navodit-artilleriyu-i-vysokotochnye-snaryady-bpla-pokazal-vysokouye-effektivnos.amp.html>

W1PROV Новый оптоволоконный дрон-камикадзе «Провод» дальностью 30 км ушел в серийное производство. www1.ru, 2026-02-05.
<https://www1.ru/news/2026/02/05/novyi-optovokonnii-dron-kanikadze-provod-dalnosti-30-km-usel-v-seriinoe-proizvodstvo.html>

W1R РСЗО 9К57 «Ураган» на новом шасси БАЗ-69092 поступила в серийное производство. www1.ru (news aggregator), 2026-02-24.
<https://www1.ru/news/2026/02/24/rszo-9k57-uragan-na-novom-shassi-baz-69092-postupila-v-rossiiskuiu-armiiu.amp.html>

W1RU Модернизированные БМД-2М с отделением «Берег» впервые поставил Ростех. www1.ru, 2026-05-29.
<https://www1.ru/news/2026/05/29/modernizirovannye-bmd-2m-s-otdeleniem-bereg-vpervye-postavil-rostex.amp.html>

W1RU2 Неуловимая птичка: российский FPV-дрон BT-40. www1.ru, 2024-07-15.
<https://www1.ru/news/2024/07/15/neulovimaia-pticka-rossiiskii-fvp-dron-vt-40-stal-kosmarom-dlia-tankov-protivnika-na-pereodovoi.html>

W1RU25 Merlin with 90x zoom directs artillery (www1.ru). www1.ru, 2025-02-19.
<https://www1.ru/news/2025/02/19/merlin-s-90-kratnym-zumom-navodit-artilleriyu-raskrytye-kharakteristiki-noveisogo-rossiiskogo-bpla.amp.html>

W1RU26 Jams hunters and flies 150 km without losing link (www1.ru). www1.ru, 2026-05-20.
<https://www1.ru/news/2026/05/20/gusit-oxotnikov-i-letit-150-km-bez-poteri-svazi-nazvaniy-polnye-kharakteristiki-drona-merlin-vp.html>

W171 Tselemekazanie za minuty: Orlan-10 zakryl petlyu mezhdu obnaruzheniem tseli i artilleriyskim udarom. www1.ru, 2026-04-04.
<https://www1.ru/news/2026/04/04/celeukazanie-za-minuty-orlan-10-zakryl-petliu-mezdu-obnaruzheniem-celi-i-artilleriyskim-udarom.html>

WEG1 ODIN WEG: 2S35 Koalitsiya-SV and 2S43 Malva Russian 152mm SPH (local extract). US Army TRADOC, 2024-12-12 (2S35); 2026-02-13 (2S43).
[https://odin.tradoc.army.mil/WEG/Asset/2S35_Koalitsiya-SV_Russian_152mm_Self-Propelled_Howitzer_\[SPH\]](https://odin.tradoc.army.mil/WEG/Asset/2S35_Koalitsiya-SV_Russian_152mm_Self-Propelled_Howitzer_[SPH])

WEG2 2S4 Tyulpan (M-1975) Russian 240mm Self-Propelled Heavy Mortar (local extract). US Army TRADOC, ODIN WEG, 2025-05-06.
[https://odin.tradoc.army.mil/WEG/Asset/2S4_Tyulpan_\(M-1975\)_Russian_240mm_Self-Propelled_Heavy_Mortar](https://odin.tradoc.army.mil/WEG/Asset/2S4_Tyulpan_(M-1975)_Russian_240mm_Self-Propelled_Heavy_Mortar)

WEG3 2S5 Giatsint-S Russian 152mm Self-Propelled Gun (SPG), WEG extract. US Army TRADOC, 2025-03-11.
[https://odin.tradoc.army.mil/WEG/Asset/2S5_Giatsint-S_Russian_152mm_Self-Propelled_Gun_\(SPG\)](https://odin.tradoc.army.mil/WEG/Asset/2S5_Giatsint-S_Russian_152mm_Self-Propelled_Gun_(SPG))

WEG4 2S7 Pion [M1975] Russian 203mm SPG, WEG extract. US Army TRADOC, 2025-04-22.

[https://odin.tradoc.army.mil/WEG/Asset/2S7_Pion_\[M1975\]_Russian_203mm_Self-Propelled_Gun_\[SPG\]](https://odin.tradoc.army.mil/WEG/Asset/2S7_Pion_[M1975]_Russian_203mm_Self-Propelled_Gun_[SPG])

WEG5 96L6E [Cheese Board] Russian Early-Warning and Acquisition Radar [WEG extract]. US Army TRADOC ODIN, 2024-10-23.

[https://odin.tradoc.army.mil/WEG/Asset/96L6E_\[Cheese_Board\]_Russian_Early-Warning_and_Acquisition_Radar](https://odin.tradoc.army.mil/WEG/Asset/96L6E_[Cheese_Board]_Russian_Early-Warning_and_Acquisition_Radar)

WEG6 WEG: ARK-1M RYS Russian Artillery Locating Radar [local extract, last modified 2024-10-22]. US Army TRADOC, 2024-10-22.

https://odin.tradoc.army.mil/WEG/Asset/ARK-1M_RYS_Russian_Artillery_Locating_Radar

WEG7 Barnaul-T Russian Air Defense Command and Control System. US Army TRADOC, ODIN WEG, 2025-06-10.

https://odin.tradoc.army.mil/WEG/Asset/Barnaul-T_Russian_Air_Defense_Command_and_Control_System

WEG8 WEG online, BAT-2 Russian Combat Engineer Vehicle [local extract]. TRADOC, 2024-10-21.

https://odin.tradoc.army.mil/WEG/Asset/WEG/BAT-2_Russian_Combat_Engineer_Vehicle

WEG9 Worldwide Equipment Guide: BMD-1 and BMD-2 Russian Airborne Amphibious IFV [local extract]. US Army TRADOC (ODIN), 2024-08-28 (BMD-1), 2025-03-03 (BMD-2).

[https://odin.tradoc.army.mil/WEG/Asset/BMD-2_Russian_Airborne_Amphibious_Infantry_Fighting_Vehicle_\[IFV\]](https://odin.tradoc.army.mil/WEG/Asset/BMD-2_Russian_Airborne_Amphibious_Infantry_Fighting_Vehicle_[IFV])

WEG10 Worldwide Equipment Guide: BMD-4 Russian Airborne Amphibious IFV [local extract, page modified 2024-08-28]. US Army TRADOC, 2024-08-28.

[https://odin.tradoc.army.mil/WEG/Asset/BMD-4_Russian_Airborne_Amphibious_Infantry_Fighting_Vehicle_\[IFV\]](https://odin.tradoc.army.mil/WEG/Asset/BMD-4_Russian_Airborne_Amphibious_Infantry_Fighting_Vehicle_[IFV])

WEG11 BMP-1 Russian Amphibious Infantry Fighting Vehicle (IFV), ODIN WEG [local extract pilot/weg_extract/bmp1.json]. US Army TRADOC, 2025-03-03.

[https://odin.tradoc.army.mil/WEG/Asset/BMP-1_Russian_Amphibious_Infantry_Fighting_Vehicle_\[IFV\]](https://odin.tradoc.army.mil/WEG/Asset/BMP-1_Russian_Amphibious_Infantry_Fighting_Vehicle_[IFV])

WEG12 WEG extract: BMP-1KSh (BMP M1978), last modified 2024-08-07. US Army TRADOC, ODIN Worldwide Equipment Guide, 2024-08-07.

[https://odin.tradoc.army.mil/WEG/Asset/BMP-1KSh_\[BMP_M1978\]_Russian_Amphibious_Command_Post_and_Staff_Infantry_Fighting_Vehicle_\[IFV\]](https://odin.tradoc.army.mil/WEG/Asset/BMP-1KSh_[BMP_M1978]_Russian_Amphibious_Command_Post_and_Staff_Infantry_Fighting_Vehicle_[IFV])

WEG13 BMP-2 Russian Amphibious Infantry Fighting Vehicle (IFV). US Army TRADOC, ODIN WEG, 2025-03-03.

[https://odin.tradoc.army.mil/WEG/Asset/BMP-2_Russian_Amphibious_Infantry_Fighting_Vehicle_\[IFV\]](https://odin.tradoc.army.mil/WEG/Asset/BMP-2_Russian_Amphibious_Infantry_Fighting_Vehicle_[IFV])

WEG14 BMPT Terminator Russian Heavy Fire Support Combat Vehicle; BMT-72 Terminator 2 Russian Tank Support Armored Fighting Vehicle. US Army TRADOC, ODIN WEG, 2024-08-29 and 2025-01-13.

https://odin.tradoc.army.mil/WEG/Asset/BMPT_Terminator_Russian_Heavy_Fire_Support_Combat_Vehicle

WEG15 Worldwide Equipment Guide: BMR-3M Russian Mine Clearing Vehicle [local extract]. US Army TRADOC, 2024-10-16.

https://odin.tradoc.army.mil/WEG/Asset/BMR-3M_Russian_Mine_Clearing_Vehicle

WEG16 Borisoglebsk-2 (RB-301B) Russian Amphibious Multipurpose Jamming Complex. US Army TRADOC, ODIN WEG, 2025-06-05.

[https://odin.tradoc.army.mil/WEG/Asset/Borisoglebsk-2_\[RB-301B\]_Russian_Amphibious_Multipurpose_Jamming_Complex](https://odin.tradoc.army.mil/WEG/Asset/Borisoglebsk-2_[RB-301B]_Russian_Amphibious_Multipurpose_Jamming_Complex)

WEG17 BRDM-2 Russian 4x4 Amphibious Armored Scout Car [Worldwide Equipment Guide online, local extract pilot/weg_extract/brdm2.json]. US Army TRADOC, 2025-03-03.

https://odin.tradoc.army.mil/WEG/Asset/BRDM-2_Russian_4x4_Amphibious_Armored_Scout_Car

WEG18 Worldwide Equipment Guide: BREM-1, BREM-2, BREM-D, RM-G [local extract]. US Army TRADOC, 2024-10-15.

https://odin.tradoc.army.mil/WEG/Asset/BREM-1_Russian_Armored_Recovery_Vehicle

WEG19 Worldwide Equipment Guide: IMR-2 Russian Combat Engineering Vehicle [local extract]. US Army TRADOC, 2024-10-16.

https://odin.tradoc.army.mil/WEG/Asset/IMR-2_Russian_Combat_Engineering_Vehicle

WEG20 WEG extract: BRM-1K Russian Amphibious Combat Reconnaissance Vehicle (CRV). TRADOC Worldwide Equipment Guide (ODIN), 2025-03-03.

[https://odin.tradoc.army.mil/WEG/Asset/BRM-1K_Russian_Amphibious_Combat_Reconnaissance_Vehicle_\[CRV\]](https://odin.tradoc.army.mil/WEG/Asset/BRM-1K_Russian_Amphibious_Combat_Reconnaissance_Vehicle_[CRV])

WEG21 WEG online: BTR-90 [GAZ-5923] Russian 8x8 Amphibious APC [local extract pilot/weg_extract/btr-90.json]. US Army TRADOC, 2025-12-10.

[https://odin.tradoc.army.mil/WEG/Asset/BTR-90_\[GAZ-5923\]_Russian_8x8_Amphibious_Armored_Personnel_Carrier_\[APC\]](https://odin.tradoc.army.mil/WEG/Asset/BTR-90_[GAZ-5923]_Russian_8x8_Amphibious_Armored_Personnel_Carrier_[APC])

WEG22 WEG: BTR-D [BMD M1979] Airborne APC [local extract pilot/weg_extract/btr-d.json]. US Army TRADOC, 2025-03-05.

[https://odin.tradoc.army.mil/WEG/Asset/BTR-D_\[BMD_M1979\]_Russian_Amphibious_Airborne_Armored_Personnel_Carrier_\[APC\]](https://odin.tradoc.army.mil/WEG/Asset/BTR-D_[BMD_M1979]_Russian_Amphibious_Airborne_Armored_Personnel_Carrier_[APC])

WEG23 WEG extract: BTR-MD Rakushka [Russian Amphibious Airborne APC]. TRADOC WEG, 2024-08-07.

[https://odin.tradoc.army.mil/WEG/Asset/BTR-MD_Rakushka_Russian_Amphibious_Airborne_Armored_Personnel_Carrier_\[APC\]](https://odin.tradoc.army.mil/WEG/Asset/BTR-MD_Rakushka_Russian_Amphibious_Airborne_Armored_Personnel_Carrier_[APC])

WEG24 Worldwide Equipment Guide: BTR-70 [BTR M1978] and BTR-80 Russian 8x8 Amphibious APC [local extract]. US Army TRADOC (ODIN), 2025-03-05.

[https://odin.tradoc.army.mil/WEG/Asset/BTR-80_Russian_8x8_Amphibious_Armored_Personnel_Carrier_\[APC\]](https://odin.tradoc.army.mil/WEG/Asset/BTR-80_Russian_8x8_Amphibious_Armored_Personnel_Carrier_[APC])

WEG25 9K37 Buk-M1 [SA-11 Gadfly] Russian Medium-Range Surface-to-Air Missile System. US Army TRADOC, ODIN Worldwide Equipment Guide, 2025-06-10.

[https://odin.tradoc.army.mil/WEG/Asset/9K37_Buk-M1_\[SA-11_Gadfly\]_Russian_Medium-Range_Surface-to-Air_Missile_System](https://odin.tradoc.army.mil/WEG/Asset/9K37_Buk-M1_[SA-11_Gadfly]_Russian_Medium-Range_Surface-to-Air_Missile_System)

WEG26 Buk-M2 [SA-17 Grizzly] Russian Medium-Range Air Defense Missile System. US Army TRADOC, ODIN Worldwide Equipment Guide, 2025-06-10.

[https://odin.tradoc.army.mil/WEG/Asset/Buk-M2_\[SA-17_Grizzly\]_Russian_Medium-Range_Air_Defense_Missile_System](https://odin.tradoc.army.mil/WEG/Asset/Buk-M2_[SA-17_Grizzly]_Russian_Medium-Range_Air_Defense_Missile_System)

WEG27 9K317M Buk-M3 [SA-27] Russian Medium-Range Air Defense Missile System. US Army TRADOC, ODIN Worldwide Equipment Guide, 2025-06-10.

[https://odin.tradoc.army.mil/WEG/Asset/9K317M_Buk-M3_\[SA-27\]_Russian_Medium-Range_Air_Defense_Missile_System](https://odin.tradoc.army.mil/WEG/Asset/9K317M_Buk-M3_[SA-27]_Russian_Medium-Range_Air_Defense_Missile_System)

WEG28 WEG entry RKhM-6 and RKhM-8 [local extract]. US Army TRADOC, 2024-10-09.

https://odin.tradoc.army.mil/WEG/Asset/RKhM-6_Russian_8x8_Amphibious_CBRN_Reconnaissance_Vehicle

WEG29 WEG extract decon.json [ARS-14KM, TMS-65 pages]. TRADOC, odin.tradoc.army.mil/WEG, 2024-10-15 and 2025-03-06.

https://odin.tradoc.army.mil/WEG/Asset/ARS-14KM_Russian_6x6_Decontamination_Truck

WEG30 Worldwide Equipment Guide: BTM-3, MDK-3, PZM-2 [local extract]. TRADOC, 2025-03-20.

https://odin.tradoc.army.mil/WEG/Asset/BTM-3_Russian_Trenching_Machine

WEG31 WEG extract: BM-21 Grad Russian 122mm MRL [last modified 2024-12-19]. US Army TRADOC, 2024-12.

[https://odin.tradoc.army.mil/WEG/Asset/BM-21_Grad_Russian_122mm_Multiple_Rocket_Launcher_\[MRL\]](https://odin.tradoc.army.mil/WEG/Asset/BM-21_Grad_Russian_122mm_Multiple_Rocket_Launcher_[MRL])

WEG32 WEG extract groza.json: R-934UM2, R-330B, R-934B pages. US Army TRADOC, 2025-06-05.

https://odin.tradoc.army.mil/WEG/Asset/R-934B_Russian_VHF_Air_Communication_Automated_Jamming_Station

WEG33 Infaua (RB-531B) Russian 8x8 Amphibious Electronic Warfare System. TRADOC WEG, 2025-06-05.

[https://odin.tradoc.army.mil/WEG/Asset/Infaua_\(RB-531B\)_Russian_8x8_Amphibious_Electronic_Warfare_System](https://odin.tradoc.army.mil/WEG/Asset/Infaua_(RB-531B)_Russian_8x8_Amphibious_Electronic_Warfare_System)

WEG34 WEG: IRM Zhuk Russian Amphibious Engineer Reconnaissance Vehicle [local extract pilot/weg_extract/irm.json]. TRADOC, 2024-10-30.

https://odin.tradoc.army.mil/WEG/Asset/IRM_Zhuk_Russian_Amphibious_Engineer_Reconnaissance_Vehicle

WEG35 Worldwide Equipment Guide, KamAZ-4310, 5350, 6350 pages. TRADOC, 2024-10.

https://odin.tradoc.army.mil/WEG/Asset/KamAZ-4310_Russian_6x6_General_Utility_Vehicle

WEG36 Krasuha Russian Mobile Electronic Warfare Jamming System [WEG extract]. US Army TRADOC, ODIN, 2025-06-05.

https://odin.tradoc.army.mil/WEG/Asset/Krasuha_Russian_Mobile_Electronic_Warfare_Jamming_System

WEG37 WEG: Leer-2 Russian 4x4 Mobile EW Vehicle [local extract]. US Army TRADOC, 2025-06-05.

[https://odin.tradoc.army.mil/WEG/Asset/Leer-2_Russian_4x4_Mobile_Electronic_Warfare_\[EW\]_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/Leer-2_Russian_4x4_Mobile_Electronic_Warfare_[EW]_Vehicle)

WEG38 WEG extract: AMN-590951 [Spartak] and Z-STs Akhmat. TRADOC WEG, 2024-08/2024-09.

[https://odin.tradoc.army.mil/WEG/Asset/AMN-590951_\[Spartak\]_Russian_4x4_Mine_Resistant_Ambush_Protected_\[JMRAP\]](https://odin.tradoc.army.mil/WEG/Asset/AMN-590951_[Spartak]_Russian_4x4_Mine_Resistant_Ambush_Protected_[JMRAP])

WEG39 Moskva-1 [1L265] Russian Electronic Warfare Vehicle. US Army TRADOC ODIN WEG, 2025-06-06.

[https://odin.tradoc.army.mil/WEG/Asset/Moskva-1_\[1L265\]_Russian_Electronic_Warfare_Vehicle](https://odin.tradoc.army.mil/WEG/Asset/Moskva-1_[1L265]_Russian_Electronic_Warfare_Vehicle)

WEG40 Worldwide Equipment Guide: 2S19 Msta-S, 2S19M1 (Msta-SM1) and 2S19M2 (Msta-SM2) 152mm SPH records. US Army TRADOC (ODIN), 2025-04-03 (2S19); 2025-04-07 (2S19M1, 2S19M2).
https://odin.tradoc.army.mil/WEG/Asset/2S19_Msta-S_Russian_152mm_Self-Propelled_Howitzr_SPH

WEG41 Worldwide Equipment Guide: MT-LB Russian Amphibious Armored Personnel Carrier [local extract pilot/weg_extract/mtlb.json]. US Army TRADOC, 2025-03-05.
https://odin.tradoc.army.mil/WEG/Asset/MT-LB_Russian_Amphibious_Armored_Personnel_Carrier_IAPC

WEG42 Murmansk-BN Russian Long-Range Communications Jamming System. US Army TRADOC, ODIN WEG, 2025-06-05.
https://odin.tradoc.army.mil/WEG/Asset/Murmansk-BN_Russian_Long-Range_Communications_Jamming_System

WEG43 48Y6-K1 Podlet K1 Russian Mobile Long-Range High-Altitude Surveillance Radar [local extract]. US Army TRADOC, ODIN WEG, 2025-04-08.
https://odin.tradoc.army.mil/WEG/Asset/48Y6-K1_Podlet_K1_Russian_Mobile_Long-Range_High-Altitude_Surveillance_Radar

WEG44 Pantsir-S1 (SA-22 Greyhound) Russian Short-Range Air Defense Gun/Missile System. US Army TRADOC ODIN WEG, 2025-06-10.
https://odin.tradoc.army.mil/WEG/Asset/Pantsir-S1_SA-22_Greyhound_Russian_Short-Range_Air_Defense_Gun_Missile_System

WEG45 PMP Russian Heavy Folding Pontoon Bridge [WEG online]. US Army TRADOC G-2, 2024-10-16.
https://odin.tradoc.army.mil/WEG/Asset/PMP_Russian_Heavy_Folding_Pontoon_Bridge

WEG46 Pole-21E Russian RF Jammer. US Army TRADOC WEG, 2025-06-05.
https://odin.tradoc.army.mil/WEG/Asset/Pole-21E_Russian_RF_Jammer

WEG47 9S52 Polyana-D4 ASU Russian 6x6 Air Defense Command and Control Vehicle [local extract]. US Army TRADOC, ODIN WEG, 2025-06-10.
https://odin.tradoc.army.mil/WEG/Asset/9S52_Polyana-D4_ASU_Russian_6x6_Air_Defense_Command_and_Control_Vehicle

WEG48 WEG: PRP-4A [Argus] Russian Amphibious Artillery Reconnaissance Vehicle. TRADOC, 2024-10-11.
https://odin.tradoc.army.mil/WEG/Asset/PRP-4A_Argus_Russian_Amphibious_Artillery_Reconnaissance_Vehicle

WEG49 WEG extract: PTS-4 and PMM-2 pages. US Army TRADOC, WEG [local extract], 2024.
https://odin.tradoc.army.mil/WEG/Asset/PTS-4_Russian_Amphibious_Cargo_Vehicle

WEG50 PU-12M7 Russian 8x8 Mobile Command and Control Vehicle; 9S737 Ranzhir Russian 6x6 Mobile Command Center. US Army TRADOC, ODIN WEG, 2024-08-26.
https://odin.tradoc.army.mil/WEG/Asset/PU-12M7_Russian_8x8_Mobile_Command_and_Control_Vehicle

WEG51 WEG BTR-60-R-145BM Command Post and Communication Center Vehicle. US Army TRADOC, 2024-08-07.
https://odin.tradoc.army.mil/WEG/Asset/BTR-60-R-145BM_Russian_8x8_Command_Post_and_Communication_Center_Vehicle

WEG52 Worldwide Equipment Guide sheets: R-149AKSh-1 [mod. 2025-04-23], R-149BMR [2025-01-08], R-149MA3 [2025-04-10]. US Army TRADOC (ODIN), 2025.
https://odin.tradoc.army.mil/WEG/Asset/R-149AKSh-1_Russian_6x6_Command_and_Signal_Vehicle

WEG53 Repellent Russian 8x8 Anti-Drone Warfare Complex. TRADOC WEG [local extract], 2025-06-05.
https://odin.tradoc.army.mil/WEG/Asset/Repellent_Russian_8x8_Anti-Drone_Warfare_Complex

WEG54 Worldwide Equipment Guide: S-300V (SA-12a Gladiator) and S-300V (SA-12b Giant) Russian Long-Range Air Defense Missile System [local extract pilot/weg_extract/s400.json]. US Army TRADOC, 2025-06-10.
https://odin.tradoc.army.mil/WEG/Asset/S-300V_SA-12a_Gladiator_Russian_Long-Range_Air_Defense_Missile_System

WEG55 Worldwide Equipment Guide: S-400 Triumph (SA-21 Growler) [local extract pilot/weg_extract/s400.json]. US Army TRADOC, 2025-11-17.
https://odin.tradoc.army.mil/WEG/Asset/S-400_Triumph_SA-21_Growler_Russian_Long-Range_Surface-to-Air_Missile_ISAM_System

WEG56 50R6 Vityaz (S-350E) Russian Medium-Range Air Defense Missile System [local extract]. US Army TRADOC ODIN WEG, 2025-06-10.
https://odin.tradoc.army.mil/WEG/Asset/50R6_Vityaz_S-350E_Russian_Medium-Range_Air_Defense_Missile_System

WEG57 Worldwide Equipment Guide: 9A52-2 (BM-30 Smerch-M); 9A52-4 Tornado; 9A53-S Tornado-S [local extract]. US Army TRADOC, 2024-10 to 2026-02.
https://odin.tradoc.army.mil/WEG/Asset/9A52-2_BM-30_Smerch-M_Russian_300mm_Multiple_Launch_Rocket_System

WEG58 WEG: TDA-3 Russian 6x6 Mobile Smokescreen Generator. US Army TRADOC, 2024-08-29.
https://odin.tradoc.army.mil/WEG/Asset/TDA-3_Russian_6x6_Mobile_Smokescreen_Generator

WEG59 WEG: SNAR-10 and SNAR-10M1 (Big Fred) entries, last modified 2024-10-22. TRADOC, 2024-10.
https://odin.tradoc.army.mil/WEG/Asset/SNAR-10M1_Big_Fred_Russian_Battlefield_Surveillance_Radar

WEG60 WEG extract: 9K35 Strela-10 and 9K33 Osa pages. US Army TRADOC, 2025-06-10.
https://odin.tradoc.army.mil/WEG/Asset/9K35_Strela-10_SA-13_Gopher_Russian_Short-Range_Surface-to-Air_Missile_System

WEG61 WEG extract t55: T-55 Russian Main Battle Tank (MBT), page last modified 2025-04-16. TRADOC WEG, 2025-04.
https://odin.tradoc.army.mil/WEG/Asset/T-55_Russian_Main_Battle_Tank_MBT

WEG62 T-62 Russian Medium Tank, Worldwide Equipment Guide [local extract pilot/weg_extract/t62.json]. US Army TRADOC, 2025-02-27.
https://odin.tradoc.army.mil/WEG/Asset/T-62_Russian_Medium_Tank

WEG63 Worldwide Equipment Guide: T-80B and T-80U Russian Main Battle Tank pages [local extract pilot/weg_extract/t80.json]. US Army TRADOC G-2 (ODIN), 2024-08-05 (T-80B), 2025-02-27 (T-80U).
https://odin.tradoc.army.mil/WEG/Asset/T-80B_Russian_Main_Battle_Tank_MBT

WEG64 -4-90 Russian Main Battle Tank [MBT]. US Army TRADOC, ODIN Worldwide Equipment Guide, 2025-02-27.
https://odin.tradoc.army.mil/WEG/Asset/T-90_Russian_Main_Battle_Tank_MBT

WEG65 GAZ-2975 Tigr and GAZ-3937 Vodnik [WEG extract, two records]. US Army TRADOC, ODIN WEG, 2025-02-11.
https://odin.tradoc.army.mil/WEG/Asset/GAZ-2975_Tigr_Russian_4x4_Combat_Support_Vehicle_-_Light_Utility_Vehicle

WEG66 Worldwide Equipment Guide online: MTU-72 AVLB and TMM-3 mobile bridging system [local extract tmm.json]. TRADOC, 2024-10 to 2025-04.
https://odin.tradoc.army.mil/WEG/Asset/TMM-3_Russian_6x6_Mobile_Bridging_System

WEG67 9K330 Tor (SA-15 Gauntlet) Russian Short-Range Surface-to-Air Missile System. US Army TRADOC, Worldwide Equipment Guide (ODIN), 2025-06-10.
https://odin.tradoc.army.mil/WEG/Asset/9K330_Tor_SA-15_Gauntlet_Russian_Short-Range_Surface-to-Air_Missile_System

WEG68 9A53-G Tornado Russian 122mm Multiple Rocket Launcher [local extract]. US Army TRADOC, ODIN WEG, 2024-08-21.
https://odin.tradoc.army.mil/WEG/Asset/9A53-G_Tornado_Russian_122mm_Multiple_Rocket_Launcher

WEG69 Worldwide Equipment Guide: TOS-1A Solntsepek, TOS-1 Buratino and TOS-2 220mm Heavy Flamethrower System records. US Army TRADOC (ODIN), 2024-09-05 (TOS-1A), 2024-10-10 (TOS-1); 2024-10-17 (TOS-2).
https://odin.tradoc.army.mil/WEG/Asset/TOS-1A_Solntsepek_Russian_220mm_Heavy_Flamethrower_System

WEG70 WEG extracts: 2S6M1 Tunguska-M1 and 2S6M Tunguska-M (SA-19 Grison). US Army TRADOC, 2025-06-10.
https://odin.tradoc.army.mil/WEG/Asset/2S6M1_Tunguska-M1_Russian_30mm_Self-Propelled_Anti-Aircraft_System

WEG71 Worldwide Equipment Guide extract: KamAZ-53949 Typhoon-K and Linza. US Army TRADOC, 2024-10.
https://odin.tradoc.army.mil/WEG/Asset/KamAZ-53949_Typhoon-K_Russian_4x4_Armored_Mine-Resistant_Ambush_Protected_IMRAP

WEG72 WEG online: UMZ Russian 6x6 Scatterable Minelaying System [local extract]. TRADOC, 2024-10-16.
https://odin.tradoc.army.mil/WEG/Asset/UMZ_Russian_6x6_Scatterable_Minelaying_System

WEG73 WEG entry: UR-77 Meteorit Russian Mine Clearing Vehicle [local extract ur77.json]. US Army TRADOC, 2024-10-24.
https://odin.tradoc.army.mil/WEG/Asset/UR-77_Meteorit_Russian_Mine_Clearing_Vehicle

WEG74 9K512 (Uragan-1M) Russian 220mm/300mm Multiple Launch Rocket System [MRL], Worldwide Equipment Guide [local extract]. US Army TRADOC G-2, 2025-03-27.
https://odin.tradoc.army.mil/WEG/Asset/9K512_Uragan-1M_Russian_220mm:300mm_Multiple_Launch_Rocket_System_MRL

- WEG75** 9P140 (BM-27 Uragan) Russian 220mm Multiple Rocket Launch (MRL). US Army TRADOC, Worldwide Equipment Guide (ODIN), local extract, 2024-10-28.
https://odin.tradoc.army.mil/WEG/Asset/9P140_BM-27_Uragan_Russian_220mm_Multiple_Rocket_Launch_MRL
- WEG76** WEG entry: Uran-6 Russian Demining UGV (local extract pilot/weg_extract/uran6.json). US Army TRADOC, 2024-10-16.
https://odin.tradoc.army.mil/WEG/Asset/Uran-6_Russian_Demining_Unmanned_Ground_Vehicle JUGV
- WEG77** ODIN WEG: GMZ-3 Russian Tracked Minelaying Vehicle; ISDM Zemledeliye Russian 8x8 Mine-Laying System (local extract). US Army TRADOC, 2024-10-16.
https://odin.tradoc.army.mil/WEG/Asset/ISDM_Zemledeliye_Russian_8x8_Mine-Laying_System
- WEG78** R-330Zh Zhitel Russian Cellular Jamming and Direction Finding System. US Army TRADOC, ODIN Worldwide Equipment Guide, 2025-09-03.
https://odin.tradoc.army.mil/WEG/Asset/R-330Zh_Zhitel_Russian_Cellular_Jamming_and_Direction_Finding_System
- WEG2015** Worldwide Equipment Guide 2015, Volume 1 Ground Systems, IV18/IV19 page and tier tables. TRADOC, 2015.
<https://archive.org/details/USArmyWorldwideEquipmentGuide2015>
- WIKRD18** Mashina komandira batarei IV18 Klen-1 [handbook text on tech.wikireading.ru]. Wikireading (reprint of Russian armour handbook).
<https://tech.wikireading.ru/7372>
- WMC1** Wikimedia Commons photographs of 96L6E (MAKS-2011, MAKS-2009, VVS 100th anniversary 2012, Patriot Park 2016). Wikimedia Commons, 2009-2016.
<https://commons.wikimedia.org/wiki/Category:96L6E>
- WMC2** MasteringTheArmament2018-13.jpg (Russian MoD photo, caption). Russian MoD via Wikimedia Commons, 2018.
<https://commons.wikimedia.org/wiki/File:MasteringTheArmament2018-13.jpg>
- WMC3** File:BMD-2 - 137AirborneRegiment22.jpg. Wikimedia Commons (photo by Vitaly V. Kuzmin), 2011-09-28.
https://commons.wikimedia.org/wiki/File:BMD-2_-_137AirborneRegiment22.jpg
- WMC4** File:ElectronicWarfareExercise2018-01.jpg [MoD caption: 74th Separate Guards MR Brigade exercise with Borisoglebsk-2]. Wikimedia Commons / Russian MoD, 2018.
<https://commons.wikimedia.org/wiki/File:ElectronicWarfareExercise2018-01.jpg>
- WMC5** File:Granat-4 UAV at Interpolitex-2015.jpg. Wikimedia Commons, 2015-10-20.
https://commons.wikimedia.org/wiki/File:Granat-4_UAV_at_Interpolitex-2015.jpg
- WMC6** File:Krasukha-2 (Kpacuxa-2) Unloaded.jpg. Wikimedia Commons (Vitaly V. Kuzmin), 2014-12-11.
[https://commons.wikimedia.org/wiki/File:Krasukha-2_\(Kpacuxa-2\)_Unloaded.jpg](https://commons.wikimedia.org/wiki/File:Krasukha-2_(Kpacuxa-2)_Unloaded.jpg)
- WMC7** File:1L269E Krasukha-2 Army-2018.jpg. Wikimedia Commons (Boevaya mashina), 2018-08-22.
https://commons.wikimedia.org/wiki/File:1L269E_Krasukha-2_Army-2018.jpg
- WMC8** File:1RL257E Krasukha-4 Army-2018.jpg. Wikimedia Commons (Boevaya mashina), 2018-08-22.
https://commons.wikimedia.org/wiki/File:1RL257E_Krasukha-4_Army-2018.jpg
- WMC9** File:1RL257E Krasukha-4 Army-2022 2022-08-20 2459.jpg. Wikimedia Commons (Mike1979 Russia), 2022-08-20.
https://commons.wikimedia.org/wiki/File:1RL257E_Krasukha-4_Army-2022_2022-08-20_2459.jpg
- WMC10** File:TB2015ExhibitionP2-47.jpg (RB-341V Leer-3 ECM system with UAV). Wikimedia Commons, 2015.
<https://commons.wikimedia.org/wiki/File:TB2015ExhibitionP2-47.jpg>
- WMC11** File:TB2015ExhibitionP2-48.jpg (RB-341V Leer-3 ECM system with UAV). Wikimedia Commons, 2015.
<https://commons.wikimedia.org/wiki/File:TB2015ExhibitionP2-48.jpg>
- WMC12** File:Развертывание P36 Палантин на учениях.jpg. Wikimedia Commons (Ministry of Defence of Russia), 2019-03.
https://commons.wikimedia.org/wiki/File:Развертывание_P36_Палантин_на_учениях.jpg
- WMC13** Open Water 2017 qualifying photographs (OpenWater2017qualifying-01 to -13). Vitaly V. Kuzmin via Wikimedia Commons, 2017.
https://commons.wikimedia.org/wiki/Category:PMP_pontoon_bridge_in_Russian_service
- WMC14** File:MAKS-2009-225.jpg (PU-12M7 on KISH1, roof sensor). Wikimedia Commons (Vitaly V. Kuzmin), 2009-08-20.
<https://commons.wikimedia.org/wiki/File:MAKS-2009-225.jpg>
- WMC15** File:Punkt ipravleniya PU-12M7.jpg [PU-12M7 at MAKS 2007]. Wikimedia Commons (Vitaly V. Kuzmin), 2007-08-23.
https://commons.wikimedia.org/wiki/File:Punkt_ipravleniya_PU-12M7.jpg
- WMC16** File:R-166-0,5 radiostation - 27th Independent Sevastopol Guards Motor Rifle Brigade (2).jpg. Wikimedia Commons (photo by Vitaly V. Kuzmin), 2011-10-29.
[https://commons.wikimedia.org/wiki/File:R-166-0,5_radiostation_-_27th_Independent_Sevastopol_Guards_Motor_Rifle_Brigade_\(2\).jpg](https://commons.wikimedia.org/wiki/File:R-166-0,5_radiostation_-_27th_Independent_Sevastopol_Guards_Motor_Rifle_Brigade_(2).jpg)
- WMC17** File:R-166 radiostation.jpg. Wikimedia Commons (photo by Vitaly V. Kuzmin), 2011-12-16.
https://commons.wikimedia.org/wiki/File:R-166_radiostation.jpg
- WMC18** File:Redut-2US - SignalExercise2017-04.jpg. Wikimedia Commons (Russian MoD photo), 2017.
https://commons.wikimedia.org/wiki/File:Redut-2US_-_SignalExercise2017-04.jpg
- WMC19** File:ConfidentReception2018-05.jpg (Category P-260T Redut-2US). Wikimedia Commons (Russian MoD photo), 2018.
<https://commons.wikimedia.org/wiki/File:ConfidentReception2018-05.jpg>
- WMC20** File:S-350E Vityaz - MAKS2013part4-26.jpg (ISOPE launcher, side view). Wikimedia Commons, 2013.
https://commons.wikimedia.org/wiki/File:S-350E_Vityaz_-_MAKS2013part4-26.jpg
- WMC21** File:S-350E Vityaz - MAKS2013firstpix01.jpg (launcher, front three-quarter). Wikimedia Commons, 2013.
https://commons.wikimedia.org/wiki/File:S-350E_Vityaz_-_MAKS2013firstpix01.jpg
- WMC22** File:Army2016demo-062.jpg (2B17M combat vehicle of 9K51M Tornado-G MLRS). Wikimedia Commons, 2016-09-11.
<https://commons.wikimedia.org/wiki/File:Army2016demo-062.jpg>
- WMC23** File:NBCProtectionExercise2019-21.jpg (TOS-1A firing, 6th NBC Protection Regiment, Sapyornoye). Wikimedia Commons (image from Mil.ru), 2019.
<https://commons.wikimedia.org/wiki/File:NBCProtectionExercise2019-21.jpg>
- WMC24** R-330Zh Zhitel ECM station (ParkPatriot2015part9-58 and -60). Vitaly V. Kuzmin, Wikimedia Commons, 2016-02-07.
<https://commons.wikimedia.org/wiki/File:ParkPatriot2015part9-58.jpg>
- WMC25** Станция помех P-330Ж Жумель (TB2015ExhibitionP2-51 and -52). Vitaly V. Kuzmin, Wikimedia Commons, 2015-08-15.
<https://commons.wikimedia.org/wiki/File:TB2015ExhibitionP2-52.jpg>
- WMC26** R-330Zh Zhitel jammer. Russian MoD (Denis Abramov), via Wikimedia Commons, 2018-07-19.
https://commons.wikimedia.org/wiki/File:R-330Zh_Zhitel_jammer.jpg
- WMC27** File:1L219M MAKS2005.jpg. Wikimedia Commons, 2005-07-20.
https://commons.wikimedia.org/wiki/File:1L219M_MAKS2005.jpg
- WMC28** File:1L260-E Army-2018.jpg. Wikimedia Commons, 2018-08-22.
https://commons.wikimedia.org/wiki/File:1L260-E_Army-2018.jpg
- WP-EN1** 2A36 Giatshint-B. English Wikipedia, 2026.
https://en.wikipedia.org/wiki/2A36_Giatshint-B
- WP-EN2** 152 mm howitzer 2A65 Msta-B. English Wikipedia, 2026.
https://en.wikipedia.org/wiki/152_mm_howitzer_2A65_Msta-B
- WP-EN3** 244th Artillery Brigade (Russia). English Wikipedia, 2026.
[https://en.wikipedia.org/wiki/244th_Artillery_Brigade_\(Russia\)](https://en.wikipedia.org/wiki/244th_Artillery_Brigade_(Russia))
- WP-EN4** 255 Giatshint-S. English Wikipedia, 2026.
https://en.wikipedia.org/wiki/255_Giatshint-S
- WP-EN5** 288th Guards Artillery Brigade (Russia). Wikipedia, 2024.
[https://en.wikipedia.org/wiki/288th_Guards_Artillery_Brigade_\(Russia\)](https://en.wikipedia.org/wiki/288th_Guards_Artillery_Brigade_(Russia))
- WP-EN6** 227th Artillery Brigade (Russia). Wikipedia, 2026.
[https://en.wikipedia.org/wiki/227th_Artillery_Brigade_\(Russia\)](https://en.wikipedia.org/wiki/227th_Artillery_Brigade_(Russia))
- WP-EN7** 2335 Koalitsiya-SV. Wikipedia (English), 2026.
https://en.wikipedia.org/wiki/2335_Koalitsiya-SV
- WP-EN8** 2343 Malva. Wikipedia (English), 2026.
https://en.wikipedia.org/wiki/2343_Malva
- WP-EN9** 234 Tyulpan (English Wikipedia). Wikipedia, 2026.
https://en.wikipedia.org/wiki/234_Tyulpan
- WP-EN10** 237 Pion. Wikipedia, 2026.
https://en.wikipedia.org/wiki/237_Pion
- WP-EN11** MT-LBu (variants list). Wikipedia, 2026.
<https://en.wikipedia.org/wiki/MT-LBu>
- WP-EN12** BTR-60 (English Wikipedia), variants list. Wikipedia, retrieved 2026-10-04.
<https://en.wikipedia.org/wiki/BTR-60>
- WP-EN13** BTR-80 (English Wikipedia), variants list. Wikipedia, retrieved 2026-10-04.
<https://en.wikipedia.org/wiki/BTR-80>
- WP-EN14** BTR-D (English Wikipedia). Wikipedia, 2026.
<https://en.wikipedia.org/wiki/BTR-D>
- WP-EN15** BMD-2. Wikipedia (en), retrieved 2026-10-03.
<https://en.wikipedia.org/wiki/BMD-2>

WP-EN16 BMD-4 [English Wikipedia]. Wikipedia, 2026.

<https://en.wikipedia.org/wiki/BMD-4>

WP-EN17 BMP-1. English Wikipedia, 2026.

<https://en.wikipedia.org/wiki/BMP-1>

WP-EN18 BMP-2 [English Wikipedia]. Wikipedia, 2026.

<https://en.wikipedia.org/wiki/BMP-2>

WP-EN19 BMPT Terminator. English Wikipedia, 2026.

https://en.wikipedia.org/wiki/BMPT_Terminator

WP-EN20 BRDM-2. English Wikipedia, retrieved 2026-10-03.

<https://en.wikipedia.org/wiki/BRDM-2>

WP-EN21 List of BMP-1 variants [BRM-1K section]. Wikipedia, retrieved 2026-10-03.

<https://en.wikipedia.org/wiki/BRM-1K>

WP-EN22 BTR-90. Wikipedia [English], 2026.

<https://en.wikipedia.org/wiki/BTR-90>

WP-EN23 247th Guards Air Assault Regiment, composition. Wikipedia [English], 2026.

https://en.wikipedia.org/wiki/247th_Guards_Air_Assault_Regiment

WP-EN24 Russian Airborne Forces, Ukraine section. Wikipedia [English], 2026.

https://en.wikipedia.org/wiki/Russian_Airborne_Forces

WP-EN25 BTR-70 [English Wikipedia]. Wikipedia, 2026.

<https://en.wikipedia.org/wiki/BTR-70>

WP-EN26 S-300V. Wikipedia [English], retrieved 2026-09-29.

<https://en.wikipedia.org/wiki/S-300V>

WP-EN27 ENICS Eleron-3. Wikipedia [en], retrieved 2026-09-30.

<https://en.wikipedia.org/wiki/Eleron-3>

WP-EN28 IAI Searcher. Wikipedia [en], retrieved 2026-10-02.

https://en.wikipedia.org/wiki/IAI_Searcher

WP-EN29 Unmanned Systems Forces of Russia. Wikipedia, 2026-09.

https://en.wikipedia.org/wiki/Unmanned_Systems_Forces_of_Russia

WP-EN30 HESA Shahed 136. Wikipedia [English], 2026.

https://en.wikipedia.org/wiki/HESA_Shahed_136

WP-EN31 Gerbera [drone]. Wikipedia [English], 2026.

[https://en.wikipedia.org/wiki/Gerbera_\(drone\)](https://en.wikipedia.org/wiki/Gerbera_(drone))

WP-EN32 BM-21 Grad. Wikipedia [English], 2026.

https://en.wikipedia.org/wiki/BM-21_Grad

WP-EN33 IMR-2. Wikipedia, 2026.

<https://en.wikipedia.org/wiki/IMR-2>

WP-EN34 1st Guards Engineer Brigade. Wikipedia, 2026.

https://en.wikipedia.org/wiki/1st_Guards_Engineer_Brigade

WP-EN35 List of BMP-1 variants, IRM Zhuk. Wikipedia, retrieved 2026-10-04.

https://en.wikipedia.org/wiki/List_of_BMP-1_variants

WP-EN36 9K720 Iskander. Wikipedia, retrieved 2026-09-29.

https://en.wikipedia.org/wiki/9K720_Iskander

WP-EN37 MZKT-7930. Wikipedia, retrieved 2026-09-29.

<https://en.wikipedia.org/wiki/MZKT-7930>

WP-EN38 K-300P Bastion-P. Wikipedia, retrieved 2026-09-29.

https://en.wikipedia.org/wiki/K-300P_Bastion-P

WP-EN39 Italmas. Wikipedia [English], 2026.

<https://en.wikipedia.org/wiki/Italmas>

WP-EN40 KamAZ-5350 [English Wikipedia]. Wikipedia, 2024.

<https://en.wikipedia.org/wiki/KamAZ-5350>

WP-EN41 KamAZ-4310 [English Wikipedia]. Wikipedia, 2026.

<https://en.wikipedia.org/wiki/KamAZ-4310>

WP-EN42 Luch Korsar. English Wikipedia, 2026.

https://en.wikipedia.org/wiki/Luch_Korsar

WP-EN43 Krasukha. Wikipedia [English], retrieved 2026-09-29.

<https://en.wikipedia.org/wiki/Krasukha>

WP-EN44 ZALA Kub-BLA. Wikipedia [English], retrieved 2026-10-03.

https://en.wikipedia.org/wiki/ZALA_KUB-BLA

WP-EN45 ZALA Lancet. Wikipedia [English], retrieved 2026-10-02.

https://en.wikipedia.org/wiki/ZALA_Lancet

WP-EN46 BPM-97. Wikipedia [English], 2026.

<https://en.wikipedia.org/wiki/BPM-97>

WP-EN47 List of equipment of the Russian Ground Forces. Wikipedia [English], 2026.

https://en.wikipedia.org/wiki/List_of_equipment_of_the_Russian_Ground_Forces

WP-EN48 Mohajer-6. Wikipedia, 2026-10-02.

<https://en.wikipedia.org/wiki/Mohajer-6>

WP-EN49 Baykar Bayraktar TB2. Wikipedia, 2026-10-03.

https://en.wikipedia.org/wiki/Baykar_Bayraktar_TB2

WP-EN50 Molniya [drone]. Wikipedia, 2026-09-30.

[https://en.wikipedia.org/wiki/Molniya_\(drone\)](https://en.wikipedia.org/wiki/Molniya_(drone))

WP-EN51 2S19 Msta-S [English Wikipedia]. Wikipedia, retrieved 2026-09-29.

https://en.wikipedia.org/wiki/2S19_Msta-S

WP-EN52 2S3 Akatsiya [English Wikipedia]. Wikipedia, retrieved 2026-09-29.

https://en.wikipedia.org/wiki/2S3_Akatsiya

WP-EN53 MT-LB [English Wikipedia]. Wikipedia, retrieved 2026-10-03.

<https://en.wikipedia.org/wiki/MT-LB>

WP-EN54 Kronstadt Orion. Wikipedia [en], 2026.

https://en.wikipedia.org/wiki/Kronstadt_Orion

WP-EN55 Orlan-10. Wikipedia [English], 2026.

<https://en.wikipedia.org/wiki/Orlan-10>

WP-EN56 Pantsir missile system. English Wikipedia, retrieved 2026-09-29.

https://en.wikipedia.org/wiki/Pantsir_missile_system

WP-EN57 Penicillin [counter-artillery system]. Wikipedia, 2026.

[https://en.wikipedia.org/wiki/Penicillin_\(counter-artillery_system\)](https://en.wikipedia.org/wiki/Penicillin_(counter-artillery_system))

WP-EN58 Aistyonok. Wikipedia, 2026.

<https://en.wikipedia.org/wiki/Aistyonok>

WP-EN59 PMP Floating Bridge. English Wikipedia, retrieved 2026-09-29.

https://en.wikipedia.org/wiki/PMP_Floating_Bridge

WP-EN60 Polanya-D4. Wikipedia [English], 2026.

<https://en.wikipedia.org/wiki/Polanya-D4>

WP-EN61 Ural-375D. Wikipedia [English], 2026.

<https://en.wikipedia.org/wiki/Ural-375D>

WP-EN62 PTS (vehicle) [en.wikipedia]. Wikipedia, 2026.

[https://en.wikipedia.org/wiki/PTS_\(vehicle\)](https://en.wikipedia.org/wiki/PTS_(vehicle))

WP-EN63 288th Guards Artillery Brigade [en.wikipedia]. Wikipedia, 2026.

https://en.wikipedia.org/wiki/288th_Guards_Artillery_Brigade

WP-EN64 4th Guards Tank Division. English Wikipedia, 2026.

https://en.wikipedia.org/wiki/4th_Guards_Tank_Division

WP-EN65 Repellent-1. Wikipedia, 2026.

<https://en.wikipedia.org/wiki/Repellent-1>

WP-EN66 S-300VM missile system [English Wikipedia]. Wikipedia, retrieved 2026-09-29.

https://en.wikipedia.org/wiki/S-300VM_missile_system

WP-EN67 S-350 missile system. Wikipedia, 2026.

https://en.wikipedia.org/wiki/S-350_missile_system

WP-EN68 List of NATO reporting names for surface-to-air missiles. Wikipedia, retrieved 2026-09-29.

https://en.wikipedia.org/wiki/List_of_NATO_reporting_names_for_surface-to-air_missiles

WP-EN69 S-400 missile system. Wikipedia [en], retrieved 2026-09-29.

https://en.wikipedia.org/wiki/S-400_missile_system

WP-EN70 BM-30 Smerch. English Wikipedia, 2026.

https://en.wikipedia.org/wiki/BM-30_Smerch

WP-EN71 79th Guards Rocket Artillery Brigade. English Wikipedia, 2026.

https://en.wikipedia.org/wiki/79th_Guards_Rocket_Artillery_Brigade

WP-EN72 9K35 Strela-10. Wikipedia, 2026.

https://en.wikipedia.org/wiki/9K35_Strela-10

WP-EN73 9K33 Osa. Wikipedia, 2026.

https://en.wikipedia.org/wiki/9K33_Osa

WP-EN74 T-54/T-55 [English Wikipedia]. Wikipedia, retrieved 2026-10-02.

<https://en.wikipedia.org/wiki/T-54/T-55>

WP-EN75 T-64 [Wikipedia]. Wikipedia, retrieved 2026-10-03.

<https://en.wikipedia.org/wiki/T-64>

WP-EN76 T-90. English Wikipedia, 2026.

<https://en.wikipedia.org/wiki/T-90>

WP-EN77 Tigr [military vehicle]. Wikipedia, 2026.

https://en.wikipedia.org/wiki/GAZ_Tigr

WP-EN78 Vodnik [Russian military vehicle]. Wikipedia, 2026.

[https://en.wikipedia.org/wiki/Vodnik_\(Russian_military_vehicle\)](https://en.wikipedia.org/wiki/Vodnik_(Russian_military_vehicle))

WP-EN79 English Wikipedia: T-72 operators and variants (MTU-72 entry). Wikipedia, retrieved 2026-09-29.

<https://en.wikipedia.org/wiki/T-72>

WP-EN80 English Wikipedia: TMM-6. Wikipedia, retrieved 2026-09-29.

<https://en.wikipedia.org/wiki/TMM-6>

WP-EN81 TOS-1. Wikipedia.

<https://en.wikipedia.org/wiki/TOS-1>

- WP-EN82** TOS-2. Wikipedia.
<https://en.wikipedia.org/wiki/TOS-2>
- WP-EN83** 2K22 Tunguska. Wikipedia, 2026.
https://en.wikipedia.org/wiki/2K22_Tunguska
- WP-EN84** KamAZ-63968 [Kamaz Typhoon], English Wikipedia, 2026.
<https://en.wikipedia.org/wiki/KamazAZ-63968>
- WP-EN85** UR-77 Meteorit. Wikipedia, 2026.
https://en.wikipedia.org/wiki/UR-77_Meteorit
- WP-EN86** 9K512 Uragan-1M [English Wikipedia]. Wikipedia, 2026.
https://en.wikipedia.org/wiki/9K512_Uragan-1M
- WP-EN87** BM-27 Uragan. English Wikipedia, 2026.
https://en.wikipedia.org/wiki/BM-27_Uragan
- WP-EN88** 338th Guards Rocket Artillery Brigade. English Wikipedia, 2026.
https://en.wikipedia.org/wiki/338th_Guards_Rocket_Artillery_Brigade
- WP-EN89** Ural-4320. English Wikipedia, retrieved 2026-10.
<https://en.wikipedia.org/wiki/Ural-4320>
- WP-EN90** Zemledeliye (minelaying system). Wikipedia.
[https://en.wikipedia.org/wiki/Zemledeliye_\(minelaying_system\)](https://en.wikipedia.org/wiki/Zemledeliye_(minelaying_system))
- WP-EN91** GMZ-3. Wikipedia.
<https://en.wikipedia.org/wiki/GMZ-3>
- WP-RU1** 1B13; 1B14; MT-A6y [Russian Wikipedia articles]. Wikipedia, retrieved 2026-09-29.
<https://ru.wikipedia.org/wiki/1%D0%92B13>; <https://ru.wikipedia.org/wiki/1%D0%92B14>;
<https://ru.wikipedia.org/wiki/%D0%9C%D0%A2-%D0%98%D0%91%D1%83>
- WP-RU2** 152-мм ныука 2A36. Russian Wikipedia, 2026.
- WP-RU3** 152-мм зayбуца 2A65. Russian Wikipedia, 2024.
<https://ru.wikipedia.org/wiki/2A65>
- WP-RU4** 2C35. Wikipedia [Russian], 2026.
<https://ru.wikipedia.org/wiki/2C35>
- WP-RU5** 2C43. Wikipedia [Russian], 2026.
<https://ru.wikipedia.org/wiki/2C43>
- WP-RU6** 2C4 [Russian Wikipedia]. Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/2C4>
- WP-RU7** 2C5 [Russian Wikipedia]. Wikipedia.
<https://ru.wikipedia.org/wiki/2C5>
- WP-RU8** 2C7 «Пуо». Russian Wikipedia, 2026.
https://ru.wikipedia.org/wiki/2C7_«Пуо»
- WP-RU9** C-400 [article, raw wikitext]. Wikipedia [Russian], retrieved 2026-09-29.
<https://ru.wikipedia.org/wiki/C-400>
- WP-RU10** ГАЗ-3344. Wikipedia [Russian], retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/ГАЗ-3344>
- WP-RU11** 1V119 [Russian Wikipedia]. Wikipedia, retrieved 2026-10-04.
<https://ru.wikipedia.org/wiki/1%D0%92B19>
- WP-RU12** Баpная-T. Russian Wikipedia, retrieved 2026-09-29.
<https://ru.wikipedia.org/wiki/Баpная-T>
- WP-RU13** БАТ-2. Russian Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/БАТ-2>
- WP-RU14** БМД-1КШ [Russian Wikipedia]. Wikipedia, 2025.
<https://ru.wikipedia.org/wiki/БМД-1КШ>
- WP-RU15** БТП-А [Russian Wikipedia]. Wikipedia, 2025.
<https://ru.wikipedia.org/wiki/БТП-А>
- WP-RU16** БМД-1 [Russian Wikipedia]. Wikipedia, 2025.
<https://ru.wikipedia.org/wiki/БМД-1>
- WP-RU17** БМД-2. Wikipedia [ru], retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/БМД-2>
- WP-RU18** БМД-4 [Russian Wikipedia]. Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/БМД-4>
- WP-RU19** 2A70 [Russian Wikipedia]. Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/2A70>
- WP-RU20** БМД-3 [Russian Wikipedia]. Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/БМД-3>
- WP-RU21** БМО-T. Russian Wikipedia, retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/БМО-T>
- WP-RU22** БМП-1. Russian Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/БМП-1>
- WP-RU23** 2A28. Russian Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/2A28>
- WP-RU24** БМП-1КШ. Wikipedia [ru], retrieved 2026-10-01.
<https://ru.wikipedia.org/wiki/БМП-1КШ>
- WP-RU25** БМП-2 [Russian Wikipedia]. Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/БМП-2>
- WP-RU26** БМПТ «Терминатор». Russian Wikipedia, 2026.
https://ru.wikipedia.org/wiki/БМПТ_«Терминатор»
- WP-RU27** БМПТ-72. Russian Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/БМПТ-72>
- WP-RU28** БМП-3М. Russian Wikipedia, retrieved 2026-10-04.
<https://ru.wikipedia.org/wiki/БМП-3М>
- WP-RU29** БМП-3. Russian Wikipedia, retrieved 2026-10-04.
<https://ru.wikipedia.org/wiki/БМП-3>
- WP-RU30** P-330. Wikipedia [Russian].
<https://ru.wikipedia.org/wiki/P-330>
- WP-RU31** БРДМ-2. Russian Wikipedia, retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/БРДМ-2>
- WP-RU32** Wikipedia [ru]: БРДМ-1. Wikipedia, retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/БРДМ-1>
- WP-RU33** Wikipedia [ru]: БРДМ-2. Wikipedia, retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/БРДМ-2>
- WP-RU34** Wikipedia [ru]: БРДМ-А. Wikipedia, retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/БРДМ-А>
- WP-RU35** Wikipedia [ru]: БРДМ-Д. Wikipedia, retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/БРДМ-Д>
- WP-RU36** Wikipedia [ru]: БРДМ-К. Wikipedia, retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/БРДМ-К>
- WP-RU37** Wikipedia [ru]: БТС-4. Wikipedia, retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/БТС-4>
- WP-RU38** Wikipedia [ru]: МТ-А6. Wikipedia, retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/МТ-А6>
- WP-RU39** БРМ-1К. Wikipedia [ru], retrieved 2026-10-03.
<https://ru.wikipedia.org/wiki/БРМ-1К>
- WP-RU40** БТР-90. Wikipedia [Russian], 2026.
<https://ru.wikipedia.org/wiki/БТР-90>
- WP-RU41** БТР-МДМ. Russian Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/БТР-МДМ>
- WP-RU42** БТР-80 [Russian Wikipedia]. Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/БТР-80>
- WP-RU43** БТР-82 [Russian Wikipedia]. Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/БТР-82>
- WP-RU44** БТР-70 [Russian Wikipedia]. Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/БТР-70>
- WP-RU45** 2A72 [Russian Wikipedia]. Wikipedia, 2026.
<https://ru.wikipedia.org/wiki/2A72>
- WP-RU46** Бук-М2. Wikipedia [Russian], retrieved 2026-09-29.
<https://ru.wikipedia.org/wiki/Бук-М2>
- WP-RU47** РХМ-4. Russian Wikipedia, retrieved 2026-10-04.
<https://ru.wikipedia.org/wiki/РХМ-4>
- WP-RU48** РХМ-6. Russian Wikipedia, retrieved 2026-10-04.
<https://ru.wikipedia.org/wiki/РХМ-6>
- WP-RU49** ТМС-65. Russian Wikipedia, retrieved 2026-10-04.
<https://ru.wikipedia.org/wiki/ТМС-65>
- WP-RU50** БТМ-3. Wikipedia [ru], retrieved 2026-10-04.
<https://ru.wikipedia.org/wiki/БТМ-3>
- WP-RU51** МАК-3. Wikipedia [ru], retrieved 2026-10-04.
<https://ru.wikipedia.org/wiki/МАК-3>
- WP-RU52** ПЗМ-2. Wikipedia [ru], retrieved 2026-10-04.
<https://ru.wikipedia.org/wiki/ПЗМ-2>
- WP-RU53** Эерон-3. Wikipedia [ru], retrieved 2026-09-30.
<https://ru.wikipedia.org/wiki/Эерон-3>
- WP-RU54** Эерон-10. Wikipedia [ru], retrieved 2026-09-30.
<https://ru.wikipedia.org/wiki/Эерон-10>
- WP-RU55** IAI Searcher (Дорноч [БЛА]). Wikipedia [ru], retrieved 2026-10-02.
[https://ru.wikipedia.org/wiki/Дорноч_\[БЛА\]](https://ru.wikipedia.org/wiki/Дорноч_[БЛА])
- WP-RU56** Герань-2. Wikipedia [Russian], 2026.
<https://ru.wikipedia.org/wiki/Герань-2>
- WP-RU57** Гербеба [БЛА]. Russian Wikipedia, retrieved 2026-10-02.
[https://ru.wikipedia.org/wiki/Гербеба_\[БЛА\]](https://ru.wikipedia.org/wiki/Гербеба_[БЛА])
- WP-RU58** БМ-21 «Граг». Wikipedia [Russian], 2026.
https://ru.wikipedia.org/wiki/БМ-21_«Граг»
- WP-RU59** Гранам-4. Russian Wikipedia, 2026-10.
<https://ru.wikipedia.org/wiki/Гранам-4>
- WP-RU60** Гранам-1. Russian Wikipedia, 2026-10.
<https://ru.wikipedia.org/wiki/Гранам-1>

WP-RU124 Russian Wikipedia: MTY-72. Wikipedia, retrieved 2026-09-29.

<https://ru.wikipedia.org/wiki/MTY-72>

WP-RU125 Top (зенитный ракетный комплекс). Russian Wikipedia, retrieved 2026-09-29.

[https://ru.wikipedia.org/wiki/Top_\(зенитный_ракетный_комплекс\)](https://ru.wikipedia.org/wiki/Top_(зенитный_ракетный_комплекс))

WP-RU126 Торнадо-Г. Russian Wikipedia, retrieved 2026-09-29.

<https://ru.wikipedia.org/wiki/%D0%A2%D0%BE%D1%80%D0%8D%D0%80%D0%B4%D0%BE-%D0%93>

WP-RU127 Slovesnye nazvaniya rossiyskogo oruzhiya/R. Russian

Wikipedia, 2026.

https://ru.wikipedia.org/wiki/Словесные_названия_российского_оружия/P

WP-RU128 Slovesnye nazvaniya rossiyskogo oruzhiya/B. Russian

Wikipedia, 2026.

https://ru.wikipedia.org/wiki/Словесные_названия_российского_оружия/B

WP-RU129 Туныуска (зенитный ракетно-пушечный комплекс).

Russian Wikipedia, 2026.

https://ru.wikipedia.org/wiki/2K22_«Туныуска»

WP-RU130 КамАЗ-63968 «Таифун-К», Russian Wikipedia. Wikipedia,

2026.

<https://ru.wikipedia.org/wiki/Таифун-К>

WP-RU131 КамАЗ-53949 «Линза», Russian Wikipedia. Wikipedia, 2026.

<https://ru.wikipedia.org/wiki/КамАЗ-53949>

WP-RU132 Таифун (семейство броневых автомобилей), Russian

Wikipedia. Wikipedia, 2026.

[https://ru.wikipedia.org/wiki/Таифун_\(семейство_броневых_автомобилей\)](https://ru.wikipedia.org/wiki/Таифун_(семейство_броневых_автомобилей))

WP-RU133 Урал-63095, Russian Wikipedia. Wikipedia, 2026.

<https://ru.wikipedia.org/wiki/Урал-63095>

WP-RU134 2C41 «Дпок», Russian Wikipedia. Wikipedia, 2026.

<https://ru.wikipedia.org/wiki/2C41>

WP-RU135 YP-77. Russian Wikipedia, 2026.

<https://ru.wikipedia.org/wiki/YP-77>

WP-RU136 Uragan-1M (Russian Wikipedia). Wikipedia, 2026.

<https://ru.wikipedia.org/wiki/Ураган-1М>

WP-RU137 Ураган (РСЗО). Russian Wikipedia, 2026.

[https://ru.wikipedia.org/wiki/Ураган_\(РСЗО\)](https://ru.wikipedia.org/wiki/Ураган_(РСЗО))

WP-RU138 Урал-6370 (Урал-6370A redirects here). Russian Wikipedia,

retrieved 2026-10.

<https://ru.wikipedia.org/wiki/Урал-6370А>

WP-RU139 Уран-6. Russian Wikipedia, 2026.

<https://ru.wikipedia.org/wiki/Уран-6>

WP-RU140 Bird-Eye 400 (Russian Wikipedia, redirect from Засмава

(БПЛА)). Wikipedia, 2026.

[https://ru.wikipedia.org/wiki/Засмава_\(БПЛА\)](https://ru.wikipedia.org/wiki/Засмава_(БПЛА))

WP-RU141 Зоопарк (радиолокационный комплекс). Russian Wikipedia,

retrieved 2026-09-29.

[https://ru.wikipedia.org/wiki/Зоопарк_\(радиолокационный_комплекс\)](https://ru.wikipedia.org/wiki/Зоопарк_(радиолокационный_комплекс))

WP-UK1 P5-3015 «Борисоглебск-2». Wikipedia (Ukrainian), retrieved

2026-09-29.

https://uk.wikipedia.org/wiki/P5-3015_«Борисоглебск-2»

WP-UK2 R-340RP Pole-21. Ukrainian Wikipedia, 2026-09-30.

https://uk.wikipedia.org/wiki/%D0%A0-340%D0%A0%D0%9F_%C2%AB%D0%9F%D0%BE%D0%BB%D0%85-21%C2%8B

WPBR60 BTR-60. Wikipedia, 2026.

<https://ru.wikipedia.org/wiki/BTR-60>

WSN1 152mm 2A36 Giatint-B. WeaponSystems.net.

<https://weaponsystems.net/system/340-152mm-2A36-Giatint-B>

WSN2 2S35 Koalitsiya-SV. weaponsystems.net.

<https://weaponsystems.net/system/936-2S35-Koalitsiya-SV>

WSN3 BAT-2. WeaponSystems.net, 2026.

<https://weaponsystems.net/system/306-BAT-2>

WSN4 BRM-1K Korshun. weaponsystems.net.

<https://weaponsystems.net/system/346-BRM-1K-20Korshun>

WSN5 TMM (weaponsystems.net). WeaponSystems.net.

<https://weaponsystems.net/system/1276-TMM>

WSPOT1 WarSpotting API, Russian losses of 2022-03-03 (records 1948,

1949, 11098). WarSpotting, 2022-03-03.

<https://ukr.warspotting.net/api/losses/russia/2022-03-03>

WSPOT2 WarSpotting, Russian losses on 2022-03-01 (records 1428,

1570, 1571, 1599). WarSpotting, 2022-03-01.

<https://ukr.warspotting.net/api/losses/russia/2022-03-01>

WSPOT3 WarSpotting, Russian losses on 2022-09-10 (records 8219,

8282, 8938). WarSpotting, 2022-09-10.

<https://ukr.warspotting.net/api/losses/russia/2022-09-10>

WSPOT4 WarSpotting, Russian losses on 2022-11-05 (record 11225).

WarSpotting, 2022-11-05.

<https://ukr.warspotting.net/api/losses/russia/2022-11-05>

WSPOT5 WarSpotting, Russian losses on 2023-02-10 (record 14048,

BMP-2M Berezhek). WarSpotting, 2023-02-10.

<https://ukr.warspotting.net/api/losses/russia/2023-02-10>

WSPOT6 WarSpotting, Russian losses on 2023-06-12 (records 17054,

17055, 17367). WarSpotting, 2023-06-12.

<https://ukr.warspotting.net/api/losses/russia/2023-06-12>

WSPOT7 WarSpotting #3046, R-330BMV jamming station (for RB-301B

Borisoglebsk-2). WarSpotting, 2022-03-12.

<https://ukr.warspotting.net/view/3046/39064/>

WSPOT8 WarSpotting #8541, R-330BMV Borisoglebsk-2B. WarSpotting,

2022-09-13.

<https://ukr.warspotting.net/view/8541/20284/>

WSPOT9 WarSpotting losses, Russia, 2 March 2022 (records 2078, 2151,

2152). WarSpotting, 2022-03-02.

<https://ukr.warspotting.net/api/losses/russia/2022-03-02>

WSPOT10 WarSpotting losses, Russia, API page from id 8997 (records

9025, 9141). WarSpotting, 2022.

<https://ukr.warspotting.net/api/losses/russia/8997>

WSPOT11 WarSpotting losses, Russia, API page from id 8565 (records

8632, 8633, 8638, 8640). WarSpotting, 2022.

<https://ukr.warspotting.net/api/losses/russia/8565>

WSPOT12 WarSpotting losses, Russia, 15 October 2024 (record 30831).

WarSpotting, 2024-10-15.

<https://ukr.warspotting.net/api/losses/russia/2024-10-15>

WSPOT13 WarSpotting loss record #2439: R-149AKSh(-), 2nd Guards

Motor Rifle Division, 2022-03-08. WarSpotting, 2022-03-08.

<https://ukr.warspotting.net/api/losses/russia/2439>

WSPOT14 WarSpotting loss record #2440: R-149AKSh(-), 60th Control

Brigade, 2022-03-27. WarSpotting, 2022-03-27.

<https://ukr.warspotting.net/api/losses/russia/2440>

WSPOT15 WarSpotting loss record #13822: R-149MA3, 1st Guards

Motor Rifle Regiment, 2022-03-01. WarSpotting, 2022-03-01.

<https://ukr.warspotting.net/api/losses/russia/13822>

WSPOT16 WarSpotting loss record #9001: R-149MA3 (tagged

Turretless), 1st Guards Motor Rifle Regiment, 2022-09-11. WarSpotting,

2022-09-11.

<https://ukr.warspotting.net/api/losses/russia/9001>

WSPOT17 WarSpotting loss record #23939: R-149MA3, 1st Tank

Regiment, 2022-09-29. WarSpotting, 2022-09-29.

<https://ukr.warspotting.net/api/losses/russia/23939>

WSPOT18 WarSpotting loss record #8999: R-149MA1, 27th Separate

Guards Motor Rifle Brigade, 2022-09-12. WarSpotting, 2022-09-12.

<https://ukr.warspotting.net/api/losses/russia/8999>

WSPOT19 WarSpotting loss record 2442: R-166-0.5 'Artek' radio station

based on BTR-80. WarSpotting, 2022-03-20.

<https://ukr.warspotting.net/api/losses/russia/2442>

WSPOT20 WarSpotting loss record 2448: R-166-0.5 'Artek' radio

station based on BTR-80. WarSpotting, 2022-03-01.

<https://ukr.warspotting.net/api/losses/russia/2448>

WSPOT21 WarSpotting loss record 2453: R-166-0.5 'Artek' radio station

based on BTR-80. WarSpotting, 2022-02-26.

<https://ukr.warspotting.net/api/losses/russia/2453>

WSPOT22 WarSpotting loss record 38683: 9A83M2 TELAR (for 9K81M2

S-300V4), Oleksiivka, Polohy raion, 2025-08-20. WarSpotting, 2025-08-20.

<https://ukr.warspotting.net/api/losses/russia/38683>

WSPOT23 WarSpotting loss record 47009: 9A83M2 TELAR (for 9K81M2

S-300V4), Volnukhyne, Luhansk raion, 2026-08-14. WarSpotting,

2026-08-14.

<https://ukr.warspotting.net/api/losses/russia/47009>

WSPOT24 WarSpotting losses, Russia, 2022-11-09 (record 11376).

WarSpotting, 2022-11-09.

<https://ukr.warspotting.net/api/losses/russia/2022-11-09>

WSPOT25 WarSpotting loss records, Russia, 2022-03-04 (records 574,

575). WarSpotting, 2022-03-04.

<https://ukr.warspotting.net/api/losses/russia/2022-03-04>

WWW1 Bezbashennyye samokhodki Z57M Malka nachali ispolzovat po novomu. www1.ru, 2025-04-24.

<https://www1.ru/news/2025/04/24/bezbashennyye-samokhodki-z57m-malka-nachali-ispolzovat-po-novomu-biut-na-dalnosti-50-km.html>

WWW1A Hasta la vista, baby: чмн хорош российский БМПТ «Терминатор» на передовой. www1.ru, 2024-07-27.

<https://www1.ru/news/2024/07/27/hasta-la-vista-baby-chem-xoros-rossiiskii-bmpt-terminator-na-pereodoi-amp.html>

WWW1A2 «Форпосты» обнаружили опорные пункты условного противника в Калининградской области. www1.ru, 2026-05-08.

<https://www1.ru/news/2026/05/08/forposty-obnaruzhili-opornye-punkty-uslovnogo-protivnika-v-kaliningradskoi-oblasti-amp.html>

WWW1A3 Kalashnikov vypolnil goskazak po postavke barazhiruschego boeprispa KUB. Ouzhie Rossii (www1.ru), Kalashnikov press service, 2023-12-19.

<https://www1.ru/news/2023/12/19/kalashnikov-vypolnil-goskazak-po-postavke-barazhiruschego-boeprispa-kub-minobrony-rf.html>

WWW1B Рота «Терминаторов» усилила танковую дивизию ЦВО в 2024 году. www1.ru, 2025-01-07.

<https://www1.ru/news/2025/01/07/rota-terminatorov-usilila-tankovuyu-diviziyu-cvo-v-2024-godu-amp.html>

WWW1B2 Разработчики комплекса «Форпост» создали новую версию БЛА с радиолокатором целеуказания. www1.ru, 2025-02-20.

<https://www1.ru/news/2025/02/20/razrabotchiki-kompleksa-forpost-sozdali-novuiu-versiu-bpla-s-radiolokatorom-celeukazaniya-amp.html>

WWW1B3 Kalashnikov отправил в voyska pervuyu в 2025 godu partiyu UB KUB. Ouzhie Rossii (www1.ru), Kalashnikov press service, 2025-03-17.

<https://www1.ru/news/2025/03/17/kalashnikov-otpravil-v-voyska-pervuyu-v-2025-godu-partiyu-upravlyaemyx-boeprispaov-kub.html>

WWW1C Smertonsnaya ptitsa v zmyla в nebo: Kalashnikov nazval glavnye plyusy KUB. Ouzhie Rossii (www1.ru), Kalashnikov press service, 2025-03-26.

<https://www1.ru/news/2025/03/26/smertonsnaya-ptitsa-v-zmyla-v-nebo-kalashnikov-nazval-glavnye-plyusy-krilatogo-boeprispa-kub.html>

WWW1D Postavki Kalashnikovym KUBa-2. Ouzhie Rossii (www1.ru), Kalashnikov press service, 2025-12-29.

<https://www1.ru/news/2025/12/29/postavki-kalashnikovym-kuba-2.html>

WWW1E Kontsern Kalashnikov postavil voennym partiyu KUBov povyshennogo mogushchestva. Ouzhie Rossii (www1.ru), Kalashnikov press service, 2025-05-01.

<https://www1.ru/news/2025/05/01/kontsern-kalashnikov-postavil-voennym-partiyu-kubov-povyshennogo-mogushchestva.html>

WWW2 Poyavilis pervye kadry redkogo drona-kamikadze Italmas perez boeyum zapuskom. www1.ru, 2025-02-09.

<https://www1.ru/news/2025/02/09/poiavilis-pervye-kadry-redkogo-drona-kamikadze-italmas-perez-boeyum-zapuskom.html>

WWW3 Drony-kamikadze Геран 3 ... nachali zapuskat s pikapov. www1.ru, 2025-03-15.

<https://www1.ru/news/2025/03/15/drony-kamikadze-geran-3-s-moshnostiu-krilatoi-raket-y-nacali-zapuskat-s-pikapov-unkalnoe-video.html>

WWW12 Vezdekhod-razvedchik ili ognegvaya platforma: Plastun-SN. www1.ru, 2023-10-30.

<https://www1.ru/news/2023/10/30/vezdekhod-razvedchik-ili-ognegvaya-platforma-cto-mozet-armeiskii-plastun-sn-amp.html>

WWW13 Новейшую машину разминирования БМР-3М «Вепрь» на базе Т-90А применили на фронте. www1.ru, 2024-09-24.

<https://www1.ru/news/2024/09/24/novyeisuiu-masinu-razminirovaniya-bmr-3m-vepr-na-baze-t-90a-primenili-na-fronte.html>

WWW14 Фото обновленной российской разведывательной БРДМ-2МС засветились в сети [English version]. www1.ru, 2023-11-08.

<https://www1.ru/en/news/2023/11/08/foto-obnovlennoi-rossiiskoi-razvedvatelnoi-brdm-2-ms-zasvetilis-v-seti-amp.html>

WWW15 Redkiy dron Italmas zasvetilsya na foto, sposoben proleat 200 km. www1.ru, 2025-01-21.

<https://www1.ru/news/2025/01/21/redkiy-dron-italmas-zasvetilsia-na-foto-sposoben-proleat-200-km.html>

WWW16 Bronemashina Sudnogo dnaya: vysokozashchishchennuyu Ladogu na shassi T-80 zametili u rossiiskikh voennosluzhashchikh. www1.ru, 2024-03-27.

<https://www1.ru/news/2024/03/27/bronemashina-sudnogo-dnaya-vysokozashchishchennuyu-ladogu-na-sassi-itaiushchego-tank-a-t-80-zametili-u-ross-amp.html>

WWW17 Raschety ZALA (Lancet strikes at almost 100 km). www1.ru (quoting Russian MoD), 2026-01-22.

<https://www1.ru/en/news/2026/01/22/raschety-zala-minobrony.html>

WWW18 Omsktransmash presents floating transporter PTS-4 [Spanish edition]. www1.ru, 2024-06-06.

<https://www1.ru/en/news/2024/06/06/proizvoditel-tos-la-solnepek-predstavil-plavaiushii-transporter-pts-4-dlia-perevozki-kamaz-amp.html>

WWW19 Российские радисты займут вышки подвжных радиостанции Р-166-05 в Р-149АКШ в Таджикистане. www1.ru, 2025-08-07.

<https://www1.ru/news/2025/08/07/rossiiskie-radisty-zadeistvovali-podviznye-radiostancii-r-166-05-i-r-149aks-v-tadzhikistane-amp.html>

WWW22 Плавающий транспортёр ПТС-2 впервые попал на фото: получил мэнгал и комплекс РЗБ. www1.ru, 2025-10-09.

<https://www1.ru/news/2025/10/09/plavaiushii-transporter-pts-2-vpervye-popal-na-foto-poluchil-mangal-i-kompleks-reb.html>

WWW110 Newest R-416GM radio stations delivered to Central Military District. www1.ru (citing Russian MoD), 2023-12-08.

<https://www1.ru/news/2023/12/08/noviseie-radiostancii-strategicheskogo-upravleniya-r-416-gm-postupili-na-voruzenie-cvo-amp.html>

WWW111 Могослобная броня и защита «Реликт»: 23 танка Т-90М «Прорыв» отправили мотострелкам ЦВО. www1.ru, 2024-05-17.

<https://www1.ru/news/2024/05/17/mogosloinaia-bronia-i-zashhita-relikt-23-tank-a-t-90-m-proryv-otpravili-motostrelkam-cvo.html>

WWW112 Раскрыт потенциал модернизированной РСЗО «Торнадо-Г». www1.ru, 2024-12-08.

<https://www1.ru/news/2024/12/08/raskryt-potencial-modernizirovannoi-rszo-tornado-g-ekipaz-unichozit-protivnika-daze-ne-vykhodia-iz.html>

WWW113 Подорывае 100-метровый шнур: «Змей Горыныч» помог российским саперам проделат проходы в минных полях. www1.ru, relaying the Russian MoD press service, 2024-10-13.

<https://www1.ru/news/2024/10/13/podryvaet-100-metrovyy-shnur-zmei-gorynych-pomog-rossiiskim-saperam-prodelat-prohody-v-minnyx-polia.html>

WWW114 Умные поля «Zemledeleya»: rossiskim voyskam передan novyeshiy kompleks. www1.ru, 2023-10-13.

<https://www1.ru/news/2023/10/13/umnye-polia-zemledeleya-rossiiskim-voyskam-predan-noviseii-kompleks-distancionnogo-minirovaniya-amp.html>

ZAIF Чмо из себя представляет разведывательный БЛА ZALA 421-16E? Argumenty i Fakty (aif.ru).

https://aif.ru/society/army/chto_iz_sebya_predstavlyayet_razvedvatelnuy_bla_zala_421-16e

ZALA1 The ZALA Z-16 50th ... Brigade Varyag is coordinating strikes. ZALA Aero (Kalashnikov Group), undated (c. 2026).

<https://zala-aero.com/en/news/raschet-zala-z-16-50-obr-%c2%abvaryag%c2%bb-kontroluiet-udary-po-tylovoy-infrastrukture-vsu/>

ZALA2 An enemy UAV command post has been destroyed under the control of the ZALA Z-16 of the Rubicon Centre. ZALA Aero (Kalashnikov Group), undated (c. 2026).

<https://zala-aero.com/en/news/punkt-upravleniya-bpla-protivnika-unichozhen-pod-kontrolom-zala-z-16-czentra-%c2%abrubikon%c2%bb/>

ZALA3 Night reconnaissance with ZALA Z-16: Marines disrupted the rotation of AFU forces. ZALA Aero (Kalashnikov Group), 2025-01-29.

<https://zala-aero.com/en/news/nochnaja-razvedka-s-zala-z-16-morpehi-sorvali-rotaciju-sil-vsu/>

ZALA4 New generation ZALA Z-16. ZALA Aero (Kalashnikov Group), 2025-05-06.

<https://zala-aero.com/en/news/novoe-pokolenie-zala-z-16/>

ZALA6 Effective combination of ZALA Z-16 and Lancet against AFU equipment. ZALA Aero (Kalashnikov Group), 2024-12-06.

<https://zala-aero.com/en/news/effektivnaja-svjazka-zala-z-16-i-lancet-protiv-voennoj-tehniki-vsu/>

ZALA7 The Eyes of Our Army: Vesti report on the work of a ZALA Z-16 crew. ZALA Aero (Kalashnikov Group), 2026-01-23.

<https://zala-aero.com/en/news/glaza-nashej-armii-reportazh-vestej-o-rabote-raschyota-ra-zvedvatelnogo-bpla-zala-z-16/>

ZALA8 ZALA Lancet: UAV units have been established in artillery regiments and artillery brigades. ZALA Aero (Kalashnikov Group), 2024-12-07.

<https://zala-aero.com/en/news/zala-lancet-in-artpalkah-i-artbrigadah-sozdany-podrazdeleniya-bpla/>

ZALAP ZALA Z-16E. ZALA Aero (Kalashnikov Group), 2024-03-02 (modified 2026-04-23).

<https://zala-aero.com/en/product/zala-z-16/>

ZAXID На Донбасі вперше зафіксували новітній російський комплекс «Наводчук-2». zaxid.net, 2020-11.

https://zaxid.net/ua_donbasi_vperше_zafiksuvali_novitnij_rossiyskij_kompleks_navodchuk_2_n1510831/amp

ZR Nove osoby vezezkhody Plastun-SN. Za Rulem, 2023.

<https://www.zr.ru/content/news/948692-nove-osoby-vezdekhody-plastu/>

ZR2 СВОшный апгрейд: как БМД-2 получила защиту от дронов и современный модуль берег. За рулём, 2026.

<https://www.zr.ru/content/articles/982867-svozhnyj-apgrejd-kak-bmd2-pol/>

ZR3 Avtomobili v rogonakh (За рулём). Za Rulem.

<https://www.zr.ru/content/articles/966328-voennye-gruzoviki/>

ZR4 Компоновка — старая, машина — новая: современные версии «мотолыги». Za Rulem.

<https://www.zr.ru/content/articles/963485-mt-lb-motolyga/>

ZR5 Катюши нового поколения: Ураган и Торнадо работают в зоне СВО. За рулём, undated, after 2022.

<https://www.zr.ru/content/articles/942236-katyushi-novogo-pokoleniya-uraga>

ZR6 В зоне СВО появились бронеавтомобили нового поколения. Za Rulem, 2024.

<https://www.zr.ru/content/articles/941588-tajfun-vdy-broneavtomobil-no/>

ZR16 Geroi inzhenernykh voysk. Za Rulem.

<https://www.zr.ru/content/articles/616786-zashhitniki-otechestva/>

ZR24 «Zemledeliye»: kak miniruyut polya po tu storonu linii fronta. Za Rulem, 2024 or later (undated page).

<https://www.zr.ru/content/articles/962576-isd-m-zemledeliie/>

ZRRU БТР-МДМ Пакушка: без башни. За рулём (zr.ru), 2015.

<https://www.zr.ru/content/articles/824306-btr-mdm-rakushka-bez-bashni/>

ZV22 Snaryady samokhodnykh minometov 2S4 Tyulpan probivayut dazhe trekhmetrovye betonnye steny. TV Zvezda, 2022-12-19.

<https://tvzvezda.ru/news/20221219528-saWQS.html>

ZVEZ Bezoblastnoe Nebo-M v Krymu (Sokirko). TV Zvezda (Russian MoD media), 2017-04-24.

<https://tvzvezda.ru/news/201704240824-op30.htm>

ZVEZDA Использование БПЛА «Тахион» позволит повысить боевые возможности разведывательного подразделения ЦВО. TV Zvezda (MoD channel), 2020-12-28.

<https://tvzvezda.ru/news/202012281555-vnVpf.html>

Table 6. Picture credits

Photographs are reproduced under the licences named, from Wikimedia Commons. None is cropped or altered other than scaled. Licence texts: creativecommons.org/licenses.

1V12 artillery fire control complex [1V13, 1V14, 1V15, 1V16] p. 67
Simon Q from United Kingdom. CC BY 2.0 (<https://creativecommons.org/licenses/by/2.0>).
[https://commons.wikimedia.org/wiki/File:ACRV_M_1974\[21_1V15_17528060300\].jpg](https://commons.wikimedia.org/wiki/File:ACRV_M_1974[21_1V15_17528060300].jpg)

2A36 Giatint-B 152 mm towed gun p. 32
Ministry of Defense of Ukraine. CC BY-SA 2.0 (<https://creativecommons.org/licenses/by-sa/2.0>).
https://commons.wikimedia.org/wiki/File:Армудеяпстм_ВМС_Буковувала_практуични_стрельбу_з_зармат_«Гіацинт-Б»_11.jpg

2A65 Msta-B p. 32
One half 3544. Public domain (https://commons.wikimedia.org/wiki/File:2A65_Msta-B.jpg).
https://commons.wikimedia.org/wiki/File:2A65_Msta-B.jpg

2S35 Koalitsiya-SV p. 30
Kirill Borisenko. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
[https://commons.wikimedia.org/wiki/File:Koalitsiya-SV_during_the_«Armiya_2020»_exhibition_\[bright\].jpg](https://commons.wikimedia.org/wiki/File:Koalitsiya-SV_during_the_«Armiya_2020»_exhibition_[bright].jpg)

2S4 Tyulpan p. 31
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:ParkPatriot2015part5-53.jpg>

2S5 Giatint-S p. 29
One half 3544. Public domain (<https://commons.wikimedia.org/wiki/Template:PD-self>).
https://commons.wikimedia.org/wiki/File:2S5_Giatint-S.jpg

2S7 Pion / 2S7M Malka p. 31
Boevaya mashina. CC BY-SA 3.0 de (<https://creativecommons.org/licenses/by-sa/3.0/de/deed.en>).
https://commons.wikimedia.org/wiki/File:2S7M_Malka_ready_to_fire_at_ARMY-2022.JPG

96L6 all-altitude acquisition radar p. 42
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:96L6_radar_radar_-_100th_Anniversary_VVS-R_-_01.jpg

Aleut and Plastun articulated and light tracked carriers p. 19
Kirill Borisenko. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
[https://commons.wikimedia.org/wiki/File:GAZ-3344-20_«Aleut»_during_the_«Armiya_2023»_exhibition_\[front_view\].jpg](https://commons.wikimedia.org/wiki/File:GAZ-3344-20_«Aleut»_during_the_«Armiya_2023»_exhibition_[front_view].jpg)

ARK-1M Rys p. 43
ShinePhantom. CC BY-SA 3.0 (<https://creativecommons.org/licenses/by-sa/3.0>).
https://commons.wikimedia.org/wiki/File:1RL239_in_togliatti_Technical_museum.jpg

Artillery command vehicles 1V18/1V19, 1V119, 1V152, 1V1003 p. 67
Mike1979 Russia. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:1V119_Reostat_Army-2022_2022-08-20_2665.jpg

Barnaul-T p. 69
Doomych. Public domain (https://commons.wikimedia.org/wiki/File:9S932-1_at_the_MAKS-2011_%2801%29.jpg).
https://commons.wikimedia.org/wiki/File:9S932-1_at_the_MAKS-2011_011.jpg

BAT-2 p. 58
Kirill Borisenko. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:Track_laying_vehicle_BAT-2M_during_the_«Armiya_2021»_exhibition.jpg

BMD-1KSh-A p. 66
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:SpB-Museum-artillery-64.jpg>

BMD-2 p. 14
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:106th_Guards_Airborne_Division_1394-1.jpg

BMD-4M p. 13
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Alabino05042017-63.jpg>

BMD-T p. 18
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:6MO-T.jpg>

BMP-1 p. 11
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:467th_Guards_District_Training_Center_1414-01.jpg

BMP-1KSh command and staff vehicle p. 65
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:ParkPatriot2015part11-131.jpg>

BMP-2 p. 12
Nickel nitride. CCO (<http://creativecommons.org/publicdomain/zero/1.0/deed.en>).
<https://commons.wikimedia.org/wiki/File:Army-2020-174.JPG>

BMPT Terminator p. 11
Nickel nitride. CCO (<http://creativecommons.org/publicdomain/zero/1.0/deed.en>).
https://commons.wikimedia.org/wiki/File:Terminator_tank_support_combat_vehicle_on_Army_forum_2020-06-08.jpg

BMR-3M p. 59
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:Russia_Arms_Expo_2013_531-10.jpg

Borisoglebsk-2 p. 48
Russian Ministry Of Defense. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
<https://commons.wikimedia.org/wiki/File:Borisoglebsk-2.png>

BRDM-2 p. 22
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:BRDM-2_-_ETIF-2010_131.jpg

BREM and BTS armoured recovery vehicles p. 62
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Army2016-308.jpg>

BRM-1K p. 13
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:4thTankDivisionOpenDay17p1-51.jpg>

BTR-90 Rostok p. 21
MsgevansCO. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:BTR-90_Kubinka_tank_museum.jpg

BTR-D p. 16
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Army2016-249.jpg>

BTR-MDM Rakushka p. 15
KonanRus. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:5TP-MAM.jpg>

BTR-80 p. 20
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:BTR-82A.jpg>

Buk-M1 / Buk-M2 / Buk-M3, Buk-M2 TELAR p. 37
Boevaya mashina. CC BY-SA 3.0 (<https://creativecommons.org/licenses/by-sa/3.0>).
https://commons.wikimedia.org/wiki/File:Buk_TELAR_Moscow_2015.jpg

Buk-M1 / Buk-M2 / Buk-M3, Buk-M3 TELAR p. 37
Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
https://commons.wikimedia.org/wiki/File:Расчеты_3ПК_«Бук-М3»_отразили_удар_условного_противника_по_пункту_базирования_кораблей_Каспийского_флотаили_в_Давгеми

Buk-M1 / Buk-M2 / Buk-M3, 9S36M radar on mast p. 37
Boevaya mashina. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:9S36M_Armia-2018.jpg

CBRN reconnaissance vehicles (RKhM family) p. 62
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Army2016-495.jpg>

Decontamination vehicles [ARS-14, TMS-65, TZ-74 family] p. 63
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:ARS-14_decontamination_vehicle_-_Rehersal-Alabino-2012-04.jpg

Military earthmoving and trenching machines p. 60
Aanome. CCO (<https://creativecommons.org/publicdomain/zero/1.0/>).
[https://commons.wikimedia.org/wiki/File:BTM-3_\[side\]_-_UMMC_Museum_Complex.jpg](https://commons.wikimedia.org/wiki/File:BTM-3_[side]_-_UMMC_Museum_Complex.jpg)

Eleron-3SV p. 80
Nickel nitride. CCO (<https://creativecommons.org/publicdomain/zero/1.0/>).
<https://commons.wikimedia.org/wiki/File:Army-2020-242.JPG>

Forpost p. 83
Mike1979 Russia. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:Forpost_UAV_Army-2022_2022-08-20_2284.jpg

- FPV drones** p. 89
National Police of Ukraine. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
[https://commons.wikimedia.org/wiki/File:Russian_drones_with_explosive_in_Dnipropetrovsk_Oblast_2024-02-11_\(2\).jpg](https://commons.wikimedia.org/wiki/File:Russian_drones_with_explosive_in_Dnipropetrovsk_Oblast_2024-02-11_(2).jpg)
- Geran-2** p. 87
S. Hossein Alaei. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
[https://commons.wikimedia.org/wiki/File:National_Aerospace_Park_in_September_2023_\(13\).jpg](https://commons.wikimedia.org/wiki/File:National_Aerospace_Park_in_September_2023_(13).jpg)
- Gerbera** p. 88
State Border Guard Service of Ukraine. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
https://commons.wikimedia.org/wiki/File:tep6epa_konia.jpg
- BM-21 Grad 122 mm MRL** p. 26
Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
<https://commons.wikimedia.org/wiki/File:CombatArtilleryExercise2018-01.jpg>
- Granat-1 / Granat-4** p. 79
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:Granat-4_UAV_at_InterpoliteX-2015.jpg
- R-330B Mandat-B and R-934BMV** p. 47
Денис Абрамов (Denis Abramov). CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
https://commons.wikimedia.org/wiki/File:ECM_station_2.jpg
- IMR-2 / IMR-2M / IMR-3M combat armoured vehicle** p. 58
Kirill Borisenko. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:Obstacle_clearing_vehicle_IMR-3_during_the_Armya_2021_exhibition.jpg
- Infaua** p. 54
Игорь Ругенко. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
<https://commons.wikimedia.org/wiki/File:JointPeacekeeperExercise2018-20.jpg>
- IRM Zhuk** p. 57
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
<https://commons.wikimedia.org/wiki/File:Army2016demo-102.jpg>
- Iskander-M** p. 33
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
<https://commons.wikimedia.org/wiki/File:Army2016demo-073.jpg>
- Iskander-M**, launcher with 9M728 cruise missiles p. 33
Boevaya mashina. CC BY-SA 3.0 (<https://creativecommons.org/licenses/by-sa/3.0/>).
https://commons.wikimedia.org/wiki/File:Iskander-K_TEL_Army-2022.jpg
- KamAZ military trucks (4310 and Mustang family)** p. 75
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:KAMAZ-5350_military_truck_of_Russia.jpg
- Korsar** p. 83
Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
<https://commons.wikimedia.org/wiki/File:VictoryDayParade2018-26.jpg>
- Krasukha-2 / Krasukha-4, Krasukha-4** p. 46
The Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
https://commons.wikimedia.org/wiki/File:1RL257E_Krasukha-4.jpg
- Krasukha-2 / Krasukha-4, Krasukha-2** p. 46
The Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
https://commons.wikimedia.org/wiki/File:1L269_Krasukha-2.jpg
- KUB-BLA** p. 86
Mztourist. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:KUB-E_Joitering_munition_at_IDEX_2023.jpg
- Lancet** p. 85
Nickel nitride. CCO (<http://creativecommons.org/publicdomain/zero/1.0/deed.en>).
<https://commons.wikimedia.org/wiki/File:Army-2020-314.JPG>
- Leer-2** p. 51
Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
[https://commons.wikimedia.org/wiki/File:85Ya6_Leer-2_\(2\).jpg](https://commons.wikimedia.org/wiki/File:85Ya6_Leer-2_(2).jpg)
- Leer-3** p. 52
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
<https://commons.wikimedia.org/wiki/File:T82015ExhibitionP2-47.jpg>
- Leer-3, Orlan-10 UAV** p. 52
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
<https://commons.wikimedia.org/wiki/File:T82015ExhibitionP2-48.jpg>
- Other armoured cars and utility vehicles (family)** p. 24
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:BPM-97_-_InnovationDay2013part1-34.jpg
- Merlin-VR** p. 82
Ministry of Defence of the Russian Federation (Commons lists Unknown author). CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
[https://commons.wikimedia.org/wiki/File:Merlin-VR_\(1\).png](https://commons.wikimedia.org/wiki/File:Merlin-VR_(1).png)
- Mohajer-6** p. 84
Hamid Vakil. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
[https://commons.wikimedia.org/wiki/File:Mohajer-6_\(2\).jpg](https://commons.wikimedia.org/wiki/File:Mohajer-6_(2).jpg)
- Molniya** p. 88
Unknown author (Russian Ministry of Defence). CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
<https://commons.wikimedia.org/wiki/File:Molniya-2.png>
- Moskva-1** p. 47
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
<https://commons.wikimedia.org/wiki/File:MAKS2015part6-37.jpg>
- 2S19 Msta-S** p. 30
Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
https://commons.wikimedia.org/wiki/File:2S19M1_Msta-S.jpg
- MT-LB** p. 17
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
[https://commons.wikimedia.org/wiki/File:MT-LBVMK_\(1\).jpg](https://commons.wikimedia.org/wiki/File:MT-LBVMK_(1).jpg)
- Murmansk-BN** p. 45
Министерство обороны России. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
https://commons.wikimedia.org/wiki/File:Комплек_P36_Мурманск-БН.png
- Nebo-M, Nebo-M RLM-M** p. 41
Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
https://commons.wikimedia.org/wiki/File:Nebo-M_03.jpg
- Nebo-M, RLM-D and KU-RLK control cabin** p. 41
Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).
https://commons.wikimedia.org/wiki/File:Nebo-M_01.jpg
- Orion** p. 82
Boevaya mashina. CC BY-SA 3.0 (<https://creativecommons.org/licenses/by-sa/3.0/>).
https://commons.wikimedia.org/wiki/File:Kronstadt_Orion_left_side_ARMY-2022.jpg
- Orlan-10** p. 78
Mike1979 Russia. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:Orlan-10_UAV_Army-2022_2022-08-20_2287.jpg
- Palantin** p. 49
Ministry of Defence of Russia. CC BY 4.0 (also CC BY-SA 4.0) (<https://creativecommons.org/licenses/by/4.0/>).
https://commons.wikimedia.org/wiki/File:Погрузка_П36_Палантин_на_марше.jpg
- Pantsir-S1** p. 39
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
<https://commons.wikimedia.org/wiki/File:Army2016-384.jpg>
- 1875 Penicillin** p. 43
Allocer. CC BY-SA 3.0; GFDL (<https://creativecommons.org/licenses/by-sa/3.0/>).
https://commons.wikimedia.org/wiki/File:Aistnok_radar_maks2009.jpg
- PMP heavy folding pontoon bridge park** p. 56
Zinnsoldat. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:KrAZ-255_PMP_Pontoon_Bridge.jpg
- Polyana-D4** p. 68
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:Polyana-D4M1_-_TM2012ch3p1photo068.jpg
- PRP-4A Argun / PRP-4M Deytery** p. 68
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
[https://commons.wikimedia.org/wiki/File:Russia_Arms_Expo_2013_\(531-23\).jpg](https://commons.wikimedia.org/wiki/File:Russia_Arms_Expo_2013_(531-23).jpg)
- PTS-4 tracked amphibious transporter and PMM-2 self-propelled amphibious ferry** p. 57
Алексе́й Ку́маёв (Aleksey Kitaev). CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:PTS-4_RAE-2013_1.jpg
- PU-12M7** p. 69
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:Punkt_1pravleniya_PU-12M7.jpg
- R-142N Deimos-N** p. 71
Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0/>).
https://commons.wikimedia.org/wiki/File:R-142N_command_vehicle.jpg

R-1458M Chayka command and staff vehicle p. 65

VargaA. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:R-1458M_Berekfudo_2018_01.jpg

R-149 command and staff vehicles p. 64

Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
https://commons.wikimedia.org/wiki/File:Внобв_сформированную_мотострелковую_бригаду_ЦВО_оснастили_новыми_бронешасси.jpg

File:Внобв_сформированную_мотострелковую_бригаду_ЦВО_оснастили_новыми_бронешасси.jpg

R-166 Artek p. 70

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
[https://commons.wikimedia.org/wiki/File:R-166-0_5_radiostation_-_27th_Independent_Sevastopol_Guards_Motor_Rifle_Brigade_\(4\).jpg](https://commons.wikimedia.org/wiki/File:R-166-0_5_radiostation_-_27th_Independent_Sevastopol_Guards_Motor_Rifle_Brigade_(4).jpg)

R-416GM radio relay station p. 72

Владислав Белогруд (Vladislav Belogrud). CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
https://commons.wikimedia.org/wiki/File:Радиорелейный_комплекс_связи_R-416FM_«Гранит-М»_jpg

File:Радиорелейный_комплекс_связи_R-416FM_«Гранит-М»_jpg

R-439 Legenda-2 p. 73

Ministry of Defence of the Russian Federation (Министерство обороны Российской Федерации). CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
[https://commons.wikimedia.org/wiki/File:International_Mine_Action_Center_in_Syria_\(Aleppo\)_17.jpg](https://commons.wikimedia.org/wiki/File:International_Mine_Action_Center_in_Syria_(Aleppo)_17.jpg)

(Aleppo)_17.jpg

Redut-2US p. 74

Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
https://commons.wikimedia.org/wiki/File:Redut-2US_-_SignalExercise2017-04.jpg

https://commons.wikimedia.org/wiki/File:Redut-2US_-_SignalExercise2017-04.jpg

S-300V4, TELAR p. 36

Alexey Ivanov. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
<https://commons.wikimedia.org/wiki/File:Army-2018-05.jpg>

https://commons.wikimedia.org/wiki/File:Army-2018-05.jpg

S-300V4, radar p. 36

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
[https://commons.wikimedia.org/wiki/File:S-300V_-_Engineering_technologies_2012_\(11\).jpg](https://commons.wikimedia.org/wiki/File:S-300V_-_Engineering_technologies_2012_(11).jpg)

S-350 Vityaz, TELAR p. 36

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:S-350U_Vityaz_-_MAKS2013firstpix01.jpg

S-350 Vityaz, radar p. 36

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:MAKS2013firstpix04.jpg>

S-400 Triumph p. 35

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Alabino05042017-68.jpg>

S-400 Triumph, radar p. 35

vitaly kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:92N6A_radar_for_S-400.jpg

BM-30 Smerch p. 28

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Army2016demo-065.jpg>

Smoke generator vehicles (TDA-2K, TDA-3, TDM-2K) p. 63

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Army2016-499.jpg>

SNAR-10 / SNAR-10M1 Leopard p. 44

Yevgeny Kel. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
<https://commons.wikimedia.org/wiki/File:ArtilleryExercise2014-07.jpg>

Strela-10 and Osa short-range SAM p. 40

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Army2016demo-081.jpg>

Strelets p. 74

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:4thTankDivisionOpenDay17p2-07.jpg>

Supercam S350 p. 79

Supercamera. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:68C_Supercam_S350.jpg

T-54 / T-55 p. 10

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:ParkPatriot2015part11-019.jpg>

T-62 p. 9

Nickel nitride. CCO (<http://creativecommons.org/publicdomain/zero/1.0/deed.en>).
https://commons.wikimedia.org/wiki/File:T-62M_-_Army-2023-14.jpg

T-64A/B/BV p. 9

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:T-64BV_mod1.1987.jpg

T-80B/BV/BVM, T-80U p. 8

Mike1979 Russia. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:T-80BVM_MBT_Army-2022_2022-08-20_2400.jpg

T-90 p. 7

Nickel nitride. CCO (<http://creativecommons.org/publicdomain/zero/1.0/deed.en>).
https://commons.wikimedia.org/wiki/File:T-90M_01.jpg

https://commons.wikimedia.org/wiki/File:T-90M_01.jpg

Takhion p. 81

Nickel nitride. CCO (<https://creativecommons.org/publicdomain/zero/1.0/>).
<https://commons.wikimedia.org/wiki/File:Army-2020-265.JPG>

Tigr and Vodnik armoured cars p. 22

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:12aprilrehearsal_23.jpg

TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge, TMM-3 p. 55

Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
<https://commons.wikimedia.org/wiki/File:EngineerFieldRide2017-02.jpg>

TMM-3 truck-mounted scissors bridge and MTU-72 armoured vehicle-launched bridge, MTU-72 p. 55

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Army2016demo-097.jpg>

Tor p. 38

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Army2016demo-095.jpg>

Tornado-G p. 26

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Army2016demo-061.jpg>

TOS-1A Solntsepek p. 29

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:Army2016-487.jpg>

Tropospheric scatter radio stations (R-410-75, R-412A/AM, R-417) p. 72

ShinePhantom. CC BY-SA 3.0 (<https://creativecommons.org/licenses/by-sa/3.0>).
https://commons.wikimedia.org/wiki/File:Part_of_R-410-75_tropospheric_radio_relay_communication_station_Technical_museum_Togliatti-0480.JPG

Tunguska p. 40

Vitaly V. Kuzmin. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:T256M1_Tunguska-M1_SAM-system_at_MAKS-2011_-_side.jpg

https://commons.wikimedia.org/wiki/File:T256M1_Tunguska-M1_SAM-system_at_MAKS-2011_-_side.jpg

Typhoon family (Typhoon-K, Typhoon-U, Typhoon-VDV, Linza) p. 23

Alexey Vasilenko. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
<https://commons.wikimedia.org/wiki/File:KAMA3-63968.jpg>

https://commons.wikimedia.org/wiki/File:KAMA3-63968.jpg

UR-77 Meteorit p. 59

Dmitry Golovko. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
<https://commons.wikimedia.org/wiki/File:EngineeringFormula2017-01.jpg>

Uragan-1M p. 27

Ministerstwo Obrony Rosjskiej Federacji (Министерство обороны Российской Федерации / Ministry of Defence of the Russian Federation). CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
https://commons.wikimedia.org/wiki/File:Ураган-1М_-_Тренировка_курсантов_Михайловской_Военной_артиллерийской_академии_02.jpg

https://commons.wikimedia.org/wiki/File:Ураган-1М_-_Тренировка_курсантов_Михайловской_Военной_артиллерийской_академии_02.jpg

BM-27 Uragan p. 27

Unknown author. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
[https://commons.wikimedia.org/wiki/File:Uragan_MLRS_of_the_52nd_Artillery_Brigade_\(2\).png](https://commons.wikimedia.org/wiki/File:Uragan_MLRS_of_the_52nd_Artillery_Brigade_(2).png)

https://commons.wikimedia.org/wiki/File:Uragan_MLRS_of_the_52nd_Artillery_Brigade_(2).png

Ural-6370 / 63706 / 532301 trucks (newer Ural family) p. 76

Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
https://commons.wikimedia.org/wiki/File:Урал_Торнадо-У.jpg

https://commons.wikimedia.org/wiki/File:Урал_Торнадо-У.jpg

Uran-6 p. 60

Kirill Borisenko. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
[https://commons.wikimedia.org/wiki/File:MultiFunctional_robotic_mine_sweeping_system_\"Uran-6\"_during_the_\"Armia_2020\"_exhibition.jpg](https://commons.wikimedia.org/wiki/File:MultiFunctional_robotic_mine_sweeping_system_\)

ZALA 421-16E p. 77

Mike1979 Russia. CC BY-SA 4.0 (<https://creativecommons.org/licenses/by-sa/4.0>).
https://commons.wikimedia.org/wiki/File:ZALA_421-16E_Army-2022_2022-08-20_2390.jpg

https://commons.wikimedia.org/wiki/File:ZALA_421-16E_Army-2022_2022-08-20_2390.jpg

Zastava p. 81

Ministry of Defence of the Russian Federation. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).
https://commons.wikimedia.org/wiki/File:Bird-Eye_400.jpg

https://commons.wikimedia.org/wiki/File:Bird-Eye_400.jpg

ISDM Zemledeliye remote mine-laying system; GMZ-3 tracked minelayer p.

61

Ministry of Defense of Russia. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).

https://commons.wikimedia.org/wiki/File:Engineering_reconnaissance_and_firing_complex_Zemledeliye_on_the_march.jpg

R-330Zh Zhitel p. 50

Denis Abramov. CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>).

https://commons.wikimedia.org/wiki/File:R-330Zh_Zhitel_jammer.jpg

Zoopark-1 / Zoopark-1M p. 42

Boevaya mashina. CC BY-SA 3.0 de (<https://creativecommons.org/licenses/by-sa/3.0/de/deed.en>).

https://commons.wikimedia.org/wiki/File:1L219_Zoopark-1_radar_Tula_2016.JPG